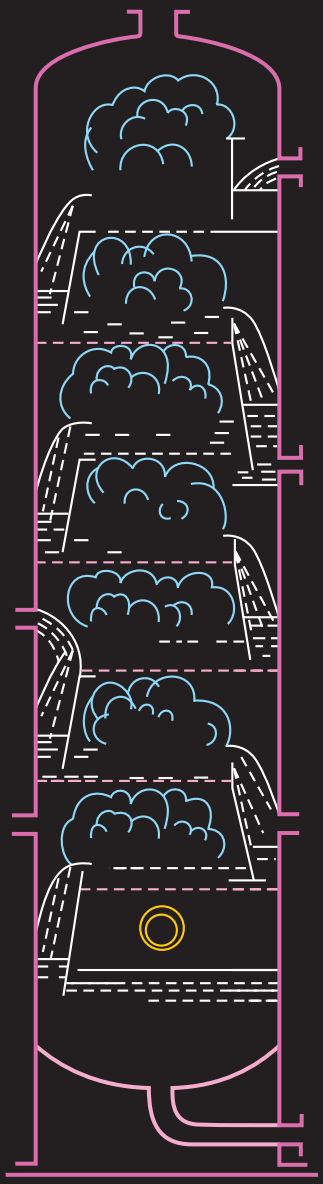
Eastern Economy Edition



Mass Transfer

Theory and Practice

N. Anantharaman
K.M. Meera Sheriffa Begum

PHI

MASS TRANSFER
Theory and Practice

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N. Anantharaman and K.M. Meera Sheriffa Begum

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Dedicated to my Mother

—N. Anantharaman

Dedicated to my Parents

—K.M. Meera Sheriffa Begum

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PREFACE

A chemical engineer is involved in the design and operation of a process plant. He has to handle the operations involving changes in composition of solutions, solids and gases. A thorough knowledge on the phase separation techniques will be very useful in handling operations involving separation of components from a reaction product mixture. They have to deal with gas–liquid, gas–solid, liquid–liquid and liquid–solid operations depending on the nature of the components to be separated. In this regard a chemical engineer has to equip himself/herself with the application of various mass transfer operations.

These operations are of great importance in the process industry as it has a direct impact on the cost of final product. An industrial operation always aims to produce a product at minimum cost of operations. Hence, it is imperative that a chemical engineer or a process engineer should have sound knowledge about the basics of mass transfer and its application. This book will certainly enable one to acquire sufficient knowledge about mass transfer operations and face the challenges ahead. The book contains twelve chapters and each one deals with the concepts/theories of mass transfer and its applications.

The objective of this book is to teach a budding chemical engineer, the principles involved in analyzing a process and apply desired mass transfer operations to separate the components involved. This book deals with operations involving diffusion, interphase mass transfer, humidification, drying, crystallization, absorption, distillation, extraction, leaching and adsorption.

Chapter 1 is an introductory chapter which deals with the different mass transfer operations and the field of applications.

Chapter 2 is devoted to diffusional mass transfer operations. The topics discussed indicate the phenomenon of gas diffusion, liquid diffusion and solid diffusion. Use of empirical equations to estimate the diffusivity has also been discussed. Numerical examples have been given at the end of the chapter,

which will help the reader apply the concept of the theory discussed. Exercise problems will enable one to work independently and gain confidence.

Chapter 3 is more on the theoretical side and deals with mass transfer coefficients and interphase mass transfer. A chemical engineer has to handle flow systems. The rate of transfer depends on the nature of fluid motion. This chapter gives definitions on various forms of mass transfer co-efficient and their interrelations. A quantitative presentation has been made on the estimation of mass transfer coefficient in laminar flow condition. As far as the flow under turbulent conditions are concerned, empirical methods and the application of various theories of mass transfer to determine the mass transfer coefficient have also been discussed. A section on the analogies between the transfer operations of heat, mass and momentum has been included. This will enable the reader to understand the significance and determine the characteristics of one with the known characteristics of the other two transfer operations. A brief presentation has been made on the analysis of different types of operations. Numerical examples have been added at the end, applying the concepts discussed earlier in this chapter.

Chapter 4 gives glimpses of the commonly used various pieces of equipment for mass transfer operations.

Chapter 5 is on humidification. A good knowledge about humidification is essential to obtain conditioned air with specified temperature and humidity for applications like drying, conditioning of space, etc. Definitions of various terms associated with humidification operation and the use of psychometric chart have been discussed. A lucid presentation has been made on the principle of wet bulb thermometry and adiabatic saturation curves. Principles and steps involved in the design of a cooling tower along with the description of different types of cooling towers are also discussed. This chapter is concluded with a series of numerical examples using the concepts discussed. Exercise problems will enable one to apply the concepts of humidification and solve them independently.

Drying, one of the final operations in a product formation, is discussed in Chapter 6. This chapter begins with a presentation on the phenomena of drying, factors affecting drying operations and theories of moisture movement. A detailed analysis of continuous dryers has also been presented. The classification of equipment used for drying is made and commonly used dryers and their applications have also been discussed. Numerical examples involving the analysis of drying data, the effect of operating parameters and also on the estimation of drying time are presented. Exercise problems included at the end of the chapter will enable one to apply their knowledge gained.

Chapter 7 deals with crystallization. This chapter commences with classification of crystals, principle of crystallization and moves over to methods of achieving supersaturation. Theory of nucleation and crystal growth have been discussed in detail. Principles involved in the design of crystallizers are also included. Different industrial crystallizers have been discussed along with their application. Examples on the estimation of yield and design of

crystallizers have been included. Exercise problems given at the end will enable the reader to apply the knowledge gained in this chapter.

Chapter 8 deals with absorption operation. The topics discussed include the principle of absorption, estimation of solvent requirement for a specific operation, design of isothermal and adiabatic absorbers and design of packed towers. Numerical examples given at the end of the chapter will enable one to apply these concepts and get thorough with the theory discussed. The problems given in exercises will enable the reader to work themselves and get familiarized with the theory.

Chapter 9 is devoted to distillation operations. In most of the chemical industries this forms an essential operation. The topics discussed include the computation of VLE data, characteristics of VLE data, types of distillation, design of packed and plate distillation columns. The features of accessories like reboilers and condensers are also discussed. Numerical examples have been given at the end of the chapter, which will help the reader to apply the concepts of the theory discussed. Exercise problems will enable one to work independently and gain confidence.

Chapter 10 deals with liquid–liquid extraction operations. A chemical engineer, depending on the industry or the type of system, comes across a partially miscible or immiscible systems. This chapter includes the equilibria of ternary systems, the effect of temperature and pressure on the equilibrium. Principles involved in the crosscurrent and countercurrent operations have been discussed. The determination of the number of stages needed for a specific operation in both partially miscible and immiscible systems have been discussed. A brief presentation has been made on some of the commonly used equipment. Numerical examples have been added at the end using the concepts discussed earlier in this chapter.

Chapter 11 deals with leaching operations. This operation also comes under the basic classification of extraction but deals more precisely with solid–liquid operations. As a process engineer, one has to be very clear with the concepts of solid–liquid extraction as he/she comes across ore processing and solvent extraction of oil from oil seeds. This chapter deals with the various types of leaching operations and principles involved in leaching. Various types of equilibria and types of operations are also discussed. This chapter is concluded with a series of numerical examples using the concepts discussed earlier. Exercise problems will enable one to apply the concepts of leaching and solve them independently.

Adsorption is one of the most important unit operations in the process industry, especially when we have to remove colour, odour and other impurities present in a very small quantity. This is discussed in Chapter 12 which is the last chapter in this book. Details on preparation of adsorbents, adsorption isotherms and different types of adsorption have been discussed. Numerical examples involving the analysis of basic data, and estimation of number of stages or the amount of adsorbent needed for a specific operation for different types of operations have been discussed. Exercise problems included at the end of the chapter will enable one to apply the knowledge gained.

We strongly feel that once the student becomes well conversant with the various topics discussed in this book, he/she will gain sufficient knowledge to face the problems that he/she is likely to encounter in the academic environment and industry.

The book is designed for a two-semester programme as a four-credit course and written as per the syllabus on Mass Transfer of most of the universities in India.

N. Anantharaman
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N. Anantharaman
K.M. Meera Sheriffa Begum

1

INTRODUCTION TO MASS TRANSFER

1.1 INTRODUCTION

A number of unit operations are carried out in Chemical Engineering applications which do not involve chemical reactions. These operations are carried out for separating either a component by mechanical means like screening, filtration and salting or increasing its concentration in a mixture. The latter is called mass transfer operation.

Frequently, these mass transfer operations are used for the separation of a product from the by-products formed and also from the unreacted raw materials. The separation technique plays a vital role in fixing the cost of final product.

1.2 CLASSIFICATION OF MASS TRANSFER OPERATIONS

It is classified as gas-liquid, liquid-liquid and fluid-solid operations.

1.2.1 Gas–Liquid

Absorption: Transfer of a solute from a gas mixture to a solvent is known as absorption. For example, (i) removal of ammonia gas from by-product coke ovens using water, (ii) removal of H_2S from naturally occurring hydrocarbon gases by alkali solutions.

Desorption: This is reverse of absorption, i.e. removal of a solute in a solution using a gas. For example, removal of NH_3 from NH_3-water solution using air.

Humidification: Transfer of a liquid to a gas phase containing one or more components by contacting dry gas with pure liquid is known as humidification.

Dehumidification: Transfer of a vapour component from a gas-vapour mixture to a liquid phase by contacting them is known as dehumidification. For example, transferring water vapour from air-water vapour mixture to liquid water.

Distillation: Method of separating the components in a liquid mixture by the distribution of substances between a gas and a liquid phase is known as distillation. The method of separation depends on their relative volatility and applied to cases where all the components are present in both the phases. Here a new phase is created from the original solution itself. For example, separation of petroleum fractions by the application of heat, separation of high boiling water insoluble mixtures using steam.

1.2.2 Liquid–Liquid

Extraction: Separation of a component (solute) from a liquid mixture using another insoluble or partially miscible solvent is known as extraction. The separation depends on the distribution of solute between the two phases based on its physico-chemical characteristics. The two phases are solvent rich phase (extract) and residual liquid phase (raffinate). For example, (i) separation of acetic acid from acetic acid-water mixture using isopropyl ether as solvent, (ii) separation of dioxane from waterdioxane solution using benzene as solvent.

1.2.3 Solid–Liquid/Gas

Leaching: Separating a soluble solute from a solid mixture by contacting it with a solvent is known as leaching. For example, (i) separation of oil from oil seeds using hexane, (ii) separation of sugar from sugar beets using hot water, (iii) removal of copper from its ore using sulphuric acid.

Adsorption: Adsorption involves contact of solid with either a liquid or a gaseous mixture in which a specific substance from the mixture concentrates on the solid surface. For example, (i) removal of colour from solutions using activated carbon, (ii) removal of moisture from air by silica gel.

Desorption: It is the reverse of adsorption operation.

Drying: Drying refers to the removal of moisture from a substance. For example, (i) removal of water from a cloth, wood or paper, (ii) removal of water from solution (manufacture of spray dried milk).

Crystallization: The process of forming solid particles within a homogeneous phase is called crystallization. For example, (i) the homogenous phase could be a vapour as in the formation of snow, (ii) the formation of crystals of sugar from a concentrated sugar solution.

2

DIFFUSION

2.1 INTRODUCTION

Separation of components in a mixture is achieved by contacting it with another insoluble phase. When transfer of the component from one phase to the other occurs due to concentration gradient, the phenomenon is called diffusion. The diffusion stops once equilibrium is attained. There are two types of diffusion,

1. Molecular diffusion
2. Eddy diffusion or Turbulent diffusion

2.2 MOLECULAR DIFFUSION AND EDDY DIFFUSION

Molecular diffusion can be defined as the movement of individual molecules in a highly zigzag manner through another fluid. The movement of molecules is imagined to be in a straight line at uniform velocity. However, the velocity and direction change when they are bombarded with other molecules. Molecular diffusion can also be called as Random-Walk process since the molecular movement is in a random path.

The phenomenon of molecular diffusion can be explained by a simple illustration, i.e. if a coloured solution is introduced in a pool of water, it begins slowly to diffuse into the entire liquid which is termed as molecular diffusion. To enhance its rate of mixing, a mechanical agitation is provided and this will cause a turbulent motion. This method of mass transfer is known as eddy or turbulent or convective diffusion.

2.3 DIFFUSIVITY OR DIFFUSION COEFFICIENT

Diffusion mainly depends upon the concentration gradient. In other words, the driving force for diffusion to occur is concentration gradient. This mass transfer

phenomenon is defined by Fick's first law of diffusion, which states that *molar flux is directly proportional to the concentration gradient.* Mathematically,

$$J_A = -D_{AB}\left(\frac{\partial C_A}{\partial Z}\right) = -C\,D_{AB}\left(\frac{\partial x_A}{\partial Z}\right) \tag{2.1}$$

where J_A is molar flux in moles/(area)(time), D_{AB} is diffusion coefficient or diffusivity in area/time, $\partial C_A/\partial Z$ is concentration gradient, C is molar concentration of constituents A and B in moles/vol. and x_A is mole fraction of A in the mixture. The –ve sign indicates the drop in concentration with respect to distance (the movement from high concentration to low concentration).

Consider two gases A and B of equal volume placed in two boxes connected by a tube and maintained at a constant total pressure. Now molecular diffusion of both gases occurs. Since the total pressure P remains constant throughout the process, the net moles of A diffused in one direction must be equal to the net moles of B diffused in opposite direction. So,

$$J_A = -J_B \tag{2.2}$$

Since the pressure is constant,

$$P = p_A + p_B = \text{constant} \tag{2.3}$$

and

$$C = C_A + C_B = \text{constant} \tag{2.4}$$

Differentiating Eq. (2.4) on both the sides,

$$dC_A = -dC_B \tag{2.5}$$

Writing Fick's law for component B,

$$J_B = -D_{BA}\frac{\partial C_B}{\partial Z} \tag{2.6}$$

Substituting for flux in Eq. (2.2) gives

$$-D_{AB}\left(\frac{\partial C_A}{\partial Z}\right) = -(-)D_{BA}\frac{\partial C_B}{\partial Z} \tag{2.7}$$

Substituting Eq. (2.5) in Eq. (2.7) and on simplification

$$D_{AB} = D_{BA} \tag{2.8}$$

This shows that the diffusivity is same for diffusion of A in B or B in A.

2.4 STEADY STATE MOLECULAR DIFFUSION IN FLUIDS

In the above discussion, we considered Fick's law for diffusion in a stationary fluid, i.e. there is no convective flow or bulk flow of the mixture. A general expression for flux N_A will consider the whole fluid moving in bulk with its

average molar velocity and its diffusional flux. Hence, the molar flux N_A can be expressed as the sum of molar average velocity and diffusional flux (J_A)

$$N_A = (N_A + N_B)\, x_A - D_{AB}\left(\frac{\partial C_A}{\partial Z}\right) \tag{2.9}$$

For steady state molecular diffusion between two gases A and B, the net flux is given by

$$N = N_A + N_B \tag{2.10}$$

Applying Eq. (2.9) to the case of diffusion in Z direction between the diffusional path Z_1 and Z_2, where the concentrations are C_{A1} and C_{A2} respectively. Equation (2.9) can be also written as,

$$N_A = (N_A + N_B)\frac{C_A}{C} - D_{AB}\left(\frac{\partial C_A}{\partial Z}\right), \text{ since } x_A = \frac{C_A}{C} \tag{2.11}$$

Rearranging the above Eq. (2.11) and integrating we get

$$\int_{C_{A1}}^{C_{A2}} \frac{-dC_A}{N_A C - C_A(N_A + N_B)} = \frac{1}{CD_{AB}} \int_{Z_1}^{Z_2} dZ \tag{2.12}$$

$$\left(\frac{CD_{AB}}{N_A + N_B}\right) \ln\left\{\left[C_{A2} - \frac{C\left(\frac{N_A}{N_A + N_B}\right)}{C_{A1}}\right] - C\left(\frac{N_A}{N_A + N_B}\right)\right\} = Z_2 - Z_1 = Z \tag{2.13}$$

or

$$N_A = \left(\frac{N_A}{N_A + N_B}\right)\left(\frac{CD_{AB}}{Z}\right) \ln\left[\frac{\left(\frac{N_A}{N_A + N_B}\right) - \frac{C_{A2}}{C}}{\left(\frac{N_A}{N_A + N_B}\right) - \frac{C_{A1}}{C}}\right] \tag{2.14}$$

2.4.1 Molecular Diffusion in Gases

It is more convenient to use ideal gas law for gaseous mixtures. Hence,

$$\frac{C_A}{C} = \frac{p_A}{P_t} = y_A \tag{2.15}$$

where p_A is the partial pressure of component A, P_t is the total pressure and y_A is mole fraction of component A. Further,

$$C = \frac{n}{V} = \frac{P_t}{RT} \tag{2.16}$$

Now substituting Eq. (2.16) in Eq. (2.14) gives

$$N_A = \left[\frac{N_A}{N_A + N_B}\right]\left[D_{AB}\frac{P_t}{RTZ}\right]\ln\left[\frac{\left\{\left(\frac{N_A}{N_A + N_B}\right)P_t - p_{A2}\right\}}{\left\{\left(\frac{N_A}{N_A + N_B}\right)P_t - p_{A1}\right\}}\right] \tag{2.17}$$

or

$$N_A = \left[\frac{N_A}{N_A + N_B}\right]\left[D_{AB}\frac{P_t}{RTZ}\right]\ln\left[\frac{\frac{N_A}{N_A + N_B} - y_{A2}}{\frac{N_A}{N_A + N_B} - y_{A1}}\right] \tag{2.18}$$

2.4.1.1 Case 1—Steady state diffusion of gas A through a stagnant gas B

In this case,

N_A = constant and $N_B = 0$,

$$\text{Hence, } \frac{N_A}{N_A + N_B} = 1 \tag{2.19}$$

Substituting Eq. (2.19) in Eq. (2.17) gives

$$N_A = \left(\frac{D_{AB}P_t}{RTZ}\right)\ln\left[\frac{P_t - p_{A2}}{P_t - p_{A1}}\right] \tag{2.20}$$

Since $P_t - p_{A2} = p_{B2}$, $P_t - p_{A1} = p_{B1}$, $p_{B2} - p_{B1} = p_{A1} - p_{A2}$.

Equation (2.20) becomes

$$N_A = \left(\frac{D_{AB}P_t}{RTZ}\right)\left[\frac{p_{A1} - p_{A2}}{p_{B2} - p_{B1}}\right]\ln\left(\frac{p_{B2}}{p_{B1}}\right) \tag{2.21}$$

Let,

$$p_{B,M} = \frac{p_{B2} - p_{B1}}{\ln\left(\frac{p_{B2}}{p_{B1}}\right)} \tag{2.22}$$

Then

$$N_A = \left[\frac{D_{AB}\,P_t}{RTZ\,p_{B,M}}\right](p_{A1} - p_{A2}) \tag{2.23}$$

or

$$N_A = \left[\frac{D_{AB}P_t}{RTZ}\right]\ln\left[\frac{1 - y_{A2}}{1 - y_{A1}}\right] \tag{2.24}$$

2.4.1.2 Case 2—Equimolar counter diffusion

In this case, $N_A = -N_B$. Then Eq. (2.17) becomes indeterminate. Hence, we can consider the general expression for flux as given in Eq. (2.9)

$$N_A = (N_A + N_B)\, x_A - D_{AB} \left(\frac{\partial C_A}{\partial Z} \right)$$

or

$$N_A = - D_{AB} \left(\frac{\partial C_A}{\partial Z} \right) \tag{2.25}$$

Integrating Eq. (2.25) between the respective limits

$$N_A \int_{Z_1}^{Z_2} dZ = - D_{AB} \int_{C_{A1}}^{C_{A2}} dC_A \tag{2.26}$$

$$N_A = \left(\frac{-D_{AB}}{Z} \right) (C_{A2} - C_{A1}) \quad \text{where } Z = Z_2 - Z_1 \tag{2.27}$$

or

$$N_A = \left(\frac{D_{AB}}{RTZ} \right) (p_{A1} - p_{A2}) \tag{2.28}$$

2.4.1.3 Case 3—Steady state diffusion in multicomponent mixtures

For multicomponent mixtures, effective diffusivity ($D_{A,M}$)can be determined by using

$$D_{A,M} = \frac{N_A - y_A \sum_{i=A}^{n} N_i}{\sum_{i=A}^{n} \frac{1}{D_{Ai}} (y_i N_A - y_A N_i)} \tag{2.29}$$

where D_{Ai} are the binary diffusivities. Here $D_{A,M}$ may vary considerably from one end of the diffusion path to the other, but a linear variation with distance can be assumed. For this situation, assume all but one component is stagnant, then Eq. (2.29) becomes,

$$D_{A,M} = \frac{1 - y_A}{\sum_{i=B}^{n} y_i / D_{Ai}} = \frac{1}{\sum_{i=B}^{n} y_i / D_{Ai}} \tag{2.30}$$

where y_i is the mole fraction of component i on an A-free basis.

Substituting $D_{A,M}$ instead of $D_{A,B}$ in Eqs. (2.23) and (2.28), the mass transfer rate for multicomponent mixtures can be determined.

2.4.2 Diffusivity Prediction in Gases

Diffusion coefficient is a significant parameter which depends upon temperature, pressure and composition of the components. Diffusivity can be determined experimentally. For some of the systems it is given in Table 2.1 (more data is

Table 2.1 Diffusivities of gases at standard atmospheric pressure, 101.3 kN/m^2

System	*Temperature*, °C	*Diffusivity*, m^2/s × 10^5
H_2-CH_4	0	6.25
O_2-N_2	0	1.81
$CO-O_2$	0	1.85
CO_2-O_2	0	1.39
Air–NH_3	0	1.98
Air–H_2O	25.9	2.58
	59.0	3.05
Air–ethanol	0	1.02
Air–*n*-Butanol	25.9	0.87
	59.0	1.04
Air–ethyl acetate	25.9	0.87
	59.0	1.06
Air–aniline	25.9	0.74
	59.0	0.90
Air–chlorobenzene	25.9	0.74
	59.0	0.90
Air–toluene	25.9	0.86
	59.0	0.92

available in literature). In some cases, it is very difficult to determine experimentally. Hirschfelder-Bird-Spotz developed an empirical relation to determine the diffusivity for mixtures of non-polar or a polar with non-polar gas.

$$D_{AB} = \frac{\left[10^{-4}\left(1.084 - 0.249\sqrt{\frac{1}{M_A}+\frac{1}{M_B}}\right)T^{3/2}\sqrt{\frac{1}{M_A}+\frac{1}{M_B}}\right]}{\left[P_t(r_{AB})^2\, f\left(\frac{KT}{\varepsilon_{AB}}\right)\right]} \tag{2.31}$$

where

D_{AB} is the diffusivity, m^2/s

T is the absolute temperature, K

M_A, M_B is the molecular weight of A and B respectively, kg/kmol

P_t is the absolute pressure, N/m^2

r_{AB} is the molecular separation at collision = $\frac{r_A + r_B}{2}$, nm

ε_{AB} is the energy of molecular attraction = $\sqrt{\varepsilon_A\, \varepsilon_B}$

K is the Boltzmann's constant

$f\left(\frac{KT}{\varepsilon_{AB}}\right)$ is the collision function given by Fig. 2.1

The values of r and ε such as those listed in Table 2.2 can be calculated from other properties of gases such as viscosity. They can also be estimated empirically by

$$r = 1.18\, v^{1/3} \tag{2.32}$$

$$\varepsilon/K = 1.21\, T_b \tag{2.33}$$

where v is the molal volume of liquid at normal boiling point, m^3/kmol and T_b is the normal boiling point, K.

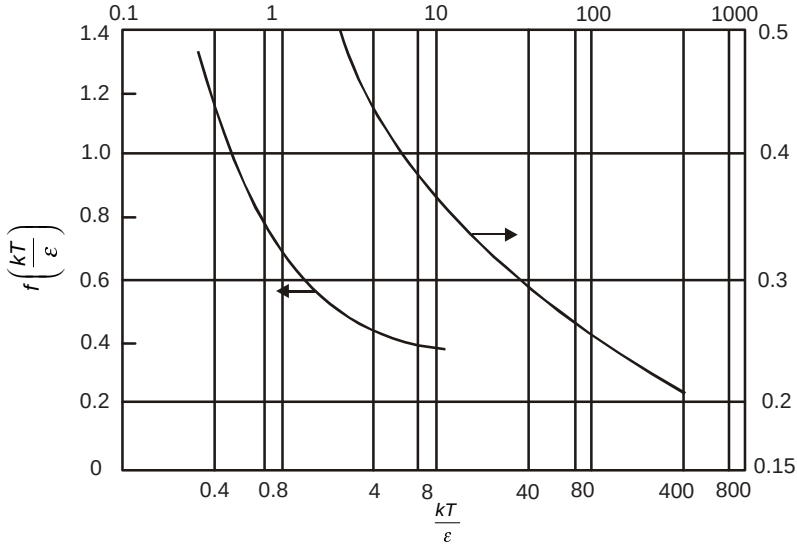


Fig. 2.1 Collision function.

Table 2.2 Force constants of gases as determined from viscosity data

Gas	ε/K, K	r, nm
Air	78.6	0.3711
CCl_4	322.7	0.5947
CH_3OH	481.8	0.3626
CH_4	148.6	0.3758
CO	91.7	0.3690
CO_2	195.2	0.3941
CS_2	467	0.4483
C_2H_6	215.7	0.4443
C_3H_8	237.1	0.5118
C_6H_6	412.3	0.5349
Cl_2	316	0.4217
HCl	344.7	0.3339
He	10.22	0.2551
H_2	59.7	0.2827
H_2O	809.1	0.2641
H_2S	301.1	0.3623
NH_3	558.3	0.2900
NO	116.7	0.3492
N_2	71.6	0.3798
N_2O	232.4	0.3828
O_2	106.7	0.3467
SO_2	335.4	0.4112

2.4.3 Molecular Diffusion in Liquids

In the case of diffusion in liquids, C and D_{AB} may vary considerably with respect to process conditions. Hence, Eq. (2.14) can be modified to,

$$N_A = \left(\frac{N_A}{N_A + N_B}\right)\left(\frac{D_{AB}}{Z}\right)\left(\frac{\rho}{M}\right)_{av} \ln\left[\frac{\frac{N_A}{N_A + N_B} - x_{A2}}{\left(\frac{N_A}{N_A + N_B}\right) - x_{A1}}\right] \tag{2.34}$$

where ρ is solution density and M is solution molecular weight.

2.4.3.1 Case 1—Diffusion of liquid A through a stagnant liquid B

In this case,

$N_B = 0$ and N_A = constant. Hence,

$$N_A = \left(\frac{D_{AB}}{Z x_{B,M}}\right)\left(\frac{\rho}{M}\right)_{av} (x_{A1} - x_{A2}) \tag{2.35}$$

where

$$x_{B,M} = \left[\frac{x_{B2} - x_{B1}}{\ln\left(\frac{x_{B2}}{x_{B1}}\right)}\right] \tag{2.36}$$

or

$$N_A = \left(\frac{D_{AB}}{Z}\right)\left(\frac{\rho}{M}\right)_{av} \ln\left[\frac{1 - x_{A2}}{1 - x_{A1}}\right] \tag{2.37}$$

2.4.3.2 Case 2—Equimolar counter-diffusion

In this case,

$$N_A = -N_B$$

Hence,

$$N_A = \left(\frac{D_{AB}}{Z}\right)(C_{A1} - C_{A2}) = \left(\frac{D_{AB}}{Z}\right)\left(\frac{\rho}{M}\right)_{av} (x_{A1} - x_{A2}) \tag{2.38}$$

2.4.4 Diffusivity Prediction in Liquids

Diffusivity has the dimension of area/time similar to that of gases. A few typical data are listed in Table 2.3 and more are available in literature. For some cases such as dilute solution of non-electrolytes, the diffusivity can be estimated by using Wilke and Chang empirical correlation.

$$D_{AB} = \frac{(117.3 \times 10^{-18})(\varphi M_B)^{0.5} T}{\mu v_A^{0.6}} \tag{2.39}$$

where

D_{AB} is the diffusivity of A in very dilute solution in solvent B, m^2/s

M_B is the molecular weight of solvent, kg/kmol.

T is the absolute temperature, K

μ is the solution viscosity, kg/m-s

v_A is the solute molal volume at normal boiling point, m^3/kmol.

= 0.0756 for water as solute.

φ is the association factor for solvent.

= 2.26 for water as solvent

= 1.90 for methanol as solvent

= 1.50 for ethanol as solvent

= 1.00 for unassociated solvents, e.g. benzene and ethyl ether.

Table 2.3 Liquid diffusivities

Solute	*Solvent*	*Temperature*, K	*Solute concentration*, kmol/m^3	*Diffusivity*, $m^2/s \times 10^9$
Cl_2	Water	289	0.12	1.26
HCl	Water	273	9	2.7
			2	1.8
		283	9	3.3
			2.5	2.5
		289	0.5	2.44
NH_3	Water	278	3.5	1.24
		288	1.0	1.77
CO_2	Water	283	0	1.46
		293	0	1.77
NaCl	Water	291	0.05	1.26
			0.2	1.21
			1.0	1.24
			3.0	1.36
			5.4	1.54
Methanol	Water	288	0	1.28
Acetic acid	Water	285.5	1.0	0.82
			0.01	0.91
		291	1.0	0.96
Ethanol	Water	283	3.75	0.50
			0.05	0.83
		289	2.0	0.90
n-Butanol	Water	288	0	0.77
CO_2	Ethanol	290	0	3.2
Chloroform	Ethanol	293	2.0	

The value of v_A can be estimated from the data of atomic volumes added together. Typical data on atomic and molecular volume is available in Table 2.4.

Table 2.4 Atomic and molecular volumes

Gas	*Atomic volume,* $m^3/katom \times 10^3$	*Gas*	*Molecular volume,* $m^3/kmol \times 10^3$
Carbon	14.8	H_2	14.3
Hydrogen	3.7	O_2	25.6
Chlorine	24.6	N_2	31.2
Bromine	27.0	Air	29.9
Iodine	37.0	CO	30.7
Sulphur	25.6	CO_2	34.0
Nitrogen	15.6	SO_2	44.8
In primary amines	10.5	NO	23.6
In secondary amines	12.0	N_2O	36.4
Oxygen	7.4	NH_3	25.8
In methyl esters	9.1	H_2O	18.9
In higher esters	11.0	H_2S	32.9
In acids	12.0	COS	51.5
In methyl ethers	9.9	Cl_2	48.4
In higher ethers	11.0	Br_2	53.2
Benzene ring subtract	15	I_2	71.5
Naphthalene ring subtract	30		

2.4.5 Pseudo Steady State Diffusion

In many mass transfer operations, one of the boundaries between the fluids may move with time. If the length of the diffusion path changes over a period of time, a pseudo steady state develops. Here, the molar flux is related to the amount of A leaving the liquid by,

Flux = rate of change of liquid level × molar concentration of A in liquid phase

$$N_{AZ} = \frac{dZ}{dt} \times C_{A,L} \tag{2.40}$$

$$N_{AZ} = \left(\frac{C\, D_{AB}}{Z x_{B,M}} \right) \times (x_{B2} - x_{B1}) = \left(\frac{dZ}{dt} \right) C_{A,L} \tag{2.41}$$

Integrating Eq. (2.41) between $t = 0$, $Z = Z_{t0}$ and $t = t$, $Z = Z_t$

$$\int_{Z_{t0}}^{Z_t} z dz = \left[\frac{C\, D_{AB} (x_{B2} - x_{B1})}{x_{B,M}\, C_{A,L}} \right] \int_0^t dt \tag{2.42}$$

After integration and simplification,

$$\frac{t}{Z_t - Z_{t0}} = \frac{x_{B,M}\, C_{A,L} (Z_t + Z_{t0})}{2\, C\, D_{AB} (x_{A1} - x_{A2})} \tag{2.43}$$

i.e.

$$\frac{t}{Z_t - Z_{t0}} = \frac{x_{B,M}\, C_{A,L}\, (Z_t + Z_{t0} + Z_{t0} - Z_{t0})}{2\, C\, D_{AB}\, (x_{A1} - x_{A2})} \tag{2.44}$$

i.e.

$$\frac{t}{Z_t - Z_{t0}} = \frac{x_{B,M} C_{A,L} (Z_t - Z_{t0})}{2\ C\ D_{AB} (x_{A1} - x_{A2})} + \frac{2\ Z_{t0} x_{B,M} C_{A,L}}{2\ C\ D_{AB} (x_{A1} - x_{A2})} \tag{2.45}$$

Equation (2.45) is of the form,

$$y = mx + C \tag{2.46}$$

$$\text{where } y = \frac{t}{Z_t - Z_{t0}} \text{ and } x = (Z_t - Z_{t0})$$

$$\text{Slope, } m = \frac{x_{B,M} C_{A,L}}{2\ C\ D_{AB} (x_{A1} - x_{A2})}$$

and

$$\text{Constant } C = \frac{Z_{t0}\ x_{B,M}\ C_{A,L}}{C\ D_{AB} (x_{A1} - x_{A2})}$$

Since Eq. (2.45) is linear, by plotting $t/(Z_t - Z_{t0})$ against $(Z_t - Z_{t0})$, from the slope of line, D_{AB} can be calculated, as the other parameters of Eq. (2.45) are all known. This equation is called as Winkelmann's relation.

2.5 DIFFUSION IN SOLIDS

Fick's law of diffusion can be applied to the system which is under steady state condition. It is applicable when diffusivity is independent of concentration and when there is no bulk flow. So, *the rate of diffusion of substance A per unit cross section of solid is proportional to the concentration gradient in the direction of diffusion.*

$$N_A = - D_A \left(\frac{dC_A}{dZ} \right) \tag{2.47}$$

where D_A is the diffusivity of A through the solid. When the diffusion is taking place through a flat slab of thickness Z, then Eq. (2.47) becomes

$$N_A = \frac{D_A (C_{A1} - C_{A2})}{Z} \tag{2.48}$$

Here C_{A1} and C_{A2} are concentrations at opposite sides of the slab. For solids of varying transfer area, the diffusional rate is given by,

$$W = N_A S_{av} = \frac{D_A S_{av} (C_{A1} - C_{A2})}{Z} \tag{2.49}$$

S_{av} is average mass transfer area of respective solid surfaces. Hence, for radial diffusion through a solid cylinder of inner and outer radii r_1 and r_2 respectively and its length l,

$$S_{av} = 2\pi r l \tag{2.50}$$

$$W = -D_A\, 2\pi r l \left(\frac{dC}{dr}\right) \tag{2.51}$$

On integrating

$$W \int_{r_1}^{r_2} \frac{dr}{r} = -D_A\, 2\pi l \int_{C_{A1}}^{C_{A2}} dC \tag{2.52}$$

or

$$W \ln\left(\frac{r_2}{r_1}\right) = -D_A 2\pi l (C_{A2} - C_{A1}) \tag{2.53}$$

or

$$W = \frac{[D_A\, 2\pi l\, (C_{A1} - C_{A2})}{\ln\left(\frac{r_2}{r_1}\right)}\left(\frac{r_2 - r_1}{r_2 - r_1}\right) \tag{2.54}$$

or

$$W = \frac{D_A S_{av} (C_{A1} - C_{A2})}{Z} \tag{2.55}$$

where $S_{av} = \dfrac{2\pi l (r_2 - r_1)}{\ln\left(\dfrac{r_2}{r_1}\right)}$ and $Z = (r_2 - r_1)$.

Similarly for radial diffusion through a spherical shell of inner and outer radii r_1 and r_2, the surface is

$$S_{av} = 4\pi r_1 r_2 \text{ and } Z = (r_2 - r_1) \tag{2.56}$$

2.5.1 Types of Solid Diffusion

The nature of solids and its interaction with the diffusing substance influence the rate of mass transfer. Different types of solid diffusion are discussed below.

2.5.1.1 Diffusion through polymers

Diffusion through polymeric membranes, e.g. gaseous separation through a membrane, mainly depends on the pressure gradient as the driving force. *Diffusion takes place from high pressure region to low pressure region*. A particular activation energy is needed for diffusion to take place and the temperature dependency of diffusivity is given by Arrhenius type relation,

$$D_A = D_o \exp\left[\frac{-H_D}{RT}\right] \tag{2.57}$$

where H_D is the energy of activation and D_o is a constant. For simple gases, D_A is independent of concentration but for permanent gases, diffusivity is strongly

dependent on solute concentration in the solid. The diffusional flux is given by

$$V_A = \frac{D_A S_A (p_{A1} - p_{A2})}{Z} \tag{2.58}$$

where

V_A is the diffusional flux, $cm^3 \cdot$ gas (STP)/cm^2s

D_A is the diffusivity of A, cm^2/s

p_A is the partial pressure of diffusing gas, cm Hg

S_A is the solubility coefficient, $cm^3 \cdot$ gas (STP)/cm^3 solid $\cdot$ cm Hg

Z is the thickness of polymeric membrane, cm

Permeability can be defined as

$$P = D_A S_A \tag{2.59}$$

where P is the permeability, cm^3 gas $\cdot$ (STP)/$cm^2 \cdot$ s (cm Hg/cm)

The solubility is related to concentration in SI units as,

$$c_A \text{ (kmol/m}^3 \text{ solid)} = Sp_A = 22.414 \tag{2.60}$$

and in CGS system as c_A (g mol/cm^3 solid) = Sp_A = 22414 (2.61)

2.5.1.2 Diffusion through crystalline solids

Solute nature and crystalline structure are the important parameters in this type of diffusion. Some of the mechanisms followed for diffusion through crystal geometry are given below:

1. Interstitial mechanism—Solute atoms diffuse from one interstitial site to the next in the crystal lattice.
2. Vacancy mechanism—If lattice sites are vacant, an atom in an adjacent site may jump into the vacant site.
3. Interstitialcy mechanism—A large atom occupying in an interstitial site pushes the neighbouring lattice into an interstitial position and moves into the vacancy produced.
4. Crowded-ion mechanism—An extra atom in a chain of close-packed atoms can displace several atoms in the line from their equilibrium position.
5. Diffusion along grain boundaries—Diffusion takes place in crystal interfaces and dislocations.

2.5.1.3 Diffusion in porous solids

The solid may be porous in nature such as adsorbents or membrane and the diffusion takes place either by virtue of concentration gradient or by hydrodynamic flow behaviour because of pressure difference. In steady state diffusion of gases, there are two types of diffusive movement, depending on the ratio of pore diameter d, to the mean free path of the gas molecules, λ.

If the ratio $d/\lambda > 20$, molecular diffusion predominates

$$N_A = \left(\frac{N_A}{N_A + N_B}\right)\left(\frac{D_{AB,eff} P_t}{RTZ}\right) \ln \left[\frac{\frac{N_A}{N_A + N_B} - y_{A2}}{\frac{N_A}{N_A + N_B} - y_{A1}}\right] \tag{2.62}$$

If $d/\lambda < 0.2$, the rate of diffusion is governed by the collisions of the gas molecules within the pore walls and follows Knudsen's law.

$$N_A = \frac{D_{K,A}\ (p_{A1} - p_{A2})}{RTl} \tag{2.63}$$

where

$D_{K,A}$ is the Knudsen diffusivity, cm^2/s

l is the length of the pore, cm

p_A is the partial pressure of diffusing substance, cmHg

Knudsen diffusivity can be determined by using an empirical relation,

$$D_{K,A} = \left(\frac{d}{3}\right)\left(\frac{8\, g_c\, RT}{\pi M_A}\right)^{1/2} \tag{2.64}$$

The mean free path λ can be estimated by

$$\lambda = \left(\frac{3.2\mu}{P_t}\right)\left(\frac{RT}{2\pi g_c M_A}\right)^{1/2} \tag{2.65}$$

If $0.2 < d/\lambda < 20$, both molecular and Knudsen diffusion take place

$$N_A = \left(\frac{N_A}{N_A + N_B}\right)\left(\frac{D_{AB,eff}\, P_t}{ZRT}\right) \ln\left[\frac{\left(\frac{N_A}{N_A + N_B}\right)\left(1 + \frac{D_{AB,eff}}{D_{KA,eff}}\right) - y_{A2}}{\left(\frac{N_A}{N_A + N_B}\right)\left(1 + \frac{D_{AB,eff}}{D_{KA,eff}}\right) - y_{A1}}\right] \tag{2.66}$$

2.5.2 Unsteady State Diffusion

Since solids are not readily transported, as fluids, unsteady state diffusional conditions arise more frequently in solids than in fluids. For unsteady state diffusion, Fick's second law is applied,

$$\frac{\partial C}{\partial t} = D_{AB}\left[\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2}\right] \tag{2.67}$$

WORKED EXAMPLES

1. Estimate the diffusivities of the following gas mixtures:
(a) Nitrogen—carbon dioxide, 1 Standard atm., 25ºC.
(b) Hydrogen chloride—air, 200 kN/m^2, 25ºC.

Solution.

(a) System: N_2 and CO_2 at 1 Standard atm., 25°C

Let A denote nitrogen and B denote carbon dioxide

$$r_A = 0.3798 \text{ nm}, \qquad r_B = 0.3941 \text{ nm}$$

$$r_{AB} = (0.3798 + 0.3941)/2 = 0.38695 \text{ nm}$$

$$\left\{\frac{\varepsilon}{K}\right\}_A = 71.4, \qquad \left\{\frac{\varepsilon}{K}\right\}_B = 195.2$$

$$\left\{\frac{\varepsilon}{K}\right\}_{AB} = \sqrt{[71.4 \times 195.2]} = 118.056$$

$$\frac{KT}{\varepsilon_{AB}} = \frac{298}{118.056} = 2.52$$

$$f\left(\frac{KT}{\varepsilon_{AB}}\right) = 0.5 \text{ (from Fig. 2.1)}$$

$$\sqrt{\left[\left(\frac{1}{M_A}\right)+\left(\frac{1}{M_B}\right)\right]} = \sqrt{\left[\left(\frac{1}{28}\right)+\left(\frac{1}{44}\right)\right]} = 0.242$$

$$D_{AB} = \frac{10^{-4}\left\{1.084 - 0.249\sqrt{\left[\left(\frac{1}{M_A}\right)+\left(\frac{1}{M_B}\right)\right]}\right\} T^{3/2} \sqrt{\left[\left(\frac{1}{M_A}\right)+\left(\frac{1}{M_B}\right)\right]}}{P_t (r_{AB})^2 f\left(\frac{KT}{\varepsilon_{AB}}\right)}$$

$$D_{AB} = \frac{10^{-4}\left\{1.084 - 0.249\sqrt{\left[\left(\frac{1}{M_A}\right)+\left(\frac{1}{M_B}\right)\right]}\right\} T^{3/2} \sqrt{\left[\left(\frac{1}{M_A}\right)+\left(\frac{1}{M_B}\right)\right]}}{P_t (r_{AB})^2 f\left(\frac{KT}{\varepsilon_{AB}}\right)}$$

$$= \frac{10^{-4}\{1.084 - (0.249 \times 0.242)\} \times (298)^{3/2} \times (0.242)}{1.013 \times 10^5 \times (0.38695)^2 \times 0.5}$$

$D_{AB} = 1.6805 \times 10^{-5}$ m^2/s **Ans.**

(b) System: HCl and Air at 200 kN/m^2, 25°C
Let A denote HCl and B denote air

$r_A = 0.3339$ nm, $r_B = 0.3711$ nm

$$r_{AB} = \frac{0.3339 + 0.3711}{2} = 0.3525 \text{ nm}$$

$$\left\{\frac{\varepsilon}{K}\right\}_A = 344.7, \qquad \left\{\frac{\varepsilon}{K}\right\}_B = 78.6$$

$$\left\{\frac{\varepsilon}{K}\right\}_{AB} = \sqrt{[344.7 \times 78.6]} = 164.6$$

$$\left(\frac{KT}{\varepsilon_{AB}}\right) = \frac{298}{164.6} = 1.81$$

$$f\left(\frac{KT}{\varepsilon_{AB}}\right) = 0.62 \text{ (from Chart of Fig. 2.1)}$$

$$\sqrt{\left[\left(\frac{1}{M_A}\right) + \left(\frac{1}{M_B}\right)\right]} = \sqrt{\left[\left(\frac{1}{36.5}\right) + \left(\frac{1}{29}\right)\right]} = 0.249$$

$$D_{AB} = \frac{10^{-4}\left\{1.084 - 0.249\sqrt{\left[\left(\frac{1}{M_A}\right) + \left(\frac{1}{M_B}\right)\right]}\right\} T^{3/2} \sqrt{\left[\left(\frac{1}{M_A}\right) + \left(\frac{1}{M_B}\right)\right]}}{P_t (r_{AB})^2 f\left(\frac{KT}{\varepsilon_{AB}}\right)}$$

$$D_{AB} = \frac{10^{-4}\{1.084 - (0.249 \times 0.249)\} \times (298)^{3/2} \times (0.249)}{200 \times 10^3 \times (0.3525)^2 \times 0.62}$$

$D_{AB} = 8.496 \times 10^{-6}$ m^2/s **Ans.**

2. Estimate the diffusivity of isoamyl alcohol ($C_5H_{12}O$) at infinite dilution in water at 288 K.

Solution.

Viscosity of water = 1.145 cp,

V_A (by Kopp's law) = $5 \times 0.0148 + 12 \times 0.0037 + 1 \times 0.0074$

= 0.1258 m^3/kmol

ϕ (Association parameter for solvent-water) = 2.26

$$D_{AB} = \frac{(117.3 \times 10^{-18})(\varphi M_B)^{0.5} T}{\mu V_A^{0.6}}$$

$$= \frac{(117.3 \times 10^{-18})(2.26 \times 18)^{0.5} \times 288}{(0.001145) \times (0.1258)^{0.6}}$$

$= 0.653 \times 10^{-9}$ m^2/s **Ans.**

3. The diffusivity of carbon tetrachloride, CCl_4 through oxygen O_2, was determined in a steady state Arnold evaporating cell. The cell, having a cross sectional area of 0.82 cm^2, was operated at 273 K and 755 mmHg pressure. The average length of the diffusion path was 17.1 cm. If 0.0208 cc of CCl_4 was evaporated in 10 hours of steady state operation, what should be the value of the diffusivity of CCl_4 through oxygen?

Solution.

Vapour pressure of CCl_4 at 273 K = 33 mm Hg
Density of liquid CCl_4 = 1.59 g/cm^3
Considering O_2 to be non-diffusing and with T = 273 K, P_t = 755 mm Hg, Z = 17.1 cm
0.0208 cc of CCl_4 is evaporating in 10 hours.

$$\text{i.e.} \quad \frac{0.0208 \times 1.59}{154 \times 10} = 2.147 \times 10^{-5} \text{ g mol/h}$$

$$\text{Flux } N_A = \frac{2.147 \times 10^{-5} \times 10^{-3}}{3600 \times 0.82 \times 10^{-4}} = 7.27 \times 10^{-8} \text{ kmol/m}^2 \text{ s}$$

$$N_A = \frac{D_{AB} P_t}{ZRT} \ln \left[\frac{(P_t - p_{A2})}{(P_t - p_{A1})} \right]$$

$$D_{AB} = \frac{N_A \times Z \times RT}{P_t \ln \left[\dfrac{P_t - p_{A2}}{P_t - p_{A1}} \right]}$$

$$= \frac{7.27 \times 10^{-8} \times 17.1 \times 10^{-2} \times 8314 \times 273}{\left[\dfrac{755}{760} \times 1.013 \times 10^5 \right] \ln \left[\dfrac{\dfrac{755}{760} \times 1.013 \times 10^5 - 0}{\dfrac{755}{760} \times 1.013 \times 10^5 - \dfrac{33}{760} \times 1.013 \times 10^5} \right]}$$

$D_{AB} = 6.355 \times 10^{-6}$ m^2/s **Ans.**

4. A crystal of copper sulphate $CuSO_4 \cdot 5H_2O$ falls through a large tank of pure water at 20ºC. Estimate the rate at which the crystal dissolves by calculating the flux of $CuSO_4$ from the crystal surface to the bulk solution. Molecular diffusion occurs through a film of water uniformly 0.0305 mm thick surrounding the crystal. At the inner side of the film, adjacent to the crystal surface, the concentration of $CuSO_4$ is 0.0229 mole fraction $CuSO_4$ (solution density = 1193 kg/m^3). The outer surface of the film is pure water. The diffusivity of $CuSO_4$ is 7.29×10^{-10} m^2/s. Temperature = 293 K. Molecular weight of $CuSO_4$= 160.

Solution.

$$Z = 0.0305 \times 10^{-3} \text{ m}$$

$$M_{av} = \frac{(0.0229 \times 160) + (0.9771 \times 18)}{1} = 21.2518$$

$$\left(\frac{\rho}{M}\right) = \frac{1193}{21.2518} = 58.136$$

For pure water,

$$\left(\frac{\rho}{M}\right)_2 = \frac{1000}{18} = 55.56$$

$$\left(\frac{\rho}{M}\right)_{av} = \frac{58.136 + 55.56}{2} = 56.848$$

$$D_{AB} = 7.29 \times 10^{-10} \text{ m}^2/\text{s}$$

Assuming water to be non-diffusing

$$N_A = \frac{D_{AB}\left(\frac{\rho}{M}\right)_{av} \ln\left[\frac{1 - x_{A2}}{1 - x_{A1}}\right]}{Z}$$

$$N_A = \left[\frac{7.29 \times 10^{-10}}{0.0305 \times 10^{-3}}\right] \times 56.848 \times \ln\left[\frac{1 - 0}{1 - 0.0229}\right]$$

$$N_A = 3.15 \times 10^{-5} \text{ kmol/m}^2 \text{ s.} \qquad \textbf{Ans.}$$

5. Alcohol vapour is diffusing through a layer of water vapour under equimolar counter diffusion at 35ºC and 1 atm. pressure. The molal concentration of alcohol on the two sides of the gas film (water vapour) 0.3 mm thick are 80% and 10% respectively. Assuming the diffusivity of alcohol–water vapour to be 0.18 cm^2/s, (i) calculate the rate of diffusion of alcohol and water vapour in kg/hr through an area of 100 cm^2 (ii) if the water vapour layer is stagnant, estimate the rate of diffusion of alcohol vapour.

Solution.

(i) Equimolar counter diffusion

$$T = (273 + 35) = 308 \text{ K}, \qquad P_t = 1 \text{ atm}$$

$$Z = 0.3 \text{ mm}, \qquad D_{AB} = 0.18 \times 10^{-4} \text{ m}^2/\text{s}$$

(Position 1)		moles	mol fraction
	Air	80	0.8
	Water	20	0.2
(Position 2)		moles	mol fraction
	Air	10	0.1
	Water	90	0.9

$$N_A = \frac{D_{AB}}{ZRT} \times [p_{A1} - p_{A2}]$$

$$N_A = \frac{D_{AB} \times P_t}{ZRT} [y_{A1} - y_{A2}]$$

$$N_A = \left[\frac{0.18 \times 10^{-4} \times 1.013 \times 10^{-5}}{0.3 \times 10^{-3} \times 8314 \times 308} \right] \times [0.8 - 0.1]$$

$$N_A = 1.66 \times 10^{-3} \text{ kmol/m}^2 \text{ s}$$

$$\text{Rate} = N_A \times 100 \times 10^{-4} \times 3600 \times 46 \text{ kg/h}$$

$$= 1.66 \times 10^{-3} \times 100 \times 10^{-4} \times 3600 \times 46$$

$$= 2.749 \text{ kg/h} \quad \textbf{Ans.}$$

(ii) Diffusion through a stagnant film

$$N_A = \frac{D_{AB} \times P_t}{ZRT} \ln \left[\frac{1 - y_{A2}}{1 - y_{A1}} \right]$$

$$N_A = \left[\frac{0.18 \times 10^{-4} \times 1.013 \times 10^{5}}{0.3 \times 10^{-3} \times 8314 \times 304} \right] \ln \left[\frac{1 - 0.1}{1 - 0.8} \right]$$

$$N_A = 3.5706 \times 10^{-3} \text{ kmol/m}^2 \text{ s}$$

$$\text{Rate} = N_A \times 100 \times 10^{-4} \times 3600 \times 46 \text{ kg/h}$$

$$= 3.5706 \times 10^{-3} \times 100 \times 10^{-4} \times 3600 \times 46$$

$$= 5.9129 \text{ kg/h}$$

$$= 1.6425 \times 10^{-3} \text{ kg/s.} \quad \textbf{Ans.}$$

6. Hydrogen gas at 1 standard atm. and 25ºC flows through a pipe made of unvulcanised neoprene rubber with ID and OD of 25 and 50 mm respectively. If the concentration of hydrogen at the inner surface of the pipe is 2.37×10^{-3} kmol hydrogen/m^3 and the diffusivity of hydrogen gas through the rubber is 1.8×10^{-6} cm^2/s, estimate the rate of loss of hydrogen by diffusion through a pipe of 2 m length. The outside air may be assumed to be free from hydrogen.

Solution.

Given: T = 298 K, P_t = 1 atm, ID = 25 mm, OD = 50 mm,

$C_1 = 2.37 \times 10^{-3}$ kmol/m^3, $D_{AB} = 1.8 \times 10^{-6}$ cm^2/s, L = 2 m

This is the case of diffusion through polymers, so

$$V_A = \frac{D_A S_A [p_{A1} - p_{A2}]}{Z}$$

$$Z = \frac{50 - 25}{2} = 12.5 \text{ mm}$$

As per Eq. (2.60), we have

$$V_A = \frac{D_A(C_{A1} - C_{A2})}{Z}$$

$$= 1.8 \times 10^{-10} \ (2.37 \times 10^{-3} - 0)/12.5 \times 10^{-3}$$

$$= 0.3413 \times 10^{-10} \text{ kmol/m}^2 \text{ s}$$

$$S_{av} = \frac{2\pi L \ (\text{OD} - \text{ID})}{2 \ln \left[\frac{\text{OD}}{\text{ID}} \right]}$$

$$= \frac{2\pi \times 2 \times 25 \times 10^{-3}}{2 \ln \left(\frac{50}{25} \right)} = 0.2266 \text{ m}^2$$

$$\text{Rate} = V_A \times S_{av}$$

$$= 0.3413 \times 10^{-10} \times 0.2266$$

$$= 7.734 \times 10^{-12} \text{ kmol/s} \qquad \textbf{Ans.}$$

7. Ammonia diffuses through nitrogen gas under equimolal counter diffusion at a total pressure of 1.013×10^5 Pa and at a temperature of 298 K. The diffusion path is 0.15 m. The partial pressure of ammonia at one point is 1.5×10^4 Pa and at the other point is 5×10^3 Pa. Diffusivity under the given condition is 2.3×10^{-5} m^2/s. Calculate the flux of ammonia.

Solution.

Equimolal counter diffusion

$P_t = 1.013 \times 10^5$ Pa, $T = 298$ K, $Z = 0.15$ m, $p_{A1} = 1.5 \times 10^4$ Pa, $p_{A2} = 5 \times 10^3$ Pa, $D_{AB} = 2.3 \times 10^{-5}$ m^2/s

$$N_A = \frac{D_{AB}}{ZRT} \times [p_{A1} - p_{A2}]$$

$$N_A = \frac{2.3 \times 10^{-5} \times [1.5 - 0.5] \times 10^4}{0.15 \times 8314 \times 298}$$

$N_A = 6.19 \times 10^{-7}$ kmol/m^2 s **Ans.**

8. An ethanol–water solution is in contact at 20ºC with an organic liquid of film thickness 0.4 cm in which water is insoluble. The concentration of ethanol at the interface is 6.8 wt% and at the other side of film it is 10.8 wt%. The densities are 0.9881 g/cc and 0.9728 g/cc respectively for 6.8 wt% and 10.8 wt% ethanol solutions. Diffusivity of ethanol is 0.74×10^{-5} cm^2/s. Calculate the steady state flux in kmol/m^2 s.

Solution.

(Position 1)		weight	moles	mol fraction
	Ethanol	6.80	0.1478	0.02775
	Water	93.20	5.18	0.9722

(Position 2)		weight	moles	mol fraction
	Ethanol	10.8	0.235	0.0453
	Water	89.20	4.96	0.9547

$$M_{av} \text{ (position 1)} = \frac{(0.02775 \times 46) + (0.9722 \times 18)}{1} = 18.776$$

$$M_{av} \text{ (position 2)} = \frac{(0.0453 \times 46) + (0.9547 \times 18)}{1} = 19.268$$

$$\left(\frac{\rho}{M}\right)_1 = \frac{0.9881 \times 10^3}{18.776} = 52.626$$

$$\left(\frac{\rho}{M}\right)_2 = \frac{972.8}{19.268} = 50.488$$

$$\left(\frac{\rho}{M}\right)_{av} = \frac{52.626 + 50.488}{2} = 51.557$$

Assuming the organic liquids is stagnant

$$N_A = \frac{D_{AB}\left(\frac{\rho}{M}\right)_{av} \ln\left[\frac{1 - x_{A2}}{1 - x_{A1}}\right]}{Z}$$

$$= \frac{0.74 \times 10^{-5} \times 10^{-4}}{0.4 \times 10^{-2}} \; 51.557 \times \ln\left[\frac{1 - 0.02775}{1 - 0.0453}\right]$$

$N_A = 1.737 \times 10^{-5}$ kmol/m^2 s. **Ans.**

9. Calculate the rate of diffusion of acetic acid (A) across a film of non-diffusing water (B) solution 2 mm thick at 17ºC, when the concentrations (by weight) on opposite sides of the film are 10% and 4% acid. The diffusivity of acetic acid in the solution is 0.95×10^{-9} m^2/s. Density of 10% and 4% acid (by weight) are 1013 kg/m^3 and 1004 kg/m^3 respectively.

Solution.

Z = 2 mm, T = 290 K, Basis: 100 kg of mixture

(Position 1)	weight, kg	kmol	mole fraction
CH_3COOH	10	0.167	0.0323
H_2O	90	5	0.9677

(Position 2)	weight, kg	kmol	mole fraction
CH_3COOH	4	0.067	0.0124
H_2O	96	5.33	0.9876

$$M_{av} \text{ (position 1)} = \frac{(0.0323 \times 60) + (0.9677 \times 18)}{1} = 19.3566$$

$$M_{av} \text{ (position 2)} = \frac{(0.0124 \times 60) + (0.9876 \times 18)}{1} = 18.5208$$

$$\left(\frac{\rho}{M}\right)_1 = \frac{1013}{19.3566} = 52.3335$$

$$\left(\frac{\rho}{M}\right)_2 = \frac{1004}{18.5208} = 54.209$$

$$\left(\frac{\rho}{M}\right)_{av} = \frac{52.3335 + 54.209}{2} = 53.2714$$

Assuming water to be non-diffusing

$$N_A = \frac{D_{AB}}{Z}\left(\frac{\rho}{M}\right)_{av} \ln\left[\frac{1 - x_{A2}}{1 - x_{A1}}\right]$$

$$= \frac{0.95 \times 10^{-9}}{2 \times 10^{-3}} \times 53.2714 \times \ln\left[\frac{1 - 0.0124}{1 - 0.0323}\right]$$

$$N_A = 0.515 \times 10^{-6} \text{ kmol/m}^2\text{ s} \qquad \textbf{Ans.}$$

10. Carbon dioxide and oxygen experience equimolal counter diffusion in a circular tube whose length and diameter are 1 m and 50 mm respectively. The system is at a total pressure of 10 atm. and a temperature of 25°C. The ends of the tube are connected to large chambers in which the species concentrations are maintained at fixed values. The partial pressure of CO_2 at one end is 190 mm Hg while at the other end is 95 mm Hg. (i) Estimate the rate of mass transfer. (ii) Find the partial pressure of CO_2 at 0.75 m from the end where the partial pressure is 190 mm Hg. Diffusivity under given condition is 2.1×10^{-5} m^2/s.

Solution.

Diffusivity of $CO_2 - O_2 = 2.1 \times 10^{-5}$ m^2/s.

$L = 1$ m, diameter = 50 mm, $P_t = 10$ atm, $T = 298$ K,

$p_{A1} = 190$ mm Hg, $p_{A2} = 95$ mm Hg, $D_{AB} = 2.1 \times 10^{-5}$ m^2/s

(i) $N_A = \dfrac{D_{AB}[p_{A1} - p_{A2}]}{ZRT}$

$$R = \frac{P_o V_o}{T_o} = \frac{760 \times 22.414}{273} = 62.4 \text{ (mm Hg)(m}^3\text{)/(K)(kmol)}$$

$$N_A = \frac{2.1 \times 10^{-5}}{1 \times 62.4 \times 298} (190 - 95)$$

$N_A = 1.073 \times 10^{-7}$ kmol/m^2 s.

Rate of mass transfer $= 1.073 \times 10^{-7} \times \pi r^2$

$$= 1.073 \times 10^{-7} \times \pi \left[\frac{50 \times 10^{-3}}{2}\right]^2$$

$= 2.107 \times 10^{-10}$ kmol/s. **Ans.**

(ii) $$\frac{y_A - y_{A1}}{y_{A2} - y_{A1}} = \frac{Z - Z_1}{Z_2 - Z_1}$$

$$\frac{p_A - p_{A1}}{p_{A2} - p_{A1}} = \frac{Z - Z_1}{Z_2 - Z_1} \qquad \left[\because y = \frac{p_A}{p_t}\right]$$

$$\frac{p_A - 190}{95 - 190} = \frac{0.75 - 0}{1 - 0} = p_A = 118.75 \text{ mm Hg}$$ **Ans.**

11. In an oxygen–nitrogen gas mixture at 1 atm. 25°C, the concentrations of oxygen at two planes 0.2 cm apart are 10% and 20% (by volume) respectively. Calculate the flux of oxygen when (i) nitrogen is non-diffusing and (ii) there is equimolar counter diffusion. Diffusivity of oxygen in nitrogen is 0.215 cm^2/s.

Solution.

$P_t = 1$ atm., $T = 298$ K, $Z = 0.2$ cm, $y_{A1} = 0.2$, $y_{A2} = 0.1$, $D_{AB} = 0.215$ cm^2/s

(i) When N_2 is non-diffusing,

$$N_A = \frac{D_{AB} \times P_t}{ZRT} \ln \left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{0.215 \times 10^{-4} \times 1.013 \times 10^5}{0.2 \times 10^{-2} \times 8314 \times 298} \ln \left[\frac{1 - 0.1}{1 - 0.2}\right]$$

$N_A = 5.18 \times 10^{-5}$ kmol/m^2 s **Ans.**

(ii) For equimolar counter diffusion

$$N_A = \frac{D_{AB} \times P_t}{ZRT} [y_{A1} - y_{A2}]$$

$$N_A = \frac{0.215 \times 10^{-4} \times 1.013 \times 10^5}{0.2 \times 10^{-2} \times 8314 \times 298} [0.2 - 0.1]$$

$N_A = 4.395 \times 10^{-5}$ kmol/m^2 s **Ans.**

12. Ammonia is diffusing through an inert air film 2 mm thick at a temperature of 20ºC and a pressure of 1 atm. The concentration of ammonia is 10% by volume on one side of the film and zero on the other side. Determine the mass flux. Estimate the effect on the rate of diffusion if the pressure is increased to 10 atm. The diffusivity of NH_3 in air at 20°C and 1 atm. is 0.185 cm^2/s.

Solution.

$P_t = 1$ atm, $T = 293$ K, $Z = 2$ mm, $y_{A1} = 0.1$, $y_{A2} = 0$, $D_{AB} = 0.185$ cm^2/s

Assuming air to be stagnant and non-diffusing,

$$N_A = \frac{D_{AB} \times P_t}{ZRT} \ln \left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{0.185 \times 10^{-4} \times 1.013 \times 10^5}{2 \times 10^{-3} \times 8314 \times 293} \ln \left[\frac{1 - 0}{1 - 0.1}\right]$$

$N_A = 4.05 \times 10^{-5}$ kmol/m^2 s

Mass flux = $N_A \times$ Molecular weight

$= 4.05 \times 10^{-5} \times 17 = 6.89 \times 10^{-4}$ kg/m^2 s

When pressure is increased to 10 atm.,

For gases, $D_{AB} \propto \frac{1}{P_t}$

$$\frac{(D_{AB})_1}{(D_{AB})_2} = \frac{(P_t)_2}{(P_t)_1}$$

$$\frac{0.185}{(D_{AB})_2} = \left(\frac{10}{1}\right) \qquad (D_{AB})_2 = 0.0185 \text{ cm}^2/\text{s}$$

$$N_A = \frac{D_{AB}\, P_t}{ZRT} \ln \left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{0.0185 \times 10^{-4} \times 10^5 \times 10 \times 1.013}{2 \times 10^{-3} \times 8314 \times 293} \ln \left[\frac{1 - 0}{1 - 0.1}\right]$$

$N_A = 4.05 \times 10^{-5}$ kmol/m^2 s.

So, rate of diffusion remains same on increasing the pressure. **Ans.**

13. Calculate the rate of diffusion of acetic acid (A) across a film of non-diffusing water (B) solution 2 mm thick at 17ºC, when the concentrations on the opposite sides of the film are 9% and 3% acid (by weight). The diffusivity of acetic acid in the solution is 0.95×10^{-9} m^2/s. Density of 9% and 3% by weight acid are 1012 kg/m^3 and 1003 kg/m^3 respectively.

$$Z = 2 \text{ mm}, \quad T = 290°\text{C}, \quad D_{AB} = 0.95 \times 10^{-9} \text{ m}^2/\text{s}$$

Solution.

(Position 1)		weight	moles	mol fraction
	CH_3COOH	9	0.15	0.0288
	H_2O	91	5.056	0.9712
(Position 2)		weight	moles	mol fraction
	CH_3COOH	3	0.05	0.0092
	H_2O	97	5.389	0.9908

$$M_{av} \text{ (Position 1)} = \frac{(0.0288 \times 60) + (0.9712 \times 18)}{1} = 19.2096$$

$$M_{av} \text{ (Position 2)} = \frac{(0.0092 \times 60) + (0.9908 \times 18)}{1} = 18.3864$$

$$\left(\frac{\rho}{M}\right)_1 = \frac{1012}{19.2096} = 52.682$$

$$\left(\frac{\rho}{M}\right)_2 = \frac{1003}{18.3364} = 54.699$$

$$\left(\frac{\rho}{M}\right)_{av} = \frac{52.682 + 54.699}{2} = 53.691$$

Assuming water to be stagnant

$$N_A = \frac{D_{AB}\left(\frac{\rho}{M}\right)_{av}}{Z} \ln\left[\frac{1 - x_{A2}}{1 - x_{A1}}\right]$$

$$= \frac{0.95 \times 10^{-9}}{2 \times 10^{-3}} \times 53.691 \times \ln\left[\frac{1 - 0.0092}{1 - 0.0288}\right]$$

$$N_A = 5.0956 \times 10^{-7} \text{ kmol/m}^2 \text{ s} \qquad \textbf{Ans.}$$

14. In an oxygen-nitrogen gas mixture at 1 atm., 25°C, the concentrations of oxygen at two planes 0.2 cm apart are 10% and 20% volume respectively. Calculate the rate of diffusion of oxygen expressed as g mol/cm^2 s for the case where

(i) the nitrogen is non-diffusing.
(ii) there is equimolar counter diffusion of the two gases.

Diffusivity of oxygen in nitrogen at 25°C and 1 atm. is 0.206 cm^2/s.

Solution.

The value of gas constant is 82.06 cm^3 atm./(g mol) (K).

P_t = 1 atm., T = 293 K, Z = 2 cm, y_{A1} = 0.2, y_{A2} = 0.1, D_{AB} = 0.206 cm^2/s

(For ideal gases, volume fraction = mole fraction)

(i) N_2 is non-diffusing

$$N_A = \frac{D_{AB}P_t}{ZRT} \ln\left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{0.206\times10^{-4}\times1.013\times10^5}{0.2\times10^{-2}\times8314\times293} \ln\left[\frac{1-0.1}{1-0.2}\right]$$

$N_A = 4.96 \times 10^{-5}$ kmol/m^2 s.

(ii) In equimolar diffusion,

$$N_A = \frac{D_{AB}P_t}{ZRT} [y_{A1} - y_{A2}]$$

$$N_A = \frac{0.206\times10^{-4}\times1.013\times10^5}{0.2\times10^{-2}\times8314\times293} [0.2 - 0.1]$$

$N_A = 4.283 \times 10^{-5}$ kmol/m^2 s. **Ans.**

15. Benzene is stored in a tank of diameter 10 m and open at the top. A stagnant air film 10 mm thick is covering the surface liquid beyond which benzene is absent. If the atmospheric temperature is 25°C and the corresponding vapour pressure is 150 mm Hg, estimate the rate of loss of benzene. Diffusivity of benzene is 0.02 m^2/h. Total pressure is 1.0 atm.

Solution.

P_t = 1 atm., T = 298 K, p_{A1} = 150 mm Hg = 0.2×10^5 N/m^2, p_{A2} = 0, D_{AB} = 0.02 m^2/h

Assuming air layer to be stagnant

$$N_A = \frac{D_{AB}\,P_t}{ZRT} \ln\left[\frac{P_t - p_{A2}}{P_t - p_{A1}}\right]$$

$$= \frac{0.02\times1.013\times10^5}{3600\times10^{-3}\times10\times8314\times298} \ln\left[\frac{1.013\times10^5 - 0}{1.013\times10^5 - 0.2\times10^5}\right]$$

$= 4.996 \times 10^{-6}$ kmol/m^2 s.

Rate = $4.996 \times 10^{-6} \times \pi(10/2)^2 = 3.925 \times 10^{-4}$ kmol/s **Ans.**

16. Alcohol vapour is being absorbed from a mixture of alcohol vapour and water vapour by means of a nonvolatile solvent in which alcohol is soluble but water is not. The temperature is 97°C and the total pressure is 760 mm Hg. The

alcohol vapour can be considered to be diffusing through a film of alcohol–water–vapour mixture 0.1 mm thick. The mole % of alcohol in the vapour at the outside of the film is 80%, and that on the inside, next to the solvent is 10%. The diffusivity of alcohol–water vapour mixtures at 25°C and 1 atm. is 0.15 cm^2/s. Calculate the rate of diffusion of alcohol vapour in kg per hour if the area of the film is 10 m^2.

Solution.

P_t = 760 mm Hg, T = 370 K, Z = 0.1 mm, y_{A1} = 0.8, y_{A2} = 0.1,

D_{AB} = 0.15 × 10^{-4} m^2/s at 25°C and 1 atm., area of film = 10 m^2

For gases,

$$D_{AB} \propto T^{3/2}$$

$$\left(\frac{D_{AB1}}{D_{AB2}}\right) = \left(\frac{T_1}{T_2}\right)^{3/2}$$

$$\left[\frac{0.15\times10^{-4}}{D_{AB2}}\right] = \left(\frac{298}{370}\right)^{3/2}$$

$$D_{AB2} \text{ (at 97°C)} = 2.075 \times 10^{-5} \text{ m}^2/\text{s}$$

Water is insoluble in solvent and thus non-diffusing

$$N_A = \frac{D_{AB}P_t}{ZRT} \ln\left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{2.075\times10^{-5}\times1.013\times10^{5}}{0.1\times10^{-3}\times8314\times370} \ln\left[\frac{1-0.1}{1-0.8}\right]$$

$$N_A = 1.0278 \times 10^{-2} \text{ kmol/m}^2 \text{ s.}$$

$$\text{Rate} = 1.0278 \times 10^{-2} \times 10 \times 3600 \times 46 = 1.70 \times 10^{-4} \text{ kg/hr.}$$

17. Ammonia is diffusing through an inert air film 2 mm thick at a temperature of 20°C and a pressure of 1 atm. The concentration of ammonia is 10% by volume on one side of the film and zero on the other side. D_{AB} at 0°C and 1 atm. 0.198 cm^2/s. Estimate rate of diffusion if the temperature is 20°C and pressure is raised to 5 atm.

Solution.

P_t = 1 atm., T = 293 K, Z = 2 mm, y_{A1} = 0.1, y_{A2} = 0,

D_{AB} = 0.198 cm^2/s at 0°C and 1 atm.

(Volume% = mole%, for ideal gases)

$$\left(\frac{D_{AB1}}{D_{AB2}}\right) = \left(\frac{T_1}{T_2}\right)^{3/2}$$

$$\left[\frac{0.198 \times 10^{-4}}{D_{AB2}}\right] = \left(\frac{273}{293}\right)^{3/2}$$

$$D_{AB2} = 2.2015 \times 10^{-5} \text{ m}^2/\text{s}$$

Assuming air film to be stagnant

$$N_A = \frac{D_{AB}P_t}{ZRT} \ln\left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{2.2015 \times 10^{-5} \times 1.013 \times 10^5}{2 \times 10^{-3} \times 8314 \times 293}\left[\frac{1-0}{1-0.1}\right]$$

$$N_A = 4.823 \times 10^{-5} \text{ kmol/m}^2\text{ s}.$$

Now pressure is increased to 5 atm.

$$D_{AB} \propto \left(\frac{T^{3/2}}{P_t}\right)$$

$$\frac{D_{AB1}}{D_{AB2}} = \left(\frac{T_1}{T_2}\right)^{3/2}\left(\frac{P_2}{P_1}\right)$$

$$\frac{0.198 \times 10^{-4}}{D_{AB2}} = \left(\frac{273}{293}\right)^{3/2} \times \left(\frac{1}{5}\right)$$

$$D_{AB2} = 0.198 \times 10^{-4} \times \left(\frac{293}{273}\right)^{3/2} \times \left(\frac{1}{5}\right)$$

$$D_{AB2} = 4.403 \times 10^{-6} \text{ m}^2/\text{s}$$

$$N_A = \frac{D_{AB}P_t}{ZRT} \ln\left[\frac{1 - y_{A2}}{1 - y_{A1}}\right]$$

$$N_A = \frac{4.403 \times 10^{-6} \times 5 \times 1.013 \times 10^5}{2 \times 10^{-3} \times 8314 \times 293}\left[\frac{1-0}{1-0.1}\right]$$

$$N_A = 4.823 \times 10^{-5} \text{ kmol/m}^2\text{ s}.$$

So, there is no change in flux when pressure is changed. **Ans.**
$(N_A)_{new} = (N_A)_{initial}$, where P gets cancelled with the P_t term in the equation in numerator.

18. An open bowl 0.2 m in diameter contains water at 350 K evaporating into the atmosphere. If the currents are sufficiently strong to remove the water vapour as it is formed and if the resistances to its mass transfer in air is equivalent to

that of a 2 mm layer for condition of molecular diffusion, what will be the rate of evaporation? Diffusivity is 0.2 cm^2/s, vapour pressure is 41.8 kN/m^2.

Solution.

P_t = 1 atm. T = 350 K, Z = 2 mm, P_{A1} = 0.1, P_{A2} = 0 (pure air), D_{AB} = 0.2 cm^2/s

Assuming air to be non-diffusing and a stagnant layer of air of 2 mm

$$N_A = \frac{D_{AB}\, P_t}{ZRT} \ln\left[\frac{P_t - p_{A2}}{P_t - p_{A1}}\right]$$

$$= \frac{0.2\times10^{-4}\times1.013\times10^{5}}{2\times10^{-3}\times 8314\times350} \ln\left[\frac{101.3-0}{101.3-41.8}\right]$$

$$= 1.852 \times 10^{-4} \text{ kmol/m}^2 \text{ s.}$$

Rate of evaporation $= N_A \times$ area

$= 1.852 \times 10^{-4} \times \pi \times (0.2/2)^2$

$= 5.82 \times 10^{-6}$ kmol/s

$= 5.82 \times 10^{-6} \times 18$

$= 1.048 \times 10^{-4}$ kg/s **Ans.**

19. In an experimental determination of diffusivity of toluene in air, Stefan's method is being used. A vertical glass tube 3 mm in diameter is filled with liquid toluene to a depth of 20 mm from the top open end. After 275 hrs at 39.4ºC and a total pressure of 1 atm., the level has dropped to 80 mm from the top. Neglecting counter diffusion of air to replace the liquid, estimate the diffusivity.

Solution.

Density of liquid toluene = 850 kg/m^3

Vapour pressure of toluene at 39.4ºC = 7.64 kN/m^2

Gas law constant, R = 8314 Nm/kmol. K

T = 312.4 K, t = 275 hrs, P_t = 1 atm., $Z_{to} = 20 \times 10^{-3}$ m, $Z_t = 80 \times 10^{-3}$ m

Air is assumed to be stagnant or non-diffusing

$$\rho_L = 850 \text{ kg/m}^3, \qquad p_A = 7.64 \text{ kN/m}^2$$

$$C_{A,L} = \frac{\rho_{A,L}}{M_L} = \frac{850}{92} = 9.24 \text{ kmol/m}^3$$

$$C_A = P_t/RT = \left[\frac{1.013 \times 10^5}{8314 \times 312.4}\right] = 0.039 \text{ kmol/m}^3$$

$$x_{A1} = \frac{p_A}{P_t} = \frac{7.64}{101.3} = 0.0754$$

$x_{B1} = 1 - 0.0754 = 0.9246$

$x_{A2} = 0, \qquad x_{B2} = 1$

$$x_{B,M} = \frac{x_{B2} - x_{B1}}{\ln\left(\frac{x_{B2}}{x_{B1}}\right)} = \frac{(1 - 0.9246)}{\ln\left(\frac{1}{0.9246}\right)} = 0.9618$$

$$\frac{t}{(Z_t - Z_{t0})} = \frac{X_{B,M} C_{A,L} (Z_t + Z_{t0})}{2CD_{AB}(x_{A1} - x_{A2})}$$

$$D_{AB} = \frac{x_{B,M} C_{A,L} (Z_t^2 - Z_{t0}^2)}{2 \times C \times (x_{A1} - x_{A2}) \times t}$$

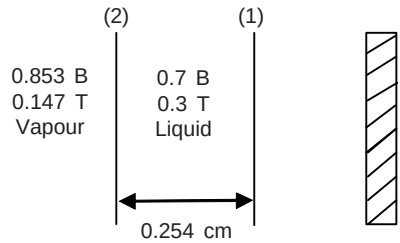
$$D_{AB} = \frac{0.9613 \times 11.81 \times [(80 \times 10^{-3})^2 - (20 \times 10^{-3})^2]}{275 \times 3600 \times 2 \times 0.039 \times (0.0754 - 0)}$$

$D_{AB} = 0.916 \times 10^{-5}$ m^2/s. **Ans.**

20. A mixture of benzene and toluene is distilled in distillation unit. At one plane in the vertical tube where both benzene and toluene are condensing the vapour contains 85.3 mole% benzene and the adjacent liquid film contains 70 mole % benzene. The temperature is 360 K. Gas layer is 0.254 cm thick. The molal latent heat of vaporisation of both benzene and toluene are very close to each other. Vapour pressure of toluene is 368 mm Hg at 360 K. The system is assumed to behave ideal in liquid phase. Calculate the rate of interchange of benzene and toluene between vapour and liquid at atmospheric pressure. The diffusion coefficient is 0.0506×10^{-4} m^2/s.

Solution.

This is a case of equimolal counter diffusion as the latent heat of vaporisation are very close to each other.



$$N_A = \frac{D_{AB}}{ZRT}[p_{A1} - p_{A2}]$$

The partial pressure, $p_{Tol,1} = x_{Tol} \times$ Vapour pressure$_{Tol}$

$= 0.3 \times 368/760 = 0.145$ atm

The partial pressure of toluene in vapour phase, $p_{Tol,2}$

= Mole fraction of toluene × Total pressure

$$= 0.147 \text{ atm}$$

$$N_B = \frac{D_{BA}}{ZRT}[p_{B1} - p_{B2}]$$

$$= \frac{0.0506 \times (0.145 - 0.147)}{0.254 \times 82.06 \times 360}$$

$$= -1.331 \times 10^{-8} \text{ g mol/cm}^2 \text{ s} \qquad \textbf{Ans.}$$

The negative sign indicates that the toluene is getting transferred from gas phase to liquid phase. (Hence, the transfer of benzene is from liquid to gas phase.)

21. A vertical glass tube 3 mm in diameter is filled with toluene to a depth of 2 cm from the top open end. After 275 hours of operation at 303 K and at a total pressure of 1 atm., the level dropped to 7.75 cm from the top. The density of the liquid is 820 kg/m^3 and its vapour pressure is at 57 mm Hg under the given operating conditions. Neglecting the counter diffusion of air to replace the liquid, calculate the diffusivity of toluene in air.

Solution.

This is a case of pseudo steady state diffusion as there is a significant change in the length of diffusion path.

$$Z_{t0} = 0.02 \text{ m}$$

$$Z_t = 0.0775 \text{ m}$$

$$t = 275 \text{ hrs.}$$

$$\text{Vapour pressure} = 57 \text{ mm Hg}$$

Molal density of liquid,

$$C_{AL} = 820/92 = 8.913 \text{ kmol/m}^3$$

$$x_{A1} = 57/760 = 0.075,\ x_{B1} = 1 - 0.075 = 0.925$$

$$x_{A2} = 0.0;\ x_{B2} = 1.0$$

$$= (x_B)_{lm} = \frac{[x_{B2} - x_{B1}]}{\ln\left(\frac{x_{B2}}{x_{B1}}\right)} = \frac{[1 - 0.925]}{\ln\left(\frac{1}{0.9250}\right)} = 0.962$$

$$= C = \frac{P}{RT} = \frac{1.0132 \times 10^{-5}}{8314 \times 303} = 0.04022 \text{ kmol/m}^3$$

$$D_{AB} = \frac{C_{AL}(x_B)_{lm}(Z_t^2 - Z_{t0}^2)}{C(x_{A1} - x_{A2})2t}$$

$$= \frac{8.913 \times 0.962 \times (0.0775^2 - 0.02^2)}{0.04022 \times (0.075 - 0.0) \times 2 \times (275 \times 3600)}$$

$$= 0.805 \times 10^{-5} \text{ m}^2/\text{s} \qquad \textbf{Ans.}$$

22. The diffusivity of the vapour of CCl_4 is determined by Winklemann method in which the level of liquid contained in a narrow tube maintained at a constant temperature of 321 K is continuously measured. At the top of the tube air is flowing and the partial pressure of the vapour at the top of the tube may be taken as zero at any instant. Assuming molecular mass transport, estimate the diffusivity of CCl_4 in air. Vapour pressure of CCl_4 282 mm Hg and density of CCl_4 is 1540 kg/m^3.

The variation in liquid level with respect to time is given below:

Time t, min	0	26	185	456	1336	1958	2810	3829	4822	6385
Liquid level, $(Z - Z_{t0})$ cm	0	0.25	1.29	2.32	4.39	5.47	6.70	7.38	9.03	10.48

Compute $t/(Z_t - Z_{t0})$ and plot it against time, t

Time t, min	0	26	185	456	1336	1958	2810	3829	4822	6385
Liquid level, $(Z_t - Z_{t0})$ cm	0	0.25	1.29	2.32	4.39	5.47	6.70	7.38	9.03	10.48
$t/(Z_t - Z_{t0})$	–	104	143.5	190.5	304	357.5	418.5	514	533.5	610

Solution.

$$\text{Slope} = 51.4385 \times 60 = 3086 \text{ s}^{-1}/\text{cm}^2$$

$$3086 = \frac{C_{A,L}(x_B)_{lm}}{2D_{AB}C(x_{A1} - x_{A2})}$$

$$C_{A,L} = \frac{1540}{154} = 10 \text{ kmol/m}^3$$

$$C = \left(\frac{P}{RT}\right) = \frac{1}{82.06 \times 321} = 3.8 \times 10^{-5} \text{ g mol/cm}^3$$

$$(x_B)_{lm} = \frac{[x_{B2} - x_{B1}]}{\ln\left(\frac{x_{B2}}{x_{B1}}\right)} = \frac{[1 - 0.629]}{\ln\left(\frac{1}{0.629}\right)} = 0.8$$

$$D_{AB} = \frac{10 \times 0.8}{3086} \times 10^4 \times 2 \times 3.8 \times 10^{-2} \times (0.371 - 0) = 9.2 \times 10^{-6} \text{ m}^2/\text{s}$$

Ans.

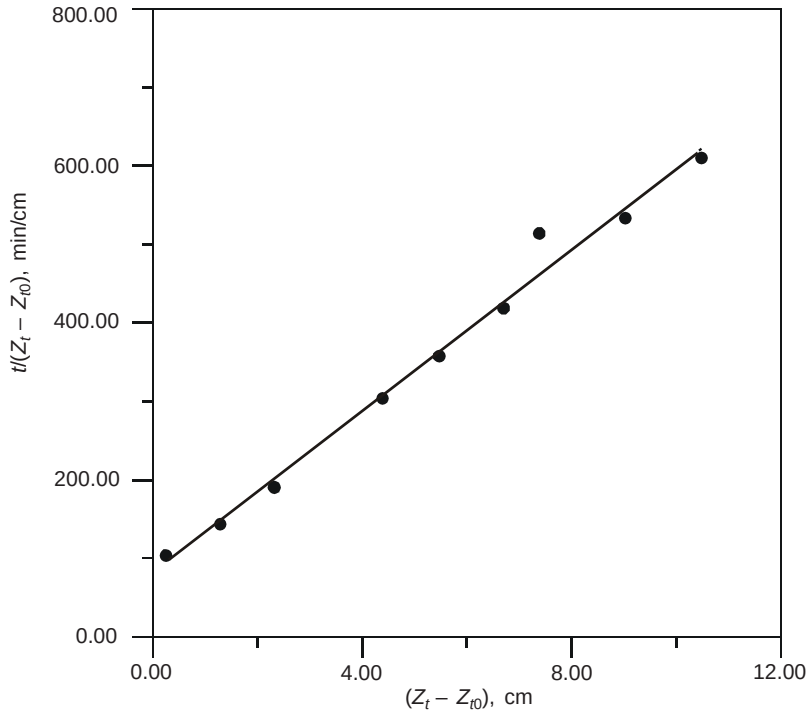


Fig. 2.2 Example 22.

23. There are two bulbs connected by a straight tube 0.001 m in diameter and 0.15 m in length. Initially the bulb at end '1' contains nitrogen and the bulb at end '2' contains hydrogen. The pressure and temperature are maintained constant at 25°C and 1 standard atm. At a certain time after allowing the diffusion to occur between the two bulbs, the nitrogen content of the gas at end '1' of the tube is 80 mole % and at the other end is 25 mole %. If the diffusion coefficient is 0.784 cm^2/s, determine the rates and direction of transfer of hydrogen and nitrogen.

Solution.

It is a case of equimolal counter diffusion as the tube is perfectly sealed to two bulbs at the end and the pressure throughout is constant.

$$\text{Cross-sectional area of tube} = \frac{\pi}{4}D^2 = \frac{22 \times (0.001)^2}{7 \times 4} = 7.85 \times 10^{-7}\ \text{m}^2$$

$$C = \frac{P}{RT} = \frac{1.013 \times 10^5}{(8314)(298)} = 0.0409\ \text{kmol/m}^3$$

$$x_{A1} = 0.8 \text{ and } x_{A2} = 0.25$$

$$\text{Rate of transfer} = \text{Area} \times N_A = \frac{\text{Area} \times D_{AB} \times C \times [x_{A1} - x_{A2}]}{Z}$$

$$= \frac{7.85 \times 10^{-7} \times 0.78 \times 10^{-4} \times 0.0409 \times (0.8 - 0.25)}{0.15}$$

$$= 0.923 \times 10^{-11} \text{ kmol/s} \qquad \textbf{Ans.}$$

24. Estimate the diffusivity of methanol in carbon tetrachloride at 15°C.

Solution.

$$D_{AB} = \frac{(117.3 \times 10^{-18})(\varphi M_B)^{0.5} T}{\mu v_A^{0.6}}$$

M_B = Molecular weight of methanol 32

φ = 1.9

v_A = solute volume at normal BP, m^3/kmol

v_A for CCl_4 = $[14.8 + (4 \times 24.6)] \times 10^{-3}$

= 0.1132 m^3/kmol

μ = 0.6 cP = 0.006 P = 0.0006 kg/m s

$$D_{AB} = \frac{(117.3 \times 10^{-18})(1.9 \times 32)^{0.5} (288)}{(0.0006)(0.1132)^{0.6}}$$

$$= 1.623 \times 10^{-9} \text{ m}^2/\text{s} \qquad \textbf{Ans.}$$

25. Estimate the diffusivity of methanol in water at 15°C.

Solution.

M_B = 18, ϕ = 2.26, T = 288, μ = 0.001 kg/m·s

v_A for C_2H_5OH = $[(2 \times 14.8) + (6 \times 3.7) + 7.4] \times 10^{-3}$

= 0.0592 m^3/kmol

$$D_{AB} = \frac{(117.3 \times 10^{-18})(2.26 \times 18)^{0.5} (288)}{(0.001)(0.0592)^{0.6}}$$

$$= 1.175 \times 10^{-9} \text{ m}^2/\text{s} \qquad \textbf{Ans.}$$

26. An unglazed porcelain plate 5 mm thick has an average pore diameter of 0.2 μm. Pure oxygen gas at 20 mm Hg (abs) 373 K on one side of the plate passes through at a rate of 0.093 cc (20 mm Hg, 373 K)/cm^2 s. When the pressure on the downstream side was so low as to be considered negligible. Estimate the rate of passage of Hydrogen at 298 K and 10 mm Hg abs. with negligible downstream pressure.

Solution.
Viscosity of oxygen = 0.02 *cp* = 20×10^{-6} Ns/m^2
Total pressure = 20 mmHg = $20 \times 133.3 = 2666$ N/m^2
Molecular weight of oxygen = 32
Then

$$\lambda = \frac{3.2\mu}{P_t}\left[\frac{RT}{2\pi g_c M}\right]^{0.5}$$

$$= \frac{3.2 \times 20 \times 10^{-6}}{2666}\left[\frac{8314 \times 373}{2 \times \pi \times 1 \times 32}\right]^{0.5}$$

$$= 2.98 \times 10^{-6} \text{ m}$$

Pore diameter d = 0.2 μm = 0.2×10^{-6} m

Therefore,

$$\frac{d}{\lambda} = \frac{0.2 \times 10^{-6}}{2.98 \times 10^{-6}} = 0.067$$

Hence, Knudsen diffusion occurs.
Now,

N_A = 0.093 cc (20 mm Hg, 373 K)/cm^{2}s

$$= \frac{0.093 \times 20 \times 273}{760 \times 373} = 1.79 \times 10^{-3} \text{ cm}^3/\text{cm}^2\text{s}$$

$$= \frac{1.79 \times 10^{-3}}{22414} = 7.99 \times 10^{-8} \text{ g mol/cm}^2\text{ s} = 7.99 \times 10^{-7} \text{ kmol/m}^2\text{ s}$$

$$D_{KA} = \left[\frac{d}{3}\right]\left[\frac{8gc\,RT}{\pi M_A}\right]^{0.5}$$

$$= \left[\frac{0.2 \times 10^{-6}}{3}\right]\left[\frac{8 \times 1 \times 8314 \times 373}{\pi \times 32}\right]^{0.5}$$

$$= 33.11 \times 10^{-6} \text{ m}^2/\text{s}$$

$$N_A = \frac{D_{K,A}}{RTl}\left[\frac{p_{A1} - p_{A2}}{RTl}\right]$$

$$= 7.99 \times 10^{-7} = 33.11 \times 10^{-6} \times \left[\frac{2666 - 0}{8314 \times 373 \times l}\right]$$

Therefore,

l = 0.0356 m

For the diffusion with Hydrogen, viscosity is 0.0085 cp, pressure is 1333 N/m^2, molecular weight is 2 and temperature is 298 K.
Then

$$\lambda = \frac{3.2\mu}{P_t}\left[\frac{RT}{2\pi g_c M}\right]^{0.5}$$

$$= \frac{3.2 \times 8.5 \times 10^{-6}}{1333}\left[\frac{8314 \times 298}{2 \times \pi \times 1 \times 2}\right]^{0.5}$$

$$= 9.06 \times 10^{-6} \text{ m}$$

Proe diameter d = 0.2 μm = 0.2 × 10^{-6} m
Therefore,

$$\frac{d}{\lambda} = \frac{0.2 \times 10^{-6}}{9.06 \times 10^{-6}} = 0.022$$

Hence, Knudsen diffusion occurs.
Now,

$$[D_{K,A}]_{\text{new (Hydrogen)}} = \left[\frac{0.2 \times 10^{-6}}{3}\right] \times \left[\frac{8 \times 1 \times 8314 \times 298}{\pi \times 2}\right]^{0.5}$$

$$= 1.184 \times 10^{-4} \text{ m}^2/\text{s}$$

$$N_A = \frac{D_{K,A}}{RTl}(p_{A1} - p_{A2})$$

$$= \frac{1.184 \times 10^{-4}}{8314 \times 298 \times 0.0356}(1333 - 0)$$

$$= 1.789 \times 10^{-6} \text{ kmol/m}^2 \text{ s} \qquad \textbf{Ans.}$$

EXERCISES

Note: *Any missing data may be taken from literature.*

1. Estimate the diffusivities of the following gas mixtures:
   (i) Acetone–air at STP
   (ii) Toluene–air, 1 Standard atm., 30°C.
   (iii) Aniline–air, STP.

(**Ans:** (i) 9.2838 × 10^{-6} m^2/s (ii) 8.3186 × 10^{-6} m^2/s (iii) 6.8596 × 10^{-6} m^2/s)

2. Estimate the diffusivity of ethanol in water at 10°C.
(**Ans:** 1.008 × 10^{-9} m^2/s)

3. Ethanol is diffusing through a layer of water of thickness 3 mm at 20°C. Diffusivity of alcohol in water is 0.52×10^{-9} m^2/s. The concentrations on opposite sides of water film are 4% and 10% (by weight) of alcohol respectively are 0.99 and 0.98 g/cm^3. Assuming that water film is stagnant, estimate (i) the flux of alcohol and (ii) concentration of alcohol in the middle of water film.

(**Ans:** (i) 2.432×10^{-7} $kmol/m^2 s$ (ii) 0.029 (mole fraction))

4. Through the accidental opening of a valve, water has been spilled on the floor of an industrial plant in a remote, difficult to reach area. It is desired to estimate the time required to evaporate the water into the surrounding quiescent air. The water layer is 1 mm thick and may be assumed to remain at a constant temperature of 24°C. The air is also at 24°C and 1 atm pressure with an absolute humidity of 0.002 kg water vapour/kg of dry air. The evaporation is assumed to take place by molecular diffusion through a gas film 0.5 cm thick. Diffusion coefficient for water vapour in air is 0.259 cm^2/s.

(**Ans:** 13.67 hours)

5. Calculate the rate of diffusion of NaCl at 18°C through a stagnant film of NaCl-water mixture 1 mm thick when the concentrations are 20% and 10% (by weight) respectively on either side of the film. Diffusivity of NaCl in water is 1.26×10^{-9} m^2/s. The densities of 20% and 10% NaCl solutions are 1149 and 1067 kg/m^3 respectively.

(**Ans:** 2.81×10^{-6} $kmol/m^2$ s)

6. In an O_2–N_2 gas mixture at 1.01325 bar and 20°C, the concentration of O_2 at two planes 0.002 m apart are 20% and 10% volume respectively. (i) Calculate the rate of diffusion of O_2 expressed as kg moles of oxygen/m^2 s for the case where N_2 is non diffusing $D_{O_2-N_2} = 0.181 \times 10^{-4}$ m^2/s. (ii) Calculate the rate of diffusion of oxygen in kmol/m^2 s assuming equimolal counter diffusion.

(**Ans:** (i) 4.4326×10^{-5} $kmol/m^2$ s (ii) 3.763×10^{-5} $kmol/m^2$ s)

7. A vertical glass tube of diameter 0.3 cm is filled with benzene at 30°C to a depth of 2 cm from top end. After 24 hours, the liquid level in the tube had fallen to 2.5 cm from the top end. Estimate the diffusivity of benzene into air if the air above the liquid surface in the tube is stagnant. The vapour pressure and density of benzene at 30°C are 60 mm Hg and 800 kg/m^3 respectively.

(**Ans:** 0.4×10^{-5} m^2/s)

8. A vertical glass tube 1 cm in diameter is filled with liquid acetone to a depth of 5 cm from the top open end. After 4 hours of operation at 303 K and at a total pressure of 1 atm., the level dropped by 2 mm. The density of the liquid is 790 kg/m^3 and its vapour pressure is at 288 mm Hg under the given operating conditions. Neglecting the counter diffusion of air to replace the liquid, calculate the diffusivity of acetone in air.

(**Ans:** 0.49×10^{-5} m^2/s)

9. A gas mixture containing 1/5 hydrogen and 4/5 methane by volume is prepared through which oxygen is allowed to diffuse. The total pressure is 1×10^5 N/m^2 and temperature is 2°C. Estimate the rate of diffusion of O_2 through the gas film of thickness 3 mm when concentration change across the film is 12 to 7% by volume. Diffusivity data at 1 atm., 0°C is

(i) $D_{O_2-H_2} = 7.1 \times 10^{-5}$ m^2/sec.

(ii) $D_{O_2-CH_4} = 1.88 \times 10^{-5}$ m^2/sec.

(**Ans:** 1.82×10^{-5} kmol/m^2 s)

10. A volatile organic compound costing Rs. 6.50 per kg is stirred in a tank 8 m in diameter and open to the atmosphere. A stagnant air film of thickness 10 mm is covering the surface of the compound, beyond which the compound is absent. If the atmospheric temperature is 27°C, vapour pressure of the compound is 160 mmHg and its diffusivity is 0.02 m^2/h, calculate the loss in rupees per day. Molecular weight of the organic compound is 78.

(**Ans:** Rs. 11,745/day)

11. Estimate the rate of diffusion of chloropicrin ($CCl_3\ NO_2$) into air, which is stagnant at 25°C and 1 atm. pressure. Diffusivity = 0.088 cm^2/s, vapour pressure at 25°C of $CCl_3\ NO_2$ is 23.81 mm Hg, Density of chloropicrin = 1.65 g/cm^3. Surface area of liquid exposed for evaporation = 2.29 cm^2. Distance from liquid level to top of tube is 11.14 cm.

(**Ans:** 0.28×10^{-11} kmol/s)

12. A mixture of alcohol and water vapour is rectified by contact with alcohol–water liquid solution. Alcohol is transferred from gas phase to liquid phase and water from liquid to gas phase. The model flow rates are maintained equal but in opposite directions. The temperature 80°C and pressure of 1 atmosphere are maintained constant. Both components diffuse through a gas film of 0.15 mm thick. The molal concentration of alcohol on outer and inner sides of the film is 85% and 10% respectively. Calculate (i) the rate of diffusion of alcohol, (ii) rate of diffusion of water in kg per hour through a film area of one cm^2. The diffusivity is 0.184 cm^2/s.

(**Ans:** (i) 3.17×10^{-3} kmol/s (ii) 11.435 kg/h)

13. Ammonia is diffusing through an inert air film 2 mm thick at a temperature of 20°C and a pressure of 1 atmosphere. The concentration of NH_3 is 10% by volume on one side of the film and zero on the other side. Estimate the effect on the rate of diffusion of raising the total pressure to 5 atmospheres. The diffusivity of NH_3 in air at 0°C and 1 atm. is 0.198 cm^2/s.

(**Ans:** 48.18×10^{-6} kmol/m^2 s)

14. Alcohol is diffusing from gas to liquid and water from liquid to gas under conditions of equimolal counter diffusion at 35°C and 1 atmosphere pressure. The molal concentrations of alcohol on the two sides of a gas film 0.3 mm thick are 80% and 10% respectively. Assuming the diffusivity of alcohol–water vapour to be 0.18 cm^2/s, calculate the rate of diffusion of

alcohol and water in kilograms per hour through an area of 100 cm^2. Molecular weight of alcohol = 74.1; R = 82.06 $cm^3 \cdot$ atm g mol K.

(**Ans:** 4.43 kg/h)

15. Oxygen is diffusing through a stagnant layer of methane 5 mm thick. The temperature is 0°C and the pressure of 1 atmosphere. Calculate the rate of diffusion of oxygen in kilograms per hour through 1 m^2 of methane film when the concentration change across the film is 15% to 5% oxygen by volume. The value of diffusivity may be taken as 0.184 cm^2/s. R = 82.06 cm^3 atm./g mol K.

(**Ans:** 1.05 kg/hr)

3

MASS TRANSFER COEFFICIENT AND INTERPHASE MASS TRANSFER

3.1 INTRODUCTION

In the previous chapter, we have emphasised molecular diffusion in stagnant fluids or fluids at laminar flow. It is well known that the rate of diffusion under molecular diffusion is very slow. In order to increase the fluid velocity for introducing turbulence, the fluid has to flow past a solid surface. When a fluid flows past a solid surface, three regions for mass transfer can be visualized.

There is a region of laminar or thin viscous sub layer very adjacent to the surface where most of the mass transfer occurs by molecular diffusion due to which a sudden concentration drop is seen. Next, a gradual change in concentration of the diffusing substance is obtained in transition region. In the third region called turbulent region, a very small variation in the concentration is observed since the eddies present, which tend to make the fluid in more uniform concentration. The above trend of concentration distribution with distance from the solid surface is shown in Fig. 3.1.

3.2 MASS TRANSFER COEFFICIENT

As the turbulent flow mechanism is yet to be understood, it is better to express the turbulent diffusion in a similar manner as that of molecular diffusion. Molecular diffusion is characterised by the term $D_{AB}\ C/Z$ as in Eq. (2.14) which is modified by '*F*', a mass transfer coefficient for binary system. Here the flux depends upon the cross sectional surface area which may vary, the diffusional path which is not specifically known and the bulk average concentration difference. Hence, flux can be written using a convective mass transfer coefficient.

Flux = (coefficient) (concentration difference)

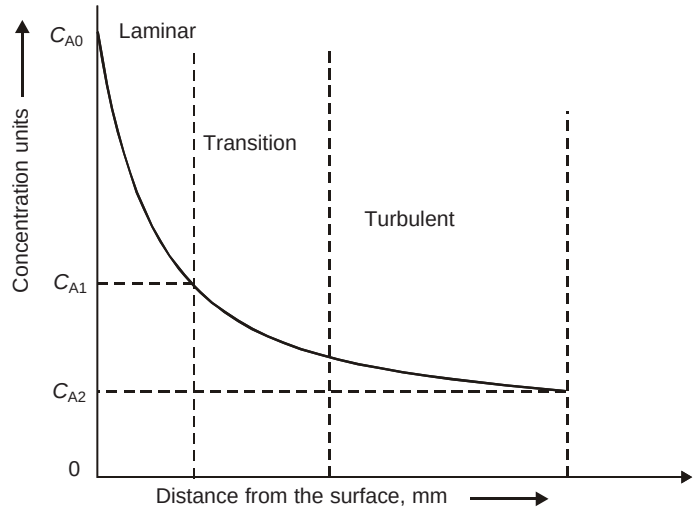


Fig. 3.1 Concentration distribution flow past a solid surface.

Since concentration can be expressed in many ways, different types of equations are possible as mentioned below.

1. For transfer of A through stagnant B [$N_B = 0$]
   For gases: $N_A = k_G\,(p_{A1} - p_{A2}) = k_y\,(y_{A1} - y_{A2}) = k_C\,(C_{A1} - C_{A2})$ (3.1)
   For liquids: $N_A = k_x\,(x_{A1} - x_{A2}) = k_L\,(C_{A1} - C_{A2})$ (3.2)
   where k_G, k_y, k_C, k_x and k_L are individual mass transfer coefficients.
2. For equimolar counter transfer [$N_A = -N_B$]
   For gases: $N_A = k'_G\,(p_{A1} - p_{A2}) = k'_y(y_{A1} - y_{A2}) = k'_C\,(C_{A1} - C_{A2})$ (3.3)
   For liquids: $N_A = k'_x\,(x_{A1} - x_{A2}) = k'_L\,(C_{A1} - C_{A2})$ (3.4)

Thus, k_C is a replacement of $D_{AB}\,C/Z$ of Eq. (2.14) used for low mass transfer rates. F can be used for high mass transfer rates and it can be related to k's as $F = k_G(\overline{p}_B)_{lm}$. The other relations of mass transfer coefficients and flux equations are given in Table 3.1. The mass transfer coefficient can also be correlated as a dimensionless factor J_D by

$$J_D = \left(\frac{k'_c}{V}\right)(N_{SC})^{2/3} \tag{3.5}$$

where V is the mass average velocity of the fluid, and

N_{SC} is Schmidt number i.e. $\left(\dfrac{\mu}{\rho D_{AB}}\right)$

where ρ and μ are the density and viscosity of the mixture respectively.

Table 3.1 Relations between mass transfer coefficient

Gases:

$$F = k_G \cdot (\bar{p}_B)_{\text{lm}} = k_y \cdot \frac{(\bar{p}_B)_{\text{lm}}}{P_t} = k_C \frac{(p_B)_{\text{lm}}}{RT} = \frac{k_Y}{M_{\text{B}}} = k'_G P_t = k'_C . \frac{P_t}{RT} = k'_C C$$

$$= k_y(y_{\text{BM}}) = k'_y$$

Liquids:

$$F = k_x (x_B)_{\text{lm}} = k_L (x_B)_{\text{lm}} \; C = (k'_L) \;\; C = (k'_L)\left(\frac{\rho}{M}\right) = k'_x$$

Units of Mass Transfer Coefficient:

$$k'_G \text{ and } k_G = \frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{pressure})}$$

$$k'_x, k'_y, k_x \text{ and } k_y = \frac{\text{Moles transferred}}{(\text{Area})(\text{time})(\text{mole fraction})}$$

$$k'_C, k_C, k'_L \text{ and } k_L = \frac{\text{Mole transferred}}{(\text{Area})(\text{time})(\text{mol/vol})}$$

$$k_Y = \frac{\text{Mass transferred}}{(\text{Area})(\text{time})(\text{mass A/mass B})}$$

3.3 MASS TRANSFER COEFFICIENTS IN LAMINAR FLOW

When mass transfer occurs in a fluid flowing in laminar flow, it follows the same phenomena of heat transfer by conduction in laminar flow. However, both heat and mass transfer are not always analogous since mass transfer involves multi-component transport. Thus, it needs some simplications to manipulate the mathematical equations for conditions of laminar flow in many complex situations. One simplified situation is illustrated in this section.

3.3.1 Mass Transfer from a Gas into a Falling Liquid Film

Consider the absorption of solute from a gas into a falling liquid film as in wetted wall column, shown in Fig. 3.2. Here the laminar flow of liquid and diffusion occur in such conditions that the velocity field can be virtually unaffected by the diffusion.

The component, A from gas is slightly soluble in liquid B, and hence the viscosity of liquid is not changing appreciably. In addition to that diffusion takes place very slowly in the liquid film and A will not penetrate more into B. Thus, the penetration distance is very small when compared with film thickness.

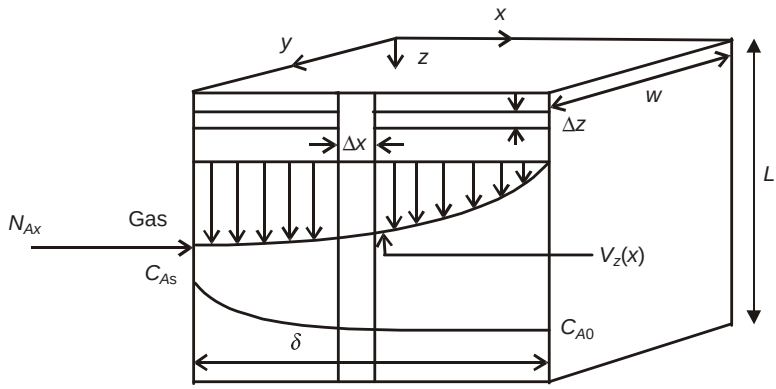


Fig. 3.2 Falling liquid film.

Let the concentration of A in the inlet liquid be C_{A0}, the concentration of A at the surface of the liquid is in equilibrium with the concentration of A in the gas phase which is constant throughout at C_{Ai} and the film thickness is δ. First by solving the momentum transfer equation we obtain the velocity profile $V_z(x)$ for the film as

$$V_z(x) = V_{max}\left[1-\left(\frac{x}{\delta}\right)^2\right] \tag{3.6}$$

Having seen the momentum transfer let us analyse the mass transfer. Let us make a mass balance for component A in the elemental volume (W) (Δx) (Δz).
For steady state,

Rate of input = Rate of output

Hence,

$$N_{Az}\big|_z W\Delta x - N_{Az}\big|_{z+\Delta z} W\Delta x + N_{Ax}\big|_x W\Delta z - N_{Ax}\big|_{x+\Delta x} W\Delta z = 0 \tag{3.7}$$

where W is width of the film
Let us divide Eq. (3.7) by $W\,\Delta x\,\Delta z$ and let Δx and $\Delta z \to 0$, we get

$$\left(\frac{\partial}{\partial x}\right) N_{Ax} + \left(\frac{\partial}{\partial z}\right) N_{Az} = 0 \tag{3.8}$$

Now let us substitute in Eq. (3.8) the flux components in terms of diffusional and convectional fluxes of A.

The one-directional molar fluxes are defined by,

$$N_{Ax} = -D_{AB}\left(\frac{\partial C_A}{\partial x}\right) + x_A(N_{Ax} + N_{Bx}) \tag{3.9}$$

$$N_{Az} = -D_{AB}\left(\frac{\partial C_A}{\partial z}\right) + x_A(N_{Az} + N_{Bz}) \tag{3.10}$$

In Eq. (3.9), A is transported in X-direction by diffusion and not by convective transport because of very slight solubility of A in B. Similarly in Eq. (3.10), A moves in the Z-direction because of the flow of the film, and thereby diffusive contribution is negligible. Hence, Eqs. (3.9) and (3.10) become

$$N_{Ax} = -D_{AB}\left(\frac{\partial C_A}{\partial x}\right) \tag{3.11}$$

$$N_{Az} = x_A(N_{A,z} + N_{B,z}) = C_A V_z(x) \tag{3.12}$$

Substituting Eqs. (3.11) and (3.12) into Eq. (3.8) we get,

$$V_z\left(\frac{\partial C_A}{\partial z}\right) = \frac{\partial}{\partial z}\,[V_z\ C_A] = D_{AB}\left(\frac{\partial^2 C_A}{\partial x^2}\right) \tag{3.13}$$

In this case, solute has penetrated only a very short distance and it gives the impression that it is carried along with the film with a velocity equal to V_{max}. Hence, at $x = 0$, V_z is replaced by V_{max} in Eq. (3.13)

$$V_{max}\left(\frac{\partial C_A}{\partial z}\right) = D_{AB}\left(\frac{\partial^2 C_A}{\partial x^2}\right) \tag{3.14}$$

The Eq. (3.14) has been solved by Laplace transform using the boundary conditions,

B.C.1; at $z = 0$, $C_A = 0$
B.C.2; at $x = 0$, $C_A = C_{A0}$
B.C.3; at $x = \infty$, $C_A = 0$

Since A does not penetrate very far, the distance, δ becomes infinite in view of A. The solution of Eq. (3.14) is given as,

$$\frac{C_A}{C_{A0}} = erfc\left[x\Big/\sqrt{\frac{4\,D_{AB}Z}{V_{max}}}\right] = 1 - erf\left[x\Big/\sqrt{\frac{4D_{AB}Z}{V_{max}}}\right] \tag{3.15}$$

where *erf* is the error function.
The flux at the surface, $x = 0$ as a function of position Z is given by

$$(N_{Ax})_{x=0} = -D_{AB}\left(\frac{\partial C_A}{\partial x}\right)_{x=0} = C_{A0}\sqrt{\frac{D_{AB}V_{max}}{\pi Z}} \tag{3.16}$$

The rate of transfer of A to the fluid over the length $Z = L$ is given by

$$N_A\,[L\cdot W] = W\int_0^L (N_{Ax})\,|_{x=0}\ dZ \tag{3.17}$$

$$= W\int_0^L C_{A0}\sqrt{\frac{D_{AB}V_{max}}{\pi Z}}\ dZ \tag{3.18}$$

$$N_A[L\cdot W] = L\cdot W\ C_{A0}\sqrt{\frac{4(D_{AB}V_{max})}{\pi L}} \tag{3.19}$$

$$N_A = C_{A0}\sqrt{\frac{4(D_{AB}V_{max})}{\pi L}} \tag{3.20}$$

This shows that the liquid mass transfer coefficient is proportional to $D_{AB}^{0.5}$ for short contact times.

3.4 MASS TRANSFER THEORIES

Various theories have been used as models for explaining the turbulent mass transfer. These models can be used for predicting the mass transfer coefficients and they can be correlated with experimental data to obtain the design parameters of process equipments.

3.4.1 Film Theory

Consider turbulent flow of liquid over a solid surface and a simultaneous mass transfer is taking place. The film theory postulates that there is a stagnant film of thickness, Z_f adjacent to the interface, where the concentration difference is attributed to molecular diffusion as shown in Fig. 3.3. As the molecular diffusion is occurring only in Z_f, the flux equation can be written as

$$N_A = k_C (C_{A1} - C_{A2}) = \left(\frac{D_{AB}}{Z_f}\right)(C_{A1} - C_{A2}) \tag{3.21}$$

where $(C_{A1} - C_{A2})$ is the concentration difference. Hence, $k_C = (D_{AB}/Z_f)$, the mass transfer coefficient is proportional to $D_{AB}^{1.0}$. However, the J_D factor is given by,

$$J_D = \left(\frac{k_C}{V}\right)(N_{Sc})^{2/3} = \left(\frac{k_C}{V}\right)\left(\frac{\mu}{\rho D_{AB}}\right)^{2/3} \tag{3.22}$$

Hence, the film theory deviates from the actual turbulent mass transfer.

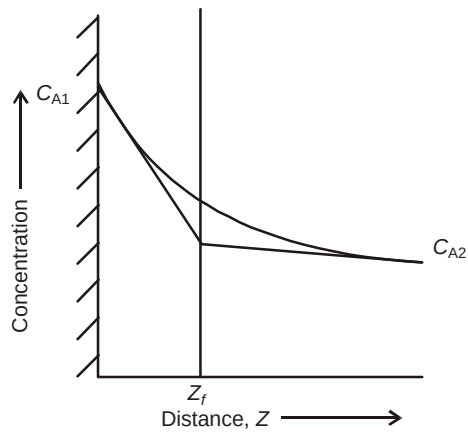


Fig. 3.3 Concentration distribution in film theory.

3.4.2 Penetration Theory

This theory explains the mass transfer at fluid surface and was proposed by Higbie. In many situations, the time of exposure for mass transfer is too short and hence, there may not be sufficient time for the steady state concentration gradient of film theory to develop. This theory has been described in Fig. 3.4. An eddy *b*, rising

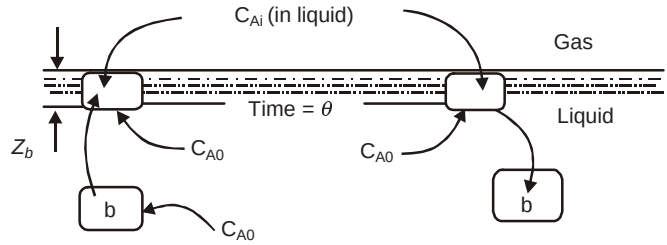


Fig. 3.4 Higbie's theory.

from the turbulent liquid is exposed for a short time, θ at the interface for absorption. In this situation, the exposure time is assumed to be constant for all the eddies or particles of liquid.

Initially the eddy concentration is C_{A0} and when it comes to the surface, the interfacial concentration is C_{Ai}. Since the exposure time is less, molecules of solute from gas never reach the depth Z_b, which is nothing but the thickness of eddy. The liquid particle is subjected to unsteady state diffusion and hence Fick's second law is applicable, i.e.

$$\frac{\partial C_A}{\partial \theta} = D_{AB}\left[\frac{\partial^2 C_A}{\partial Z^2}\right] \tag{3.23}$$

From the solute point of view, the depth Z_b is considered to be infinite.
The boundary conditions are as follows:

$$C_A = C_{A0} \text{ at } \theta = 0 \text{ for all } Z$$
$$C_A = C_{Ai} \text{ at } Z = 0 \quad \theta > 0$$
$$C_A = C_{A0} \text{ at } Z = \infty \text{ for all } \theta$$

By solving the above Eq. (3.23), the average flux can be obtained as described in falling film

Hence,

$$N_{A,av} = 2(C_{Ai} - C_{A0})\sqrt{\frac{D_{AB}}{\pi\theta}} \tag{3.24}$$

$$k_{L,av} = \sqrt{\frac{D_{AB}}{\pi\theta}} \tag{3.25}$$

Thus, in penetration theory k_L is proportional to $D_{AB}^{0.5}$. However, the exponent on D_{AB} varies from zero to 0.8 or 0.9.

3.4.3 Surface Renewal Theory

In reality, the time of exposure of all eddies as proposed in penetration theory is not constant. Hence, Danckwerts modified the penetration theory to account for varying lengths of time of exposure.

If S is the fractional rate of replacement of elements,

Then,
$$N_{A,av} = (C_{Ai} - C_{A0})\sqrt{D_{AB}S} \tag{3.26}$$

Hence, $k_{L,av}$ is proportional to $D_{AB}^{0.5}$ in this theory.

3.4.4 Combination of Film–Surface Renewal Theory

Film theory is meant for steady state diffusion where $k_L \propto D_{AB}$ and in surface renewal theory, $k_L \propto D_{AB}^{0.5}$. So k_L is proportional to D_{AB}^n with 'n' dependent upon circumstances. In this theory Dobbins replaced the third boundary condition of Eq. (3.23) by $C_A = C_{A0}$ at $Z = Z_b$, where Z_b is of finite depth. Finally he obtained

$$k_{L,av} = \sqrt{D_{AB}S} \coth \sqrt{\frac{SZ_b^2}{D_{AB}}} \tag{3.27}$$

3.4.5 Surface–Stretch Theory

Lightfoot and his coworkers explained this theory where they found that the mass transfer at the interface varies with time periodically. When mass transfer is proceeding for a particular system the central portion of the drop is thoroughly turbulent and resistance to mass transfer resides in a surface layer with varying thickness and the drop is elongated as shown in Fig. 3.5. According to this theory,

$$k_{L,av} = \frac{\left[(A/A_r)\sqrt{\frac{D_{AB}}{\pi\theta_r}}\right]}{\sqrt{\int_0^{\theta/\theta r} (A/A_r)^2 d\theta}} \tag{3.28}$$

where A is time dependent interfacial surface, A_r is reference value of A, defined for every situation and θ_r is constant with dimensions of time or drop formation time.

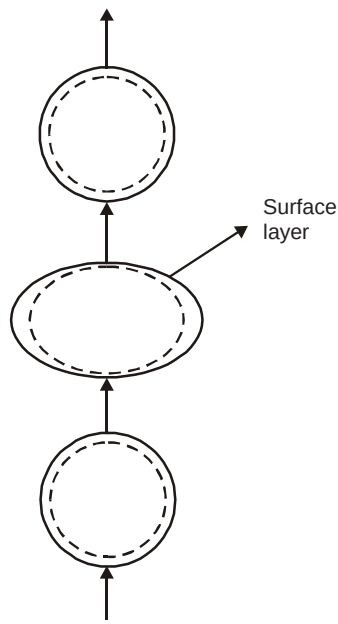


Fig. 3.5 Surface–stretch theory.

3.5 ANALOGIES

As flow past solid surface occurs, at a uniform velocity u_0, the curve ABCD separates the region of velocity u_0 from a region of lower velocity. The curve ABCD which separates these two regions is called boundary layer.

In the same way as mass transfer takes place, a similar concentration boundary layer also occurs. In understanding the analogies between momentum and mass transfer it is worth having a review of universal velocity distribution which is shown in Fig. 3.6.

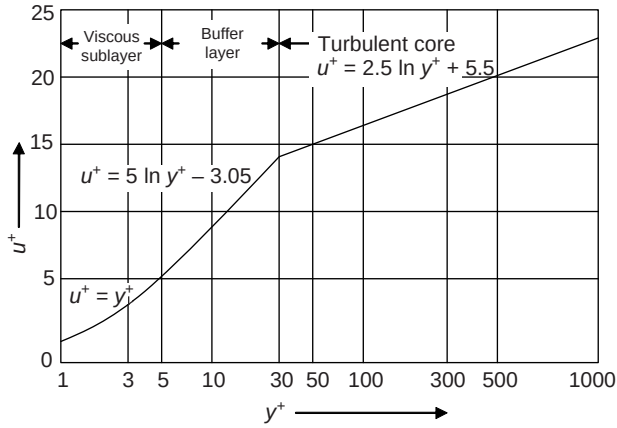


Fig. 3.6(a) Universal velocity profile.

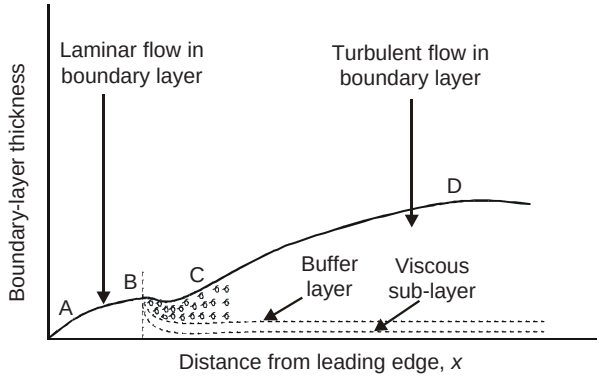


Fig. 3.6(b) Turbulent boundary layer.

The similarity between the transfer processes of momentum, heat and mass lead to the possibility of determining mass transfer characteristics for different situations from the knowledge of other two processes. Let us now deal with the various analogies and the assumptions involved in each analogy.

3.5.1 Reynolds Analogy

In this analogy, the assumptions considered are:

(i) Only turbulent core is present.
(ii) Velocity, temperature and concentration profiles are perfectly matching.
(iii) All diffusivities are same.

Hence, $$(\alpha) = (D_{AB}) = \left(\frac{\mu}{\rho}\right) \tag{3.29}$$

When all the three diffusivities are equal, then

Prandtl Number (N_{Pr}) = Schmidt number (N_{Sc}) = 1.

The basic flux equations of heat, mass and momentum can be written as follows:

$$q = h(t_i - t_0) = -\alpha \frac{\partial}{\partial z} (\rho C_P t) \tag{3.30}$$

$$N_A = k_c (C_{Ai} - C_{A0}) = -D_{AB} \frac{\partial C_A}{\partial z} \tag{3.31}$$

$$\tau_i = \left(\frac{\mu}{g_c}\right) \frac{\partial u}{\partial z} \tag{3.32}$$

Let us consider heat and momentum transfer and from Eq. (3.30)

$$h(t_i - t_0) = -\alpha \frac{\partial}{\partial z} (\rho C_P t) = -\left(\frac{K}{\rho C_P}\right) (\rho C_P) \frac{dt}{dz} \text{ since } \alpha = \left(\frac{K}{\rho C_P}\right) \tag{3.33}$$

$$h (t_i - t_0) = -K\left(\frac{d}{dz}\right) (t - t_i) \tag{3.34}$$

$$\therefore \frac{h}{K} = \left(\frac{d}{dz}\right)\left[\frac{t - t_i}{t_0 - t_i}\right] \tag{3.35}$$

As per assumption (ii), velocity and temperature profiles match and hence,

$$\left[\frac{d}{dz}\right]\left[\frac{u_x}{u_0}\right] = \left(\frac{d}{dz}\right)\left[\frac{(t - t_i)}{(t_0 - t_i)}\right] \tag{3.36}$$

Multiplying by $C_P \mu$ on both sides of Eq. (3.36), we get

$$\left(\frac{C_P \mu}{u_0}\right)\left(\frac{du_x}{dz}\right) = K\left(\frac{d}{dz}\right)\left[\frac{(t - t_i)}{(t_0 - t_i)}\right] \tag{3.37}$$

Since $K = C_P \mu$ by assumption (iii), and rearranging gives

$$\left(\frac{C_P \mu}{K u_0}\right)\left(\frac{du_x}{dz}\right) = \left(\frac{d}{dz}\right)\left[\frac{t - t_i}{t_0 - t_i}\right] \tag{3.38}$$

Combining Eqs. (3.35) and (3.38), we get

$$\left(\frac{C_P \mu}{K u_0}\right)\left(\frac{du_x}{dz}\right) = \frac{h}{K} \tag{3.39}$$

Therefore,
$$\mu\left(\frac{du_x}{dz}\right) = \left(\frac{h u_0}{C_P}\right) = \left(\frac{f}{2}\right)\rho u_0^2 \tag{3.40}$$

i.e.
$$\left(\frac{f}{2}\right) = \frac{h}{\rho\, C_P u_0} \tag{3.41}$$

Similarly, by considering mass and momentum transfer, we get

$$\left(\frac{f}{2}\right) = \left(\frac{k_c}{u_0}\right) \tag{3.42}$$

Hence, Reynolds analogy equation is

$$\left(\frac{f}{2}\right) = \frac{h}{\rho C_P u_0} = \frac{k_c}{u_0} \tag{3.43}$$

3.5.2 Chilton–Colburn Analogy

In this analogy, the assumptions considered are,

(i) only turbulent core is present.
(ii) Velocity, temperature and concentration profiles are same.
(iii) N_{Pr} and N_{Sc} are not equal to unity.

In this analogy, the equation obtained is

$$\left(\frac{f}{2}\right) = \left(\frac{k_c}{u_0}\right)(\text{Sc})^{2/3} = \frac{h}{(\rho C_P u_0)}(\text{Pr})^{2/3} \tag{3.44}$$

3.5.3 Taylor–Prandtl Analogy

In this analogy, the assumptions considered are,

(i) assumes the presence of turbulent core and laminar sublayer.
(ii) N_{Pr} and N_{Sc} are not equal to unity.

In this Analogy, the equation obtained is

$$\frac{k_c}{u_0} = \frac{h}{\rho C_P u_0} = \frac{\left(\frac{f}{2}\right)}{\left[1 + 5\sqrt{\frac{f}{2}}(\text{Sc} - 1)\right]} = \frac{\left(\frac{f}{2}\right)}{\left[1 + 5\sqrt{\frac{f}{2}}(\text{Pr} - 1)\right]} \tag{3.45}$$

3.5.4 Von–Karman Analogy

In this analogy, the assumptions considered are,

(i) assumes the presence of turbulent core, laminar sublayer and buffer layers.

(ii) universal velocity profile equations are applicable.

(iii) N_{Pr} and N_{Sc} are not equal to unity.

In this Analogy, the equation obtained is

$$\frac{k_c}{u_0} = \frac{\frac{f}{2}}{1 + 5\sqrt{\frac{f}{2}}\left[(Sc - 1) + \ln\left(\frac{5Sc + 1}{6}\right)\right]}$$

$$= \frac{\frac{f}{2}}{1 + 5\sqrt{\frac{f}{2}}\left[(Pr - 1) + \ln\left(\frac{5Pr + 1}{6}\right)\right]} \quad (3.46)$$

Equations (3.44), (3.45) and (3.46) lead to Reynolds analogy when Sc = Pr = 1.

3.6 INTERPHASE MASS TRANSFER

3.6.1 Equilibrium

To generalise the equilibrium characteristics, consider that an amount of solute from a gaseous mixture is dissolved in solvent. After sufficient time, the system will attain equilibrium with respect to a particular temperature and pressure. The concentration of solute in both gas and liquid phase may not be equal but the chemical potential of solute will be equal at equilibrium. At the same temperature and pressure, if some more amount of solute is introduced, then once again a new equilibrium will be attained in the same system. The equilibrium curve can be represented for any system as shown in Fig. 3.7. The net rate of diffusion is zero

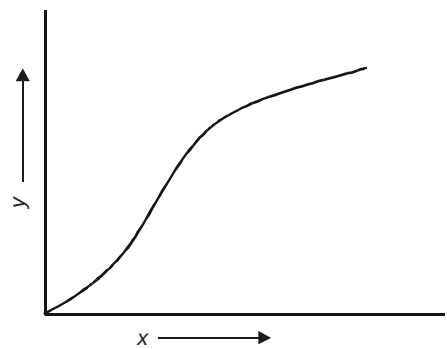


Fig. 3.7 Equilibrium curve.

at equilibrium. Conventionally, the concentration of solute in liquid phase is expressed by mole fraction, x and the concentration of solute in gas phase by mole fraction, y.

3.6.2 Two-phase Mass Transfer

Generally the two-phase systems occur in most of the mass transfer operations. Suppose the two phases are immiscible with each other, then an interface is seen between the two phases. Consider a solute A which is in bulk gas phase G and diffusing into the liquid phase L. There should be a concentration gradient within each phase to cause diffusion through the resistances and is shown in Fig. 3.8.

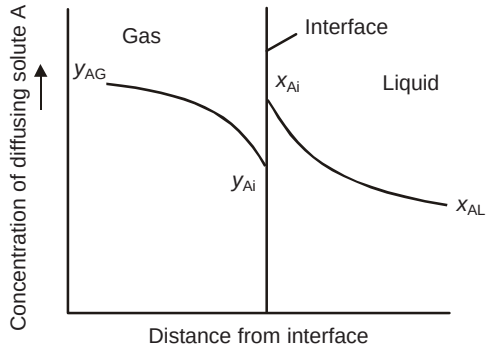


Fig. 3.8 Two-phase mass transfer.

In Fig. 3.8, y_{AG} is the concentration of A in bulk gas phase, y_{Ai} is the concentration at interface, x_{AL} is the concentration of A in bulk liquid phase and x_{Ai} is the concentration at interface. The bulk phase concentrations, y_{AG} and x_{AL} are certainly not at equilibrium. This enables diffusion to occur. At the interface, there is no resistance to transfer of solute and the concentrations y_{Ai} and x_{Ai} are in equilibrium and they are related by the equilibrium distribution relation as

$$y_{Ai} = f\,(x_{Ai}) \tag{3.47}$$

The concentration driving forces can be shown graphically as in Fig. 3.9. If we consider a steady state mass transfer, the rate at which molecules reach the interface will be the same rate at which the molecules are transferred to the liquid phase. Since interface has no resistance, the flux for each phase can be expressed in terms of mass transfer coefficient.

$$N_A = k_y\,(y_{AG} - y_{Ai}) = k_x\,(x_{Ai} - x_{AL}) \tag{3.48}$$

where k_y and k_x are local gas and liquid mass transfer coefficients.

i.e.
$$\frac{y_{AG} - y_{Ai}}{x_{AL} - x_{Ai}} = -\left(\frac{k_x}{k_y}\right) \tag{3.49}$$

Hence, the interface compositions can be determined if k_x, k_y, y_{AG} and x_{AL} values are known.

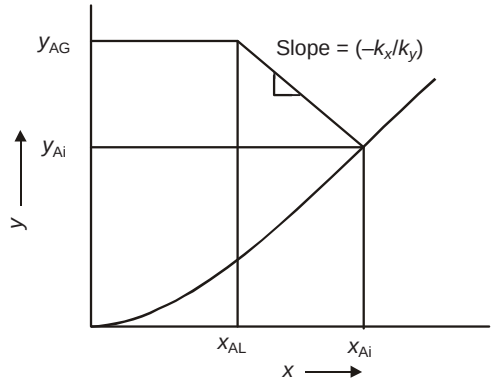


Fig. 3.9 Concentration driving force.

3.6.3 Overall Mass Transfer Coefficient

Experimentally determining the rate of mass transfer is very difficult since it is not possible to evaluate the interface compositions. However, bulk concentrations are easily measured and measuring x_{AL} is as good as measuring y_A^* because both have the same chemical potential. Similarly y_{AG} is as good as measuring x_A^*.The concentration driving forces can be shown as in Fig. 3.10.

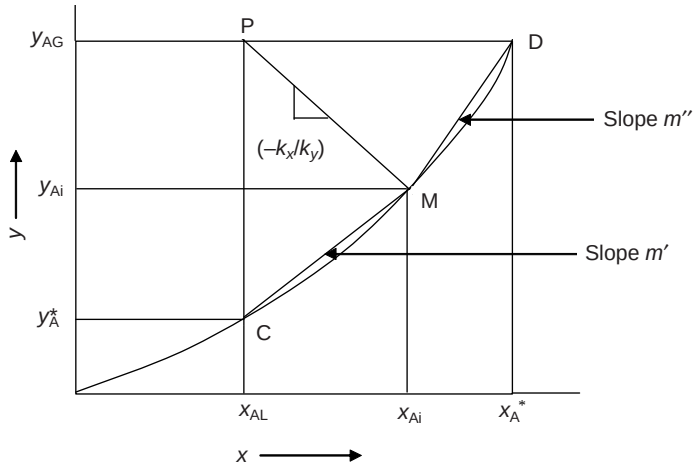


Fig. 3.10 Concentration driving force.

The flux can be written in terms of overall mass transfer coefficient for each phase.

$$N_A = K_y\,(y_{AG} - y_A^*) \tag{3.50}$$

where K_y is overall mass transfer coefficient.

From the geometry of Fig. 3.10

$$(y_{AG} - y_A^*) = (y_{AG} - y_{Ai}) + (y_{Ai} - y_A^*) = (y_{AG} - y_{Ai}) + m'\,(x_{Ai} - x_{AL}) \tag{3.51}$$

where m' is the slope of the chord CM in Fig. 3.10. Substituting for the concentration differences as given by Eqs. (3.48) and (3.50)

$$\left(\frac{N_A}{K_y}\right) = \left[\left(\frac{N_A}{k_y}\right) + \left(\frac{m' N_A}{k_x}\right)\right] \tag{3.52}$$

i.e.
$$\frac{1}{K_y} = \frac{1}{k_y} + \frac{m'}{k_x} \tag{3.53}$$

Similarly, for the liquid side

$$N_A = K_x (x_A^* - x_{AL}) \tag{3.54}$$

On simplification, we get

$$\frac{1}{K_x} = \frac{1}{m'' k_y} + \frac{1}{k_x} \tag{3.55}$$

where m'' is the slope of the chord MD in Fig. 3.10. The two Eqs. (3.53) and (3.55) show the relationship between the individual and overall mass transfer coefficients. These equations also lead to the following relationships between the mass transfer resistances.

$$\frac{\text{Resistance in gas phase}}{\text{Total resistance}} = \frac{1/k_y}{1/K_y} \tag{3.56}$$

$$\frac{\text{Resistance in liquid phase}}{\text{Total resistance}} = \frac{1/k_x}{1/K_x} \tag{3.57}$$

Assuming that k_x and k_y are same and if m' is small so that solute A is highly soluble in liquid (i.e. equilibrium curve will be flat), the term m'/k_x will be negligible when compares to $1/k_y$.

Hence,
$$\frac{1}{K_y} \approx \frac{1}{k_y} \tag{3.58}$$

This condition says that overall resistance lies only in the gas phase, conversely when m'' is very large, then solute A is relatively insoluble in liquid. Under this condition, the term $(1/m'' k_y)$ will be negligible compared to $1/k_x$. Then

$$\frac{1}{K_x} \approx \frac{1}{k_x} \tag{3.59}$$

In this case the entire rate of mass transfer is controlled by liquid phase. For cases where k_x and k_y are not nearly equal, then it will be the relative size of the ratio (k_x/k_y) and of m' or m'' which will determine the location of the controlling mass transfer resistance.

3.7 TYPES OF OPERATIONS

The mass transfer operations take place both on batch and continuous basis. However, due to various demands in an industry most of the operations are carried

out on continuous basis. Such continuous operations are carried out on co-current and countercurrent basis. In these operations the concentration of each phase changes with position whereas in batch process the concentration changes with time.

3.7.1 Co-current Process

Schematic diagram for a co-current process is shown in Fig. 3.11.

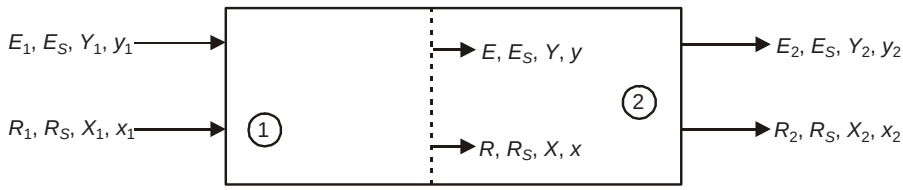


Fig. 3.11 Co-current process.

where E_1, E_2 are mass or molar flow rates of E stream at ① and ② position respectively, R_1, R_2 are mass or molar flow rates of R stream at ① and ② position respectively, E_S, R_S are solute free flow rates of streams, x_1, x_2, y_1, y_2 are concentration of solute in mass or mole fraction of streams at ① and ② position respectively and X_1, X_2, Y_1, Y_2 are mass or mole ratio of solute in streams at ① and ② position respectively.

Making a component balance for solute, we get

$$R_1x_1 + E_1y_1 = R_2x_2 + E_2y_2 \tag{3.60}$$

$$R_SX_1 + E_SY_1 = R_SX_2 + E_SY_2 \tag{3.61}$$

i.e.
$$R_S\,(X_1 - X_2) = E_S\,(Y_2 - Y_1) \tag{3.62}$$

$$\left(\frac{-R_S}{E_S}\right) = \frac{Y_2 - Y_1}{X_2 - X_1} \tag{3.63}$$

This indicates a line passing through the points (X_1, Y_1) and (X_2, Y_2) which is called as operating line in the X vs. Y plot. The operating line also indicates the material balance in the operation.

Also,

$$R_SX_1 + E_SY_1 = R_SX + E_SY \tag{3.64}$$

$$R_S\,(X_1 - X) = E_S\,(Y - Y_1) \tag{3.65}$$

This represents the general equation of operating line in a co-current process. Graphically the operation can be represented for transfer from R to E as shown in Fig. 3.12.

3.7.2 Counter-current Process

Schematic diagram for a counter-current process is shown in Fig. 3.13.

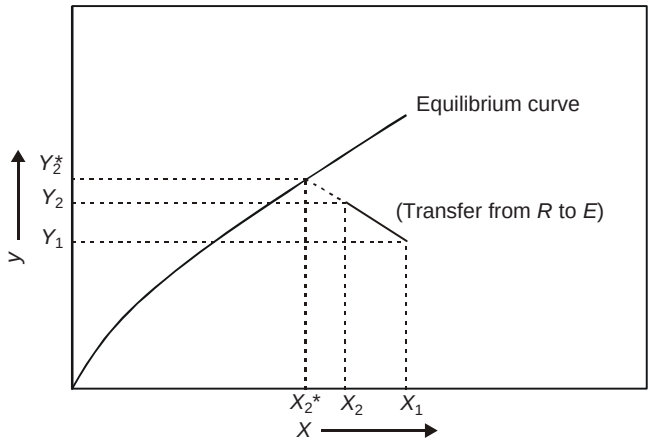


Fig. 3.12 Equilibrium curve and operating line for a co-current process.

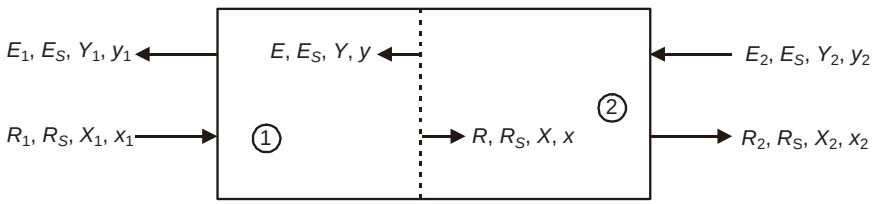


Fig. 3.13 Counter-current process.

Making a component balance for solute yields

$$E_2y_2 + R_1x_1 = E_1y_1 + R_2x_2 \tag{3.66}$$

$$E_SY_2 + R_SX_1 = E_SY_1 + R_SX_2 \tag{3.67}$$

$$R_S\,(X_1 - X_2) = E_S\,(Y_1 - Y_2) \tag{3.68}$$

i.e.
$$\frac{R_S}{E_S} = \frac{Y_1 - Y_2}{X_1 - X_2} \tag{3.69}$$

This represents the equation of a line passing through the coordinates (X_1, Y_1) and (X_2, Y_2) with a slope of R_S/E_S in a plot of X vs Y.

Similarly another balance gives,

$$E_SY + R_SX_1 = E_SY_1 + R_S\,X \tag{3.70}$$

i.e.
$$\frac{R_S}{E_S} = \frac{Y_1 - Y}{X_1 - X} \tag{3.71}$$

This equation is a generalized equation representing the operating line in a counter-current process. The following Fig. 3.14 curve graphically shows the operating line and equilibrium for a counter-current process.

The advantage of the counter-current process over the co-current is the higher driving force, which results in reduced size of equipment for a specified transfer condition or lesser flow rates for a given equipment.

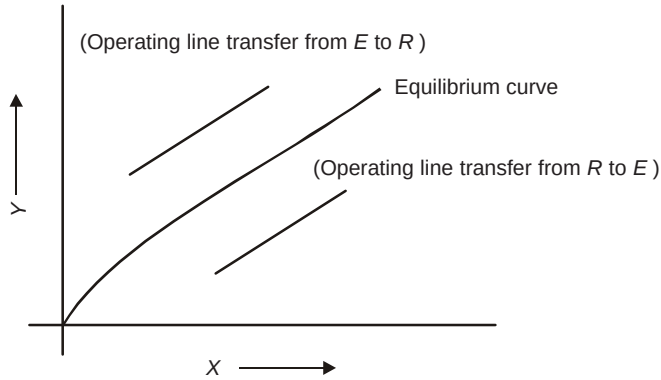


Fig. 3.14 Equilibrium curve and operating line for a counter-current process.

3.7.3 Stages

A stage is defined as any device or combination of devices in which two insoluble phases are brought into intimate contact, where mass transfer occurs between phases leading them to equilibrium and subsequently the phases are separated. A process carried out in this manner is a single stage process. An ideal theoretical or equilibrium stage is one in which the leaving streams are in equilibrium. However, in reality there is a shortfall in reaching the equilibrium and more number of actual stages are needed to effect a desired separation.

3.7.4 Stage Efficiency

It is defined as the fractional approach to equilibrium, which a real stage produces. *Murphree stage efficiency:* It is defined as the fractional approach of one leaving stream to equilibrium with the actual concentration in the other leaving stream

i.e. $$E_{ME} = \frac{(Y_2 - Y_1)}{(Y_2^* - Y_1)} \quad \text{and} \quad E_{MR} = \frac{(X_1 - X_2)}{(X_1 - X_2^*)} \tag{3.72}$$

3.7.5 Cascade

It is one which has a group of interconnected stages, in which the streams from one stage flows to the other. Cascades are of cross-flow and counter-flow types. A typical cross-current cascade is shown in Fig. 3.15.

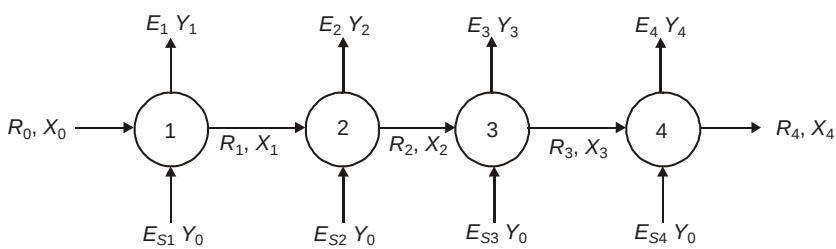


Fig. 3.15 Cross-current cascade.

A typical countercurrent cascade is shown in Fig. 3.16.

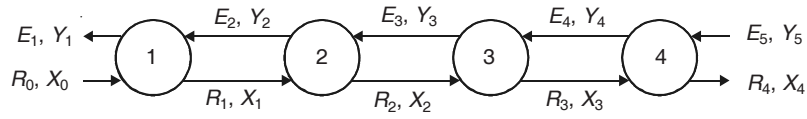


Fig. 3.16 Countercurrent cascade.

The number of stages N, required for a countercurrent cascade can be estimated analytically for cases where both the equilibrium relationship and operating line are linear.

If m is the slope of the equilibrium curve and $A = R_S/mE_S$, the absorption factor then **for absorption**, (transfer from E to R)

For, $A \neq 1$

$$\frac{(Y_{N+1} - Y_1)}{(Y_{N+1} - mX_0)} = \frac{(A^{N+1} - A)}{(A^{N+1} - 1)} \tag{3.73}$$

For, $A = 1$

$$N = \frac{(Y_{N+1} - Y_1)}{(Y_1 - mX_0)} = \frac{(Y_{in} - Y_{out})}{(Y_{out} - mX_{in})} \tag{3.74}$$

For *desorption*, (transfer from R to E)

For $A \neq 1$

$$\frac{(X_0 - X_N)}{\left[X_0 - \left(\frac{Y_{N+1}}{m}\right)\right]} = \frac{\left[\left(\frac{1}{A}\right)^{N+1} - \left(\frac{1}{A}\right)\right]}{\left[\left(\frac{1}{A}\right)^{N+1} - 1\right]} \tag{3.75}$$

For $A = 1$

$$N = \frac{(X_0 - X_N)}{\left[X_N - \left(\frac{Y_{N+1}}{m}\right)\right]} = \frac{(X_{in} - X_{out})}{\left[X_{out} - \left(\frac{Y_{in}}{m}\right)\right]} \tag{3.76}$$

The above four equations are called Kremser–Brown–Souders equation.

WORKED EXAMPLES

1. Calculate the rate of sublimation from a cylinder of naphthalene 0.075 m ID. by 0.6 m long into a stream of pure CO_2 flowing at a velocity of 6 m/s at 1 atm. and 100ºC. The vapour pressure of naphthalene at 100ºC and 1 atm. maybe taken as 10 mm Hg and the diffusivity of naphthalene in CO_2 as 8.3×10^{-6} m^2/s. Density and viscosity of CO_2 are: 0.946 kg/m^3 and 0.021 cp respectively at operating condition. $C_f = 0.023\ (Re)^{-0.2}$. Use analogy.

Solution.

$u_0 = 6$ m/s, $P = 1$ atm, $T = 373$ K, $p_A = 10$ mm Hg, $D_{\text{napth-CO}_2} = 8.3 \times 10^{-6}$ m^2/s, $\rho_{CO_2} = 0.946$ kg/m^3, $\mu_{CO_2} = 0.021$ *cp*, $C_f/2 = 0.023(\text{Re})^{-0.2}$

$$N_{\text{Re}} = \frac{D\bar{V}\rho}{\mu} = \frac{0.075 \times 6 \times 0.946}{0.021 \times 10^{-3}} = 20271.43$$

$$\frac{f}{2} = 3.1648 \times 10^{-3}$$

$$N_{\text{Sc}} = \frac{\mu}{\rho D_{\text{AB}}} = \frac{0.012 \times 10^{-3}}{0.946 \times 8.3 \times 10^{-6}} = 1.528$$

$$k_c = \frac{f \times u_o}{2(N_{\text{Sc}})^{2/3}} \quad (\because u_0 \simeq V \text{ and } C_f \simeq f)$$

$$k_c = \frac{3.1648 \times 10^{-3} \times 6}{1.3266} = 0.0143 \text{ m/s}$$

$$N_{\text{A}} = k_c (C_{\text{A1}} - C_{\text{A2}}) = \frac{k_c}{RT} (p_{\text{A1}} - p_{\text{A2}})$$

$$= \frac{0.0143 \times 10^5}{8314 \times 373} \left[\left(\frac{10}{760} \right) - 0 \right]$$

$$N_{\text{A}} = 6.073 \times 10^{-6} \text{ kmol/m}^2 \text{ s}$$

Rate of sublimation $= N_{\text{A}} \times 2\pi rl$

$= 6.073 \times 10^{-6} \times 2 \times \pi \times (0.075/2) \times 6$

$= 8.59 \times 10^{-6}$ kmol/s **Ans.**

2. A 1 m^2 thin plate of solid naphthalene is oriented parallel to a stream of air flowing at 30 cm/s. The air is at 300 K and 1 atm pressure. The plate is also at 300 K. Determine the rate of sublimation from the plate. The diffusivity of naphthalene in air at 300 K and 1 atm is 5.9×10^{-4} m^2/s. Vapour pressure of naphthalene at 300 K is 0.2 mm Hg.

Solution.

$u_0 = 30$ cm/s, $T = 300$ K, $P_t = 1$ atm, $D_{\text{AB}} = 5.9 \times 10^{-4}$ m^2/s,

$p_{\text{A}} = 0.2$ mm Hg

$\rho_{\text{air}} = 1.15 \times 10^{-3}$ g/cc, $\mu_{\text{air}} = 0.0185$ *cp*, $D = 1$ m (Length)

$$N_{\text{Sc}} = \frac{\mu}{\rho D_{\text{AB}}} = \frac{0.0185 \times 10^{-3}}{1.15 \times 10^{-3} \times 10^3 \times 5.9 \times 10^{-4}} = 0.0273 \neq 1$$

So we will use Chilton analogy,

$$N_{Re} = \frac{D\bar{V}\rho}{\mu} = \frac{1.15\times10^{-3}\times10^{3}\times1\times0.3}{0.0185\times10^{-3}} = 18648.65$$

So flow is turbulent, so Chilton–Colburn analogy can be used.

$$\frac{f}{2} = \frac{k_c}{u_0}(N_{Sc})^{2/3}$$

$$f = 0.072 \times (N_{Re})^{-0.25}$$

$$f = 6.161 \times 10^{-3}$$

$$k_c = \frac{f \times u_0}{2(N_{Sc})^{2/3}}$$

$$k_c = \frac{6.161\times10^{-3}\times0.3}{2(0.0273)^{2/3}} = 0.0102 \text{ m/s}$$

$$N_A = k_c\,(C_{A1} - C_{A2}) = k_c\frac{(p_{A1} - p_{A2})}{RT}$$

$$= \frac{0.0102\times1.0133\times10^{5}}{8314\times300}\left[\left(\frac{0.2}{760}\right) - 0\right]$$

$$N_A = 1.09 \times 10^{-7} \text{ kmol/m}^2 \text{ s.} \qquad \textbf{Ans.}$$

3. In a wetted wall column carbon dioxide is being absorbed from air by water flowing at 2 atm. pressure and 25°C. The mass transfer coefficient k'_y has been estimated to be 6.78×10^{-5} kmol/m^2 s (mole fraction). Calculate the rate of absorption if the partial pressure of carbon dioxide at the interface is 0.2 atm. and the air is pure. Also determine k_y and k_G.

Solution.

$$p_{A1} = 0.2 \text{ atm}, \quad y_{A1} = 0.1, \; y_{B1} = 0.9$$

$$p_{A2} = 0, \; y_{A2} = 0.0 \text{ and } y_{B2} = 0.0$$

$$(y_B)_{lm} = \frac{(y_{B2} - y_{B1})}{\ln\left(\frac{y_{B2}}{y_{B1}}\right)} = 0.95$$

Also,

$$k_y\,(y_B)_{lm} = k_y' = k_G(y_B)_{lm}\,P$$

Hence,

$$k_y = k_y'/(y_B)_{lm} = 6.78 \times 10^{-5}/0.95$$

$$= 7.138 \times 10^{-5} \text{ kmol/m}^2 \text{ s (mole fraction)}$$

$$k_G = \frac{k_y}{P} = 3.569 \times 10^{-5} \text{ kmol/m}^2 \text{ s atm}$$

$$N_A = k_y (y_{A1} - y_{A2}) = 7.138 \times 10^{-5} (0.1 - 0.0)$$

$$= 7.138 \times 10^{-6} \text{ kmol/m}^2 \text{ s} \quad \textbf{Ans.}$$

$$N_A = k_G (p_{A1} - p_{A2}) = 3.569 \times 10^{-5} \times (0.2 - 0.0)$$

$$= 7.138 \times 10^{-6} \text{ kmol/m}^2 \text{ s} \quad \textbf{Ans.}$$

4. Sulphur dioxide is absorbed from air into water in a packed absorption tower. At a certain location in the tower, the mass transfer flux is 0.027 kmol SO_2/m^2h and the liquid phase concentrations in mole fraction are 0.0025 and 0.0003 respectively at the two-phase interface and in the bulk liquid. If the diffusivity of SO_2 in water is 1.7×10^{-9} m^2/s, determine the mass transfer coefficient, k_c and film thickness.

Solution.

$$\text{Mass flux of } SO_2 = \frac{0.027 \times 1000}{3600 \times 100 \times 100} = 7.5 \times 10^{-7} \text{ g mol/cm}^2 \text{ s}$$

$$C = \frac{\text{Density}}{\text{Molecular weight}} = \frac{1}{18.02} = 5.55 \times 10^{-2} \text{ g mol/cm}^3$$

$$N_A = k_c(C_{A1} - C_{A2}) = D_{AB} \times \frac{(C_{A1} - C_{A2})}{\delta}$$

Therefore,

$$k_c = \frac{D_{AB}}{\delta} = \frac{N_A}{C(x_{A1} - x_{A2})}$$

$$= \frac{7.5 \times 10^{-7}}{(5.55 \times 10^{-2}) \times (0.0025 - 0.0003)}$$

$$= 0.00614 \text{ cm/s}$$

$$\delta = \frac{D_{AB}}{k_c} = \frac{1.7 \times 10^{-5}}{0.00614} = 0.0028 \text{ cm} \quad \textbf{Ans.}$$

5. In an experimental study of absorption of ammonia by water in a wetted wall column the overall gas phase mass transfer coefficient, K_G was estimated as 2.72×10^{-4} kmol/m^2 s atm. At one point in the column the gas contained 10 mol % ammonia and the liquid phase concentration was 6.42×10^{-2} kmol NH_3/m^3 of solution. Temperature is 293 K and the total pressure is 1 atm. 85% of the resistance to mass transfer lies in gas phase. If Henry's law constant is 9.35×10^{-3} atm. m^3/kmol, calculate the individual film coefficient and the interfacial composition.

Solution.

$$K_G = 2.72 \times 10^{-4} \text{ kmol/m}^2 \text{ s atm.}$$

Resistance $1/K_G = 1/2.72 \times 10^{-4} = 3676.47$ m^2 s atm./kmol

$$\frac{1}{k_G} = 0.85 \times \frac{1}{K_G}$$

$$= 3125 \text{ m}^2 \text{ s atm./kmol}$$

We know that,

$$\frac{1}{K_G} = \frac{1}{k_G} + \frac{m}{k_L}$$

$$m = 9.35 \times 10^{-3} \text{ atm.m}^3\text{/kmol}$$

Hence $k_L = 1.695 \times 10^{-5}$ m/s

We shall now calculate the interfacial concentration from the following relation:

$$N_A = K_G(p_{Ag} - p^*_A)$$
$$= k_G(p_{Ag} - p_{Ai})$$
$$= k_L(C_{Ai} - C_{AL})$$
$$y_{AG} = 0.1$$
$$C_{AL} = 6.42 \times 10^{-2} \text{ kmol } NH_3/\text{m}^3 \text{ of solution.}$$
$$p_{Ag} = y_{Ag} \times P_t$$
$$= 0.1 \times 1.0 = 0.1 \text{ atm.}$$
$$C_{AL} = 6.42 \times 10^{-2} \text{ kmol } NH_3/\text{m}^3 \text{ of solution.}$$
$$p_{Ai} = m\ C_{Ai}$$
$$N_A = k_G(p_{Ag} - p_{Ai}) = k_L\ (C_{Ai} - C_{AL})$$
$$\frac{k_G}{k_L} = \frac{(C_{Ai} - C_{AL})}{(p_{Ag} - p_{Ai})}$$
$$\frac{k_G}{k_L} = \frac{(C_{Ai} - C_{AL})}{(p_{Ag} - mC_{Ai})}$$
$$18.88 = \frac{(C_{Ai} - 6.42 \times 10^{-2})}{(0.1 - 9.35 \times 10^{-3} C_{Ai})}$$

On solving,

$$C_{Ai} = 1.6593 \text{ kmol/m}^3$$
$$p_{Ai} = 0.0155 \text{ atm.} \qquad \textbf{Ans.}$$

6. At 293 K the solubility of ammonia in water is given by Henry's law $p = 0.3672\ C$, where p is in atmosphere and C is in kmol/m^3. A mixture of 15% ammonia and 85% air by volume at 1 atm is in contact with an aqueous solution containing 0.147 g mol/lit. The air velocity is such that $k_G/k_L = 0.9$. Find the concentration of ammonia and partial pressure at interface.

Solution.

We have, $p = 0.3672\ C$, where p is in atmosphere and C is in kmol/m^3.

$$N_A = k_G\ (p_{AG} - p_{Ai})$$
$$= k_L\ (C_{Ai} - C_{AL})$$

where p_{Ai} and C_{Ai} indicate the interfacial pressure and composition and p_{AG} and C_{AL} indicate the bulk phase compositions.

$$p_{AG} = 1 \times 0.15 = 0.15 \text{ atm.}$$
$$C_{AL} = 0.147 \text{ g mol/lit} = 0.147 \text{ kmol/m}^3$$
$$\frac{k_G}{k_L} = \frac{(C_{Ai} - C_{AL})}{(p_{AG} - p_{Ai})}$$
$$0.9 = \frac{C_{Ai} - 0.147}{0.15 - 0.3672\ C_{Ai}}$$

Solving for C_{Ai} we get

$$C_{Ai} = 0.212 \text{ kmol/m}^3$$
$$p_{Ai} = 0.078 \text{ atm.} \qquad \textbf{Ans.}$$

7. Pure gas is absorbed in a laminar liquid jet. The volumetric flow rate of the liquid was 4 cc/s and the diameter and length of the jet were 1 mm and 3 mm respectively. The rate of absorption of A at atmospheric pressure was 0.12 cc/s at 303 K. The solubility of gas at 303 K is 0.0001 g mol/cc. atm. Estimate the diffusivity of gas. If the diameter of the jet is reduced to 0.9 mm, under otherwise the same conditions how would it affect the rate of evaporation. Assume the validity of Higbie's penetration theory.

Solution.

We know,

$$N_A = k_L\ A\ (C_A^* - C_A)$$
$$(k_L)_{av} = 2\ (D_{AB}/\pi t)^{0.5}$$
$$\text{Molar rate of absorption} = 0.12 \times 1 \times 273/(22414 \times 303)$$
$$= 0.482 \times 10^{-5} \text{ g mol/s}$$
$$A = \pi DL = \pi(0.1)(3) = 0.942 \text{ cm}^2$$
$$C_A^* = 0.0001 \text{ g mol/cc} \cdot \text{atm}$$
$$C_A = 0$$
$$k_L = N_A/[A(C_A^* - C_A)]$$
$$= 0.482 \times 10^{-5}/[0.942 \times (0.0001 - 0)]$$
$$= 0.051 \text{ cm/s}$$

$$\text{Time of contact } t = \frac{\text{Bubble length}}{\text{Linear velocity}}$$

$$\text{Linear velocity} = \frac{Q}{A} = \frac{4}{\left[\frac{\pi \times 0.1 \times 0.1}{4}\right]}$$

$$= 509 \text{ cm/s}$$

$$t = \frac{3}{509} = 0.006 \text{ s}$$

$$(k_L)_{av} = 2\left(\frac{D_{AB}}{\pi t}\right)^{0.5}$$

$$0.051 = 2\left(\frac{D_{AB}}{\pi} \times 0.006\right)^{0.5}$$

$$D_{AB} = 1.23 \times 10^{-5} \text{ cm}^2/\text{s}$$

Revised diameter = 0.09 cm

$$\text{Area} = \pi DL = \pi(0.09)\,(3) = 0.848 \text{ cm}^2$$

$$\text{Velocity} = \frac{Q}{A} = \frac{4}{\pi \times 0.09} \times \frac{0.09}{4}$$

$$= 628.8 \text{ cm/s}$$

$$\text{Time of contact } t = \frac{\text{Bubble length}}{\text{Linear velocity}}$$

$$= 3/628.8$$

$$= 0.00477 \text{ cm/s}$$

$$(k_L)_{av} = 2\left(\frac{D_{AB}}{\pi t}\right)^{0.5}$$

$$= 2\left(\frac{1.23 \times 10^{-5}}{\pi \times 0.00477}\right)^{0.5}$$

$$= 0.0573 \text{ cm/s.}$$

$$N_A = k_L A\,(C_A^* - C_A)$$

$$= 0.0573 \times 0.848 \times (0.0001 - 0)$$

$$= 4.86 \times 10^{-6} \text{ g mol/s} \qquad \textbf{Ans.}$$

8. In an apparatus for the absorption of SO_2 in water at one point in the column the concentration of SO_2 in gas phase was 10% SO_2 by volume and was in contact with a liquid containing 0.4% SO_2 by weight. Pressure and temperature are 1 atm. and 323 K respectively. The overall gas phase mass transfer coefficient is 7.36×10^{-10} kmol/m^2 s. (N/m^2). Of the total resistance 45% lies in gas phase and 55% in the liquid phase.
Equilibrium data:

kg SO_2/100 kg water	0.2	0.3	0.5	0.7
Partial pressure of SO_2, mm Hg	29	46	83	119

(i) Estimate the film coefficients and overall mass transfer coefficient based on liquid phase.

(ii) Estimate the molar flux based on film coefficients and overall transfer coefficients.

Solution.

kg SO_2/100 kg water	0.2	0.3	0.5	0.7
Partial pressure SO_2 mm Hg	29	46	83	119
x (mole fraction of SO_2 in liquid phase) $\times 10^4$	5.63	8.46	14.11	19.79
y (mole fraction of SO_2 in gas phase) $\times 10^2$	3.82	6.05	10.92	15.66

$K_G = 7.36 \times 10^{-10}$ kmole/m^2 s (N/m^2).

$K_y = K_G P$ (based on Eq. 3.3)

$= 7.36 \times 10^{-10} \times 1.013 \times 10^5 = 7.456 \times 10^{-5}$ kmol/m^2 s. (mole fraction).

Resistance in gas phase ($1/k_y$) is 0.45 of total resistance

Resistance in liquid phase is 0.55 of total resistance

We know that,

$$\frac{1}{K_y} = \frac{1}{k_y} + \frac{m'}{k_x}$$

Resistance in gas phase $= 0.45 \times \left(\frac{1}{7.456} \times 10^{-5} \right)$

$= 6035.4$ m^2 s (mole fraction)/kmol

Therefore,

$k_y = 1.657 \times 10^{-4}$ kmol/m^2 s (mole fraction).

Resistance in liquid phase (m'/k_x) $= 0.55 \times \left(\frac{1}{7.456} \times 10^{-5} \right)$

$= 7376.6$ m^2 s (mole fraction)/kmol

The equilibrium relationship indicates a linear behaviour in the range of x from 0.0008 to 0.0012 and the value of slope of the equilibrium curve line (m') is 86.45.

Therefore,

$$k_x = 0.0117 \text{ kmol/m}^2 \text{ s (mole fraction)}$$

$$y_{A,G} = 0.1$$

The liquid phase composition is 0.4 wt% of SO_2

$$x_{A,L} = \frac{\left(\frac{0.4}{64}\right)}{\left(\frac{99.6}{18}\right)+\left(\frac{0.4}{64}\right)} = 0.001128$$

Slope of the line (to determine the interfacial compositions) $\left(-\frac{k_x}{k_y}\right) = -70.61$

It is also clear from the graph that the slope, m'' is same as m' in the range under consideration.

Hence,

$$m' = m'' = 86.45$$

$$\frac{1}{K_x} = \frac{1}{k_x} + \frac{1}{m''k_y}$$

$$= \frac{1}{0.0117} + \frac{1}{86.45 \times 1.657 \times 10^{-4}}$$

$$K_x = 6.44 \times 10^{-3} \text{ kmol/m}^2 \text{ s (mole fraction).}$$

$$y_A^* = 0.083,\ x_A^* = 0.00132,\ y_{A,i} = 0.0925 \text{ and } x_{A,i} = 0.00123$$

Flux based on overall coefficient:

Flux based on gas phase $= K_y(y_{A,G} - y_A^*)$

$= 7.456 \times 10^{-5} (0.1 - 0.083)$

$= 1.268 \times 10^{-6} \text{ kmol/m}^2 \text{ s.}$

Flux based on liquid phase $= K_x(x_A^* - x_{A,L})$

$= 6.44 \times 10^{-3} (0.00132 - 0.001128)$

$= 1.236 \times 10^{-6} \text{ kmol/m}^2 \text{ s.}$

Flux based on film coefficient:

Flux based on gas phase $= k_y(y_{A,G} - y_{A,i})$

$= 1.657 \times 10^{-4} (0.1 - 0.0925)$

$= 1.243 \times 10^{-6} \text{ kmol/m}^2 \text{ s.}$

Flux based on liquid phase $= k_x(x_{A,i} - x_{A,L})$

$= 0.0117 (0.00123 - 0.001128)$

$= 1.193 \times 10^{-6} \text{ kmol/m}^2 \text{ s.}$

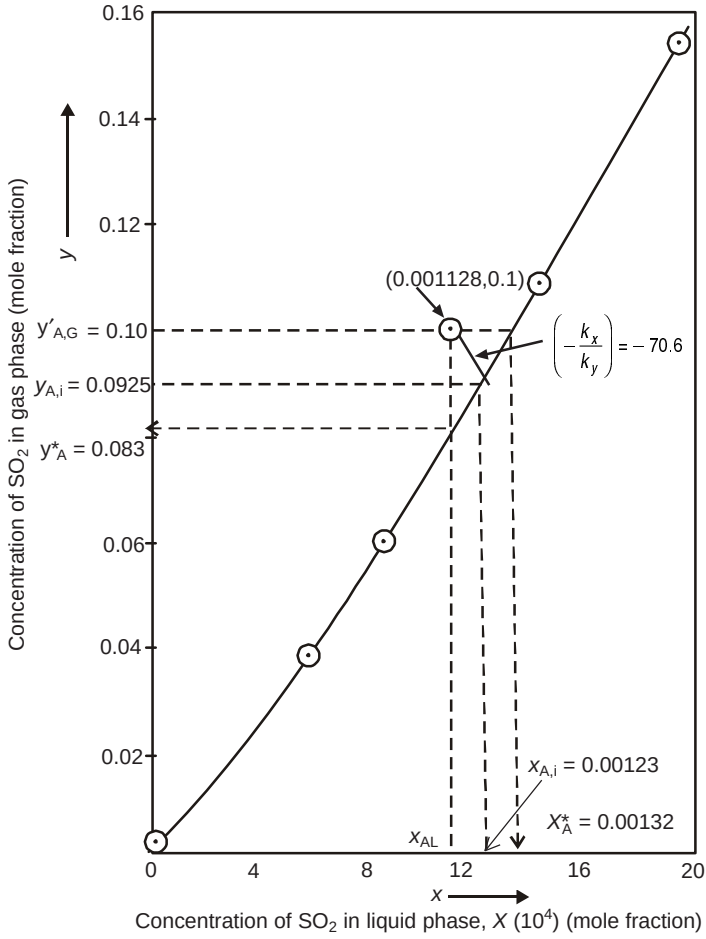


Fig. 3.17 Example 8.

9. Air at 27°C is flowing at a velocity of 1525 cm/s through a tube coated with an acid of 25.4 mm in diameter. The length of the tube is 183 cm. Calculate the concentration of acid at the outlet. Take

$$\mu = 1.786 \times 10^{-4} \text{ P}, \quad \rho = 1.25 \text{ g/lit}$$

$$D_{AB} = 0.0516 \text{ cm}^2/\text{s}, \; C_{As} = 1.521 \times 10^{-7} \text{ g mol/cc}$$

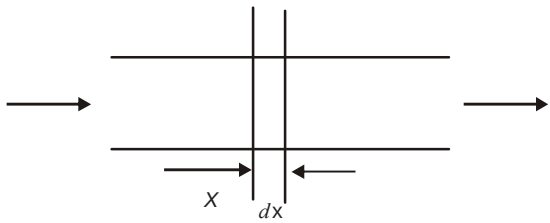
Solution.

$$\text{Reynold's number} = \frac{DV\rho}{\mu} = \frac{2.54 \times 1525 \times 1.25 \times 10^{-3}}{1.786 \times 10^{-4}} = 27110$$

$$\text{Schmidt number} = \frac{\mu}{\rho D_{AB}} = \frac{1.786 \times 10^{-4}}{1.25 \times 10^{-3} \times 0.0516} = 2.77$$

$$\frac{C_f}{2} = 0.036 \times (27110)^{-0.25}$$

$$= 2.806 \times 10^{-3}$$



$$\frac{C_f}{2} = \frac{k_c}{u_0}(\text{Sc})^{2/3}$$

$$k_c = 2.806 \times 10^{-3} \times (2.77)^{-2/3} \times 1525 = 2.17 \text{ cm/s}$$

Consider an elemental section of length dx at a distance of x from the point of entry of air. Let the concentration of diffusing component be C in this section and let it leave with a concentration of $C + dC$ from this section. A mass balance across this elemental section gives,

Rate of mass transfer = (Cross sectional area) (Air flow velocity) (dC)

$$= \left(\frac{\pi D^2}{4}\right)(V)\,(dC)$$

Flux for mass transfer from the surface = $k_c\,[C_{\text{As}} - C]$

Therefore,

Rate of mass transfer = (Flux) (Mass transfer area)

$$= k_c\,[C_{\text{As}} - C]\;\pi\;dx\;D$$

Equating the above two expressions for rate of mass tansfer, we get

$$(\pi D^2/4)\;(V)\;(dC) = k_c\,[C_{\text{As}} - C]\;\pi\;dx\;D$$

Rearranging,

$$\frac{(dC)}{[C_{\text{As}} - C]} = \left[\frac{k_c \pi D}{(\pi D^2/4)}(V)\right] dx$$

$$\frac{(dC)}{[C_{\text{As}} - C]} = \left[\frac{4k_c}{DV}\right] dx$$

On integrating between $x = 0$, $C = C_{\text{in}} = 0$

and

$$x = 183,\ C = C_{\text{final}}$$

On solving

$$\text{In}\left[\frac{C_{\text{As}} - C_{\text{final}}}{C_{\text{As}} - C_{\text{in}}}\right] = \frac{4k_c}{DV} \times x$$

We get

$$C_{\text{final}} = 5.117 \times 10^{-8} \text{ g mol/cc}$$

Then Rate of mass transfer = (Cross sectional area) (Air flow velocity) $(C_{\text{final}} - C_{\text{in}})$

$$= \left(\frac{\pi D^2}{4}\right)(V)\,(C_{\text{final}} - C_{\text{in}})$$

$$= \left[\frac{\pi \times (2.54)^2}{4}\right][1525][5.117 \times 10^{-8}]$$

$$= 3.95 \times 10^{-4} \text{ g mol/s} \qquad \textbf{Ans.}$$

EXERCISES

1. Air at 25°C and 50% relative humidity flows over water surface measuring 12 m × 6 m at a velocity of 2 m/s. Determine the water loss per day considering flow direction is along the 12 m side. Diffusivity of water in air is 0.26×10^{-4} m^2/s. Sc = 0.6 and Kinematic viscosity is 15.7×10^{-6} m^2/s.

(**Ans:** 361.84 kg/day)

2. The absorption of solute A from a mixture is done in a wetted wall column by a solvent at 1 atm. and 25°C. The value of mass transfer coefficient is 9.0×10^{-4} m/s. At a point, the mole fraction of A in the liquid gas interface is 2.0×10^{-5} in the liquid phase. Partial pressure of A in the gas phase is 0.08 atm. Henry's law relation is

$$p_A = (600)\ x_A \text{ in atm.}$$

Calculate the rate of absorption of A.

(**Ans:** 2.5×10^{-6} kmol/m^2 s)

3. Pure water at 27°C is flowing at a velocity of 3.5 m/s through a tube coated with benzoic acid of 6 mm in diameter. The length of the tube is 1.25 m. Calculate the concentration of benzoic acid at the outlet. Take μ = 0.871 *cp*; ρ = 1 g/cc, $D_{AB} = 1.3 \times 10^{-5}$ cm^2/s, C_{As} = 0.03 g mol/lit.

(**Ans:** 1.017×10^{-3} kmol/m^3)

4

EQUIPMENT FOR GAS–LIQUID OPERATIONS

4.1 INTRODUCTION

The equipment used for gas–liquid operations are classified under two types, 1. stage contactors (bubble cap, valve trays and sieve tray columns) and 2. continuous contactors (packed towers and spray towers).

4.2 TRAY TOWERS

A typical tray tower is shown in Fig. 4.1. These are cylindrical towers with trays or plates with a downspout to facilitate the flow of liquid from one tray to the other by gravity. The gas passes upward through the openings of one sort or another, in the trays and then passes through the liquid to form froth and subsequently discharges from it and then passes on to the next tray located above. Each tray of the tower acts as a stage, since there is an intimate contact between the gas phase and liquid phase in each tray. In order to provide a longer contact time, the liquid pool on each tray should be deep so that the gas bubbles will require relatively a longer time to rise through the column of the liquid. When the gas velocity is relatively high, it is dispersed very thoroughly into the liquid, which in turn is agitated into froth. This provides large interfacial areas.

However, these lead to certain operational difficulties like entrainment of droplets of liquid in the rising gas stream and a high-pressure drop for the gas in flowing through the trays. The higher pressure drop also results in high pumping cost and hence a higher operating cost. Especially, in the case of distillation, one may need to maintain a higher pressure in the reboiler, which also results in a higher boiling point that may lead to decomposition of heat sensitive compounds.

Sometimes a higher pressure drop also leads to a condition of *flooding* in which there will be a gradual build-up of liquid in each tray and may ultimately fill the entire space between the trays. The tower is said to be flooded and the liquid

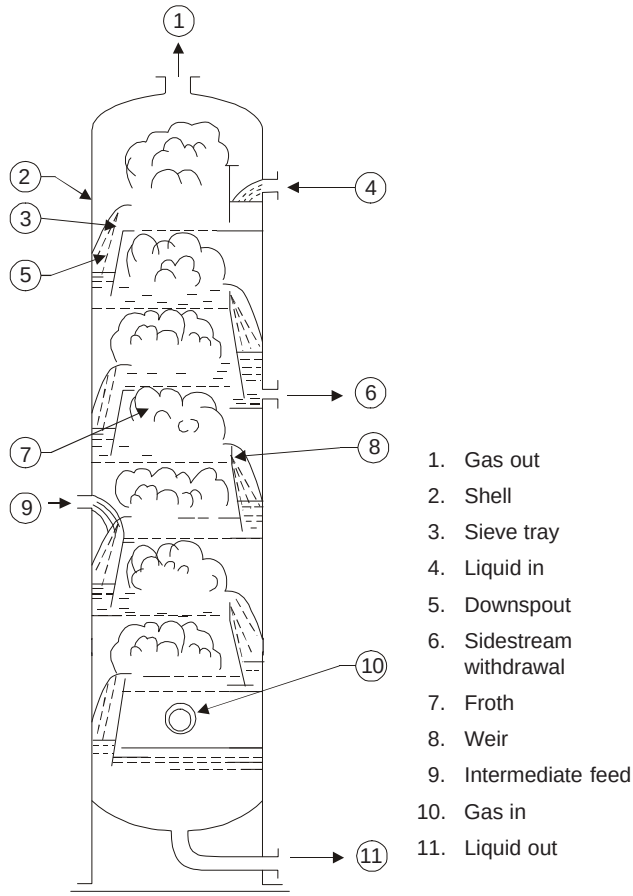


Fig. 4.1 Schematic section of a sieve-tray tower.

may escape from the top position of the column through the gas exit, which results in lowering the efficiency.

In the case of gas–liquid systems, which tend to foam excessively, high gas velocities may lead to a condition of *priming.* In such case the foam is present in the space between trays and there is a great deal of liquid getting entrained with the gas. The liquid carried is recirculated between trays and the added liquid handling load gives rise to an increase in pressure drop leading to flooding.

If the liquid rates are too low, the gas rising through the openings of the tray may push the liquid away, a phenomenon called *coning* resulting in poor gas-liquid contact. When the gas rate is too low, much of the liquid may rain down through the opening of tray, called *weeping*, thus failing to obtain the benefit of complete flow over the trays. At very low gas rates, none of the liquid reaches the downspouts and this is known as *dumping.*

4.2.1 General Features

Generally the towers are made of metals depending on the nature of gas and liquid being handled. Some of them are made of glass or at times glass lined or made of

plastics. To facilitate the maintenance work, smaller towers are fitted with hand holes and larger towers with manways. Trays are also made of metals or alloys and are fastened suitably to the shell to prevent their movement owing to surges of gas.

Tray spacing is chosen on the basis of expediency in construction, maintenance cost, flooding and entrainment. It varies from 15 cm. Tower diameter should be sufficiently large to handle the gas and liquid rates under satisfactory operating conditions. It can also be decreased by the use of increased tray spacing. Hence, the cost of tower, which depends also on the height, can be optimized with suitable tray spacing.

The liquid is drawn to the next lower tray by means of downcomers or downspouts. These may be circular pipes or portion of the tower cross section set aside for liquid flow by vertical plates. Since the liquid is agitated into froth on the tray, sufficient time must be provided in the downspout, so that the gas gets detached from the liquid and the liquid also flows down to the next lower tray. The legs of the downcomer will normally dip in the liquid in the next lower tray, which prevents short-circuiting of gas.

The depth of liquid on the tray required for gas contacting is maintained by overflow weir, which may or may not be a continuation of the downspout plate. Though straight weirs are common, V-notch weirs and circular weirs are also used. Weir length varies from 60 to 80% of tower diameter.

Having seen some of the constructional features of the towers let us now discuss the constructional features of trays.

4.3 TYPE OF TRAYS

4.3.1 Bubble Cap Trays

In these trays, chimneys or risers lead the gas through the tray and underneath caps surrounding the risers. The gas passes through a series of slots cut into the rim or skirt of each cap. The liquid depth is such that the caps are fully covered by them.

4.3.2 Sieve Trays

These are trays with perforations and the gas flows through them. The gas dispersed by the perforations, expands the liquid into a turbulent froth and results in providing enormous interfacial area for mass transfer. These trays are subject to flooding because of backup of liquid in the downspouts or excessive entrainment. In comparison to bubble caps these are quite simple and are also cost effective.

4.3.3 Linde Trays

These are slotted trays which show an alteration in the perforation pattern to influence the flow of liquid. The slots distributed all over the tray, not only reduce the hydraulic gradient in large trays but also influence the direction of liquid flow and eliminate stagnant areas. Thus, the efficiency of these trays are very high.

4.3.4 Valve Trays

These are sieve trays with large variable openings for gas flow. The perforations are covered with movable caps, which rise as the gas flow rate increases. Though the gas pressure drop is low, it is higher than sieve trays. Due to small openings the tendency to weep is also reduced.

4.3.5 Counter-flow Trays

In this liquid and vapour flow counter-currently through the same openings. Downspout is absent in these trays. They are more suited for vacuum distillation as the pressure drop is low.

4.4 TRAY EFFICIENCY

It is the fractional approach to an equilibrium stage, which is attained by a real tray. The conditions at various locations on the tray differ and are not the same. Hence, the efficiency varies at various locations.

$$\text{Point efficiency} = \left[\frac{(y_{n,\text{local}} - y_{n+1,\text{ local}})}{(y^{*}_{n,\text{local}} - y_{n+1,\text{ local}})}\right]$$

(This depends on the particular place in the tray.)

Murphree tray efficiency: This is based on the bulk average concentrations of all the local pencils of gas $= \left[\frac{(y_n - y_{n+1})}{(y^{*}_n - y_{n+1})}\right]$

(n is tray under consideration, $n + 1$ is tray below the nth tray)

$$\text{Overall tray efficiency} = \frac{\text{Number of ideal trays required}}{\text{Number of real trays required}}.$$

4.5 VENTURI SCRUBBER

This is similar to ejectors. Here a stream of absorbing liquid sprayed in the convergent duct suction draws the gas into the throat of a venturi. These devices will be useful when the liquid contains suspended solids, which may plug the plate/packed towers. The pressure drop is also low.

4.6 WETTED-WALL TOWERS

In these towers as shown in Fig. 4.2, a thin film of liquid flows down the inner wall of the empty vertical tube with the gas flowing co-currently or counter-currently. Generally the flow of gas is countercurrent to liquid flow. These are normally used for the measurement of mass transfer coefficient.

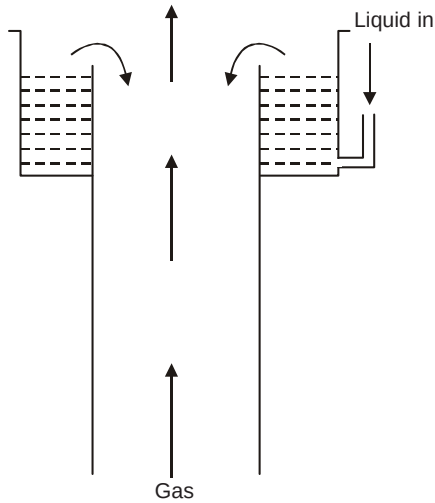


Fig. 4.2 Wetted wall column.

4.7 SPRAY TOWERS AND SPRAY CHAMBERS

In these units the liquid is sprayed into a gas stream by means of a nozzle as fine droplets. The flow may be counter-current as in vertical towers with the liquid flowing downward and gas upward. It can also be co-current as in the case of horizontal spray chambers. Their main feature is the low pressure drop for gas. However, it suffers from the disadvantage of high pumping cost for liquids as it has to flow out through fine nozzles and also a very high entrainment of liquids with gas, which necessitates the use of mist eliminators.

4.8 PACKED TOWERS

These are towers filled with packings and are used for continuous contact of liquid and gas either co-currently or countercurrently. The presence of packing gives enormous gas–liquid contact area. The liquid is distributed over the packings and trickles down through the packed bed. A typical tower is shown in Fig. 4.3.

4.8.1 Characteristics of Packings

1. Should provide large interfacial surface between liquid and gas.
2. Should possess desirable fluid flow characteristics like low pressure drop for gas and good enough to give high value mass transfer coefficients.
3. Chemically inert to the fluids being processed.
4. Should have good structural strength to permit easy handling and installation.
5. Cost effective.

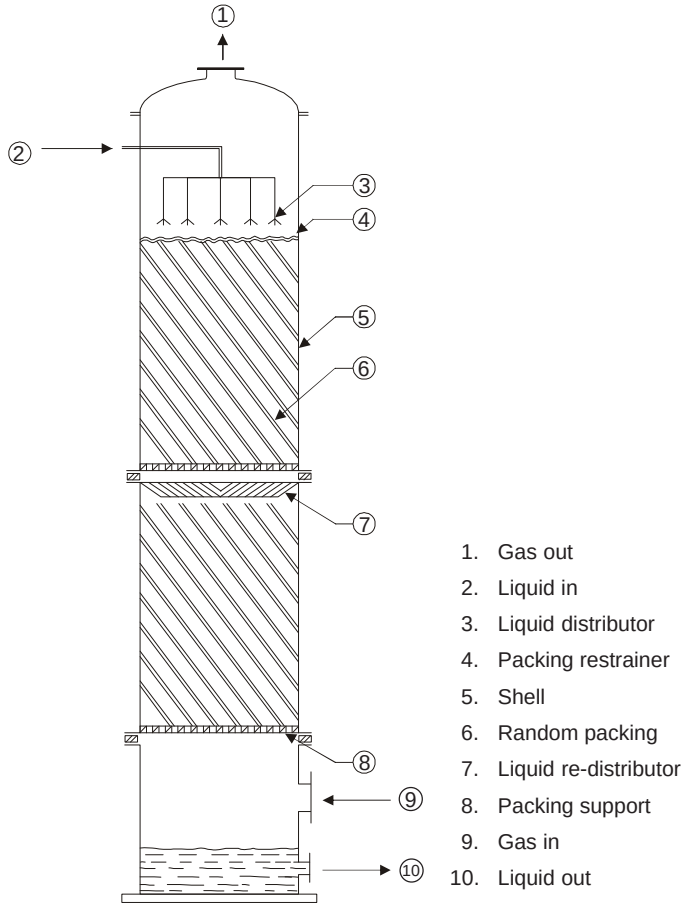


Fig. 4.3 Packed tower.

4.9 TYPES OF PACKINGS

There are two types of packings,

1. Random or dumped packing
2. Regular or stacked packing

4.9.1 Random Packing

In this type, packings are simply allowed to fall randomly. Earlier these were materials like broken stone, gravel or coke. However, due to their poor surface characteristics, they are now replaced by regular materials like Raschig rings, Berl saddles, Pall rings etc. These are made of ceramics, metals or plastics. The material of construction for these depends on the nature of fluid being handled. Ceramics are good except when alkalis and hydrofluoric acid are being used. Metals are good except in oxidising and corrosive atmospheres. Plastics deteriorate in presence of

organic solvents and also at high temperatures. Advantages with thin walled metal and plastic packings over ceramics are the lightness in weight. With smaller size, random packings offer large specific surface and hence large pressure drop. However, with larger packing sizes the cost per unit volume is less. Packings in the range of 25 mm to 50 mm are used for gas rates of 0.25 m^3/s and 50 mm or larger are used for gas rates of 1 m^3/s. During installation, the tower is filled with water and the packings are allowed to fall randomly. This prevents the disintegration of packing materials during their fall. However, when the packings are made of metals or plastics, one can drop them randomly. Some of the commonly used packings are shown in Figs. 4.4(a–d).

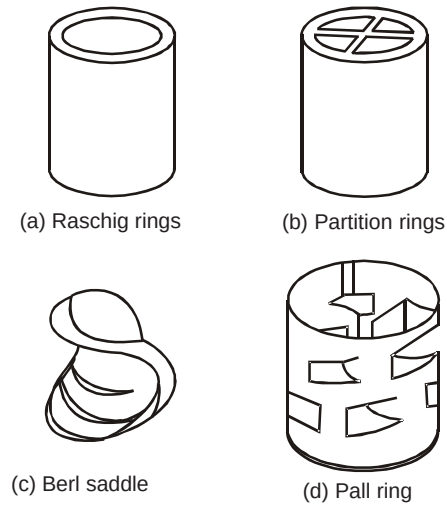


Fig. 4.4 Random tower packing.

4.9.2 Regular Packing

In these packings there is an organised manner in which the packings are arranged in the tower. The main feature of this is the low pressure for gas and higher fluid flow rates compared to random packings. Stacked packings like stacked Raschig rings, wood grids and woven wire screens are some of the examples for regular packings. Some of the commonly used regular packings are shown in Fig. 4.5.

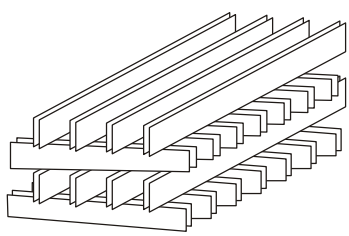


Fig. 4.5 Regular packing.

4.9.3 Shell

Tower shell is made of wood, metal, stoneware, acid proof brick, glass, plastic and metals lined with glass or plastic used as material of construction depending on the corrosive nature of the liquid or gas. They are generally circular in cross-section. In most of the instances it is made of metal because of their strength and ease of operation.

4.9.4 Packing Supports

The packing material is normally supported in the tower by means of supports. The objective of these supports is not only that they should carry the weight of packings but also ensure a proper flow of gas and liquid with minimum restriction. They are also made of different materials like metals, ceramics and plastics.

4.9.5 Packing Restrainer

In order to prevent the lifting of packings, restrainers are provided at the top of the packings. Heavy screens or bars can be used for this purpose. In the case of heavy packing materials especially of ceramics, heavy bar plates resting freely on top of the packing may be used. For plastics and other light weight packings, restrainer is attached to the tower.

4.9.6 Liquid Distributors

The liquid distribution at the top of the tower must be uniform so that the wetting of packing is uniform. A uniformly wetted packing is essential to have effectiveness in mass transfer. With non-uniform distribution of liquid one has dry packing which is ineffective for mass transfer. A ring of perforated pipe can be used in small towers. For larger diameters it is necessary to provide a number of liquid introduction points so that distribution is uniform.

4.9.7 Entrainment Eliminators

At high gas velocities the gas leaving the column may carry droplets of liquid as a mist. The mist can be removed by passing the gas through a layer of mesh made of wire or polymeric materials with about 99% voids or cyclones.

4.9.8 Channeling

As liquid flows down over the packings as thin film, the films tend to grow thicker in some places and thinner in others and liquid collects into small rivulets and flows along some localised paths. At low liquid rates much of the packing surface may be dry or, at the most, covered by a stagnant film of liquid. This effect is known as channeling. Channeling is more severe in stacked packings than in dumped packings.

4.9.9 Loading

Pressure drop in a packed bed is basically due to fluid friction. As the gas flow rate is increased, the pressure drop per unit length of packing increases. Pressure drop is low when the packing is dry. With increase in liquid flow rate, pressure drop increases as it reduces the space available for gas flow. When the packing is gradually wetted with a constant flow of liquid, initially there is a linear relationship between pressure drop and gas flow rate and is parallel to that of dry packing as shown in Fig. 4.6. The linear line becomes steeper at moderate gas velocities since the gas flow retards the down flowing liquid resulting in an increase in liquid hold up. The point at which the liquid holdup starts to increase, as indicated by a change in slope of the pressure drop–gas flow rate relationship is called the loading point.

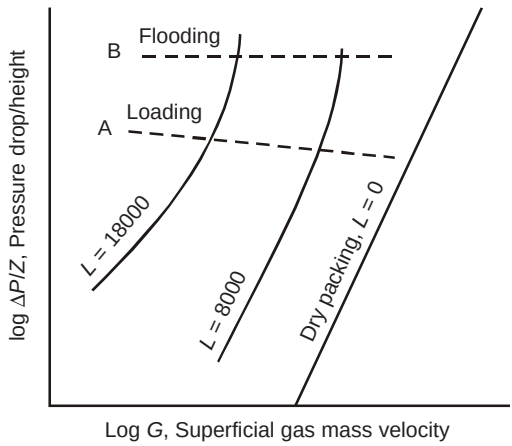


Fig. 4.6 Loading and flooding.

4.9.10 Flooding

With further increase in velocity of gas (beyond loading point) the pressure drop increases rapidly and pressure drop–gas flow rate relationship becomes almost vertical. At some portions of the column, the liquid becomes the continuous phase and the flooding point is said to be reached, and the accumulation of liquid is rapid and the entire column may be filled with liquid. Hence, while a bed is being operated, the gas velocity must be lesser than the flooding velocity and as flooding is approached, most or the entire packing surface is wetted, maximising the gas–liquid contact area.

As we design a column, we must choose a velocity lower than the flooding velocity. This will lead to a larger column diameter. The flooding velocity depends on the type and size of packing, liquid velocity and properties of liquid and gas. Several correlations are available in literature relating the pressure drop with flooding velocity for the design of packed columns.

4.10 COMPARISON OF PACKED TOWERS WITH PLATE TOWERS

Sl. No.	*Criterion*	*Packed towers*	*Plate towers*
1.	Gas pressure drop	Low–very useful in vacuum distillation	High
2.	Holdup	Low and hence very useful in handling heat sensitive materials (especially in distillation)	High
3.	Liquid/gas ratios	High values are preferred	Low values are preferred
4.	Liquid cooling	Not so easy	Heat evolved can easily be removed.
5.	Foaming systems	Can easily handle	Not so easy
6.	Removal as side streams	Not so easy	More easy
7.	Handling of corrosive systems	Not preferred	More preferred as trays can easily be replaced.
8.	Cleaning	Not easy as plate trays	Easier
9.	Thermal strain	Fragile packings tend to crush. However, metallic packings withstand the strain	Trays are more satisfactory under these circumstances.
10.	Presence of solids in gas or liquid stream	Not suitable	Not suitable
11.	Floor loading	Plastic packed towers are lighter than tray towers. Ceramic and metal packed towers are heavier than tray towers. (They are designed for the complete filling of tower with liquid)	Intermediate (They are designed for the complete filling of tower with liquid)
12.	Cost	Not easy to predict	Not easy to predict.

5

HUMIDIFICATION

5.1 INTRODUCTION

Humidification operation is a classical example for an interphase transfer of mass and energy, when a gas and a pure liquid are brought into intimate contact. The term humidification is used to designate a process where the liquid is transferred to gas phase and dehumidification indicates a process where the transfer is from gas phase to liquid phase. The matter transferred between phases in both the cases is the substance which constitutes liquid phase and it either vapourises or condenses indicating either humidification or dehumidification process.

5.2 DEFINITIONS

The substance that is transferred (vapour) is designated by A and the main gas phase is designated by B.

5.2.1 Molal Absolute Humidity (Y)

It is defined as the moles of vapour carried by a unit mole of vapour free gas.

$$Y = \left(\frac{y_A}{y_B}\right) = \left(\frac{p_A}{p_B}\right) = \left[\frac{p_A}{(P_t - p_A)}\right] = \frac{\text{moles of A}}{\text{moles of B}} \tag{5.1}$$

where y_A, y_B are moles of A and B respectively, p_A is the partial pressure of A and P_t is total pressure.

When the quantities y_A and y_B are expressed in mass, then it is called mass absolute humidity (Y') or Grosvenor humidity.

$$Y' = Y \cdot \left(\frac{M_A}{M_B}\right) = \left[\frac{p_A}{(P_t - p_A)}\right]\left(\frac{M_A}{M_B}\right) = \frac{\text{mass of A}}{\text{mass of B}} \tag{5.2}$$

5.2.2 Saturated Absolute Humidity (Y_S)

When the vapour-gas mixture is saturated, then the partial pressure becomes equal to the vapour pressure of that substance.

$$Y_S = \frac{P_A}{P_B} = \left[\frac{P_A}{(P_t - P_A)}\right] \tag{5.3}$$

When the quantities are expressed in mass,

$$Y'_S = \left(\frac{P_A}{P_B}\right)\left(\frac{M_A}{M_B}\right) = \left[\frac{P_A}{(P_t - P_A)}\right]\left(\frac{M_A}{M_B}\right) \tag{5.4}$$

5.2.3 Dry Bulb Temperature (DBT)

The temperature indicated by the thermometer by ordinary immersion in the vapour–gas mixture is called dry bulb temperature.

5.2.4 Relative Humidity or Relative Saturation (% RH)

It is normally expressed as a percentage. If p_A is the partial pressure under a given condition and P_A is the vapour pressure at any dry bulb temperature (DBT) of the mixture then,

$$\% \text{ RH} = \left(\frac{p_A}{P_A}\right) \times 100 \tag{5.5}$$

5.2.5 Percentage Saturation or Percentage Humidity (H_p)

It is defined as the percentage of humidity under given condition to the humidity under the saturated condition.

$$H_p = \left(\frac{Y}{Y_s}\right) \times 100 \tag{5.6}$$

5.2.6 Dew Point

This is the temperature t_{DP} at which a vapour–gas mixture becomes saturated when cooled at constant total pressure out of contact with a liquid. Moment the temperature is reduced below dew point, vapour will condense as a liquid dew.

5.2.7 Humid Heat

The humid heat C_S is the heat required to raise the temperature of unit mass of gas and its accompanying vapour by one degree at constant pressure.

$$C_S = C_A Y' + C_B \tag{5.7}$$

where C_A and C_B are specific heats of vapour and gas respectively.

5.2.8 Enthalpy

The enthalpy of a vapour–gas mixture is the sum of the enthalpies of the gas and of the vapour content. For a gas at a DBT of t_G, with a humidity of Y', the enthalpy relative to the reference state t_0 is,

$$\begin{aligned} H' &= \text{Enthalpy of gas + Enthalpy of vapour component} \\ &= C_B (t_G - t_0) + Y'[C_A (t_G - t_{DP}) + \lambda_{DP} + C_{A,L} (t_{DP} - t_0)] \end{aligned} \tag{5.8}$$

where λ_{DP} is latent heat of vaporisation at dew point and $C_{A,L}$ is specific heat of component A (vapour) in liquid phase.

This expression can further be simplified as low pressures are normally encountered in humidification operations. Let us consider the point P in Fig. 5.1, which actually lies on a line of constant pressure corresponding to the partial pressure of the vapour in the mixture and, for all practical purposes can be considered as lying on the line whose pressure is the saturation pressure of the vapour at the reference temperature or at P'.

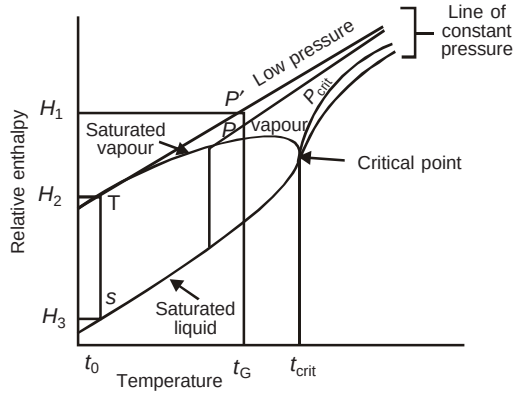


Fig. 5.1 Typical enthalpy—temperature diagram for a pure substance.

The vapour enthalpy can then be computed by the following path P′TS and becomes enthalpy per unit mass of vapour, $C_A (t_G - t_0) + \lambda_0$, where λ_0 is the latent heat of vaporisation at the reference temperature. The enthalpy of the mixture, per unit mass of dry gas is then,

$$H' = C_B (t_G - t_0) + Y' [C_A(t_G - t_0) + \lambda_0]$$
$$= C_S (t_G - t_0) + Y'\lambda_0 \qquad (5.9)$$

5.2.9 Humid Volume

The humid volume, V_H of a vapour gas mixture is the volume of unit mass of dry gas and it is accompanying vapour at the prevailing temperature and pressure. The expression for humid volume in m^3/kg of dry gas is

$$V_H = \left[\frac{1}{M_B} + \frac{Y'}{M_A}\right] 22.41 \left[\frac{(t_G + 273)}{273}\right]\left[\frac{1.013 \times 10^5}{P_t}\right]$$
$$= 8315\left[\frac{1}{M_B} + \frac{Y'}{M_A}\right]\left[\frac{t_G + 273}{P_t}\right] \qquad (5.10)$$

where P_t is the total pressure in N/m^2.

A typical psychrometric chart is shown in Fig. 5.2 from which the various properties of air–water system can be obtained. Alternatively, the equations given above can be used.

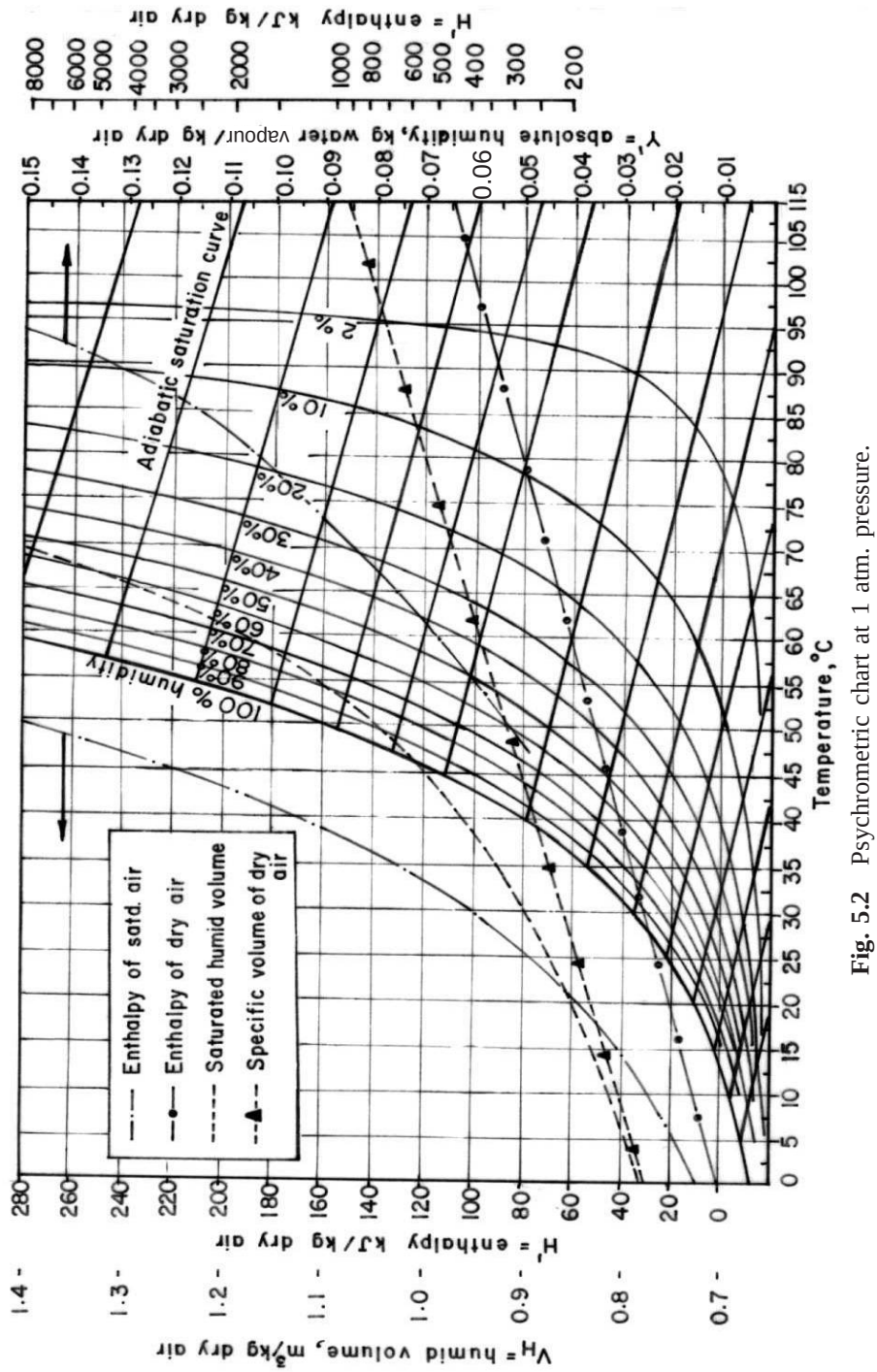


Fig. 5.2 Psychrometric chart at 1 atm. pressure.

5.3 ADIABATIC SATURATION CURVES

Consider a system as shown in Fig. 5.3. A feed stream of gas is contacted with a liquid spray and as a result of diffusion and heat transfer between the gas and liquid, the gas leaves the system humidified. Assuming the operation to be adiabatic, we obtain the following.

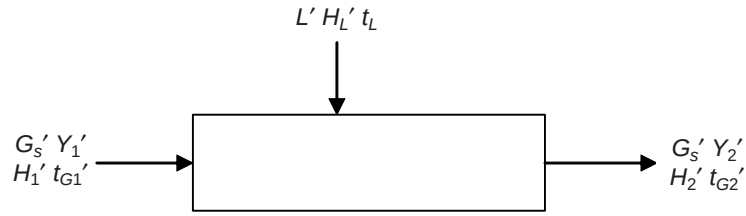


Fig. 5.3 Adiabatic chamber.

A mass balance for vapour transferred yields,

$$L' = G_S' \, (Y_2' - Y_1') \tag{5.11}$$

Enthalpy balance yields

$$G_S' \, H_1' + L' \, H_L = G_S' \, H_2' \tag{5.12}$$

i.e.

$$H_1' + (Y_2' - Y_1') \, H_L = H_2' \tag{5.13}$$

Using the definition for enthalpy, Eq. (5.13) is modified as

$$C_{S1}(t_{G1} - t_0) + Y_1'\lambda_0 + (Y_2' - Y_1')C_{A,L}(t_L - t_0) = C_{S2}(t_{G2} - t_0) + Y_2'\lambda_0 \tag{5.14}$$

If the gas mixture leaving the system is fully saturated and hence the various quantities are denoted as t_{as}, Y_{as}', H_{as} and if the liquid enters at t_{as}, Eq. (5.14) becomes

$$[C_B \, (t_{G1} - t_0) + Y_1' \, C_A \, (t_{G1} - t_0)] + Y_1'\lambda_0 + [Y_{as}' - Y_1'] \, C_{A,L} \, (t_{as} - t_0) = [C_B \, (t_{as} - t_0) + Y_{as}' \, C_A \, (t_{as} - t_0)] + Y_{as}' \, \lambda_0 \tag{5.15}$$

By subtracting $Y_1'C_A \, t_{as}$ from both sides and further simplifying, we get Eq. (5.15) as

$$(C_B + Y_1' \, C_A) \, (t_{G1} - t_{as}) = C_{S1} \, (t_{G1} - t_{as}) = (Y_{as}' - Y_1') \, [C_A(t_{as} - t_0) + \lambda_0 - C_{A,L}(t_{as} - t_0)] \tag{5.16}$$

i.e.

$$C_{S1} \, (t_{G1} - t_{as}) = (Y_{as}' - Y_1' \, (\lambda_{as}) \tag{5.17}$$

or

$$(t_{G1} - t_{as}) = (Y_{as}' - Y_1') \left(\frac{\lambda_{as}}{C_{S1}} \right) \tag{5.18}$$

Equation (5.18) is the adiabatic saturation curve on the psychromatic chart which passes through ($Y_{as}' \; t_{as}$) and ($Y'_1 \; t_{G1}$) on the 100% curve. Since C_{S1} contains the term Y_1', in Eq. (5.18) represents a curve. Thus for any vapour–gas mixture, there is an adiabatic saturation temperature t_{as}, such that if contacted with liquid at t_{as}, the gas will get cooled and humidified. If the contact time is large enough, the gas will become saturated at ($Y_{as}' \; t_{as}$) and if the contact time is insufficient, it will leave at ($Y_2' \; t_{G2}$).

5.4 WET BULB TEMPERATURE (WBT)

It is the steady state temperature attained by a small amount of liquid evaporating into a large amount of unsaturated vapour–gas mixture. To measure the wet bulb temperature, a thermometer or an equivalent temperature measuring device such as a thermocouple is covered by a wick which is saturated with pure liquid and immersed in a stream of moving gas having a definite temperature and humidity. The ultimate temperature attained is called wet bulb temperature and will be less than the dry bulb temperature (if the gas is unsaturated). If the wet bulb temperature is to be measured accurately, the following precautions will have to be followed.

(i) The wick must be completely wet.
(ii) The velocity of air should be fairly large.
(iii) The make up liquid, if supplied to the bulb should be at the wet bulb temperature.

5.4.1 Theory of Wet Bulb Thermometry

Consider a drop of liquid immersed in a rapidly moving stream of unsaturated vapour–gas mixture shown below in Fig. 5.4. If the liquid is initially at a temperature higher than the dew point of vapour, the vapour pressure of the liquid will be higher at the drop surface than the partial pressure of vapour in the gas and hence the liquid will evaporate and diffuse into the gas. The latent heat required for evaporation is supplied at the expense of sensible heat of liquid drop due to which the temperature of the droplet will reduce. As the liquid temperature is reduced below the dry bulb temperature of the gas, heat will flow from gas to the liquid at an increasing rate due to large temperature difference. Ultimately, the rate of heat transfer from the gas to the liquid will be equal to the rate of heat required for evaporation and ultimately the temperature of the liquid will remain constant at some low value, t_w.

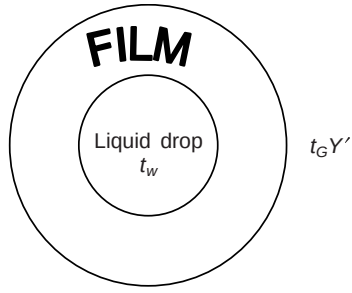


Fig. 5.4 The wet bulb temperature.

Here both heat and mass transfer occur simultaneously. The heat lost by gas to the liquid per hour, Q, is given by,

$$Q = h\ A\ (t_G - t_w) \tag{5.19}$$

where h is the heat transfer coefficient in kJ/hr m^2 K, A is the area of heat transfer in m^2, t_G is the dry bulb temperature in K and t_w is the wet bulb temperature in K.

The mass of liquid evaporated per hour,

$$W = k_C A\ [C_{A1} - C_{A2})\ M_w] \tag{5.20}$$

where k_C is mass transfer coefficient, m^2/hr, A is area of mass transfer, m^2, C_{A1} and C_{A2} are concentrations of water vapour at the liquid–gas film and surrounding air respectively in kmole/m^3 and M_w is molecular weight of water in kg/kmole.
Equation (5.20) can be modified in terms of pressure and temperature as,

$$W = k_C A \frac{[(M_w)(p_{A1} - p_{A2})]}{RT} \tag{5.21}$$

where T is temperature, K, p_{A1} and p_{A2} are partial pressure of water vapour at the interface and in the surrounding air respectively, N/m^2 and R is gas constant J/kmole K. Humidity at the interface is given by

$$Y_1' = \left[\frac{P_w}{(P_t - P_w)}\right]\left[\frac{M_w}{M_A}\right] \tag{5.22}$$

where P_w is vapour pressure of water N/m^2, P_t is total pressure of water in N/m^2 and M_A is molecular weight of air in kg/kmole.

Humidity in the surrounding air is given by,

$$Y_2' = \left[\frac{p_w}{(P_t - p_w)}\right]\left[\frac{M_w}{M_A}\right] \tag{5.23}$$

where p_w is partial pressure of water in N/m^2, P_t is total pressure of water in N/m^2 and M_A is molecular weight of air in kg/kmole.
By substituting Eqs. (5.22) and (5.23) in Eq. (5.21) is modified as

$$W = k_C \cdot \frac{A M_w}{RT}\left[\left(\frac{M_A}{M_w}\right)(P_t - P_w)Y_1' - \left(\frac{M_A}{M_w}\right)Y_2'\ (P_t - p_w)\right] \tag{5.24}$$

$$= k_C \cdot A\left(\frac{M_A}{RT}\right)[(P_t - P_w)Y_1' - Y_2'(P_t - p_w)] \tag{5.25}$$

In humidification operations, especially when the temperatures are low, P_w and p_w are small in comparison to P_t and hence neglected. Now the Eq. (5.25) gets simplified to

$$W = k_C \cdot A\left(\frac{M_A P_t}{RT}\right)(Y_1' - Y_2') \tag{5.26}$$

If λ is the latent heat of vaporisation in kJ/kg, then the heat required for evaporation is given by,

$$Q_{Evap} = W\lambda = k_C A\lambda\left(\frac{M_A P_t}{RT}\right)[Y_1' - Y_2'] \tag{5.27}$$

By the principle of wet bulb thermometry, we know

Heat lost by the gas = Heat required for evaporation (5.28)

Substituting in Eq. (5.28) from Eqs. (5.19) and (5.27), we get

$$h.A\ (t_G - t_w) = k_C\ A\lambda\left(\frac{M_A P_t}{RT}\right)\ [Y_1' - Y_2'] \tag{5.29}$$

$$(t_G - t_w) = \left(\frac{k_C}{h}\right)\lambda\left(\frac{M_A P_t}{RT}\right)[Y_1' - Y_2'] = \left(\frac{k_C}{h}\right)\lambda M_A C\ [Y_1' - Y_2'] \tag{5.30}$$

By the definition of mass transfer coefficient

$$k_y = k_C \cdot C \tag{5.31}$$

$$\therefore \qquad (t_G - t_w) = \left[k_y \lambda \cdot \frac{M_A}{h}\right][Y_1' - Y_2'] \tag{5.32}$$

From Chilton–Colburn analogy, we have

$$\left(\frac{h}{\rho C_P u_0}\right)(\mathrm{Pr})^{2/3} = \left(\frac{k_C}{u_0}\right)(\mathrm{Sc})^{2/3} = \left(\frac{k_y}{C u_0}\right)(\mathrm{Sc})^{2/3} \tag{5.33}$$

On simplification,

$$\left(\frac{\mathrm{Sc}}{\mathrm{Pr}}\right)^{2/3} = [\mathrm{Le}]^{2/3} = \left[\frac{h}{\rho C_p k_c}\right] = \left[\frac{hC}{\rho C_p k_y}\right] \tag{5.34}$$

where Le is Lewis number. Substituting Eq. (5.34) in Eq. (5.32), gives

$$(t_G - t_w) = \left[\frac{(Y_1' - Y_2')\lambda}{(\mathrm{Le})^{2/3} C_p}\right]\left[\frac{C M_A}{\rho}\right];\ \text{since } CM_A = \rho$$

Then,

$$(t_G - t_w) = \frac{(Y_1' - Y_2')\lambda}{(\mathrm{Le})^{2/3} C_p} \tag{5.35}$$

The quantity $(t_G - t_w)$ is called the wet bulb depression. For the system of air–water at ordinary conditions, the humid heat C_S is almost equal to the specific heat C_p and the Lewis number is approximately unity.

Therefore,

$$(t_G - t_w) = \frac{(Y_1' - Y_2')\lambda}{C_S} \tag{5.36}$$

Now it can be inferred on comparing Eq. (5.18) with Eq. (5.36), both the equations are identical. The adiabatic saturation curve (line) and the wet bulb temperature (line) merge in the case of air–water system only since Lewis number is unity for that system. For the other systems, adiabatic saturation curve and wet bulb temperature lines are different.

5.5 GAS–LIQUID OPERATIONS

Some of the following examples are under adiabatic operations and non-adiabatic operations:

5.5.1 Adiabatic Operations

(i) Cooling a liquid: It occurs by transfer of sensible heat and also by evaporation. Its main application is in cooling towers where the water is cooled.

(ii) Cooling a hot gas: By providing a direct contact between hot gas and liquid, cooling of gas is effected. Fouling in heat exchangers is avoided. However, this is used only when vapours which come out from liquid are not objectionable.

(iii) Humidifying a gas: For controlling the humidity in air, this is more desirable—used in preparing air for drying.

(iv) Dehumidifying a gas: By direct contact with a cold liquid, vapours are condensed and removed. This finds application in drying and recovery of solvent vapours from gases.

5.5.2 Non-adiabatic Operations

(i) Evaporative cooling: A liquid or gas inside a pipe is cooled by water flowing in a film over the outside of the pipe and the latter in turn is cooled by direct contact with air.

(ii) Dehumidifying a gas: A gas–vapour mixture is brought into contact with pipes through which a refrigerant flows and the vapour condenses on the surface of pipes.

5.6 DESIGN OF COOLING TOWER

The operation of prime importance in industries is cooling hot water from heat exchangers, condensers and the like by direct contact with air for re-use. As the latent heat of vaporisation is so large, even a small amount of vaporisation will produce a very large cooling effect. This principle is used in the design and operation of a cooling tower. These are usually carried out in some sort of packed tower and generally a countercurrent flow of gas and liquid will be adopted. The schematic arrangement with various streams and their properties are shown in Fig. 5.5.

Consider a differential section of height dZ from the bottom of the tower ① wherein the change in humidity of air is dY'.

The amount of moisture transferred to air = $G_S\ [dY']$ (5.37)

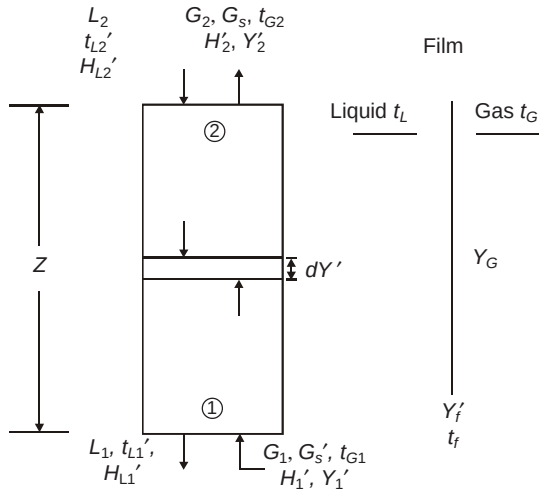


Fig. 5.5 Flow of streams in a counter-current cooling tower.

Making an enthalpy balance for the water and air stream we get,

Heat gained by gas = Heat lost by the liquid.

$$G_S \cdot dH_G = L \cdot dH_L \tag{5.38}$$

We know,

$$dH_G = C_S\, dt_G + \lambda_0\, dY'$$

$$dH_L = C_L\, dt_L$$

Integrating Eq. (5.38) between junctions (1) and (2), yield

$$G_S\,[H_{G2} - H_{G1}] = L[H_{L2} - H_{L1}]$$

i.e.
$$G_S\,[H_{G2} - H_{G1}] = LC_L\,[t_{L2} - t_{L1}] \tag{5.39}$$

Heat transfer rate per unit cross-sectional area of the bed for the liquid side is

$$h_L\, a\, dZ\, (t_L - t_f) = L\, C_L\, dt_L \tag{5.40}$$

Heat flux for the air is

$$h_G\, a\, dZ\, (t_f - t_G) = G_S\, C_S\, dt_G \tag{5.41}$$

Mass flux from the gas side is

$$k_y\, a\, dZ\, (Y_f' - Y_G') = Gs\, dY' \tag{5.42}$$

We know for air–water system, Lewis number is unity.

i.e.
$$\frac{h}{\rho C_p k_c} = \frac{h}{C_S k_y} = 1.0$$

i.e.
$$h_G = k_y C_S \tag{5.43}$$

Substituting Eq. (5.43) in Eq. (5.41), we get

$$k_y\, a\, C_S\, dZ\, (t_f - t_G) = G_S\, C_S\, dt_G \tag{5.44}$$

Multiplying both sides of Eq. (5.42) by λ, gives

$$k_y\, a dZ\, [Y_f' - Y'_G]\, \lambda = G_S\, \lambda\, dY' \tag{5.45}$$

Adding Eq. (5.44) and (5.45) gives and rearranging, we get

$$G_S\, [C_S\, dt_G + \lambda\, dY'] = k_y\, a\, dZ\, [C_S\, (t_f - t_G) + \lambda(Y_f' - Y_G')]$$

$$= k_y\, a\, dZ\, [(C_S\, t_f + \lambda Y_f') - (C_S\, t_G + \lambda Y_G')] \tag{5.46}$$

Modifying Eq. (5.46) in terms of enthalpy gives,

$$G_S\, dH_G = k_y\, a\, dZ\, [H_f - H_G] \tag{5.47}$$

Integrating Eq. (5.47) yields

$$\int_1^2 \frac{dH_G}{(H_f - H_G)} = \left(\frac{k_y a}{G_S}\right) \int_0^Z dZ = \left(\frac{k_y a}{G_S}\right) Z \tag{5.48}$$

or

$$Z = N_{tG} \cdot H_{tG} \tag{5.49}$$

where

$$N_{tG} = \int_1^2 \frac{dH_G}{(H_f - H_G)} \quad \text{and} \quad H_{tG} = \left(\frac{G_S}{k_y\, a}\right)$$

Equation (5.48) can be used to estimate the height of cooling tower. Already, we have deduced Eqs. (5.38) and (5.40)

$$G_S\, dH_G = L\, dH_L = L\, C_L\, dt_L \text{ (by definition of } dH_L)$$

According to Eqs. (5.40) and (5.47), $h_L\, a\, dZ\, (t_L - t_f) = L\, C_L\, dt_L$

$$G_S\, dH_G = k_y\, a\, dZ\, (H_f - H_G)$$

Combining Eqs. (5.38) and (5.40) and (5.47), we finally have

$$h_L\, a\, dZ\, (t_L - t_f) = k_y\, a\, dZ\, (H_f - H_G) \tag{5.50}$$

i.e.

$$\left[\frac{(H_f - H_G)}{(t_L - t_f)}\right] = \frac{h_L}{k_y} \text{ or } \left(\frac{(H_f - H_G)}{(t_f - t_L)}\right) = -\left(\frac{h_L}{k_y}\right) \tag{5.51}$$

Equation (5.51) gives the interfacial conditions which can be used in L.H.S. of Eq. (5.48) for estimating the height of the cooling tower. However, if the overall driving force and the bulk fluid properties are used, the film conditions are replaced by equilibrium properties and Eq. (5.49) takes the form

$$Z = N_{tOG} \cdot H_{tOG} \tag{5.52}$$

where

Number of overall gas transfer units, $N_{tOG} = \int_1^2 dH_G/(H^* - H_G)$

and

Height of overall gas transfer units, $H_{tOG} = G_S/K_y a$

Steps involved in the use of above procedure for the design of cooling tower.

***Step 1*:** Construct equilibrium curve
Draw the temperature—Enthalpy diagram.

***Step 2*:** Draw the operating line.
Heat lost by liquid = Heat gained by gas.

$$LC_L\ (t_{L2} - t_{L1}) = G_S\ (H_{G2} - H_{G1})$$

$$\left[\frac{(H_{G2} - H_{G1})}{(t_{L2} - t_{L1})}\right] = \left(\frac{LC_L}{G_S}\right)$$

The minimum air flow requirement is obtained by drawing a tangent to the temperature–enthalpy curve from (t_{L1}, H_{G1}) point.

***Step 3*:** Interfacial conditions.
Determine the interfacial conditions using Eq. (5.51)

$$\left[\frac{H_f - H_G}{t_f - t_L}\right] = -\left(\frac{h_L}{k_y}\right)$$

by drawing lines with a slope of $-(h_L/k_y)$ from operating line to equilibrium curve
or

$$\left[\frac{(H^* - H_G)}{(t - t_L)}\right] = -\left(\frac{h_L}{k_y}\right)$$

***Step 4*:** Find interfacial properties or equilibrium properties as the case may be and determine

$$(H_f - H_G) \text{ or } (H^* - H_G)$$

***Step 5*:** Graphically determine

$$\int \frac{dH_G}{(H_f - H_G)} = N_{tG}$$

or

$$\int \frac{dH_G}{(H^* - H_G)} = N_{tOG}$$

and find the height of the tower by $N_{tG} \times H_{tG}$ or $N_{tOG} \times H_{tOG}$

The difference between the temperature of liquid at the exit and the wet bulb temperature of entering air is called the wet bulb temperature approach. In the design of cooling towers, this is ordinarily specified to be from 2.5°C to 5°C.

Make up fresh water in re-circulating water system must be added to replace losses from entrainment (drift or windage), evaporation losses and blow down. Windage losses can be estimated as 0.1 to 0.3 per cent of re-circulation rate for induced draft towers. If the make up water also contains dissolved salts, a small amount of water is discarded to keep the salt concentration at a specified level.

5.7 RE-CIRCULATING LIQUID–GAS HUMIDIFICATION–COOLING

This is a case where the liquid enters the tower at the adiabatic saturation temperature of the entering gas mixture. This is normally achieved by re-circulating the exit liquid back to the tower. The gas not only gets cooled but also humidified in the process along the adiabatic saturation curve which passes through the entering gas conditions. Depending upon the degree of contact, the gas will approach the equilibrium conditions. A typical arrangement is shown in Fig. 5.6. Based on the temperature and humidity changes which lie in the gas phase, the mass balance is

$$G_s dY' = k_y a(Y'_{as} - Y')dZ \tag{5.53}$$

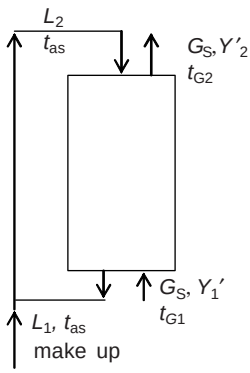


Fig. 5.6 Schematic arrangement of re-circulation humidifier.

i.e.
$$\frac{dY'}{(Y'_{as} - Y')} = \frac{k_y a}{G_s} \int_0^Z dZ \tag{5.54}$$

and since Y'_{as} is constant, integration yields,

$$\ln \left[\frac{(Y'_{as} - Y'_1)}{(Y'_{as} - Y'_2)} \right] = \left(\frac{k_y a}{G_S} \right) Z \tag{5.55}$$

i.e.
$$Z = \left(\frac{G_s}{k_y a} \right) \left\{ \ln \left[\frac{Y'_{as} - Y'_1}{Y'_{as} - Y'_2} \right] \right\} \tag{5.56}$$

$$Z = (H_{tG})\ (N_{tG}) \tag{5.57}$$

where

$$H_{tG} = \frac{G}{k_y a}$$

and

$$N_{tG} = \ln \left[\frac{(Y'_{as} - Y'_1)}{(Y'_{as} - Y'_2)} \right]$$

$$\text{Let } (\Delta Y')_{av} = \frac{[(Y'_{as} - Y'_1) - (Y'_{as} - Y'_2)]}{\ln \left[\frac{(Y'_{as} - Y'_1)}{(Y'_{as} - Y'_2)} \right]} \tag{5.58}$$

$$(\Delta Y')_{av} = \frac{(Y'_2 - Y'_1)}{\ln \left[\frac{(Y'_{as} - Y'_1)}{(Y'_{as} - Y'_2)} \right]} \tag{5.59}$$

$$\therefore \quad \ln \left[\frac{(Y'_{as} - Y'_1)}{(Y'_{as} - Y'_2)} \right] = \left[\frac{(Y'_2 - Y'_1)}{(\Delta Y')_{av}} \right] \tag{5.60}$$

Substituting Eq. (5.60) in Eq. (5.56) gives

$$Z = \left[\frac{G_S}{k_y a} \right] \left[\frac{(Y'_2 - Y'_1)}{(\Delta Y')_{av}} \right] \tag{5.61}$$

$$Z = (H_{tG})\ (N_{tG})$$

where

$$H_{tG} = \left(\frac{G_S}{k_y a} \right)$$

and

$$N_{tG} = \left[\frac{(Y'_2 - Y'_1)}{(\Delta Y')_{av}} \right]$$

Since the humidity of gas in equilibrium with liquid is Y'_{as} Murphree gas-phase stage efficiency is

$$\left[\frac{(Y'_2 - Y'_1)}{(Y'_{as} - Y'_1)} \right]$$

$$\eta = \left[\frac{(Y'_2 - Y'_1)}{(Y'_{as} - Y'_1)} \right] = 1 - \frac{(Y'_{as} - Y'_2)}{(Y'_{as} - Y'_1)} \tag{5.62}$$

From Eq. (5.57) $$\eta = 1 - \exp(-N_{tG}) \tag{5.63}$$

5.8 EQUIPMENTS

There are various types of equipments available in industries for humidification operations and they are discussed below.

5.8.1 Packed Cooling Towers

As with any other operations, the cost of installation and operation will have to be minimum. As the operation involves almost a continuous contact with water, the framework and packing materials should posses good durability in such environment. Red wood is impregnated suitably with coal tar creosote. Pentachlorophenols and the like is commonly used for framework. Nowadays, towers made of plastic are also used. The internal packing, which is a staggered arrangement of horizontal slats, could be made of wood or plastic materials. Normally, more than 90% of the tower will have voids such that the pressure drop is very low. Tray towers are not used for these operations.

Typical arrangements of cooling tower is shown in Fig. 5.7.

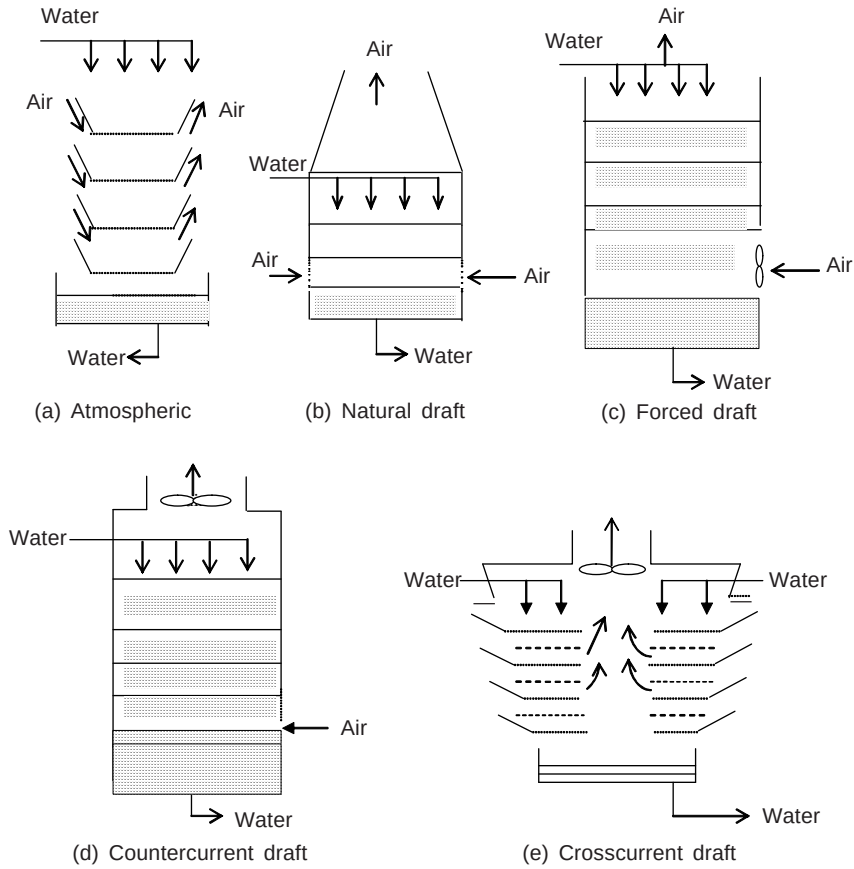


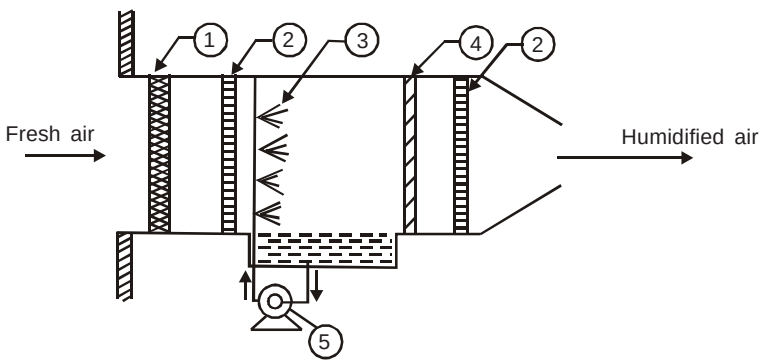
Fig. 5.7 Cooling tower arrangement.

The types, shown in Fig. 5.7(a) and (b), are atmospheric towers depend on air movement. In natural draft towers (b) depend upon displacement of warm inside air by the cool air from outside. Chimney is needed in this type of towers. These find application in places where the humidity and air temperatures are low.

In types (c), (d) and (e) due to the provision of air by fans more uniform distribution of air can be expected. Chimneys are not needed in these arrangements. Whenever fogging is common, finned type heat exchangers can be used to evaporate the fog by heat from the hot water to be cooled.

5.8.2 Spray Chambers

A typical spray chamber is shown in Fig. 5.8. They are generally used for humidification and cooling operations under adiabatic conditions. They can also be used for dehumidification process. They are provided with heaters both at the inlet and outlet points of air. Preheating of air is necessary when large humidity changes are needed. Same thing can be achieved by using hot water in the spray chamber. Similarly by using water at a low temperature, dehumidification can be achieved. Operations of this nature is useful in providing conditioned air for different applications.



1. Filter 2. Heater 3. Nozzle manifold 4. Entrainment eliminators 5. Pump

Fig. 5.8 Schematic arrangement of a spray chamber.

The process is shown in a typical psychrometric chart of Fig. 5.9.

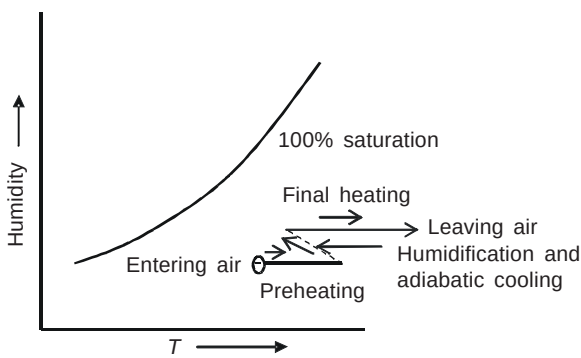


Fig. 5.9 Process in a spray chamber.

5.8.3 Spray Ponds

They are also used for cooling water where close approach to the air-wet bulb temperature is not required. Examples for this are the water fountains, where the water is thrown up as a spray in air and the liquid falls back into the collection basin. Generally, these operations show high windage losses of water.

WORKED EXAMPLES

1. An air (B) – water (A) sample has a dry bulb temperature of 50°C and a wet bulb temperature of 35°C. Estimate its properties at a total pressure of 1 atm. 1 atm = 1.0133×10^5 N/m^2

Average molecular weight of air = 28.84

Solution.

(i) Y' (Chart) = 0.03 kg water vapour/kg dry air

= 0.0483 kmol/kmol

(ii) % Humidity (Chart) = 35%

(iii) % Relative saturation = Partial pressure/Vapour pressure

Partial pressure under the given condition is given by

$$\text{Molal humidity} = \frac{p_A}{(P_t - p_A)}$$

$$0.0483 = \frac{p_A}{[1.0133 \times 10^5 - p_A]}$$

Hence, partial pressure, $p_A = 0.04672 \times 10^5$ N/m^2

Vapour pressure of water (steam tables) at 50°C = 92.55 mm = 0.1234×10^5 N/m^2

$\therefore$ % R.H. = 37.86%

(iv) Dew point = 31.5°C

(v) Humid heat = $C_S = C_B + C_A Y'$

= 1.005 + 1.884 (0.03) = 1.062 kJ/kg dry air °C

(vi) Enthalpy (for a reference temperature of 0°C)

(a) $H = C_S (t_G - t_0) + Y' \lambda_0$

$\lambda_0 = 2502$ kJ/kg

= 1.062 (50 – 0) + (0.03) (2502) = 128.16 kJ/kg

(b) Enthalpy of saturated air = 274 kJ/kg

Enthalpy of dry air = 50 kJ/kg

$\therefore$ Enthalpy of wet air = 50 + (274 – 50) (0.35) = 128.4 kJ/kg

(vii) Humid volume (V_H)

(a) $$V_H = 8315\left[\left(\frac{1}{M_B}\right)+\left(\frac{Y'}{M_A}\right)\right]\left[\frac{(t_G+273)}{P_t}\right]$$

$$= 8315\left[\left(\frac{1}{28.84}\right)+\left(\frac{Y'}{18}\right)\right]\left[\frac{325}{(1.0133\times 10^5)}\right]$$

$= 0.969$ m^3 mixture/kg of dry air

(b) Specific volume of saturated air = 1.055 m^3/kg

Specific volume of dry air = 0.91 m^3/kg

By interpolation $v_H = 0.91 + (1.055 - 0.91)\ (0.35)$

$= 0.961$ m^3/kg of dry air **Ans.**

2. Air is entering into a cooling tower with characteristics as follows: Dry bulb temperature = 25°C, Wet bulb temperature = 22°C and Pressure = 1 atm. Find (i) humidity, (ii) % humidity, (iii) % relative humidity, (iv) dew point and (v) enthalpy.

Solution.

From psychrometric chart,

(i) Humidity = 0.0145 kg water/kg dry air **Ans.**

(ii) % humidity = 61%. **Ans.**

(iii) Y_S' (Saturation humidity) = 0.0255 kg water/kg dry air

$$Y_S' = \left[\frac{P_A}{(P_t - P_A)}\right]\times\left[\frac{18}{28.84}\right]$$

$$0.0255 = \left[\frac{P_A}{(1 - P_A)}\right]\times\left[\frac{18}{28.84}\right]$$

Vapour pressure, $P_A = 0.0393$ atm.

$$Y' = \left[\frac{p_A}{(p_t - p_A)}\right]\times\left[\frac{18}{28.84}\right]$$

$$0.0145 = \left[\frac{p_A}{(1 - p_A)}\right]\times\left[\frac{18}{28.84}\right]$$

$$p_A = 0.0227 \text{ atm.}$$

$$\text{R.H.} = \left(\frac{p_A}{P_A}\right)\times 100$$

R.H. = (0.0227/0.0393) × 100 = 57.77% **Ans.**

(iv) Dew point = 19.5°C **Ans.**

(v) Humid heat, $C_S = 1005 + 1884\ Y'$

$= 1005 + 1884 \times 0.0145 = 1032.32$ J/kg °C

Enthalpy, $H = C_S\ t_G + 2502300\ Y'$

$= (1032.32 \times 25) + (2502300 \times 0.0145)$

$= 65188.25$ J/kg dry air. **Ans.**

3. A mixture of nitrogen–acetone vapour at 800 mm Hg and 25°C has percentage saturation of 80%. Calculate (i) absolute humidity (ii) partial pressure of acetone (iii) absolute molal humidity and (iv) volume percent of acetone. Assume vapour pressure of acetone at 25°C as 190 mm Hg.

Solution.

(i) $Ys' = \dfrac{P_A}{(P_t - P_A)} \times \left(\dfrac{58}{28}\right)$

$= \left[\dfrac{190}{(800-190)}\right] \times \left(\dfrac{58}{28}\right)$

$= 0.645$ kg acetone/kg nitrogen

% Saturation $= \dfrac{Y'}{Y_S'} \times 100$

$80 = \left(\dfrac{Y'}{0.645}\right) \times 100$

$Y' = 0.516$ kg acetone/kg N_2 **Ans.**

(ii) $Y' = \dfrac{p_A}{(P_t - p_A)} \times \left(\dfrac{58}{28}\right)$

$p_A = 159.54$ mm Hg **Ans.**

(iii) $Y = \dfrac{p_A}{(P_t - p_A)} = 0.249$ kmol acetone/kmol N_2 **Ans.**

(iv) Volume of 0.249 k mole acetone vapour at NTP $= 0.249 \times 22.414 = 5.581\ m^3$

Volume of 1 kmol of N_2 at NTP $= 22.414\ m^3$

Calculating volume of acetone and N_2 at 25°C, using Ideal gas law,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Volume of acetone at 25°C $= \dfrac{(298 \times 5.581 \times 760)}{(800 \times 273)} = 5.787\ m^3$

Volume of N_2 at 25°C $= \dfrac{(298 \times 22.414 \times 760)}{(800 \times 273)} = 23.243\ m^3$

Hence,

Total volume of mixture $= 5.787 + 23.243 = 29.03\ m^3$

Thus,

$$\% \text{ volume of acetone} = \left(\frac{5.787}{29.03}\right) \times 100 = 19.93\% \qquad \textbf{Ans.}$$

4. Partial pressure of water vapour in a mixture of air–water vapour at a total pressure of 106.6 kPa and a temperature of 60°C is 13.3 kPa. Express the concentration of water vapour in (i) absolute humidity (ii) mole fraction (iii) volume fraction (iv) relative humidity and (v) g water/m^3 mixture. Assume vapour pressure is 20.6 kPa at 60°C.

Solution.

(i) $Y = \dfrac{p_A}{(P_t - p_A)} = \dfrac{13.3}{(106.6 - 13.3)}$

= 0.14255 kmol water vapour/kmol dry air

$Y' = Y \times \left(\dfrac{18}{28.84}\right)$ = 0.08897 kg water vapour/kg dry air **Ans.**

(ii) Mole fraction = $\dfrac{p_A}{P_t}$ = 0.1248 **Ans.**

(iii) Volume fraction = Mole fraction = 0.1248 **Ans.**

(iv) Relative humidity = $\left(\dfrac{p_A}{P_A}\right) \times 100$ = 64.6%

Humid volume, $V_H = 8315\left(\dfrac{1}{M_B} + \dfrac{Y'}{M_A}\right)\left[\dfrac{(t_G + 273)}{P_t}\right]$

$= 8315\left(\dfrac{1}{28.84} + \dfrac{0.08897}{18}\right)\left(\dfrac{333}{106.6 \times 10^3}\right)$

= 1.029 m^3 mixture/kg dry air **Ans.**

(v) g water/m^3 mixture $= \dfrac{Y'}{V_H} = \dfrac{0.08897}{1.029}$

= 0.0865 kg water/m^3 mixture

= 86.5 g water/m^3 mixture **Ans.**

5. Air is available at a DBT and WBT of 30°C and 25°C respectively. Find its humidity, percentage saturation, humid volume, enthalpy and dew point.

Solution.

From Psychrometric chart,

Y' = 0.0183 kg water vapour/kg dry air

% Saturation = 67%

$$\text{Humid volume} = 8315\left(\frac{1}{M_B} + \frac{Y'}{M_A}\right)\left[\frac{(t_G + 273)}{P_t}\right]$$

= 0.883 m^3 mixture/ kg dry air

Humid heat, $C_s = 1005 + 1884\ Y'$

= 1039.48 J/kg dry air °C

Enthalpy = $C_s t_G + 2502300\ Y'$

= 76976.49 J/kg dry air

Dew point = 23.5°C **Ans.**

6. Air–water vapour mixture has a DBT of 55°C with humidity of 0.048 kmol water vapour/kmol dry air and 1 standard atmospheric pressure. Find absolute humidity, % humidity, humid volume, humid heat and total enthalpy.

Solution.

$$Y' = Y \times \left(\frac{18}{28.84}\right) = 0.03 \text{ kg water vapour/kg dry air}$$

% humidity = 25.5%

$$\text{Humid volume} = 8315\left(\frac{1}{M_B} + \frac{Y'}{M_A}\right)\left[\frac{(t_G + 273)}{P_t}\right]$$

= 0.978 m^3 mixture/kg dry air

Humid heat, $C_s = 1005 + 1884\ Y'$

= 1061.52 J/kg air °C

Enthalpy = $C_s t_G + 2502300\ Y'$

= 133452.6 J/kg dry air **Ans.**

7. Air at 85°C and absolute humidity of 0.03 kg water vapour/kg dry air at 1 standard atmosphere is contacted with water at an adiabatic saturation temperature and it is thereby humidified and cooled to 70% saturation. What are the final temperature and humidity of air?

Solution.

From psychrometric chart,

Final temperature = 46°C **Ans.**

Y' = 0.0475 kg water vapour/kg dry air. **Ans.**

8. Air at a temperature of 30°C and a pressure of 100 kPa has a relative humidity of 80%.

(i) Calculate the molal humidity of air.

(ii) Calculate the molal humidity of this air if its temperature is reduced to 15°C and its pressure increased to 200 kPa, condensing out some of the water.

(iii) Calculate the weight of water condensed from 100 m^3 of the original wet air in cooling to 15°C and compressing to 200 kPa.

(iv) Calculate the final volume of the wet air of part (iii).

Solution.

Data: Vapour pressure of water at 30°C = 4.24 kPa
Vapour pressure of water at 15°C = 1.70 kPa

(i) Saturated molal humidity of air, $Y'_S = \dfrac{P_A}{(P_t - P_A)}$

$= 4.24/(100 - 4.24)$

$= 0.04428$ kmol/kmol dry air.

Relative humidity $= \dfrac{p_A}{P_A}$

$0.8 = \dfrac{p_A}{4.24}$

Hence,

$p_A = 3.392$ kPa

Molal humidity $= \dfrac{p_A}{(P_t - p_A)} = \dfrac{3.392}{(100 - 3.392)}$

$= 0.0351$ kmol/kmol dry air.

$= 0.0351 \times 18/28.84 = 0.0219$ kg/kg dry air **Ans.**

(ii) Under these conditions the air will be saturated at 15°C as some water is condensed.

Saturated molal humidity of air, $Y_s = \dfrac{P_A}{(P_t - P_A)}$

$= 1.7/(200 - 1.7)$

$= 0.00857$ k mol/kmol dry air

$Y'_s = 0.00857 \times 18/28.84$

$= 0.00535$ kg/kg dry air **Ans.**

(iii) Humid volume: $V_{H,\text{Original}} = 8315\left(\dfrac{1}{M_B} + \dfrac{Y'}{M_A}\right)\left[\dfrac{(t_G + 273)}{P_t}\right]$

$$= 8315\left(\frac{1}{28.84} + \frac{0.0219}{18}\right)\left[\frac{(30 + 273)}{100000}\right]$$

$$= 0.9042 \text{ m}^3\text{/kg dry air}$$

100 m^3 of original mixture contains, $\frac{100}{0.9042}$ = 110.6 kg dry air

Therefore, water present in original air = 110.6 × 0.0219 = 2.422 kg

Water present finally = 110.6 × 0.00535 = 0.5918 kg

Water condensed from 100 m^3 of original mixture = 2.422 – 0.5918
= 1.830 kg **Ans.**

(iv) $V_{H,\text{final}} = 8315\left[\frac{1}{28.84} + \frac{0.00535}{18}\right]\left[\frac{(15 + 273)}{200000}\right]$

$= 0.4187 \text{ m}^3\text{/kg dry air}$

Final volume of mixture = 110.6 × 0.4187 = 46.329 m^3 **Ans.**

9. An air–water vapour sample has a dry bulb temperature of 55°C and an absolute humidity 0.030 kg water/kg dry air at 1 standard atm pressure. Using humidity chart, if vapour pressure of water at 55°C is 118 mm Hg, calculate the relative humidity, the humid volume in m^3/kg dry air, enthalpy in J/kg dry air and the heat required if 100 m^3 of this air is heated to 110°C.

Solution.

DBT = 55°C and Humidity = 0.030 kg water/kg dry air

$$Y = \frac{Y'}{(18/28.84)} = \left[\frac{0.03}{18}\right] \Big/ \left(\frac{1}{28.84}\right)$$

= 0.04807 kmol of water vapour/kmol of dry air

$$0.04807 = \left[\frac{p_A}{P_t - p_A}\right] = \left[\frac{p_A}{760 - p_A}\right]$$

$$p_A = 34.86 \text{ mm Hg}$$

(i) Relative humidity = $\frac{p_A}{P_A} = \frac{34.86}{118}$ = 29.5%

Saturated humidity = $\left[\frac{118}{760 - 118}\right]$

= 0.184 kmol of water vapour/kmol of dry air

% Humidity = $\left[\frac{\text{Humidity at given condition}}{\text{Humidity at saturated condition}}\right] \times 100$

$$= \left[\frac{0.04807}{0.184}\right] \times 100 = 0.261 \qquad \textbf{Ans.}$$

(ii) $V_H = 8315\left(\frac{1}{M_B} + \frac{Y'}{M_A}\right)\left[\frac{(t_G + 273)}{P_t}\right]$

$$= 8315\left[\frac{1}{28.84} + \frac{0.03}{18}\right]\left[\frac{55 + 273}{101300}\right] = 0.978 \text{ m}^3\text{/kg dry air}$$

$V_H = V_{H,\text{Dry air}} + (V_{H,\text{Sat. Air}} - V_{H,\text{Dry air}}) \times \%$ saturation (from chart)

$= 0.93 + (1.1 - 0.93)\ 0.261 = 0.974 \text{ m}^3\text{/kg dry air}$ **Ans.**

(iii) Humid heat, $C_S = 1005 + 1884\ Y'$

$= 1005 + 1884 \times 0.03$

$= 1061.52$ J/kg dry air

Enthalpy, $H = C_S\ (t_G - t_0) + Y'\lambda_0$

$= 1061.52\ (55 - 0) + 0.03 \times 2502300$

$= 133452$ J/kg dry air $= 133.452$ kJ/kg dry air

From chart: $H = H_{dry} + (H_{sat} - H_{dry}) \times 0.261$

$= 56 + (350 - 56) \times 0.261$

$= 132.7$ kJ/kg dry air **Ans.**

(iv) Heat needed if volume of air = 100 m^3

$$\text{Mass of dry air} = \frac{\text{Volume}}{\text{Humid volume}}$$

$= 100/0.978 = 102.25$ kg dry air

Enthalpy, $H_{final} = C_S\ (t_G - t_0) + Y'\lambda_0$

$= 1061.52\ (110 - 0) + 0.03 \times 2502300$

$= 191836$ J/kg dry air

$= 191.836$ kJ/kg dry air

Heat added $= (H_{final} - H_{initial}) \times$ Mass of dry air

$= (191.836 - 133.452) \times 102.25 = 5969.76$ kJ **Ans.**

10. A plant requires 2,000 kg/min of cooling water to flow through its distillation equipment condensers. The water will leave the condensers at 50°C. It is planned to design a countercurrent cooling tower in order to cool this water to 30°C from 50°C for reuse, by contact with air. Air is available at 30°C dry bulb temperature and 24°C wet bulb temperature. 30% excess air will be used and the make up water will enter at 15°C. For the packing to be used, the value

of the mass transfer coefficient is expected to be 2500 kg/(h)(m^3)($\Delta Y'$), provided the minimum liquid rate and gas rates are 12,000 and 10,000 kg/(h)(m^2) respectively. Determine the diameter of the cooling tower and makeup water to be used.

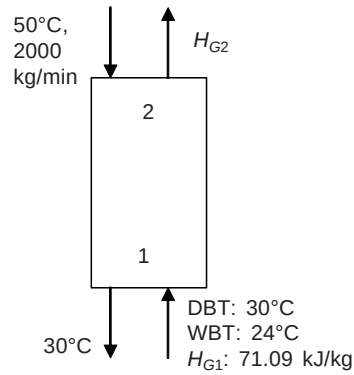


Fig. 5.10(a) Example 10.

Solution.

Flow rate of water to be cooled, L = 2000 kg/min.

Inlet temperature of water = 50°C

Outlet temperature of water = 30°C

Humidity of incoming air (DBT 30°C and WBT 24°C) = 0.016 kg/kg

Specific heat of water = 4.18 kJ/kg °C

Let us now compute the (Temperature – Enthalpy) data:

Temperature 20°C, Saturated humidity = 0.016 kg/kg (from chart)

$$\text{Enthalpy} = C_{p,air}\,(t_G - t_0) + [C_{p,w.v}.(t_G - t_0) + \lambda_0]\,Y'$$

$$= 1.005 \times 20 + [1.884 \times 0.016 \times 20] + 2502 \times 0.016$$

$$= 60.735 \text{ kJ/kg}$$

In the same manner the enthalpy for other temperatures are also estimated and have been given below:

Temp, °C	20	30	40	50	55
Enthalpy, kJ/kg	60.735	101.79	166.49	278.72	354.92

Alternatively, these values can be obtained from the psychrometric chart also corresponding to enthalpy at saturated conditions.

Enthalpy of incoming air,

$$H_{G1} = 1.005 \times 30 + [1.884 \times 0.016 \times 30] + 2502 \times 0.016$$

$$= 71.09 \text{ kJ/kg}$$

Now draw the temperature vs. enthalpy curve from the above data.

Locate (t_{L1}, H_{G1}), i.e., (30, 71.09) the operating condition at the bottom of the tower and draw the tangent to the curve.

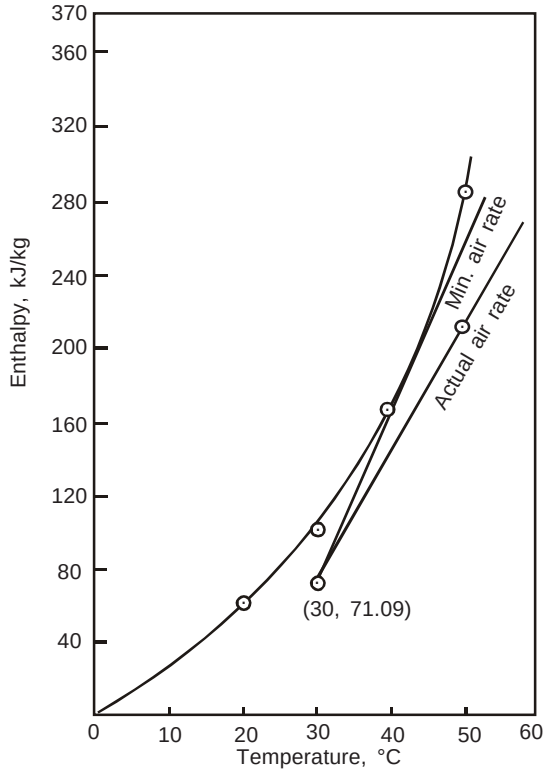


Fig. 5.10(b) Example 10, temperature–enthalpy plot.

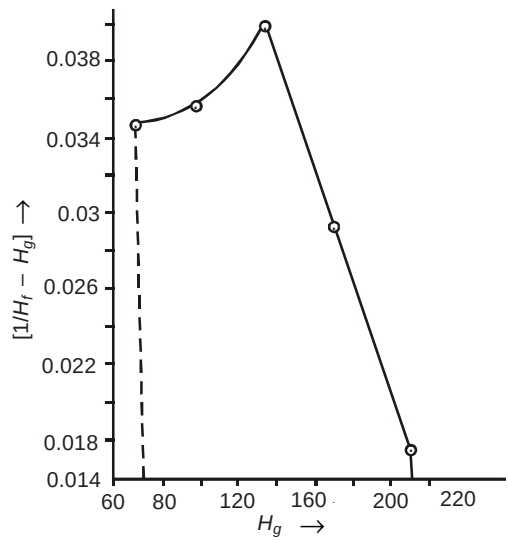


Fig. 5.10(c) Example 10, $[1/(H_f - H_G)]$ vs H_G.

Slope of the tangent = $\left(\frac{LC_L}{G_S}\right)_{\min} = \frac{(H_{G2} - H_{G1})}{(t_{L2} - t_{L1})}$

H_{G2} (from graph) = 253 kJ/kg

$$\left(\frac{LC_L}{G_S}\right)_{\min} = \frac{(253 - 71.09)}{(50 - 30)} = 9.1$$

Therefore,

$$G_{S\min} = 2000 \times 60 \times 4.18/9.1 = 55120.88 \text{ kg/h.}$$

$$G_{\text{actual}} = 1.3 \times G_{\min} = 1.3 \times 55120.88 \text{ kg/hr} = 71657.14 \text{ kg/h.}$$

Slope of the operating line = $\left(\frac{LC_L}{G_S}\right)_{\text{act}} = \frac{2000 \times 60 \times 4.18}{71657.14} = 7$

i.e $$\frac{(H_{G2} - H_{G1})}{(t_{L2} - t_{L1})} = 7$$

Therefore,

$$H_{G2}, \text{ act} = [7 \times 20] + 71.09 = 211.09 \text{ kJ/kg}$$

The minimum gas rate is to be 10000 kg/h · m^2.
Therefore, the maximum area of tower (based on gas) is given by

$$\frac{\text{Actual gas rate}}{\text{Minimum gas rate}} = \frac{71657.14}{10000} = 7.1657 \text{ m}^2$$

The minimum liquid rate is to be = 12000 kg/h m^2. Therefore, the maximum area of tower (based on liquid) is given by,

$$\frac{\text{Actual liquid rate}}{\text{Minimum liquid rate}} = \frac{2000 \times 60}{12000} = 10 \text{ m}^2.$$

The minimum area of the tower, among the ones estimated (from gas and liquid flow rates) will be chosen, as this will alone meet the expected minimum flow rates of gas and that of liquid to ensure the mass transfer coefficient of 2500 kg/(h)(m^3)($\Delta y'$). Hence, the area of tower is 7.1657 m^2. Therefore, the diameter is

$$\left[\frac{7.1657 \times 4}{3.14}\right]^{0.5} = 3.02 \text{ m}$$

The line joining points (30, 71.09) and (50, 211.09) is the operating line.

Let us assume that the resistances to mass transfer lies basically in gas phase. Hence, the interfacial conditions and the equilibrium conditions are one and the same. From the vertical lines drawn between the operating line and the equilibrium curve we get the conditions of gas and that of equilibrium conditions. The values are tabulated as follows:

Temperature, °C	*Enthalpy*, kJ/kg		$1/[H^* - H_G]$, kg/kJ
	H^*	H_G	
30	101.79	71.09	0.0326
35	133.00	103.00	0.0333
40	166.49	140.00	0.0378
45	210.00	173.00	0.0270
50	278.72	211.09	0.0148

$$N_{OG} = \int \frac{dH_G}{[H^* - H_G]}$$

The above integral is evaluated graphically by plotting $1/[H^* - H_G]$ against H_G from the above table

Area under the curve = N_{OG} = 4.26

The gas flow rate is 10000 kg/(h)(m^2) based on an area of tower of 7.1657 m^2

$$H_{OG} = \frac{G_s}{K_y a} = \frac{10000}{2500} = 4 \text{ m}$$

Height of the tower is

$$= H_{OG} \times N_{OG} = 4 \times 4.26 = 17.04 \text{ m}$$

Makeup water (M) is based on the evaporation loss (E), blow down loss (B) and windage loss (W)

$$\therefore \quad M = E + B + W$$

Windage loss = 0.2% of circulation rate

= 0.002 × 2000 = 4 kg/min = 240 kg/h

Blow down loss = Neglected

Evaporation loss is calculated assuming that the outlet air leaves fully saturated (based on the enthalpy of leaving gas 211.09 kJ/kg) and is equal to 0.064 kg/kg

Evaporation loss = 71657.14 × (0.064 – 0.016) = 3439.5 kg/h

Total makeup water = 240 + 3439.5 = 3679.5 kg/h. **Ans.**

11. A cooling tower is used to cool 1,00,000 kg/h of water from 30°C to 17°C with air entering at 8°C and a humidity of 0.004 kg/kg. Air leaves the tower at 19°C at fully saturated condition. The cross-sectional area of the tower is 14.4 m^2. Calculate air velocity in kg/h · m^2 and quantity of makeup water needed.

Solution.

Humidity of incoming air = 0.004 kg/kg dry air

Humidity of leaving air = 0.015 kg/kg dry air

Enthalpy of incoming air = 18.11 kJ/kg dry air

Enthalpy of leaving air = 57.16 kJ/kg dry air

Let W be the water evaporated

$$W = m_{\text{dry air}} (0.015 - 0.004)$$

$$= 0.011 \, m_{\text{dry air}} \qquad (1)$$

Making an energy balance,

Total heat in = Total heat out

Heat in entering water + Heat in entering air = Heat in leaving water + Heat in leaving air

$$(100000)(4.18)(30 - 0) + (m_{\text{dry air}})(H_{\text{in air}}) = (100000 - W)(4.18)(17 - 0) + (m_{\text{dry air}})(H_{\text{out air}})$$

Substituting for W from Eq. (1), we get

$$30 \times 10^5 + 18.11 m_{\text{dry air}} = (100000 - 0.011 m_{\text{dry air}})(4.18)17 + 57.16\, m_{\text{dry air}}$$

$$(30 \times 10^5 - 17 \times 10^5) = m_{\text{dry air}} [(57.16 - 18.11 - (17 \times 0.011 \times 4.18)]$$

$$m_{\text{dry air}} = 141{,}997.3 \text{ kg/h}$$

Air velocity = 9860.9 kg/h · m^2 **Ans.**

Makeup water = 141997.3 × 0.011 = 1561.97 kg/h **Ans.**

12. A horizontal spray chamber with recirculated water is used for adiabatic humidification and cooling of air. The chamber has a cross-section of 2 m^2 with air rate of 3.5 m^3/s at dry bulb temperature of 65°C and absolute humidity of 0.017 kg water/kg dry air, the air is cooled and humidified to dry bulb temperature of 34°C and leaves at 90% saturation. For the system the volumetric mass transfer coefficient may be taken as 1.12 kg/m^3 s (mole fraction). The density of the air is 1.113 kg/m^3. Determine the length of the chamber for the requirements.

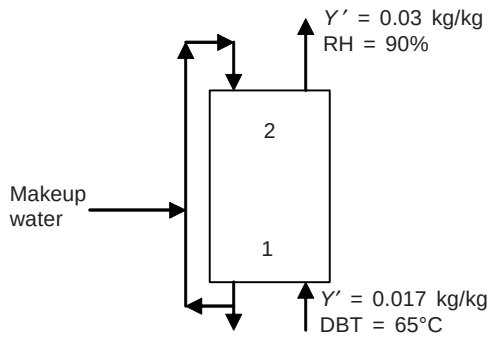


Fig. 5.11 Example 12.

Solution.

Cross-sectional area of chamber 2 m^2

Air flow rate: 3.5 m^3/s

Humidity of incoming air: 0.017 kg/kg

$$\text{Mass flow rate of air} = 3.5 \times 1.113 = 3.8955 \text{ kg/s}$$

$$G'_S = 3.8955/1.017 = 3.83 \text{ kg/s}$$

$$\text{Humidity of leaving air} = 0.03 \text{ kg/kg}$$

$$Y'_{\text{as}} = 0.032 \text{ kg/kg}$$

For recirculation humidifier

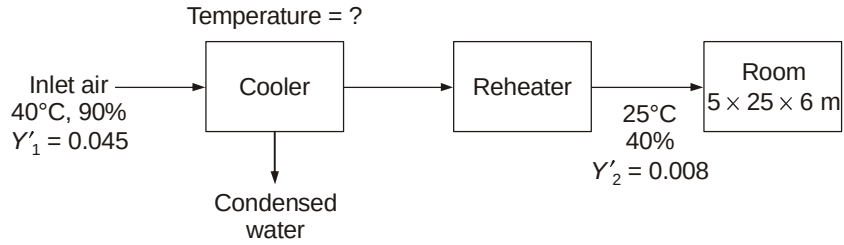
$$\ln\left[\frac{(Y'_{as} - Y_1)}{(Y'_{as} - Y'_2)}\right] = \frac{K_{ya}Z}{G_S}$$

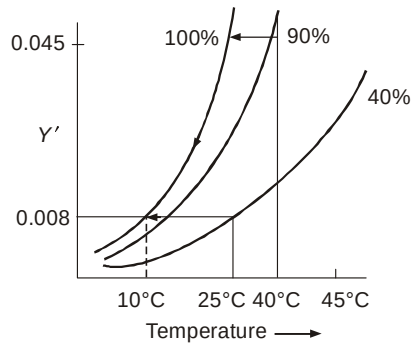
$$\ln\left[\frac{(0.032 - 0.017)}{(0.032 - 0.030)}\right] = 1.12 \times \frac{Z}{3.83}$$

$$Z = 6.89 \text{ m}$$

13. Atmospheric air at 40°C with 90% saturation is cooled and separated out the condensed water. Then it is reheated in a heat exchanger for conditioning at 25°C with 40% saturation using steam at 1 atm pressure. This conditioned air is supplied to a conference room of size 5 × 25 × 6 m without any facility for recirculation. Determine the temperature at which it is cooled and the volume of outside air at entry conditions.

Solution.





Temperature at cooler = 10°C.

Volume of the conditioned,

$$\text{air} = 5 \times 25 \times 6$$

$$= 750 \text{ m}^3.$$

Humid volume of entering air

$$V_{H_1} = \left[\frac{1}{28.84} + \left(\frac{1}{18} \times 0.045\right)\right] 22.414 \left[\frac{273 + 40}{273}\right]$$

$$= 0.9553 \text{ m}^3/\text{kg dry air.}$$

Humid volume of conditioned air

$$V_{H_2} = \left[\frac{1}{28.84} + \left(\frac{1}{18} \times 0.008\right)\right] 22.414 \left[\frac{273 + 25}{273}\right]$$

$$= 0.8592 \text{ m}^3/\text{kg dry air.}$$

$$1 \text{ kg dry air} = 0.8592 \text{ m}^3$$

$$\therefore 750 \text{ m}^2 \text{ occupies} = \frac{750}{0.8592}$$

$$= 872.875 \text{ kg dry air}$$

$$1 \text{ kg dry air} = 0.9553 \text{ m}^3$$

$$\text{then } 872.875 \text{ kg dry air} = 0.9553 \times 872.875$$

$$= 833.86 \text{ m}^3$$

Volume of entering air = 833.86 m^3.

EXERCISES

1. An air water system has DBT of 65°C with a humidity of 0.042 kg water vapour/kg dry air. What is the WBT and % saturation?

(**Ans:** 41°C and 20%)

2. A horizontal spray chamber with recirculated water is used for adiabatic humidification and cooling of air. The chamber has a cross-section of 3 m^2 with an air rate of 5 m^3/s at a dry bulb temperature of 60°C and absolute humidity of 0.017 kg water/kg dry air, the air is cooled and humidified to a dry bulb temperature of 34°C and leaves at 80% RH. For the system volumetric mass transfer coefficient may be taken as 1.12 $kg/m^3 \cdot s$ ($\Delta Y'$). The density of the air is 1.113 kg/m^3. Determine the length of the chamber for the requirements.

(**Ans:** 4.59 m)

3. A drier is used to remove 100 kg of water per hour from the material being dried. The available air has a humidity of 0.010 kg per kg of bone dry air and a temperature of 23.9°C and is heated to 68.3°C before entering the drier. The air leaving the drier has a wet bulb temperature of 37.8°C and a dry bulb temperature of 54.4°C. Calculate the (i) consumption rate of wet air, (ii) humid volume of air before and after preheating, (iii) wet bulb temperatures of air before and after preheating, and (iv) dew point of the air leaving the drier.

(**Ans:** (i) 3.672 kg/h, (ii) 0.8586 m^3/kg dry air and 0.987 m^3/kg dry air, (iii) 300 K and 303 K and (iv) 308.5 K)

4. A spray chamber of 2 metres length and 2.5 m^2 area has been used for adiabatic humidification and cooling of air. Water is sprayed into the chamber through nozzles and the coefficient of heat transfer is found to be

1360 kcal/h m^3°C. Air is passed at a rate of 3,000 cubic metres per minute at 70°C, containing water vapour 0.014 kg per every kg dry air.

(i) What exit temperature and humidity can be expected for the air?
(ii) What is the amount of make–up water?

(**Ans:** (i) 32°C and 0.032 kg/kg, assuming fully saturated (ii) 54.108 kg/min)

5. A tray drier contains 10 trays in a tier on racks at 10 cm apart. Each tray is 3.5 cm deep and 90 cm wide and there are 16 m^2 of drying surface. It is desired that the material on trays is dried by blowing a part of recycled air with fresh air, drawn through a heater kept in the drier itself. Atmospheric air enters at 26.6°C having a humidity content of 0.01. Further, it is desired that the air entering trays have a dry bulb temperature of 93.3°C and humidity 0.05 kg water/kg dry air. The air velocity at the entrance of trays is to be 3.3 m/s. The material looses water at a constant rate of 30 kg of water per hour. Determine:

(i) Percentage recirculation of air.
(ii) Heat load.

(**Ans:** (i) 63.5%, (ii) 82697.2 kJ/h)

6. Calculate the cross-sectional area and depth of packing required in wooden slats packed water-cooling tower. The tower is required to cool 37,735 kg/h of water initially at 54.4°C to 32.2°C, by counter current contact with air at atmospheric pressure having a dry bulb temperature of 25°C and a wet bulb temperature of 21.0°C. The air rate will be 30% more than the minimum air flow rate, and the superficial velocity will be 6,970 kg/h (air). H_{tOG} = 2.65 metres (enthalpy based).

Data:

Temperature, °C	*Enthalpy*, kcal/kg
4.4	8.47
10	11.30
15.5	14.72
21	18.96
26.07	24.3
32.2	31.1
37.7	39.9
43.3	51.36
48.8	66.5
54.4	86.7
57.2	99.52

(**Ans:** Area = 2.2623 m^2, Height = 10.878 m)

7. The air supply for a drier has dry bulb temperature of 26°C and a wet bulb temperature of 17°C. It is heated to 85°C by heating coils and introduced into the drier. In the drier, it cools along the adiabatic cooling line and leaves the drier fully saturated.

(i) What is its humidity initially and after heating?
(ii) What is the dew point of the initial air?

(iii) How much water will be evaporated per 100 m^3 of entering air?
(iv) How much heat is needed to heat 100 m^3 air to 85°C?
(v) At what temperature does the air leave the drier?

(**Ans:** (i) 0.01 kg/kg and 0.033 kg/kg, (ii) 285 K, (iii) 2.659 kg, (iv) 6984 2 kJ, and (v) 306.5 K)

8. 350 m^3/min of air at 70°C and 1 atmosphere pressure having a wet bulb temperature of 30°C is to be adiabatically humidified and cooled in a chamber using recirculated water. The chamber is 1.5 m. wide, 1.5 m high and 2.0 m long. The coefficient of heat transfer has been estimated to be 1200 kcal/(h)(m^3)(°C). The specific volume and specific heat of the entering air are 0.85 m^3/kg and 0.248 kcal/(kg)(°C) respectively. Determine the (i) temperature, (ii) humidity of the exit air, and (iii) Estimate the number of transfer units.

(**Ans:** (i) 46.6°C, (ii) 0.02 kg/kg, (iii) 0.754)

9. Fresh air at 25°C in which partial pressure of water vapour is 15 mm Hg is blown at a rate of 215 m^3/h first through a preheater and then adiabatically saturated in a spray chamber to 100% saturation and again reheated. This reheated air has a humidity of 0.024 kg water vapour/kg dry air. It is assumed that the fresh air and the air leaving the reheaters have the same percentage humidity. Determine (i) the temperature of air after preheater, spray chamber and reheater and (ii) heat requirements for preheating and reheating.

(**Ans:** (i) 53.5°C, 28°C and 32.5°C (ii) 7287.5 kJ and 1174.65 kJ)

10. Air is to be cooled and dehumidified by counter-current contact with water in a packed tower. The tower is to be designed for the following conditions, DBT and WBT are 28°C and 25°C respectively. Flow rate of inlet air 700 kg/h of dry air. Inlet and outlet temperatures of water are 10°C and 18°C respectively. For the entering air estimate (i) humidity (ii) %R.H. (iii) dew point and (iv) enthalpy.

(**Ans:** (i) 0.019 kg/kg, (ii) 88.08%, (iii) 23.5°C and (iv) 79.22 kJ/kg dry air)

11. Air is available at a DBT of 30°C and a WBT of 25°C respectively. Determine (i) humidity, (ii) percentage saturation, (iii) humid volume, (iv) enthalpy, and (v) dew point.

(**Ans:** (i) 0.013 kg/kg, (ii) 59%, (iii) 0.874 m^3/kg dry air, (iv) 63.4 kJ/kg, and (v) 16°C)

12. Air–water sample mixture has a DBT of 50°C and a humidity of 0.03 kg water vapour/kg dry air. If the pressure is 1 atmosphere, find (i) % humidity (ii) humid volume (iii) dew point and (iv) enthalpy.

(**Ans:** (i) 36%, (ii) 138 m^3/kg dry air, (iii) 31.75°C, and (iv) 42 kJ/kg)

13. 1.5 m^3/s of air is required for a specific operation at 65°C and 20% humidity. This is prepared from air available at a DBT of 27°C and a WBT of 18°C by direct spray of water into the air stream followed by passage over steam heated finned tube. Estimate the water and heat needed per second.

(**Ans:** 0.0496 kg/s and 128.07 kJ/kg dry air)

14. Air at a DBT of 40°C and a WBT of 30°C is to be dried by first cooling to 16°C to condense water vapour and then reheating to 25°C. Calculate (i) the initial humidity and % humidity and (ii) the final humidity and % humidity of air.

(**Ans:** (i) 0.023 kg/kg, 48% (ii) 0.012 kg/kg, 50%)

15. Air at a temperature of 20°C and a pressure of 760 mm Hg has a relative humidity of 80%.

(i) Calculate the humidity of air.

(ii) Calculate the molal humidity of this air if its temperature is reduced to 10°C and its pressure increased to 1900 mm Hg, condensing out some of the water.

Data: VP of water at 20°C = 17.5 mm Hg

VP of water at 10°C = 9 mm Hg

(**Ans:** (i) 0.01171 kg/kg, (ii) 4.759×10^{-3} kmol/kmol)

16. Air at a DBT of 35°C and WBT of 30°C and at 1 atm is passed into an evaporator. The DBT and WBT of air at the outlet of evaporator are 45°C and 38°C respectively. Determine (i) humidity and relative humidity (ii) percent saturation of air at the exit of evaporator and (iii) weight of water evaporated.

(**Ans:** (i) 0.0255 kg/kg, 70.7% (ii) 65% and (iii) 0.0155 kg)

6

DRYING

6.1 INTRODUCTION

Drying refers to the removal of relatively small amounts of moisture from a substance which is generally a solid. However, in some specific cases, it includes the removal of moisture from liquids and gases as well. Drying is generally a final step in the production process and the product from the dryer is often sent for final packaging.

6.2 DEFINITIONS OF MOISTURE AND OTHER TERMS ON DRYING

When an insoluble solid is dried, the moisture is lost to the surrounding air and the solid attains an equilibrium moisture content depending on the relative saturation of air. The different moisture contents exhibited by a substance when exposed to air of different saturation levels is shown in Fig. 6.1.

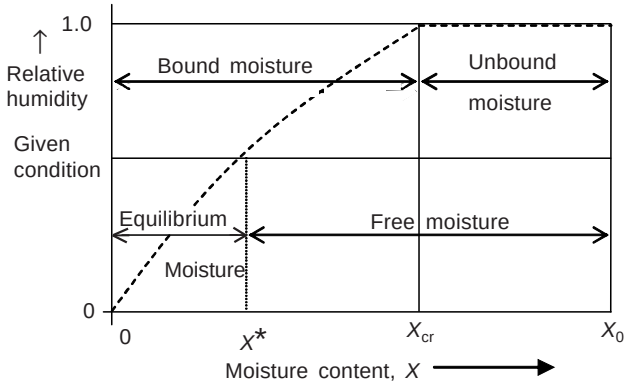


Fig. 6.1 Moisture contents at different humidity conditions.

6.2.1 Moisture Content (Wet Basis), X

This is defined as the weight of moisture per unit weight of wet substance.

6.2.2 Moisture Content (Dry Basis), X

This is defined as the weight of moisture per unit weight of bone dry substance.

6.2.3 Equilibrium Moisture, X*

This is the moisture content of a substance when it is at equilibrium with a given partial pressure of the vapour. It is the limiting moisture content to which a given material can be dried under specific conditions of air temperature and humidity.

6.2.4 Bound Moisture

This refers to the moisture contained by a substance which exerts an equilibrium vapour pressure less than that of the pure liquid at the same temperature. Liquid may become bound by retention in small capillaries, by solution in cell or fibre walls or by adsorption on solid surface.

6.2.5 Unbound Moisture

This refers to the moisture contained by a substance which exerts an equilibrium vapour pressure equal to that of the pure liquid at the same temperature.

6.2.6 Free Moisture (X–X*)

This is the moisture contained by a substance in excess of equilibrium moisture. Only free moisture can be removed with air of given temperature and humidity. It may include both bound and unbound moisture.

6.2.7 Critical Moisture Content

It is the moisture content when the constant rate drying period ends and falling rate drying period starts.

6.2.8 Fibre–Saturation Point

It is the moisture content of cellular materials (e.g. wood, fibre) at which the cell walls are completely saturated while the cavities are liquid free. It may be defined as the equilibrium moisture content, as the humidity of the surrounding atmosphere approaches saturation.

6.2.9 Constant Rate Drying Period

Constant rate drying period is that drying period during which the rate of water removed per unit area of drying surface is constant.

6.2.10 Falling Rate Drying Period

It is a drying period during which the instantaneous drying rate continually decreases. Some of the substances show a linear behaviour and some show non-linear behaviour during falling rate drying period. In certain instances we observe both. The linear one is due to unsaturated surface drying wherein one sees certain dry spots on the drying surface and the non-linear one is observed when the moisture movement is controlled by diffusion mechanism.

6.2.11 Funicular State

It is a condition that occurs in drying a porous body when capillary suction results in air being sucked into the pores. This generally indicates the first falling rate drying period. The drying rate varies linearly with free moisture content in this period.

6.2.12 Pendular State

As the water is progressively removed from the solid, the fraction of the pore volume that is occupied by air increases. When the fraction reaches a certain limit, there is insufficient water left to maintain continuous film across the pores, the interfacial tension in the capillaries breaks, and the pores filled with air, which now becomes continuous phase. The left out water is relegated to small isolated pores and interstices of the pores. This state is called the pendular state and it generally refers to the second falling rate drying period. During this period, the variation of drying rate with free moisture content is non-linear.

6.3 HYSTERESIS

Many substances exhibit different equilibrium moisture relationships during the adsorption and desorption of moisture as shown in Fig. 6.2. This phenomenon of following different paths is known as hysteresis in drying.

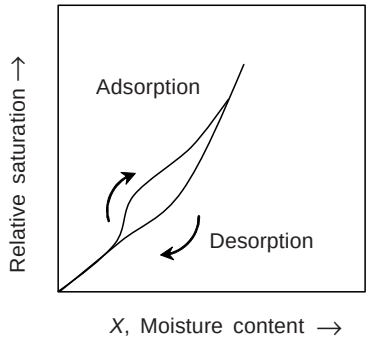


Fig. 6.2 Hysteresis in drying.

6.4 DRYING OF SOLUBLE SOLIDS

Soluble solids show insignificant equilibrium moisture content when exposed to gases whose partial pressure of vapour is less than that of the saturated solution of the solid. A typical trend is shown in Fig. 6.3.

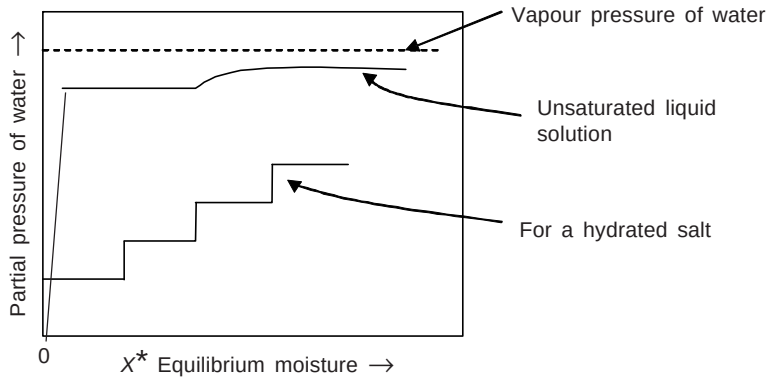


Fig. 6.3 Equilibrium moisture content of soluble solids.

6.5 CLASSIFICATION OF DRYING OPERATIONS

Drying operations can be broadly classified as,

(i) Batch drying
(ii) Continuous drying

6.5.1 Batch Drying

Here the material to be dried is fed to a drier and exposed to drying media under unsteady state conditions.

6.5.1.1 Drying test

In order to determine the drying schedule and also the size of drying equipments, it is necessary to know the time required for drying a substance. The rate of drying is determined by suspending a substance in a chamber in a stream of air and measuring its weight periodically. The operation is carried out under constant drying condition by maintaining the same temperature, humidity and air flow rate. A typical drying curve is shown in Fig. 6.4. This is drawn by estimating weight itself or determining moisture content on dry basis and plotting against time.

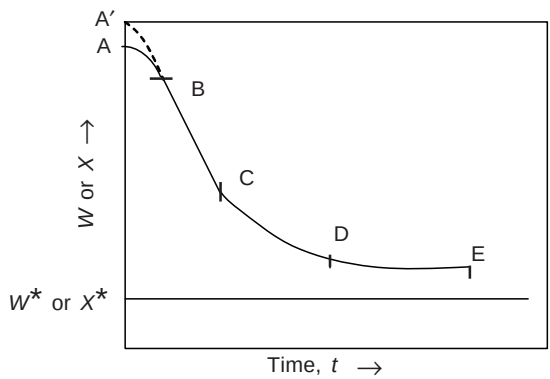


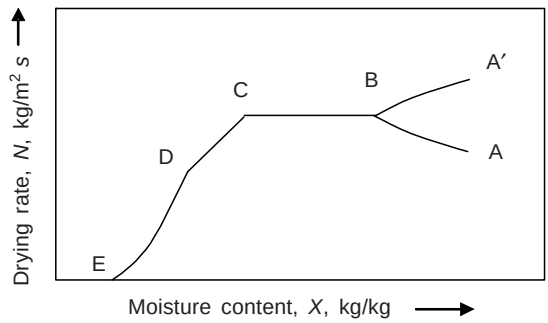
Fig. 6.4 Drying curve.

From the drying curve, the rate of drying, N is calculated as,

$$N = \left(-\frac{L_S}{A}\right)\left(\frac{dX}{dt}\right) = -\frac{1}{A}\cdot\frac{dw}{dt} \tag{6.1}$$

where L_S is the mass of bone dry solid; A is the drying surface, from which drying takes place.

However, in the case of through circulation drying, A is defined as the cross-sectional area of the bed, perpendicular to the direction of air flow. When the drying rate is plotted against moisture content (on dry basis), the rate curve is obtained and a typical rate curve is shown in Fig. 6.5.



ED is internal movement of moisture controls (second falling rate period)
DC is unsaturated surface drying (first falling rate)
CB is constant rate drying period
A′B and AB are unsteady drying (initial stage)

Fig. 6.5 Rate curve.

When the solid to be dried is fully wet, the surface will be covered with a thin film of liquid and will have unbound moisture. If the air is unsaturated with a humidity of Y and if the gas at the liquid surface is Y_S (saturated humidity), the rate of drying at constant rate period is expressed as,

$N_C = k_y(Y_S - Y)$, where k_y is the mass transfer coefficient.

6.5.1.2 Time of drying

From Eq. (6.1) we have,

$$N = \left(\frac{-L_S}{A}\right)\left(\frac{dX}{dt}\right)$$

Rearranging and integrating Eq. (6.1) to determine the time needed to dry the material from X_1 to X_2, we get,

$$t = \int_0^t dt = \left(\frac{-L_S}{A}\right)\int_{X_1}^{X_2}\frac{dx}{N} \tag{6.2}$$

(a) The constant rate period: The drying period is said to be constant rate period when both X_1 and X_2 are greater than critical moisture content X_c. Under such conditions, drying rate remains constant and $N = N_C$.

The Eq. (6.2) can now be rearranged as $\int_0^t dt = \left(\frac{-L_S}{AN_C}\right) \int_{X_1}^{X_2} dX$ and on integration this yields,

$$t = L_S \frac{(X_1 - X_2)}{AN_C} \tag{6.3}$$

(b) The falling rate period: If X_1 and X_2 are both less than X_C, the drying rate N decreases with decrease in moisture content. Equation (6.2) can be integrated graphically by plotting $(1/N)$ in y-axis against moisture content X in x-axis or by using a numerical technique. However, when N varies linearly with X in the region CE, the drying rate can be expressed mathematically as,

$$N = aX + b \tag{6.4}$$

where a is the slope of the line and b is a constant. The Eq. (6.2) can be integrated between the limits

$t = 0,\ x = X_1$

$t = t,\ x = X_2$ and we get

$$t = \int_0^t dt = \frac{L_S}{A} \int_{X_2}^{X_1} \frac{dX}{(aX + b)} = \left(\frac{L_S}{aA}\right) \ln \left[\frac{(aX_1 + b)}{(aX_2 + b)}\right] \tag{6.5}$$

However, $N_1 = a\ X_1 + b$, $N_2 = a\ X_2 + b$ and $a = \left[\frac{(N_1 - N_2)}{(X_1 - X_2)}\right]$

Substituting these in Eq. (6.5) gives

$$t = \left[\frac{L_S(X_1 - X_2)}{A(N_1 - N_2)}\right] \ln \left[\frac{N_1}{N_2}\right] = \frac{L_S(X_1 - X_2)}{A \cdot N_m} \tag{6.6}$$

where N_m is the logarithmic mean rate of drying.
In a specific case of drying from X_c to X^*

$$N_C = aX_C + b \tag{6.7}$$

$$N^* = 0 = aX^* + b \tag{6.8}$$

Subtracting Eq. (6.8) from Eq. (6.7) gives,

$$N_C = a\ (X_C - X^*) \tag{6.9}$$

Also, subtracting Eq. (6.8) from Eq. (6.4), we get

$$N = a\ [X - X^*] \tag{6.10}$$

Eliminating a in Eq. (6.10), using Eq. (6.9)

$$N = N_C \frac{(X - X^*)}{(X_C - X^*)} \tag{6.11}$$

Replacing N_1 and N_2 in Eq. (6.6) in terms of N_C and N^* and also X_1 and X_2 in terms of X_C and X^*, we get,

$$t = \left[\frac{L_S(X_C - X^*)}{AN_C}\right] \ln \left\{\frac{[N_C(X_1 - X^*)(X_C - X^*)]}{[(X_C - X^*)\, N_C(X_2 - X^*)]}\right\}$$

$$t = \left[\frac{L_S(X_C - X^*)}{AN_C}\right] \ln \left[\frac{(X_1 - X^*)}{(X_2 - X^*)}\right] \tag{6.12}$$

6.6 PARAMETERS AFFECTING DRYING RATE DURING CONSTANT RATE DRYING PERIOD

Gas velocity, gas temperature, gas humidity and thickness of the drying solid are the parameters which affect drying rate.

6.6.1 Effect of Gas Velocity (G)

When radiation and conduction effects are present, the effect of gas rate will be less significant. However, when they are negligible, then drying rate N_C is proportional to $G^{0.71}$ for parallel flow of gas and to $G^{0.37}$ for perpendicular flow of gas.

6.6.2 Effect of Gas Temperature

Increased air temperature, T_G increases the driving force, $(T_G - T_S)$ for heat transfer and hence N_C is directly proportional to $(T_G - T_S)$. T_S is the surface temperature of drying solid and is assumed to be at the WBT condition during constant rate drying.

6.6.3 Effect of Gas Humidity

As the humidity of air decreases, the driving force $(Y_S - Y)$ available for mass transfer increase, and hence N_C is proportional to $(Y_S - Y)$. Y_S is the saturation humidity of air corresponding to T_S.

6.6.4 Effect of Thickness of Drying Solid

When heat transfer occurs through the solid, N_C increases with decrease in solid thickness. However, if drying occurs from all surfaces, N_C is independent of thickness.

6.7 MOISTURE MOVEMENT IN SOLIDS

When drying takes place, moisture moves from the inner core to the external surface and evaporates. The nature of movement influences the drying during the

falling rate period and the following theories have been proposed to explain the moisture movement in solids.

6.7.1 Liquid Diffusion

Due to concentration gradients between the higher concentration in depths of the solid and the low concentration at the surface, moisture movement takes place. This type of phenomenon is exhibited by substances like soap, glue, gelatin, textiles and paper.

During constant rate period, the rate of moisture movement from inner core and the rate of removal of moisture from the surface balance each other. However, after sometime, dry spots appear on the surface resulting in unsaturated surface drying and then the moisture movement from the solid takes place which is entirely controlled by the diffusion rates within the solid. Whenever, the constant drying rates are very high, the drying substance may exhibit only diffusion controlled falling rate drying.

6.7.2 Capillary Movement

In some of the porous solids, moisture moves through the capillaries in them which is quite similar to the burning of lamp with wick. These capillaries extend from water reservoir to the drying surface. As the drying process is initiated, the moisture starts moving by capillarity to the drying surface and maintains a uniformly wetted surface, which corresponds to constant rate drying period. Subsequently air replaces the water and the wetted area at the surface also decreases leading to unsaturated surface drying. After sometime, when the sub-surface water also dries up, the liquid surface recedes into capillaries and water evaporates from there setting in second falling rate period. This phenomenon is exhibited by clays, paints and pigments.

6.7.3 Vapour Diffusion

When one surface of a wet solid is heated and the other surface allows the drying to take place, the moisture gets vapourised from the hot surface and diffuses outward as a vapour from the other surface.

6.7.4 Pressure Diffusion

When bound moisture is removed from a colloidal non-porous solid, it tends to shrink when the substance is dried very rapidly. The moisture present on the surface is removed very quickly and the moisture movement from the inner core to the outer surface will not be equal to the rate of removal of moisture from the surface. During this process, an impervious membrane forms and prevents the movement of moisture under such circumstances from the inner core to the surface. The outer surface will be fully dry whereas the inner core will be wet under such conditions. This phenomenon is called **case hardening**.

However, under certain circumstances, the shrinkage of outside layers of solid may also squeeze out moisture to the surface.

6.8 SOME MORE ASPECTS ON FALLING RATE DRYING

The moisture movement during the falling rate period is governed by either unsaturated surface drying or internal diffusion controlling mechanism.

6.8.1 Unsaturated Surface Drying

In this phase, the rate of drying will vary linearly with moisture content. The moisture removal mechanism is same as that in the constant rate period and the general effects of temperature, humidity, gas flow rate and thickness of the solid are the same as for constant rate drying.

6.8.2 Internal Diffusion Controlling

In this period of drying, the moisture movement is controlled by the pores in the drying substance. The drying rate decreases with decrease in moisture content.

6.9 THROUGH CIRCULATION DRYING

When a gas passes through a bed of solids, the drying zone varies as shown in Fig. 6.6. At the point, where the gas enters, maximum drying occurs and a zone of drying of bound moisture forms. In this zone there is a gradual rise in temperature. This zone is followed by a zone of drying unbound moisture. The temperature in this zone remains constant and the particles are at their wet bulb temperature. The zone of drying unbound moisture is followed by a zone of initial moisture concentration where the solids also remain at their initial temperature. The gas leaves the system fully saturated.

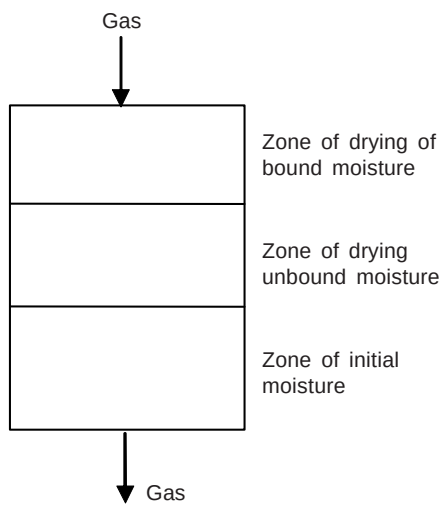


Fig. 6.6 Through circulation drying.

6.9.1 The Rate of Drying of Unbound Moisture

Let us consider that a gas of humidity Y_1 enters the bed at a moisture free flow rate of G_S, kg/m^{2}hr. The maximum drying rate N_{max} will occur if the gas leaving the bed is saturated at adiabatic saturation temperature and hence at the humidity Y_{as},

$$N_{max} = G_S (Y_{as} - Y_1) \tag{6.13}$$

However, if Y_2 is the humidity of leaving air and N is the drying rate, then

$$N = G_S (Y_2 - Y_1) \tag{6.14}$$

For a differential section of the bed, where the change in humidity for incoming air of humidity Y_1 is dY and leaves at a humidity of "Y", the rate of drying dN is given by

$$dN = G_S \, dY = k_y \, dS \, (Y_{as} - Y) \tag{6.15}$$

where S is the interfacial surface per unit area of bed cross-section. If a is the interfacial area per unit volume of bed and if Z_S is the bed thickness, then

$$dS = a \cdot dZs \tag{6.16}$$

Substituting Eq. (6.16) in Eq. (6.15) and integrating it yields,

$$\int_{Y_1}^{Y_2} \left(\frac{dY}{Y_{as} - Y} \right) = \int_{0}^{Zs} \left(\frac{k_y a dZ_S}{G_S} \right) \tag{6.17}$$

$$\ln \left[\frac{(Y_{as} - Y_1)}{(Y_{as} - Y_2)} \right] = N_{tG} = \left(\frac{k_y a Z_S}{G_S} \right) \tag{6.18}$$

where N_{tG} is the number of gas phase transfer units in the bed. From Eqs. (6.13), (6.14) and (6.15), we get

$$\frac{N}{N_{max}} = \left[\frac{(Y_2 - Y_1)}{(Y_{as} - Y_1)} \right] = 1 - \left[\frac{(Y_{as} - Y_2)}{(Y_{as} - Y_1)} \right]$$

$$= 1 - \exp^{-N_{tG}} = 1 - \exp^{-\left(\frac{k_y a Z_s}{Gs} \right)} \tag{6.19}$$

Using Eq. (6.19) we can predict drying rate when k_y a is known.

6.9.2 Drying of Bound Moisture

In order to analyse the drying rate in this regime, it is preferable to make experimental investigation as both particle and gas characteristics play a significant role.

6.10 CONTINUOUS DIRECT HEAT DRIER

It is the one in which there is a continuous flow of solid and gas through the drier. The drying takes place as the solid moves through the drier.

6.10.1 Material and Energy Balance

Consider a continuous drier with gas flowing countercurrent to the flow of solids which is dried from a moisture content of X_1 to X_2. The flow rate of dry solids is L_S. The gas enters at a moisture free flow rate of G_S and at a temperature of t_{G_2} and a humidity of Y_2. It leaves at a temperature of t_{G_1} and at a humidity of Y_1.

A schematic diagram shown in Fig. 6.7 indicates the flow of various streams and their conditions.

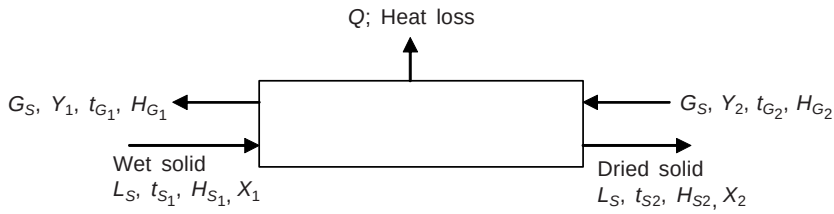


Fig. 6.7 Schematic diagram of a continuous direct heat drier.

Balance for moisture gives,

$$L_S X_1 + G_S Y_2 = L_S X_2 + G_S Y_1 \tag{6.20}$$

i.e.
$$L_S (X_1 - X_2) = G_S (Y_1 - Y_2) \tag{6.21}$$

Enthalpy of the wet solids, H_S is given by

$$H_S = C_S (t_S - t_0) + X(C_A)\ (t_S - t_0) + \Delta H \tag{6.22}$$

where C_S is specific heat of solids, C_A is the specific heat of moisture, X is the moisture content and ΔH is the integral heat of wetting.
An energy balance gives,

Heat entering = Heat leaving + Heat loss

$$L_S H_{S_1} + G_S H_{G_2} = L_S H_{S_2} + G_S H_{G_1} + Q \tag{6.23}$$

For an adiabatic operation, where there is no heat loss, $Q = 0$. Sometimes the solids are conveyed through drier in some supports in which case the heat carried in and out by these materials should also be accounted.

6.10.2 Rate of Drying for Continuous Direct Heat Driers

Direct heat driers fall under two categories, namely, whether the high temperature or low temperature prevails in the drier.

6.10.2.1 Drying at high temperature

Figure 6.8 shown below indicates the temperature profile of both solid and gas.

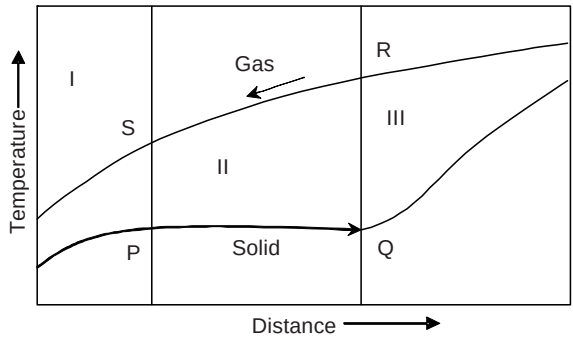


Fig. 6.8 Temperature profile of gas and solid in a continuous drier.

For analysis, let us divide the drier into three zones. The zone I is a preheating zone, where the solid is heated and very little drying takes place. In zone II, the temperature of solid remains fairly constant and the surface and unbound moisture are evaporated. The point Q corresponds to the critical moisture content. In zone III, unsaturated surface drying and removal of bound moisture takes place. There is a sharp increase in the temperature of the solid. The zone II represents a major portion of the drier and it will be interesting to see how the gas temperature and humidity varies in this section.

The point R (see Fig. 6.9) denotes the condition of air entering zone II. When the drying takes place adiabatically without any heat loss, the variation of gas temperature and humidity is along the adiabatic saturation line RS_1. The solid temperature could be A (corresponding to point Q) or A′ (corresponding to point P).

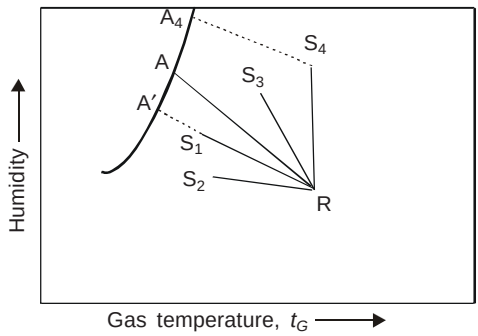


Fig. 6.9 Temperature–humidity variation in a continuous drier.

However, if heat losses are there, then the path could be RS_2. In case the heat is added within the drier then the path would be RS_3. The path will be RS_4 if the gas temperature is kept constant and the solid temperature is likely to be A_4.

Now, let us make an energy balance. The heat lost q_G by the gas is transferred partially to the solid q and partially lost as Q. For a drier of differential length, dz

$$dq_G = dq + dQ \tag{6.24}$$

i.e.
$$dq = dq_G - dQ = U \cdot ds\ (t_G - t_S) \tag{6.25}$$

$$= U \cdot adz\ (t_G - t_S) \tag{6.26}$$

where S is the interfacial surface/cross-sectional area, a is the interfacial surface/drier volume, U is the overall heat transfer coefficient, t_G is the gas temperature and t_S is the solid temperature.

$$dq = G_S \cdot C_S \cdot dt_G \tag{6.27}$$

where G_S is the dry gas flow rate, C_S is the humid heat of gas and dt_G is drop in gas temperature.

i.e. $$dq = G_S \cdot C_S \cdot dt_G = Ua\ (t_G - t_S)\ dz \tag{6.28}$$

$$dN_{toG} = \frac{dt_G}{(t_G - t_S)} = \left(\frac{Ua}{G_S C_S}\right) \tag{6.29}$$

i.e. $$N_{toG} = \frac{\Delta t_G}{\Delta t_m} \tag{6.30}$$

i.e. $$H_{toG} = \frac{G_S C_S}{Ua} \tag{6.31}$$

Δt_m is logarithm of average temperature difference from gas to solid.

6.10.2.2 Drying at low temperatures

Since the drying takes place at low temperature the preheating of solid is not a major factor. The preheating zone merges with zone II (Refer Fig. 6.10). In zone II unbound and surface moisture are removed and the moisture content of leaving solid reaches critical moisture content, X_C as in drying at high temperatures. The unsaturated surface drying and evaporation of bound moisture occurs in zone III. The humidity of incoming gas increases from Y_2 to Y_C as it leaves zone III.

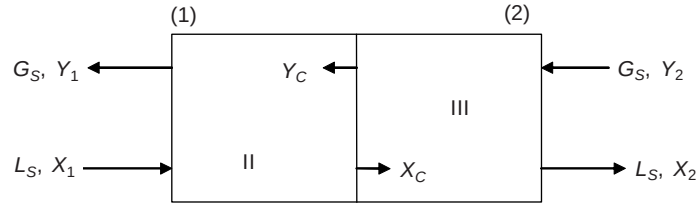


Fig. 6.10 Continuous countercurrent drier of low temperature operation.

The retention time can be calculated using Eq. (6.1)

$$t = t_{II} + t_{III} \tag{6.32}$$

$$= \frac{L_S}{A}\left[\int_{X_C}^{X_1} \frac{dX}{N} + \int_{X_2}^{X_C} \frac{dX}{N}\right] \tag{6.33}$$

where (A/L_S) is the specific exposed drying surface.

In zone II, $X > X_C$, the rate of drying $N = N_C$ is given by Eq. (6.2)

$$t_{II} = \frac{L_S}{A}\int_{X_C}^{X_1} \frac{dX}{[k_y(Ys - Y)]} \tag{6.34}$$

Also a material balance yields,

$$G_S \cdot dY = L_S \cdot dX \tag{6.35}$$

Substituting for dX in Eq. (6.34) from Eq. (6.35), we get

$$t_{III} = \frac{L_S}{A} \cdot \frac{G_S}{L_S} \int_{Y_C}^{Y_1} \frac{dY}{[k_y(Y_s - Y)]} \tag{6.36}$$

Assuming Y_S to be constant (which will correspond to saturation humidity at the wet bulb temperature of incoming air and there is no heat loss).

$$t_{II} = \left(\frac{L_S}{A}\right)\left(\frac{G_S}{L_S}\right)\left(\frac{1}{k_y}\right) \cdot \ln \left[\frac{(Y_S - Y_C)}{(Y_S - Y_1)}\right] \tag{6.37}$$

In zone III, $X < X_C$ we have from Eq. (6.11)

$$N = \frac{[N_C(X - X^*)]}{(X_C - X^*)}$$

Applying Eq. (6.2) to constant rate period, we get

$$N_C = k_y (Y_S - Y) \tag{6.38}$$

Substituting for N_C in Eq. (6.11) from Eq. (6.38), we get,

$$N = \frac{k_y(Y_S - Y)(X - X^*)}{(X_C - X^*)} \tag{6.39}$$

Substituting in Eq. (6.33) from Eq. (6.39), we get

$$t_{III} = \frac{L_S(X_C - X^*)}{A \cdot k_y} \int_{X_2}^{X_C} \frac{dX}{(Y_S - Y)(X - X^*)} \tag{6.40}$$

This cannot be integrated directly as Y also varies with X.
A simple material balance for moisture across any section yields

$$G_S[Y - Y_2] = L_S\ [X - X_2]$$

$$Y = Y_2 + (X - X_2)\left(\frac{S_S}{G_S}\right) \tag{6.41}$$

The humidity Y_S can be determined from the humidity Y_1 of inlet gas. X^* can be determined experimentally for the given humidity Y_2. By substituting for Y in Eq. (6.40) from Eq. (6.41) we can estimate $t_{III.}$
However, for the case where $X^* = 0$, we shall analyse Eq. (6.40).
By differentiating Eq. (6.41), we get,

$$G_S \cdot dY = L_S \cdot dX \tag{6.42}$$

From Eq. (6.40) using Eq. (6.42), we get

$$t_{III} = \left(\frac{L_S}{A}\right)\left(\frac{X_C}{k_y}\right)\left(\frac{G_S}{L_S}\right)\int_{Y_2}^{Y_C} \frac{dY}{(Y_S - Y)\left[(Y - Y_2)\left(\frac{G_S}{L_S}\right) + X_2\right]}$$

$$= \left(\frac{L_S}{A}\right)\left(\frac{X_C}{k_y}\right)\left(\frac{G_S}{L_S}\right)\frac{1}{\left[(Y_S - Y_2)\frac{G_S}{L_S} + X_2\right]} \ln\left[\frac{X_C\,(Y_S - Y_2)}{X_2(Y_S - Y_C)}\right] \qquad (6.43)$$

The above Eq. (6.43) cannot be applied when the internal diffusion controls the drying process. Since we have assumed that in the falling rate period, drying rate varies linearly with free moisture content. Whenever the drying is controlled by internal diffusion, t_{III} can be determined experimentally. In the case of parallel flow driers,

$$t_{II} = \left(\frac{L_S}{A}\right)\left(\frac{G_S}{L_S}\right)\left(\frac{1}{k_y}\right)\int_{Y_1}^{Y_C} \frac{dY}{(Ys - Y)}$$

$$= \left(\frac{G_S}{L_S}\right)\left(\frac{L_S}{A}\right)\left(\frac{1}{k_y}\right) \ln\left[\frac{(Y_S - Y_1)}{(Y_S - Y_C)}\right] \qquad (6.44)$$

$$t_{III} = L_S \frac{(X_C - X^*)}{A \,.\, ky} \int_{X_2}^{X_C} \frac{dX}{[(Y_S - Y)(X - X^*)]}$$

$$= \left(\frac{G_S}{L_S}\right)\left(\frac{L_S}{A}\right)\left(\frac{X_C}{k_y}\right)\frac{1}{\left[(Y_S - Y_C)\frac{G_S}{L_S} - X_2\right]} \ln\left[\frac{X_C(Y_S - Y_2)}{X_2(Y_S - Y_C)}\right] \qquad (6.45)$$

6.11 DRYING EQUIPMENTS

Drying equipments are classified on different basis, as mentioned below.

6.11.1 Based on Contact between Drying Substance and Drying Material

(a) Direct contact: In these dryers, there is a direct contact between hot gas and the drying substance. For example, rotary dryer, spray dryer, etc.

(b) Indirect contact: There is no direct contact between gas and drying substance. For example, drum dryer, mechanically agitated dryer, etc.

6.11.2 Based on the Type of Operation

(a) Batch dryer: Tray dryer, freeze dryer, etc.

(b) Continuous dryer: rotary dryer, mechanically agitated dryer and tunnel dryer.

6.11.3 Based on the Nature of Substance being Dried

(a) Materials in sheets or masses carried on conveyors or trays,
  (i) Batch dryers: Atmospheric tray and vacuum tray.
  (ii) Continuous dryers: Tunnel.

(b) Materials which are granular or loose,
  (i) Rotary dryers: rotary and roto–Louvre
  (ii) Turbo dryers
  (iii) Conveyors
  (iv) Filter–dryer combinations

  (all these dryers operate on continuous basis)

(c) Materials in continuous sheets,
  (i) Cylinder dryer
  (ii) Festoon dryers

(d) Materials in the form of pastes or sludges or caking crystals,
  (i) Atmospheric agitator dryers (mechanically agitated)
  (ii) Vacuum dryers

(e) Materials in the form of solution,
  (i) Drum dryers: Atmospheric and vacuum
  (ii) Spray dryers

(f) Special dryers,
  (i) Freeze dryers
  (ii) Infrared dryers
  (iii) Dielectric dryers

Some of the dryers used in industries have been described as follows.

6.11.4 Atmospheric Compartment Dryers

This dryer consists of a rectangular chamber with insulating walls and a door. The chamber is partitioned to enable the loading of materials in these compartments. In some of the designs, trucks or cars may run into the dryer with provisions for closing doors. These dryers also have provisions for heating the air inside and also circulating it over the trays/shelves. Dampers are also provided to regulate the flow of air.

6.11.5 Vacuum Compartment Dryer

It consists of a rectangular chamber with a number of shelves. These shelves are hollow and during operation they are filled with steam or hot water. Steam enters through a steam inlet manifold and there is a provision to remove the condensate and also non condensables. The material to be dried is loaded in trays and placed on shelves. The door is closed after placing the material to be dried and vacuum is created in the chamber by means of a vacuum pump. The water released from the substance is condensed in a condenser placed between the dryer and the vacuum pump. If the vapour from the drying substance has a value, these dryers enable us in the recovery of these vapours.

These dryers are used for materials that cannot withstand high temperatures, as in the case of atmospheric compartmental dryer, such as pharmaceuticals. They are also suitable for systems where in the contact between air or other oxidizing gases are to be avoided.

6.11.6 Tunnel Dryers

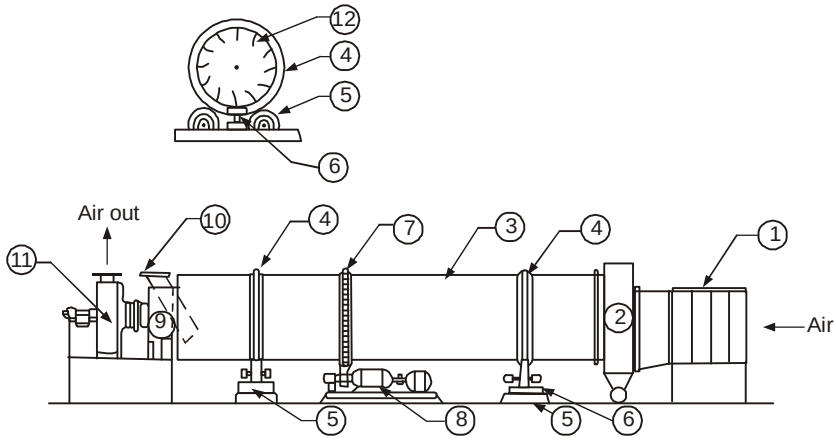
The compartmental dryers operate on batch basis. The dryer is in the form of a long tunnel. However, if a continuous operation is desired the material to be dried can be loaded in trucks or cars and sent through the tunnel. The air flow could be either co-current or countercurrent at right angles to the path of travel of trucks. Heaters are also provided in the dryer in different sections so that the air may be sent through the trucks, taken to a re-heater and sent back again to the trucks in the same section.

These are used for drying bricks, ceramic products and other material which have to be dried rather slowly but in large quantities.

6.11.7 Rotary Dryers

It consists of a cylindrical shell slightly inclined to the horizontal and mounted so that it can be rotated. The feed material enters at the elevated end of the dryer and due to the rotation of the shell, the material slowly moves towards the lower end and finally leaves the dryer. Inside the shell, flights are present which help in lifting the solids and showering them over air stream. Heat can be supplied from outside to the shell of the dryer or through the hot air which flows either co-currently or countercurrently. The rotating shell carries forged tires which ride on rolls. Thrust rolls prevent the endwise travel of the shell. The shell is driven by a gear arrangement which is driven by a motor. The shell rotates at a peripheral speed of 20 to 25 m/min.

This dryer is used for drying granular or crystalline substances which has to be handled in bulk and cannot be used for sticky material. A typical drier is shown in Fig. 6.11.



1. Air Heater 2. Stationary hood 3. Dryer shell 4. Tires 5. Supporting rolls 6. Thrust rolls
7. Drive gear 8. Motor and speed reducer 9. Air discharge hood 10. Feed chute
11. Discharge 12. Flights

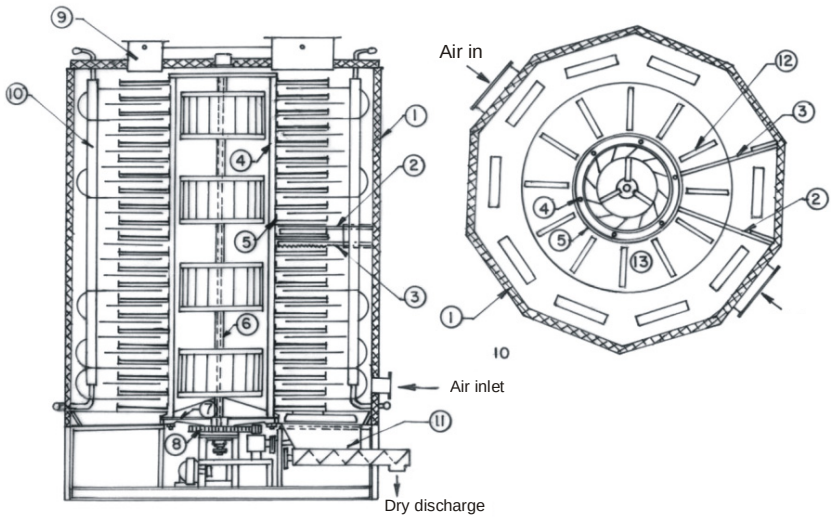
Fig. 6.11 Rotary dryer.

6.11.8 Roto–Louvre Dryer

This is a modified form of rotary dryer described in Section 6.11.7. The dryer consists of an outer cylindrical shell and a tapered inner shell made of number of overlapping plates. The space between the shell and the overlapping plates is divided into longitudinal channels by ribs. These channels are open at the large end to receive hot air and are closed at the smaller end. The louvres are inclined against the flow of rotation so that they do not lift the solid but merely serve to keep the solid from dropping down the channels. The air from these channels flow through the bed of solids. As the material entering the dryer has a higher moisture content, the tapering provides a thinner bed to start with and hence a lower resistance for gas/air flow. As the air flows through a bed of solids, the air comes more nearly in equilibrium with the material and drying rate is faster and hence a shorter size of the shell. Further, as the material is not lifted and showered down, the tendency for degradation of material is less. Here, the material rolls along the bottom of inner surface, and reaches the product outlet region.

6.11.9 Turbo Dryer

This dryer, as shown in Fig. 6.12, is of vertical orientation with a cylindrical or polygonal shell. At the bottom we have a base plate which is driven by a gear coupled to a motor assembly. From this base plate vertical rods rise to the top, which are suitably connected to a guide bearing. Around these rods are cylindrical bands of sheet metal to which are attached wedge shaped trays. The whole assembly rotates as a single unit. Feed enters through the feed opening, fills the trays and as they rotate they pass under a fixed leveling scraper. After one revolution they pass under another scraper that scrapes the charge on the tray through the transfer slots to fall to the tray below. Every tray is provided with a



1. Casing 2. Scraper 3. Leveler blade 4. Rods 5. Metal Bands 6. Fan shaft 7. Base casting
8. Drive gears 9. Feed opening 10. Re-heaters 11. Chute for dried product 12. Transfer slots 13. Trays

Fig. 6.12 Turbo dryer.

leveler scraper and a scraper to push the solids to next lower tray. The dried material leaving the last tray is finally scraped into a hopper and discharged out of the dryer by a screw conveyor. Air enters the drier at the bottom of the shell. With the help of fans attached to the central shaft, air is circulated over the material on trays. As it picks up moisture, its temperature drops and the saturation level goes up. In order to maintain a fairly steady temperature, the circulating air is continuously reheated with the finned tube heaters. The humidified air finally leaves the top of the dryer. The major advantage of this dryer is, it occupies a lesser floor space and low power consumption compared to rotary dryer and Roto–Louvre dryers.

6.11.10 Conveyor Dryers

It is a type of tunnel dryer in which the granular solids to be dried is loaded in a conveyor belt of wire screen of a suitable mesh. The belt screen retains the solid but permits the flow of air through the conveyor and also the solid. In some cases the flow of air could be downwards also.

6.11.11 Filter Dryer Combination

When the material to be dried is in suspension in a liquid, a rotary continuous filter is used. It consists of a drum which is fed at the top surface of the drum. After the filtration, hot air is blown through the filtered material, so that filtration and drying take place at the same time.

6.11.12 Cylinder Dryers

A typical cylinder drier is shown in Fig. 6.13. This consists of steam heated rolls with facilities to remove the condensate. The rolls are driven by motors. The material to be dried in the form of continuous sheets of paper or textiles passes over these rolls and gets spooled to the final product spool. The speed of the train of rolls must suitably be adjusted, through it is complicated, to compensate the shrinkage of drying material. A typical dryer assembly in a paper manufacturing unit may have 50 to 75 rolls.

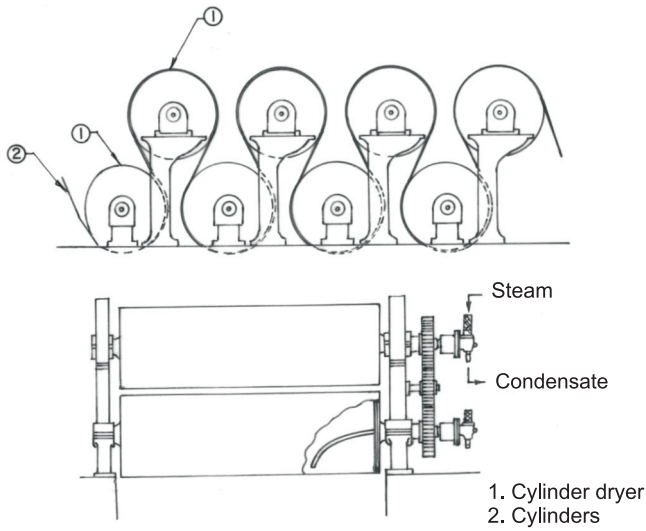


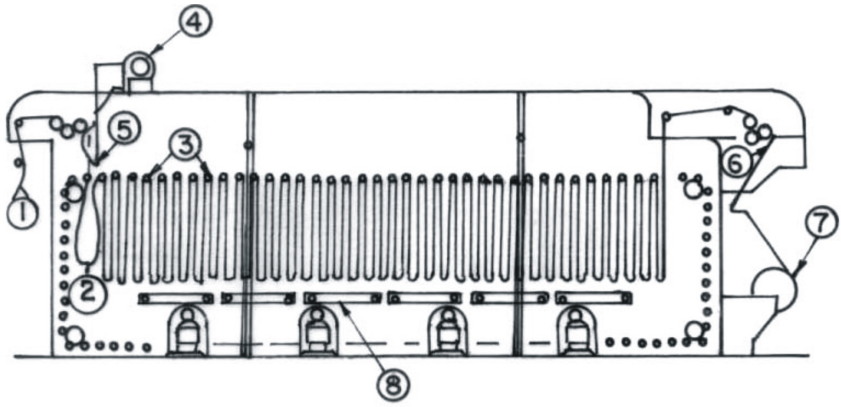
Fig. 6.13 Cylinder dryer.

6.11.13 Festoon Dryers

In this dryer, as shown in Fig. 6.14, wet sheet of material passes over a series of rolls and drops down to form a series of loops. A continuous chain conveyor carrying cross bars is so timed with respect to the speed of the sheet that the loops drop down to a certain predetermined length pass as the next roll comes along to catch the next loop. The proper loop formation is ensured by the loop blower. The material as it leaves the dryer is fully dried and rolls on to the product roll. Hot air needed for drying is obtained by passing air through a bank of finned tube heaters. The moist air leaves through air discharge port.

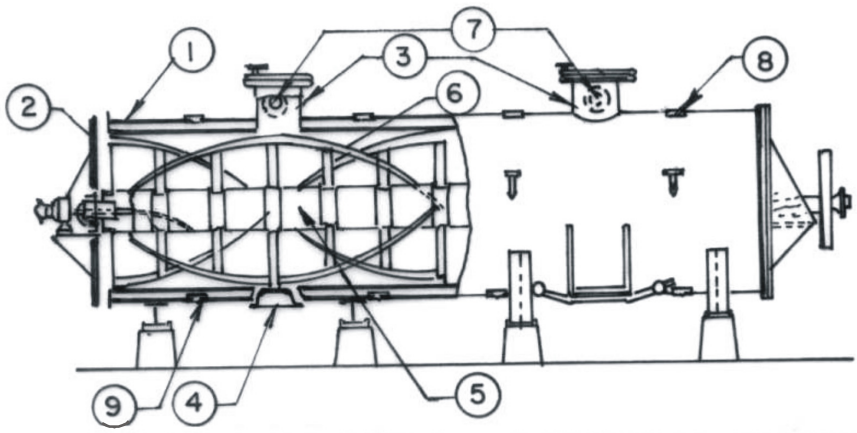
6.11.14 Mechanically Agitated Dryer

A mechanically agitated dryer, as shown in Fig. 6.15, consists of a horizontal jacketed cylindrical shell. Inside is a central shaft carrying agitator blades



1. Entering sheet 2. Festoons 3. Cross bars 4. Loop blower
5. Air nozzle 6. Exit sheet 7. Product roll 8. Heaters.

Fig. 6.14 Festoon dryer.



1. Jacketed shell 2. Heads 3. Feed inlet 4. Discharge door 5. Agitator shaft
6. Agitator blades 7. Vapour outlets 8. Steam inlets 9. Condensate outlets

Fig. 6.15 Mechanically agitated dryer.

arranged in a way, that one set of blades moves the material in one direction and the other set in the opposite direction. Heat needed for drying is sent through the jacket. The central shaft is rotated by a motor. The feed enters through the feed inlet point and leaves from the product outlet point.

6.11.15 Drum Dryer

A typical double roll drum dryer consists of two large steam heated cast iron rolls with a smooth external surface. The rolls rotate toward each other and the liquid to be dried is fed directly into the V-shaped space between the rolls. In small units generally there is no special feeding device whereas in larger dryers swinging pipe or a traveling discharge pipe is used to keep the feed

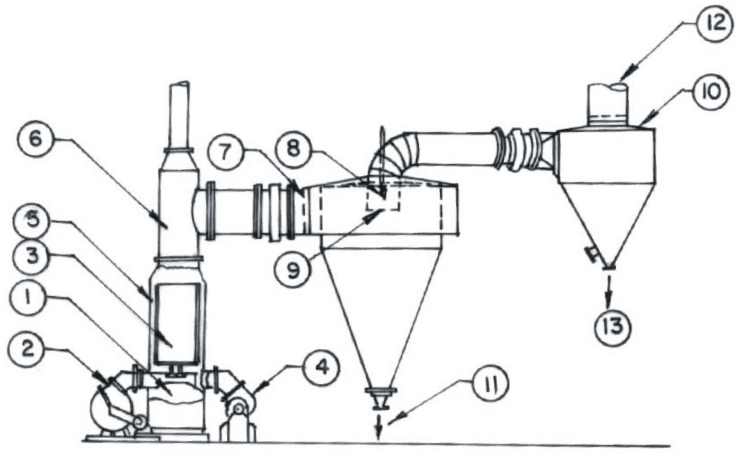
uniform. Thickness of the material deposited on the rolls is determined by the space between them. Doctor knife placed near the top of the rolls on the outside, removes the dried product which is taken away with a conveyor. In very small capacity driers, only one roll will be present which will be dipping, in a feed trough. The doctor knife at the lower part of the drum removes the dried solid.

6.11.16 Vacuum Drum Dryers

When the substance being dried is heat sensitive, the drying can be accomplished under vacuum. A typical vacuum drum dryer consists of a single drum made of cast iron. The drum is heated with steam with facility to remove the condensate. The pool of feed liquid is maintained at the bottom of the unit and is pumped up to the spreader trough which is very close to the bottom section of the drum. The dried product is removed by a doctor knife and taken out subsequently by a conveyor. The entire unit is maintained under vacuum.

6.11.17 Spray Dryers

Spray dryers are extensively used for drying solutions, pharmaceuticals, detergent products, fruit juices and milk. Many designs are available. In a typical design shown in Fig. 6.16, the liquid to be dried is atomised and introduced into the large drying chamber with a conical bottom. The droplets are dispersed into a stream of hot air. The particles of liquid evaporate rapidly and dry before they can be carried to the sides of the chamber and the bulk of the dried powder which results, falls to the bottom of the chamber from where they are removed by a stream of air to the dust collector. For atomisation, either spray nozzles or rapidly rotating disks are used. Nozzles



1. Burner 2. Primary air blower 3. Combustion chamber 4. Secondary air blower
5. Secondary air passage 6. Hot air pipe 7. Hot air value 8. Spray nozzle
9. Air discharge pipe 10. Dust collector 11. Main product discharge
12. Air discharge 13. Dust discharge

Fig. 6.16 Spray dryer.

are relatively inflexible in their operating characteristics and do not permit even moderate variation in liquid flow rates with out large changes in droplet size. They are also subjected to rapid erosion and wear. Rotating disks are about 30 cm in diameter and rotate at speeds of 3000 to 12000 rpm. They also easily handle variations in liquid flow rates.

6.11.18 Freeze Drying

Substances which cannot be heated even to moderate temperatures, are frozen by exposure to very cold air and placed in a vacuum chamber, where the moisture sublimes and is pumped off by steam jet ejectors or mechanical vacuum pumps. This is used for drying fish, vegetables like peas, vitamins and other heat sensitive materials.

6.11.19 Infrared Drying

It has been used in the drying of paint films on objects such as automobile bodies. The radiation is usually supplied by infrared lamps and the material to be dried travels in a tunnel lined with banks of such lamps. This process is suitable only for the drying of thin films on the surface of the material to be dried and never for cases where the water or solvent to be removed is deep inside the solid. It is a very expensive drying operation.

6.11.20 Dielectric Drying

In this operation the object to be dried is passed through a very high frequency (2 to 100 $\times$ 10^6 cycles) electrostatic field. This generates heat uniformly throughout the object. Its only important field of application is in polymerising the resin that forms the bond between layers of plywood which is a rare drying operation. However, some people will disagree in calling it as a very expensive drying operation.

WORKED EXAMPLES

1. Air containing 0.005 kg of water vapour per kg of dry air is preheated to 52°C in a dryer and passed to the lower shelves. It leaves these shelves at 60% relative humidity and is reheated to 52°C and passed over another set of shelves, again leaving at 60% relative humidity. This is again repeated for the third and fourth sets of shelves, after which the air leaves the dryer. On the assumption that the material in each shelf has reached the wet bulb temperature and heat loss is negligible, estimate: (i) the temperature of the material on each tray; (ii) the amount of water removed, in kg/hr, if 300 m^3/min of moist air leaves the dryer.

Solution.

(i) Air leaves the pre-heater of the dryer at 325 K
Humidity of incoming air = 0.005 kg water/kg dry air

It enters the first shelf. So, the wet bulb temperature = 25°C

Moisture is removed along wet bulb temperature line till 60% R.H. is reached. This gives the exit condition of air from first shelf.

From the chart, Humidity of air leaving first shelf = 0.016 kg water/kg dry air. Dry bulb temperature of exit air is at 27°C and is at a humidity of 0.016 kg water/kg dry air. This air is again heated to 52°C dry bulb temperature in second heater. So, air leaves heater at 52°C and at a humidity of 0.016 kg water/kg dry air. When it leaves the second shelf, the corresponding dry bulb temperature is 34°C and the humidity is 0.023 kg water/kg dry air. This air enters the third shelf after preheating to 52°C. Similarly for third shelf, exit air has a humidity of 0.028 kg water/kg dry air and has a dry bulb temperature of 39°C. The air leaving the fourth shelf has a humidity of 0.032 kg water/kg dry air and a dry bulb temperature of 42°C. (The figure is only indicative and does not correspond to actual one.)

The solid temperatures correspond to WBT and they are 23°C, 27°C, 32°C and 34°C respectively. **Ans.**

(ii) Final moist air conditions: (Y') = 0.032 kg water/kg dry air

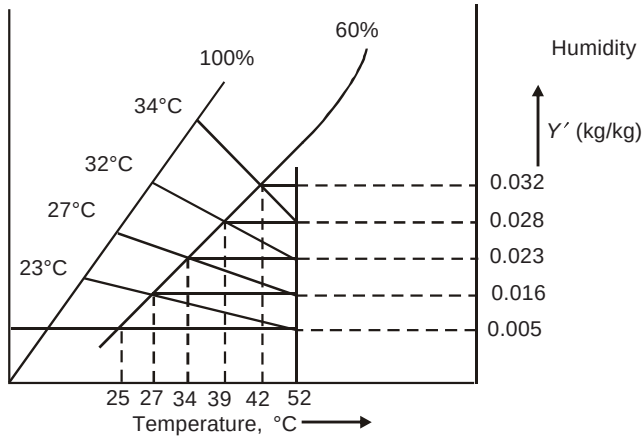


Fig. 6.17 Example 1 Humidity vs temperature.

Dry bulb temperature = 42°C

$$V_H = 8315\left(\frac{1}{M_{\text{air}}}\right) + \left(\frac{Y'}{M_{\text{water}}}\right)\left[\frac{(t_G + 273)}{P_t}\right]$$

$$V_H = 8315\left(\frac{1}{28.84}\right) + \left(\frac{0.032}{18}\right)\left[\frac{(42 + 273)}{1.013 \times 10^5}\right]$$

V_H = 0.945 m^3/kg dry air.

$$\text{Amount of dry air leaving/hr} = \frac{(300 \times 60)}{0.945} = 1.905 \times 10^4 \text{ kg}$$

Water removed/hr = 1.905×10^4 (0.032 – 0.005) = 514.35 kg/hr. **Ans.**

2. A batch of the solid, for which the following table of data applies, is to be dried from 25 to 6 percent moisture under conditions identical to those for which the data were tabulated. The initial weight of the wet solid is 350 kg, and the drying surface is 1 m^2/8 kg dry weight. Determine the time for drying.

$X \times 100$, $\frac{\text{kg moisture}}{\text{kg dry solid}}$	$N \times 100$, $\frac{\text{kg moisture evaporated}}{\text{hr} \cdot \text{m}^2}$
35	30
25	30
20	30
18	26.6
16	23.9
14	20.8
12	18
10	15
9	9.7
8	7
7	4.3
6.4	2.5

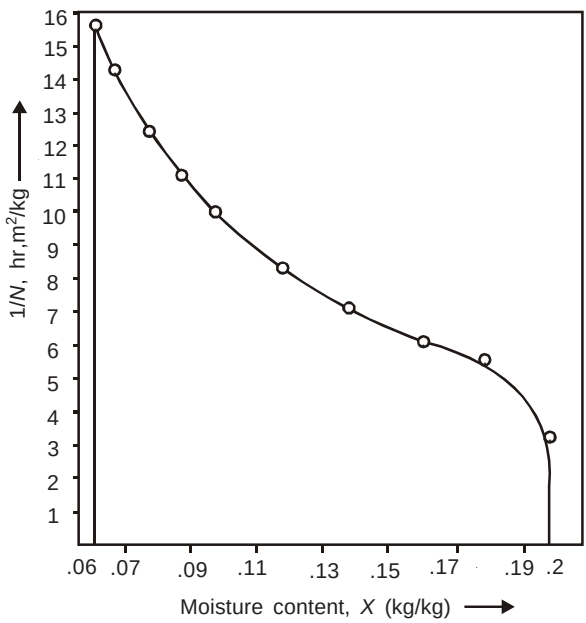


Fig. 6.18 Example 2 $1/N$ vs X for falling rate period.

Solution.

$$X_1 = \frac{0.25}{(1 - 0.25)} = 0.333, \qquad X_2 = \frac{0.06}{(1 - 0.06)} = 0.0638,$$

Initial weight of wet solid = 350 kg

Initial moisture content = 0.333 kg moisture/kg dry solid

So, total moisture present in wet solid (initially) = $350 \times 0.25 = 87.5$ kg moisture

Weight of dry solid, $L_S = 262.5$ kg

$$A = \frac{262.5}{8} = 32.8125 \text{ m}^2, \qquad \text{or } \frac{L_S}{A} = 8 \text{ kg/m}^2$$

$$X_{\text{Cr}} = 0.20, \qquad N_C = 0.3 \text{ kg/m}^2\text{hr}$$

So for constant rate period, drying time is

$$t_{\text{I}} = \frac{L_S}{AN_C}[X_1 - X_{Cr}] = \left[\frac{262.5}{(32.8125 \times 0.3)}\right][0.333 - 0.2] = 3.55 \text{ hr.}$$

For falling rate period, we are finding drying time graphically,

X	0.2	0.180	0.16	0.14	0.120	0.100	0.090	0.080	0.07	0.064
$1/N$	3.33	5.56	6.25	7.14	8.32	10.00	11.11	12.5	14.29	15.625

Area = 1.116,

$\therefore$ Time = Area under the curve $\times \frac{L_S}{A} = 1.116 \times \frac{L_S}{A} = 1.116 \times 8 = 8.928$ hr.

$\therefore$ Total time = 8.928 + 3.55 = 12.478 hr. **Ans.**

3. A wet slab of material weighing 5 kg originally contains 50 percent moisture on wet basis. The slab is 1 m × 0.6 m × 7.5 cm thick. The equilibrium moisture is 5 per cent on wet basis. When in contact with air, the drying rate is given in the table below. Drying takes place from one face only.

(i) Plot the drying rate curve and find the critical moisture content.

Wet slab wt, kg	5.0	4.0	3.6	3.5	3.4	3.06	2.85
Drying rate, kg/(hr)(m^2)	5.0	5.0	4.5	4.0	3.5	2.00	1.00
X, Dry basis	1.00	0.6	0.44	0.4	0.36	0.224	0.14

(ii) How long will it take to dry the wet slab to 15 percent moisture on wet basis?

Solution. Weight of wet solid = 5 kg

Moisture content = 0.50 moisture/kg wet solid

$$= \frac{0.5}{[(0.5 \text{ moisture}) + (0.5 \text{ dry solid})]} = \frac{0.5}{(1 - 0.5)}$$

$\therefore X_1 = 1$ = moisture/dry solid

For 5 kg wet solid, moisture = 5 × 0.5 = 2.5 kg
Weight of dry solid = 5 – 2.5 = 2.5 kg

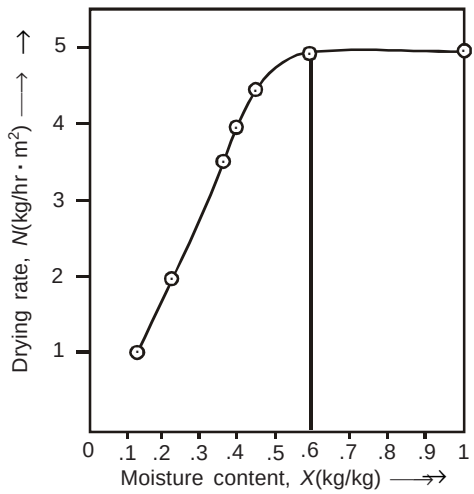


Fig. 6.19(a) Example 3 Drying rate curve.

$$x^* = 0.05, \qquad X^* = \frac{0.05}{(1-0.05)} = 0.0526$$

$$\text{Moisture content in dry basis} = \frac{\text{weight of wet solid} - \text{weight of dry solid}}{\text{weight of dry solid}}$$

Ans.

(i) X_{Cr} = 0.6 kg moisture/kg dry solid.
(ii) From X = 0.6 to 0.44 the falling rate curve is non-linear and from X = 0.44 to 0.14, falling rate period is linear.

$$X_2 = \frac{0.15}{(1 - 0.15)} = 0.1765.$$

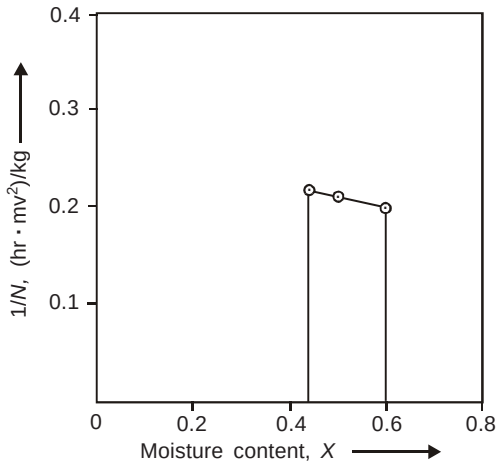


Fig. 6.19(b) Example 3 1/N vs X.

So, we can find time for drying from 0.6 to 0.44 graphically and then for X = 0.44 to 0.1765, we can go in for analytical solution as the 'N' vs 'X' relation is linear.

Time taken for constant rate drying period (From X = 1 to X = 0.6)

$$t_I = \left[\frac{L_S}{AN_C}\right] [X_1 - X_{Cr}] = \left[\frac{2.5}{(5 \times 0.6)}\right] [1 - 0.6] = 0.333 \text{ hr}$$

(from X = 0.44 to 0.1765)

$$t_{II} = \frac{L_S}{AN_C} \times \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_1 - X^*)}{(X_2' - X^*)}\right]\right\}$$

$$(X^* = \frac{0.05}{(1 - 0.05)} = 0.0526)$$

$$t_{II} = \left[\frac{2.5}{(5 \times 0.6)}\right] \times \left\{(0.6 - 0.0526) \times \ln \left[\frac{(0.44 - 0.0526)}{(0.176 - 0.0526)}\right]\right\}$$

= 0.522 hr **Ans.**

From graph, t_{III} (From X = 0.6 to X = 0.44) = $\frac{(0.0336 \times 2.5)}{0.6}$ = 0.14 hr.

Total time = $t_I + t_{II} + t_{III}$ = 0.333 + 0.522 + 0.14 = 0.995 hr or 59.58 min. **Ans.**

4. Data on drying rate curve of a particular solid is given below. The weight of the dry material in the solid is 48.0 kg/m^2. Calculate the time required to dry the material from 25% to 8% moisture (dry basis).
Data:

X	0.30	0.20	0.18	0.15	0.14	0.11	0.07	0.05
N	1.22	1.22	1.14	0.90	0.80	0.56	0.22	0.05

where X is the moisture content in kg water/kg dry solid and N is the drying rate in kg/(hr) (m^2).

Solution.

N_C = 1.22 kg/m^{2}hr, X_{Cr} = 0.2, X_1 = 0.25, X_2 = 0.08, L_S/A = 48 kg/m^2

Time taken for constant rate drying period,

$$t_I = \left[\frac{L_S}{AN_C}\right] [X_1 - X_{Cr}]$$

$$t_I = 48 \frac{[0.25 - 0.2]}{1.22} = 1.967 \text{ hr.}$$

X	0.18	0.15	0.14	0.11	0.07	0.05
$1/N$	0.8772	1.111	1.25	1.7857	4.545	20

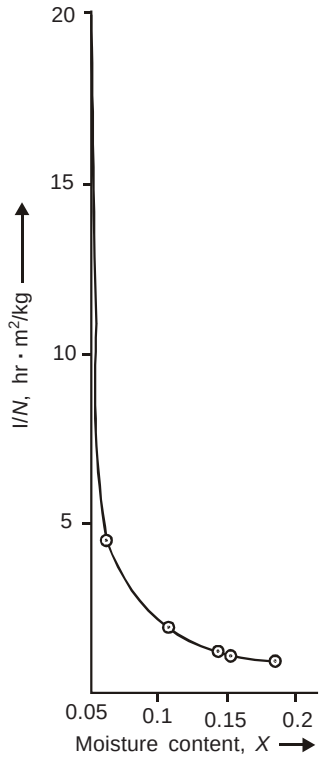


Fig. 6.20 Example 4 $1/N$ vs X for falling rate.

Area under the curve = $14 \times 0.025 \times 1 = 0.35$

$t_{II} = 0.35 \times 48 = 16.8$ hr

Total time taken = $t_I + t_{II} = 1.967 + 16.8 = 18.767$ hr **Ans.**

5. In a drying experiment, a tray drier, containing a single tray of 1 m^2 area is used, to dry crystalline solids. The following data have been collected:

Sl. No	*Time* (hr)	*Weight of wet material*, kg
1	0.0	5.314
2	0.4	5.238
3	0.8	5.162
4	1.0	5.124
5	1.4	5.048
6	1.8	4.972
7	2.2	4.895
8	2.6	4.819
9	3.0	4.743
10	3.4	4.667
11	4.2	4.524
12	4.6	4.468
13	5.0	4.426
14	6.0	4.340
15	Infinity	4.120

Sl. No.	*Time,* hr	*Weight of wet material,* kg	*Moisture content dry basis*	*Drying rate,* kg/hr · m^2
1	0	5.314	0.29	—
2	0.4	5.238	0.271	0.19
3	0.8	5.162	0.253	0.19
4	1.0	5.124	0.244	0.19
5	1.4	5.048	0.225	0.19
6	1.8	4.972	0.206	0.19
7	2.2	4.895	0.188	0.1925
8	2.6	4.819	0.169	0.19
9	3.0	4.743	0.151	0.19
10	3.4	4.667	0.133	0.19
11	4.2	4.524	0.098	0.179
12	4.6	4.468	0.084	0.14
13	5	4.426	0.074	0.105
14	6	4.340	0.053	0.086
15	Infinity	4.120	0.0	—

(i) Calculate and plot drying rates. Find the critical moisture content.

(ii) If dry air is available at 40°C with an absolute humidity of 0.01 kg/kg dry air and the drier is maintained at 90°C, calculate the amount of air required in first 2 hours. Assume the air is heated up to 90°C and the dry air leaves the drier at 90°C with 5% saturation.

(iii) Test the consistency of the falling rate period. (Choose critical moisture content and any one point in falling rate period.)

Solution.

From the above data after getting the rate curve it is clear that $X_{Cr} = 0.11$. The loss in weight is due to moisture evaporated. After two hours the weight is

$$(4.972 + 4.895)/2 = 4.934 \text{ kg}$$

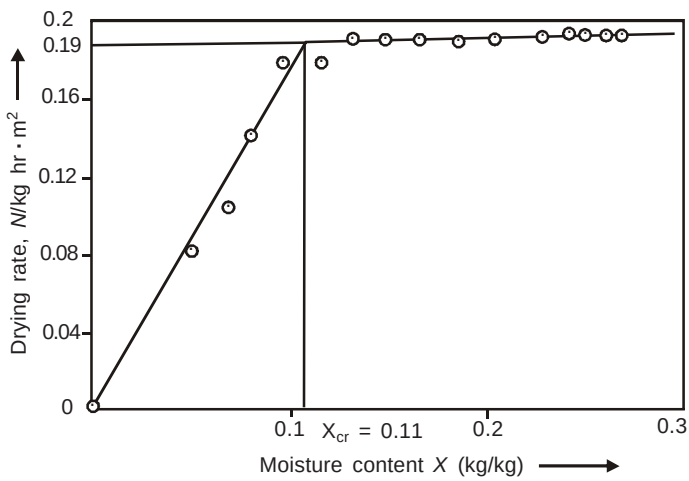


Fig. 6.21 Example 5 Drying rate curve.

The water evaporated in 2 hours is

$$5.314 - 4.934 = 0.38 \text{ kg}$$

Humidity of incoming air = 0.01 kg/kg

Humidity of leaving air = 0.03 kg/kg (for 90°C with 5% saturation)

Water carried away by air = $G_s(Y_{out} - Y_{in})$

$$0.38 = G_s\ (0.03 - 0.01)$$

Therefore,

$$G_s \text{ is } = \frac{0.38}{0.02} = 19 \text{ kg of air for 2 hours.}$$

$$t_{falling} = \frac{L_S}{AN_C} \times \left\{(X_{Cr} - X^*) \times \ln \left[\frac{(X_1 - X^*)}{(X_2 - X^*)}\right]\right\}$$

Let us choose readings (11) and (13) to check the consistency

$$= \left[\frac{4.12}{(1) \times (0.19)}\right] \times (0.11 - 0) \times \ln \left[\frac{0.098}{0.074}\right] = 0.67 \text{ hours}$$

Here, X^* is taken as 0. Actual time is 0.8 hour. **Ans.**

6. A woolen cloth is dried in a hot air dryer from an initial moisture content of 100% to a final moisture content of 10%. If the critical moisture content is 55% and the equilibrium moisture content is 6% (at dryer condition), calculate the saving in drying time if the material is dried to 16% instead of 10%. All other drying conditions remain the same.
All moisture contents are on the dry basis.

$X_1 = 100\%$, $X_{Cr} = 55\%$, $X_2 = 10\%$, $X^* = 6\%$ all are on dry basis.

Solution.

$X_2' = 16\%$

$$t_{T1} = \frac{L_S}{AN_C}[X_1 - X_{Cr}] + \left\{\frac{L_S}{AN_C}(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}$$

$$t_T = \frac{L_S}{AN_C}[X_1 - X_{Cr}] + \left\{\frac{L_S}{AN_C}(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\} \quad \ldots (1)$$

$$t_T' = \frac{L_S}{AN_C}[X_1 - X_{Cr}] + \left\{\frac{L_S}{AN_C}(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2' - X^*)}\right]\right\} \quad \ldots (2)$$

i.e. (1)/(2) is,

$$\frac{t_T}{t'_T} = \frac{[X_1 - X_{Cr}] + \left\{(X_{Cr} - X^*) \ln\left[\frac{X_{Cr} - X^*}{X_2 - X^*}\right]\right\}}{[X_1 - X_{Cr}] + \left\{(X_{Cr} - X^*) \ln\left[\frac{X_{Cr} - X^*}{X'_2 - X^*}\right]\right\}}$$

$$\frac{t_T}{t_{T'}} = \frac{[1 - 0.55] + \left\{(0.55 - 0.06) \ln\left[\frac{(0.55 - 0.06)}{(0.1 - 0.06)}\right]\right\}}{[1 - 0.55] + \left\{(0.55 - 0.06) \ln\left[\frac{(0.55 - 0.06)}{(0.16 - 0.06)}\right]\right\}}$$

$$\frac{t_T}{t'_T} = \frac{1.678}{1.2287}$$

$$\frac{t_T}{t'_T} = 1.3657$$

$$\frac{t'_T}{t_T} = 0.7322$$

$$t'_T = 0.7322\ t_T$$

The reduction in drying time is,

$$\frac{(t_T - t'_T)}{t'_T} = 0.2678$$

Thus, the time reduces by 26.78%. **Ans.**

7. A filter cake is dried for 5 hours from an initial moisture content of 30% to 10% (wet basis). Calculate the time required to dry the filter cake from 30% to 6% (wet basis).

Equilibrium moisture content = 4% on dry basis

Critical moisture content = 14% on dry basis

Assume that the rate of drying in the falling rate period is directly proportional to the free moisture content.

Solution.

$x_1 = 0.3$, $x_f = 0.10$, $X_1 = \frac{0.3}{(1-0.3)} = 0.4286$, $X_{Cr} = 0.14$

$X_f = X_2 = \frac{0.1}{(1 - 0.1)} = 0.111$, $X^* = 0.04$,

Then, $X_2' = \frac{0.06}{(1-0.06)} = 0.064$

$$t_T = \left[\frac{L_S}{AN_C}\right][(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln\left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}$$

$$5 = \left[\frac{L_S}{AN_C}\right](0.4286 - 0.14) + \left\{(0.14 - 0.04)\ln\left[\frac{(0.14 - 0.04)}{(0.111 - 0.04)}\right]\right\}$$

Thus, $L_S/AN_C = 15.487$ hrs

$$t_T' = \frac{L_S}{AN_C}\left[(X_i - X_{Cr}) + \left\{(X_{Cr} - X^*)\ln\left[\frac{(X_{Cr} - X^*)}{(X_2' - X^*)}\right]\right\}\right]$$

$$t_T' = 15.487\left[(0.4286 - 0.14) + \left\{(0.14 - 0.04)\ln\left[\frac{(0.14 - 0.04)}{(0.064 - 0.04)}\right]\right\}\right]$$

$t_T' = 6.68$ hrs **Ans.**

8. 1000 kg dry weight of non-porous solid is dried under constant drying conditions with an air velocity of 0.75 m/s, so that the surface area of drying is 55 m^2. The critical moisture content of the material may be taken as 0.125 kg water/kg dry solids?

(i) If the initial rate of drying is 0.3 $g/m^2 \cdot s$. How long will it take to dry the material from 0.15 to 0.025 kg water/kg dry solid?

(ii) If the air velocity were increased to 4.0 m/s, what would be the anticipated saving in time if surface evaporation is controlling.

Solution.

$L_S = 1000$ kg, Air velocity = 0.75 m/s, $A = 55\ m^2$, $X_{Cr} = 0.125$, $X_1 = 0.15$, $X_2 = 0.025$. Assume $X^* = 0$, $N_C = 0.3\ g/m^2 \cdot s$ or 0.3×10^{-3} kg/m^2 s

$$t_T = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*)\ln\left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right]$$

$$t_T = \left[\frac{1000}{(55 \times 0.3 \times 10^{-3})}\right] \times \left[(0.15 - 0.125) + \left\{(0.125 - 0)\ln\left[\frac{(0.125 - 0)}{(0.025 - 0)}\right]\right\}\right]$$

$= 13708$ s

$t_T = 3.8077$ hr

(ii) Assuming only surface evaporation, and assuming air moves parallel to surface

$\because$ $N_C \propto G^{0.71}$ and $G = V \times \rho$

i.e. $N_C \propto V^{0.71}$

$\therefore$ $$\frac{N_{C1}}{N_{C2}} = \frac{(V_1)^{0.71}}{(V_2)^{0.71}}$$

$$\frac{(0.3 \times 10^{-3})}{N_{C2}} = \frac{(0.75)^{0.71}}{(4)^{0.71}}$$

$N_{C2} = 0.985 \times 10^{-3}$ kg/m^2 s.

$$t_T = \left[\frac{1000}{(55 \times 0.985 \times 10^{-3}}\right] \times \left[(0.15 - 0.125) + \left\{(0.125 - 0) \ln \left[\frac{(0.125 - 0)}{(0.025 - 0)}\right]\right\}\right] \times \frac{1}{3600}$$

$= 1.1597$ hrs.

So, time saved $= 3.8077 - 1.1597 = 2.648$ hr **Ans.**

9. A plant wishes to dry a certain type of fibreboard. To determine drying characteristics, a sample of 0.3×0.3 m size with edges sealed was suspended from a balance and exposed to a current of hot dry air. Initial moisture content was 75%. The sheet lost weight at the rate of 1×10^{-4} kg/s until the moisture content fell to 60%. It was established that the equilibrium moisture content was 10%. The dry mass of the sample was 0.90 kg. All moisture contents were on wet basis. Determine the time for drying the sheets from 75% to 20% moisture under the same drying conditions.

Solution.

$x_1 = 0.75$, $x^* = 0.1$, $x_{Cr} = 0.6$ $L_S = 0.90$ kg,

$A = (0.3 \times 0.3) \times 2$ (both upper and lower surfaces are exposed) $= 0.18$ m^2.

$A \times N_C = 10^{-4}$ kg/s,

$x_1 = 0.75$, $x_2 = 0.2$, $X_{Cr} = \dfrac{0.6}{0.4} = 1.5$,

$X_1 = 3$, $X_2 = 0.25$, $X^* = \dfrac{0.1}{0.9} = 0.111$

$$t_T = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right]$$

$$t_T = \frac{0.90}{10^{-4}} \times \left[(3 - 1.5) + \left\{(1.5 - 0.111) \ln \left[\frac{(1.5 - 0.111)}{(0.25 - 0.111)}\right]\right\}\right]$$

$= 11.74$ hr **Ans.**

10. A commercial drier needed 7 hours to dry a moist material from 33% moisture content to 9% on bone dry basis. The critical and equilibrium moisture content were 16% and 5% on bone dry basis respectively. Determine the time needed to dry the material from a moisture content of 37% to 7% on bone dry basis if the drying conditions remain unchanged.

Solution.

$X_1 = 0.33$, $X^* = 0.05$, $X_{Cr} = 0.16$, $X_2 = 0.09$, $X_1' = 0.37$,

$X_2' = 0.07$, $t_T = 7$ hrs

$$t_T = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right]$$

$$7 = \frac{L_S}{AN_C} \times \left[(0.33 - 0.16) + \left\{(0.16 - 0.05) \ln \left[\frac{(0.16 - 0.05)}{(0.09 - 0.05)}\right]\right\}\right]$$

$$\frac{L_S}{AN_C} = 24.8866 \text{ h}$$

Now, $X_1 = 0.37$, $X_2 = 0.07$

$$t_T = 24.8866 \times \left[(0.37 - 0.16) + \left\{(0.16 - 0.05) \ln \left[\frac{(0.16 - 0.05)}{(0.07 - 0.05)}\right]\right\}\right]$$

$= 9.893$ h **Ans.**

11. A slab of paper pulp 1.5 m × 1.5 m × 5 mm, is to be dried under constant drying conditions from 65% to 30% moisture (wet basis) and the critical moisture is 1.67 kg free water/kg dry pulp. The drying rate at the critical point has been estimated to be 1.40 kg/(m^2)(hr). The dry weight of each slab is 2.5 kg. Assuming drying to take place from the two large faces only, calculate the drying time to be provided.

Solution.

$x_1 = 0.65$, $L_S = 2.5$ kg,

$A = (1.5 \times 1.5) \times 2$ (drying takes place from both the larger surface only) $= 4.5$ m^2.

$N_C = 1.4$ kg/m^2hr, $x_2 = 0.3$, $X_{Cr} = 1.67$,

$$X_1 = \frac{0.65}{(1 - 0.65)} = 1.857, \quad X_2 = \frac{0.3}{(1 - 0.3)} = 0.4286,$$

Assuming $X^* = 0$

$$t_T = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right]$$

$$t_T = \left(\frac{2.5}{4.5 \times 1.4}\right) \times [1.857 - 1.67] + \left\{(1.67 - 0) \ln \left[\frac{(1.67 - 0)}{(0.4286 - 0)}\right]\right\}$$

$t_T = 0.976$ h **Ans.**

12. A slab of paper pulp 1.5 m × 1.5 m × 5 mm, thick is to be dried under constant drying conditions from 15% to 8.5% moisture (dry basis). The equilibrium moisture is 2.5% (dry basis) and the critical moisture is 0.46 kg free water kg dry pulp. The drying rate at the critical point has been estimated to be 1.40 kg/(m^2)(hr). Density of dry pulp is 0.22 gm/cc. Assuming drying to take place from the two large faces only, calculate the drying time to be provided.

Solution.

$X^* = 0.025$, $N_C = 1.4$ kg/m^2hr, $X_{Cr} = 0.46$, $X_1 = 0.15$, $X_2 = 0.085$,

Density of dry pulp = 0.22 g/cc, $A = (1.5 \times 1.5) \times 2 = 4.5$ m^2,

Volume of material = $1.5 \times 1.5 \times 0.5 = 1.125 \times 10^{-2}$ m^3,

$\therefore \quad L_S = 1.125 \times 10^{-2} \times 0.22 \times 10^3 = 2.475$ kg

$$t_T = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right]$$

But here initial moisture is less than the X_{Cr}. So there is no constant rate drying period and only falling rate period is observed.

$$\therefore \quad t_T = \frac{L_S}{AN_C} [X_{Cr} - X^*] \ln \left[\frac{(X_1 - X^*)}{(X_2 - X^*)}\right]$$

$$t_T = \left\{\left[\frac{2.475}{4.5 \times 1.4}\right] \times [0.46 - 0.025] \ln \left[\frac{(0.15 - 0.025)}{(0.085 - 0.025)}\right]\right\}$$

= 0.125 hr or 7.526 min **Ans.**

13. Under constant drying conditions, a filter cake takes 5 hours to reduce its moisture content from 30% to 10% on wet basis. The critical moisture is 14% and the equilibrium moisture 4%, both on dry basis. Assuming the rate of drying in the falling rate period to be directly proportional to the free moisture content, estimate the time required to dry the cake from 30% to 6% moisture on wet basis.

Solution.

$x_1 = 0.3, \quad x_2 = 0.1,$

$X_1 = 0.3/0.7 = 0.4286, \quad X^* = 0.04, \quad X_{Cr} = 0.14,$

$X_2 = 0.1/0.9 = 0.111, \quad t_T = 5$ hrs,

$x_1' = 0.3$ and hence, $X_1' = 0.4286. \quad x_2' = 0.06$ and $X_2' = 0.0638,$

$$5 = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right] \quad \ldots (1)$$

$$t_T' = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1' - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln \left[\frac{(X_{Cr} - X^*)}{(X_2' - X^*)}\right]\right\}\right] \quad \ldots (2)$$

Dividing (1)/(2)

$$\frac{5}{t_T'} = \frac{[0.4286 - 0.14] + \left\{0.14 - 0.04) \ln \left[\frac{(0.14 - 0.04)}{(0.111 - 0.04)}\right]\right\}}{[0.4286 - 0.14] + \left\{(0.14 - 0.04) \ln \left[\frac{(0.14 - 0.04)}{(0.0638 - 0.04)}\right]\right\}}$$

$$\frac{5}{t_T'} = 0.3228/0.4321$$

Hence,

$t_T' = 6.69$ h. **Ans.**

14. Sheet material, measuring 1 m^2 and 5 cm thick, is to be dried from 45% to 5% moisture under constant drying conditions. The dry density of the

material is 450 kg/m^3 and its equilibrium moisture content is 2%. The available drying surface is 1 m^2. Experiments showed that the rate of drying was constant at 4.8 kg/(hr)(m^2) between moisture contents of 45% and 20% and thereafter the rate decreased linearly. Calculate the total time required to dry the material from 45% to 5%. All moisture contents are on wet basis.

Solution.

A = 1 m^2, 5 cm thick, x_i = 0.45, x^* = 0.02, x_{Cr} = 0.2, x_2 = 0.05,

$$X^* = \frac{0.02}{(1 - 0.02)} = 0.02041, \quad N_C = 4.8 \text{ kg/m}^2\text{hr}, \quad X_{Cr} = 0.25,$$

$$X_1 = \frac{0.45}{(1 - 0.45)} = 0.818, \quad X_2 = 0.0526.$$

Density of dry pulp = 450 kg/m^3

Volume of material = $1 \times 5 \times 10^{-2}$ = 0.05 m^3

$L_S = 450 \times 0.05 = 22.5$ kg

$$t_T = \left(\frac{L_S}{AN_C}\right) \times \left[(X_1 - X_{Cr}) + \left\{(X_{Cr} - X^*) \ln\left[\frac{(X_{Cr} - X^*)}{(X_2 - X^*)}\right]\right\}\right]$$

$$t_T = \left[\frac{22.5}{(1 \times 4.8)}\right] \times \left[(0.818 - 0.25) + \left\{(0.25 - 0.02041) \ln\left[\frac{(0.25 - 0.02041)}{(0.0526 - 0.02041)}\right]\right\}\right]$$

= 4.78 hr **Ans.**

15. Wet solids containing 120 kg/hour of dry stuff are dried continuously in a specially designed drier, cross-circulated with 2,000 kg per hour of dry air under the following conditions:

Ambient air temperature	= 20°C
Exhaust air temperature	= 70°C
Evaporation of water	= 150 kg/hr
Outlet solids moisture content	= 0.25 kg/hr
Inlet solids temperature	= 15°C
Outlet solids temperature	= 65°C
Power demand	= 5 kW
Heat loss	= 18 kW

Estimate heater load per unit mass of dry air and fraction of this heat used in evaporation of moisture.

Data:

Mean specific heat of dry air	= 1 kJ kg^{-1} K^{-1}
Enthalpy of saturated water vapour	= 2,626 kJ per kg
Mean specific heat of dry materials	= 1.25 kJ kg^{-1} K^{-1}
Mean specific heat of moisture	= 4.18 kJ kg^{-1} K^{-1}.

Solution.

Total quantity of solid 120 kg dry stuff

Air used is 2000 kg/hr dry air

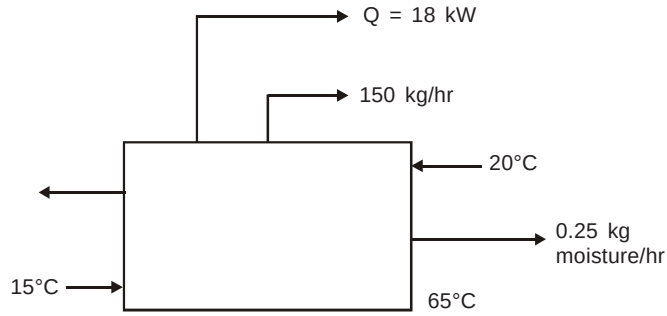


Fig. 6.22 Example 15.

Basis: 1 hour

Heat required for heating 150 kg water from 15°C to 65°C

$$150 \times 4.18 \times (65 - 15) = 31350 \text{ kJ}$$

Heat required for 150 kg water evaporation

$$150 \times 2626 = 393900 \text{ kJ}$$

Heat required for heating air from 20°C to 70°C

$$2000 \times 1 \times (70 - 20) = 100000 \text{ kJ}$$

Heat required for heating moisture in solid from 15°C to 65°C

$$0.25 \times 4.18 \times (65 - 15) = 52.25 \text{ kJ}$$

Heat required for heating dry solid from 15°C to 65°C

$$120 \times 1.25 \times (65 - 15) = 7500 \text{ kJ}$$

Heat lost = $18 \times 3600 = 64800$ kJ

So, total heat required/hr

393900 + 100000 + 52.25 + 7500 + 64800 = 566252.25 kJ/hr. = 157.3 kW

i.e. 166 kW of heat is needed for 2000 kg/hr of dry air.

$$\text{Heat required/mass of dry air} = \frac{157.3}{2000} = 0.0787 \text{ kW}$$

(i) Heat needed for evaporation = [393900 + 31350]/3600 = 118.13 kW **Ans.**

(ii) Fraction of this heat needed for evapouration = $\frac{118.13}{157.3} = 0.751$ or 75.1%. **Ans.**

16. A drum drier is being designed for drying of a product from an initial total moisture content of 12% to final moisture content of 4%. An overall heat transfer coefficient (U) 1700 W/m^{2}C is being estimated for the product. An average temperature difference between the roller surface and the product of 85°C will be used for design purpose. Determine the surface area of the roller required to provide a production rate of 20 kg product per hour.

Solution.

Initial moisture content = 12%
Final moisture content = 4%
Production rate = 20 kg final product/hour
4 kg moisture is present in 100 kg product

In 20 kg product, weight of moisture = $\frac{(4 \times 20)}{100}$ = 0.8 kg

Dry solid weight = 20 – 0.8 = 19.2 kg

Total initial moisture content = $19.2 \times \left[\frac{0.12}{(1 - 0.12)}\right]$ = 2.6182 kg

Water evaporated = 2.6182 – 0.8 = 1.8182 kg/hour
λ_S at 85°C = 2296.1 kJ/kg
Heat required = $W \times \lambda_S$ (Assuming the solid mixture enters at 85°C and only moisture removal by evaporation is alone considered)
= 1.8182 × 2296.1 = 4174.73 kJ/hour
$U \times A \times \Delta T = W \times \lambda s$
$1700 \times A \times 85 = 4174730/3600$
$\therefore \quad A = 8.025 \times 10^{-3}$ m^2 or 80.25 cm^2. **Ans.**

17. A sample of porous sheet material of mineral origin is dried from both sides by cross circulation of air in a laboratory drier. The sample was 0.3 m × 0.3 m and 6 mm thick and edges were sealed. The air velocity is 3 m/s. DBT and WBT of air were 52°C and 21°C respectively. There was no radiation effect. Constant rate drying was 7.5×10^{-5} kg/s until critical moisture content of 15% (on wet basis) was obtained. In the falling rate period, rate of drying fell linearly with moisture content until the sample was dry. The dry weight of the sheet was 1.8 kg. Estimate the time needed for drying similar sheets 1.2 m × 1.2 m × 12 mm thick from both sides from 25% to 2% moisture on wet basis using air at a DBT of 66°C but of the same absolute humidity and a linear velocity of 5 m/s. Assume the critical moisture content remains the same.

Solution.

Constant drying rate 7.5×10^{-5} kg/s
Area of the specimen = $0.3 \times 0.3 \times 2 = 0.18$ m^2

Drying rate = $N_{C1} = \frac{7.5 \times 10^{-5}}{0.18} = 4.167 \times 10^{-4}$ kg/m^2 s

$$X_C = \frac{0.15}{0.85} = 0.1765$$

$$X_O = \frac{0.25}{0.75} = 0.3333$$

$$X_{\text{final}} = \frac{0.02}{0.98} = 0.02041$$

Area of new solid = $1.2 \times 0.6 \times 2 = 1.44$ m^2
Bone dry Weight of new solid = 28.8 kg
(Based on total volume of old and new solid)
Volume of old solid = $0.3 \times 0.3 \times 0.006 = 54 \times 10^{-5}$ m^3
Volume of new solid = $0.6 \times 1.2 \times 0.012 = 864 \times 10^{-5}$ m^3
Weight of old solid = 1.8 kg

Weight of new Bone dry solid = $(864 \times 10^{-5}) \times \frac{1.8}{54} \times 10^{-5} = 28.8$ kg

$N_c \propto (T - T_s)$
$\propto (Y_s - Y)$
$\propto (G)^{0.71}$ where, G is the mass flow rate of air
Old velocity = 3 m/s
Old DBT = 52°C and WBT = 21°C T_s = 21°C, Hence T_G–T_S = 31°C
Humidity: 0.002 Saturated humidity: 0.015 kg/kg
New velocity = 5 cm/s
New DBT: 66°C but of same humidity as before, T_s = 24°C
Hence $T_G - T_s$ = 41°C
Humidity: 0.002 Saturated humidity: 0.018 kg/kg
Hence, Drying rate of air under new conditions is =

$$4.167 \times 10^{-4} \left[\left(\frac{5}{3}\right) \times \left(\frac{325}{339}\right)\right]^{0.71} \times \left[\frac{(0.018 - 0.002)}{(0.015 - 0.002)}\right] \times \left[\frac{4}{3}\right] = 0.933 \times 10^{-3} \text{ kg/m}^2 \text{ s}$$

$$\text{Drying time} = \left[\frac{L_S}{AN_C}\right] \times [X_o - X_{\text{Cr}}] + (X_{\text{Cr}} - X^*) \times \ln\left[\frac{(X_{\text{Cr}} - X^*)}{(X_f - X^*)}\right]$$

$$= \left[\frac{28.8}{1.44 \times 0.933 \times 10^{-3}}\right]$$

$$\times \left[(0.3333 - 0.1765) + (0.176 - 0.0) \times \ln\left\{\frac{0.1765 - 0}{0.0204 - 0.0}\right\}\right]$$

= 11524 s
= 3.20 hrs

EXERCISES

1. Sheet material, measuring 1 m^2 and 5 cm thick, is to be dried from 50% to 2% moisture under constant drying conditions. The dry density of the material is 400 kg/m^3 and its equilibrium moisture content is negligible. The available drying surface is 1 m^2. Experiments showed that the rate of drying was constant at 4.8 kg/(hr)(m^2) between moisture contents of 50% and 25% and thereafter the rate decreased linearly. Calculate the total time required to dry the material from 50% to 2%. All moisture contents are on wet basis.

(**Ans:** 6.653 hrs)

2. Calculate the critical moisture content and the drying rate during the constant rate period for drying a wet slab of size 20 cm × 75 cm × 5 cm, whose dry weight is 16 kg. Both the sides are used for drying. The steam used was at 3 atm. pressure and was consumed at the rate of 0.135 g/s cm^2 of the contact surface. The following drying data is available for the sample. Assume equilibrium moisture content is negligible.

Drying time, hrs	0	0.25	1.0	1.5	2.0	2.5	3.0	4.0	6.0	8.0	10.0	12.0
Sample weight, kg	19.9	19.7	19.2	18.9	18.6	18.3	18.1	17.65	16.92	16.4	16.15	16.05

(**Ans:** 0.14375 kg/kg, 2 kg/hr m^2)

3. The following data are available for drying a substance. Estimate the drying time needed to dry a similar sample under similar drying conditions from 40% to 12% moisture content, on wet basis. The drying surface is 1 m^2/4 kg of dry weight and the initial weight of the wet sample is 80 kg.

X (dry basis)	0.35	0.25	0.2	0.18	0.16	0.14	0.10	0.08	0.065
N, kg/hr·m^2	0.3	0.3	0.3	0.266	0.24	0.21	0.15	0.07	0.05

(**Ans:** 7.281 hrs)

4. 175 kg of wet material with 25% moisture is to be dried to 10% moisture. Air enters at 65ºC DBT and a WBT of 25ºC. The velocity of air is 150 cm/s. Drying area equals 1 m^2/40 kg dry weight.

X (dry basis)	0.26	0.22	0.20	0.18	0.16	0.14	0.12	0.1	0.08
N, kg/hr·m^2	1.5	1.5	1.5	1.3	1.2	1.04	0.9	0.75	0.6

(**Ans:** 6.687 hrs)

5. A wet solid is dried from 35% to 8% moisture in 5 hrs under constant drying condition. The critical moisture content is 15% and equilibrium moisture content is 5%. All the moisture contents are reported as percentage on wet basis. Calculate how much longer it would take place under the similar drying conditions to dry from 8% to 6% moisture on wet basis.

(**Ans:** 1.3115 hrs)

6. A certain material was dried under constant drying conditions and it was found that 2 hours are required to reduce the free moisture from 20% to 10%. How much longer would it require to reduce the free moisture to 4%? Assume that no constant rate period is encountered.

(**Ans:** 4.643 hrs)

7. It is desired to dry sheets of material from 73% to 4% moisture content (wet basis). The sheets are 2 m × 3 m × 5 mm. The drying rate during constant rate period is estimated to be 0.1 kg/hr · m^2. The bone-dry density of the material is 30 kg/m^3. The material is dried from both the sides. The critical

moisture content is 30% on wet basis and equilibrium moisture content is negligible. The falling rate period is linear. Determine the time needed for drying.

(**Ans:** 2.798 hrs)

8. A slab of paper pulp 1 m × 1 m × 5 mm is to be dried under constant drying conditions from 60% to 20% moisture (wet basis) and the critical moisture is 1.5 kg water/kg dry pulp. The drying rate at the critical point has been estimated to be 1.40 kg/(m^2)(hr). The dry weight of each slab is 2.5 kg. Assuming that drying rate is linear in falling rate period and drying takes place from the two large faces only, calculate the drying time needed.

(**Ans:** 4.8 hrs)

9. 50 kg of batch of granular solids containing 25% moisture is to be dried in a tray dryer to 12% moisture by passing a stream of air at 92°C tangentially across its surface at a velocity of 1.8 m/s. If the constant rate of drying under these conditions is 0.0008 kg moisture/m^2s and critical moisture content is 10%. Calculate the drying time if the surface available is 1.0 m^2 (all moisture contents are on wet basis).

(**Ans:** 2.565 hrs)

10. A plant wishes to dry a certain type of fibre board in sheets 1.2 m × 2 m × 12 mm. To determine the drying characteristics a 0.3 m × 0.3 m board with the edge sealed, so that drying takes place only from two large faces only, was suspended from a balance in a laboratory dryer and exposed to a current of hot dry air. Initial moisture content is 75%, critical moisture content is 60% and equilibrium moisture content is 10%. Dry mass of the sample weighs 0.9 kg. Constant drying rate 0.0001 kg/m^2·s. Determine the time for drying large sheets from 75% to 20% moisture under the same drying conditions (all moisture contents are on wet basis).

(**Ans:** 65.24 hrs)

11. A batch of wet solid was dried on a tray drier using constant drying conditions and the thickness of material on the tray was 25 mm. Only the top surface was exposed for drying. The drying rate was 2.05 kg/m^2 · hr during constant rate period. The weight of dry solid was 24 kg/m^2 exposed surface. The initial free moisture content was 0.55 and the critical moisture content was 0.22. Calculate the time needed to dry a batch of this material from a moisture content of 0.45 to 0.30 using the same drying condition but the thickness of 50 mm with drying from the top and bottom surfaces.

(**Ans:** 1.756 hrs)

12. A pigment material, which has been removed wet from a filter press, is to be dried by extending it into small cylinders and subjecting them to through circulation drying. The extrusions are 6 mm in diameter, 50 mm long and are to be placed in screens to a depth of 65 mm. The surface of the particles is estimated to be 295 m^2/m^3 of bed and the apparent density is 1040 kg dry solid/m^3. Air at a mass velocity of 0.95 kg dry air/m^2 · s will flow through

the bed entering at 120°C and a humidity of 0.05 kg water/kg dry air, estimate the constant drying rate to be expected.

(**Ans:** 0.002906 g/cm^2 s)

13. It is necessary to dry a batch of 160 kg of wet solid from 30% to 5% moisture content under constant rate and falling rate period. The falling rate is assumed to be linear. Calculate the total drying time considering an available drying surface of 1 m^2/40 kg of dry solid. The flux during constant rate period is 0.0003 kg/m^2 s. The critical and equilibrium moisture contents are 0.2 and 0.02 respectively. If the air flow rate is doubled, what is the drying time needed? The critical and equilibrium moisture contents do not change with velocity of air but N_C varies as $G^{0.71}$, where G is the mass flow rate of air.

(**Ans:** 23.33 hrs and 14.265 hrs)

14. A rotary dryer using counter-current flow is to be used to dry 12000 kg/hr of wet salt containing 5% water (wet basis) to 0.1% water (wet basis). Heated air at 147°C with a WBT of 50°C is available. The specific heat of the salt is 0.21 kcal/kg °C. The outlet temperatures of air and salt are 72°C and 93°C respectively. Calculate the diameter of the dryer required.

(**Ans:** 2.55 m)

15. During the batch drying test of a wet slab of material 0.35 m^2 and 7 mm thick, the falling rate N was expressed as 0.95 (X – 0.01) where N, is the drying rate in kg/m^2 s and X is the moisture content in kg moisture/kg dry solid. The constant drying rate was 0.38 kg/m^2 s and slab was dried from one side only with the edges sealed. Density of the dry material is 1200 kg/m^3. It is desired to reduce the moisture content from 35% to 5% on wet basis. What is the time needed for drying?

(**Ans:** 22.63 s)

7

CRYSTALLIZATION

7.1 INTRODUCTION

Crystallization is a process in which the solid particles are formed from a homogeneous phase. During the crystallization process, the crystals form from a saturated solution. The mixture of crystals and the associated mother liquor is known as *magma*. The advantages of crystals are given below:

(i) uniform size and shape
(ii) ease in filtering and washing
(iii) caking tendency is minimised
(iv) high purity
(v) they do not crumble easily

7.2 CRYSTAL GEOMETRY

A crystal is the most highly organised type of non-living matter. It is characterised by the fact that its constituent particles like atoms or molecules or ions are arranged in an orderly three-dimensional arrays called *space lattices*. The angles made by corresponding faces of all crystals of the same material are equal and characteristic of that material, although the size of the faces and edges may vary.

7.2.1 Classification of Crystals

The classification of crystals based on the interfacial angle and lengths of axes is as follows

Cubic: Three equal rectangular axes.

Hexagonal: Three equal coplanar axes inclined to 60° to each other and a fourth axis different in length from the other three and perpendicular to them.

Trigonal: Three equal and equally inclined axes.

Tetragonal: Three rectangular axes, two of which are equal and different in length from the third.

Orthorhombic: Three unequal rectangular axes.

Monoclinic: Three unequal axes, two of which are inclined but perpendicular to the third.

Triclinic: Three mutually inclined and unequal axes, all angles unequal and other than 30°, 60° and 90°.

Crystals can also be classified based on the type of bond needed to hold the particles in place in the crystal lattice. The crystals are classified as indicated below.

Metals: These are electropositive elements bonded together by coulomb electrostatic forces from positive ions. The mobility of electrons is responsible for their excellent thermal and electrical conductivity in metals. Metal alloys do not follow valence rules.

Ionic crystals: They are the combinations of highly electronegative and highly electropositive ions, such as the ordinary inorganic salts. They are held together by strong coulomb forces and obey valence rules. It can be thought of as a single giant molecule.

Valence crystals: They are formed from the lighter elements in the right half of the periodic table. The atomic particles are held together by sharing electrons. The forces are strong and valence crystals are extremely hard with high melting points. They follow valence rules. For example, diamond and silicon carbide.

Molecular crystals: They are bonded by weak Vander-waals forces. They are soft, weak and have low melting points. They do not follow valence rules and many organic crystals belong to this category.

Hydrogen bonded crystals: Substances such as ice and hydrogen are held together by special bonds originating in the electron spins of the orbital electrons of the hydrogen atom.

Semiconductors: Substances such as silicon and germanium, when they contain very small amounts of impurities, have lattice deficiencies with 'holes' where electrons are missing and positive charges are in excess or with extra electrons present.

7.3 INVARIANT CRYSTAL

Under ideal conditions, a growing crystal maintains its geometric similarity during growth and such a crystal is called invariant. A single dimension can be used as the measure of the size of an invariant crystal of any definite shape. The ratio of the total surface area of a crystal, S_P to the crystal volume, V_P is

$$\frac{S_P}{V_P} = \frac{6}{(\varphi_s d_p)} \tag{7.1}$$

where φ_s is the sphericity and d_p is particle size. If the characteristic length L of the crystal is defined as equal to $\varphi_s\ d_p$ then,

$$L = \varphi_s\ d_p = 6\left(\frac{V_P}{S_P}\right) \tag{7.2}$$

7.4 PRINCIPLES OF CRYSTALLIZATION

The following factors govern the principles of crystallization.

7.4.1 Purity of Product

Although a crystal is as such pure, it retains mother liquor when removed from the final magma. If the crop contains crystalline aggregates, considerable amount of mother liquor may be occluded within the solid mass. When the product dries under such conditions, the impurity of the mother liquor is retained by the crystals. Hence, in practice, the retained mother liquor is separated from the crystals by filtration or centrifuging and the balance is removed by washing with fresh solvent to improve the purity of product.

7.4.2 Equilibria and Yield

Equilibrium in crystallization process is reached when the solution is saturated and it is normally indicated by the solubility curve. The solubility is normally indicated as a function of temperature. The characteristics of the solubility curve changes with the solute involved as indicated below in Fig. 7.1.

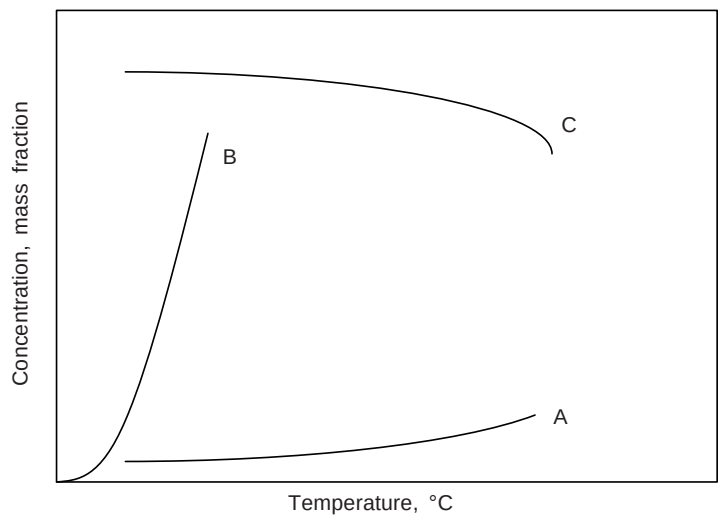
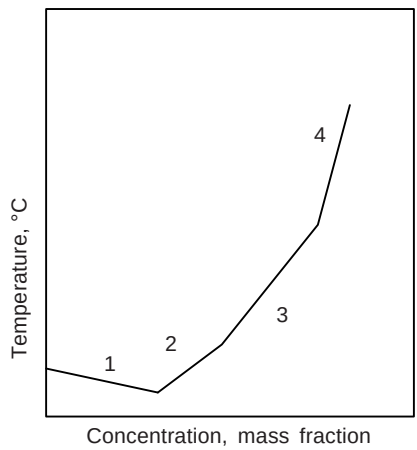


Fig. 7.1 Solubility curve.

In Fig. 7.1, A—flat solubility curve (NaCl), B—Steep solubility curve (KNO_3), and C—inverted solubility curve ($MnSO_4 \cdot H_2O$).

Substances like NaCl show a flat solubility curve (curve A) i.e. the increase in solubility with temperature is very less. The curve B is exhibited by most of the materials like KNO_3. In such systems the solubility increases very rapidly with temperature. Some of the substances like $MnSO_4 \cdot H_2O$ indicate an inverted solubility curve (curve C) in which solubility decreases with increase in temperature. Many important inorganic substances crystallise with water during crystallization. In some systems, as in the case of $MgSO_4 + H_2O$, several hydrates are formed depending on various levels of concentration and temperature. A typical phase diagram is shown in Fig. 7.2.



1: salt. a H_2O, 2: salt. b H_2O, 3: salt. c H_2O, 4: salt. d H_2O

Fig. 7.2 Phase diagram for hydrated salt.

7.4.3 Yield

The yield in the crystallization process is calculated by knowing the original concentration of solution and the solubility of solute at the final temperature. When the rate of crystal growth is slow, considerable time is required to reach equilibrium. This generally occurs in viscous solutions and when the crystals have a tendency to sink to the bottom. In such cases, there is little crystal surface exposed to the super-saturated solution and hence the actual yield is less than that calculated from solubility curve. Estimating the yield of anhydrous crystals is simple as the solid phase is free from solvent. However, when the crop contains water of crystallization, as in the case of hydrated salt, that must also be considered in estimating the yield.

7.4.4 Enthalpy Balance

In heat balance calculations for crystallizers, the heat of crystallization is important. This is the latent heat evolved when solid forms from solution. The

heat of crystallization varies with both temperature and concentration. This is also equal to the heat absorbed by crystals dissolving in a saturated solution. This data is available in literature. The enthalpy balance enables us to determine the quantity of heat that should be removed from a solution during crystallization process.

7.5 SUPER-SATURATION

Crystallization from a solution is an example of the creation of a new phase within a homogeneous phase. It occurs in two stages, viz., nucleation and crystal growth. The driving force for these is super-saturation and without this crystallization does not occur. This is nothing but the concentration difference between that of the super-saturated solution in which the crystal is growing and that of a solution in equilibrium with the crystal. Super-saturation is generated by one or more of the following methods,

(i) by evaporation of solvent
(ii) by cooling
(iii) by adding a third component which may combine with the original solvent to form a mixed solvent, where the solubility of the solute is highly reduced is called *salting*. A new solute may be created chemically by the addition of third component by the reaction between the original solute and the new third component added is called *precipitation*.

The super-saturation is defined as follows

$$\Delta y = y - y_s$$
$$\Delta c = C - C_s$$

where

Δy = super-saturation, mole fraction of solute
y = mole fraction of solute in super-saturated solution
y_s = mole fraction of solute in saturated solution
Δc = molar super-saturation, moles per unit volume
C = molar concentration of solute in super-saturated solution
C_s = molar concentration of solute in saturated solution

The super-saturation ratio, s is defined as y/y_s.

Since the effect of super-saturation differs in nucleation process and crystal growth process, separate treatment of both the processes is necessary.

7.6 NUCLEATION

Crystal nuclei may form from molecules, atoms or ions. Due to their rapid motion, these particles are called *kinetic units*. These kinetic units join together and also break frequently. When they are held together, they can also be joined by a third particle. Combinations of this sort are called *clusters*. These clusters are also liable for break in their formation. When the number of kinetic units in a cluster is very large, it is called an *embryo*. However, they can also break into clusters or kinetic units. Depending on the level of super-saturation, the embryo grows and forms

nucleus. The number of kinetic units in a nucleus is of the order of hundred and when the nuclei gains kinetic units, it results in the formation of a crystal. However, nuclei can also loose units and dissolve in solution. Hence, the stages of crystal growth may be indicated as,

Kinetic units → Cluster → Embryo → Nucleus → Crystal units

The factors which influence nucleation are,

(i) super-saturation
(ii) it is stimulated by an input of mechanical energy—by the action of agitators and pumps
(iii) the presence of solid particles—microscopic or macroscopic
(iv) the effect of particle size

7.6.1 Theory of Homogeneous Nucleation

The solubility data in literature applies only to large crystals. A small crystal can be in equilibrium with a super-saturated solution. Such an equilibrium is unstable, because if a large crystal is also present in the solution, the smaller crystal will dissolve and the larger one will grow until the smaller one disappears. The surface energy of a particle is given by

$$U_S = \gamma \cdot A_p \tag{7.3}$$

where U_S is surface energy in ergs, A_p is area of particle in cm^2 and γ is interfacial tension in ergs/cm^2.

The interfacial tension γ is the work required to increase the area of the particle by 1 cm^2 in the absence of other energy effects. Interfacial tension depends on temperature but not on the shape or size of the particle.

Although the crystals are polyhedra nuclei, the embryos are assumed to be spherical in shape. The particle diameter is taken as D cm. A nucleus has a definite size and it depends on super-saturation $(y - y_s)$ and the diameter of the nucleus is D_C. The volume of a particle is V_P in cm^3 and the mass is N_P in g-moles. Nucleation is assumed to take place at constant temperature and volume.

The formation of embryo is due to two kinds of work, viz., the one to form surface and the other to form volume. The surface work is given by Eq. (7.3). The work required to form the volume is $-(\mu - \mu_\alpha)\, N_P$, where μ is the chemical potential of the solute in the super-saturated solution and μ_α is the chemical potential of the solid, based on a crystal sufficiently large to reach equilibrium with a saturated solution. The unit of μ is ergs/g-mole.

The total work is equated to increase of work function, $\Delta\Omega$, and is shown below

$$\Delta\Omega = \gamma A_p - (\mu - \mu_\alpha)\, N_p \tag{7.4}$$

If V_m is the molal volume of the solid phase in cc/g mole then,

$$N_p = \frac{V_P}{V_m} \tag{7.5}$$

For spherical particles $A_P = \pi D^2$ (7.6)

$$V_P = \left(\frac{\pi}{6}\right) D^3 \tag{7.7}$$

Substituting Eqs. (7.5), (7.6) and (7.7) in (7.4), we get

$$\Delta\Omega = \pi D^2 \left[\gamma - \left(\frac{D}{6V_M}\right) \cdot (\mu - \mu_\alpha)\right] \tag{7.8}$$

Under equilibrium for a definite value of $(\mu - \mu_\alpha)$, $\frac{d}{dD}\ [\Delta\Omega] = 0$

On differentiating Eq. (7.8) and equating to zero, we get

$$\frac{d}{dD}(\Delta\Omega) = 0 = 2\pi\gamma D - \left(\frac{\pi}{6V_m}\right)(\mu - \mu_\alpha)\ (3D^2) = 0$$

i.e. $$D = \frac{4\gamma V_m}{[\mu - \mu_\alpha]} \tag{7.9}$$

The diameter under the above equilibrium condition is the critical diameter D_C for nucleus and the work function becomes $(\mu - \mu_\alpha)_C$. Therefore,

$$D_C = \frac{4\gamma V_m}{[\mu - \mu_\alpha]_C} \tag{7.10}$$

The chemical potential difference is related to the concentrations of the saturated and super-saturated solutions by

$$(\mu - \mu_\alpha)_C = nR_oT \ln\left(\frac{y}{y_s}\right) = n\ R_oT \ln s \tag{7.11}$$

where n is the number of ions per molecule of solute and for molecular crystals, $n = 1$.

Substituting $(\mu - \mu_\alpha)_C$ from Eq. (7.11) in Eq. (7.10), we get

$$\ln s = \ln\left(\frac{y}{y_s}\right) = \frac{4\gamma V_m}{n\ R_oT \cdot D_C} \tag{7.12}$$

Equation (7.12) is the Kelvin equation which relates solubility of a substance to its particle size.

The work required for nucleation $\Delta\Omega_C$ is found by substituting for D_C from Eq. (7.10) and for D from Eq. (7.9) and then $(\mu - \mu_\alpha)_C$ from Eq. (7.11) into the resulting equation.

$$\Delta\Omega_C = \gamma\pi\ D_C{}^2 - (\mu - \mu_\alpha)_C \left[\left(\frac{\pi}{6}\right)\left(\frac{D_C^3}{V_m}\right)\right]$$

$$= \gamma\pi\ D_C{}^2 - \left(\frac{4\gamma V_m}{D_C}\right)\left[\left(\frac{\pi}{6}\right)\left(\frac{D_C^3}{V_m}\right)\right]$$

$$= \frac{1}{3}\gamma\pi D_C^2 = \frac{1}{3}\gamma\pi\left[\frac{4\gamma V_m}{nR_oT\ln s}\right]^2$$

$$= \frac{16\pi\gamma^3 V_m^2}{3(nR_oT\ln s)^2} \tag{7.13}$$

The work of nucleation represents an energy barrier which controls the kinetics of embryo building. The rate of nucleation from the theory of chemical kinetics is

$$\tilde{N} = C\exp\left[-\frac{\Delta\Omega_C}{KT}\right] = C\exp\left[\frac{-\Delta\Omega_C\cdot N_a}{R_oT}\right] \tag{7.14}$$

Substituting for $\Delta\Omega_C$ from Eq. (7.13) in Eq. (7.14)

$$\tilde{N} = C\exp\left[\frac{-16\pi\gamma^3 V_m^2 N_a}{3n^2(R_oT)^3(\ln s)^2}\right] \tag{7.15}$$

where $\tilde{N}$ is the nucleation rate, number/cm^3 s

$\mathcal{K}$ is Boltzmann constant, 1.3805×10^{-16} erg/(g mol K)

N_a is Avogadro number, 6.0225×10^{23} molecules/g mol

R_o is gas constant, 8.3143×10^7 ergs/g mol ^oC

C is frequency factor

The frequency factor represents the rate at which individual particle strikes the surface of crystal. As far as the formation of water droplets from saturated vapour is concerned, it is of the order of 10^{25}. Its value from solutions is not known and the nucleation rate is dominated by ln s term in the exponent. Similarly, the numerical values for γ also are uncertain. They can be calculated from solid state theory and for ordinary salts, it is of the order of 80 to 100 ergs/cm^2. With these values of C and γ for a nucleation rate of one nucleus per sec, per cm^3, the value of s can be calculated and it is found to be very high and it is highly impossible for materials of usual solubility. Hence, homogeneous nucleation in ordinary crystallization from solution never occurs and that all actual nucleations in these situations are heterogeneous.

However, in precipitation reactions where y_s is very small and where large super-saturations can be generated rapidly, homogenous nucleation probably occurs.

7.6.2 Heterogeneous Nucleation

The catalytic effect of solid particles on nucleation rate is the reduction of energy barrier. Hence, the level of super-saturation needed is greatly reduced.

7.7 CRYSTAL GROWTH

Crystal growth is a diffusional process, modified by the effect of the solid surfaces on which the growth occurs. Solute diffuses through the liquid phase and reach the growing faces of a crystal and crystal starts growing.

7.7.1 ΔL—Law of Crystal Growth

If all crystals in magma grow in a uniform super-saturation field and at the same temperature and if all crystals grow from birth at a rate governed by the super-saturation, then all crystals are not only invariant but also have the same growth rate that is independent of size. Hence, growth rate is not a function of crystal size and it remains constant.

i.e. $$\Delta L = G \cdot \Delta t \tag{7.16}$$

7.7.2 Growth Coefficients

By using the expression for overall coefficient K in terms of film coefficients, the overall coefficient is expressed in terms of film coefficient.

The overall coefficient K is defined as,

$$K = \frac{m'}{[S_P(y - y_s)]} \tag{7.17}$$

where m' is the rate of mass transfer in moles/hr, S_P is surface area of crystal and $(y - y_s)$ is the driving force.

For an invariant crystal of volume V_P, the volume is proportional to the cube of its characteristic length L

i.e. $$V_P = aL^3, \text{ where } a \text{ is a constant} \tag{7.18}$$

If ρ_m is the molar density, the mass of the crystal m is then,

$$m = V_P\ \rho_m = aL^3\ \rho_m \tag{7.19}$$

Differentiating the above Eq. (7.19), we get

$$m' = \frac{dm}{dt} = 3aL^2\ \rho_m \left(\frac{dL}{dt}\right)$$

$$= 3a\ L^2\ \rho_m G \tag{7.20}$$

We have $$\frac{S_P}{V_P} = \frac{6}{L} \text{ and } Vp = aL^3$$

$\therefore$ $$S_P = 6aL^2 \tag{7.21}$$

Substituting the above Eq. (7.20) and (7.21) in Eq. (7.17), we get

$$K = \frac{3aL^2\rho_m G}{6aL^2\ (y - y_s)} = \frac{(\rho_m G)}{2(y - y_s)}$$

$\therefore$ $$G = 2\frac{(y - y_s)\,K}{\rho_m} \tag{7.22}$$

7.8 APPLICATION TO DESIGN

Once the yield is estimated, then it is desirable to solve the problem of estimating the crystal size distribution of the product. An idealised model, called the Mixed Suspension–Mixed Product Removal (MSMPR) model has served well as a basis for identifying the kinetic parameters and hence the evaluation of the performance of such a crystallizer. The assumptions of the MSMPR model is as follows,

(i) steady state operation
(ii) no product classification
(iii) uniform super-saturation exists throughout the magma
(iv) ΔL—law of crystal growth applies
(v) no size classified withdrawal system is used
(vi) no crystals in feed
(vii) mother liquor in product magma is saturated
(viii) no disintegration of crystals

Due to the above assumptions, nucleation rate is constant at all points in the magma. The particle size distribution is independent of location in the crystallization and is identical to that of the distribution in the product.

7.8.1 Population Density Function

The crystal number density is defined as the number of crystals of size L and smaller in the magma per unit volume of mother liquor. Hence, if N is the number of crystal of size L and less in the magma, V is the volume of mother liquor in the magma and L is the size of crystals, then N/V is the crystal number density. A plot of crystal density vs crystal size is shown below in Fig. 7.3. At $L = 0$; $N = 0$ and at $L = L_T$, the largest of the crystal in the magma, then $N = N_T$. The population density n, is defined as the slope of the cumulative distribution curve at size L or it can be expressed as

$$n = \frac{d\left(\frac{N}{V}\right)}{dL} = \left(\frac{1}{V}\right)\left(\frac{dN}{dL}\right) \tag{7.23}$$

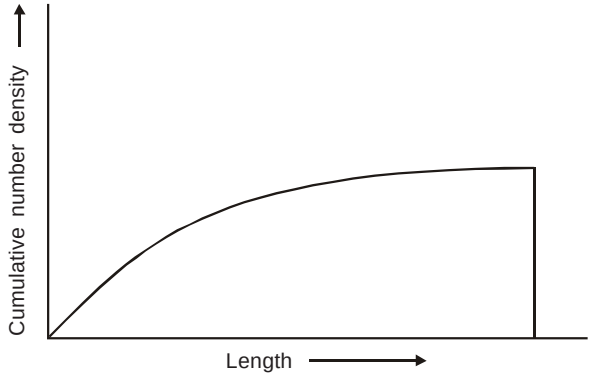


Fig. 7.3 Population density function.

At $L = 0$, the population density is the maximum, i.e. $n = n^0$ and at $L = L_T$, the population density is zero. In MSMPR model, both n and N/V are invariant with time and location.

Consider $N\,dL$ crystals between sizes L and $L + dL$ per unit volume of magma in the crystallizer. In the MSMPR model, each crystal of length L has same age $t_{m.}$

$$\therefore \qquad L = G\,t_m \tag{7.24}$$

Let $(\Delta n)\,dL$ be the amount of product withdrawn from the above magma during a time interval of Δt. Since the operation is in steady state, withdrawal of product does not affect size distribution in the product. If Q is the volumetric flow rate of liquid in the product and V_C is the total volume of liquid in the crystallizer, then

$$\frac{-\Delta n\,dL}{n\,dL} = -\left(\frac{\Delta n}{n}\right) = \left(\frac{Q\Delta t}{V_C}\right) \tag{7.25}$$

where the negative sign indicates the withdrawn product. The growth of each crystal is given by,

$$\Delta L = G\Delta t \tag{7.26}$$

Combining Eqs. (7.25) and (7.26) for eliminating Δt,

$$-\Delta n = Q\,\Delta t \cdot \frac{n}{V_C}$$

$$-\frac{\Delta n}{\Delta L} = \frac{Qn}{V_C G} \tag{7.26a}$$

Letting $\Delta L \to 0$; $-\dfrac{\Delta n}{\Delta L} = -\dfrac{dn}{dL}$

$$\Delta L \to 0$$

i.e.
$$-\left(\frac{dn}{dL}\right) = \frac{Qn}{V_C G} \tag{7.27}$$

The retention time τ of the magma in the crystallizer is defined by

$$\tau = V_C/Q$$

Hence

$$\frac{-dn}{n} = \left(\frac{1}{G\tau}\right)dL \tag{7.28}$$

Integrating Eq. (7.28), we get

$$\int_{n^0}^{n} \frac{dn}{n} = \left(-\frac{1}{G\tau}\int_0^L dL\right)$$

$$\ln\left(\frac{n^o}{n}\right) = \left(\frac{L}{G\tau}\right) \tag{7.29}$$

i.e. $$n = n^o e^{-z} \tag{7.30}$$

The quantity $L/G\tau$ is the dimensionless length and is denoted as Z.

7.8.2 Number of Crystals per Unit Mass

The number of crystals n_c in a unit volume of liquid in either magma or product is,

$$n_c = \int_0^\infty n\,dL \tag{7.31}$$

$$= \int_0^\infty G\tau\, n^o \cdot e^{-z} dz = n_c = n^o\, G\tau \tag{7.32}$$

The total mass of product crystals in a unit volume of liquid is,

$$m_c = \int_0^\infty mn \cdot dL \tag{7.33}$$

$$= a\rho_c n^o (G\tau)^3 \int_0^\infty Z^3 e^{-z} \cdot dZ = 6a\rho_c n^o (G\tau)^4 \tag{7.34}$$

where ρ_c is the density of crystals.

$\therefore$ The number of crystals per unit mass is

$$\frac{n_c}{m_c} = \frac{n^o G\tau}{6a\rho_c n^o (G\tau)^4} = \frac{1}{6a\rho_c (G\tau)^3} \tag{7.35}$$

The predominant crystal size L_{pr} in the product occurs when $L_{pr} = 3G\tau$

$\therefore$ $$\frac{n_c}{m_c} = \frac{1}{\left[6a\rho_c \left(\frac{L^3_{pr}}{27}\right)\right]} = \frac{9}{2a\rho_c L^3_{pr}} \tag{7.36}$$

7.9 CRYSTALLIZERS

The classification of crystallizers is generally based on the method by which super-saturation is achieved.

7.9.1 Super-saturation by Cooling

This is done on batch basis in a (i) tank crystallizer and in an (ii) agitated tank crystallizer and on continuous basis in Swenson–Walker crystallizer.

7.9.1.1 Tank crystallizer

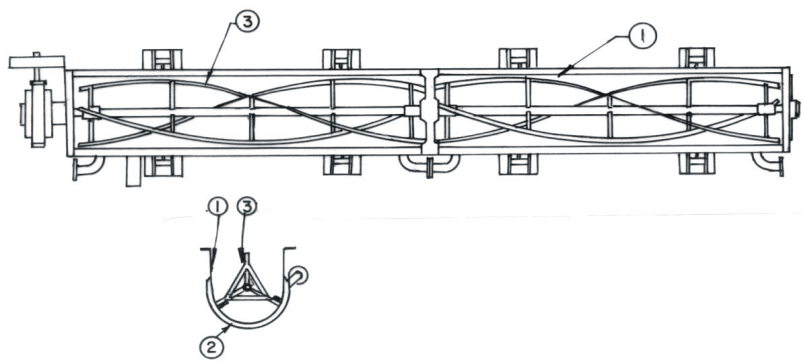
A simple tank crystallizer is an open tank containing the solution from which the crystal grows without any agitation. In this crystallizer, the crystal growth is very slow, irregular and interlocking of crystals occur. The mother liquor is also occluded due to interlocking of crystals which leads to impurities in the crystals.

7.9.1.2 Agitated tank crystallizer

This is a modified form of tank crystallizer with provisions of cooling coil and agitation. The agitation not only helps in a uniform transfer of heat but also enables a uniform growth of crystals. However, the major disadvantage is the build up of crystals on cooling coil which decreases the rate of heat transfer.

7.9.1.3 Swenson–Walker crystallizer

It is a continuous crystallizer which makes use of cooling to achieve super-saturation. A sketch of the equipment is shown in Fig. 7.4. It generally consists of an open trough, *A* which is approximately 0.6 m wide with a semi-cylindrical bottom and an external water jacket, *B*. Through the water jacket, cooling water is circulated. A slow speed, long pitch spiral agitator rotating at about 5–7 rpm, and set as close to the bottom of the trough as possible is provided inside the trough. The crystallizer comprises of several units for handling higher amount of feed. Each unit will be generally 3 m in length. The maximum length that could



1. Trough 2. Jackets 3. Agitator

Fig. 7.4 Swenson–Walker crystalliser.

be driven by one shaft is around 12 m. If it is desired to have higher lengths, then the units are arranged one above the other and the solution transferred from one set of unit to the other in the cascade.

The hot concentrated solution is fed into the trough and cooling water flows through jackets in countercurrent direction. If necessary to control crystal size, an extra amount of water can also be let into certain sections. The objective of providing the spiral stirrer is primarily to prevent the accumulation of crystals on the cooling surface and subsequently to lift the formed crystals and shower them down through the solution. This enables a uniform size of crystals which is free from inclusions or aggregations. The rotation of stirrer also helps to transfer the crystals towards the crystal discharge point for subsequent processing.

7.9.2 Super-saturation by Evaporation

Super-saturation occurs by evaporation in a Krystal crystallizer.

7.9.2.1 Krystal crystallizer

The schematic diagram of Krystal crystallizer is shown in Fig. 7.5. This consists of a vapour head, '1' and crystal growth chamber, '2'. Solution is pumped from chamber '2' by using a pump, '6' to chamber '1' through a heater '3'. Vapour

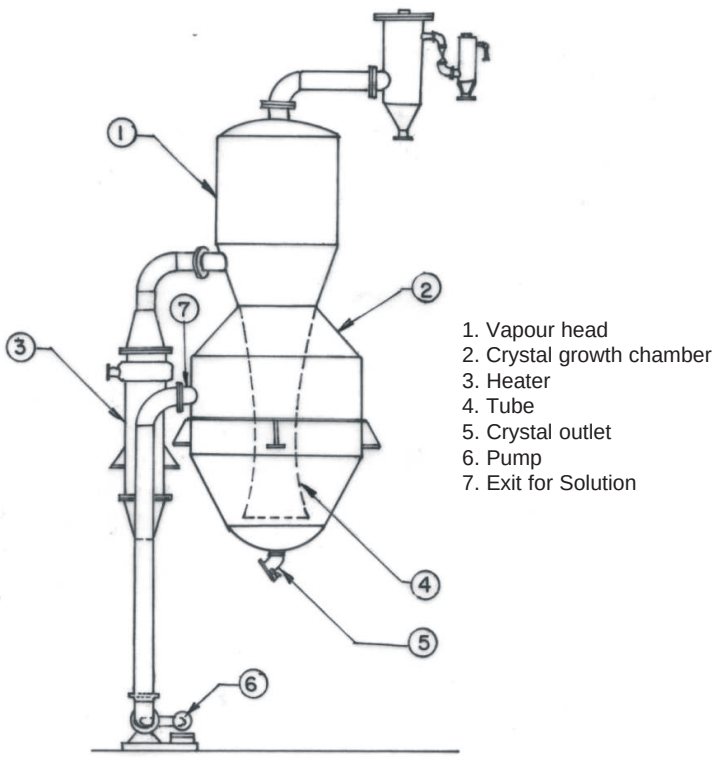


Fig. 7.5 Krystal crystallizer.

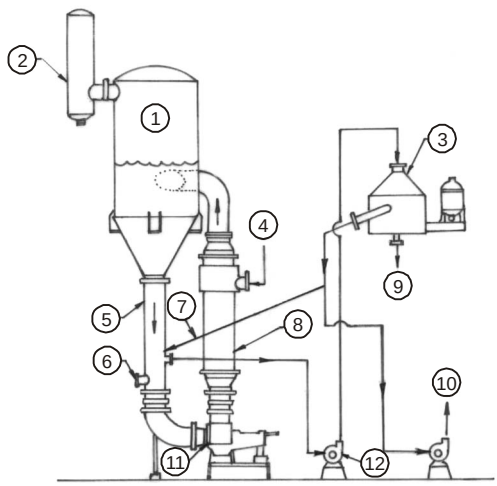
from '1' discharges into a condenser and vacuum pump. The operation is so effectively controlled that crystals do not form in '1'. The section '1' is connected to almost the bottom of chamber '2' through a tube '4'. The lower part of '2' contains a bed of crystals suspended in an upward flowing stream of liquid caused by the discharge from '4'. The super-saturated liquid formed in '1' flows over the surface of the crystals in '2'. The liquid from '2' after contributing to crystallization process leaves through '7' and recirculated. Periodically the coarse crystals are drawn out from the bottom of the vessel through '5'. There is a gradual variation in size of crystals in '2' with the coarser ones at the bottom and the finer ones at the top. Feed is usually introduced into the suction of pump '6'.

7.9.3 Super-saturation by Evaporation and Cooling

Most of the modern crystallizers achieve super-saturation by adiabatic evaporative cooling. In such crystallizers vacuum is maintained and a warm saturated solution at a temperature well above the boiling point of solution (corresponding to the pressure in crystallizer and concentration of solute) is fed in. The feed solution cools spontaneously to the equilibrium temperature, since both the enthalpy of cooling and the enthalpy of crystallization appear as enthalpy of vapourisation, a portion of the solvent evaporates. The super-saturation thus generated by both cooling and evaporation causes nucleation and crystal growth. The yield of crystals is proportional to the difference between the concentration of the feed and the solubility of the solute at equilibrium temperature.

7.9.3.1 Vacuum crystallizer

A typical continuous vacuum crystallizer using the above principle is shown in Fig. 7.6. The crystallizer consists of a body maintained under vacuum and is similar to that of single effect evaporator. From the conical bottom of the body, magma flows down through a down-pipe. It is mixed with the fresh feed from the



1. Body 2. Barometric condenser 3. Centrifuge 4. Steam 5. Down-pipe 6. Feed 7. Mother liquor recycle 8. Heater 9. Product 10. Bleed 11. Circulating pump 12. Slurry pump

Fig. 7.6 Continuous vacuum crystallizer.

feed inlet point located before the suction of the circulating pump. The mixture is sent up through a vertical tubular heater by the pump. The heated mixture enters the crystallizer body through a tangential inlet just below the level of magma surface. The swirling motion to the magma facilitates flash evaporation and cooling. The super-saturation thus generated provides the driving potential for nucleation and crystal growth.

Mother liquor is separated from the crystals in a centrifuge. Crystals are taken off as a product for further processing and the mother liquor is recycled back to the down-pipe. Some portion of the mother liquor bleeds from the system to prevent accumulation of impurities.

The drawbacks of vacuum crystallizer are as follows:

(i) The crystals tend to settle at the bottom of the crystallizer where there may be little or no super saturation.
(ii) The crystallizer will not be effective in the absence of agitation to magma.

7.9.3.2 Draft tube baffle (DTB) crystallizer

A more versatile and effective crystallizer shown in Fig. 7.7 is draft tube baffle crystallizer. In this the crystallizer body is equipped with a draft tube which also acts as a baffle to control the circulation of magma and a downward directed propeller agitator to provide a controllable circulation within the crystallizer. The DTB crystallizer is provided with an elutriation leg below the body to classify the

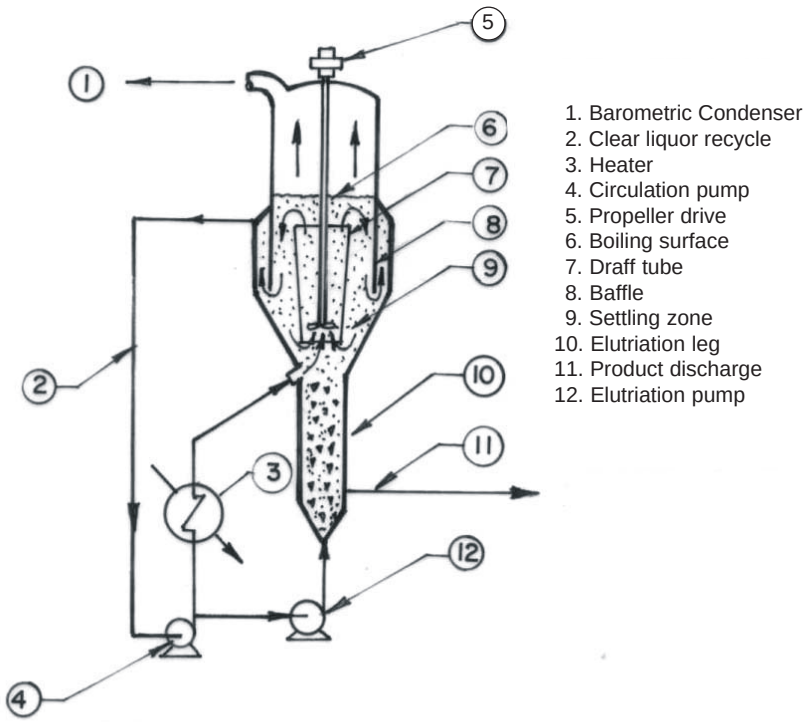


Fig. 7.7 Draft tube baffle crystalliser.

crystals by size and may also be equipped with a baffled settling zone for fines removal. There is an additional circulating pump outside the crystallizer body which circulates the recycle liquid and fresh feed through a heater. Part of the circulating liquid is pumped to the bottom of the leg and used as a hydraulic sorting fluid to carry small crystals back into crystallising zone for further growth.

Discharge slurry is withdrawn from the lower part of the elutriation leg and sent to a filter or centrifuge, and the mother liquor is returned to the crystallizer. Unwanted nuclei is removed by providing an annular space or jacket by enlarging the conical bottom and using the lower wall of the crystallizer body as a baffle. In the annular space, fines are separated from the larger ones and they float due to upward flowing stream of mother liquor. This stream of liquor along with fines of size 60 mesh and smaller, also called clean liquor, is mixed with fresh feed and sent through a heater. In the heater these tiny crystals get dissolved. The liquor is now clear and mixes with the slurry in the main body of the crystallizer.

WORKED EXAMPLES

1. Mother liquor after crystallization has a solute content of 49.8 kg of $CaCl_2$ per 100 kg of water. Find out the weight of this solution needed to dissolve 100 kg of $CaCl_2 \cdot 6H_2O$ at 25°C. Solubility at 25°C is 81.9 kg of $CaCl_2$/100 kg of water.

Solution.

Let x be the weight of water in the quantity of solution needed.

Molecular weight of $CaCl_2 = 111$,

Molecular weight of $CaCl_2 \cdot 6H_2O = 219$

$$\text{Water present in 100 kg of } CaCl_2 \cdot 6H_2O = \left(\frac{108}{219}\right) \times 100 = 49.3 \text{ kg}$$

$$CaCl_2 \text{ present in 100 kg of } CaCl_2 \cdot 6H_2O = \left(\frac{111}{219}\right) \times 100 = 50.68$$

$$\text{Total } CaCl_2 \text{ entering for solubility} = 50.68 + 0.498x$$

$$\text{Total water used for solubility} = x + 49.3$$

$$\text{Total } CaCl_2 \text{ after solubility} = \frac{[81.9 \times (x + 49.3)]}{100}$$

$$\text{Making material balance for } CaCl_2 = 50.68 + 0.498x = \left(\frac{81.9}{100}\right)(x + 49.3)$$

$$50.68 + 0.498x = 0.819x + 40.37$$

$$0.819x - 0.498x = 50.68 - 40.37$$

$$0.321x = 10.30$$

$$\therefore\ x = 32.09$$

Weight of $CaCl_2$ in mother liquor corresponding to the weight water is 32.09 kg

$$= \left(\frac{49.8}{100}\right) \times 32.09$$

$$= 15.98 \text{ kg}$$

Total weight of solution needed = 15.98 + 32.09
= 48.07 kg. **Ans.**

2. Sodium nitrate solution at 50°C contains 45% by weight of sodium nitrate.
(i) Find out the percentage saturation of this solution
(ii) Find out the weight of sodium nitrate crystal formed if 1000 kg of this solution is cooled to 10°C
(iii) Find out the percentage yield of this process.
Data: Solubility at 50°C = 104.1 kg of $NaNO_3$/100 kg of water
Solubility at 10°C = 78 kg of $NaNO_3$/100 kg of water

Solution.

(i) $NaNO_3$ weight percentage of saturated solution at 50°C

$$= \left(\frac{104.1}{204.1}\right) \times 100 = 51\%$$

$$\text{Percentage saturation at 50°C} = \left[\frac{(45/55)}{(51/49)}\right]$$

$$= \left[\frac{(45 \times 49)}{(55 \times 51)}\right] \times 100 = 78.6\% \quad \textbf{Ans.}$$

(ii) Let x be the weight of $NaNO_3$ crystal formed after crystallization
By writing material balance for $NaNO_3$
$1000 \times 0.45 = x + (1000 - x) \times (78/178)$
$450 = x + 438.2 - 0.438x$
$x = 20.99$ kg. **Ans.**

(iii) Yield = weight of $NaNO_3$ crystal formed/weight of $NaNO_3$ in original solution

$$= \frac{20.99}{450} = 0.0466$$

% Yield = 4.66% **Ans.**

3. A saturated solution, of potassium sulphate is available at a temperature of 70°C. Calculate the temperature to which this should be cooled to crystallise 50% of potassium sulphate.

Data: Solubility at 70°C = 19.75/100 g of water
Solubility at 50°C = 16.5/100 g of water
Solubility at 30°C = 12.97/100 g of water
Solubility at 10°C = 9.22/100 g of water
Solubility at 0°C = 7.34/100 g of water
Basis: 1000 kg of saturated solution

Solution.

Weight of K_2SO_4 in original solution = $1000 \times \left(\frac{19.75}{119.75}\right)$ = 164.92 kg

Weight of water = 1000 – 164.92 = 835.08 kg

After crystallisation, the weight of K_2SO_4 in solution is = 164.92 × 0.5
= 82.46 kg

Weight percentage of K_2SO_4 in solution after crystallization

$$= \left[\frac{82.46}{(835.08 + 82.46)}\right] \times 100 = 8.98\%$$

From the solubility data, it is found that temperature corresponding to 8.98% of K_2SO_4 is 15°C (by linear interpolation between 10 to 30°C). **Ans.**

4. Sodium acetate solution is available at a temperature of 70°C with a solute content of 58%. Find out (i) percentage saturation (ii) yield of crystal if 2000 kg of this solution is cooled to 10°C (iii) percentage yield.
Data: Solubility at 70°C = 146 gms of sodium acetate/100 gms of water
Solubility at 10°C = 121 gms of sodium acetate/100 gms of water

Solution.

(i) Weight percentage of solute at 70°C at saturation condition

$$= \left(\frac{146}{246}\right) \times 100 = 59.34\%$$

$$\text{Percentage saturation} = \left[\frac{(58/42)}{(59.34/40.66)}\right]$$

$$= \left[\frac{(58 \times 40.66)}{(42 \times 59.34)}\right] \times 100 = 94.62\% \quad \textbf{Ans.}$$

(ii) Weight of solute in 2000 kg of solution = 2000 × 0.58 = 1160 kg
Let x be the weight of crystal formed making solute balance

$$1160 = x + (2000 - x)\ \frac{121}{221}$$

$$1160 = x + 1055.02 - 0.547x$$

$$x = 231.74 \text{ kg} \quad \textbf{Ans.}$$

(iii) % Yield = $\left(\frac{231.74}{1160}\right) \times 100 = 19.97\%$ **Ans.**

5. A saturated solution of sodium sulphate solution is available at a temperature of 30°C. Find out the weight of $Na_2SO_4 \cdot 10H_2O$ formed, if 1000 kg of this solution cooled to 10°C.
Data: Solubility at 30°C = 40.8 g of Na_2SO_4/100 g of water
Solubility at 10°C = 9.0 g of Na_2SO_4/100 g of water

Solution.

Weight percent of solute in $Na_2SO_4 \cdot 10H_2O = \frac{142}{322} = 44.2\%$

Let x be the quantity of crystal formed, by making material balance for solute

$$1000 \times \left(\frac{40.8}{140.8}\right) = 0.442x + (1000 - x) \times \frac{9}{109}$$

$$x = 576.07 \text{ kg}$$

Weight of crystals formed = 576.07 kg **Ans.**

6. A solution of sodium carbonate available at a temperature of 40°C with a solute content of 30%. Find out the weight of $Na_2CO_3 \cdot 10H_2O$ crystal formed if 2000 kg of this solution is cooled to 10°C. Also find out the yield.

Data: Solubility at 10°C = 12.5 gms of Na_2CO_3/100 gms of water

Solution.

Molecular weight of $Na_2CO_3 \cdot 10H_2O = 286$

Weight percent of solute in $Na_2CO_3 \cdot 10H_2O = \frac{106}{286} = 0.3706$

Let x be the quantity of $Na_2CO_3 \cdot 10H_2O$ crystal formed

$$2000 \times 0.3 = 0.3706x + (2000 - x) \times \frac{12.5}{112.5}$$

$x = 1455.86$ kg **Ans.**

Weight of crystals present in the original solution

$$= \left(\frac{286}{106}\right) \times 2000 \times 0.3 = 1618.87$$

$$\% \text{ Yield} = \left(\frac{1455.87}{1618.87}\right) \times 100 = 89.93\% \quad \textbf{Ans.}$$

7. A saturated solution of K_2CO_3 is available at a temperature of 80°C. If it is cooled to 20°C, find the weight of crystal ($K_2CO_3 \cdot 2H_2O$) formed and yield for 500 kg of solution.

Data: Solubility of K_2CO_3 at 80°C = 139.8 g of K_2CO_3/100 g of water
Solubility of K_2CO_3 at 20°C = 110.5 g of K_2CO_3/100 g of water

Solution.

Molecular weight of $K_2CO_3 \cdot 2H_2O = 174.2$

$$\text{Percentage solute in } K_2CO_3 \cdot 2H_2O = \left(\frac{138}{174.2}\right) \times 100 = 79.21\%$$

Let x be the quantity of crystal formed, making material balance for solute

$$500 \times \left(\frac{139.8}{239.8}\right) = (0.7921x) + (500 - x) \times \left[\frac{110.5}{210.5}\right]$$

$$x = 108.62 \text{ kg}$$

Weight of crystals formed = 108.62 kg **Ans.**

Weight of K_2CO_3 present in the original solution

$$= \left(\frac{174.7}{138}\right) \times 500 \times \left(\frac{139.8}{239.8}\right) = 369.013$$

$$\text{Percentage yield} = \left(\frac{108.62}{369.01}\right) \times 100 = 29.4\% \quad \textbf{Ans.}$$

8. 900 kg of ferrous sulphate solution with a solute content of 40% is available. If it is cooled to 10°C, find out the weight of crystal formed and yield of the crystal with the crystal of the form $FeSO_4 \cdot 7H_2O$.

Data: Solubility of $FeSO_4$ at 10°C = 20.51 g of $FeSO_4$/100 g of water

Solution.

Molecular weight of $FeSO_4 \cdot 7H_2O$ = 277.85

Percentage solute in $FeSO_4 \cdot 7H_2O$ = 151.85/277.85 = 0.5465

Let x be the quantity of $FeSO_4 \cdot 7H_2O$ crystal formed

by making material balance for solute

$$900 \times 0.4 = 0.5465x + (900 - x) \times \left(\frac{20.51}{120.51}\right)$$

x = 549.34 kg **Ans.**

Weight of $FeSO_4 \cdot 7H_2O$ in original solution

$$= \left(\frac{277.85}{151.85}\right) \times 900 \times 0.4 = 658.71 \text{ kg}$$

$$\text{Percentage yield} = \left(\frac{549.34}{658.71}\right) \times 100 = 83.39\% \quad \textbf{Ans.}$$

9. Cesium chloride solution with a solute content of 68% is at 60°C. Find out (i) percentage saturation (ii) weight $CsCl_2$ crystal formed if 1000 kg of solution is cooled (iii) percent yield of solution if cooled to 10°C.

Data: Solubility at 60°C = 229.7 g of $CsCl_2$/100 g of water
Solubility at 10°C = 174.7 g of $CsCl_2$/100 g of water

Solution.

(i) Weight percent of solute at saturation condition at 60°C

$$= \left(\frac{229.7}{329.7}\right) \times 100 = 69.66\%$$

$$\text{Percentage saturation} = \left[\frac{(68/32)}{(69.66/30.34)}\right] \times 100 = 92.5\% \quad \textbf{Ans.}$$

(ii) Let x be the weight of $CsCl_2$ formed
Making material balance gives,

$$1000 \times 0.68 = x + (1000 - x) \times \left(\frac{174.7}{274.7}\right)$$

$$x = 120.97 \text{ kg}$$

Weight of crystal formed = 120.97 kg **Ans.**

(iii) Percentage yield = $\left(\frac{120.97}{680}\right) \times 100 = 17.8\%$ **Ans.**

10. A solution of sodium carbonate in water is saturated at a temperature of 10°C. Calculate the weight of $Na_2CO_3 \cdot 10H_2O$ needed to dissolve in 200 kg of original solution at 30°C.

Data: Solubility at 30°C = 38.8 gms of Na_2CO_3/100 g of water
Solubility at 10°C = 12.5 gms of Na_2CO_3/100 g of water

Solution.

Molecular weight of $Na_2CO_3 \cdot 10H_2O = 286$

$$\text{Percentage solute in } Na_2CO_3 \cdot 10H_2O = \left(\frac{106}{286}\right) \times 100 = 37.06\%$$

Let x be the weight of $Na_2CO_3 \cdot 10H_2O$ needed to dissolve
Weight of Na_2CO_3 originally present in 200 kg of solution

$$= 200 \times \left(\frac{12.5}{112.5}\right) = 22 \text{ kg}$$

Weight of water = 200 – 22 = 178 kg
Weight of Na_2CO_3 after dissolution = $22 + 0.3706x$
Weight fraction of solute after dissolution at 30°C

$$= \frac{38.8}{138.8} = 0.279$$

For the total solution after dissolution

$$= \frac{(22 + 0.3706x)}{[(22 + 0.3706x) + (178 + 0.6294x)]} = 0.279$$

Solving for x,

x = 369 kg **Ans.**

11. A 35% solution of Na_2CO_3 weighing 6000 kg is cooled to 20°C to yield crystals of $Na_2CO_3 \cdot 10H_2O$. During cooling 4% by weight of original solution is lost due to vapourisation. Find out the weight of crystal formed.

Data: Solubility at 20°C = 21.5 g of Na_2CO_3/100 g of water

Solution.

Molecular weight of $Na_2CO_3.10H_2O = 286$

$$\text{Percentage of solute in hydrated salt} = \left(\frac{106}{286}\right) \times 100 = 37.06\%$$

Weight of solute = 6000 × 0.35 = 2100 kg

Weight of solution lost by vapourisation = 6000 × 0.04 = 240 kg

Let x be the weight of $Na_2CO_3 \cdot 10H_2O$ formed

Making material balance on solute

$$2100 = 0.3706x + (6000 - 240 - x) \times \left(\frac{21.5}{121.5}\right)$$

$$x = 5580 \text{ kg} \qquad \textbf{Ans.}$$

12. How much feed is required when 10,000 kg of crystal as $FeSO_4 \cdot 7H_2O$ is produced per hour by a simple vacuum crystallizer. The feed containing 40 parts of $FeSO_4$ per 100 parts of total water, enters the crystallizer at 80°C. The crystallizer vacuum is such that crystallizer temperature of 30°C can be produced.

Data: Saturated solution at 30°C contains 30 parts of $FeSO_4$ per 100 parts of total water vapour enthalpy is 612 cal/g (neglect superheat). The enthalpies of saturated solution, the crystals leaving the crystallizer and feed are: –1.33, –50.56 and 26.002 cal/g.

Solution.

$FeSO_4 \cdot 7H_2O$

Crystals formed = 10000 kg

Enthalpy of feed h_F at 80°C = 26.002 cal/g

Enthalpy of saturated solution at 30°C = h_L = –1.33 cal/g

Enthalpy of crystals h_C = – 50.56

$$x_F = \frac{40}{(100 + 40)} = 0.286$$

$$x_M = \frac{30}{(100 + 30)} = 0.231$$

$$x_C = \frac{151.85}{277.85} = 0.547$$

Component balance,

$$F(x_F) = (M)\ (x_M) + C(x_C)$$

$$0.286F = (M)\ (0.231) + (10000)\ (0.547)$$

$$F = M + 10000 + V$$

$$F \cdot H_F = V \cdot H_V + M \cdot H_M + C \cdot H_C$$

$$H_F = 26.002 \text{ cal/g}$$

$$H_V = 612 \text{ cal/g}$$

$$H_M = -1.33 \text{ cal/g}$$

$$H_C = -50.56 \text{ cal/g}$$

$$F = M + V + 10000 \quad (1)$$

$$0.286\ F = 0.231\ M + 5470 \quad (2)$$

$$(26.002)\ (F) = (612)\ V + (-1.33)\ (M) + (-50.56)\ (10000) \quad (3)$$

Solving Eq. (1) × 0.286

$$0.286\ F = 0.286\ M + 0.286\ V + 2860 \quad (4)$$

$$0 = 0.055\ M + 0.286\ V - 2610$$

$$2610 = 0.055\ M + 0.286\ V \quad (5)$$

Equation (1) × 26.002

$$26.002\ F = 26.002\ M + 26.002\ V + 260200 \quad (6)$$

Equation (6) – Eq. (3)

$$0 = 27.332\ M - 585.998\ V + 765800 \quad (7)$$

$$765800 = -27.33\ M + 585.998\ V$$

Equation (5) × 496.9 gives

$$1296909 = +27.33\ M + 142.113\ V \quad (8)$$

Equation (7) + Eq. (8) $2062709 = 728.111\ V$

$$\therefore\ V = 2832.96 \text{ kg/h}$$

$$M = 32723.16 \text{ kg/h}$$

$$F = 45556.12 \text{ kg/h} \quad \textbf{Ans.}$$

13. A Swenson–Walker crystallizer has to produce 800 kg per hour of $FeSO_4 \cdot 7H_2O$ crystals. The saturated solution enters the crystallizer at 49°C and the slurry leaves at 27°C. Cooling water enters the crystallizer jacket at 15°C and leaves at 21°C. The overall heat transfer co-efficient has been estimated to be 175 kcal/(hr)(m^2)(°C). There are 1.3 m^2 of cooling surface per metre of crystallizer length.

(i) Estimate the cooling water requirement in kg/h.

(ii) Determine the number of crystallizer sections, each section being 3 metre long.

Data: Saturated solutions of $FeSO_4$ at 49°C and 27°C contain 140 parts and 74 parts of $FeSO_4 \cdot 7H_2O$ per 100 parts of free water respectively. Average specific heat of the initial solution is 0.70 and the heat of crystallization is 15.8 kcal/kg.

Solution.

Crystals produced = 800 kg/h

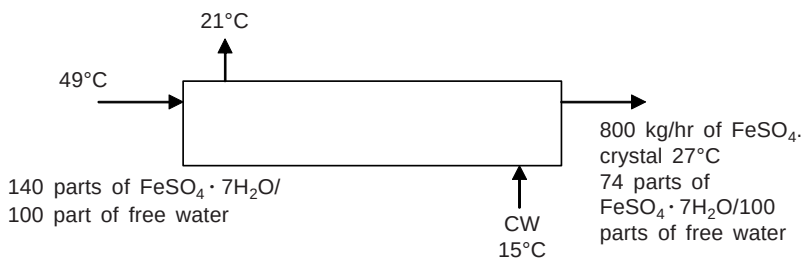


Fig. 7.8 Example 13 Material and energy balance schematic diagram.

$$U = 175 \text{ kcal/(h)(m}^2\text{)(°C)}$$

Feed concentration = 140 parts of $FeSO_4 \cdot 7H_2O$/100 parts of free water

$$140 \times \frac{151.85}{277.85} FeSO_4$$

i.e., 140 kg of $FeSO_4 \cdot 7H_2O$ = 76.51 kg of $FeSO_4$

Concentration in feed solution = $\frac{76.51}{240}$ = 0.319

Product concentration in leaving solution/100 parts of free water =

$$74 \times \frac{151.85}{277.85} = 40.44$$

Concentration of $FeSO_4$ in product = $\frac{40.44}{174}$ = 0.2324

$$x_L = \frac{151.85}{277.85} = 0.547$$

$\therefore$ Total feed solution, F entering by mass balance is

$$\text{Feed} = \text{Mother liquor} + \text{Crystals}$$

$$F = M + C$$

Making a solute balance

$$F \cdot x_F = M \cdot x_M + C \cdot x_C$$

$$(F)(0.319) = (M)(0.2324) + (C)\ (0.5465)$$

$$= (F - C)\ (0.2324) + (C)\ (0.5465)$$

$$F\ (0.0866) = C\ (0.3141)$$

$$F = \frac{800 \times 0.3141}{0.0866} = 2901.62 \text{ kg/h}$$

$$M = 2101.616 \text{ kg/h}$$

$$F = 2901.62 \text{ kg/h}$$

$$M = 2101.62 \text{ kg/h}$$

$$C = 800.00 \text{ kg/h}$$

Making an energy balance,

Heat to be removed by cooling water 'Q' = (Heat to be removed from solution + Heat of Crystillisation) = $F \cdot C_p \cdot (\Delta T) + (\Delta H_C)_C$

$$Q = (2901.62)\ (0.7)\ (49 - 27) + (15.8)(800)$$

$$= 57324.95 \text{ kcal/h}$$

Cooling water needed = $m_W C_p \Delta T$

$$= \frac{57324.95}{(1)(6)} = 9554.16 \text{ kg/h}$$

$$Q = UA\ (\Delta T)\ l_m$$

$$(\Delta T)\ l_m = \frac{(49-21)-(27-15)}{\ln\left[\frac{49-21}{27-15}\right]} = \frac{(28-12)}{\ln\left(\frac{28}{12}\right)} = \frac{16}{0.847} = 18.88$$

$$A = \frac{57324.94}{175(18.88)} = 17.35 \text{ m}^2$$

$$\text{Length needed} = \frac{17.35}{1.3} = 13.35 \text{ m} \qquad \textbf{Ans.}$$

EXERCISES

1. A solution of sodium nitrate in water at a temperature of 40°C contains 45% $NaNO_3$ by weight.

(a) Calculate the percentage saturation of this solution

(b) Calculate the weight of $NaNO_3$ that may be crystallised from 500 kg of solution by reducing the temperature to 10°C

(c) Calculate the percentage yield of the process.

Solubility of $NaNO_3$ at 40°C = 51.4% by weight.
Solubility of $NaNO_3$ at 10°C = 44.5% by weight.

(**Ans:** (a) 87.55%, (b) 4.5 kg, (c) 2%)

2. A solution of $K_2Cr_2O_7$ in water contains 10% by weight. From 1000 kg of this solution are evaporated 600 kg of water. The remaining solution is cooled to 20°C. Calculate the amount and the percentage yield of $K_2Cr_2O_7$ crystals formed.

Solubility at 20°C = 0.39 kmole/1000 kg H_2O.

(**Ans:** 65.58 kg, 65.58%)

3. 1000 kg of a 25% aqueous solution of Na_2CO_3 is slowly cooled to 20°C. During cooling 10% water originally present evaporates. The crystal is $Na_2CO_3 \cdot 10H_2O$. If the solubility of anhydrous Na_2CO_3 at 20°C is 21.5 kg/100 kg of water, what weight of salt crystallises out?

(**Ans:** 445.64 kg)

4. A batch of saturated Na_2CO_3 solution of 100 kg is to be prepared at 50°C. The solubility is 4.48 g moles/1000 g H_2O at 50°C.

(i) If the monohydrate were available, how many kg of water would be required to form the solution?

(ii) If the decahydrate is available how many kg of salt will be required?

(**Ans:** 62.33 kg, 54.68 kg)

5. A crystallizer is charged with 10000 kg of aqueous solution at 104°C containing 30% by weight of anhydrous Na_2SO_4. The solution is then cooled

to 20°C. During this operation 4% of water is lost by evaporation. Glauber salt crystallises out. Find the yield of crystals.

Solubility at 20°C = 19.4 g Na_2SO_4/100 g water.

(**Ans:** 74.98%)

6. 2500 kg of KCl are present in a saturated solution at 80°C. The solution is cooled to 20°C in an open tank. The solubilities of KCl at 80°C and 20°C are 55 and 35 parts per 100 parts of water.

(a) Assuming water equal to 5% by weight of solution is lost by evaporation, calculate the weight of crystals obtained.

(b) Calculate the yield of crystals neglecting loss of water by evaporation KCl crystallises without any water of crystals.

(**Ans:** 908.755 kg and 36.37%)

7. A crystallizer is charged with 6400 kg of an aqueous solution containing 29.6% of anhydrous sodium sulphate. The solution is cooled and 10% of the initial water is lost by evaporation. $Na_2SO_4 \cdot 10H_2O$ crystallises out. If the mother liquor (after crystallization) is found to contain 18.3% Na_2SO_4, calculate the weight of the mother liquor.

(**Ans:** 2826.7 kg)

8. A hot solution containing 2000 kg of $MgSO_4$ and water at 330 K and with a concentration of 30 wt% $MgSO_4$ is cooled to 293 K and $MgSO_4 \cdot 7H_2O$ crystals are removed. The solubility at 293 K is 35.5 kg $MgSO_4$ per 100 kg total water. The average heat capacity of feed solution is 2.93 kJ/kg K. The heat of solution at 293 K is—13.31×10^3 kJ/K · mol $MgSO_4 \cdot 7H_2O$. Calculate the yield of crystals and make a heat balance. Assume no water is vaporised. Molecular weight of $MgSO_4$ = 120.35.

(**Ans:** 27.29%)

9. A hot solution containing 5000 kg of Na_2CO_3 and water with a concentration of 25 wt% Na_2CO_3 is cooled to 293 K and crystals of $Na_2CO_3 \cdot 10H_2O$ are precipitated. At 293 K, the solubility is 21.5 kg anhydrous Na_2CO_3/100 kg of water. Calculate the yield of crystals obtained if 5% of the original water in the system evaporates on cooling. Molecular weight of Na_2CO_3 = 106.

(**Ans:** 60.98%)

8

ABSORPTION

8.1 INTRODUCTION

Absorption is one of the important gas–liquid contact operations in which a gaseous mixture is contacted with a solvent to dissolve one or more components of the gas preferentially and provide a solution of them in the solvent. Some of the applications of this operation are as follows:

(i) Ammonia is removed from coke-oven gas with water

(ii) Benzene and toluene vapours are removed using hydrocarbon oil from the coke-oven gas.

(iii) Hydrogen sulfide is removed from naturally occurring hydrocarbon gases with alkaline solutions.

(iv) Ammonia and other water soluble harmful gases from air are removed using water.

8.2 GAS SOLUBILITY IN LIQUIDS AT EQUILIBRIUM

The equilibrium characteristics of gas solubility in liquids are generally represented as partial pressure of solute in gas (p^*) vs mole fraction of solute in liquid (x). A typical gas solubility curve drawn at a particular temperature and pressure for different gases is shown in Fig. 8.1. If the gas solubility is low, then the equilibrium pressure for that particular system is very high. The solubility of gas is significantly affected by the temperature. Generally absorption processes are exothermic and if the temperature is increased at equilibrium, the solubility of gases, but not always, will be decreased due to evolution of heat.

8.3 IDEAL AND NON-IDEAL LIQUID SOLUTIONS

In an ideal solution, all the components present in the solution approach similarity with regard to their chemical nature. When the gas mixture is in equilibrium with an ideal solution, then it follows Raoult's law.

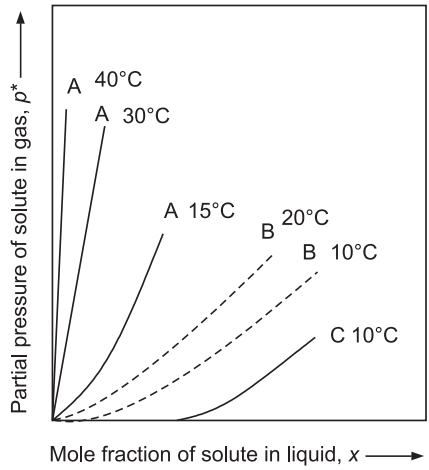


Fig. 8.1 Solubility of gas in liquid.

$$p^* = Px \tag{8.1}$$

where p^* is the partial pressure of solute, P is the vapour pressure of solute at the same temperature and x is the mole fraction of solute in liquid.

For non-ideal solutions, Henry's law can be applied and is given by,

$$y^* = \frac{p^*}{P_t} = m \cdot x \tag{8.2}$$

where m is Henry's constant, P_t is the total pressure and y^* is the mole fraction of solute in gas.

8.4 CHOICE OF SOLVENT FOR ABSORPTION

The following properties are to be considered while choosing a particular solvent in any absorption system.

(i) ***Gas solubility:*** Solubility of the solute to be absorbed in solvent should be relatively high, as it will decrease the quantum of solvent requirement.

(ii) ***Chemical nature:*** Generally solvent should be chemically similar in structure to that of the solute to be absorbed as it will provide good solubility.

(iii) ***Recoverability:*** The solvent should be easily recovered and as it will help in reusing it.

(iv) ***Volatility:*** The solvent should have a low vapour pressure, i.e. less volatile.

(v) ***Corrosiveness:*** The solvent should not be corrosive to the material of construction equipment.

(vi) ***Cost and availability:*** The solvent should be inexpensive and readily available.

(vii) ***Viscosity:*** The solvent should have low viscosity as it will reduce pumping and transportation costs.

(viii) ***Toxic, flammability and stability:*** The solvent should be non-toxic, inflammable, chemically stable and non-reactive.

8.5 DESIGN OF ISOTHERMAL ABSORPTION TOWERS

The design of isothermal absorption towers is based on material balance in them. The flow of streams could be either co-current or countercurrent. The operation is either carried out as a single stage operation or as a multistage operation.

8.5.1 Single Stage—One Component Transferred—Countercurrent and Isothermal Operation

Consider a single stage isothermal absorber as shown in Fig. 8.2, where (1) and (2) refer to the bottom and top sections of the equipment respectively. Gaseous mixture entering the absorber at the bottom is contacted countercurrently with liquid solvent, entering from the top.

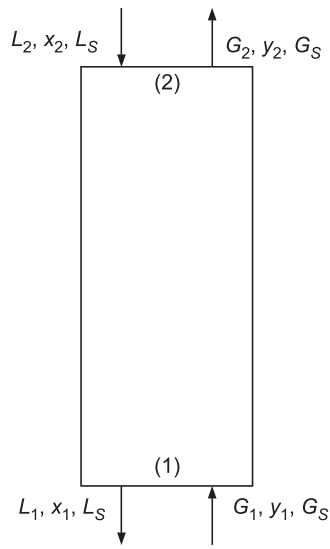


Fig. 8.2 Flow in a countercurrent absorber.

Let G_1 and L_2 be the molar flow rates of entering binary gaseous mixture and liquid respectively in moles / (area) (time).

Let G_2 and L_1 be the molar flow rates of leaving gaseous mixture and liquid respectively in moles / (area) (time).

Let G_S and L_S be the molar flow rates of inert gas and pure liquid respectively in moles/ (area) (time).

Let x, y be the mole fractions of solute in liquid and gas phases respectively.

Let X, Y be the mole ratios of solute to inert component in liquid and gas phases respectively.

In the gas phase, only one component is transferred and the other component remains as inert. Similarly, in the liquid phase, solvent is the inert component. It is more convenient to represent the concentrations of solute in liquid and gas phases in terms of mole ratios (X and Y) of solute to inert component. So

$$X = \frac{x}{(1-x)} \text{ and } Y = \frac{y}{(1-y)} \tag{8.3}$$

Likewise,
$$x = \frac{X}{(1+X)} \text{ and } y = \frac{Y}{(1+Y)} \tag{8.4}$$

$$G_S = G_1\,(1 - y_1) \text{ or } G = \frac{G_S}{(1-y)} \tag{8.5}$$

Writing the material balance on solute basis for the above countercurrent operation, we get

$$G_S Y_1 + L_S X_2 = G_S Y_2 + L_S X_1 \tag{8.6}$$

$\therefore$
$$Gs(Y_1 - Y_2) = L_S(X_1 - X_2) \tag{8.7}$$

i.e.
$$\left[\frac{L_S}{G_S}\right] = \left[\frac{(Y_1 - Y_2)}{(X_1 - X_2)}\right] \tag{8.8}$$

Equation (8.8) represents the operating line for a single stage countercurrent absorber. The operating line is linear which passes through the coordinates (X_1, Y_1) and (X_2,Y_2) with a slope of (L_S/G_S). Since the solute transfer is taking place from gas to liquid phase, the operating line always lies above the equilibrium curve, which is shown in Fig. 8.3.

Suppose, if the flow rates of gas and liquid streams are not considered on inert basis, i.e. when considered on mole fraction basis, then the operating line would be a non-linear one passing through the coordinates (x_1, y_1) and (x_2, y_2) as shown in Fig. 8.3. It is highly impossible to know the intermediate concentrations which will enable one to draw this operating curve passing through the terminal points (x_1, y_1) and (x_2, y_2). Hence, it is more preferable to obtain the linear operating line with the known terminal concentrations of the system by considering mole ratio basis as shown in Fig. 8.4.

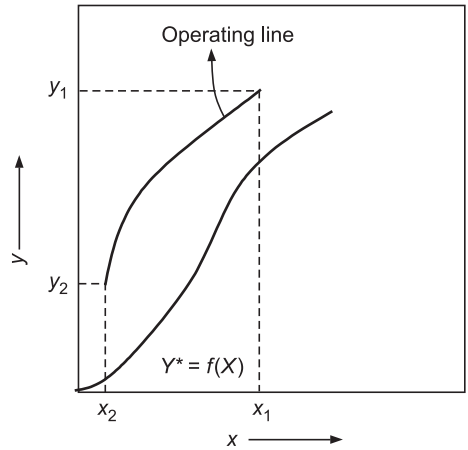


Fig. 8.3 Equilibrium curve and operating line on mole fraction basis.

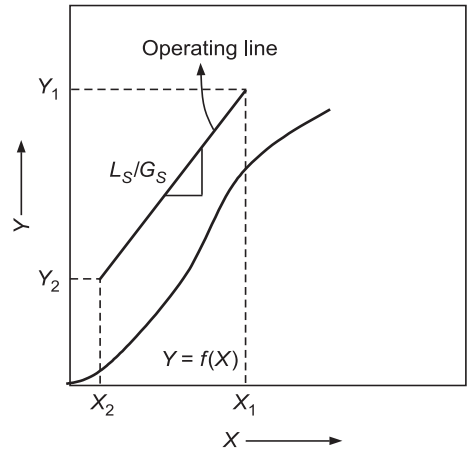


Fig. 8.4 Equilibrium Curve and Operating Line in Mole Ratio Basis.

8.5.2 Determination of Minimum (L_S/G_S) Ratio

In absorption, minimum (L_S/G_S) ratio indicates a slope for operating line, at which the maximum amount of solute concentration is obtained in the final liquid. It will be achieved only in the presence of infinite number of stages for a desired level of absorption of solute. When the operating line is tangent to the equilibrium curve, then there is no net driving force and the required time of contact for the concentration change desired is infinite and an infinitely tall tower will result. This is highly uneconomical. So, the tower is operated at the (L_S/G_S) ratio of 1.2 to 2.0 times the minimum (L_S/G_S) ratio.

8.5.3 Steps Involved in Determining $(L_S/G_S)_{min}$

1. Plot X and Y data to draw the equilibrium curve.
2. Locate point $A(X_2,Y_2)$.
3. From point A draw a tangent to the equilibrium curve.
4. Determine the slope of this line which will be $(L_S/G_S)_{\ min}$.
5. Extend the line from Y_1 to intersect this operating line which corresponds to the point $B'[(X_1)_{\ max,}Y_1]$.
6. Determine $(L_S/G_S)_{\ actual}$ and find the slope.
7. Using the operating line equation, obtain $(X_1)_{\ actual}$ and draw the actual operating line AB as shown in Fig. 8.5

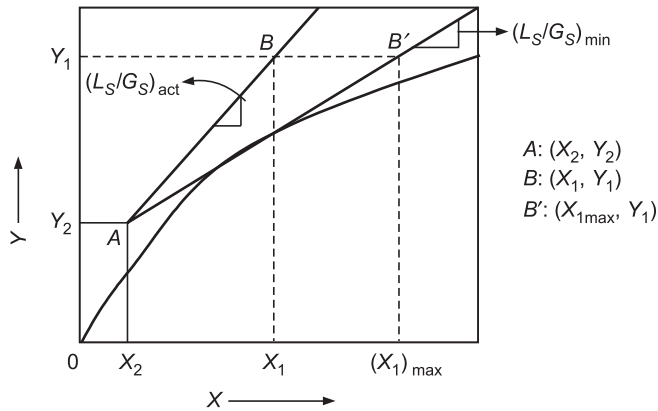


Fig. 8.5 Minimum L/G ratio.

In some cases, the equilibrium curve will be more or less a straight line or concave upward. In such cases the minimum (L_S/G_S) ratio can be determined as shown in Figs. 8.6(a) and (b).

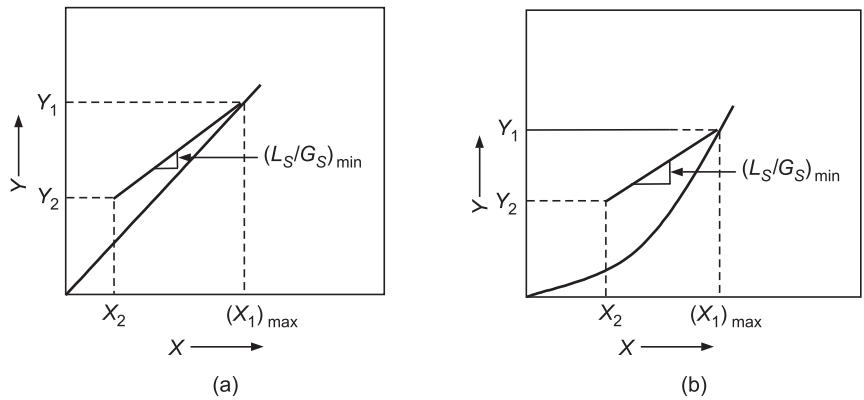


Fig. 8.6 Equilibrium curve and operating line for special cases.

Therefore,
$$(L_S/G_S)_{\min} = \left[\frac{(Y_1 - Y_2)}{(X_1)_{\max} - X_2}\right] \tag{8.9}$$

Since $(L_S/G_S)_{\min}$ is known, $(X_1)_{\max}$ can be determined as all the other quantities in Eq. (8.9) are known.

8.5.4 Multistage Countercurrent Isothermal Absorption

Let us consider a multistage tray tower containing N_p number of stages as shown in Fig. 8.7, where the suffix p represents the tray number. The operation is isothermal and it is assumed that the average composition of gas leaving from a tray is in equilibrium with the average composition of liquid leaving from the same tray. The flow of streams is countercurrent. The liquid flows downwards and the gas upwards and only one component is transferred. The number of theoretical or ideal stages required for the desired operation in the tower is determined as follows:

The material balance on inert basis gives,

$$G_S Y_{Np+1} + L_S X_0 = G_S Y_1 + L_S X_{Np} \tag{8.10}$$

$$G_S(Y_{Np+1} - Y_1) = L_S (X_{Np} - X_0) \tag{8.11}$$

i.e.
$$\left[\frac{L_S}{G_S}\right] = \left[\frac{(Y_{Np+1} - Y_1)}{(X_{Np} - X_0)}\right] \tag{8.12}$$

Equation (8.12) represents a linear operating line for a multistage countercurrent absorber which passes through the coordinates (X_0, Y_1) and (X_{Np}, Y_{Np+1}) with a slope (L_S/G_S). Between the equilibrium curve and the operating line, a stepwise construction is made to obtain the number of theoretical trays. The stepwise construction is started from (X_0, Y_1) since it represents operating condition in plate number 1 (as per our convention). This is illustrated in Fig. 8.8.

8.5.5 Analytical Method to Determine the Number of Trays

In some special cases such as dilute gaseous mixtures or solutions, the equilibrium curve is a straight line, the number of trays can be determined analytically by using Kremser-Brown–Souders equation given in Eq. (8.13) without going in for a graphical method.

$$N_p = \frac{\log\left[\left(\frac{y_{Np+1} - mx_0}{y_1 - mx_0}\right)(1 - 1/A) + 1/A\right]}{\log A} \tag{8.13}$$

where A is the absorption factor given by L/mG and m is the slope of the equilibrium curve.

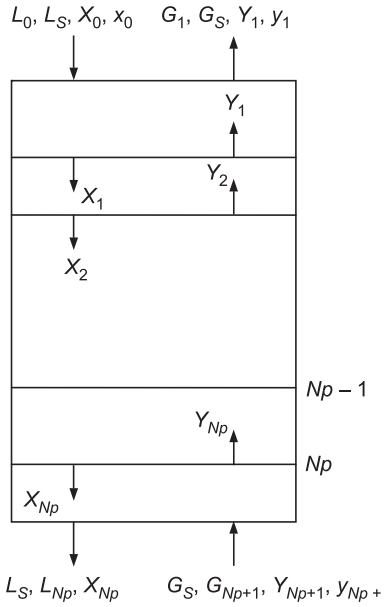


Fig. 8.7 Various streams in a countercurrent multistage tray tower.

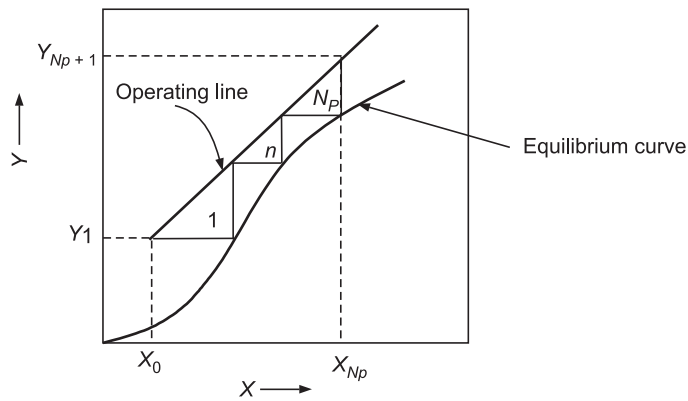


Fig. 8.8 Stepwise construction for estimating the number of plates/stages.

Absorption factor, *A*, is defined as the ratio of the slope of the operating line to that of the equilibrium curve. If '*A*' varies due to small changes in *L*/*G* from bottom to top of the tower, then the geometric mean value of *A* has to be considered.

Hence, geometric mean value of $A = \sqrt{A_1 A_2}$ (8.14)

where, $$A_1 = \frac{L_1}{mG_1} \approx \frac{L_0}{mG_1}$$

$$A_2 = \frac{L_{Np}}{mG_{Np}} \approx \frac{L_{Np}}{mG_{Np+1}}$$

where A_1 is the absorption factor at the top of the tower and A_2 is the absorption factor at the bottom of the tower. For larger variations in A, graphical computations must be followed.

8.5.6 Significance of Absorption Factor

If $A < 1$, the operating line and equilibrium curve converge at the lower end of the tower indicating that the solubility of solute is limited even when large number of trays are provided.

If $A > 1$, any degree of separation is possible with adequate number of trays. However, as A increases beyond 1.0 for a fixed quantity of gas and a given degree of absorption, the absorbed solute is dissolved in a larger quantity of liquid (solvent) and hence becomes less valuable. In addition to that, the number of trays also decreased leading to a lower cost of equipment. This leads to a variation in total cost of operation which will pass through a minimum. Hence, for an economical operation, the value of A has been estimated for various systems and found to be in the range of 1.2 to 2.0.

8.6 DESIGN OF MULTISTAGE NON-ISOTHERMAL ABSORBER

Generally the absorption operations are exothermic in nature. Hence, the solubility of gas decreases as temperature of the liquid increases which in turn decreases the capacity of the absorber. When concentrated gaseous mixtures are to be absorbed in solvent then the temperature effects have to be taken into account. If the heat liberated is more, then cooling coils should be provided for an efficient absorption operation. Since the temperature is varying from tray to tray, it influences the concentration changes as well as the flow rate of streams. Hence, energy balance should also be incorporated along with material balance to determine the number of trays. It is very difficult to compute manually the tray to tray calculations. A simple algorithm is developed for one ideal tray involving trial and error calculations and then the programming is extended to other trays for the determination of the number of trays.

Consider a stagewise tray tower operating non-isothermally as shown in Fig. 8.9.

Total mass balance around the entire tower gives

$$G_{Np+1} + L_0 = L_{Np} + G_1 \tag{8.15}$$

Component balance gives,

$$[G_{Np+1}]\, y_{Np+1} + L_0 x_0 = L_{Np} x_{Np} + G_1 y_1 \tag{8.16}$$

Energy balance gives,

$$[G_{Np+1}] H_{G,Np+1} + L_0 H_{L_0} = L_{Np} H_{L_{Np}} + G_1 H_{G_1} \tag{8.17}$$

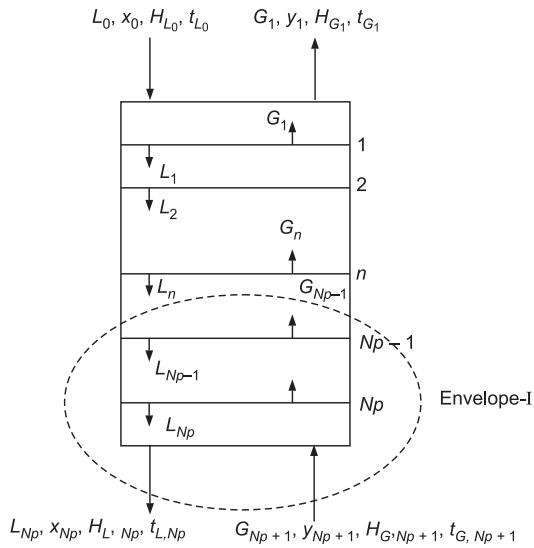


Fig. 8.9 Streams in a countercurrent multistage tray tower and envelope-I.

where, H is the molal enthalpy of streams. Enthalpies can be determined using the available literature data with reference to some base temperature say, t_0, (Pure state).

$$H_G = C_{pG,\,\text{inert}}\,(t_G - t_0)(1 - y) + y[C_{pG,\,\text{solute}}\,(t_G - t_0) + \lambda_0] \tag{8.18}$$

$$H_L = C_{pL,\,\text{inert}}\,(t_L - t_0)(1 - x) + x[C_{pL,\,\text{solute}}\,(t_L - t_0)] \tag{8.19}$$

where C_p is the specific heat of the component and λ_0 is the latent heat of vapourization at reference temperature, t_0.

Now let us consider the envelope-I.

Mass and energy balance in envelope-I follow,

$$L_n + G_{Np+1} = G_{n+1} + L_{Np} \tag{8.20}$$

$$L_n x_n + [G_{Np+1}]\,y_{Np+1} = G_{n+1} y_{n+1} + L_{Np} x_{Np} \tag{8.21}$$

$$L_n H_{L_n} + [G_{Np+1}]\,H_{G_{Np+1}} = G_{n+1}\,H_{G_{n+1}} + L_{Np} H_{L_{Np}} \tag{8.22}$$

Let $n = N_p - 1$,

$$L_{Np-1} + G_{Np+1} = G_{Np} + L_{Np} \tag{8.23}$$

$$L_{Np-1} x_{Np-1} + [G_{Np+1}] y_{Np+1} = G_{Np} y_{Np} + L_{Np} x_{Np} \tag{8.24}$$

$$L_{Np-1}\,H_{L,\,Np-1} + [G_{Np+1}]\,H_{G,\,Np+1} = G_{Np} H_{G,\,Np} + L_{Np} H_{L,\,Np} \tag{8.25}$$

To solve the above system of equations and determine the number of trays, the following procedure is used.

1. Assume the top tray temperature, t_{G1}. The other values like G_{Np+1}, y_1, y_{Np+1}, L_0, x_0, t_{L_0} and $t_{G\,Np+1}$ are known.
2. Calculate G_S from the relationship, $G_S = G_{Np+1}(1 - y_{Np+1})$
3. Calculate G_1 from the relationship, $G_1 = \dfrac{G_S}{1 - y_1}$
4. Using Eq. (8.15), Calculate L_{Np}.
5. Find x_{Np} from Eq. (8.16).

6. Calculate $H_{G_{Np+1}}$, H_{L_0} and H_{G_1} using Eqs. (8.18) and (8.19).
7. Find $H_{L_{Np}}$ from Eq. (8.17).
8. Determine $t_{L_{Np}}$ making use of Eq. (8.19).
9. With this knowledge of the temperature of the last tray N_p, the compositions can be determined by $y^* = (\text{V.P/T.P})x$ or $y^* = m.x$, where V.P. is vapour pressure, T.P. is total pressure and m is equilibrium constant. Hence $y_{Np} = (m)\ x_{Np}$
10. Now for the last tray, x_{Np}, y_{Np}, $t_{L_{Np}}$ are known.
11. Find $G_{Np} = \dfrac{G_s}{(1 - y_{Np})}$
12. Now calculate $L_{Np\,-\,1}$ using Eq. (8.23).
13. Find x_{Np-1} from Eq. (8.24).
14. Calculate $H_{L_{Np-1}}$ using Eq. (8.25).
15. Find $t_{L_{Np-1}}$ from Eq. (8.19).
16. Now determine the composition, y_{Np-1} and G_{Np-1} as mentioned in step (9) and step (11) respectively.
17. Similarly calculate for the next tray by taking $n = N_{p-2}$ and starting from step (12), by making use of material and enthalpy balances.
18. Finally, the computation is stopped on reaching the value of y_1 and also satisfying the assumed t_{G_1}. If these two values namely, the assumed composition and computed composition y_1 and the assumed temperature and the calculated temperature t_0 are not satisfied together, once again the iteration has to be started fresh by assuming a new temperature t_{G_1}. However, when both the values of y_1 and t_{G_1}, are satisfied, the number of trays are known from the computation values.

8.7 DESIGN OF COCURRENT ABSORBER

In a cocurrent absorber both gas and solvent streams enter at the same end and flow in the same direction as shown in Fig. 8.10.

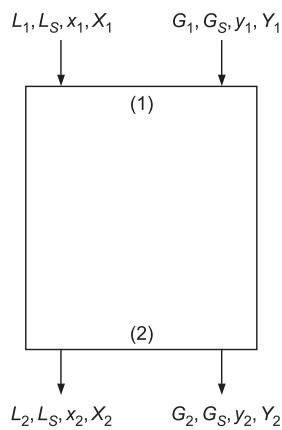


Fig. 8.10 Cocurrent absorber.

Making material balance,

$$L_S X_1 + G_S Y_1 = L_S X_2 + G_S Y_2 \tag{8.26}$$

$$\therefore \quad L_S(X_1 - X_2)] = G_S(Y_2 - Y_1) \tag{8.27}$$

$$\text{i.e.} \quad -\left[\frac{L_s}{G_s}\right] = \left[\frac{(Y_1 - Y_2)}{(X_1 - X_2)}\right] \tag{8.28}$$

Equation (8.28) is the equation of operating line for cocurrent absorption operation with the slope – (L_S/G_S) and this is presented in the X–Y diagram of Fig. 8.11. If the leaving streams are in equilibrium with each other, then the compositions are represented by (X'_2, Y'_2) and for a typical liquid phase composition of X_2, the gas phase composition will be Y_2.

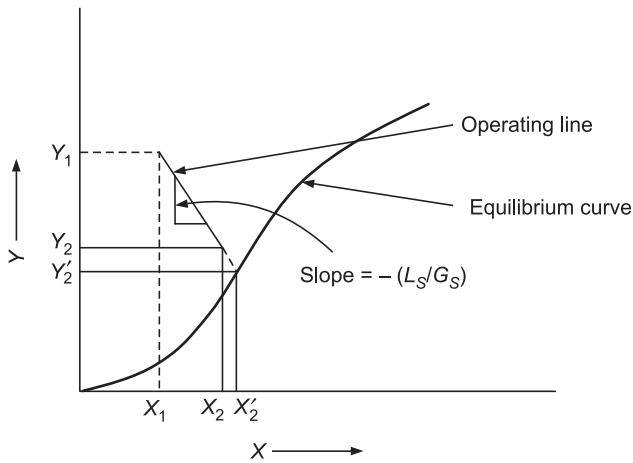


Fig. 8.11 Equilibrium curve and operating line in a cocurrent absorber.

8.8 DESIGN OF CONTINUOUS CONTACT EQUIPMENT FOR ABSORPTION

Packed columns and spray towers fall in the category of continuous contact or differential contact towers. They are different from stagewise contactors in the sense that the fluids are in continuous contact throughout the tower. So the liquid and gas compositions change continuously with respect to the height of the tower.

Consider a packed tower of unit cross sectional area as shown in the Fig. 8.12. The characteristics of inlet and outlet streams are also indicated.

Let Z be the total height of the tower and dZ be the differential height which is same as the differential volume. S is the total effective interfacial surface per unit tower cross section. Hence,

$$S = \frac{\text{Interfacial area}}{\text{Area of tower}} = \frac{(a)\cdot[A\cdot Z]}{A} \tag{8.29}$$

$$\therefore \quad dS = a \cdot dZ \tag{8.30}$$

where dS is the differential interfacial surface in the differential volume of packing.

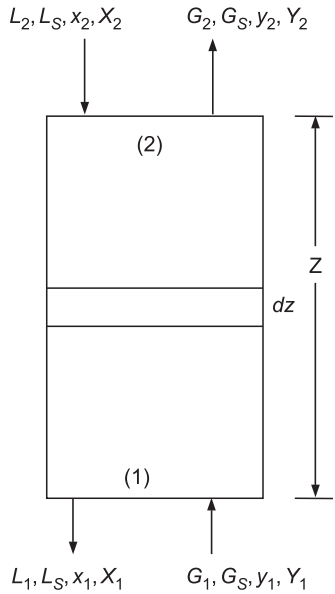


Fig. 8.12 Continuous countercurrent absorber.

As shown in Fig. 8.12, the quantity of solute A passing through the differential section is $G{\cdot}y$ moles/(area) (time). The rate of mass transfer is $d(G{\cdot}y)$ mole A/(differential volume) (time). Since $N_B = 0$, $N_A/(N_A + N_B) = 1.0$. The molar flux of A is obtained by applying the original basic flux equation,

$$N_A = \frac{\text{Rate of absorption of solute } A}{\text{Interfacial area}}$$

$$= \frac{d(Gy)}{adZ} = F_G \ln \left[\frac{(1-y_i)}{(1-y)}\right] \tag{8.31}$$

$d(Gy)$ can be written as

$$d(Gy) = d\left[\frac{G_s y}{(1-y)}\right] \tag{8.32}$$

Since one component is transferred, G and y vary throughout the tower.

i.e. $$d\left[\frac{G_s y}{(1-y)}\right] = \frac{G_s dy}{(1-y)^2} = \frac{G dy}{(1-y)} \tag{8.33}$$

Substituting Eq. (8.33) in Eq. (8.31), rearranging and integrating, we get

$$Z = \int_0^Z dZ = \int_{y_2}^{y_1} \frac{G dy}{F_G a(1-y)\ln[(1-y_i)/(1-y)]} \tag{8.34}$$

It is more convenient to write, $y - y_i = [(1 - y_i) - (1 - y)]$ (8.35)

The numerator and denominator of Eq. (8.34) can be multiplied by the right and left hand sides of Eq. (8.35) respectively to obtain

$$Z = \int_{y_2}^{y_1} \frac{G(1-y)_{iM}\,dy}{F_G a(1-y)(1-y_i)} \tag{8.36}$$

where $(1 - y)_{iM}$ is logarithmic mean of $(1 - y_i)$ and $(1 - y)$.

$$Z = \frac{G}{F_G a}\int_{y_2}^{y_1} \frac{(1-y)_{iM}\,dy}{(1-y)(y-y_i)} \approx H_{tG} \cdot N_{tG} \tag{8.37}$$

where H_{tG} is the height of a gas transfer unit and N_{tG} is the number of gas transfer units.

Thus, $$H_{tG} = \frac{G}{F_G a} = \frac{G}{k_y a(1-y)_{iM}} = \frac{G}{k_G a P_t (1-y)_{iM}} \tag{8.38}$$

in terms of other individual mass transfer coefficients.
N_{tG} is simplified further by substituting the arithmetic average instead of logarithmic average of $(1 - y)_{iM}$
Hence,

$$(1-y)_{iM} = \frac{(1-y_i)-(1-y)}{\ln\left[\dfrac{(1-y_i)}{(1-y)}\right]} \approx \frac{(1-y_i)+(1-y)}{2} \tag{8.39}$$

$$N_{tG} = \int_{y_2}^{y_1} \frac{(1-y)_{iM}\;dy}{[(1-y)(y-y_i)} = \int_{y_2}^{y_1} \frac{dy}{(y-y_i)} + \frac{1}{2}\ln\frac{(1-y_2)}{(1-y_1)} \tag{8.40}$$

Similarly, when the above mentioned relations have been applied for liquid compositions, we obtain

$$Z = \frac{L}{F_L a}\int_{x_2}^{x_1} \frac{(1-x)_{iM}\,dx}{[(1-x)(x_i-x)]} \approx H_{tL} \cdot N_{tL} \tag{8.41}$$

where H_{tL} is the height of liquid transfer unit, N_{tL} is the number of liquid transfer units and $(1 - x)_{iM}$ is logarithmic mean of $(1 - x)$ and $(1 - x_i)$
On simplification, we get

$$H_{tL} = \frac{L}{F_L a} = \frac{L}{k_x a(1-x)_{iM}} \tag{8.42}$$

and

$$N_{tL} = \int_{x_2}^{x_1} \frac{dx}{(x_i-x)} + \frac{1}{2}\ln\frac{(1-x_1)}{(1-x_2)} \tag{8.43}$$

Equations (8.38), (8.40), (8.42) and (8.43) can be used to determine the height of the tower.

With the known quantities, H_{tG} or H_{tL} can be easily determined. But N_{tG} and N_{tL} can be determined only through the graphical method. For this, plot of $1/(y - y_i)$ against y is drawn and the area under the curve will give N_{tG}. The values of y and y_i can be evaluated by drawing a line between equilibrium curve and operating line with the slope $(-k_x a/k_y a)$ where y and y_i are points of intersection of this line on operating line and equilibrium curve respectively.

8.8.1 Overall Transfer Units

In some cases where the equilibrium curve is straight and the ratio of mass transfer coefficients is constant, it is more convenient to make use of overall mass transfer coefficients. The height of the tower can be expressed in such cases as

$$Z = N_{toG} \cdot H_{toG} \tag{8.44}$$

$$N_{toG} = \int_{y_2}^{y_1} \frac{(1-y)_{*M}\, dy}{(1-y)(y-y^*)} \approx \int_{y_2}^{y_1} \frac{dy}{(y-y^*)} + \frac{1}{2} \ln \frac{(1-y_2)}{(1-y_1)} \tag{8.45}$$

$$H_{toG} = \frac{G}{F_{OG} a} = \frac{G}{K_y a (1-y)_{*M}} = \frac{G}{K_G a P_t (1-y)_{*M}} \tag{8.46}$$

$$N_{toL} = \int_{x_2}^{x_1} \frac{dx}{(x^* - x)} + \frac{1}{2} \ln \frac{(1-x_1)}{(1-x_2)} \tag{8.47}$$

$$H_{toL} = \frac{L}{F_{oL} a} = \frac{L}{K_x a (1-x)_{*M}} \tag{8.48}$$

8.8.2 Dilute Solutions

For dilute solutions or gaseous mixtures, the above equations become much simpler. The second term in Eq. (8.45) and in Eq. (8.47) become negligible. Hence,

$$N_{toG} = \int_{y_2}^{y_1} \frac{dy}{(y-y^*)} \quad \text{or} \quad N_{toL} = \int_{x_2}^{x_1} \frac{dx}{(x^* - x)} \tag{8.49}$$

If the equilibrium curve in terms of mole fractions is also linear over the entire range of x, then

$$y^* = m \cdot x + C \tag{8.50}$$

If the solutions are dilute, there won't be variations in L/G ratio throughout, and the operating line can be considered as a straight line so that the driving force $(y - y^*)$ is also linear. In such cases, Eq. (8.43) is simplified to

$$N_{toG} = \frac{(y_1 - y_2)}{(y - y^*)_M} \tag{8.51}$$

where $(y - y^*)_M$ is logarithmic average of the concentration differences at the terminals of the tower.
Therefore,

$$(y - y^*)_M = \frac{(y_1 - y_1^*) - (y_2 - y_2^*)}{\ln \dfrac{(y_1 - y_1^*)}{(y_2 - y_2^*)}} \tag{8.52}$$

and

$$H_{toG} = \frac{G}{K_y a} \quad \text{or} \quad \frac{G}{K_G a P_t} \tag{8.53}$$

8.8.3 Dilute Solutions Using Henry's Law

In dilute solutions, if Henry's law is applied, then

$$y^* = m \cdot x \tag{8.54}$$

The operating line can be written in a linear form as

$$(y - y_2) = \left(\frac{L}{G}\right)(x - x_2) \tag{8.55}$$

Eliminating x between Eqs. (8.54) and (8.55) and substituting y^* in Eq. (8.49), we get

$$N_{toG} = \frac{\ln\left\{\left[\dfrac{y_1 - mx_2}{y_2 - mx_2}\right]\left[1 - \dfrac{1}{A}\right] + \left[\dfrac{1}{A}\right]\right\}}{\left[1 - \dfrac{1}{A}\right]} \tag{8.56}$$

where A is the absorption factor = L/mG
The overall height of transfer units can also be expressed in terms of individual phases,

$$H_{toG} = H_{tG} + \left(\frac{mG}{L}\right)H_{tL} \quad \text{or} \quad H_{toL} = H_{tL} + \left(\frac{L}{mG}\right)H_{tG} \tag{8.57}$$

8.9 STRIPPING OR DESORPTION

When mass transfer occurs from liquid to gas, i.e. the solute is removed from the liquid solution by contacting with a gas, then the operation is called Desorption or Stripping.

8.9.1 Operating Line for Stripper

The schematic representation of operating lines for both countercurrent and co-current operations of a stripper are shown in Fig. 8.13 and Fig. 8.14.

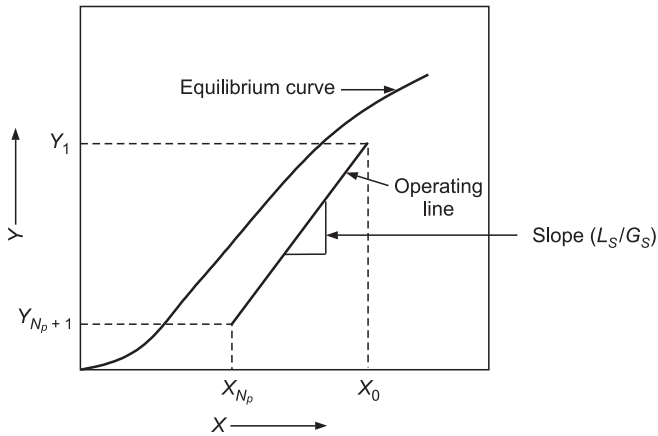


Fig. 8.13 Equilibrium curve and operating line in a countercurrent stripper.

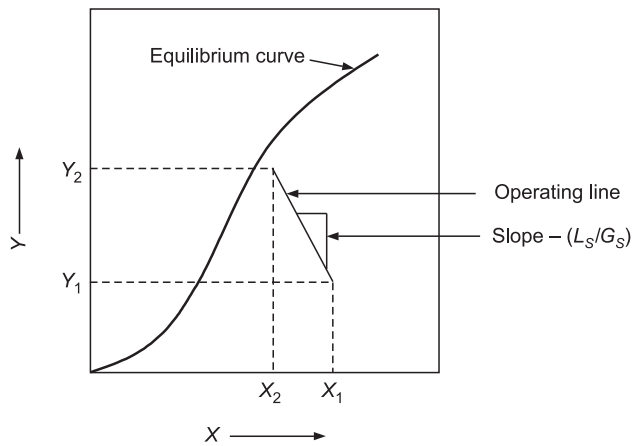


Fig. 8.14 Equilibrium curve and operating line in a cocurrent stripper.

8.9.2 Analytical Relation to Determine Number of Plates

From Kremser–Brown–Souder's Eq. (8.13) on rearranging for desorption we get,

$$N_p = \frac{\log\left\{\left[\frac{x_0 - y_{Np+1}/m}{x_{Np} - y_{Np+1}/m}\right]\left[1 - \frac{1}{S}\right] + \left[\frac{1}{S}\right]\right\}}{\log S} \qquad (8.58)$$

where S is the stripping factor, $S = \dfrac{mG}{L}$

For dilute solutions, if Henry's law is applied,

$$N_{toL} = \frac{\ln\left\{\left[\frac{x_2 - y_1/m}{x_1 - y_1/m}\right](1 - A) + A\right\}}{1 - A} \tag{8.59}$$

WORKED EXAMPLES

1. An air–NH_3 mixture containing 5% NH_3 by volume is absorbed in water using a packed tower at 20°C and 1 atm pressure to recover 98% NH_3. Gas flow rate is 1200 kg/h m^2. Calculate (a) Minimum mass flow rate of liquid; (b) NTU using 1.25 times the minimum liquid flow rate; (c) Height of packed column using $K_G a$ = 128 kg/h m^2 atm. The equilibrium relation is y = 1.154x where, x, y are expressed in mole fraction units.

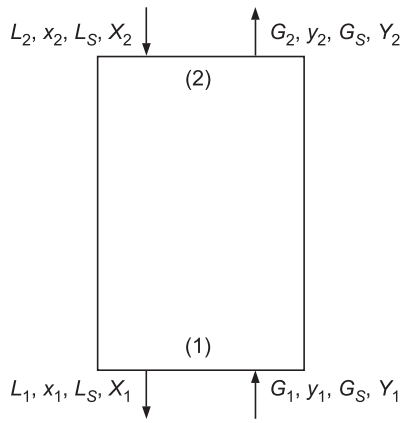


Fig. 8.15(a) Example 1.

Solution.

(a) Given that

$$y_1 = 0.05,\ P_t = 1 \text{ atm},\ T = 20^{\circ}\text{C and } X_2 = 0$$

Gas flow rate = 1200 kg/h m^2

Average molecular weight of mixture = (0.05 × 17) + (0.95 × 28.84) = 28.25

$$G_1 = \frac{1200}{28.25} = 42.478 \text{ kmol/h m}^2$$

$$G_s = G_1(1 - y_1) = 42.478\ (1 - 0.05) = 40.354 \text{ kmol/h m}^2$$

$$Y_2 = 0.02 \times 0.0526 = 0.001052$$

$$Y_1 = \frac{y_1}{1 - y_1} = \frac{0.05}{1 - 0.05} = 0.0526 \text{ kmol } NH_3\text{/kmol dry air}$$

$$Y_2 = 0.001052$$

$$y_2 = \frac{Y_2}{1+Y_2} = \frac{0.001052}{1.001052} = 0.00105$$

$$G_2 = \frac{G_s}{1-y_2} = \frac{40.354}{1-0.00105} = 40.396 \text{ kmol/h m}^2$$

$$y = 1.154x$$

$$\frac{Y}{1+Y} = \frac{1.154X}{1+X}$$

$$Y = \frac{1.154X}{1-0.154X}$$

X	0.01	0.02	0.03	0.04	0.05
Y	0.0116	0.0232	0.0348	0.0464	0.058

For minimum liquid flow rate

$$y = 1.154x, \text{ then } y_1 = 1.154x_1$$

$$0.05 = 1.154x_1, \text{ so } x_1 = 0.0433$$

$$X_1 = \frac{x_1}{1-x_1} = 0.04526 \text{ kmol NH}_3\text{/kmol water}$$

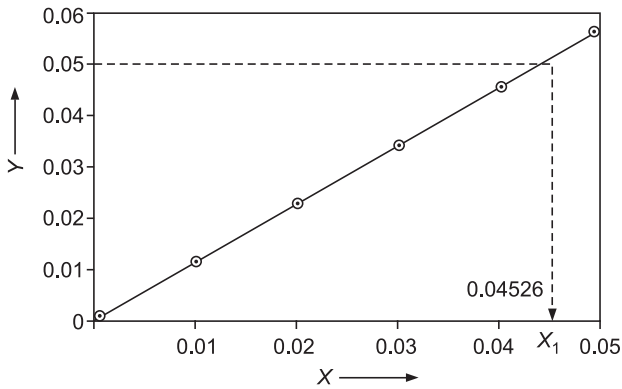


Fig. 8.15(b) Example 1.

(This can also be obtained from graph as shown in Fig. 8.15(b))

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.0526 - 0.001052)}{(0.04526 - 0)} = 1.139$$

$$(L_S)_{\min} = 40.354 \times 1.139 = 45.969 \text{ kmole/h m}^2$$

Mass flow rate of minimum water required = $45.969 \times 18 = 827.44$ kg/h m^2

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = 1.25\left(\frac{L_S}{G_S}\right)_{\min} = 1.25 \times 1.139 = 1.42375$$

(b) Again, $\left(\frac{L_S}{G_S}\right)_{\text{actual}} = \left(\frac{(Y_1 - Y_2)}{(X_{1,\text{actual}} - X_2)}\right) = 1.42375 = \frac{(0.0526 - 0.001052)}{(X_{1,\text{actual}} - 0)}$

$$X_{1,\text{ actual}} = 0.0361.$$

Hence,

$$x_1 = \frac{X_1}{(1 + X_1)} = 0.0349$$

$$y_1^* = m\, x_1 = 1.154 \times 0.0349 = 0.0403$$

$$y_2^* = m\, x_2 = 0$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}}$$

$$(y - y^*)_{lm} = \frac{[(y_1 - y_1^*) - (y_2 - y_2^*)]}{\ln\left[\frac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{[(0.05 - 0.0403) - (0.001052 - 0)]}{\ln\frac{(0.05 - 0.0403)}{(0.001052)}} = 3.89 \times 10^{-3}$$

$$\text{NTU} = \frac{(0.05 - 0.001)}{3.89 \times 10^{-3}} = 12.581 \approx 13$$

(c) Average gas flow rate

$$= \frac{(G_1 + G_2)}{2} = \frac{(42.478 + 40.396)}{2} = 41.437 \text{ kmol / h/m}^2$$

$$HTU = \frac{G}{K_G a P_t} = \frac{41.437}{128} = 0.3237 \text{ m}$$

Height of the tower, Z = NTU × HTU = 12.581 × 0. 3237 = 4.073 m

2. Air containing methanol vapour (5-mole %) is scrubbed with water in a packed tower at 26ºC and 760 mm Hg pressure to remove 95% of the methanol. The entering water is free of methanol. The gas-phase flow rate is 1.22 kmol/m^2 s and the liquid-phase flow rate is 0.631 kmol/m^2 s. If the overall height of a transfer unit based on the liquid phase resistance is 4.12 m, determine NTU and the overall liquid phase mass transfer coefficient. The equilibrium relation is $p = 0.280\, x$, where p is the partial pressure of methanol in atmospheres and x is the mole fraction of methanol in liquid.

Solution.

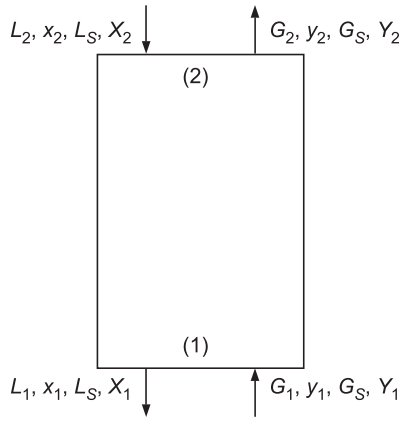


Fig. 8.16 Example 2.

$y_1 = 0.05$, $T = 26°C$, pressure = 760 mm Hg

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.05}{0.95} = 0.0526$$

$$Y_2 = 0.05 \times 0.0526 = 0.00263$$

Gas flow rate = 1.22 kmol/m^2 s

Liquid flow rate = 0.631 kmol/m^2 s

H_{toL} = 4.12 m

Equilibrium relationship is: $p = 0.280x$

where, p = partial pressure

x = mole fraction of methanol in liquid.

$$L_2 = L_S = \frac{0.631}{18} = 0.0351 \text{ kmol/m}^2 \text{ s}$$

(Assuming entering water is pure)

$$\text{Average molecular weight of entering gas} = \frac{(0.05 \times 32) + (0.95 \times 28.84)}{1}$$

$$= 2.8.998$$

$$G_1 = \frac{1.22}{28.998} = 0.0421 \text{ kmol/m}^2 \text{ s}$$

$$G_S = G_1(1 - y_1) = 0.0421(1 - 0.05) = 0.04 \text{ kmol/m}^2 \text{ s}$$

Equilibrium relation is: $p = 0.280x$

$p_t y = 0.280x$ (p_t = total pressure, x = mole fraction in liquid phase, y = mole fraction in gas phase)

$1 \times y = 0.280x$

$y = 0.280x$

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_1 - X_2)}$$

$$y_2 = \frac{Y_2}{1 + Y_2} = \frac{0.00263}{(1 + 0.00263)} = 0.00262$$

$X_2 = 0$ (assuming pure water enters the absorber)

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_1 - X_2)}$$

i.e. $$\frac{0.0351}{0.04} = \frac{(0.0526 - 0.00263)}{(X_1 - 0)}$$

Therefore, $X_1 = 0.0569$

$L_1 = L_S(1 + X_1) = 0.0351\ (1.0569) = 0.0371$ kmol/m^2 s

$L_{\text{Avg}} = (L_1 \times L_2)^{0.5} = (0.0371 \times 0.0351)^{0.5} = 0.0361$ kmol/m^2 s

$$x_1 = \frac{X_1}{(1 + X_1)} = \frac{0.0569}{(1 + 0.0569)} = 0.0539$$

We have, $y_1^* = 0.280x_1$

$x_1 = 0.0539$

$x_2 = 0.0$

$y_1 = 0.05$

$y_2 = 0.00262$

$$x_1^* = \frac{y_1}{0.280} = \frac{0.05}{0.280} = 0.1786$$

$$x_2^* = \frac{y_2}{0.280} = \frac{0.00262}{0.280} = 0.00936$$

$$(x^* - x)_{lm} = \frac{\left[(x_1^* - x_1) - (x_2^* - x_2)\right]}{\ln\left[\frac{(x_1^* - x_1)}{(x_2^* - x_2)}\right]}$$

$$= \frac{\left[(0.1786 - 0.0539) - (0.00936 - 0)\right]}{\ln\left[\frac{(0.1786 - 0.0539)}{(0.00936 - 0)}\right]} = 0.04455$$

$$N_{toL} = \frac{(x_1 - x_2)}{(x^* - x)_{lm}} = \frac{(0.0539 - 0)}{0.04455} = 1.21$$

$$H_{toL} = 4.12 \text{ m}$$

$$H_{toL} = \frac{L_{avg}}{K_{La}}$$

Therefore, $K_{La} = \dfrac{0.0361}{4.12} = 8.76 \times 10^{-3}$ kmol/m^2 s (Δx)

3. An air-NH_3 mixture containing 20-mole % NH_3 is being treated with water in a packed tower to recover NH_3. The incoming gas rate is 1000 kg/h m^2. The temperature is 35ºC and the total pressure is 1 atm. Using 1.5 times the minimum water flow rate, 95% of NH_3 is absorbed. If all the operating conditions remain unchanged, how much taller should the tower be to absorb 99% of NH_3? Henry's law is valid and $y_e = 0.746x$. Variations in gas flow rate may be neglected.

Solution.

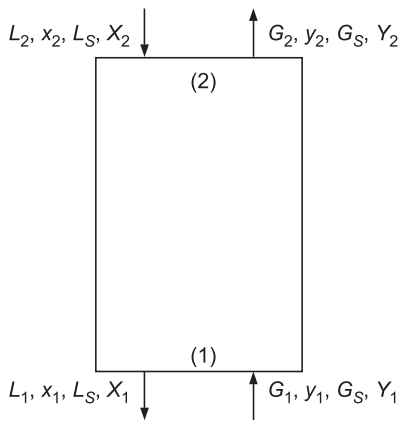


Fig. 8.17(a) Example 3.

Given that

$$y_1 = 0.2$$

Gas flow rate (incoming) = 1000 kg/h m^2
Temperature = 35°C, pressure = 1 atm

$$\text{HTU} = 1 \text{ m}$$

$$(L_S)_{\text{actual}} = 1.5 \times (L_S)_{\min}$$

Assuming, incoming water to be pure, its flow rate L_2 is L_S kmol/h m^2
Equilibrium relation = $y_e = 0.746x$

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.2}{(1 - 0.2)} = 0.25$$

95% Ammonia is absorbed

$$Y_2 = (1 - 0.95) \times 0.25 = 0.0125$$

$$y_2 = \frac{Y_2}{1+Y_2} = \frac{0.0125}{(1+0.0125)} = 0.0123$$

Average molecular weight of incoming gas mixture
$= [(0.2 \times 17) + (0.8 \times 28.84)] = 26.472$

$$G_1 = \frac{1000}{26.472} = 37.776 \text{ kmol/h m}^2$$

$$G_S = G_1 (1 - y_1) = 37.776(1 - 0.2) = 30.221 \text{ kmol/h m}^2$$

For minimum liquid flow rate.

$$y_1^* = 0.746x_1$$

$$x_1 = \frac{0.2}{0.746} = 0.2681$$

$$X_1 = \frac{x_1}{1 - x_1} = 0.3663$$

$$\left(\frac{L_s}{G_s}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.25 - 0.0125)}{(0.3663 - 0)} = 0.648$$

(Assuming pure water enters, $X_2 = 0$)
We can also obtain this graphically for which X–Y data has to be computed.

$$y_1^* = 0.746x_1$$

$$\frac{Y}{(1+Y)} = 0.746\frac{X}{(1+X)}$$

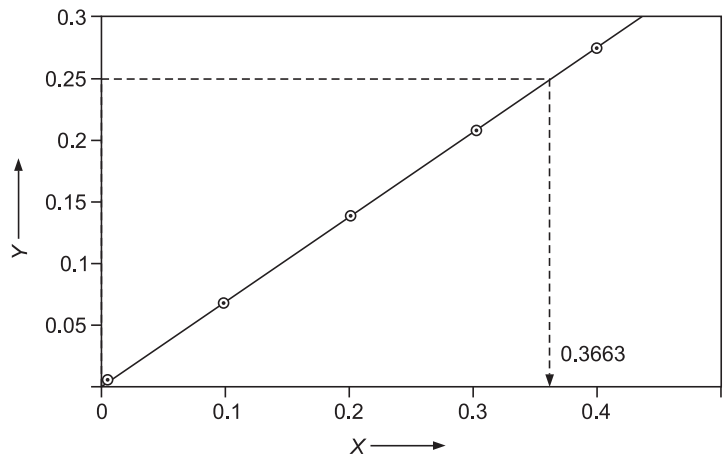


Fig. 8.17(b) Example 3.

$$Y = \frac{0.746X}{(1 + 0.254X)}$$

X	0	0.1	0.2	0.3	0.4
Y	0	0.0728	0.142	0.208	0.271

From the graph $X_{1,max}$ = 0.3663 (which is also the same as obtained from calculation)

$$\left(\frac{L_S}{G_S}\right)_{min} = 0.648$$

$$\left(\frac{L_S}{G_S}\right)_{actual} = 1.5\left(\frac{L_S}{G_S}\right)_{min} = 1.5 \times 0.648 = 0.972$$

$$\left(\frac{L_S}{G_S}\right)_{actual} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.25 - 0.0125)}{(X_1 - 0)}$$

$$X_1 = 0.2443$$

$$X_1 = \frac{X_1}{(1 + X_1)} = 0.1963$$

$$y_1^* = 0.746x_1$$

$$y_1^* = 0.746 \times 0.1963 = 0.1464$$

$$(y - y^*)_{lm} = \frac{[(y_1 - y_1^*) - (y_2 - y_2^*)]}{\ln\left[\frac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{[(0.2 - 0.1464) - (0.0123 - 0)]}{\ln\frac{(0.2 - 0.1464)}{(0.0123 - 0)}} = 0.0281$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}} = \frac{(0.2 - 0.0123)}{0.0281} = 6.68$$

$$Z = \text{HTU} \times \text{NTU} = 1 \times 6.68 = 6.68 \text{ m}$$

Now if 99% of NH_3 is absorbed,

$$Y_2 = 0.25 \times 0.01 = 0.0025$$

$$y_2 = \frac{Y_2}{1 + Y_2} = 0.0025$$

For, $\left(\frac{L_S}{G_S}\right)_{min}$

$$y_1^* = 0.746x_1$$

$$x_1 = \frac{X_1}{(1 + X_1)} = \frac{0.2}{0.746} = 0.2681$$

$$X_1 = \frac{x_1}{1 - x_1} = 0.3663$$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.25 - 0.0025)}{(0.3663 - 0)} = 0.6755$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = 1.5 \times 0.6755 = 1.013$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.25 - 0.0025)}{(X_1 - 0)}$$

$$X_1 = 0.2443$$

$$x_1 = \frac{X_1}{(1 + X_1)} = 0.1963$$

$$y_1^* = 0.746 \times 0.1963 = 0.1464$$

$$(y - y^*)_{lm} = \frac{\left[(y_1 - y_1^*) - (y_2 - y_2^*)\right]}{\ln\left[\dfrac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{[(0.2 - 0.1464) - (0.0025 - 0)]}{\ln \dfrac{(0.2 - 0.1464)}{(0.0025 - 0)}} = 0.01667$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}} = \frac{(0.2 - 0.0025)}{0.01667} = 11.847$$

$$Z = \text{NTU} \times \text{HTU} = 11.847 \times 1 = 11.847 \text{ m.}$$

In the first case, when 95% of NH_3 was absorbed, Z = 6.68 m
Increase in length of tower = 11.847 – 6.68 = 5.168 m
So, when 99% of NH_3 is to be absorbed, the tower should be 5.168 m taller than that needed for 95% NH_3 absorption, or 77.36% taller.

4. An effluent gas containing 12% C_6H_6 is to be scrubbed in a packed column, operating at 43ºC and 1 atm. pressure. The column is to be designed for treating 15 m^3 of entering gas per hour per m^2 of column cross-section, such that the exit gas will contain 1% benzene. The solvent for scrubbing is mineral oil which will enter the top of the column at a rate of 28 kg/h m^2 and a benzene content of 1%. Determine the height of the column assuming height of transfer unit to be 0.75 m. The equilibrium concentration at the operating conditions is given by $y^* = 0.263x$, where x and y are in mole fraction units.

Solution.

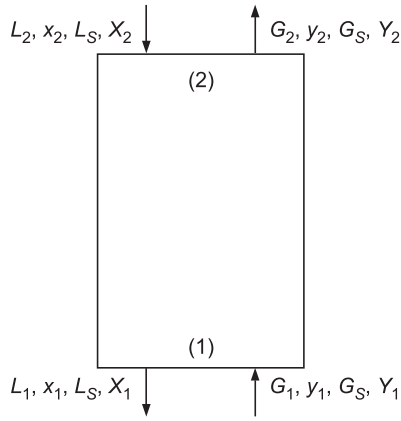


Fig. 8.18 Example 4.

$y_1 = 0.12$, $T = 43°C$, pressure = 1 atm

Gas flow rate = 15 $m^3/h\ m^2$

$y_2 = 0.01$

Solvent is mineral oil

$L_2 = 28$ kmol/h m^2, $x_2 = 0.01$,

HTU = 0.75 m

The equilibrium relation is $y^* = 0.263x$

Assuming the gas mixture to be ideal,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \frac{(1 \times 15)}{(273 + 43)} = \frac{(1 \times V_2)}{273}$$

$V_2 = 12.9589\ m^3$ (at N.T.P)

or Molar flow rate = $\frac{12.9589}{22.414} = 0.5782$ kmol

$G_1 = 0.5782$ kmol/h m^2

$G_S = G_1\ (1 - y_1) = 0.5782(1 - 0.12) = 0.5088$ kmol/h m^2

$L_S = L_2\ (1 - x_2) = 28(1 - 0.01) = 27.72$ kmol/h m^2

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_1 - X_2)}$$

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.12}{(1 - 0.12)} = 0.1364$$

$$Y_2 = \frac{y_2}{(1-y_2)} = \frac{0.01}{(1-0.01)} = 0.0101$$

$$X_2 = \frac{x_2}{(1-x_1)} = \frac{0.01}{(1-0.01)} = 0.0101$$

$$\left(\frac{L_S}{G_S}\right) = \frac{27.72}{0.5088} = \frac{(0.1364-0.0101)}{(X_1-0.0101)}$$

$$X_1 = 0.01242$$

$$x_1 = \frac{X_1}{(1+X_1)} = 0.0123$$

$$y_1^* = mx_1$$

$$y_1^* = 0.263 \times 0.0123 = 0.00323$$

$$y_2^* = 0.263 \times 0.01 = 0.00263$$

$$(y-y^*)_{lm} = \frac{[(y_1-y_1^*)-(y_2-y_2^*)]}{\ln\left[\frac{(y_1-y_1^*)}{(y_2-y_2^*)}\right]}$$

$$= \frac{[(0.12-0.00323)-(0.01-0.00263)]}{\ln\frac{(0.12-0.00323)}{(0.01-0.00263)}} = 0.0395$$

$$\text{NTU} = \frac{(y_1-y_2)}{(y-y^*)_{lm}} = \frac{(0.12-0.01)}{0.0395} = 2.786 \approx 3$$

Height of tower, Z = NTU × HTU = 2.786 × 0.75 = 2.0895 m.

5. An air–NH_3 mixture containing 5% NH_3 is being scrubbed with water in a packed tower to recover 95% NH_3. G_1= 3000 kg/h m^2, L_s = 2500 kg/h m^2. Tower is maintained at 25ºC and 1 atm pressure. Find NTU and height of the tower. The equilibrium relation is given by $y^* = 0.98x$, where x and y are mole fraction units. K_Ga = 65 kmol/h m^3 atm

Solution.

$$y_1 = 0.05$$

$$Y_1 = \frac{0.05}{(1-0.05)} = 0.0526$$

$$Y_2 = 0.05 \times 0.0526 = 0.00263$$

$$y_2 = \frac{Y_2}{1+Y_2} = \frac{0.00263}{(1+0.00263)} = 0.00262$$

Entering gas flow rate = 3000 kg/h m^2

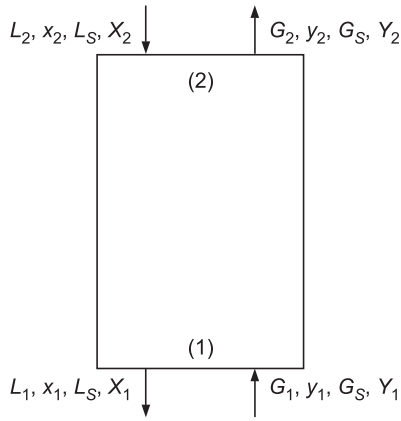


Fig. 8.19 Example 5.

L_S = 2500 kg/h m^2, T = 25°C, Pressure = 1 atm,

$K_G a$ = 65 kmol/h m^3 atm

Equilibrium relation = $y^* = 0.98x$

Average molecular weight of incoming gas mixture

$$= \frac{(0.05 \times 17) + (0.95 \times 28.84)}{1} = 28.248$$

$$G_1 = \frac{3000}{28.248} = 106.20 \text{ kmol/h m}^2$$

$$L_S = \frac{2500}{18} = 138.89 \text{ kmol/h m}^2$$

$$G_S = G_1(1 - y_1) = 106.2(1 - 0.05) = 100.89 \text{ kmol/h m}^2$$

$$G_2 = \frac{G_S}{1 - y_2} = \frac{100.89}{(1 - 0.00262)} = 101.16 \text{ kmol/h m}^2$$

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{138.89}{100.89} = \frac{(0.0526 - 0.00263)}{(X_1 - 0)}$$

Therefore, X_1 = 0.0363

$$x_1 = \frac{X_1}{(1 + X_1)} = \frac{0.0363}{(1 + 0.0363)} = 0.035$$

$$y_1^* = 0.98x_1$$

$$y_1^* = 0.98 \times 0.035 = 0.0343$$

$$x_2 = 0;\ y_2^* = 0$$

$$(y - y^*)_{lm} = \frac{\left[(y_1 - y_1^*) - (y_2 - y_2^*)\right]}{\ln\left[\frac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{\left[(0.05 - 0.0343) - (0.00262 - 0)\right]}{\ln\frac{(0.05 - 0.0343)}{(0.00262 - 0)}} = 0.0073$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}} = \frac{(0.05 - 0.00262)}{0.0073} = 6.486$$

$$G_{\text{avg}} = \frac{(G_1 + G_2)}{2} = \frac{(106.2 + 101.16)}{2} = 103.68 \text{ kmol/m}^2 \text{ h}$$

$$\text{HTU} = \frac{G_{\text{avg}}}{K_{Ga} P_t} = \frac{103.68}{(65 \times 1)} = 1.595 \text{ m}$$

$$Z = \text{NTU} \times \text{HTU} = 6.486 \times 1.595 = 10.346 \text{ m}.$$

6. An air-C_6H_6 mixture containing 5% benzene enters a countercurrent absorption tower where it is absorbed with hydrocarbon oil. $G_s = 600$ kmol/h. The solubility follows Raoult's law. Temperature at 26.7°C and 1 atm pressure are the operating conditions. The average molecular weight of oil is 200. The vapour pressure of benzene at 26.7°C is 103 mm Hg.
Find:

(i) $(L_S)_{\min}$ to recover 90% of entering C_6H_6.

(ii) The number of theoretical stages if 1.5 times the minimum liquid rate used.

(iii) The concentration of solute in liquid learning the absorber for condition (ii).

Solution.

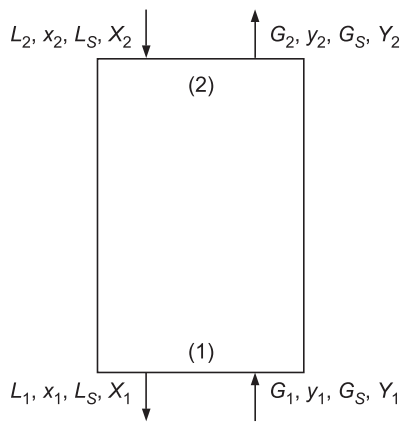


Fig. 8.20(a) Example 6.

Given that

$y_1 = 0.05$, $G_S = 600$ kmol/h $T = 26.7°C$, Pressure = 1 atm

Average molecular weight of oil = 200, $p_A = 103$ mm Hg
According to Raoult's law,

$$p^* = p_A x_A$$

$$y^* = \frac{p^*}{P_t} = \frac{(p_A x_A)}{P_t} = \frac{103}{760} x = 0.1355x$$

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.05}{(1 - 0.05)} = 0.0526$$

$$Y_2 = (0.1 \times 0.0526) = 0.00526$$

$$y_2 = \frac{Y_2}{1 + Y_2} = \frac{0.00526}{(1 + 0.00526)} = 0.00523$$

$X_2 = 0$. (Assuming pure oil enters)

We have, $y^* = 0.1355x$

$$\frac{Y}{(1+Y)} = 0.1355 \frac{X}{(1+X)}$$

Therefore,
$$Y = \frac{0.1355x}{(1 + 0.8645x)}$$

X	0	0.1	0.2	0.3	0.4	0.5	0.6
Y	0	0.0125	0.023	0.03228	0.0403	0.0473	0.0535

From the graph, we can get, $X_{1,\,\max} = 0.54$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_{1,\max} - X_2)}$$

For minimum flow rate of oil,

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.0526 - 0.00526)}{(0.54 - 0)} = 0.0877$$

$$(L_S)_{\min} = 0.0877 \times 600 = 52.62 \text{ kmol/h}$$

$$(L_S)_{\text{actual}} = 1.5 (L_S)_{\min}$$

$$(L_S)_{\text{actual}} = 1.5 \times 52.62 = 78.93 \text{ kmol/h}$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{78.93}{600} = \frac{(0.0526 - 0.00526)}{(X_1 - 0)}$$

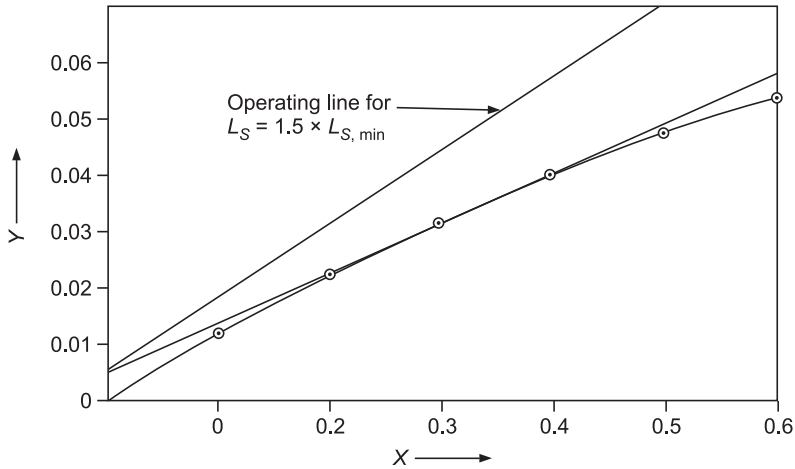


Fig. 8.20(b) Example 6.

$X_1 = 0.36$ (which is the same as obtained from graph)

$$x_1 = \frac{X_1}{(1 + X_1)} = \frac{0.36}{(1 + 0.36)} = 0.265$$

The number of stages by stepwise construction is 6.

7. It is desired to absorb 95% of acetone from feed mixture of acetone and air containing 2 (mole) % of acetone using a liquid flow rate of 20 % more than the minimum. Gas flow rate is 450 kg/h. The gas mixture enters at 25ºC and 1 atm pressure, which is the operating condition. The equilibrium relation is $y^* = 2.5x$. Find (i) Flow rate of water, and (ii) Number of theoretical plates, when the operation is carried out countercurrently.

Solution.

$$y_{Np+1} = 0.02 \qquad Y_{Np+1} = \frac{y_{Np+1}}{(1 - y_{Np+1})} = 0.0204$$

$$Y_1 = 0.0204 \times 0.05 = 0.00102,\ y_1 = 0.00102$$

$(L_S)_{\text{actual}}$ = 20% more than $(L_S)_{\text{min}}$

Gas flow rate (entering) = 450 kg/h

T = 25°C, Pressure = 1 atm, $y^* = 2.5x$

Average molecular weight of feed mixture

$$= \frac{(0.02 \times 58) + (0.98 \times 28.84)}{1} = 29.42$$

$$G_{Np+1} = \frac{450}{29.42} = 15.296 \text{ kmol/h}$$

$$G_S = G_{Np+1}(1 - y_{Np+1}) = 15.296\ (1 - 0.02) = 14.99 \text{ kmol/h}$$

Equilibrium relation is $y^* = 2.5x$

$$\text{(i.e.)} \quad \frac{Y}{(1+Y)} = 2.5\frac{X}{(1+X)}$$

$$Y = \frac{2.5X}{(1-1.5X)}$$

X	0.0	0.002	0.004	0.006	0.008	0.01
Y	0.0	0.005	0.01	0.0151	0.0202	0.0254

$$X_{Np} = \frac{x_{Np}}{(1-x_{Np})} = \frac{0.008}{(1-0.008)} = 0.00806$$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_{Np+1} - Y_1)}{(X_{Np} - X_0)} = \frac{(0.0204 - 0.001)}{(0.00806 - 0)}$$

(Assuming pure water enters, $X_0 = 0$) $\left(\frac{L_S}{G_S}\right)_{\min} = 2.4069$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = \left(\frac{L_S}{G_S}\right)_{\min} \times (1 + 0.20) = 2.4069 \times 1.2 = 2.888$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = \frac{(Y_{Np+1} - Y_1)}{(X_{Np} - X_0)} = \frac{(0.0204 - 0.001)}{(X_{Np} - 0)}$$

$$X_{Np} = 0.00672$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = 2.888$$

$$(L_S)_{\text{actual}} = 2.888 \times 14.8205 = 42.802 \text{ kmol/h}$$

(ii) $X_{Np} = 0.00672$, $x_{Np} = \dfrac{0.00672}{(1+0.00672)} = 0.00667$

$$(L_S) = L_{Np}(1 - x_{Np})$$

$$L_{Np} = \frac{42.802}{(1-0.00667)} = 43.089 \text{ kmol/h}$$

$$G_S = G_1\,(1 - y_1)$$

$$G_1 = \frac{14.8205}{(1-0.01)} = 14.835 \text{ kmol/h}$$

(Assuming pure water enters, $L_0 = L_S$)

$$A_1 = \frac{L_0}{mG_1} = \frac{42.802}{(2.5 \times 14.835)} = 1.154$$

$$A_2 = \frac{L_{Np}}{mG_{Np+1}} = \frac{43.089}{(2.5 \times 15.213)} = 1.133$$

$$A = \sqrt{A_1 A_2} = \sqrt{(1.154 \times 1.133)} = 1.143$$

$$N_p = \frac{\log\left[\frac{(y_{Np+1} - m_{x0})}{(y_1 - m_{x1})} \times \left(1 - \frac{1}{A}\right) + \left(\frac{1}{A}\right)\right]}{\log A}$$

$$= \frac{\log\left[\frac{(0.02 - (2.5 \times 0))}{(0.001 - 0)} \times \left(1 - \frac{1}{1.143}\right) + \left(\frac{1}{1.143}\right)\right]}{\log 1.143} = 9.1 \approx 10$$

8. A soluble gas is absorbed in water using a packed tower. The equilibrium relation is $Y_e = 0.06X_e$. $H_x = 0.24$ m, $H_y = 0.36$ m. Find H_{toG}.

Solution.

Given

$X_2 = 0$, $X_1 = 0.08$, $Y_2 = 0.0005$, $Y_1 = 0.009$, where X and Y are mole ratios.

$Y_e = 0.06X_e$.

X	0	0.02	0.04	0.06	0.08
$x = \frac{X}{(1+X)}$	0	0.0196	0.038	0.057	0.074
$Y = 0.06X$	0	0.0012	0.0024	0.0036	0.0048
$y = \frac{Y}{(1+Y)}$	0	0.0012	0.0024	0.0036	0.0048
$m = \frac{y}{x}$	—	0.0612	0.0632	0.0632	0.0649

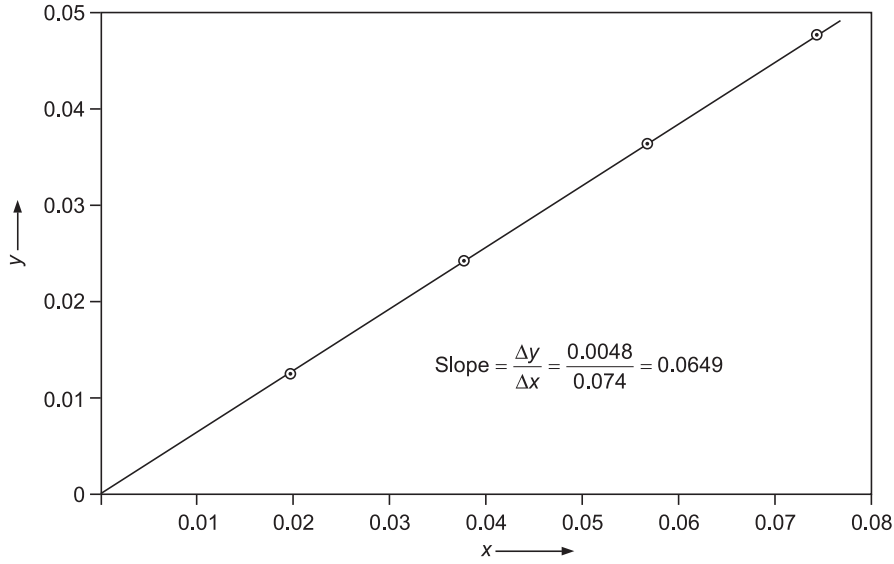


Fig. 8.21 Example 8.

Average m = 0.063

$$H_{toG} = H_y + \left(\frac{mG}{L}\right)H_x$$

H_x = 0.24 m, H_y = 0.36 m, X_1 = 0.08, Y_1 = 0.10, X_2 = 0, Y_2 = 0.005

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.1 - 0.005)}{(0.08 - 0)} = 1.1875$$

$$H_{toG} = H_y + \left(\frac{mG}{L}\right)H_x = 0.36 + \left(\frac{0.063 \times 1}{1.1875}\right) \times 0.24 = 0.3727 \text{ m}$$

(Since absolute flow rates are not available, we have taken the flow rates on solute free basis.)

9. Acetone is to be recovered from a 5% acetone air mixture by scrubbing with water in a packed tower using countercurrent flow. Both liquid and gas rates are 0.85 kg/m^2 s and 0.5 kg/m^2 s respectively. $K_G a$ =1.5 $\times 10^{-4}$ kmol/m^2 s (kN/m^2) partial pressure difference and the gas film resistance controls the process. What should be the height of the tower to remove 98 % acetone? The equilibrium data in mole fractions are as follows:

x	0.0099	0.0196	0.036	0.04
y	0.0076	0.0156	0.0306	0.0333

Solution.

$y_1 = 0.05;$

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.05}{(1 - 0.05)} = 0.05263$$

$L_2 = 0.85$ kg/m^2 s, $K_G a = 1.5 \times 10^{-4}$ kmol/m^2 s (kN/m^2)

Gas flow rate = 0.5 kg/m^2 s

$$Y_2 = 0.05263 \times 0.02 = 0.001053$$

$$y_2 = \frac{0.001053}{(1 + 0.001053)} = 0.00105$$

x	0.0099	0.0196	0.036	0.04
y	0.0076	0.0156	0.0306	0.0333
$m = \frac{y}{x}$	0.7677	0.7959	0.85	0.8325
$X = \frac{x}{(1 - x)}$	0.01	0.02	0.037	0.042
$Y = \frac{y}{(1 - y)}$	0.0077	0.0158	0.0316	0.0344

$$m_{\text{average}} = \frac{0.7677 + 0.7959 + 0.85 + 0.8325}{4} = 0.8115$$

Hence, the equilibrium relation will be $y^* = 0.8115x$

Average molecular weight of gas feed mixture

$$= \frac{(0.05 \times 58) + (0.95 \times 28.84)}{1} = 30.298$$

$$G_1 = \frac{0.5}{30.298} = 0.0165 \text{ kmol/m}^2 \text{ s}$$

$G_S = G_1 (1 - y_1) = 0.0165(1 - 0.05) = 0.0157$ kmol/m^2 s

$G_2 = Gs(1 + Y_2) = 0.0157 \times (1 + 0.001053) = 0.01572$ kmol/m^2 s

Assuming pure water enters, so $L_2 = L_S = \frac{0.85}{18} = 0.0472$ kmol/m^2 s

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{0.0472}{0.0157} = \frac{(0.05263 - 0.001053)}{(X_1 - 0)}; \qquad X_1 = 0.01716$$

$$x_1 = \frac{X_1}{(1 + X_1)} = \frac{0.01716}{(1 + 0.01716)} = 0.01687$$

$$y_1^* = mx_1 = 0.8115 \times 0.01687 = 0.01369,\ y_2^* = 0 \text{ (since } x_2 = 0)$$

$$(y - y^*)_{lm} = \frac{\left[(y_1 - y_1^*) - (y_2 - y_2^*)\right]}{\ln\left[\frac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{\left[(0.05 - 0.01369) - (0.00105 - 0)\right]}{\ln\left[\frac{(0.05 - 0.01369)}{(0.00105 - 0)}\right]} = 0.00995$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}} = \frac{(0.05 - 0.00105)}{0.00995} = 4.92$$

$$G_1 = 0.0165 \text{ kmol/m}^2 \text{ s};\ G_2 = 0.01572 \text{ kmol/m}^2 \text{ s}$$

$$G_{\text{average}} = G = \sqrt{G_1 G_2} = 0.01611 \text{ kmol/m}^2 \text{ s}$$

$$\text{HTU} = \frac{G}{K_{Ga}} = \frac{0.01611}{\left[(1.5 \times 10^{-4}) \times (1.013 \times 10^2)\right]} = 1.06$$

$$Z = \text{NTU} \times \text{HTU} = 4.92 \times 1.06 = 5.216 \text{ m}$$

(Alternative method)

We can also draw the equilibrium curve and operating line (on mole ratio basis) and evaluate $\int \frac{dY}{(Y - Y^*)}$ between the limits $Y_1 = 0.001$ and $Y_2 = 0.0525$ graphically. The values of Y and Y^* have been presented below.

Y^*	Y	$\frac{1}{(Y - Y^*)}$
0.000	0.001	1000
0.001	0.005	250
0.0025	0.01	133.3
0.005	0.0175	80
0.0075	0.0275	50
0.01	0.03625	38.1
0.01125	0.04125	33.33
0.0125	0.04625	29.6
0.014	0.0525	25.97

The N_{OG} thus calculated is 4.9, which is in close agreement with the value reported above.

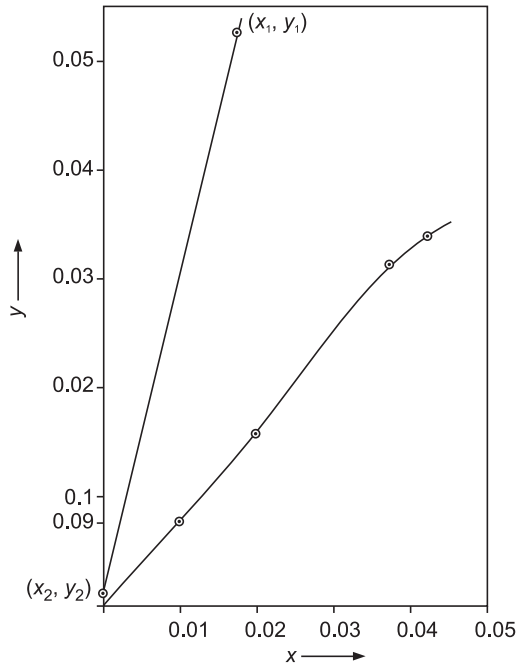


Fig. 8.22(a) Example 9.

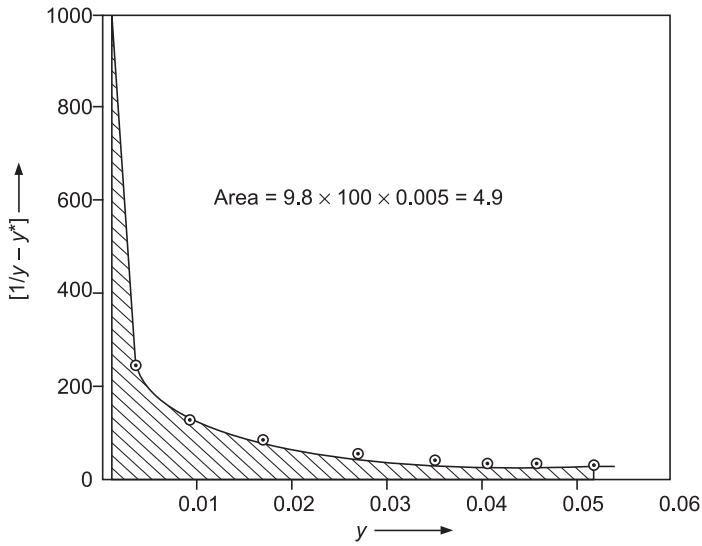


Fig. 8.22(b) Example 9.

10. A countercurrent packed absorption tower is to be designed to handle a gas containing 5% C_6H_6, 95% air at 26.5ºC and 1 atm. At the top of the tower, a non-volatile oil is to be introduced containing 0.2% C_6H_6 by weight. The other data are as follows

$$L_S = 2000 \text{ kg/h}$$

Molecular weight of oil = 230
Vapour pressure of C_6H_6 at 26.5ºC = 106 mm Hg.
Volumetric flow rate of inlet gas = 1140 m^3/h at 26.5ºC and 1 atm.

$$K_y a = 34.8 \text{ kmol/h m}^3 \text{ (mole fraction).}$$

Mass velocity of entering gas = 1460 kg/h m^2.
Calculate the height and the diameter of packed tower for 90% C_6H_6 recovery. Raoult's law is valid.

Solution.

Given that

y_1 = 0.05, gas flow rate = 1140 m^3/h at 26.5°C, 1 atm

Pressure (p_t) = 1 atm

Liquid flow rate = 2000 kg/h

Molecular weight of oil = 230

Vapour pressure of C_6H_6 = 106 mm Hg

$$K_y a = 34.8 \text{ kmol/h m}^3 \text{ (mole fraction)}$$

Mass velocity of inert gas = 1460 kg/h m^2

$$y_2 = 0.05 \times 0.1 = 0.005$$

According to Raoult's law:

$$p = p_A x_A$$

$$P_t \times y = p_A \times x$$

$$y = \frac{p}{P_t} = \frac{(p_A x_A)}{P_t} = \frac{106}{706} x = 0.1395x$$

$$\text{Average moleculor weight of incoming gas} = \frac{(0.05 \times 78) + (0.95 \times 28.84)}{1}$$

$$= 31.3$$

$$\text{Mass velocity of incoming gas in moles} = \frac{1460}{31.3} = 46.645 \text{ kmol/h m}^2$$

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.05}{(1 - 0.05)} = 0.0526$$

$$Y_2 = (0.1 \times 0.0526) = 0.00526$$

$$y_2 = \frac{Y_2}{1+Y_2} = \frac{0.00526}{(1+0.00526)} = 0.00523$$

$$x_2 = 0.002,\ X_2 = \frac{x_2}{(1-x_1)} = \frac{0.002}{(1-0.002)} = 0.00204$$

Mass velocity of incoming gas = $\frac{1460}{31.3}$ = 46.645 kmol/h m^2

Volumetric flow rate of incoming gas = 1140 m^3/h at 26.5°C and 1 atm

Assume that mixture follows ideal gas law,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \frac{(1\times 1140)}{(299.5)} = \frac{(1\times V_2)}{273}$$

$$V_2 = 1039.132 \text{ m}^3\text{/h}.$$

$$\text{Molar flow rate} = \frac{1039.132}{22.414} = 46.361 \text{ kmol/h}$$

$$G_1 = 46.361 \text{ kmol/h}.$$

We know that $y = 0.1395x$

$$\frac{Y}{(1+Y)} = 0.1395\frac{X}{(1+X)}$$

$$Y = \frac{0.1395X}{(1+0.8605X)}$$

$$\text{Area of cross section} = \frac{\text{Volumetric flow rate}}{\text{Mass velocity}}$$

$$\frac{\pi D^2}{4} = \frac{46.361}{46.645} \qquad D = 1.1249 \text{ m}$$

$$L_S = \frac{2000}{230} = 8.696 \text{ kmol/h}$$

$$G_S = G_1(1-y_1) = 46.361(1-0.05) = 44.043 \text{ kmol/h}$$

$$G_2 = \frac{G_S}{1-y_2} = \frac{44.043}{(1-0.00523)} = 44.275 \text{ kmol/h}$$

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1-Y_2)}{(X_1-X_2)} = \frac{8.696}{44.043} = \frac{(0.0526-0.00526)}{(X_1-0.00204)}$$

Therefore, $X_1 = 0.242$

$$x_1 = \frac{X_1}{(1 + X_1)} = \frac{0.242}{(1 + 0.242)} = 0.1948$$

$$y_1^* = mx_1 = 0.1395 \times 0.1948 = 0.0272$$

$$y_2^* = 0.1395 \times 0.002 = 0.000279$$

$$(y - y^*)_{lm} = \frac{\left[(y_1 - y_1^*) - (y_2 - y_2^*)\right]}{\ln\left[\frac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{\left[(0.05 - 0.0272) - (0.00523 - 0.00279)\right]}{\ln\frac{(0.05 - 0.00272)}{(0.00523 - 0.00279)}} = 0.01133$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}} = \frac{(0.05 - 0.005)}{0.01133} = 3.95 \approx 4$$

$$G_{\text{average}} = \sqrt{G_1 G_2} = 45.306 \text{ kmol/h}$$

$$\text{Cross sectional area} = \frac{\pi D^2}{4}$$

Diameter = 1.1249 m (calculated earlier)

$$\text{Cross sectional area} = \frac{\pi D^2}{4} = \frac{\pi \times (1.1249)^2}{4} = 0.9938 \text{ m}^2$$

$$\text{HTU} = \frac{G}{K_y a} = \frac{45.306}{[0.9938 \times 34.8]} = 1.31 \text{ m}$$

$$Z = \text{NTU} \times \text{HTU} = 4 \times 1.31 = 5.14 \text{ m}$$

11. It is desired to recover 98 % of NH_3 from air – NH_3 mixture containing 2% NH_3 at 20ºC and 1 atm by scrubbing with water in a tower packed with 2.54 cm stoneware Raschig rings. If the gas flow rate is 19.5 kg/min m^2 at the inlet and liquid flow rate is 1.8 times the minimum, estimate the height of the tower for a countercurrent operation. Absorption is isothermal. $y^* = 0.746x$, where x and y are mole fractions. $K_G a$ = 1.04 (kmol/min m^3 atm.)

Solution.

$y_1 = 0.02$,

$$Y_1 = \frac{y_1}{(1 - y_1)} = \frac{0.02}{(1 - 0.02)} = 0.02041$$

$$Y_1 = 0.02041 \times 0.02 = 0.00041$$

$$y_2 = \frac{Y_2}{1+Y_2} = \frac{0.00041}{1.00041} = 0.00041$$

Gas flow rate = 19.5 kg/min m^2

$$(L_S)_{\text{actual}} = 1.8 \times (L_S)_{\min}$$

Equilibrium relation = $y^* = 0.746x$

$$\frac{Y}{(1+Y)} = 0.746\frac{X}{(1+X)}$$

Therefore,

$$Y = \frac{0.746X}{(1+0.254X)}$$

X	0	0.010	0.020	0.025	0.03
$Y = \frac{0.746X}{(1+0.254X)}$	0	0.00744	0.01484	0.0185	0.0222

We can calculate minimum liquid flow rate using the equilibrium relationship or from the graph shown in Fig. 8.23.

$$y_1^* = mx_1, \quad x_1 = \frac{0.0242}{0.746} = 0.0268$$

$$X_1 = \frac{x_1}{1-x_1} = \frac{0.0268}{(1-0.0268)} = 0.0275$$

$$Y_1 = \frac{y_1}{(1-y_1)} = \frac{0.02}{(1-0.02)} = 0.02041$$

$$Y_2 = \frac{y_2}{(1-y_2)} = \frac{0.0004}{(1-0.0004)} = 0.0004002$$

From Graph also we get, $X_1 = 0.0275$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = \frac{(0.02041 - 0.00041)}{(0.0275 - 0)} = 0.7273$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = 0.7273 \times 1.8 = 1.309$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = \frac{(Y_1 - Y_2)}{(X_1 - X_2)} = 1.309 = \frac{(0.02041 - 0.00041)}{(X_1 - 0)} = X_1 = 0.01528$$

$$x_1 = \frac{X_1}{(1+X_1)} = \frac{0.01528}{(1+0.01528)} = 0.01505$$

$$y_1^* = 0.746x_1 = 0.746 \times 0.01505 = 0.01123, \qquad y_2^* = 0$$

$$(y - y^*)_{lm} = \frac{\left[(y_1 - y_1^*) - (y_2 - y_2^*)\right]}{\ln\left[\frac{(y_1 - y_1^*)}{(y_2 - y_2^*)}\right]}$$

$$= \frac{\left[(0.02 - 0.01123) - (0.00041 - 0)\right]}{\ln\left[\frac{(0.02 - 0.01123)}{(0.00041 - 0)}\right]} = 0.00273$$

$$\text{NTU} = \frac{(y_1 - y_2)}{(y - y^*)_{lm}} = \frac{(0.02 - 0.00041)}{0.00273} = 7.176$$

Average molecular weight of incoming gas

$$= \frac{(0.02 \times 17) + (0.98 \times 28.84)}{1} = 28.6$$

$$G_1 = 19.5 \text{ kg/min m}^2 = \frac{19.5}{28.6} = 0.682 \text{ kmol/min m}^2$$

$$G_s = G_1(1 - y_1) = 0.682 \times (1 - 0.02) = 0.6684 \text{ kmol/min m}^2$$

$$G_2 = \frac{G_S}{1 - y_2} = \frac{0.6684}{(1 - 0.00041)} = 0.6687 \text{ kmol/min m}^2$$

$$G_{\text{average}} = \sqrt{G_1 G_2} = 0.6753 \text{ kmol/min m}^2$$

$$\text{HTU} = \frac{G}{K_{Ga} P_c} = \frac{0.6753}{1.04 \times 1} = 0.649 \text{ m}$$

$$Z = \text{NTU} \times \text{HTU} = 7.176 \times 0.649 = 4.657 \text{ m}$$

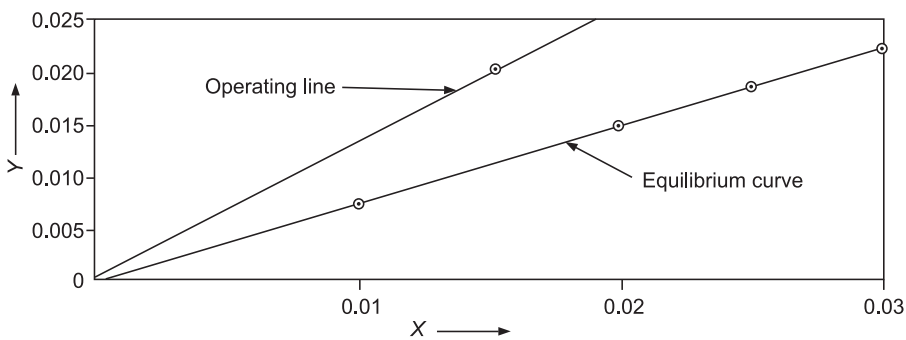


Fig. 8.23 Example 11.

12. $CS_2 - N_2$ mixture containing 7% CS_2 is to be absorbed by using absorption oil. The gas mixture enters at 24ºC and 1 atm at a rate of 0.4 m^3/s. The vapour content is to be brought down to 0.5%. The oil enters free from CS_2. Raoults law is valid. Determine:

(i) Minimum liquid/gas ratio.

(ii) For a liquid/gas ratio of 1.5 times the minimum, determine the kgs of oil entering the tower and the number of theoretical stages required.

Vapour pressure of CS_2 = 346 mm Hg, Molecular weight of oil = 180.

Solution.

Average molecular weight of feed gas = (0.07 × 32) + (0.93 × 28) = 28.28

Gas flow rate = 0.4 m^3

$$\frac{P_0V_0}{T_0} = \frac{P_1V_1}{T_1}$$

$$V_0 = \frac{V_1}{T_1} \times T_0 = \frac{0.4}{0.297} \times 273 = 0.3677 \text{ m}^3\text{/s (At NTP condition)}$$

$$G_1 = \frac{0.3677}{22.414} \text{ kmol/s}$$

$$G_1 = 59.06 \text{ kmol/h}$$

$$G_S = 59.06(1 - 0.07) = 54.93 \text{ kmol/h}$$

Now,

$$y = mx$$

$$y = \frac{346}{760} x = 0.455x$$

$$\frac{Y}{(1+Y)} = 0.455 \frac{X}{(1+X)}$$

$$\frac{(1+Y)}{Y} = \frac{(1+X)}{0.455X}$$

$$\frac{1}{Y} = \frac{1 + X - 0.455X}{0.455X} = \frac{(1 + 0.545X)}{0.455X}$$

$\therefore$

$$Y = \frac{0.455X}{(1 + 0.545X)}$$

X	0	0.05	0.1	0.15	0.2
Y	0	0.022	0.043	0.0631	0.082

$$X_{1,\,\max} = 0.1775$$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_{1,\max} - X_2)} = \frac{(0.0753 - 0.005)}{(0.1775 - 0)} = 0.396$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = (1.5 \times 0.395) = 0.594$$

$$\therefore \quad L_S = (0.594 \times 54.93) = 32.63 \text{ kmol/h}$$

$$= 32.63 \times 180 = 5873.4 \text{ kg/h}$$

$$0.594 = \frac{(Y_1 - Y_2)}{(X_{1,\,\text{Act}} - X_2)} = \frac{(0.0753 - 0.005)}{(X_{1,\text{Act}} - 0)}$$

$$\therefore \quad X_{1,\,\text{act}} = \frac{(0.0753 - 0.005)}{0.594} = 0.1184$$

Number of theoretical stages: 5 as shown in Fig. 8.24.

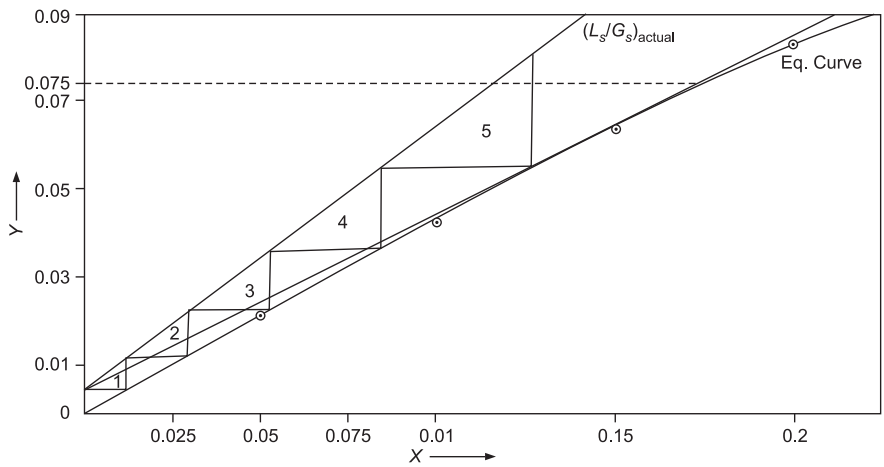


Fig. 8.24 Example 12.

13. NH_3 is absorbed from a gas by using water in a scrubber under atmospheric pressure. The initial NH_3 content in the gas is 0.04(kmol/kmol of inert gas). The recovery of NH_3 by absorption is 90 %. The water enters the tower free from NH_3. Estimate (i) the concentration of NH_3 in the exiting liquid if the actual water used is 1.5 times of the minimum. (ii) the number of theoretical stages required.

X	0.005	0.01	0.0125	0.015	0.02	0.023
Y	0.0045	0.0102	0.0138	0.0183	0.0273	0.0327

where x and y are in mole ratios.

Solution.

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{(X_{1,\max} - X_2)}$$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{(Y_1 - Y_2)}{(X_{1,\max} - X_2)} = \frac{(0.04 - 0.004)}{(X_{1,\max} - 0)}$$

$$X_{1,\max} = 0.027 \text{ (from graph in Fig. 8.25)}$$

$$\left(\frac{L_S}{G_S}\right)_{\min} = \frac{0.036}{0.027} = 1.333$$

$$\left(\frac{L_S}{G_S}\right)_{\text{actual}} = 1.5 \times \left(\frac{L_S}{G_S}\right)_{\min} = 2$$

$$2 = \frac{(0.04 - 0.004)}{(X_{1,\text{act}} - 0)}$$

$$X_{1,\text{act}} = \frac{(0.04 - 0.004)}{2} = \frac{0.036}{2} = 0.018$$

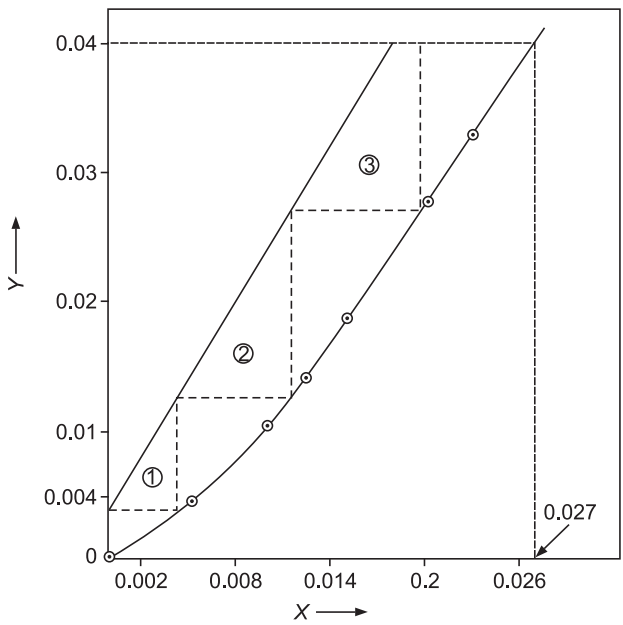


Fig. 8.25 Example 13.

The concentration of ammonia in the exiting liquid: 0.018 kmol/kmol of water. The number of theoretical stages required: 3 (from Fig. 8.25).

14. Gas from petroleum refinery has its concentration of H_2S reduced from 0.03 (kmol H_2S/kmol inert gas) to 1% of these value by scrubbing with a solvent in a countercurrent tower at 27°C and 1 atm. The equilibrium relation is $Y^* = 2X$, where X and Y^* are in mole ratios. Solvent enters free of H_2S and leaves at a concentration of 0.013 kmol H_2S/kmol of solvent. If the flow rate of incoming gas is 55.6 kmol/hr m^2, calculate the height of absorber used if the entire resistance to mass transfer lies in gas phase. Assume $K_y a$ = 0.04 kmol/(m^3 of tower volume × s × Δy.)

Solution.

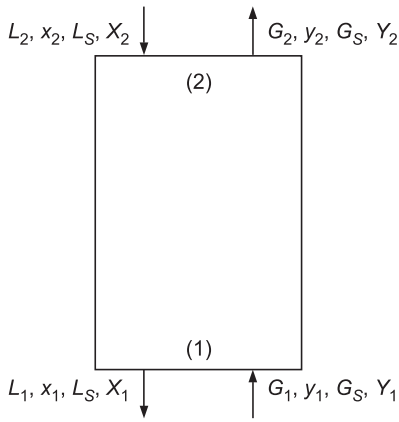


Fig. 8.26 Example 14.

$$X_1 = 0.013; \quad X_2 = 0;$$

$$Y_1 = 0.03; \quad Y_2 = 0.0003$$

$$Y_1^* = 2 \times 0.013 = 0.026; \quad Y_2^* = 0$$

$$y_1 = \frac{Y_1}{1+Y_1} = \frac{0.03}{1.03} = 0.029$$

$$y_2 = \frac{Y_2}{1+Y_2} = \frac{0.0003}{1.0003} = 0.0003$$

Inert gas flow rate = $G_s = G_1(1 - y_1) = 55.6 \times 0.971 = 54$ kmol/h m^2

$$G_2 = Gs(1 + Y_2) = 54 \times 1.0003 = 54.016 \text{ kmol/h m}^2$$

$$G = (54.016\ 55.6)^{0.5} = 54.6 \text{ kmol/h m}^2$$

$$\text{NTU} = \frac{(Y_1 - Y_2)}{\left[\dfrac{(Y_1 - Y_1^*) - (Y_2 - Y_2^*)}{\ln \dfrac{(Y_1 - Y_1^*)}{(Y_2 - Y_2^*)}}\right]} = \frac{(Y_1 - Y_2)}{(\Delta Y)_{lm}}$$

$$(\Delta Y)_{lm} = \frac{(0.03 - 0.026) - (0.0003 - 0.0)}{\ln \dfrac{(0.03 - 0.026)}{(0.0003 - 0.0)}} = 1.428 \times 10^{-3}$$

$$\text{NTU} = \frac{(0.03 - 0.0003)}{(1.428 \times 10^{-3})} = 20.79$$

$$\text{HTU} = \frac{G}{K_y a} = \frac{(54.6)}{(0.04 \times 3600)} = 0.379 \text{ m}$$

Height of tower = HTU × NTU = 0.379 × 20.79 = 7.879 m

EXERCISES

1. An air–NH_3 mixture containing 20% (mole) NH_3 is being treated with water in a packed tower to recover NH_3. Incoming gas rate = 1000 kg/h m^2. Water used is 1.5 times the minimum. The temperature is 35ºC and the pressure is 1 atm. The equilibrium relation is $y^* = 0.746x$, where x and y are mole fraction units. Find the NTU for removing 95% NH_3 in the feed.

2. An air–SO_2 mixture containing 5% SO_2 is scrubbed with water to remove SO_2 in a packed tower. 20 kmol/s of gas mixture is to be processed, to reduce SO_2 concentration at exit to 0.15%. If (L_s) actual is twice (L_s) min, and the equilibrium relationship is $y = 30x$, HTU = 30 cms, find the height of packing to be used.

3. It is desired to absorb 95% NH_3 from a feed mixture containing 10% NH_3 and rest air. The gas enters the tower at a rate of 500 kmol/h. If water is used as solvent at a rate of 1.5 times of the minimum, estimate (i) NTU, (ii) (L_s) actual.

4. An air–SO_2 mixture containing 5.5% SO_2 is scrubbed with water to remove SO_2. 500 kg/hr of gas mixture is to be processed and the SO_2 content in the exit should be brought to 0.15%. Calculate the height of packing required if the liquid used is 2.5 times the minimum liquid rate. Dilute solutions are involved in operation. The equilibrium lines are given by $y = 30x$, where x and y are mole fractions. The HTU is 30 cm.

5. An air–NH_3 mixture containing 5% NH_3 enters a packed tower at the rate of 500 kmol/h m^2. It is desired to recover 95% NH_3 using a liquid flow rate of 1.5 times the minimum. Estimate the height of the tower. HTU is 0.25 m. Fresh solvent enters the absorber. The equilibrium relation is $y^* = 1.08x$ where x and y are mole fractions.

6. A packed tower is to be designed to absorb SO_2 from air by scrubbing the gas with water. The entering gas contains 20% of SO_2 by volume and the leaving gas contains 0.5% of SO_2 by volume. The entering water is SO_2 free. The water flow to be used is twice the minimum. The airflow rate on SO_2 free basis is 975 kg/h m^2. The temperature is 30ºC and pressure is 1 atm. $y^* = 21.8x$, where x and y are mole fractions. Find the NTU.

7. NH_3 is to be absorbed from air at 20°C and 1 atm pressure in a packed tower using water as absorbent. G_S = 1500 kmol/h m^3, L_S = 2000 kmol/h m^2. y_1 = 0.0825; y_2 = 0.003. $K_y a$ = (0.3 kmol/h m^2 (Δy)). Determine the height of the tower by N_{toG} method.

X	0.0164	0.0252	0.0359	0.0455	0.072
Y	0.021	0.032	0.042	0.053	0.08

X and Y are mole ratios.

8. An air–NH_3 mixture containing 20 mole % of NH_3 is being treated with water in a packed tower to recover NH_3. The incoming gas rate is 700 kg/h m^2. The water used is 1.5 times the minimum and enters the tower free of NH_3. Under these conditions, 95% of NH_3 is absorbed from the incoming feed. If all the operating conditions remain unchanged, how much taller the tower should be to absorb 99% of NH_3, under the given conditions $y^* = 0.75x$ where x and y are mole fractions of NH_3 in liquid and gas phase respectively.

9. A packed tower is to be designed to recover 98% carbon dioxide from a gas mixture containing 10% carbon dioxide and 90% air using water. The equilibrium relationship is $Y = 14X$, where Y = (kg CO_2/kg dry air) and X = (kg CO_2/kg dry water). The water to gas rate is kept 30% more than the minimum value. Calculate the height of the tower if $(HTU)_{OG}$ is 1 metre.
(**Ans:** 11.42 m)

10. An air–NH_3 mixture containing 6% of NH_3 is being scrubbed with water to recover 90% of NH_3. The mass velocities of gas and water are 3200 kg/h m^2 and 2700 kg/h m^2 respectively. The operating conditions are 25°C and 1 atm. Find NTU and height of the tower. Given that, $K_G a$ = 65 kmol/h m^3 atm, $y^* = 0.987x$, where x and y are mole fractions.

11. 500 m^3/h of a gas at 760 mm Hg and 35°C containing 3% by volume of toluene is absorbed using a wash oil as an absorbent to remove 95% of toluene. The wash oil enters at 35°C contains 0.5% toluene and has an average molecular weight of oil 250. The oil rate used is 1.5 times the minimum. Wash oil is assumed to be ideal. Vapour pressure of toluene is 110 mm Hg. Find the amount of wash oil used and the number of theoretical stages.

12. Ammonia is recovered from a 10% NH_3–air mixture by scrubbing with water in a packed tower at 20°C and 1 atm. pressure such that 99% of the NH_3 is removed. What is the required height of the tower? Gas and water enter at the rate of 1.2 kg/m^2 s and 0.94 kg/m^2 s respectively. Assume $K_G a$ = 0.0008 kmol/m^3 s. atm. The equilibrium data is as follows:

x	0.021	0.031	0.042	0.053	0.079	0.106	0.159
p (mm Hg)	12	18.2	24.9	31.7	50	69.6	114

where x is the mole fraction of NH_3 in liquid, p is the partial pressure of NH_3 in mm Hg.

13. An air–acetone mixture, containing 5% acetone by volume, is to be scrubbed with water in a packed tower to recover 95% of the acetone. Airflow rate is 1400 m^3/h at 20ºC and 1 atmosphere. The water rate is 3000 kg/h. The equilibrium relation is $Y_e = 1.68\ X$, where Y_e and X are mole fractions of acetone in vapour and liquid respectively. The flooding velocity is 1.56 metre per second and the operating velocity is 25% of the flooding velocity. The interfacial area of the packing is 204 m^2/m^3 of packing and the overall mass transfer coefficient K_y is 0.40 kmol/h m^2 mole fraction. Estimate the diameter and packed height of the tower operating at 1 atmosphere.

14. CO_2 evolved during the production of ethanol by fermentation contains 1 mole ratio of alcohol. It is proposed to remove alcohol by absorption in water at 40ºC. The water contains 0.0001-mole ratio of alcohol. 500 moles/hr of gas is to be processed. Equilibrium data is given by $y = 1.05x$, where x and y are mole fractions. Calculate the water rate for 98% absorption of alcohol by using 1.5 times the minimum liquid rate and determine the number of plates.

15. A gas stream containing a valuable hydrocarbon (molecular weight = 44) and air is to be scrubbed with a non-volatile oil (molecular weight = 300) in a tower placed with 2.54 cm Raschig rings. The entering gas analyses 10 mole % hydrocarbon and 95% of this hydrocarbon is to be recovered. The gas stream enters the bottom of the column at 2270 kg/h and the hydrocarbon free oil used is 1.5 times the minimum. Find N_{toG} for this operation. The equilibrium data is as follows:

X	0	0.1	0.2	0.3	0.4	0.458
Y	0	0.01	0.02	0.06	0.118	0.2

where X and Y are mole ratios. (ii) If the flow rate of liquid is 4600 kg/h, estimate the number of transfer units needed and the solute concentration in mole fraction in leaving liquid?

(**Ans:** (ii) 4, 0.322)

16. A soluble gas is absorbed in water using a packed tower. The equilibrium relationship may be taken as $y = 0.06x$.

	Terminal conditions	
	Top	*Bottom*
x	0	0.08
y	0.001	0.009

(x, y: Mole fraction of solute in liquid and vapour phase respectively)
If the individual height of transfer units based on liquid and gas phase respectively are H_x = 0.24 m and H_y = 0.36 m, (i) what is the value of $(HTU)_{OG}$ and (ii) what is the height of packed section?

(**Ans:** (i) 0.511 m and (ii) 1.833 m)

17. An air–NH_3 mixture containing 20-mole % NH_3 is being treated with water in a packed tower to recover NH_3. The incoming gas rate is 1000 kg/h m^2. The temperature is 35ºC and the total pressure is 1 atm. The water flow rate is 3000 kg/h m^2. 95% of incoming NH_3 is to be absorbed. If all the operating conditions remain unchanged, how much taller should the tower be to absorb 99% of NH_3? Henry's law is valid and Henry's constant is 0.746. Variations in gas flow rates may be neglected.

(**Ans:** 58.15%)

9

DISTILLATION

9.1 INTRODUCTION

The method of separating the components from a solution depending on its distribution between a liquid phase and vapour phase is termed distillation. This is applied to mixtures which distribute in both the phases.

This can also be defined as an operation in which a liquid or vapour mixture of two or more components is separated into its component fractions of desired purity, by the application of heat. Thus, in this process, a vapour is obtained from a boiling mixture which will be richer in components that have lower boiling points.

9.2 VAPOUR LIQUID EQUILIBRIA (VLE)

The vapour liquid equilibrium data is the basis for design of distillation operation equipments.

9.2.1 Constant Pressure Equilibria

A typical VLE at constant pressure is shown in Fig. 9.1.

The upper curve is the dew point curve which provides the relationship between temperature and mole fraction of the more volatile component in vapour phase (y) and the lower curve is the bubble point curve which gives the relationship between the temperature and mole fraction of the more volatile component in liquid phase (x) at a particular pressure. The horizontal tie lines CD, EF and GH at different temperatures provide equilibrium compositions of liquid and vapour phase at each temperature. Any mixture lying on the lower (bubble point) curve will be a saturated liquid and the mixture lying on the upper (dew point) curve will be a saturated vapour. A mixture located in between the two

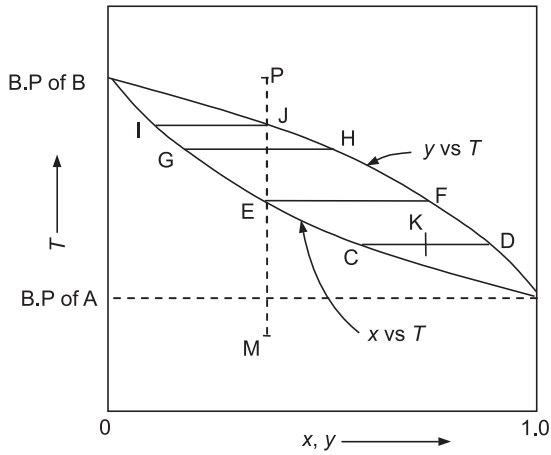


Fig. 9.1 VLE diagram at constant pressure.

curves, say K, will be a two-phase mixture of liquid and vapour with compositions C and D in liquid phase and vapour phase respectively. Their relative amounts are given by

$$\frac{\text{moles of } C}{\text{moles of } D} = \frac{\text{Length of line } KD}{\text{Length of line } KC}$$

Consider a mixture at point M. It is only a liquid. If it is kept inside a cylinder fitted with a frictionless piston and heated, its temperature will increase till it reaches 'E' when it will become a saturated liquid. The vapour in equilibrium with it will have a composition of F. As heating is further continued, more vapourization takes place, the liquid phase composition will move towards G and the associated vapour will have a composition of H. The effective composition of the entire mass comprising both liquid and vapour continues to remain at M. Finally, when the last droplet of liquid as indicated at point 'I' is vapourized, the vapour generated would have a composition of 'J'. Further application of heat results in superheating of the vapour. During the entire operation, the pressure is kept constant.

9.2.2 Effect of Pressure

As pressure is increased, the boiling points of components increase and the looped curves become more and more narrow. As the critical pressure is exceeded for one of the components, there is no longer a distinction between vapour and liquid for that component, and for mixtures the looped curves are, therefore, shorter as depicted in Fig. 9.2, for case (C). Distillation is possible only in the region where a looped curve exists. It is also clear that relative volatility, α, also changes in such cases.

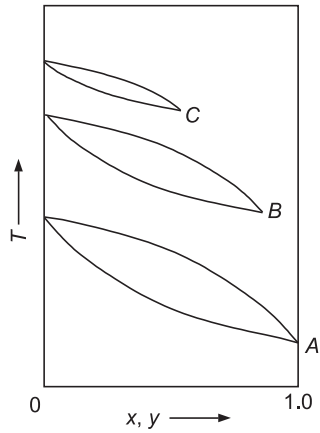


Fig. 9.2 Effect of pressure on VLE.

9.2.3 Constant Temperature Equilibria

A typical VLE at constant temperature is shown in Fig. 9.3.

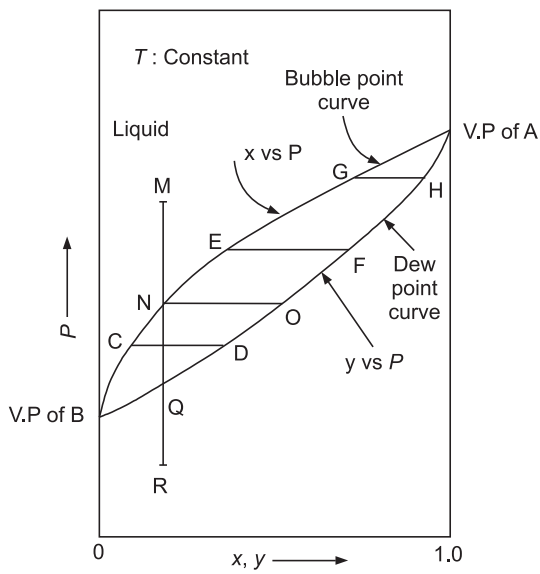


Fig. 9.3 VLE at constant temperature.

As in the case of constant temperature equilibria, lines CD, EF and GH are tie lines indicating the equilibrium compositions of liquid and vapour phase at various pressures. A liquid defined at point M is a liquid below its bubble point and as the pressure is reduced at constant temperature, at point 'N' on the upper (bubble point) curve, a saturated liquid is obtained. As the pressure is brought down further, at point Q on the lower (dew point) curve, a saturated vapour forms and a further reduction in pressure gives a fully superheated vapour as defined by R.

9.3 RELATIVE VOLATILITY (α)

This is defined as the ratio of vapour pressure of more volatile component to that of less volatile component. If P_A and P_B are the vapour pressures of A and B respectively, the relative volatility of A with respect to B, α_{AB} is defined as the ratio of vapour pressure of A to that of B.

i.e. $$\alpha_{AB} = \frac{P_A}{P_B} \tag{9.1}$$

Raoult's law states that when a gas and a liquid are in equilibrium, the partial pressure of A, p_A is equal to the product of its vapour pressure, P_A at that temperature and its mole fraction x_A in the liquid.

i.e. $$p_A = P_A \cdot x_A \tag{9.2}$$

Similarly, $$p_B = P_B \cdot x_B \tag{9.3}$$

When the gas and liquid behave ideally, Raoult's law holds good.

We know that sum of the partial pressures of components in a gas mixture is equal to the total pressure, P_T. The composition of a component y, in gas phase is given by Dalton's law,

$$y_A = \frac{p_A}{P_T} \quad \text{and} \quad y_B = \frac{p_B}{P_T} \tag{9.4}$$

$\therefore$ $$\alpha_{AB} = \frac{P_A}{P_B} = \frac{\left(\frac{p_A}{x_A}\right)}{\left(\frac{p_B}{x_B}\right)} = \frac{\left(\frac{P_T y_A}{x_A}\right)}{\left(\frac{P_T y_B}{x_B}\right)} = \frac{\left(\frac{y_A}{y_B}\right)}{\left(\frac{x_A}{x_B}\right)}$$

$$\alpha_{AB} = \frac{\left(\frac{y_A}{1-y_A}\right)}{\left(\frac{x_A}{1-x_A}\right)} \tag{9.5}$$

Rearranging, we get

$$y_A = \frac{\alpha_{AB} x_A}{1 + x_A(\alpha_{AB} - 1)} \tag{9.6}$$

and more simply as

$$y = \frac{\alpha x}{1 + x(\alpha - 1)} \tag{9.7}$$

9.4 COMPUTATION OF VLE DATA (EQUILIBRIUM DATA)

The vapour pressure of the components involved is the basis for the computation of VLE data.

From Eqs. (9.2) and (9.3), $p_A = x_A P_A$

$$p_B = x_B P_B = (1 - x_A) P_B \tag{9.8}$$

For a binary system,

$$p_A + p_B = P_T = x_A P_A + (1 - x_A) P_B = P_B + x_A (P_A - P_B) \tag{9.9}$$

$$\therefore \quad x_A = \frac{(P_T - P_B)}{(P_A - P_B)} \tag{9.10}$$

From the vapour pressure data at each temperature, x_A can be computed using Eq. (9.10). After computing x_A, the partial pressure p_A can be estimated by using Eq. (9.2). The mole fraction of A in gas phase, y_A is then determined by using Eq. (9.4). Thus, for the whole range of boiling points of components involved, the VLE data can be computed.

Whenever α lies in a narrow range, y can be computed by assuming various values of x using Eq. (9.7).

9.5 DEVIATION FROM IDEALITY

A mixture whose total pressure is either greater or lesser than that computed using Raoult's law is said to exhibit either a positive deviation or a negative deviation from ideality.

9.5.1 Positive Deviation from Ideality

When the total pressure of a mixture is greater than that for ideal mixtures computed using Raoult's law, the mixture is said to exhibit positive deviations from ideality and such mixtures are called minimum boiling azeotropes, i.e. at some composition the mixture shows minimum boiling point (at constant pressure) and maximum pressure (at constant temperature) as shown in Figs. 9.4 and 9.5. A typical x-y diagram is also shown in Fig. 9.6. Most of the azeotropic mixtures fall under this category.

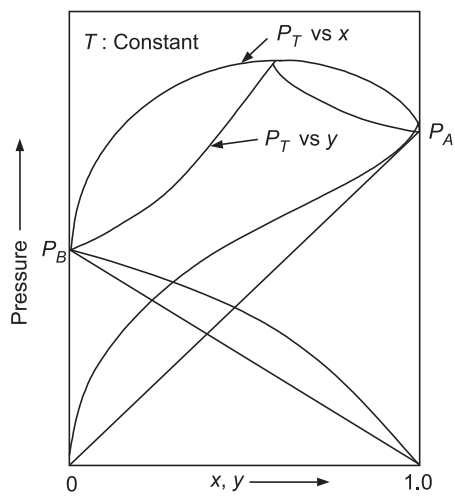


Fig. 9.4 Minimum boiling azeotrope at constant temperature.

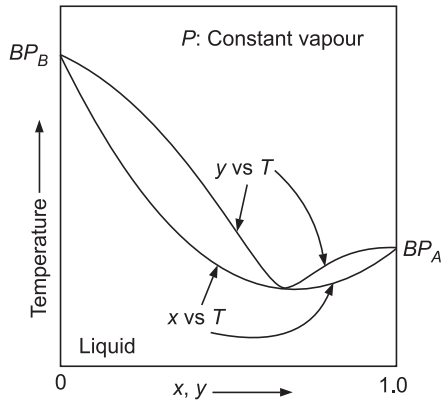


Fig. 9.5 Minimum boiling azeotrope at constant pressure.

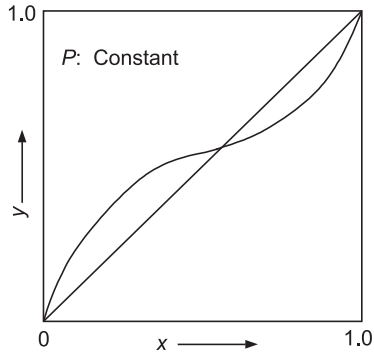


Fig. 9.6 VLE of minimum boiling azeotrope.

9.5.2 Negative Deviations from Ideality

When the total pressure of a system is less than the ideal value as computed using Raoult's law, the system is said to deviate negatively. Such systems are very rare and they are also called maximum boiling azeotropes, i.e. at some composition the mixture shows maximum boiling point. Typical *P–x–y*, *T–x–y* and *x–y* diagrams are shown in Figs. 9.7, 9.8 and 9.9.

9.6 TYPES OF DISTILLATION COLUMNS

Based on the nature of operation, distillation columns have been classified as batch and continuous columns as shown in Fig. 9.10.

9.6.1 Batch Columns

In batch columns, the feed to the column is introduced batchwise and the distillation is carried. When the desired quality is reached or when the desired quantity is distilled out, the operation is stopped and next batch of feed is introduced.

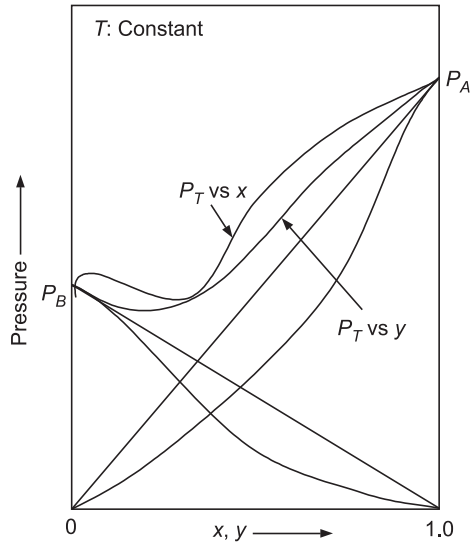


Fig. 9.7 Maximum boiling azeotrope at constant temperature.

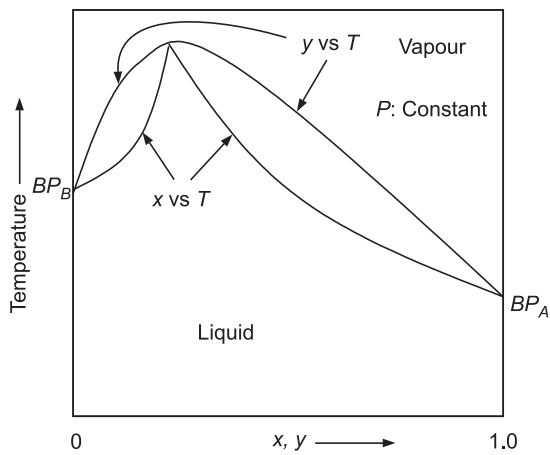


Fig. 9.8 Maximum boiling azeotrope at constant pressure.

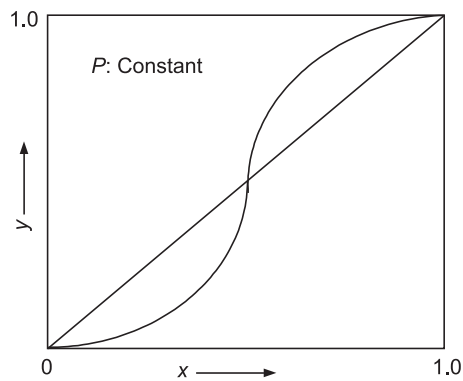


Fig. 9.9 VLE of maximum boiling azeotrope.

9.6.2 Continuous Columns

These columns have a continuous feed stream and are capable of handling high throughputs. These are further classified on the basis of,

- *The nature of the feed they are further processing*
  - ♦ Binary columns—Feed has only two components
  - ♦ Multicomponent column—Feed has more than two components
- *The number of product streams they have*
  - ♦ Two product streams
  - ♦ Multi product streams
- *The use of additional components in distillation*
  - ♦ Extractive distillation—use of solvent
  - ♦ Azeotropic distillation—use of entrainer
- *The type of columns*:
  - ♦ Tray columns—use of sieve plate columns/Bubble cap trays/Valve trays for better vapour–liquid contacting
  - ♦ Packed towers—use of packings in columns for better vapour—liquid contacting.

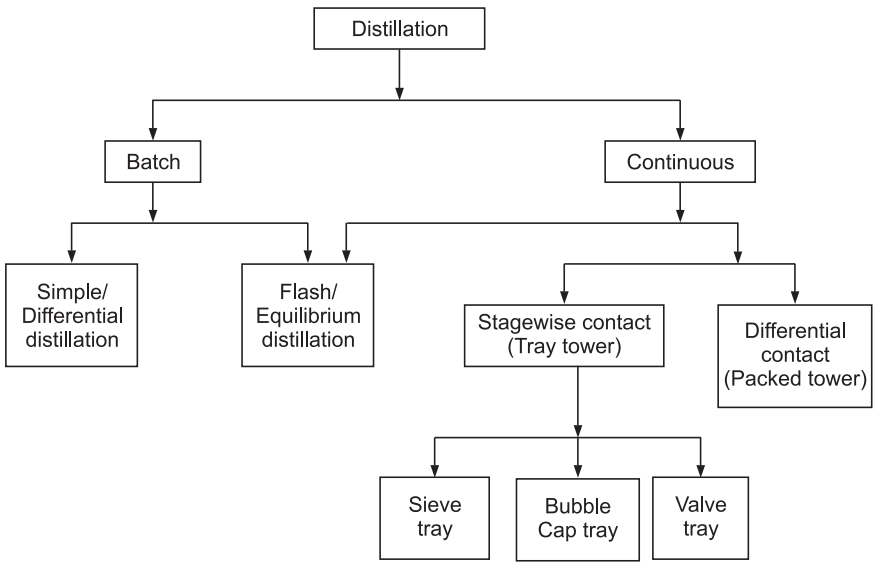


Fig. 9.10 Types of distillation and equipments.

9.7 STEAM DISTILLATION

Some systems have very high boiling points and some of these substances are unstable at high temperatures. Especially when such systems are completely insoluble with each other, steam distillation can be a useful method of separating such mixtures.

For example, consider a mixture of hydrocarbon and water which are immiscible. The vapour pressure of either component cannot be influenced by the presence of the other component and each exerts its own vapour pressure at the prevailing temperature. When the sum of the vapour pressures is equal to the total pressure, the mixture boils. With vapour pressure data of the individual components, one can also estimate the temperature at which such distillations take place.

$$P_T = P_A + P_B \tag{9.11}$$

It is clear from Fig. 9.11, that this type of distillation takes place at a temperature which will be even less than that of the boiling point water. This method suffers from poor efficiency in its operation, as large quantity of water has to be evaporated. However, one can introduce the effectiveness in such operations by

- Operating at different total pressures in which case the ratio of vapour pressure of the substances may be more favourable.
- Sparging the mixture with superheated steam or other insoluble gas.

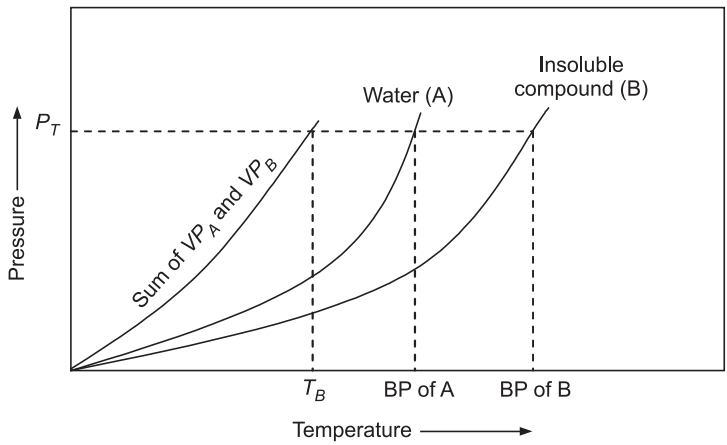


Fig. 9.11 Steam distillation.

9.8 DIFFERENTIAL OR SIMPLE DISTILLATION

Consider a feed F containing x_F mole fraction of more volatile component fed into a batch still as shown in Fig. 9.12. Let L be the total moles present in the still at any instant, t and x be the mole fraction of more volatile component. Let dL be the moles distilled out. The concentration of vapour in the leaving stream is y^*. The moles left behind in the still is $(L - dL)$. During this process the concentration of more volatile component left behind in the still is $(x - dx)$.

Total moles of more volatile component present initially is Lx

Total moles of more volatile component in distillate is y^*dL

Total moles of more volatile component in residue is $(L - dL)\ (x - dx)$

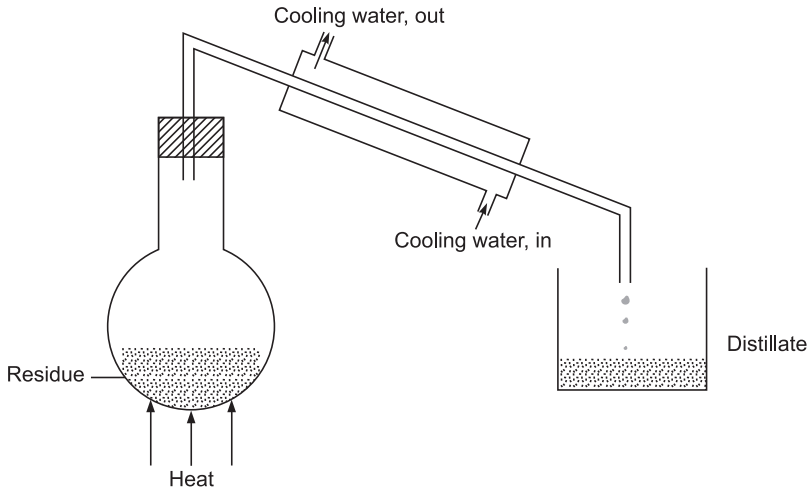


Fig. 9.12 Differential distillation.

Making a component balance, we get

$$Lx = y^* \, dL + (L - dL)(x - dx) \tag{9.12}$$

$$Lx = y^* \, dL + Lx - L\,dx - x \cdot dL + dx\,dL \tag{9.13}$$

($\because$ Product of two very small quantities)

Then, $$dL(y^* - x) = L\,dx \tag{9.14}$$

$\therefore$ $$\frac{dL}{L} = \frac{dx}{(y^* - x)} \tag{9.15}$$

Integrating between limits $x = x_F \quad L = F$

$x = x_W \quad L = W$

$\therefore$ $$\int_W^F \frac{dL}{L} = \int_{X_w}^{X_F} \frac{dx}{(y^* - x)} \tag{9.16}$$

$$\ln \frac{F}{W} = \int_{X_w}^{X_F} \frac{dx}{(y^* - x)} \tag{9.17}$$

Equation (9.17) is called *Rayleigh's equation*.

The right-hand side cannot be integrated as y* is a function of *x*.

Hence, the right-hand side of Eq. (9.17) can be evaluated either graphically or numerically with the help of *x-y* or VLE data.

For systems where the relative volatility lies in a narrow range, we can use Eq. (9.7) which states that

$$y = \frac{\alpha x}{[1 + (\alpha - 1)x]}$$

Hence, replacing y in terms of Eq. (9.7), we get

$$\ln \frac{F}{W} = \int_{x_w}^{x_F} \left[\frac{dx}{\frac{\alpha x}{\left[1+(\alpha-1)x\right]} - x} \right] \tag{9.18}$$

$$\text{RHS of Eq. (9.18)} = \int \left[\frac{dx}{\frac{\alpha x}{\left[1+(\alpha-1)x\right]} - x} \right] \tag{9.19}$$

On simplification,

$$= \int \frac{[1+(\alpha-1)x]dx}{[\alpha x - x - \alpha x^2 + x^2]} = \int \frac{[1+(\alpha-1)x]dx}{x[\alpha - 1 - \alpha x + x]}$$

$$= \int \frac{[1+(\alpha-1)x]dx}{x[1(\alpha-1) - x(\alpha-1)]} = \int \frac{[1+(\alpha-1)x]dx}{[x(1-x)(\alpha-1)]}$$

$$= \int \frac{dx}{[x(1-x)(\alpha-1)]} + \int \frac{dx}{(1-x)} = \frac{1}{(\alpha-1)} \int \left[\left(\frac{A}{x}\right) + \left(\frac{B}{1-x}\right) \right] dx - \ln(1-x)$$

$$= \frac{1}{(\alpha-1)} \int \frac{A(1-x)+B(x)}{x(1-x)} dx - \ln(1-x)$$

$$= \frac{1}{(\alpha-1)} \left[\int \frac{dx}{x} + \int \frac{dx}{(1-x)} \right] - \ln(1-x)$$

Substituting the limits for x as x_F and x_W, we get

$$= \frac{1}{(\alpha-1)} \left[\ln x - \ln(1-x) \right] - \ln(1-x)$$

$$= \frac{1}{(\alpha-1)} \left[\ln\left(\frac{x_F}{x_W}\right) - \ln\frac{(1-x_F)}{(1-x_W)} \right] - \ln\frac{(1-x_F)}{(1-x_W)}$$

i.e.

$$\ln\left(\frac{F}{W}\right) = \frac{1}{(\alpha-1)} \left[\ln\left(\frac{x_F}{x_W}\right) - \ln\left(\frac{(1-x_F)}{(1-x_W)}\right) \right] - \ln\frac{(1-x_F)}{(1-x_W)}$$

$$\ln\left(\frac{F}{W}\right) + \ln\left(\frac{(1-x_F)}{(1-x_W)}\right) = \frac{1}{(\alpha-1)} \ln\left[\frac{\frac{x_F}{x_W}}{\frac{(1-x_F)}{(1-x_W)}} \right]$$

$$\ln \frac{F(1-x_F)}{W(1-x_W)} = \frac{1}{(\alpha-1)} \ln \left\{ \left[\frac{x_F}{(1-x_F)} \right] \left[\frac{(1-x_W)}{x_W} \right] \right\}$$

$$(\alpha-1) \ln \frac{F(1-x_F)}{W(1-x_W)} = \ln \left\{ \left[\frac{x_F}{(1-x_F)} \right] \left[\frac{(1-x_W)}{x_W} \right] \right\}$$

$$\alpha \ln \frac{F(1-x_F)}{W(1-x_W)} = \ln \left[\frac{\frac{x_F}{x_W}}{\frac{(1-x_F)}{(1-x_W)}} \right] + \ln \left[\frac{F(1-x_F)}{W(1-x_W)} \right]$$

$$\alpha \ln \left[\frac{F(1-x_F)}{W(1-x_W)} \right] = \ln \left(\frac{x_F}{x_W} \right) \left(\frac{F}{W} \right)$$

$$\alpha \ln \left[\frac{F(1-x_F)}{W(1-x_W)} \right] = \ln \left(\frac{Fx_F}{Wx_W} \right)$$

i.e.
$$\frac{Fx_F}{Wx_W} = \left[\frac{F(1-x_F)}{W(1-x_W)} \right]^{\alpha} \tag{9.20}$$

Equation (9.20) is very useful in the estimation of the amount of residue (alternatively, the estimation of the quantity to be distilled) in case of systems of constant relative volatility. This is also used in the estimation of relative volatility for such systems.

9.9 EQUILIBRIUM OR FLASH DISTILLATION

Consider a feed at a flow rate of F (moles per hour), containing the more volatile component with a composition of Z_F and an enthalpy of H_F (per mole of feed) entering a preheater. Let the heat added in the preheater be Q. The mixture then enters a flash chamber where a distillate leaves at a rate of D (moles per hour) with a composition of y_D and an enthalpy of H_D (per mole of distillate). The bottoms (residue) leave at a rate of W, with a composition of x_W and an enthalpy of H_W (per mole of residue). The entire process is shown in Fig. 9.13

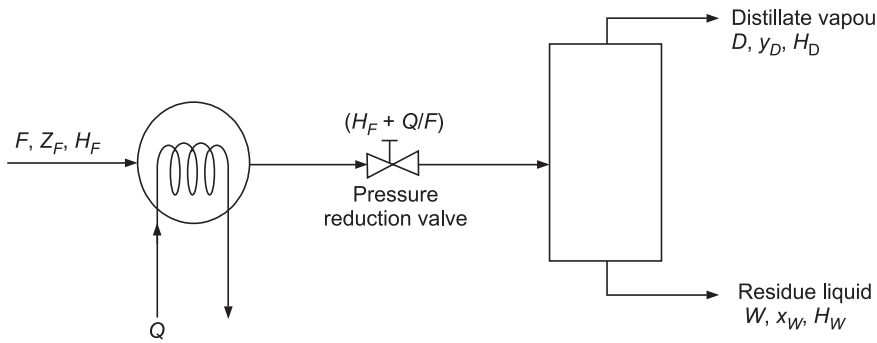


Fig. 9.13 Flash distillation.

A total material balance, gives,

$$F = W + D \tag{9.21}$$

A component balance gives,

$$FZ_F = Wx_W + Dy_D \tag{9.22}$$

An enthalpy balance gives,

$$FH_F + Q = WH_W + DH_D$$

i.e.
$$F\left(H_F + \frac{Q}{F}\right) = WH_W + DH_D \tag{9.23}$$

From Eqs. (9.21) and (9.22), we get

$$(W + D)Z_F = Wx_W + Dy_D \tag{9.24}$$

$\therefore$
$$W\,[Z_F - x_W] = -D[Z_F - y_D] \tag{9.25}$$

$\therefore$
$$-\frac{W}{D} = \frac{(Z_F - y_D)}{(Z_F - x_W)} \tag{9.26}$$

Similarly from Eqs. (9.21) and (9.23), we get

$$-\frac{W}{D} = \frac{\left[\left(H_F + \frac{Q}{F}\right) - H_D\right]}{\left[\left(H_F + \frac{Q}{F}\right) - H_W\right]} \tag{9.27}$$

Dividing Eq. (9.25) by F, we get

$$\frac{W}{F}(Z_F - x_W) = -\frac{D}{F}(Z_F - y_D) \tag{9.28}$$

Let f be the fraction of feed vaporised and subsequently condensed and removed. Hence, $(1 - f)$ is the fraction of feed left behind as residue.

$\therefore$
$$(1 - f)\,(Z_F - x_W) = f\,(y_D - Z_F) \tag{9.29}$$

$$Z_F - x_W - fZ_F + f\,x_W = fy_D - fZ_F$$

$\therefore$
$$Z_F - x_W = f\,(y_D - x_W)$$

$$Z_F + x_W\,(f - 1) = fy_D$$

$\therefore$
$$y_D = \frac{Z_F}{f} + x_W\,\frac{(f - 1)}{f} \tag{9.30}$$

So Eq. (9.30) can be called an operating line drawn with a slope of $[(f - 1)/f]$ and simplified as,

$$y = \left(\frac{Z_F}{f}\right) + x\left(\frac{f - 1}{f}\right) \tag{9.31}$$

The feed point is $x = y = Z_F$

Having seen the principles involved in flash distillation, let us now see how compositions are estimated in a flash distillation operation.

9.9.1 Steps

There are two methods available to estimate the composition of products. They are explained in detail below.

Case I

When the equilibrium data and the quantity of either the distillate or the residue and feed are available, the following procedure shall be adopted:

- Draw the equilibrium curve
- Draw the diagonal ($x = y$ line)
- Locate feed point corresponding to x_F on the diagonal ($x_F = y_F = Z_F$)
- Draw the operating line with a slope of $\left[-\frac{W}{D}\right]$
- The intersection of this line with equilibrium curve gives x_W and y_D as shown in Fig. 9.14.

Case II

When the enthalpy–concentration data (H_L vs x and H_G vs y) and heat added Q are available, the following procedure shall be adopted.

- Plot the enthalpy concentration data and also equilibrium curve below it.
- Locate the feed point corresponding to $F(Z_F, H_F + Q/F)$
- Draw a line by trial and error, passing through F such that it will be a tie line.
- The points of intersection of this line (drawn by trial and error) with enthalpy–concentration curves gives the enthalpy and concentration of both the distillate and the residue.

Figures (9.14) and (9.15) represent the procedures followed to determine the product concentrations for case I and case II respectively.

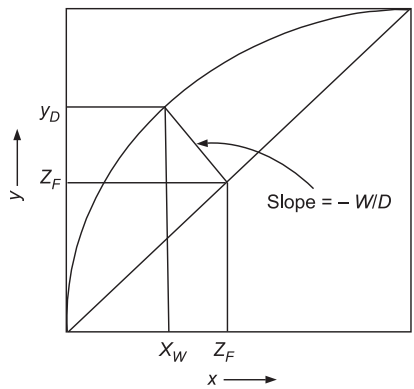


Fig. 9.14 Estimation of composition of products in flash distillation.

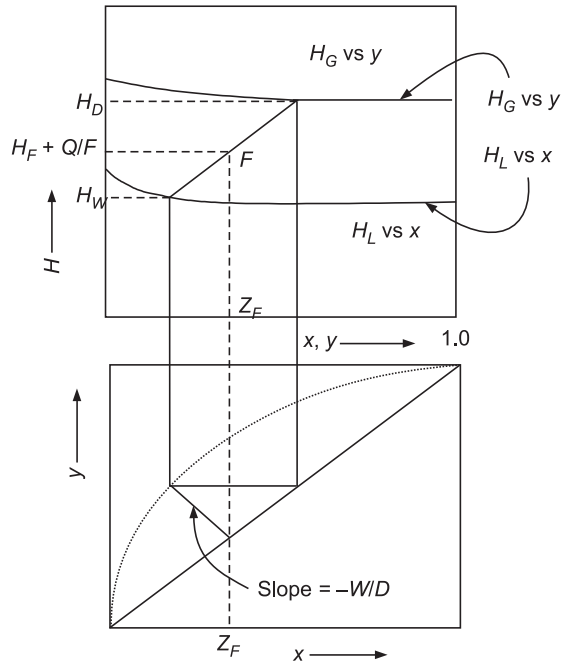


Fig. 9.15 Enthalpy–concentration diagram.

9.10 MULTICOMPONENT SIMPLE DISTILLATION

Let us consider a multicomponent mixture fed to a still. The distillate and the residue left behind will also be multicomponent mixtures. For our analysis, let us consider a three-component system wherein α remains fairly constant. Modified Rayleigh's equation can be applied for material balance,

$$\ln\left[\frac{Fx_{F,A}}{Wx_{W,A}}\right] = \ln\left[\frac{Fx_{F,B}}{Wx_{W,B}}\right]^{\alpha_{AB}} \tag{9.32}$$

$$\ln\left[\frac{Fx_{F,A}}{Wx_{W,A}}\right] = \ln\left[\frac{Fx_{F,C}}{Wx_{W,C}}\right]^{\alpha_{AC}} \tag{9.33}$$

Similarly,

$$\ln\left[\frac{Fx_{F,B}}{Wx_{W,B}}\right] = \ln\left[\frac{Fx_{F,C}}{Wx_{W,C}}\right]^{\alpha_{BC}} \tag{9.34}$$

Here, $BP_A < BP_B < BP_C$, where BP is the boiling point. We also know that

$$x_{WA} + x_{WB} + x_{WC} = 1.0 \tag{9.35}$$

In a typical feed mixture, the values of F, x_{FA}, x_{FB} and x_{FC} are known. The unknown quantities will be W, D, x_{WA}, x_{WB} and x_{WC}. To solve such problems one

has to assume W. By substituting in the first three equations, a relationship between x_{WA}, x_{WB} and x_{WC} is obtained. Then it can be solved and checked for the validity of assumed value of W using Eq. (9.35). If Eq. (9.35) is satisfied, the assumed value of W is correct. If not, a new value for W is assumed and the above calculations are repeated till Eq. (9.35) is satisfied.

Subsequently using the total material balance equation, D can be calculated and then the mole fraction of each component in the distillate phase (vapour phase) can be evaluated by making a component balance.

However, in cases where α varies significantly, the Rayleigh's equation of the form $\ln \frac{F}{W} = \int_{x_w}^{x_F} \frac{dx}{(y-x)}$ has to be used taking two components at a time. Here also one has to assume W and suitably estimate x_W. The values of x_W will have to be determined for all the components and finally checked using Eq. (9.35). If Eq. (9.35) is not satisfied, one has to make a fresh assumption of W and has to proceed till Eq. (9.35) is satisfied.

9.11 MULTICOMPONENT FLASH DISTILLATION

At low pressures almost all systems behave ideally. As flash distillation occurs generally at low pressures, ideal behaviour can be expected and Raoult's law is applicable. Hence, the equilibrium relationship for any component may be expressed as

$$y_i = m_i x_i \tag{9.36}$$

where m_i = vapour pressure of component/total pressure. The suffix i denotes the component.

i.e. $$y_{i,D} = m_i\ (x_{i,W}) \tag{9.37}$$

We know that $-\frac{W}{D} = \frac{(y_D - Z_F)}{(x_W - Z_F)}$ [from Eq. (9.26)]

$\therefore$ $$\frac{W}{D} = \frac{(y_{i,D} - Z_{i,F})}{(Z_{i,F} - x_{i,W})} \tag{9.38}$$

$$= \frac{\left[y_{i,D} - Z_{i,F}\right]}{\left[Z_{i,F} - \frac{y_{i,D}}{m_i}\right]} \tag{9.39}$$

or $$\frac{W}{D}\left[Z_{i,F} - \frac{y_{i,D}}{m_i}\right] = [y_{i,D} - Z_{i,F}] \tag{9.40}$$

i.e $$Z_{i,F}\left[\frac{W}{D} + 1\right] = y_{i,D}\left[1 + \left(\frac{1}{m_i}\right)\left(\frac{W}{D}\right)\right] \tag{9.41}$$

$$\therefore \quad y_{i,D} = Z_{i,F} \frac{\left[\frac{W}{D} + 1\right]}{\left[1 + \left(\frac{1}{m_i}\right)\left(\frac{W}{D}\right)\right]} \tag{9.42}$$

and

$$x_{i,W} = Z_{i,F} \frac{\left[\frac{W}{D} + 1\right]}{\left[m_i + \frac{W}{D}\right]} \tag{9.43}$$

$y_{i,D}$ is evaluated using Eq. (9.42) and $x_{i,W}$ is evaluated using Eq. (9.43) by assuming (W/D) value and finally checked for its validity by using

$$\Sigma x_{i,W} = 1.0; \qquad \Sigma y_{i,D} = 1.0$$

9.11.1 Steps Involved

- From vapour pressure, determine m for each component.
- Assume W/D value and determine $x_{i,W}$ and $y_{i,D}$
- Check whether $\Sigma x_{i,W}$ and $\Sigma y_{i,D}$ are 1.0.
- If they are 1.0, then the assumed W/D ratio is correct.
- If not, assume a new value for W/D and ensure that $\Sigma x_{i,W} = 1.0$; $\Sigma y_{i,D} = 1.0$ are satisfied.

9.12 CONTINUOUS RECTIFICATION

A schematic sketch of a typical distillation column with a feed stream and a distillate and residue stream is shown in Fig. 9.16 along with its main accessories.

9.12.1 Ponchon–Savarit Method

There are two methods by which the design of the continuous fractionator can be established. Let us first consider Ponchon–Savarit method where it requires both enthalpy and concentration data.

Envelope I: Condenser section

Envelope II: Full distillation unit

Envelope III: Enriching/Rectifying section

Envelope IV: Stripping/Exhausting section

The numbering of plates or trays is accounted from the top to bottom. Suffix denotes the properties of streams leaving a particular plate or tray. Let n and m, denote general plates in the enriching section and stripping section respectively.

Let G be the molar flow rate of vapour in enriching section, $\overline{G}$ the molar flow rate of vapour in stripping section, L the molar flow rate of liquid in enriching

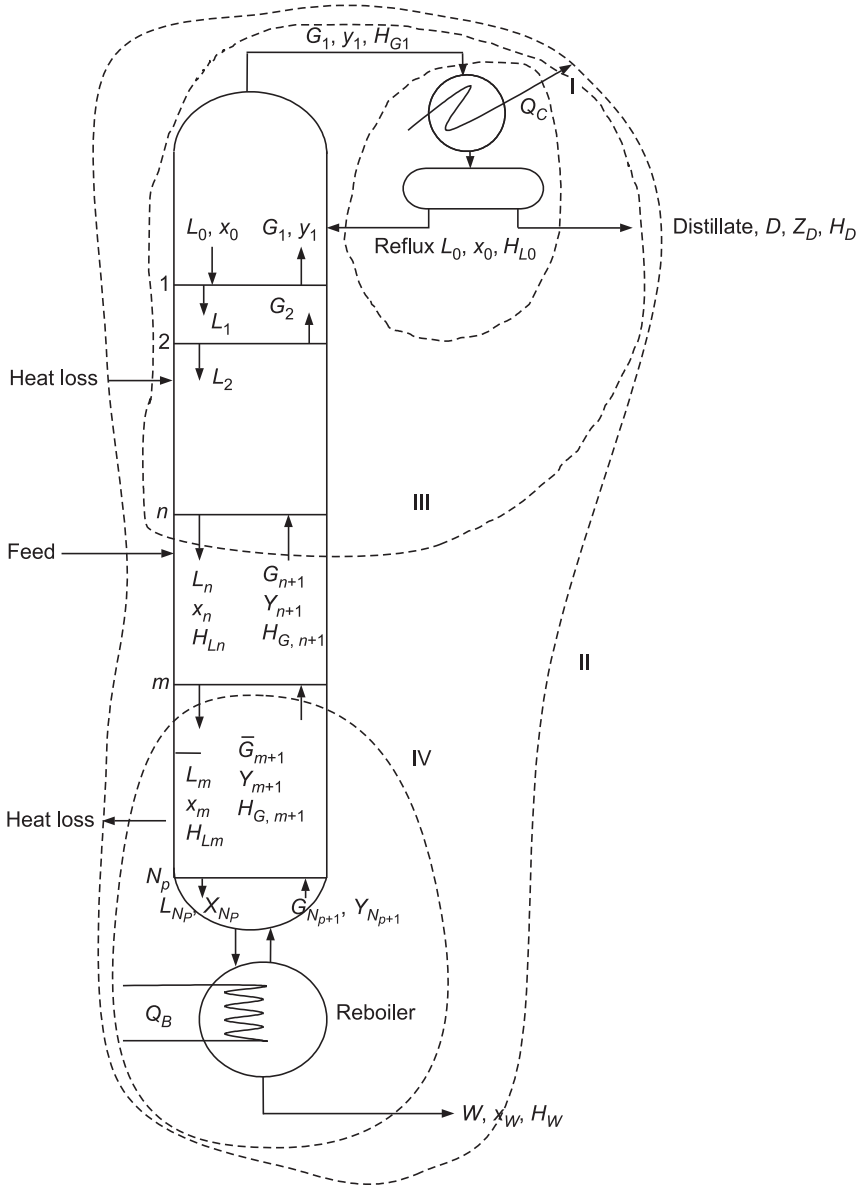


Fig. 9.16 Continuous fractionator.

section, $\bar{L}$ the molar flow rate of liquid in stripping section, H_G the Enthalpy of vapour, H_L the Enthalpy of liquid, y the mole fraction of more volatile component in vapour and x the mole fraction of more volatile component in liquid.

Let R be the *external reflux ratio* L_0/D, Q_C the load on condenser, Q_B the heat supplied in reboiler and Q_L the total heat loss.

Considering envelope I and making a mass balance,

$$G_1 = D + L_0 \tag{9.44}$$

$$G_1 = D + RD = D(R + 1) \tag{9.45}$$

A component balance gives

$$G_1 y_1 = DZ_D + L_0 x_0 \tag{9.46}$$

Making an energy balance, we get

$$G_1 H_{G_1} = L_0 H_{L_0} + DH_D + Q_C \tag{9.47}$$

$$\therefore \quad Q_C = G_1 H_{G_1} - L_0 H_{L_0} - DH_D \tag{9.48}$$

Substituting for G_1 from Eq. (9.45), we get

$$\begin{aligned} Q_C &= [D(R + 1)\, H_{G_1}] - RDH_{L_0} - DH_D \\ &= D[(R + 1)H_{G_1} - RH_{L_0} - H_D] \end{aligned} \tag{9.49}$$

Considering envelope II and making an energy balance, we have
Heat in = Heat out

$$Q_B + FH_F = DH_D + WH_W + Q_C + Q_L \tag{9.50}$$

$$\therefore \quad \text{Heat added in reboiler } Q_B = DH_D + WH_W + Q_C + Q_L - FH_F \tag{9.51}$$

Now, let us consider envelope III, the enriching section and make mass and energy balance.
A total mass balance yields,

$$G_{n+1} = L_n + D \tag{9.52}$$

A component balance gives,

$$G_{n+1} y_{n+1} = L_n x_n + DZ_D \tag{9.53}$$

An energy balance gives,

$$G_{n+1} H_{G_{n+1}} = L_n \cdot H_{L_n} + DH_D + Q_C \tag{9.54}$$

$$\text{Let } Q' = (\text{Net heat out/Net moles out}) = \left[\frac{(Q_C + DH_D)}{D}\right] \tag{9.55}$$

Then, Eq. (9.54) becomes

$$G_{n+1}\, H_{G_{n+1}} = L_n \cdot H_{L_n} + DQ' \tag{9.56}$$

Eliminating D from Eq. (9.53) using Eq. (9.52), we get

$$(G_{n+1})(y_{n+1}) - L_n x_n = (G_{n+1} - L_n) Z_D \tag{9.57}$$

$$(G_{n+1})[Z_D - y_{n+1}] = L_n (Z_D - x_n)$$

$$\frac{L_n}{G_{n+1}} = \frac{(Z_D - y_{n+1})}{(Z_D - x_n)} \tag{9.58}$$

where (L_n/G_{n+1}) is defined as *internal reflux ratio.*

Similarly, Eqs. (9.52) and (9.56) yield

$$\frac{L_n}{G_{n+1}} = \frac{(Q' - H_{G_{n+1}})}{(Q' - H_{L_n})} \tag{9.59}$$

Equating Eq. (9.58) with Eq. (9.59), we get

$$\frac{L_n}{G_{n+1}} = \frac{(Z_D - y_{n+1})}{(Z_D - x_n)} = \frac{(Q' - H_{G_{n+1}})}{(Q' - H_{L_n})} \tag{9.60}$$

Equation (9.60) represents a straight line passing through

$(H_{G_{n+1}}, y_{n+1})$ at G_{n+1}, (H_{L_n}, x_n) at L_n and (Q', Z_D) at Δ_D

where Δ_D is called the difference point and it represents

Q': Net heat, out/Net moles, out and

Z_D: Net moles of more volatile component, out/Net moles, out

Let us consider Eq. (9.59).

$$\frac{L_n}{G_{n+1}} = \frac{(Q' - H_{G_{n+1}})}{(Q' - H_{L_n})}$$

Substituting for G_{n+1} in the above expression from Eq. (9.52), we have

$$\frac{L_n}{(L_{n+D})} = \frac{(Q' - H_{G_{n+1}})}{(Q' - H_{L_n})} \tag{9.61}$$

i.e. $$\frac{(L_{n+D})}{L_n} = \frac{(Q' - H_{L_n})}{(Q' - H_{G_{n+1}})}$$

i.e. $$1 + \left[\frac{D}{L_n}\right] = \frac{(Q' - H_{L_n})}{(Q' - H_{G_{n+1}})}$$

$\therefore$ $$\frac{D}{L_n} = \frac{(H_{G_{n+1}} - H_{L_n})}{(Q' - H_{G_{n+1}})} \tag{9.62}$$

$\therefore$ $$\frac{L_n}{D} = \frac{(Q' - H_{G_{n+1}})}{(H_{G_{n+1}} - H_{L_n})} \tag{9.63}$$

When $n = 0$, it indicates the condenser and for $n = 0$, we get

$$\frac{L_0}{D} = \frac{(Q' - H_{G_1})}{H_{G_1} - H_{L_0}} \tag{9.64}$$

i.e. R, the external reflux ratio $= \dfrac{L_0}{D} = \dfrac{\text{Length of line } \Delta_D G_1}{\text{Length of line } G_1 L_0}$ (9.65)

as indicated in Fig. 9.17(a).

Hence, if the reflux ratio R is known, then it will be easy for us to locate Δ_D point (Z_D, Q').

Also, $$\frac{L_n}{D} = \frac{(Z_D - y_{n+1})}{(y_{n+1} - x_n)} \qquad \text{[from Eq. (9.58)]}$$

Let us consider envelope IV in the stripping section.
A mass balance yields

$$\therefore \qquad \bar{L}_m = \bar{G}_{m+1} + W \tag{9.66}$$

A component balance yields

$$\bar{L}_m x_m = \bar{G}_{m+1} y_{m+1} + W x_W \tag{9.67}$$

$$\therefore \qquad L_m\, x_m - \bar{G}_{m+1} y_{m+1} = W x_W \tag{9.68}$$

An energy balance yields

$$L_m \cdot H_{L_m} + Q_B = \bar{G}_{m+1} H_{G_{m+1}} + W H_W \tag{9.69}$$

Let $$Q'' = \frac{(W H_W - Q_B)}{W} \tag{9.70}$$

$$\therefore \qquad \bar{L}_m \cdot H_{L_m} - \bar{G}_{m+1} H_{G_{m+1}} = W Q'' \tag{9.71}$$

Eliminating W from Eqs. (9.66) and (9.68), we get

$$\frac{\bar{L}_m}{G_{m+1}} = \frac{(y_{m+1} - x_W)}{(x_m - x_W)} \tag{9.72}$$

Similarly eliminating W from Eqs. (9.66) and (9.71), we have

$$\frac{\bar{L}_m}{\bar{G}_{m+1}} = \frac{(H_{G_{m+1}} - Q'')}{(H_{L_m} - Q'')} \tag{9.73}$$

Hence from Eqs. (9.72) and (9.73), we get

$$\frac{\bar{L}_m}{\bar{G}_{m+1}} = \frac{(y_{m+1} - x_W)}{(x_m - x_W)} = \frac{(H_{G_{m+1}} - Q'')}{(H_{L_m} - Q'')} \tag{9.74}$$

From Eq. (9.66), we have

$$\frac{\bar{L}_m}{\bar{G}_{m+1}} = 1 + \left(\frac{W}{\bar{G}_{m+1}}\right)$$

Hence, rearrangement of Eq. (9.74) using Eq. (9.66) gives

$$\frac{W}{\bar{G}_{m+1}} = \frac{(y_{m+1} - x_W)}{(x_m - x_W)} \tag{9.75}$$

$$\therefore \quad \frac{\bar{L}_m}{W} = \frac{(H_{G_{m+1}} - Q'')}{(H_{G_{m+1}} - H_{L_m})} = \frac{y_{m+1} - x_W}{(y_{m+1} - x_m)} \tag{9.76}$$

Equation (9.76) represents a line passing through (x_W, Q''), (x_m, H_{L_m}) and (y_{m+1}, $H_{G_{m+1}}$) where Q'' represents net heat out /Net moles out and x_W denotes moles of A out/net moles out. Now let us consider the fractionator as single unit and make mass and energy balances;

Total mass balance gives

$$F = D + W \tag{9.77}$$

A component balance gives

$$FZ_F = Dy_D + Wx_W \tag{9.78}$$

An enthalpy balance gives

$$F \cdot H_F = (DQ' + WQ'') \tag{9.79}$$

(Neglecting Q_L, the heat loss)

Eliminating F from Eqs. (9.77), (9.78) and (9.79) we get

$$\frac{D}{W} = \frac{(Z_F - x_W)}{(Z_D - Z_F)} = \frac{(H_F - Q'')}{(Q' - H_F)} \tag{9.80}$$

Equation (9.80) represents a line passing through (Z_D, Q'), (Z_F, H_F) and (x_W, Q'') In other words,

$$F = \Delta_D + \Delta_W \tag{9.81}$$

The schematic representation of enthalpy concentration and distribution diagrams for determination of number of stages for a total condensation of distillate vapour is shown in Fig. 9.17(a).

Steps involved

1. Draw H vs x, y diagram and the equilibrium curve.
2. Locate z_D, y_D and x_W in both the diagrams and draw vertical lines from z_D in positive y-axis direction and from x_W in the negative y-axis direction.
3. Locate $F(x_F, H_F)$ in the H–x, y diagram.
4. Obtain Q' using the given reflux ratio, $R = \dfrac{(Q' - H_{G_1})}{(H_{G_1} - H_{L_0})}$

   where H_{G_1}, H_{L_0} indicate the enthalpy of vapour and liquid correspond to the distillate composition for a total condensation.
5. In cases where the reflux ratio is not given, an optimum reflux ranging from 1.5 to 2 times the minimum reflux can be chosen.
6. To determine the *minimum reflux*, several lines can be drawn through the feed point F in the entire range of x and projected downwards from both H vs x and H vs y curves to the x-y diagram as shown in Fig. 9.17(b) and one such horizontal line in x-y diagram will be a tie line. This line is extended to cut the vertical line drawn at z_D and this intersection point corresponds to the value of Q' at minimum reflux and the value of R estimated is $R_{\min}$.

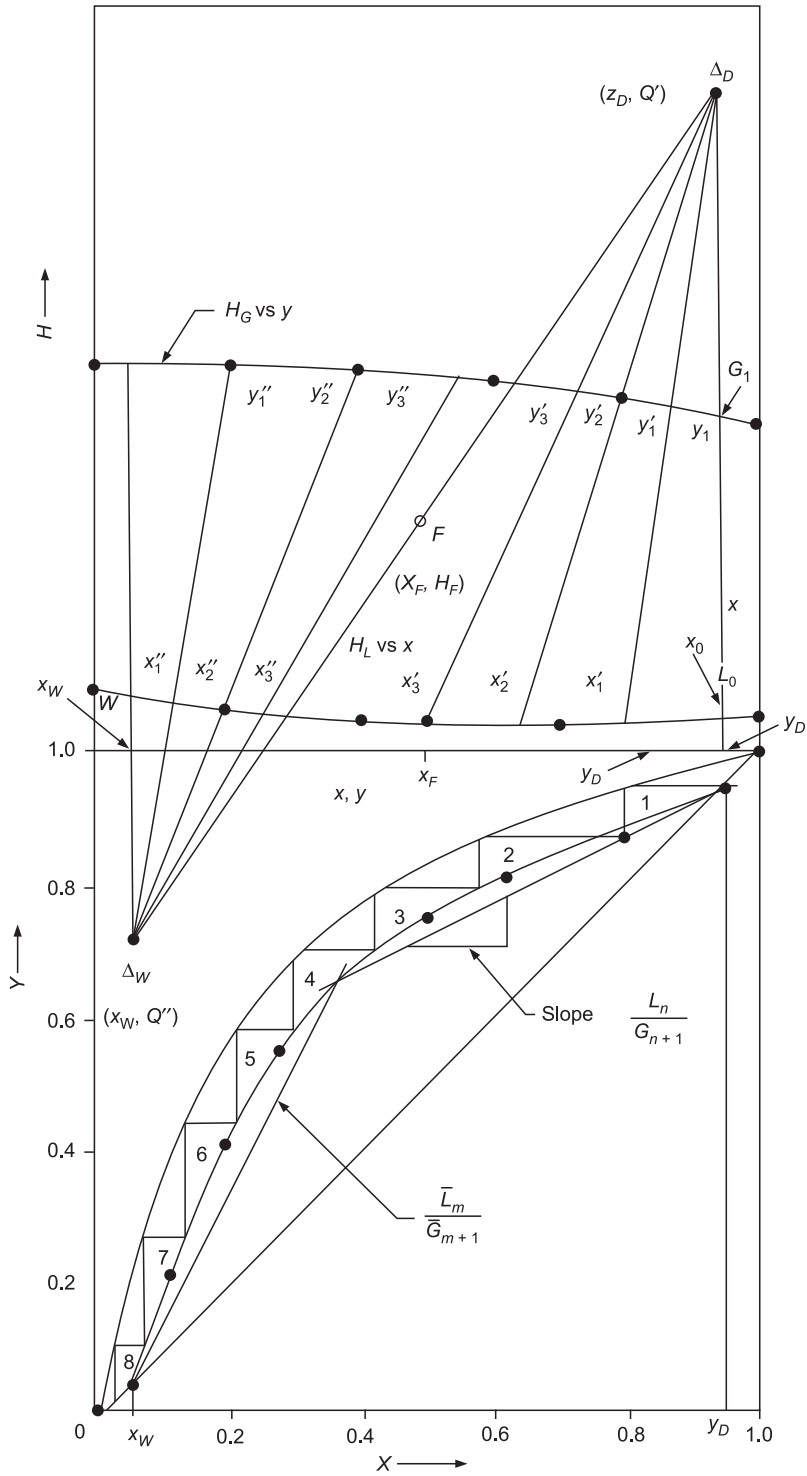


Fig. 9.17(a) Determination of number of stages by Ponchon–Savarit method.

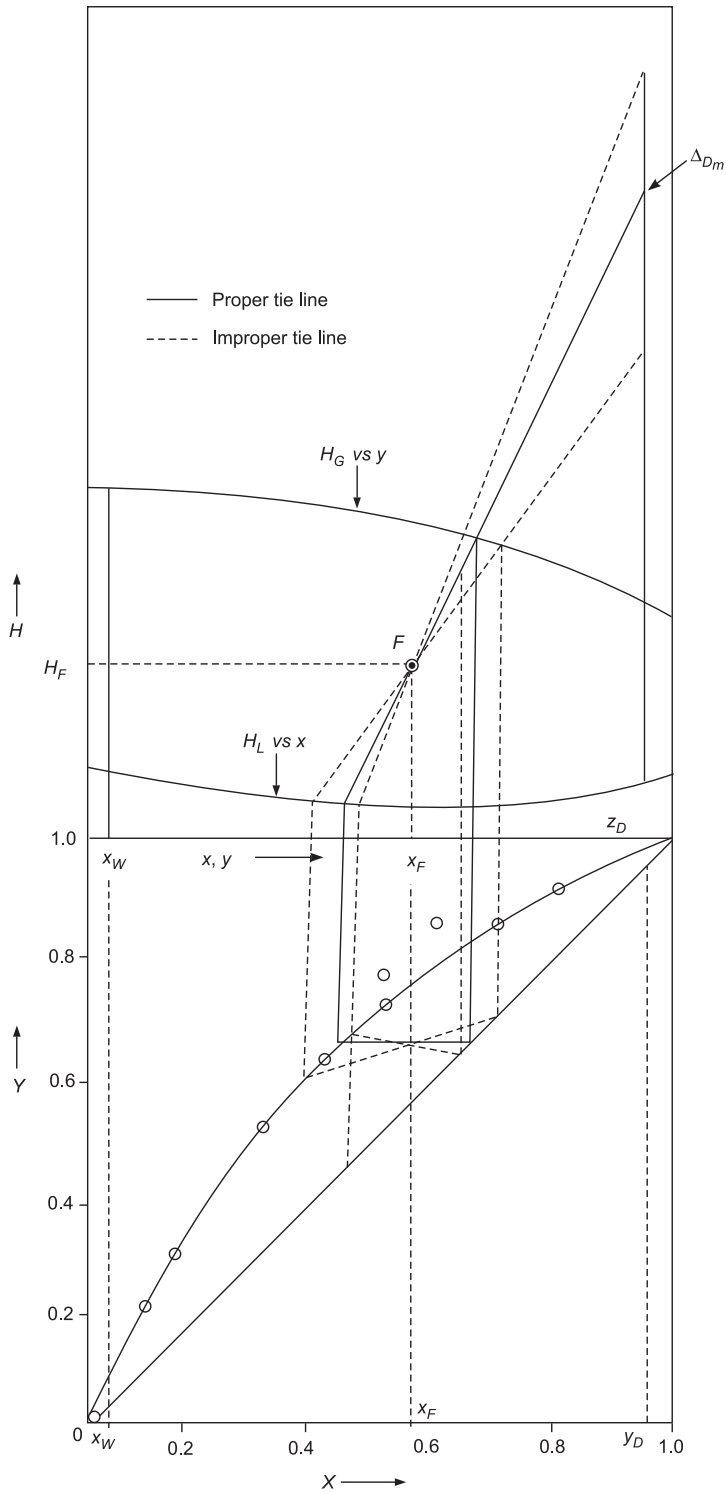


Fig. 9.17(b) Determination of minimum reflux in Ponchon–Savarit method.

7. Locate Δ_D (y_D, Q') using the H_{G_1}, H_{L_0} and reflux ratio.
8. Join Δ_D with F and project it to cut the vertical line at x_W and that point is Δ_W.
9. Draw arbitrarily several lines both from Δ_D and Δ_W to cut both the curves. The values taken from H vs y give y' and the line from H vs x gives x'. For each line drawn we will have a set of x' and y' values with which we can construct the operating line in the distribution diagram for both enriching and stripping sections.
10. Draw the equilibrium curve and plot x', y' data obtained from step (9).
11. By stepwise construction starting from point D, between the equilibrium curve and operating line up to W, the number of stages for the desired separation is determined.

9.12.2 McCabe–Thiele Method

When systems exhibit ideal behaviour, the time-consuming Ponchon–Savarit method of determining the number of ideal stages, can be replaced with the following technique. Let us begin our analysis by considering enriching section for total condensation of distillate as shown in Fig. 9.18.

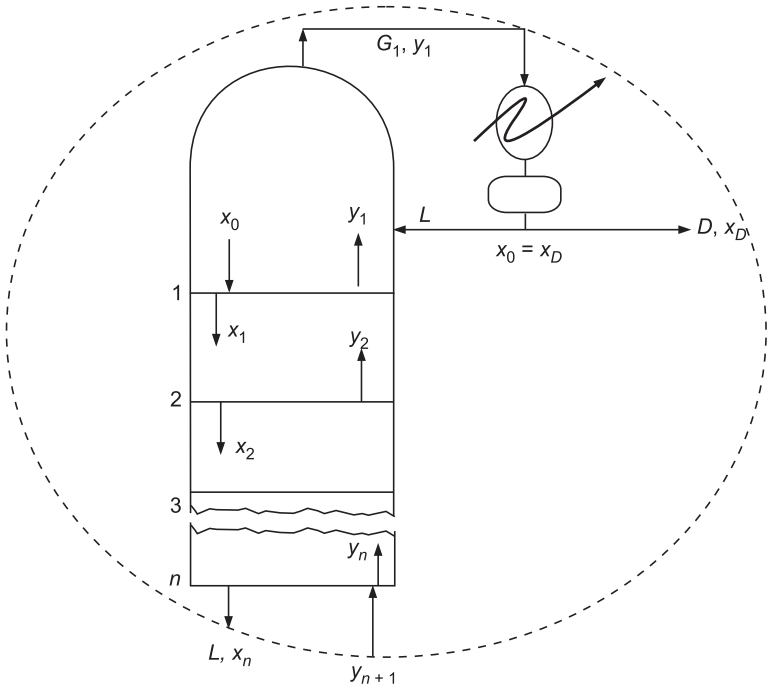


Fig. 9.18 Enriching section of a fractionator.

Assuming the application of equimolar counter diffusion, i.e. the molar flow rates are assumed to be constant for both the vapour and liquid steams irrespective of the stages, we get

$$G_1 \approx G_2 \approx G_{n+1} \approx G$$

A total material balance gives

$$G = L + D \tag{9.82}$$

Let the external reflux ratio R be given by

$$R = \frac{L_0}{D} \approx \frac{L}{D}$$

then, $$G = DR + D = D(R + 1) \tag{9.83}$$

A component balance for A in enriching section gives

$$Gy_{n+1} = Lx_n + Dy_D \tag{9.84}$$

i.e. $$y_{n+1} = \left[\frac{L}{G}\right] x_n + \left[\frac{D}{G}\right] x_D \tag{9.85}$$

$$\frac{L}{G} = \frac{L}{L+D} = \frac{\left(\frac{L}{D}\right)}{\left(\frac{L}{D}\right)+\left(\frac{D}{D}\right)} = \frac{R}{R+1} \tag{9.86}$$

$$\frac{D}{G} = \frac{D}{L+D} = \frac{1}{R+1} \tag{9.87}$$

$\therefore$ $$y_{n+1} = \left[\frac{R}{R+1}\right] x_n + \left[\frac{1}{R+1}\right] x_D \tag{9.88}$$

Equation (9.88) represents the operating line for enriching section, which has a slope of $\left[\frac{R}{R+1}\right]$ and an intercept of $\left[\frac{x_D}{R+1}\right]$. If $x_n = x_D$, then substituting in Eq. (9.88), we get

$$y_{n+1} = \left(\frac{1}{R+1}\right)(Rx_D + x_D) = y_D \tag{9.89}$$

i.e. when $x_n = x_D$, $y_{n+1} = x_D$.

Hence, this line passes through $x = y = x_D$, i.e. it lies on the diagonal. This point on the diagonal and the y-intercept $\left[\frac{x_D}{R+1}\right]$ permit us the construction of operating line for enriching section.

Let us consider the stripping section as shown in Fig. 9.19.

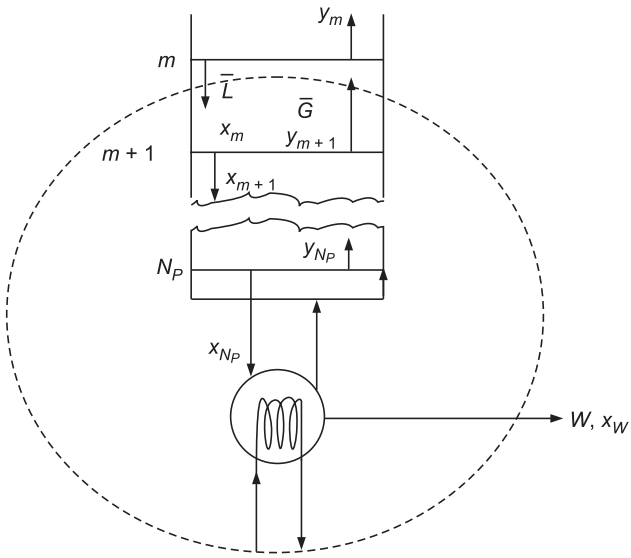


Fig. 9.19 Stripping section.

Material balance gives

$$\bar{L} = \bar{G} + W \tag{9.90}$$

A component balance for *A* gives,

$$\bar{L}x_m = \bar{G}y_{m+1} + Wx_W \tag{9.91}$$

$$\therefore \quad y_{m+1} = \left(\frac{\bar{L}}{\bar{G}}\right)x_m - \left(\frac{W}{\bar{G}}\right)x_W \tag{9.92}$$

i.e. $$y_{m+1} = \left(\frac{\bar{L}}{\bar{L}-W}\right)x_m - \left(\frac{W}{\bar{L}-W}\right)x_W \tag{9.93}$$

Equation (9.93) describes the operating line for stripping section. The operating line has a slope of $\left(\frac{\bar{L}}{\bar{L}-W}\right)$ and an intercept of $-\left(\frac{W}{\bar{L}-W}\right)$.

Let us assume that $x_m = x_W$ (Reboiler)

$$\therefore \quad y_{m+1} = \left(\frac{\bar{L}}{\bar{L}-W}\right)x_W - \left(\frac{W}{\bar{L}-W}\right)x_W = x_W \tag{9.94}$$

i.e. $$x_m = y_{m+1} = x_W$$

Hence, the operating line passes through the point $x = y = x_W$ (i.e. it lies on the diagonal). Having seen the analysis of enriching section and stripping section separately, let us analyze the feed plate, *f*, shown in Fig. 9.20.

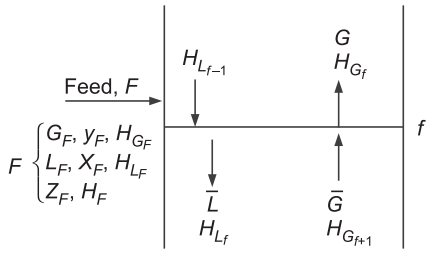


Fig. 9.20 Feed plate section.

A mass balance on feed plate gives

$$F + L + \bar{G} = G + \bar{L} \tag{9.95}$$

$$(L - \bar{L}) = (G - \bar{G}) - F \tag{9.96}$$

i.e. $$(\bar{L} - L) = (\bar{G} - G) + F \tag{9.97}$$

Enthalpy balance on feed plate gives

$$F \cdot H_F + L \cdot H_{L_{f-1}} + \bar{G} \cdot H_{G_{f+1}} = G \cdot H_{G_f} + \bar{L} \cdot H_{L_f} \tag{9.98}$$

As an approximation, $H_{G_f} \approx H_{G_{f+1}} = H_G$ and $H_{L_{f-1}} = H_{L_f} = H_L$ (9.99)

$\therefore$ $$(\bar{L} - L)H_L = (\bar{G} - G)H_G + FH_F \tag{9.100}$$

i.e. $$\frac{(\bar{L} - L)}{F} H_L = \frac{(\bar{G} - G)}{F} H_G + H_F \tag{9.101}$$

Substituting for $(\bar{L} - L)$ from Eq. (9.97) we get

$$\left[\frac{(\bar{G} - G) + F}{F}\right] H_L = \frac{(\bar{G} - G)}{F} H_G + H_F \tag{9.102}$$

i.e. $$\frac{(\bar{G} - G)}{F} H_L + H_L = \frac{(\bar{G} - G)}{F} H_G + H_F \tag{9.103}$$

$$\frac{(\bar{G} - G)}{F}(H_L - H_G) = (H_F - H_L) \tag{9.104}$$

$$\left[\frac{(\bar{G} - G)}{F}\right] = \frac{(H_F - H_L)}{(H_L - H_G)} \tag{9.105}$$

Substituting for $(\bar{G} - G)$ from Eq. (9.97), we get

$$\frac{(\bar{L} - L) - F}{F} = \frac{(H_F - H_L)}{(H_L - H_G)} \tag{9.106}$$

i.e. $$\frac{(\bar{L} - L)}{F} = \frac{(H_F - H_L)}{(H_L - H_G)} + 1 \tag{9.107}$$

$$= \frac{(H_G - H_F)}{(H_G - H_L)} \tag{9.108}$$

Let us now define $\frac{(H_G - H_F)}{(H_G - H_L)}$ as q, where q is the quantity of heat required to convert one mole of feed at its thermal condition to a saturated vapour, to the molal latent heat of vaporization.

$\therefore$ $$q = \frac{(\bar{L} - L)}{F} = \frac{(H_G - H_F)}{(H_G - H_L)} \tag{9.109}$$

Similarly, $$\frac{(\bar{G} - G)}{F} = \frac{(\bar{L} - L)}{F} - 1 = (q - 1) \tag{9.110}$$

i.e. $$(\bar{G} - G) = F(q - 1) \tag{9.111}$$

A solute balance above feed plate gives

$$Gy = Lx + Dx_D \tag{9.112}$$

A solute balance below feed plate gives

$$\bar{G}y = \bar{L}x - Wx_W \tag{9.113}$$

Subtracting Eq. (9.112) from Eq. (9.113), we have

$$(\bar{G} - G)y = (\bar{L} - L)x - (Dx_D + Wx_W) \tag{9.114}$$

Total component balance for the distillation column gives

$$FZ_F = Dx_D + Wx_W \tag{9.115}$$

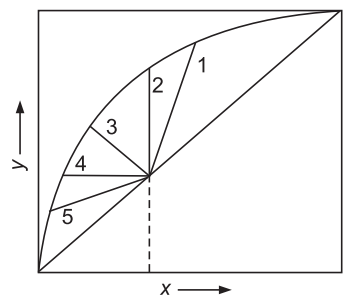
Substituting Eqs. (9.109), (9.111) and (9.115) in Eq. (9.114), we get

$$\cancel{F}(q - 1)y = \cancel{F}q \cdot x - \cancel{F}Z_F$$

$\therefore$ $$y = \frac{q}{(q - 1)}x - \frac{Z_F}{(q - 1)} \tag{9.116}$$

Equation (9.116) is the equation for feed line. It has a slope of $q/q - 1$ and passes through $y = x = Z_F$.

The various values of slope obtained under different thermal conditions of feed are given below and shown in Fig. 9.21.



1. Cold feed
2. Saturated liquid
3. Liquid + vapour
4. Saturated vapour
5. Superheated vapour

Fig. 9.21 Feed line for different thermal conditions of feed.

Feed condition	G_F	L_F	H_{G_F}	H_{L_F}	Enthalpy of feed, H_F	$q = \left[\frac{(H_G - H_F)}{(H_G - H_L)}\right]$	$\frac{q}{q-1}$
Liquid below boiling point	0	F	–	H_F	$H_F < H_L$	> 1.0	>1.0
Saturated liquid	0	F	–	H_F	$H_F = H_L$	1.0	∞
Liquid + vapour	G_F	L_F	H_G^*	H_L^*	$H_G^* > H_F > H_L^*$	1.0 to 0	$\frac{L_F}{L_F - F}$
Saturated vapour	F	0	H_F	–	$H_F = H_G$	0	0
Superheated vapour	F	0	H_F	–	$H_F > H_G$	< 0	1.0 to 0

* indicates H_G and H_L are enthalpies per mole of individual phases.

Determination of q is as follows:

(i) *Cold feed*

From Eq. (9.109), we have

$$q = \frac{(H_G - H_F)}{(H_G - H_L)}$$

Let T_b be the boiling point of mixture and T_F be the feed temperature. Let H_G and H_L be the enthalpies of saturated vapour and liquid respectively. If λ is the latent heat of vaporization, $C_{P,L}$ is the specific heat of feed liquid and T_0 is the reference temperature, then

$$H_G = C_{P,L}(T_b - T_0) + \lambda$$

$$H_F = C_{P,L}(T_F - T_0) \text{ and } (H_G - H_L) = \lambda \tag{9.117}$$

$$q = \frac{[C_{P,L}(T_b - T_0) + \lambda] - [C_{P,L}(T_F - T_0)]}{\lambda} = \frac{[C_{P,L}(T_b - \cancel{T_0} - T_F - \cancel{T_0}) + \lambda]}{\lambda} \tag{9.118}$$

i.e. $$q = 1 + \frac{C_{P,L}(T_b - T_F)}{\lambda} \tag{9.119}$$

(ii) *Saturated liquid*

$$q = \frac{(H_G - H_F)}{(H_G - H_L)} \tag{9.120}$$

For saturated liquid $H_F = H_L$, $\therefore \quad q = 1.0$

(iii) *Mixture of liquid and vapour*

Let x be the mole fraction of liquid in feed in the case of liquid + vapour mixture. Then,

$$H_F = xH_L + (1 - x)H_G \tag{9.121}$$

Therefore,
$$q = \frac{\left[\cancel{H_G} - xH_L - \cancel{H_G} + xH_G\right]}{\left(H_G - H_L\right)}$$

$$= \frac{x(H_G \overset{1.0}{\cancel{-}} H_L)}{(H_G \cancel{-} H_L)} = x \qquad (9.122)$$

(iv) *Saturated vapour*

$$q = \frac{(H_G - H_F)}{(H_G - H_L)} \qquad (9.123)$$

For saturated vapour $H_F = H_G$, $\therefore\ q = 0$

(v) *Superheated vapour*

Let $C_{P,V}$ be the specific heat of feed vapour

$$H_G = H_G$$

$$H_F = H_G + C_{P,V}\,(T_F - T_b) \qquad (9.124)$$

$$q = \frac{\cancel{H_G} - [\cancel{H_G} + C_{P,V}(T_F - T_b)]}{\lambda} = -\frac{C_{P,V}(T_F - T_b)}{\lambda} \qquad (9.125)$$

Steps involved in the determination of number of trays

The equilibrium curve along with the operating lines for both enriching and stripping sections to determine the number of stages is shown in Fig. 9.22.

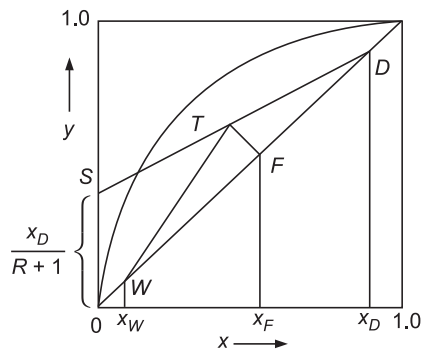


Fig. 9.22 Determination of number of stages by McCabe–Thiele method.

1. Draw the equilibrium curve and diagonal.
2. Locate *F*, *D* and *W* corresponding to feed, distillate and residue compositions based on more volatile component.
3. Estimate $x_D/(R + 1)$ and locate it on *y*-axis as *S*.
4. Join *SD*, this is the operating line for enriching section.
5. From *F* draw *q*-line depending on feed condition. Let it cut the operating line for enriching section at *T*.

6. Join *TW*-operating line for stripping section.
7. Construct stepwise from *D* to *W* and the steps so constructed will give the number of stages.

9.13 LOCATION OF FEED TRAY

For an optimal design or when a column is designed first (wherein one goes for optimal design) the feed tray is located at the intersection of operating lines of enriching and exhausting sections of the tower.

However, we may at times use a column which has been designed with some other objectives. Whenever the quality and condition of feed is fixed along with reflux ratio, x_D and x_W, the operating lines are fixed. It may so happen that in an existing column, the location of feed nozzle is fixed and it may not really lie at the optimal point as shown in Figs. 9.23(a), (b) and (c).

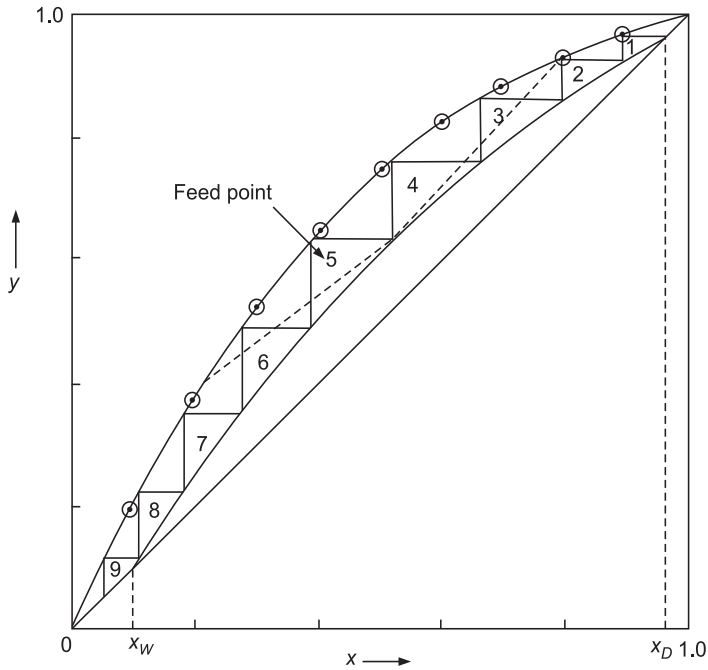


Fig. 9.23(a) Optimal feed location.

The point of intersection of the two operating lines is generally believed to be the point that demarcates enriching and exhausting sections. This normally occurs in the newly designed columns from a specific x_w, x_D, x_F and condition of feed. However, in an existing column, designed for a different utility, the feed point location is fixed and may not be at the optimum location. Further, the feed entry point will not demarcate the enriching and exhausting sections.

Generally when the reflux ratio and the x_D values are fixed, the operating line for enriching section is fixed. Further, when the x_w and the condition of the feed are fixed the operating line for exhausting section is fixed.

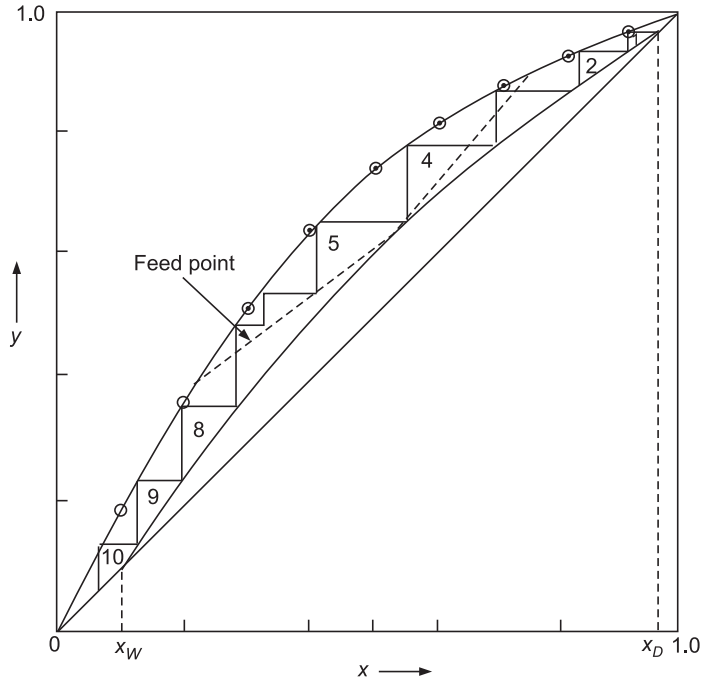


Fig. 9.23(b) Delayed feed entry.

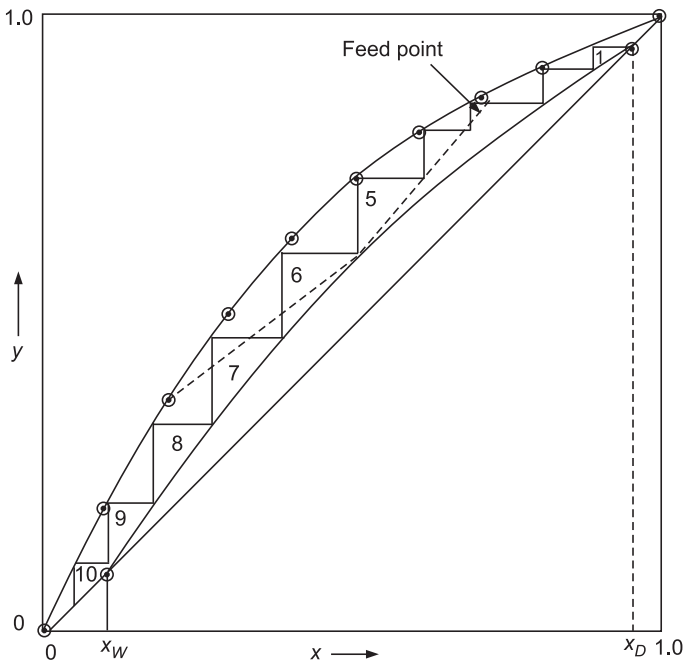


Fig. 9.23(c) Early feed entry.

Once a feed enters a specific plate, below the point of intersection of operating lines and *q*-line (in an existing column), from the top plate to feed entry point, the operating line for enriching section is to be used and subsequently the operating line for exhausting section. Such an arrangement indicates a *delayed feed entry*. If the feed enters at a specific plate, above the point of intersection of operating lines and *q*-line (in an existing column), from the top plate to feed entry point, the operating line for enriching section is to be used and subsequently the operating line for exhausting section. This arrangement indicates an *early feed entry*. In both the cases the number of stages estimated will always be more compared to the number of stages estimated with feed entering exactly at the point of intersection of operating lines and *q*-line.

Consider the above three figures,

Figure 23(a): Optimal design with 9 plates and the 5th plate is feed plate.

Figure 23(b): An existing column with 10 plates and feed enters at 7th plate.

Figure 23(c): An existing column with 10 plates and feed is introduced at 3rd plate.

9.14 REFLUX RATIO

It is one of the important operating parameters in distillation, by which the quality of the products can be changed. Let us deal with the relationship between reflux ratio and the number of trays in the tower.

9.14.1 Determination of Minimum Reflux Ratio

To determine the minimum reflux ratio, draw the *q*-line from F to cut the equilibrium curve at T'. Join DT' and extend it to intersect on *y*-axis and indicate it as S'. OS' gives $\left[\frac{x_D}{R_{\min}+1}\right]$ from which $R_{\min}$ is estimated as shown in Fig. 9.24.

Normally at $R_{\min}$ condition, the number of stages will be infinity as the equilibrium curve and operating line get pinched.

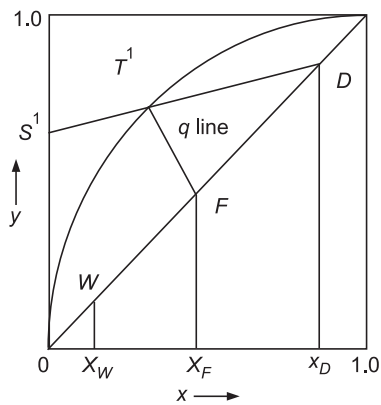


Fig. 9.24 Determination of minimum reflux ratio.

Steps involved in the determination of minimum reflux

1. Draw the equilibrium curve and diagonal.
2. Locate F, D and W corresponding to feed, distillate and residue compositions based on more volatile component.
3. Draw the q-line from F and allow it to intersect the equilibrium curve at T'.
4. Join $T'D$ and allow it to intersect the y-axis at S'.
5. OS' corresponds to $\left[\frac{y_D}{R_{\min}+1}\right]$ from which $R_{\min}$, the minimum reflux ratio is estimated.

9.14.2 Total Reflux

At total reflux, all the distillate is returned to the column and no product is taken out as distillate.

i.e. $$D = 0$$

$\therefore$ $$R = \frac{L}{D} = \infty$$

Hence, the operating line [Eq. (9.88)] for enriching section is

$$y_{n+1} = \left[\frac{R}{R+1}\right] x_n + \left[\frac{1}{R+1}\right] x_D$$

becomes, $$y_{n+1} = x_n \tag{9.126}$$

i.e. it merges with the diagonal ($x = y$ line) for both enriching and stripping sections. Under such circumstances, the minimum number of theoretical stages can be estimated by the same graphical procedure described in the Section of 9.12.2. For systems where the relative volatility is constant and under total reflux conditions the theoretical number of stages needed could be estimated analytically.

We know that for a binary system, $\alpha_{AB} = \left[\frac{\left[\frac{y_A}{x_A}\right]}{\left[\frac{y_B}{x_B}\right]}\right] = \left[\frac{\left[\frac{y_A}{y_B}\right]}{\left[\frac{x_A}{x_B}\right]}\right]$

$\therefore$ $$\frac{y_A}{y_B} = \alpha_{AB}\frac{x_A}{x_B}$$

$$\frac{y_A}{1-y_A} = \alpha_{AB}\frac{x_A}{1-x_A}$$

Let us apply this relationship to $(n + 1)$th plate

$$\therefore \quad \frac{y_{n+1}}{(1 - y_{n+1})} = \alpha \frac{x_{n+1}}{(1 - x_{n+1})} \tag{9.127}$$

At total reflux $D = 0$ and $\frac{L}{G} = \frac{R}{R+1} = 1.0$

Hence, from Eq. (9.126), we get

$$(y_{n+1}) = x_n \tag{9.128}$$

When $n = 0$, i.e. at the top of the column, $x_n = x_0$

$y_1 = x_0 = x_D$, when total condenser is used.

Substituting for y_{n+1} in terms of x_n in Eq. (9.128), we get

$$\therefore \quad \frac{x_n}{(1 - x_n)} = \alpha_{AB} \frac{x_{n+1}}{(1 - x_{n+1})} \tag{9.129}$$

When $n = 0$, $$\frac{x_0}{(1 - x_0)} = \alpha_{AB} \frac{x_1}{(1 - x_1)} \tag{9.130}$$

When $n = 1$, $$\frac{x_1}{(1 - x_1)} = \alpha_{AB} \frac{x_2}{(1 - x_2)} \tag{9.131}$$

When $n = n - 1$, $$\frac{x_{n-1}}{(1 - x_{n-1})} = \alpha_{AB} \frac{x_n}{(1 - x_n)} \tag{9.132}$$

Substituting for $x_1, x_2, \ldots, x_{n-1}$ from Eqs. (9.131) and (9.132), we get

$$\frac{x_0}{(1 - x_0)} = (\alpha_{AB})^n \frac{x_n}{(1 - x_n)} \tag{9.133}$$

Substituting $n = N_P + 1$ (last stage, i.e. reboiler), we get

$$\frac{x_0}{(1 - x_0)} = (\alpha)^{N_P+1} \frac{x_{N_P+1}}{(1 - x_{N_P+1})} \tag{9.134}$$

As $(N_P + 1)$th stage accounts for reboiler, $x_{N_P+1} = x_W$

i.e. $$\frac{y_D}{(1 - y_D)} = (\alpha)^{N_P+1} \frac{x_W}{(1 - x_W)} \tag{9.135}$$

Equation (9.135) is called *Fenske equation*. To apply this equation, α, the relative volatility must be fairly constant and the column has to be operated under total reflux conditions. This may not be possible in industries, but has theoretical importance.

9.14.3 Optimum Reflux Ratio

At minimum reflux ratio the column requires infinite number of stages or trays. However, as reflux ratio increases from minimum, for a given feed and specified

quality of distillate and residue, the number of stages or trays decrease. At minimum reflux ratio when the stages or trays are infinite, the fixed cost and the maintenance cost are also infinite. However, the operating cost for operating condenser, reboiler etc. is the least. When the reflux increases, the trays or stages reduce but the column diameter has to be increased to handle larger capacities of liquid being recycled. The size of other accessories like condenser and reboiler increase which will result in a higher requirement of cooling water or heating. Ultimately this will result in a higher operating cost. Thus, the total cost which includes both operating cost and fixed cost, vary with reflux ratio and reach a minimum value for a certain reflux ratio which is called the *optimum or economic reflux ratio*. This value is normally in the range of 1.2 R_{min} to 1.5 R_{min}. This is shown in Fig. 9.25.

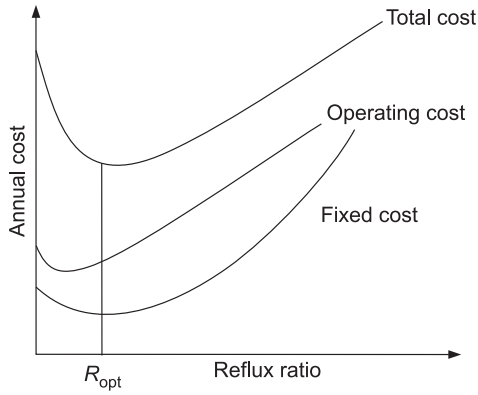


Fig. 9.25 Effect of Reflux ratio on cost.

9.15 REBOILERS

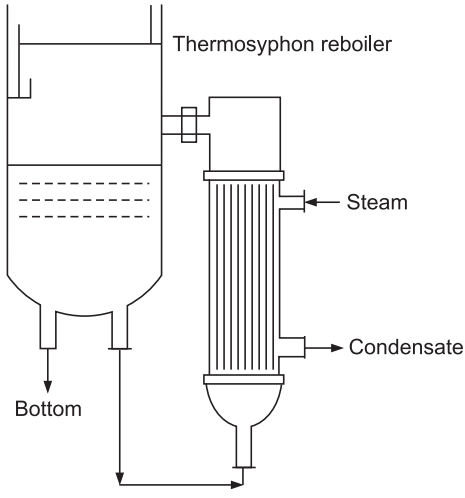


Fig. 9.26(a) Thermosyphon reboiler.

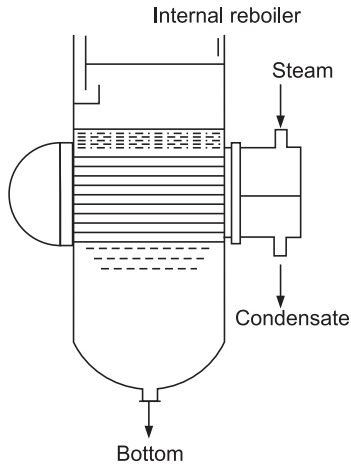


Fig. 9.26(b) Internal reboiler.

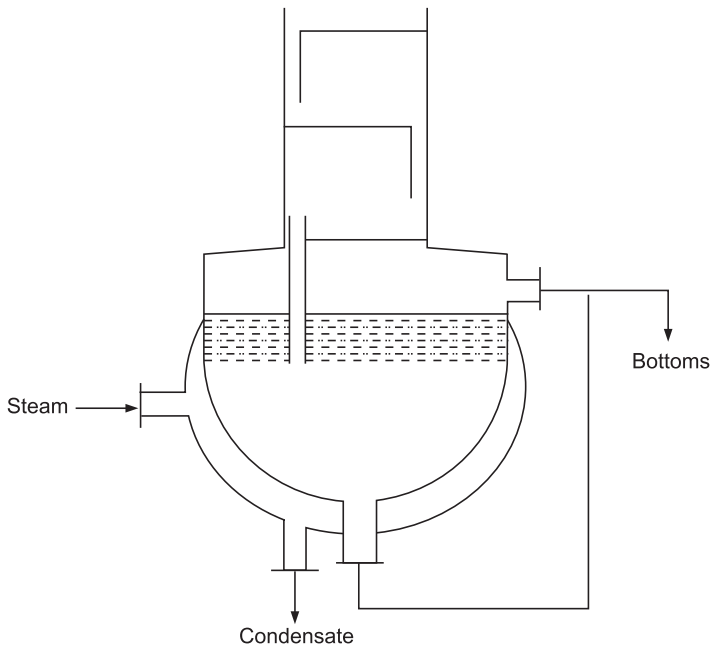


Fig. 9.26(c) Jacketed kettle reboiler.

They are heat exchangers of different configurations used to supply the heat to the liquid at the bottom of the column to vaporize them. In effect all the heat needed is basically supplied at the reboiler only.

A simple Jacketed kettle is one such reboiler which has a low heat transfer area and hence vapour generation capacity will also be poor.

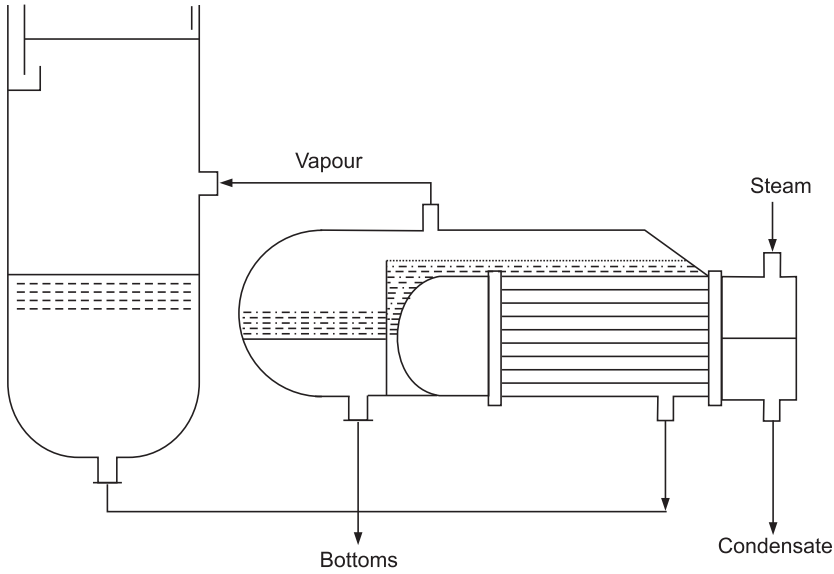


Fig. 9.26(d) External reboiler

Tubular heat exchangers (both of vertical and horizontal configurations) provide larger area of heat transfer. They can be placed inside the column or outside the column. When they are located inside, during cleaning of the exchanger, the distillation operation has to be stopped. However, when external reboilers are used, a standby exchanger is always kept which can be used during cleaning of the exchanger attached to the column. Thus, the distillation operation will proceed without any interruption. The liquid can flow either through the tube side or shell side. Reboilers can be heated by steam, oil or other hot fluids. Different types of reboilers are shown in Figs. 9.26(a)–(d).

9.16 CONDENSERS

The condensers are generally heat exchangers of horizontal orientation with coolant flowing through the tube side. However, in rare instances vertical condensers are used with the coolant flowing on either side of the tubes. They are placed above the tower in the case of laboratory scale units for gravity flow of the condensed reflux to the topmost tray. Sometimes they are placed at ground level for easy maintenance, in which case the liquid is pumped from accumulator to the top tray. The coolant is normally water. The condensers may either be a total condenser or a partial condenser. Whenever a partial condenser is used, the condensate is returned as reflux and the vapour from condenser is the main distillate product. The partial condenser itself acts as one stage for separation. In an existing distillation column, if one desires to have a highly enriched distillate (richer than the designed value) then one can resort to partial condensation and obtain an enriched product. However, when a column is being designed fresh it is always preferable to go for additional trays compared to partial condensation technique for enrichment.

9.17 USE OF OPEN STEAM

Normally the heat needed for distillation is supplied through (by heat exchangers) reboilers. However, when an aqueous solution is fractionated to give non-aqueous solute as distillate and water as residue, the heat required may be supplied by open steam in which case the reboiler is not required. The schematic arrangement is shown in Fig. 9.27 and the overall material balance is given below.

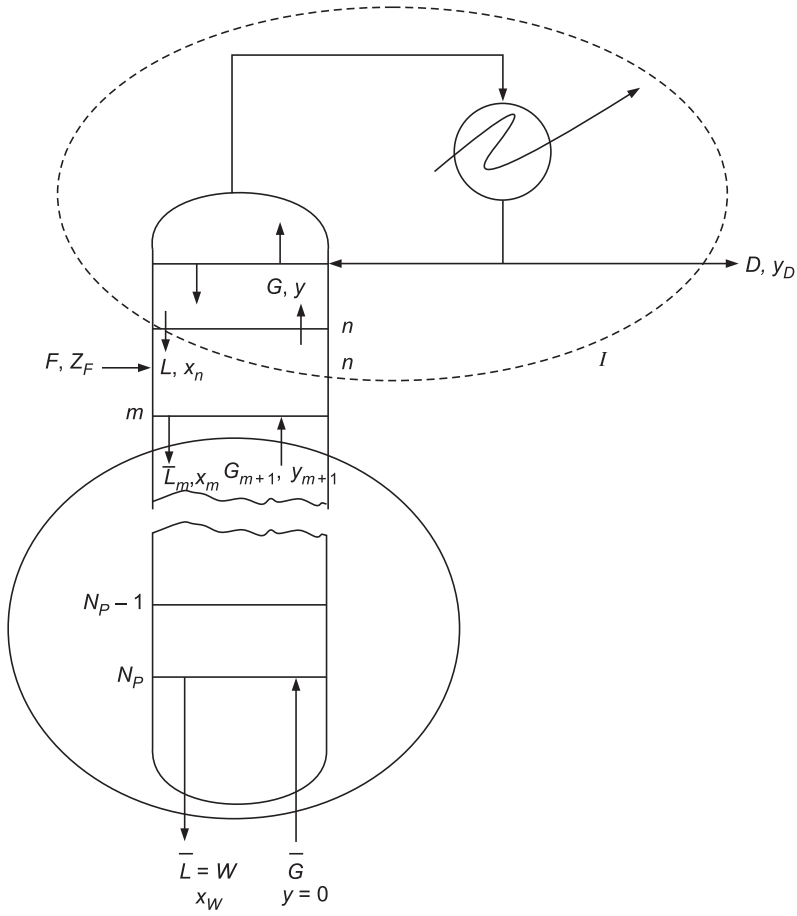


Fig. 9.27 Open steam distillation.

The equation for the operating line in enriching section is obtained as in the case of McCabe–Thiele method.

i.e. $$\text{Slope} = \frac{L}{G} = \frac{R}{R+1} \text{ and intercept} = \frac{x_D}{R+1} \tag{9.136}$$

However, let us analyze the exhausting section.

A component balance gives

$$\bar{L}x_m + \bar{G}(0) = \bar{G}y_{m+1} + \bar{L}x_W \tag{9.137}$$

$$\bar{L}x_m + \bar{G}(0) = \bar{G}y_{m+1} + Wx_w \tag{9.138}$$

i.e.
$$\frac{\bar{L}}{\bar{G}} = \frac{(y_{m+1} - 0)}{(x_m - x_W)} \tag{9.139}$$

The operating line for exhausting section passes through ($x = x_W$ and $y = 0$) and (x_m, y_{m+1}). Thus, the line passes through x_W, the point in x-axis as shown in Fig. 9.28. After constructing the equilibrium curve and operating lines, by step-wise construction the number of stages are determined as in the case of McCabe-Thiele method.

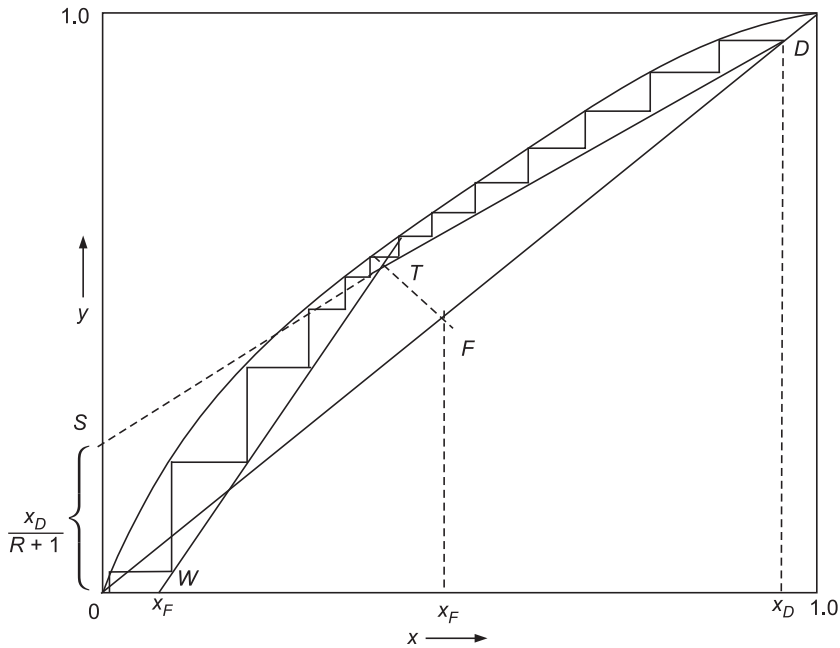


Fig. 9.28 Determination of number of stages in open steam distillation.

If the steam entering the tower is superheated ($H_{G,\,N_{P+1}}$), it will vaporize liquid on tray N_P to the extent such that the steam will reach saturation ($H_{G,\,\text{Sat}}$). An energy balance yields,

$$\bar{G} = \bar{G}_{N_{P+1}}\left[1 + \frac{\left(H_{G,N_{P+1}} - H_{G,Sat}\right)}{\lambda M}\right] \tag{9.140}$$

where λM is the molar latent heat of vaporization, and

$$\bar{L} = \bar{G} - \bar{G}_{N_{P+1}} + \bar{L}_{N_P} \tag{9.141}$$

Using Eqs. (9.140) and (9.141), the $\bar{L}/\bar{G}$ ratio is computed.

9.17.1 Determination of Number of Trays

1. Draw the equilibrium curve and diagonal.
2. Locate F and D corresponding to the composition of feed and distillate respectively on diagonal.
3. Locate W corresponding to the composition of x_W on x-axis.
4. Based on the reflux ratio and distillate composition, estimate $\frac{x_D}{R+1}$ and locate it on y-axis as S.
5. Join SD, this is the operating line for enriching section.
6. From F draw q-line depending on feed condition and allow it to cut the operating line for enriching section SD, and locate the point of intersection as T.
7. Join TW. TW is the operating line for stripping section.
8. By stepwise construction starting from D up to W, the number of stages can be determined.
9. The minimum reflux ratio needed is estimated in the same manner as in the case of McCabe–Thiele method.

9.18 CONTINUOUS DIFFERENTIAL CONTACT–PACKED TOWER DISTILLATION

Whenever we have heat sensitive materials to be distilled which require less contact time, packed towers are preferred. The pressure drop is also low and hence it is suitable for low pressure distillation operations.

The height of column can be determined in the same way as for other mass transfer operations using packed towers by making a material balance across a differential section and integrating as indicated in Fig. 9.29.

The material balance across the elemental section in enriching zone gives

$$N_A = \frac{d(Gy)}{adZ} = k'_y(y_i - y) = \frac{d(Lx)}{adZ} = k'_x(x - x_i) \tag{9.142}$$

$$\therefore \quad Z_e = \int_0^{Z_e} dZ = \int_{(Gy)_a}^{(Gy)_2} \frac{d(Gy)}{k'_y a(y_i - y)} = \int_{(Lx)a}^{(Lx)_2} \frac{d(Lx)}{k'_x a(x - x_i)} \tag{9.143}$$

Assuming that G, L, k'_y and k'_x are constants.

(However, this has to be checked before using as k_x and k_y depending on the flow rates)

$$Z_e = \frac{G}{k'_y a} \int_{y_a}^{y_2} \frac{dy}{(y_i - y)} = (H_{tG})(N_{tG}) \text{ (By definition)} \tag{9.144}$$

(Suffix a indicates the point of intersection of operating lines)

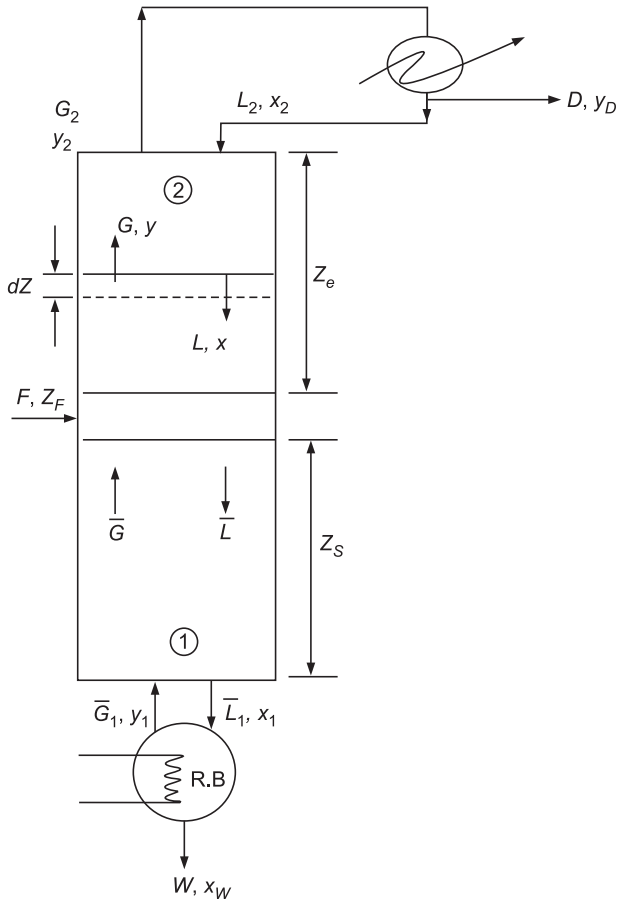


Fig. 9.29 Analysis of packed distillation column.

Based on liquid phase, Z_e can be expressed as

$$Z_e = \frac{L}{k'_x a} \int_{x_a}^{x_2} \frac{dx}{(x - x_i)} = (H_{tL})(N_{tL}) \quad \text{(By definition)} \tag{9.145}$$

Similarly, the stripping section can also be analysed and it can be shown that,

$$Z_S = (H_{tG})\ (N_{tG}) = (H_{tL})\ (N_{tL}) \tag{9.146}$$

$$\text{Total height } Z = Z_e + Z_S \tag{9.147}$$

The determination of H_{tG} or H_{tL} can be done using the flow rates of vapour or liquid and vapour or liquid mass transfer coefficients. However, to determine N_{tG} or N_{tL} one needs to find interfacial compositions. If the film coefficients are known, the interfacial concentrations can be determined from the operating line and equilibrium curve. These values are given by a line drawn with a slope of

$\left[-\frac{k'_x}{k'_y}\right]$ from the operating line to the equilibrium curve. The point at which this line cuts operating line gives x and y values and the point of intersection with the equilibrium curve gives the interfacial compositions x_i and y_i. A number of such lines can be drawn which will give various sets of $(x - x_i)$ and $(y_i - y)$ values as shown in Fig. 9.30. Using these values, N_{tL} or N_{tG} can be determined. One should use N_{tL} or N_{tG} (corresponding to H_{tL} or H_{tG}) to determine the height depending on the resistance which is controlling the mass transfer.

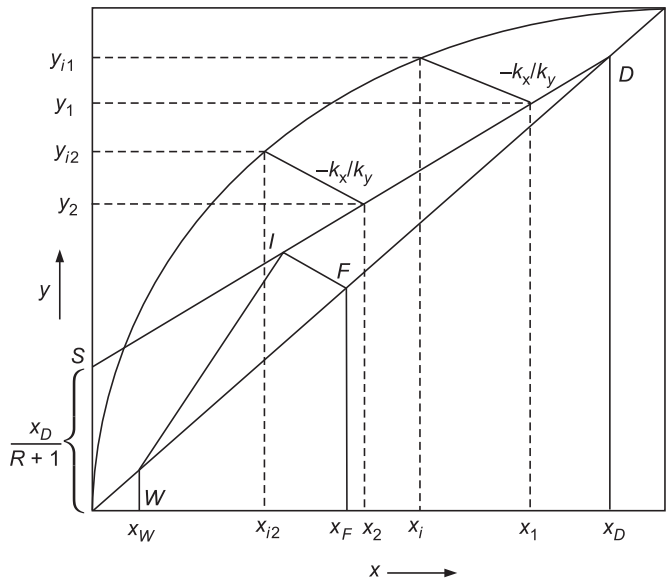


Fig. 9.30 Determination of interfacial conditions for packed distillation column.

Further, if the equilibrium curve is essentially straight in the range of our application, then the expressions are simplified as

$$Z_e = H_{t_0G} \int_{y_a}^{y_2} \frac{dy}{(y^* - y)} = (H_{t_0G})(N_{t_0G}) \tag{9.148}$$

when gas phase is controlling and

$$Z_e = H_{t_0L} \int_{x_a}^{x_2} \frac{dx}{(x - x^*)} = (H_{t_0L})(N_{t_0L}) \tag{9.149}$$

when liquid phase is controlling.

In Eqs. (9.148) and (9.149) , $(y^* - y)$ and $(x - x^*)$ are the overall driving forces in terms of vapour and liquid phase composition respectively.

Here, $$H_{t_0G} = \frac{G}{K'_y a} \quad \text{and} \quad H_{t_0L} = \frac{L}{K'_x a}$$

We know that,

$$\frac{1}{K_y'} = \frac{1}{k_y'} + \frac{m}{k_x'}$$

$$\frac{1}{K_x'} = \frac{1}{k_x'} + \frac{1}{mk_y'}$$

Similarly, H_{t_0G} and H_{t_0L} can also be written as

$$H_{t_0G} = H_{tG} + \left(\frac{mG}{L}\right) H_{tL} \tag{9.150}$$

$$H_{t_0L} = H_{tL} + \left(\frac{L}{mG}\right) H_{tG} \tag{9.151}$$

where m is the slope of the equilibrium curve.

9.18.1 Steps Involved in the Determination of the Height of Tower

1. Draw the equilibrium curve and diagonal.
2. Locate S, corresponding to $\frac{x_D}{R+1}$ in y-axis.
3. Locate F, D and W on diagonal corresponding to x_F, x_D and x_W
4. Join DS. This is the operating line for enriching section.
5. From F draw q-line. Let the point of intersection on operating line DS for enriching section be T.
6. Join TW. This is the operating line for stripping section.
7. From D to T and T to W draw lines of slope $\left[-\frac{k_x'}{k_y'}\right]$ or $\left[-\frac{K_x'}{K_y'}\right]$ (as the case may be) to obtain (x_i and y_i) or (x^* and y^*) and (x and y) values.
8. x and y values are read from operating lines and (x_i, y_i) or (x^*, y^*) values are read from equilibrium curve.
9. Evaluate $\int \frac{dx}{(x - x_i)}$ or $\int \frac{dy}{(y_i - y)}$ or $\int \frac{dx}{(x - x^*)}$ or $\int \frac{dy}{(y^* - y)}$ graphically to determine N_{tL} or N_{tG} or N_{toL} or N_{toG}.
10. H_{tL}, H_{tG}, H_{toG} or H_{toL} are determined with the help of liquid and vapour flow rates and mass transfer coefficients.
11. Height is then estimated based on the values from steps (9) and (10).
12. The tower diameter is normally set by the conditions at the top of stripping section because of large liquid flow rate at that point.

9.19 AZEOTROPIC DISTILLATION

This is a technique which is used for the separation of binary mixtures which are either difficult or impossible to separate by ordinary fractionation. This happens when either the mixture to be separated has a very low relative volatility, in which case one may require high reflux ratio and more number of trays, or when the

mixture forms an azeotrope. Under such circumstances, a third component called an 'entrainer' is added to the binary mixture to form a new low boiling hetero-azeotrope with one of the components in the original mixture whose volatility is such that it can be separated from the other original constituent.

A typical example for this operation is presented in the flow diagram of Fig. 9.31 where the separation of acetic acid (BP: 118.1°C) and water (BP: 100°C) mixture is demonstrated. This mixture has a low relative volatility and hence separation by conventional methods is not economical. Here Butyl acetate, which is slightly soluble in water, is added to the mixture from the top of the column as an entrainer. It forms a 'hetero-azeotrope' with all the water in the feed and readily distills out from the high boiling acetic acid and the acetic acid leaves as a residue product. The hetero-azeotrope on condensation forms two insoluble layers which can easily be separated. The water layer obtained is saturated with ester and vice versa. The ester layer saturated with water is returned back to the column as a source of entrainer for further separation. The aqueous layer is also sent to another column to separate water and ester. The separated ester is also sent back as entrainer.

Sometimes the new azeotrope formed contains all the three constituents. In the dehydration of ethanol water mixture, benzene is added as an entrainer which gives a ternary azeotrope containing benzene (53.9 mole %), water (3.3 mole %) and ethanol (22.8 mole %) boiling at 64.9°C as distillate and ethanol (BP: 78.4°C) as residue. Benzene is separated and sent back to the top of the column as entrainer. Since water – ethanol are equally present in distillate, the mixture should be given a preliminary rectification to produce an alcohol rich binary azeotrope. Azeotropic distillation is shown in Fig. 9.31.

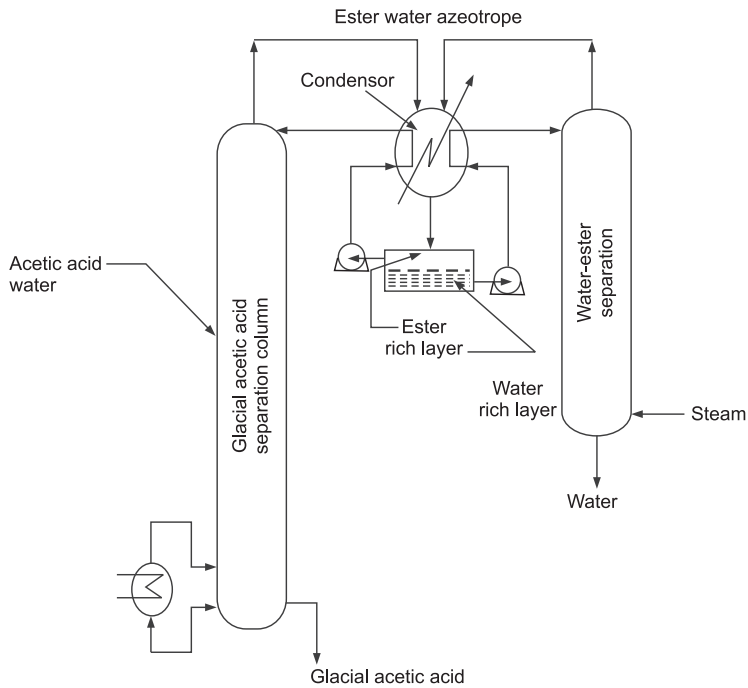


Fig. 9.31 Azeotropic distillation.

9.19.1 Desired Properties of an Entrainer for Azeotropic Distillation

(i) Should be cheap and easily available.
(ii) Chemically stable and inactive towards the solution to be separated.
(iii) Non-corrosive.
(iv) Non-toxic.
(v) Low latent heat of vaporization.
(vi) Low freezing point to facilitate storage and easy handling.
(vii) Low viscosity to provide high tray efficiency and minimum pumping cost.

9.20 EXTRACTIVE DISTILLATION

This method is also used under similar circumstances as in the case of azeotropic distillation. Here a third component called solvent is added, instead of entrainer, which alters the relative volatility of the original constituents, thus permitting the separation. The added solvent should have low volatility and not vaporized in the fractionator.

One such example is the separation of toluene (BP: 110.8°C) from isooctane (BP: 99.3°C). Their separation is relatively difficult. In the presence of Phenol (BP: 181.4°C), the relative volatility of isooctane increases, so that with an increase in phenol content, the separation becomes more and more easy. A typical flow diagram of the process is shown in Fig. 9.32.

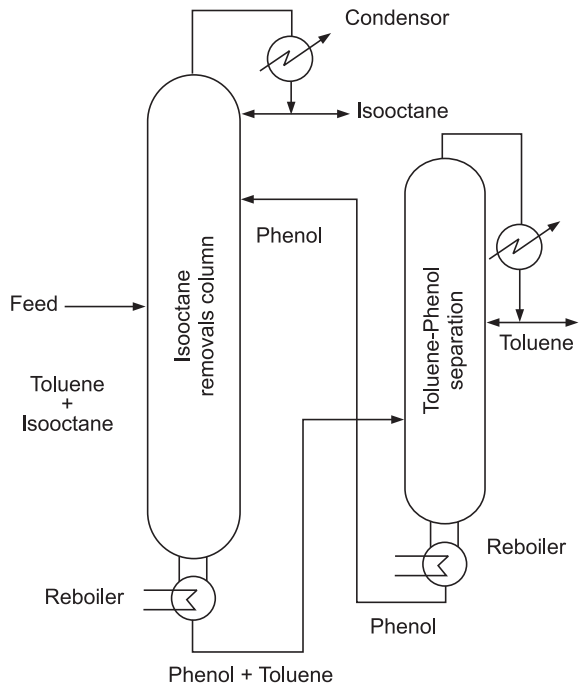


Fig. 9.32 Extractive distillation.

Here, the toluene–isooctane binary mixture is introduced in the middle of the column and phenol near the top of the column. Isooctane is readily distilled as an overhead product, while toluene and phenol are collected as residue. The residue from the tower is rectified in the auxiliary tower to separate toluene and phenol as distillate and residue respectively. Phenol is returned to the main column as solvent. Similarly a mixture of acetone (BP: 56.4°C) and methanol (BP: 64.7°C) can be separated by using Butanol (BP: 117.8°C) solvent.

9.20.1 Desired Properties of Solvent for Extractive Distillation

(i) High selectivity and capability to alter VLE for easy separation.
(ii) Ability to dissolve the components in the mixture.
(iii) Low volatility in order to prevent vaporization of solvent.
(iv) Easy separability, for easier removal of solvent.
(v) Non-corrosive.
(vi) Non-toxic.
(vii) Cheap and easily available.
(viii) Low freezing point.
(ix) Low viscosity.
(x) Chemical stability and inertness towards the components to be separated.

9.21 COMPARISON OF AZEOTROPIC AND EXTRACTIVE DISTILLATION

In both the processes an additional external agent is added, which is undesirable. Solvent to feed ratio in extractive distillation greater than 3 or 4 is found to be effective. Proper choice of material of construction and recovery technique are to be examined. However, of the two, extractive distillation is said to be more favoured than azeotropic distillation since (i) there is a greater choice of solvent, (ii) the smaller quantity of solvent to be volatilized. In spite of the above advantages, the azeotropic distillation is said to be more effective in the dehydration of ethanol from an 85.6 mole % Ethanol–water solution. In this case water is azeotroped with *n*–pentane and then separated rather than using extractive distillation with ethylene glycol as solvent.

9.22 LOW PRESSURE DISTILLATION

Whenever the heat sensitive materials are to be separated, as in the case of many organic mixtures, low pressure distillation will be effective. In this case the time of exposure of the substances to high temperature is kept minimum. Packed towers can be used for distillation under pressures of 50 to 250 mm Hg. Bubble cap and sieve trays can be used for pressure drops around 2.6 mm Hg and shower trays for pressure drops around 0.75 mm Hg. This is used in the separation of vitamins from animal and fish oils as well as the separation of plasticizers.

9.22.1 Molecular Distillation

This is a very low pressure distillation where the absolute pressure is in the range of 0.003 mm Hg to 0.03 mm Hg. On reducing the absolute pressure, the mean free path of the molecules becomes large. If the condensing surface is placed at a distance from the vaporizing liquid, surface not exceeding few cms, only a few molecules will return to the liquid. The composition of the distillate will now be different from that given by normal vaporization and the ratio of the constituents is given by

$$\frac{N_A}{N_B} = \frac{\text{moles of } A}{\text{moles of } B} = \left[\frac{\left(\frac{\bar{p}_A}{M_A{}^{0.5}} \right)}{\left(\frac{\bar{p}_B}{M_B{}^{0.5}} \right)} \right] \tag{9.152}$$

In molecular distillation this ratio is maintained by allowing the liquid to flow in a thin film over a solid surface thus renewing the surface continually and at the same time maintaining low hold-up of liquid.

A schematic arrangement of a device used for this type of distillation is shown in Fig. 9.33. The degassed liquid to be distilled is introduced at the bottom of the inner surface of the rotor, rotating at 400 to 500 rpm. A thin layer of liquid 0.05 to 0.1 mm spreads over the inner surface and travels rapidly to the upper periphery under centrifugal force. Heat is supplied to the liquid through the rotor from radiant electrical heaters. Vapors generated are condensed and collected in the collection troughs. The residue liquid is collected in the collection gutter and removed. The entire unit is maintained at low pressure, good enough for molecular distillation to occur. Normal residence time is of the order of 1 second and hence decomposition of mixture does not take place. Multiple units can be used to have multistage separation effects.

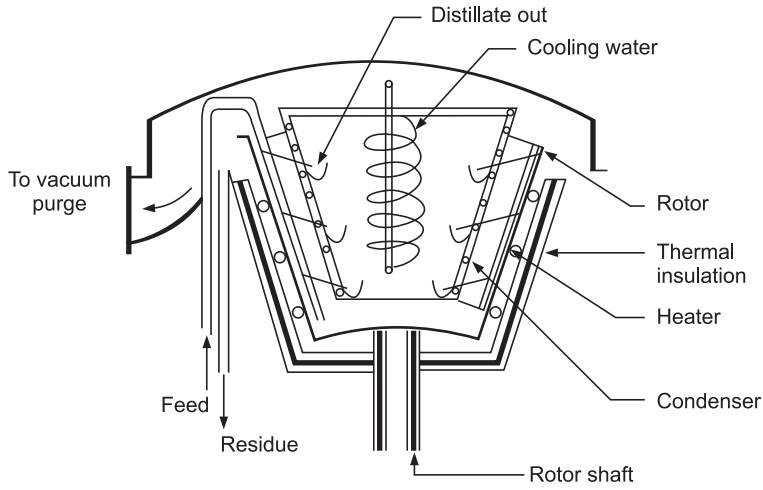


Fig. 9.33 Molecular distillation.

WORKED EXAMPLES

1. Compute the equilibrium data from the following data at 760 mm Hg pressure and calculate the relative volatility.

VP of A, mm Hg	760	830	920	1060	1200	1360
VP of B, mm Hg	200	350	420	550	690	760

P_T = 760 mm Hg

Solution.

P_A, mm Hg	760	830	920	1060	1200	1360
P_B, mm Hg	200	350	420	550	690	760

We know that $x_A = \left[\frac{(P_T - P_B)}{(P_A - P_B)}\right]$ and $y_A = \left[\frac{(P_A x_A)}{P_T}\right]$

$x_A = \left[\frac{(P_T - P_B)}{(P_A - P_B)}\right]$	1.0	0.854	0.68	0.412	0.137	0
$y_A = \left[\frac{(P_A x_A)}{P_T}\right]$	1.0	0.933	0.823	0.575	0.216	0
α_{AB} = VP of A/VP of B	3.80	2.37	2.19	1.93	1.74	1.79

Average relative volatility: 2.303

2. The vapour pressure data for *n*–Hexane –*n*–Octane system is given below. Compute the equilibrium data and relative volatility for the system at a total pressure of 101.32 kPa.

T°C	*n*-Hexane P_A, kPa (*A*)	*n*-Octane P_B, kPa (B)
68.7	101.32	16.1
79.4	136.7	23.1
93.3	197.3	37.1
107.2	284.0	57.9
125.7	456.0	101.32

Solution.

T°C	*n*-Hexane P_A,kPa (*A*)	n – Octane P_B,kPa (*B*)	$\alpha_{cal} = P_A/P_B$	$x_A = \frac{P_t - P_B}{(P_A - P_B)}$	$y_A = \frac{P_A x_A}{P_t}$	$y = \frac{\alpha x}{[1 + (\alpha - 1)x]}$
68.7	101.32	16.1	6.29	1.000	1.00	1.000
79.4	136.7	23.1	5.92	0.689	0.930	0.923
93.3	197.3	37.1	5.32	0.401	0.781	0.783
107.2	284.0	57.9	4.91	0.192	0.538	0.562
125.7	456.0	101.32	4.50	0	0	0

3. Compute *x–y* data at 1 atm. Pressure from the following data:

T	80.1	85	90	95	100	105	110.6
VP_A	760	877	1016	1168	1344	1532	1800
VP_B	–	345	405	475	577	645	760

Solution.

T	80.1	85	90	95	100	105	110.6
VP_A	760	877	1016	1168	1344	1532	1800
VP_B	—	345	405	475	577	645	760
$\alpha = \frac{VP_A}{VP_B}$	—	2.54	2.51	2.46	2.33	2.38	2.37

(**Ans:** $\alpha_{average}$ = 2.43)

$x_A = \left[\frac{(P_T - P_B)}{(P_A - P_B)}\right]$	1.0	0.78	0.58	0.411	0.239	0.13	0
$y_A = \left[\frac{(P_A x_A)}{P_T}\right]$	1.0	0.9	0.777	0.632	0.423	0.26	0

4. A solution of methanol and ethanol are substantially ideal. Compute the VLE for this system at 1 atm pressure and relative volatility.

$$\log [P, \text{ mm}]_{\text{ Methanol}} = 7.84863 - \frac{1473.11}{(230 + \text{t°C})}$$

$$\log [P, \text{ mm}]_{\text{ Ethanol}} = 8.04494 - \frac{1554.3}{(222.65 + \text{t°C})}$$

Solution.

In this problem one has to compute the vapour pressure values at different temperatures. The temperature range is fixed by keeping the pressure as

760 mm Hg for each component. Thus, in the following equation for Methanol,

$$\log [P, \text{ mm}]_{\text{Methanol}} = 7.84863 - \frac{1473.11}{(230 + t°C)}$$

Setting the vapour pressure as 760 mm Hg (at BP, vapour pressure equals the prevailing pressure), we get the temperature as 66.53°C, which is the boiling point of Methanol. Similarly, by setting P as 760 mm Hg in the equation for ethanol,

$$\log [P, \text{ mm}]_{\text{Ethanol}} = 8.04494 - \frac{1554.3}{(222.65 + t°C)}$$

we get the boiling point of Ethanol as 78.33°C. This fixes the range of temperature.

t°C	66.53	70	72	74	76	78	78.33
V.P. of Methanol, P_A, mm	760	867.5	934.94	1006.6	1082.79	1163.6	1177.4
V.P. of Ethanol, P_B, mm	467.8	541.77	588.66	638.9	692.66	750.14	760
Relative volatility, $\alpha = \frac{P_A}{P_B}$	1.625	1.601	1.588	1.576	1.563	1.551	1.549
$x_A = \frac{P_t - P_B}{(P_A - P_B)}$	1.0	0.67	0.495	0.329	0.173	0.024	0.0
$y_A = \frac{P_A x_A}{P_t}$	1.0	0.765	0.609	0.436	0.246	0.0365	0.0

Average relative volatility = 1.579

5. Methanol and Ethanol form an ideal solution. Compute the VLE data at 760 mm Hg pressure,
Vapour pressure Data:

Vapour pressure, mm Hg	200	400	760	1520
Temperature,°C, Ethanol	48.4	62.5	78.4	97.5
Temperature,°C, Methanol	34.8	49.9	64.7	84.0

Plot vapour pressure vs temperature for both the components and compute T vs. VP for Methanol and T vs. VP for Ethanol as shown in Fig. 9.34.

Solution.

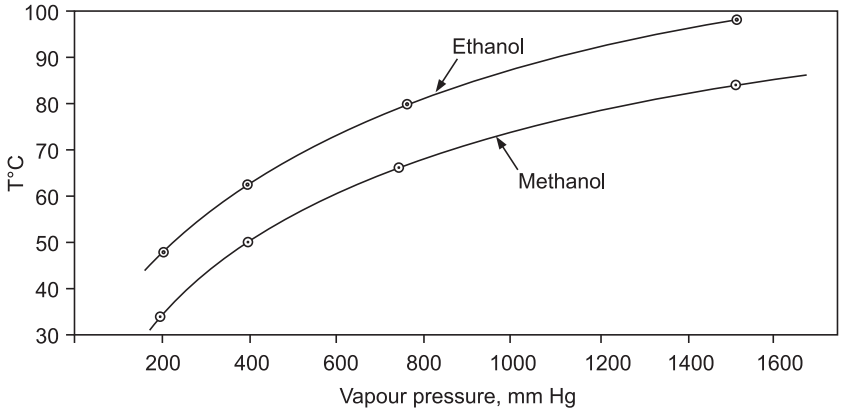


Fig. 9.34 Example 5 Vapour Pressure–temperature plot.

Temperature, °C	*V.P. of Ethanol*, mm Hg (*B*)	*V.P. of Methanol*, mm Hg (*A*)	$x_A = \frac{P_t - P_B}{(P_A - P_B)}$	$y_A = \frac{P_A x_A}{P_t}$
64.7	430	760	1.0	1.0
67.0	470	830	0.806	0.880
70.0	540	950	0.537	0.671
73.0	620	1080	0.304	0.432
76.0	700	1200	0.120	0.189
78.4	760	1300	0.0	0.0

6. It is desired to separate a feed mixture containing 40% heptane and 60% ethyl benzene, such that 60% of the feed is distilled out. Estimate the composition of the residue and distillate when the distillation process is (i) equilibrium distillation, and (ii) differential distillation.

Equilibrium Data:

x	0	0.08	0.185	0.251	0.335	0.489	0.651	0.79	0.914	1.0
y	0	0.233	0.428	0.514	0.608	0.729	0.814	0.910	0.963	1.0

x, *y*: Mole fraction of heptane in liquid and vapour phase respectively.

Solution.

(i) Plot the equilibrium data and draw the diagonal.

Draw a line with a slope of –W/D = – 0.4/0.6 = – 0.667 from a point on the diagonal corresponding to x_F = 0.4 and its intersection on the equilibrium curve and read them as x_w and y_D as shown in Fig. 9.35.

$$x_w = 0.24 \text{ and } y_D = 0.5$$

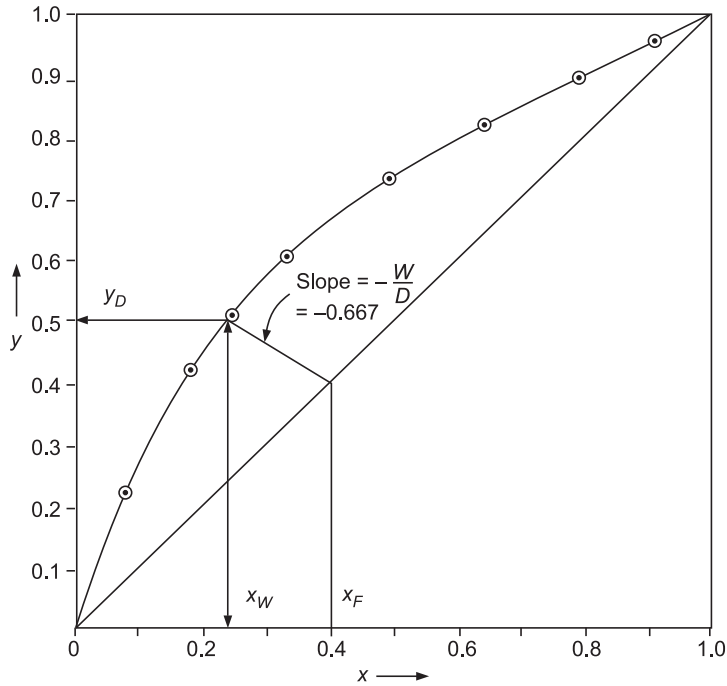


Fig. 9.35 Example 6 Solution for flash distillation.

(ii) Compute $\frac{1}{y-x}$ and plot it against x as shown in Fig. 9.36

x	0	0.08	0.185	0.251	0.335	0.489	0.651	0.79	0.914	1.0
y	0	0.233	0.428	0.514	0.608	0.729	0.814	0.91	0.963	1.0
$y-x$	0	0.153	0.243	0.263	0.273	0.240	0.163	0.12	0.049	0
$\frac{1}{y-x}$	α	6.54	4.12	3.80	3.66	4.17	6.13	8.33	20.41	α

We know that

$$\int_{x_w}^{x_F} \frac{dx}{(y-x)} = \ln\left[\frac{F}{W}\right] = \ln\left[\frac{1}{0.4}\right] = 0.916$$

By trial and error, find the x-co-ordinate which will give the area under the curve as 0.916 from x_F = 0.4. x_w = 0.2. By making component balance, y_D = 0.533.

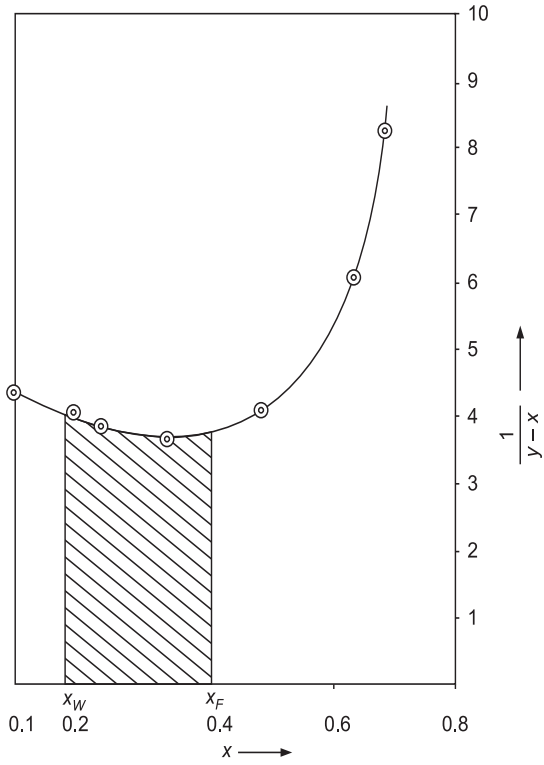


Fig. 9.36 Example 6 Solution for differential distillation.

7. A feed mixture containing 50 mole % Hexane and 50 mole % Octane is fed into a pipe still through a pressure reducing valve and flashed into a chamber. The fraction of feed converted to vapour is 0.6. Find the composition of the distillate and residue

x	0	4.5	19.2	40	69	100
y	0	17.8	53.8	78	93.2	100

x, y mole percent of Hexane in liquid and vapour phase respectively

Solution.

Draw the equilibrium curve and diagonal. From the feed point draw a line with a slope of

$$\left[-\frac{W}{D}\right] = -\frac{0.4}{0.6} = -0.667$$

From graph shown in Fig. 9.37, we get

$$x_W = 0.275, \quad y_D = 0.65$$

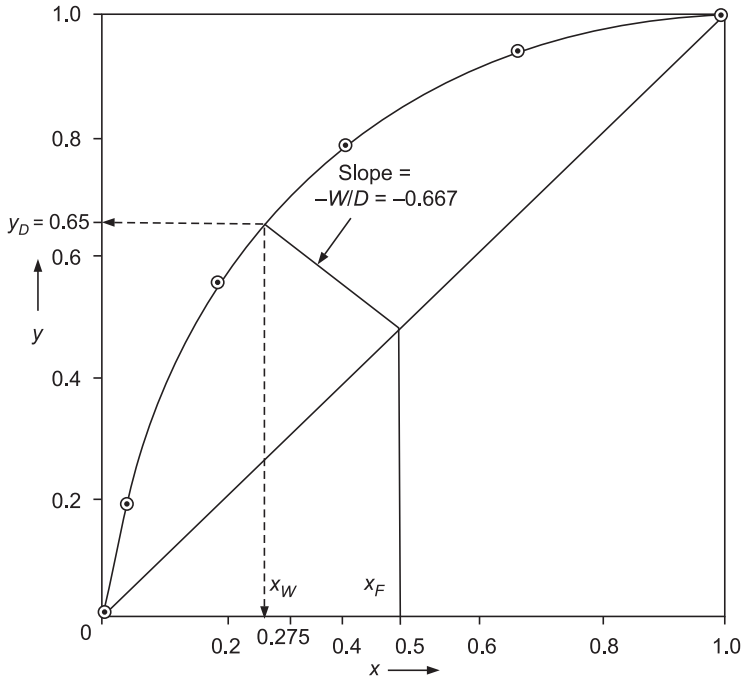


Fig. 9.37 Example 7 Flash distillation.

8. A equimolar feed mixture containing A and B is differentially distilled such that 70% of the feed is distilled out. Estimate the composition of the distillate and residue.

Equilibrium data

x	0	1	8	14	21	29	37	46	56	66	97	100
y	0	3	16	28	39	50	59	65	76	83	99	100

x, y: mole fraction of benzene in liquid and vapour phase respectively.

Solution.

x	0	0.01	0.08	0.14	0.21	0.29	0.37	0.46	0.56	0.66	0.97	1.0
y	0	0.03	0.16	0.28	0.39	0.50	0.59	0.65	0.76	0.83	0.99	1.0
$y–x$	0	0.02	0.08	0.14	0.18	0.21	0.22	0.19	0.20	0.17	0.02	0
$\frac{1}{y-x}$	α	50	12.5	7.14	5.56	4.76	4.55	5.26	5.0	5.88	50	α

Plot $\frac{1}{y-x}$ against x as shown in Fig. 9.38

We know that,

$$\int_{x_W}^{x_F} \frac{dx}{(y-x)} = \ln\left[\frac{F}{W}\right]$$

Let the feed be 100 moles
Therefore, D = 70 moles and W = 30 moles

$$\therefore \qquad \ln\left[\frac{F}{W}\right] = \ln\left[\frac{100}{30}\right] = 1.204$$

By trial and error, locate x_w such that $\int_{x_W}^{x_F} \frac{dx}{(y-x)} = 1.204$

We get, $x_w = 0.23$

Making material balance, we get

$$F = W + D$$

$$Fx_F = W\,x_W + Dy_D$$

Substituting for various quantities, we get

$$100 \times 0.5 = 30 \times 0.23 + 70 \times y_D$$

Solving, we get,

$$y_D = 0.616$$

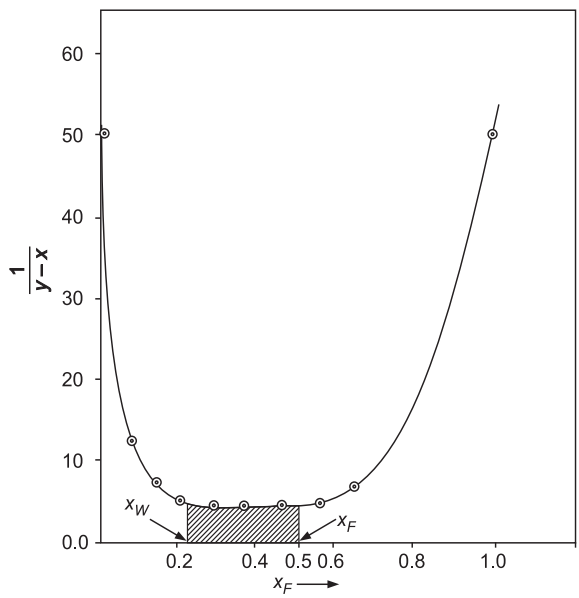


Fig. 9.38 Example 8 Solution for differential distillation.

9. A liquid mixture of components A and B containing 30 mole percent A is subjected to differential distillation. What percentage of the original mixture must be distilled off in order to increase the concentration of A in the residue to 65 mole percent?

The relative volatility of B in respect of A is 2.15.

Solution.

$$\ln\left[\frac{Fx_{F,B}}{WX_{W,B}}\right] = \alpha_{BA} \ln\left[\frac{Fx_{F,A}}{WX_{W,A}}\right]$$

$$\ln\left[\frac{F \times 0.7}{W \times 0.35}\right] = 2.15 \ln\left[\frac{F \times 0.3}{W \times 0.65}\right]$$

$$2\left[\frac{F}{W}\right] = \left[\frac{F \times 0.4615}{W}\right]^{2.15}$$

Solving, we get $\left[\frac{F}{W}\right] = 7.75$

Therefore, if F = 100 kmol, W = 12.91 kmol.
Hence, 87.09% of feed has to be distilled.

10. Nitrobenzene (NB) has to be steam distilled. If the vaporization efficiency is 85%, estimate the amount of nitrobenzene in the distillate if 100 kg of steam is present in distillate. The distillation takes place at a total pressure of 760 mm Hg.

Vapour pressure data for nitrobenzene:

T°, C	44.4	71.6	84.9	99.3	115.4	125.8	139.9	185.8	210.6
VP of NB mm Hg	1	5	10	20	40	60	100	400	760

Vapour pressure of water:

T°C	20	40	60	80	100
VP of water, mm Hg	17.5	55.3	149.4	355.1	760

T (°C)	71	78	80	82	90	96	100
p_B	5	7.5	9	10	14	17.5	21
p_A	242.5	340	355	412.5	515	605	760

Solution.

From total vapour pressure curve: Boiling point of mixture = 99.0°C

At 99°C, vapour pressure of nitrobenzene = 20 mm Hg
vapour pressure of water = 740 mm Hg

Vaporization η = [(Actual NB/Actual water)]

$$0.85 = \left[\frac{\left[\frac{\text{Actual NB}}{\text{Actual water}}\right]}{\left[\frac{\text{Theoretical NB}}{\text{Theoretical water}}\right]}\right] \text{ (all in moles)}$$

$$\frac{\text{Actual NB}}{\text{Actual water}} = 0.85 \times \left[\frac{\text{Theoretical NB}}{\text{Theoretical water}}\right]$$

$$= \frac{0.85 \times 20 \times 123}{740 \times 18} \text{ kg of NB/kg of steam}$$

$$= 0.85 \times 0.1847 = 0.157 \text{ kg of NB/kg of steam}$$

Mass of NB per 100 kg of steam = 15.7 kg

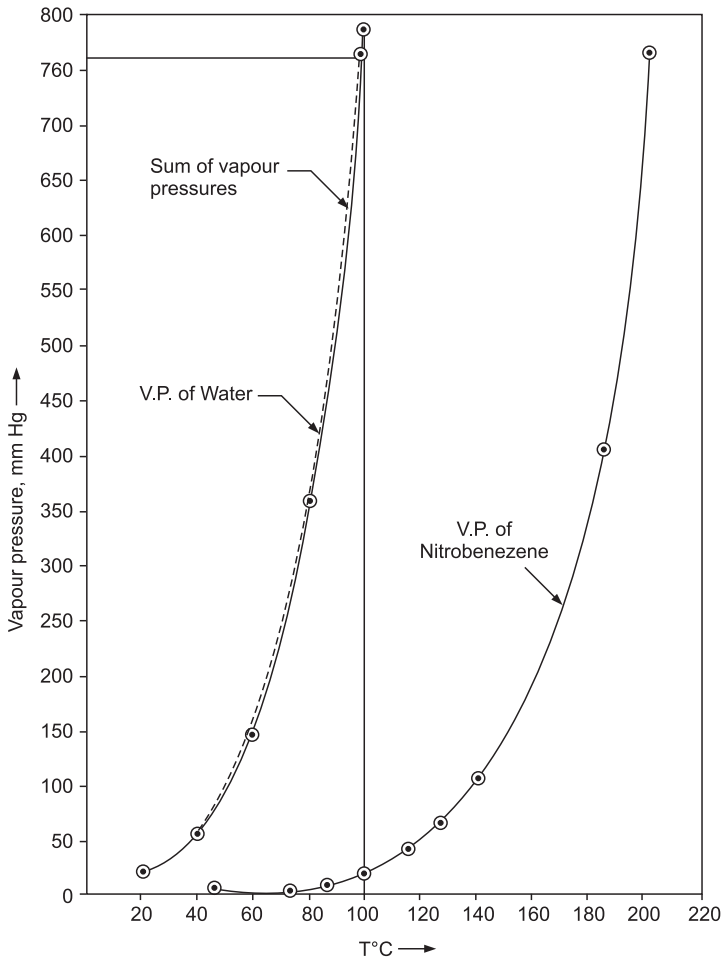


Fig. 9.39 Example 10 Determination of boiling point for steam distillation.

11. A methanol–water solution containing 36 mole % methanol at 26.7°C is continuously distilled to yield a distillate containing 91.5 mole % methanol and a residue containing 99 mole % water. The feed enters at its bubble point. Distillate is totally condensed and refluxed at its bubble point. (i) Find the minimum reflux ratio. (ii) For a reflux ratio of 3, estimate the number of plates by Ponchon–Savarit method**.**
Enthalpy data:

x or y mole fraction of methanol	*Enthalpies of saturated liquid* kJ/kmol	*Enthalpies of saturated vapour* kJ/kmol
0	8000	48000
1	7500	39000

Equilibrium data:

x, %	4	10	20	30	50	70	90	95
y, %	23	42	58	66	78	87	96	98.15

x, *y* are mole fractions of methanol in liquid and vapour phase respectively.

Solution.

(i) $x_F = 0.36$, $x_w = (1 - 0.99) = 0.01$, $x_D = 0.915$

Both feed and Reflux are at bubble point.

Plot *H-x-y* diagram and *xy* diagram as shown in Fig. 9.40.

By intrapolation, $H_{G1} = 39765$ kJ/kmol

Locate *F* corresponding to $x_F = 0.36$ on the bubble point curve.

Through *F* draw a tie line and extend it to intersect the vertical line drawn at $x_D = 0.915$

$Q'_{\min}$ (from graph) = 62500 kJ/kmol

$$R_{\min} = \frac{(Q'_{\min} - H_{G_1})}{(H_{G_1} - H_{L_0})} = \frac{(62500 - 39765)}{(39765 - 7542.5)} = 0.7056$$

Minimum reflux ratio = 0.7056

(ii) For $R = 3$

$$R = \frac{(Q' - H_{G_1})}{(H_{G_1} - H_{L_0})} = \frac{(Q' - 39765)}{(39765 - 7542.5)} = 3$$

$$Q' = 136432.5 \text{ kJ/kmol}$$

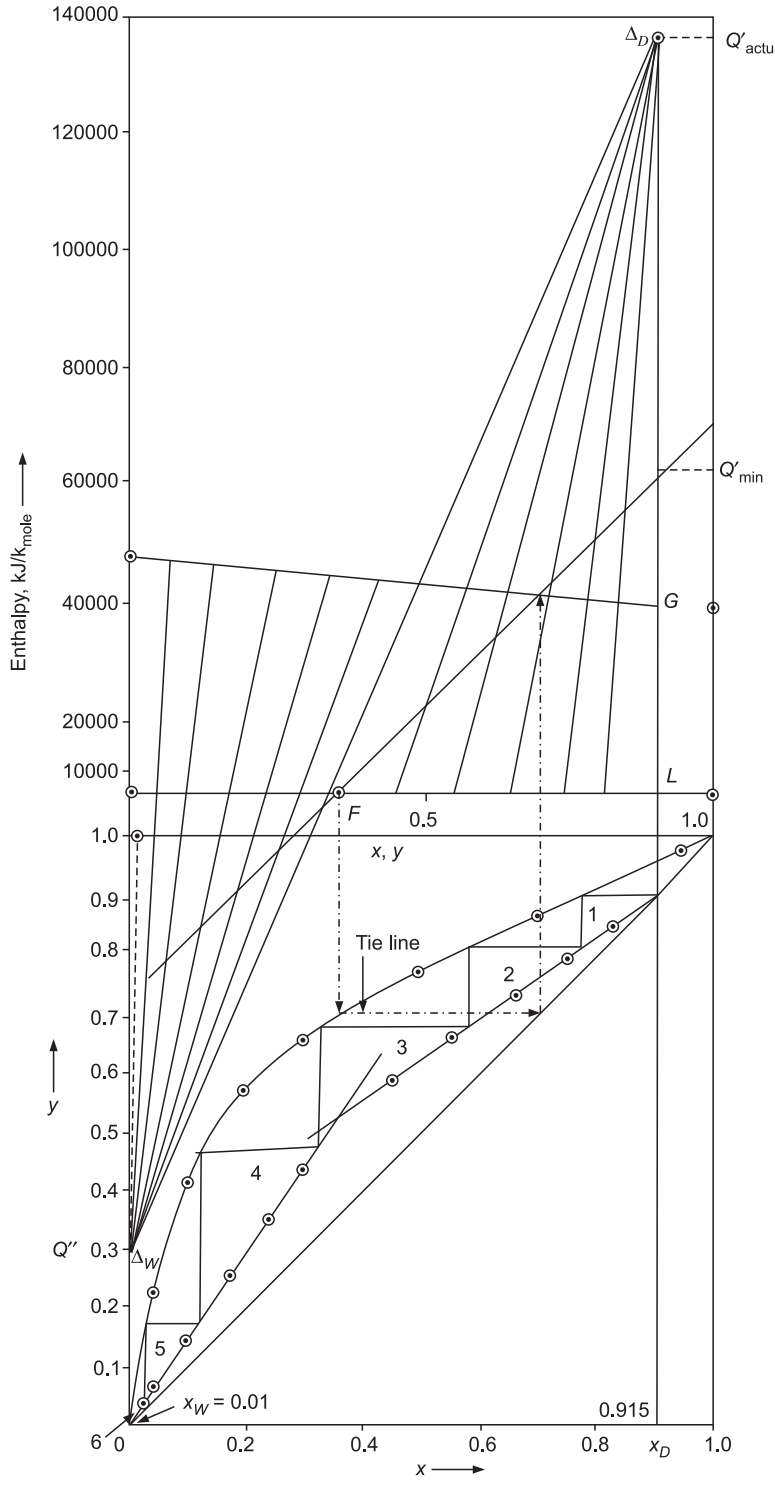


Fig. 9.40 Example 11 Ponchon–Savarit method.

We know that Δ_D, Δ_W and $F(Z_F, H_F)$ lie on a straight line

$$\frac{(Z_F - Z_w)}{(x_D - x_F)} = \frac{(H_F - Q'')}{(Q' - H_F)}$$

$$Q'' = -73004.5 \text{ kJ/kmol}$$

Locate $\Delta_D(Q', x_D)$ and $\Delta_w(Q'', x_w)$ on H_{xy} diagram. Randomly draw construction lines starting from Δ_D and Δ_w and obtain the operating curves for both sections on *xy*-diagram. Stepwise construction between equilibrium curve and operating curve will give the number of stages.

Number of stages (including reboiler) = 6

Number of plates in tower = 6 – 1 = 5

12. A continuous distillation column is used to separate a feed mixture at its boiling point, containing 24 mole % acetone and 76 mole % methanol into a distillate product containing 77 mole % acetone and a residue product containing 5 mole % acetone. A reflux ratio of twice the minimum is to be used. The overall plate efficiency is 60%. Determine the number of plates required for the separation.

Equilibrium data:

x	0.0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0.0	0.102	0.186	0.322	0.428	0.513	0.586	0.656	0.725	0.82	0.9	1.0

Solution.

x, y Mole fraction of acetone in liquid and vapour phase respectively.

$$x_F = 0.24, \qquad x_D = 0.77, \qquad x_w = 0.05$$

$R_{\text{actual}} = 2R_{\min}$

$\eta_{\text{overall}} = 60\%$

Plot *xy* diagram and draw the feed line with its corresponding slope of infinity to equilibrium curve and let it be *F*. Let the point *D* on the diagonal corresponds to x_D. Join *DF* and extend it to *y*-axis. The point of intersection is

$$\frac{x_D}{R_{\min} + 1} = 0.19 \text{ (from Fig. 9.41)}$$

$$\therefore \qquad R_{\min} = 3.053$$

$$\therefore \qquad R_{\text{actual}} = 6.106$$

$$\frac{x_D}{R_{\text{actual}} + 1} = \frac{0.77}{7.106} = 0.108$$

Locate 0.108 on *y*-axis and let it be *A*. Join *AD*. The point of intersection of *AD* with feed line is *Q*. *DQ* is the operating line for enriching section. Locate *W* on diagonal corresponding to $x_W = 0.05$. Join *W* and *Q*. *WQ* is the operating line for stripping section. By stepwise construction the number of stages = 14 (including reboiler)

$\therefore$ Number of plates is 13 (Theoretical)

Actual number of plates = $\frac{13}{0.6}$ = 21.7, i.e. 22 plates

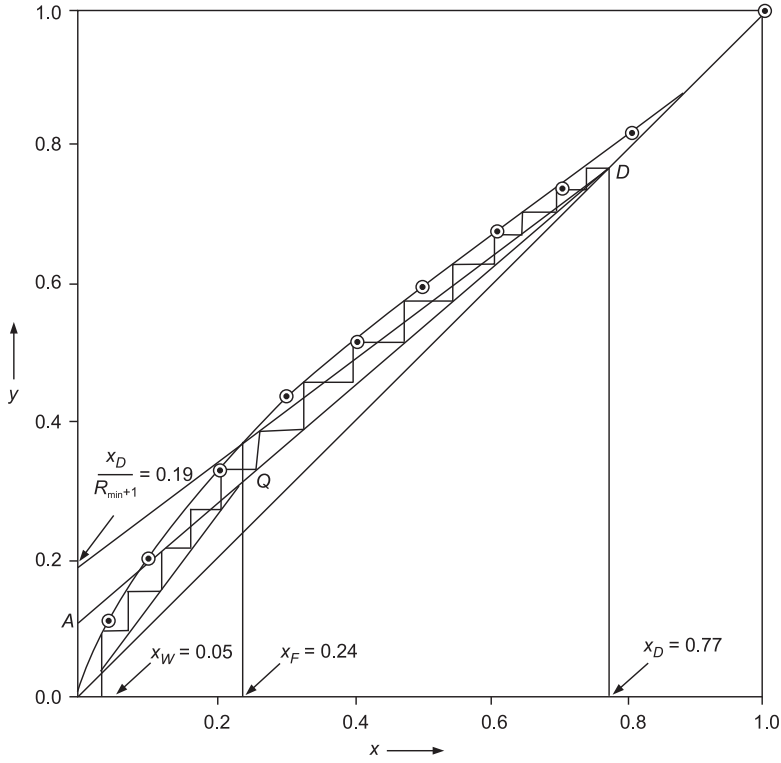


Fig. 9.41 Example 12 McCabe–Thiele method.

13. A fractionating column separates a liquid mixture entering at 5000 kmol/h containing 50 mole % A and 50 mole % B into an overhead product of 95 mole % A and a bottom product of 96 mole % B. A reflux ratio of twice the minimum will be used and the feed enters at its boiling point. Determine the number of theoretical stages required and the location of feed point.

Equilibrium data:

x	0.03	0.06	0.11	0.14	0.26	0.39	0.53	0.66	0.76	0.86	1.0
y	0.08	0.16	0.27	0.33	0.50	0.63	0.71	0.83	0.88	0.93	1.0

x, y mole fraction of A in liquid and vapour phase respectively.

Solution.

$$x_F = 0.5, \qquad x_D = 0.95, \qquad x_w = 0.04$$

Feed-saturated liquid

$$F = 5000 \text{ kmol/h}$$

Total condenser

(i) Total material balance

$$F = D + W$$

Component balance

$$Fx_F = Dx_D + Wx_W$$

$$5000 = D + W \qquad (1)$$

$$(5000 \times 0.5) = (D \times 0.95) + (W \times 0.04) \quad (2)$$

$$5000 = D + W$$

Distillate D = 2527.5 kmol/h

Residue W = 2472.5 kmol/h

(ii) $R_{min} = [(x_D - y')/(y' - x')]$

$$R_{min} = \left[\frac{(0.95 - 0.720)}{(0.720 - 0.5)}\right] = 1.045$$

Also, by graphical $x_D/(R_{min} + 1) = 0.46$ and $R_{min} = 1.065$

(iii) $R_{actual} = 2 \times R_{min}$

$R_{actual} = 2 \times R_{min} = 2 \times 1.045 = 2.09$ (Taking R_{min} value as 1.045)

$$\frac{x_D}{(R + 1)} = \frac{0.95}{(2.09 + 1)} = 0.307$$

With the above intercept, draw both enriching and stripping operating curves.
By McCabe–Thiele method,
Number of plates (including reboiler) = 11
Number of plates in tower = 11 – 1 = 10
The location of feed tray is 6th tray.

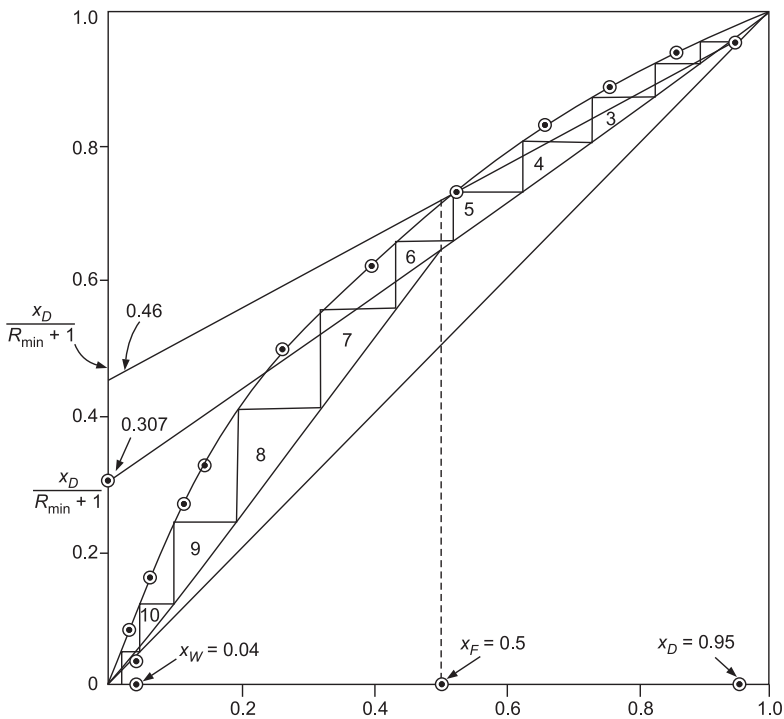


Fig. 9.42 Example 13 McCabe–Thiele method.

14. A mixture of benzene and toluene containing 38 mole % of benzene is to be separated to give a product of 90 mole % benzene at the top, and the bottom product with 4 mole % benzene. The feed enters the column at its

boiling point and vapour leaving the column is simply condensed and provide product and reflux. It is proposed to operate the unit with a reflux ratio of 3.0. Locate the feed plate and number of plates. The vapour pressures of pure benzene and toluene are 1460 and 584 mm Hg respectively. Total pressure is 750 mm Hg.

Solution.

$$\alpha = \frac{\text{vapour pressure of pure benzene}}{\text{vapour pressure of pure toluene}} = \frac{1460}{584} = 2.5$$

$$y = \frac{\alpha x}{[1 + (\alpha - 1)x]}$$

Compute equilibrium data.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.22	0.38	0.52	0.63	0.71	0.79	0.85	0.91	0.96	1.0

Draw the equilibrium curve, diagonal and locate feed, distillate and residue points as shown in Fig. 9.43.

Locate the intercept $\frac{x_D}{[R+1]} = \frac{0.9}{3+1} = 2.25$ and by stepwise construction we can get the number of stages.

No. of stages = 8 (including reboiler) and feed plate is 4

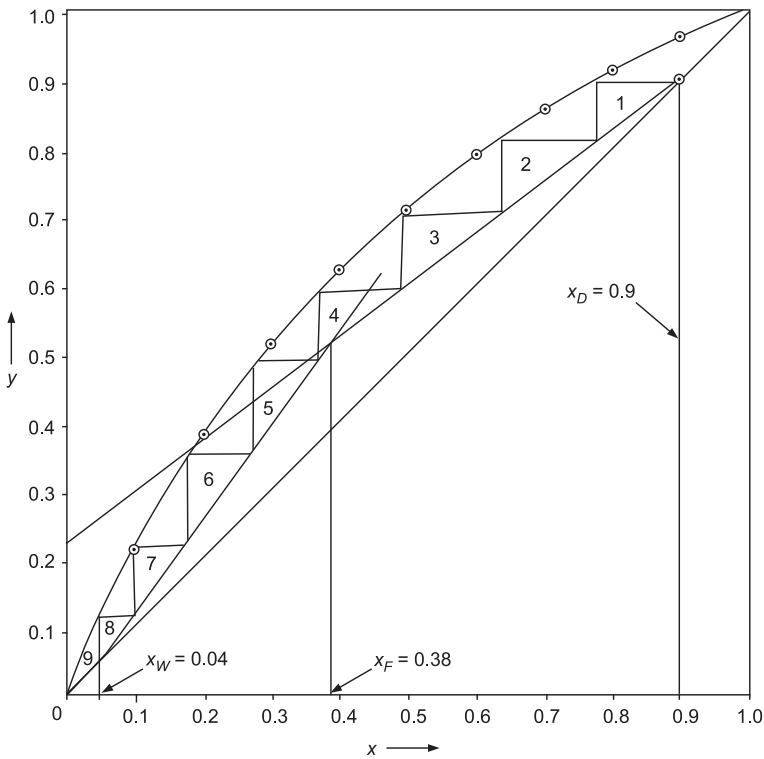


Fig. 9.43 Example 14 McCabe–Thiele method.

15. It is desired to separate a mixture of 50% vapour and 50% saturated liquid in a plate type distillation column. The feed contains 45 mole % A and the top product is to contain 96 mole % A. The bottom product is to contain 5 mole % A. Determine the minimum reflux ratio and the number of theoretical plates needed if a reflux ratio of twice the minimum is used. Eq. data:

x	0	0.1	0.16	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.215	0.30	0.52	0.625	0.725	0.78	0.89	0.89	0.95	1.0

x, y : mole fraction of A in liquid and vapour phase respectively.

Solution.

$$\left[\frac{x_D}{R_{\min}+1}\right] = 0.33$$

$$R_{\min}+1 = \frac{0.96}{0.33}$$

$$R_{\min} = 1.909$$

q = 0.5 (Fraction of liquid)

Slope of q-line $\left[\frac{q}{q-1}\right] = \frac{0.5}{0.5-1} = -1.0$

$$R_{\text{actual}} = 2.0 \times R_{\min} = 2.0 \times 1.909 = 3.818$$

$$\left[\frac{y_D}{R_{\text{actual}}+1}\right] = \frac{0.96}{3.818+1} = 0.199$$

Number of stages = 10

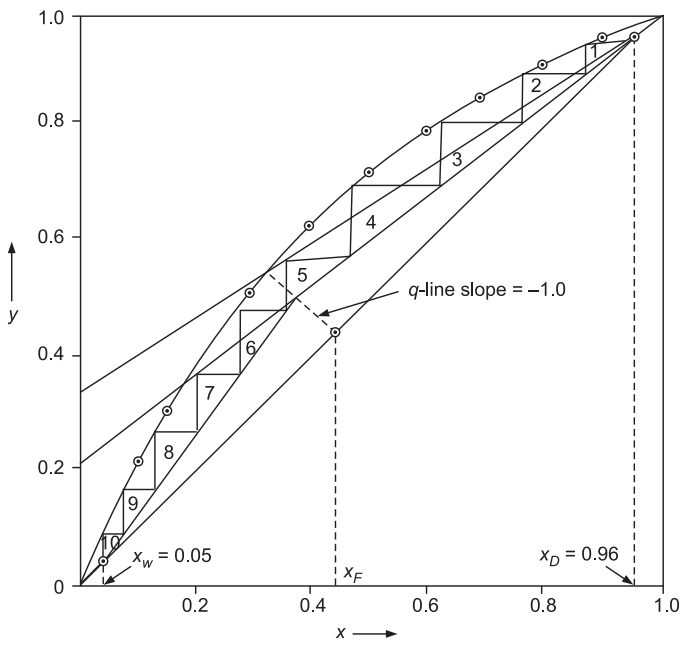


Fig. 9.44 Example 15 McCabe–Thiele method.

16. A fractionating column separates a liquid mixture containing 50 weight % chloroform and 50 weight % carbon disulphide into an overhead product of 94 weight % CS_2 and a bottom product of 95 weight % chloroform. A reflux ratio of twice the minimum will be used and the feed enters at its boiling point. Determine the number of theoretical stages required.
Equilibrium data:

x	0.03	0.06	0.11	0.14	0.26	0.39	0.53	0.66	0.76	0.86	1.0
y	0.08	0.16	0.27	0.33	0.50	0.63	0.71	0.83	0.88	0.93	1.0

x, y mole fraction of carbon disulphide in liquid and vapour phase respectively.

Solution.

Molecular weight of carbon disulphide = 76

Molecular weight of chloroform = 119.5

50 weight % of carbon disulphide,

$$x_F = \frac{\left(\frac{50}{76}\right)}{\left[\frac{50}{76} + \frac{50}{119.5}\right]} = 0.611 \text{ (in mole fraction)}$$

Similarly, the distillate and residue compositions in terms of mole fraction of carbon disulphide are $y_D = 0.961$ and $x_w = 0.076$ respectively.

From graph (Fig. 9.45), $\left(\frac{y_D}{R_{\min} + 1}\right) = 0.49$

$$R_{\min} = 0.96$$

$$R_{\text{act}} = 2 \times R_{\min} = 1.92$$

Therefore, $\left(\frac{y_D}{R_{\text{actual}} + 1}\right) = 0.329 \simeq 0.33$

Number of theoretical stages (from Fig. 9.45) including reboiler = 9

17. A laboratory rectification column is operated at atmospheric pressure and at total reflux, for benzene–chlorobenzene mixture. Samples of liquid from the condenser and reboiler analyze 95 mole percent benzene and 98 mole percent chlorobenzene respectively. Assuming a perfect reboiler, a total condenser, constant molal overflow and no heat loss from the tower, calculate the actual number of plates in the column. The average plate efficiency is 70%. The relative volatility of benzene to chlorobenzene is 4.13.

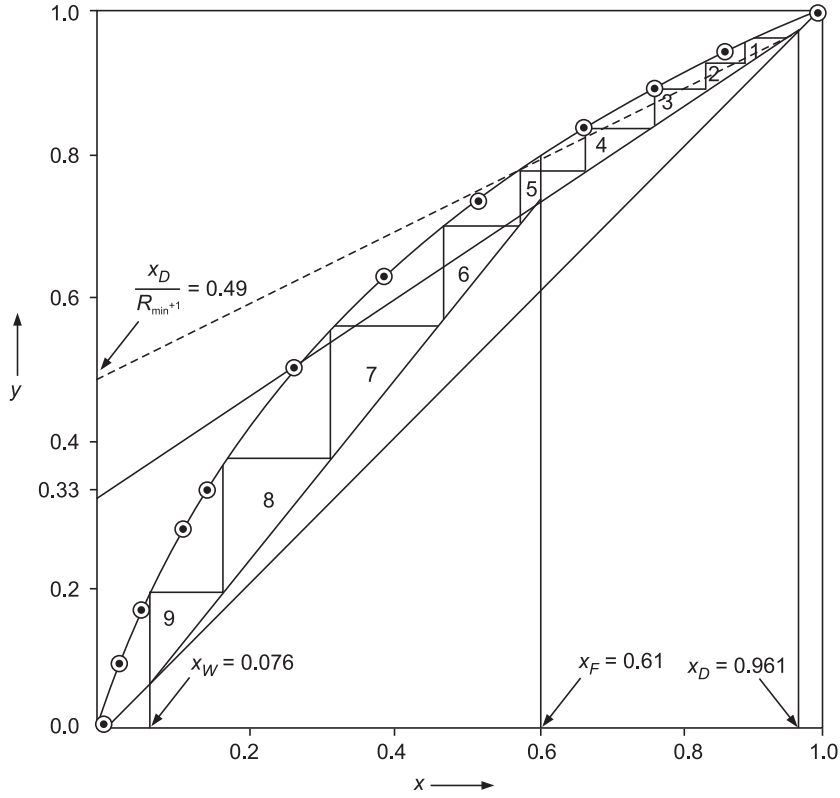


Fig. 9.45 Example 16 McCabe–Thiele method.

Solution.

$$y = \frac{\alpha x}{[1 + (\alpha - 1)x]}$$

Compute equilibrium data.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.31	0.51	0.64	0.73	0.81	0.86	0.91	0.94	0.97	1.0

Draw the equilibrium curve, diagonal and locate feed, distillate and residue points.
By stepwise construction, the number of stages determined is 5.
Hence, the theoretical plates required is 4.

Actual plates required will be $\frac{4}{0.7} = 5.71 \approx 6$.

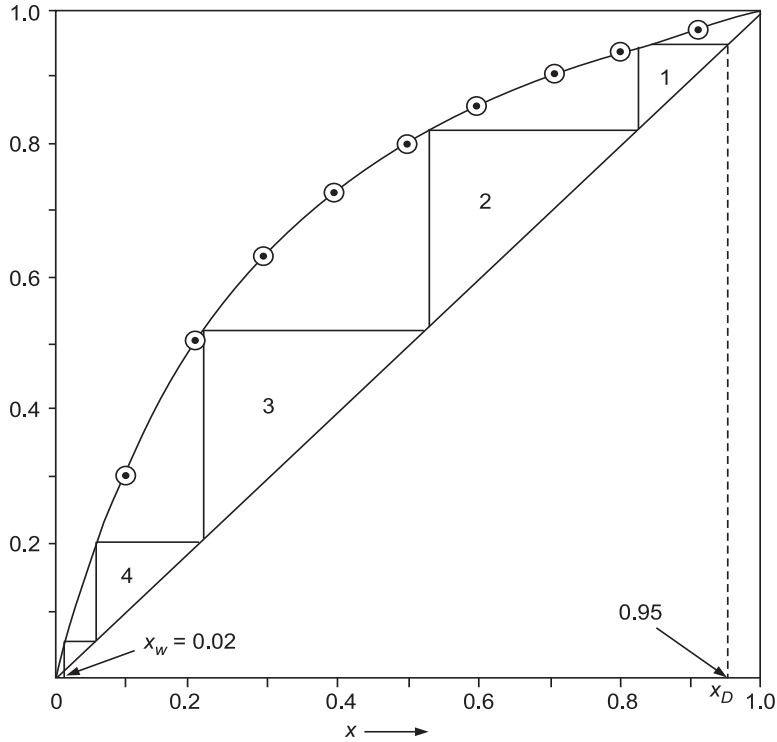


Fig. 9.46 Example 17 McCabe–Thiele method.

Alternatively, we can use the Fenske equation and determine the number of stages.

$$\frac{y_D}{(1-y_D)} = (\alpha)^{N_{P+1}} \frac{x_W}{(1-x_W)}$$

$$\frac{0.95}{(1-0.95)} = (4.13)^{N_{P+1}} \frac{0.02}{(1-0.02)}$$

$$19 = (4.13)^{N_{P+1}} \times 0.02041$$

Hence, $N_P + 1 = 4.82$ stages ≈ 5 stages
Therefore, the theoretical number of plates = 4

Actual plates required will be $\frac{4}{0.7} = 5.71 \approx 6$

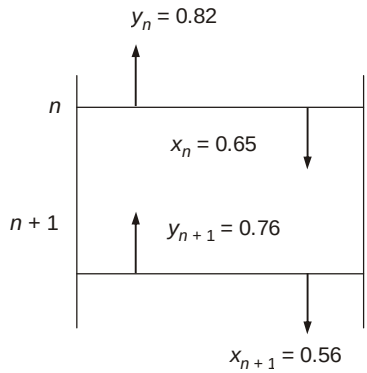
(Same as obtained from the graphical procedure)

18. A continuous rectification column is used to separate a binary mixture of A and B. Distillate is produced at a rate of 100 kmol/hr and contains 98 mole % A. The mole fractions of A in the liquid (x) and in the vapour (y) respectively from the two adjacent ideal plates in the enriching section are as follows:

x	y
0.65	0.82
0.56	0.76

The latent heat of vaporization is the same for all compositions. Feed is a saturated liquid. Calculate the reflux ratio and the vapour rate in the stripping section.

Solution.



$$\therefore \quad y_{n+1} = \left[\frac{R}{R+1}\right] x_n + \left[\frac{1}{R+1}\right] y_D$$

$$0.76 = \left[\frac{R}{R+1}\right] 0.65 + \left[\frac{1}{R+1}\right] 0.98$$

Solving, we get

$$0.76R + 0.76 = 0.65R + 0.98$$

Reflux ratio, $R = 2$

In the stripping section,

$$\bar{L} = \bar{G} + W$$

$$\therefore \quad q = \frac{(\bar{L} - L)}{F}$$

$$\frac{(\bar{G} - G)}{F} = (q - 1)$$

For a saturated feed $q = 1.0$

$$\therefore \quad q = 1.0 = \frac{(\bar{G} - G)}{F}$$

i.e. $\bar{G} = G = L + D$

$= D(R + 1) = 100(2 + 1) = 300$ kmol/h

19. A continuous rectifying column treats a mixture containing 40% benzene and 60% toluene and separates into a distillate product containing 98% benzene and a bottom product containing 98% toluene. The feed enters as a liquid at its boiling point. If a reflux ratio of 3.5 is used, estimate height of the tower. The average height of a transfer unit is 0.7 m. The overall resistance to mass transfers lies in vapour phase.

Equilibrium data:

x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
y	0.22	0.38	0.51	0.63	0.70	0.78	0.85	0.91	0.96

x, y: mole fraction of benzene in liquid and vapour phase respectively.

Solution.

y	0.98	0.92	0.81	0.74	0.655	0.57	0.44	0.318	0.2	0.1	0.02
y'	0.995	0.96	0.89	0.83	0.75	0.655	0.543	0.43	0.30	0.183	0.05
$\frac{1}{(y' - y)}$	66.67	25	12.5	11.11	10.53	11.76	9.71	8.93	10	12.05	33.33

y is from operating line and y' is obtained from equilibrium curve for a specific x value.

$x_F = 0.4$, $x_w = (1 - 0.98) = 0.02$, $R = 3.5$, $HTU = 0.7$

$[x_D/(R + 1)] = [0.98/ (3.5 + 1)] = 0.218$

Overall mass transfer lies in vapour phase. So the slope $-[(1/kx)/ (1/ky)]$ becomes vertical, y and y' values are obtained at the intersection of operating and equilibrium curves.

$$\int \left[\frac{dy}{(y' - y)}\right] = 13.175$$

$$Z = \text{HTU} \times \text{NTU} = 0.7 \times 13.175 = 9.22 \text{ m}$$

20. Feed rate to a distillation column is 400 kmol/hr. The overhead product rate is 160 kmol/h. The mole fraction of more volatile component in distillate is 94%. The residue contains 5% of more volatile component. The reflux ratio is 4. The mole fraction of vapour leaving a plate is 0.4, whereas the mole fraction of liquid coming to the same plate is 0.3. Assuming constant molal overflow, determine the condition of feed.

Solution.

Feed rate: 400 kmol/h

Distillate, D: 160 kmol/h

Therefore, flow rate of residue, W: 240 kmol/h

The composition of distillate $x_D = 0.94$

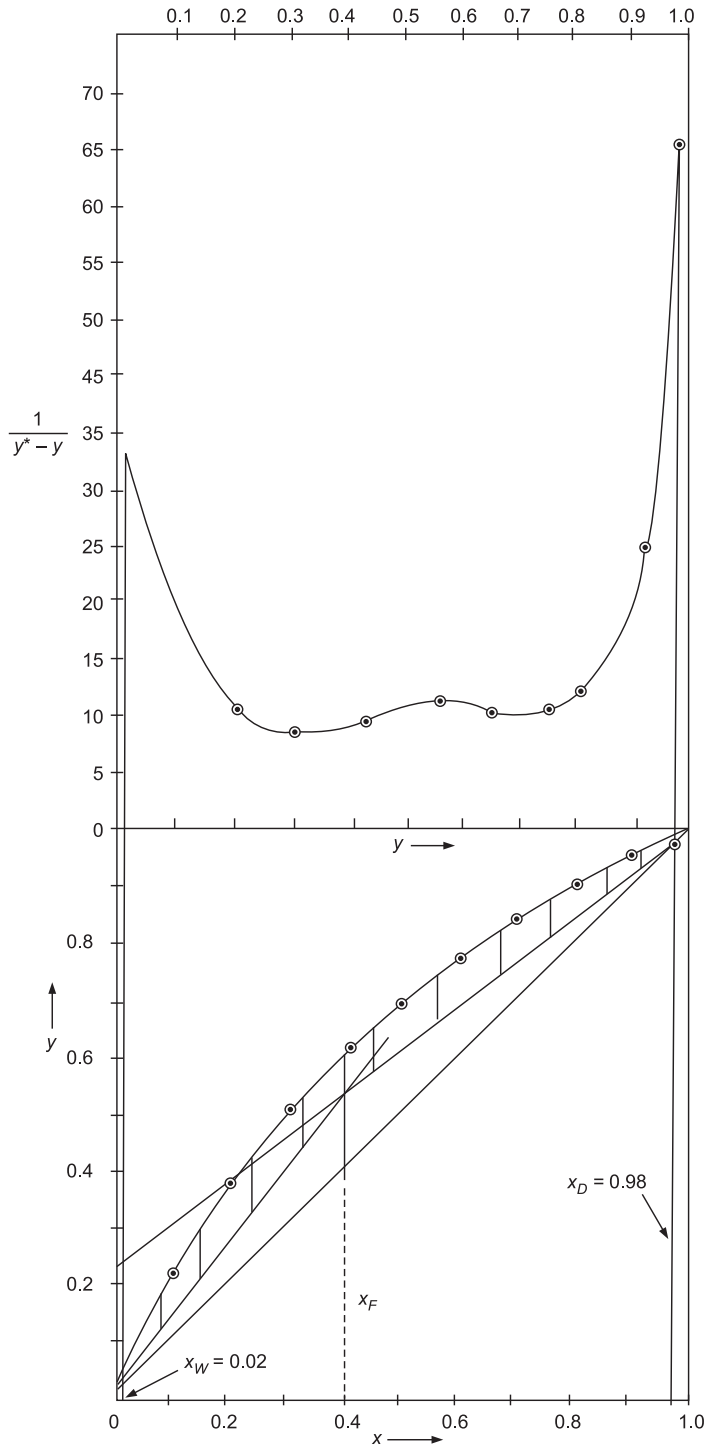


Fig. 9.47 Example 19 packed distillation.

The composition of residue $x_w = 0.05$
Reflux ratio: 4

$$y_{m+1} = 0.94$$

$$x_m = 0.05$$

We know that

$$\bar{L}_m = \bar{G}_{m+1} + W \qquad \text{[Eq. 9.67]}$$

Since, the molal overflow rate is constant, $\bar{L}_m = \bar{L}_{m+1} = \bar{L}$

$$y_{m+1} = \left(\frac{\bar{L}}{\bar{L} - W}\right) x_m - \left(\frac{W}{\bar{L} - W}\right) x_W \qquad \text{[Eq. 9.94]}$$

Substituting, we get,

$$0.4 = \left(\frac{\bar{L}}{\bar{L} - 240}\right) 0.3 - \left(\frac{240}{\bar{L} - 240}\right) 0.05$$

Solving, we get

$$\bar{L} = 880 \text{ kmol/h}$$

From Eq. (9.67) we get

$\bar{G}_{m+1} = L_m - W = 880 - 240 = 640$ kmol/h $= G_m = G$ (Due to constant molal flow rate).
Feed rate = 400 kmol/h

Reflux ratio $= \dfrac{L}{D} = 4$

Hence, $L = 4D = 640$ kmol/h.
$\bar{L} = 880$ kmol/h
We also know that

$$q = \frac{(\bar{L} - L)}{F} = \frac{(H_G - H_F)}{(H_G - H_L)} \qquad \text{(Eq. 9.110)}$$

$$\therefore \qquad \bar{L} = L + qF$$

Substituting, we get

$$q = \frac{880 - 640}{400} = 0.6 \text{ (Fraction of liquid)}$$

Hence, the feed is a mixture of 60% liquid and 40% vapour.

21. The feed rate to a binary distillation column is 200 kmol/hr and 75% of it is vaporized. Distillate flow rate is 120 kmol/h with 95% composition of more volatile component. Reboiler steam demand is 4000 kg/h. Latent heat of steam used in reboiler is 2304 kJ/kg. Latent heat of liquid to be distilled is 32000 kJ/kmol. Determine the reflux ratio.

Solution.

We know that $\bar{L} = L + qF$ (from Eq. 9.110)

$G = (R + 1)D$ (from Eq. 9.84)

From Eq. (9.111), we get

$$\frac{(\bar{G} - G)}{F} = \frac{(\bar{L} - L)}{F} - 1 = (q - 1)$$

$\therefore \quad (\bar{G} - G) = F(q - 1)$

$\therefore \quad \bar{G} = G + F(q - 1)$

i.e. $\bar{G} = (R + 1)D + F(q - 1)$

Fraction of vapour = $(1 - q) = 0.75$

Fraction of liquid = $q = 0.25$

$\therefore \quad \bar{G} = (R + 1)120 + 200(0.25 - 1)$

$= (R + 1)120 - 150 = 120R - 30$

Steam needed for the reboiler $m_s = \dfrac{\bar{G}\lambda_{\text{feed}}}{\lambda_{\text{steam}}}$

$m_s \lambda_{\text{steam}} = \bar{G}\lambda_{\text{feed}} = (120R - 30)\lambda_{\text{feed}}$

$m_s \lambda_{\text{steam}} = 4000 \times 2304 = 9.216 \times 10^6$ kJ/hr

$= (120R - 30)\lambda_{\text{feed}}$

i.e. $9.216 \times 10^6 = (120R - 30)32000$

Solving, we get

$R = 2.65.$

EXERCISES

1. Compute the VLE data from the following vapour pressure data at 760 mm Hg. Pressure assuming ideal solution.

Temperature,°C	98.4	105	110	120	125.6
Vapour pressure of *A*, mm Hg	760	940	1050	1350	1540
Vapour pressure of *B*, mm Hg	333	417	484	650	760

Ans:

x_A	1.0	0.655	0.487	0.157	0.0
y_A	1.0	0.810	0.674	0.279	0.0

2. A mixture containing benzene and toluene with 50 mole % benzene is flash distilled such that 70% of the feed is distilled out. Estimate the composition of the distillate and residue. If the same quantity of distillate is obtained by simple distillation, estimate the composition of the residue and distillate.

Equilibrium data:

x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
y	0.22	0.38	0.51	0.63	0.70	0.78	0.85	0.91	0.96

x, y: mole fraction of benzene in liquid and vapour phase respectively.

(**Ans:** (i) $-W/D = -0.3/0.7 = -0.429$

$y_D = 0.56$ and $x_w = 0.35$ (From graph)

(ii) $x_w = 0.245$ and $y_D = 0.61$)

3. A simple batch still is used to distill 1000 kg of a mixture containing 60 mass % ethyl alcohol and 40 mass % water after distillation, the bottom product contains 5 mass % alcohol. Determine the composition of the overhead product, its mass and mass of the bottom product.

The equilibrium data:

x	5	10	20	30	40	50	60
y	36	51.6	65.5	71	74	76.7	78.9

where x and y are weight percent of ethyl alcohol in liquid phase and vapour phase respectively.

(**Ans:** Residue = 192.28 kg; Distillate 807.72 kg;
$y_D = 0.731$ mass % alcohol)

4. A liquid mixture containing 50 mole % acetone and rest water is differentially distilled at 1 atm. pressure to vaporize 25% of the feed. Compute the composition of the composited distillate and residue. VLE data at 1 atm. pressure is given below.

x, mole fraction of acetone in liquid	0.1	0.2	0.3	0.4	0.6	0.7	0.9
y, mole fraction of acetone in vapour	0.76	0.82	0.83	0.84	0.86	0.87	0.94

5. A solution of 40 mole % of acetic acid in water is flash distilled at atmospheric pressure, until 60 mole % of the feed was distilled. Compute the compositions of the distillate and residue.

Equilibrium data:
Mole fraction of acetic acid in

Liquid, x	0.07	0.15	0.27	0.37	0.50	0.62	0.72	0.82	0.90	1.0
Vapour, y	0.05	0.11	0.20	0.28	0.38	0.49	0.60	0.73	0.80	1.0

(**Ans:** $x_w = 0.53$ and $y_D = 0.65$ in terms of water)

6. Feed mixture containing equimolar quantities of 'A' and 'B' is differentially distilled such that 60 mole % of feed is distilled out. Estimate the composition of distillate and residue.

x	0	0.157	0.312	0.487	0.655	1.0
y	0	0.279	0.492	0.674	0.810	1.0

x, y are mole fractions of A in liquid and vapour phase respectively.

(**Ans:** $x_w = 0.335$ and $y_D = 0.61$)

7. A equimolar feed mixture containing Benzene and Toluene is distilled such that 60% of feed is distilled out. Estimate the composition of distillate and residue by taking the relative volatility as 2.5 for (i) Simple distillation, (ii) Equilibrium distillation.

(**Ans:** (i) $x_W = 0.29$, $y_D = 0.64$ and (ii) $x_W = 0.365$, $y_D = 0.59$)

8. It is desired to separate a feed mixture of 'A' and 'B' containing 50 mole % A to a product such that 60% feed is distilled out. Estimate the composition of residue and distillate if (i) simple distillation is carried out and (ii) equilibrium distillation is carried out. VLE data:

x	0	5	10	15	20	30	40	50	60	70	80	90	100
y	0	11	21	30	38	51	63	72	78	85	91	96	100

x, y are mole % of A in liquid and vapour phase respectively.

(**Ans:** (i) $x_W = 0.3$, $y_D = 0.63$, (ii) $x_W = 0.36$, $y_D = 0.59$)

9. It is desired to separate a feed mixture of 100 kmol containing 60% heptane and 40% ethyl benzene such that 60 kmol of the feed is distilled out. Determine the composition of residue and distillate if the distillation is (i) Flash distillation and (ii) Differential distillation.

x	0	0.08	0.185	0.251	0.335	0.489	0.651	0.79	0.914	1.0
y	0	0.233	0.428	0.514	0.608	0.729	0.814	0.91	0.963	1.0

x, y is the mole fraction of heptane in liquid and vapour phase respectively.

(**Ans:** (i) $x_W = 45\%$, $y_X = 70\%$; (ii) $x_W = 37.5$, $y_D = 75\%$)

10. A liquid mixture containing 50 mole % *n*-heptane and 50 mole % *n*-octane is differentially distilled until the residue contains 33% *n*-heptane. Calculate

the % vaporization and the composition of the composited distillate. If the residue with the same composition is achieved in an equilibrium still, estimate the composition of the distillate and total moles distilled assuming $\alpha = 2.17$.

(**Ans:** y_D *n*-heptane = 0.617, total moles distilled = 60 ml/h; vaporization: 60%; (ii) y_D = 0.52% vaporization: 89.47%)

11. A mixture of 30 mole % Naphthalene and 70 mole % Dipropylene glycol is differentially distilled at 100 mm Hg until a final distillate containing 55 mole % Naphthalene is obtained. Determine the amount of residue and the composition of residue.

VLE data:

x	5.4	11.1	28.0	50.6	68.7	80.6	84.8	88
y	22.3	41.1	62.9	74.8	80.2	84.4	86.4	88

12. A mixture containing 30 mole % Hexane, 45 mole % Heptane and 25 mole % Octane is subjected to flash distillation. If 60 mole % of the feed is vaporized and condensed, calculate the composition of vapour leaving the separator.

(*m* values for Hexane, Heptane and Octane: 2.18, 0.99 and 0.46 respectively)

13. A binary mixture containing 55 mole % *n*-heptane and 45 mole % *n*-octane at 27°C is subjected to differential distillation at atmospheric pressure with 60 mole % of the feed liquid is distilled. Assuming a relative volatility of *n*-heptane with respect to *n*-octane is 2.17, determine the composition of the charge in still and that of distillate.

(**Ans:** composition of *n*-heptane $x_W = 0.38$, $y_D = 0.665$)

14. Continuous fractionating column operating at 1 atm is designed to separate 13600 kg/hr of a solution of benzene and toluene. Feed is 0.4-mole fraction benzene. Distillate contains 0.97 mole fraction benzene and residue contains 0.98 mole fraction toluene. A reflux ratio of twice the minimum is used. Feed is liquid at its saturation temperature and reflux is returned at saturation.

Determine:

(i) Quantities of products in kg/hr

(ii) Minimum reflux ratio

(iii) Number of theoretical plates

The average relative volatility for the given system is 2.56.

(**Ans:** (i) D = 5524.2 kg/h w = 8075.8 kg/h; (ii) R_m = 1.487; (iii) 13)

15. A solution of carbon tetra chloride and carbon disulfide containing 50 mole % of each is to be fractionated to get a top and a bottom product of 95% and 6% carbon disulfide respectively. The feed is a saturated liquid at its boiling point and is fed at the rate of 5000 kg/hr. A total condenser is

used and reflux returned to the top plate as a saturated liquid. The equilibrium data at 1 atm pressure is given below:

x	0	0.06	0.11	0.26	0.39	0.53	0.66	0.76	0.86	1.0
y	0	0.16	0.27	0.50	0.63	0.75	0.83	0.88	0.93	1.0

where x, y are mole fractions of carbon disulfide in liquid and vapour phase respectively.

(i) Determine the product rate in kg/hr.
(ii) What is the minimum reflux ratio?
(iii) Determine the theoretical number of plates required and the feed plate location if the tower is operated at twice the minimum reflux ratio.

16. A mixture of 35 mole % A and 65 mole % B is to be separated in the fractionating column. The concentration of A in the distillate is 93 mole % and 96% A in the feed is recovered in the distillate. The feed is half vapour and reflux ratio is to be 4.0. The relative volatility of A to B is 2.0. Calculate the number of theoretical plates in the column and locate the feed plate.

17. A continuous fractionating column, operating at atmospheric pressure, is to be designed to separate a mixture containing 30% CS_2 and 70% CCl_4 into an overhead product of 96% CS_2 and a bottom product of 96% CCl_4 (all mole percent). A reflux ratio of twice the minimum will be used and the overall efficiency of the column is estimated to be 65%. Feed enters at its boiling point. Determine the number of plates to be provided and the correct location of the feed plate.

Equilibrium data:

x	0.0296	0.0615	0.258	0.390	0.532	0.663	0.758	0.860
y	0.0823	0.1555	0.495	0.634	0.747	0.830	0.880	0.932

18. A continuous fractionating column, operating at atmospheric pressure, is to separate a mixture containing 30 mole % CS_2 and 70 mole % CCl_4 into an overhead product of 95 mole % CS_2 and a bottom product of 95 mole % CCl_4. The feed enters the column as liquid at its boiling point.
Assuming an overall plate efficiency of 70% and a reflux ratio of 3.16, determine the number of plates to be provided.
Mole fractions of CS_2 in liquid (x) in equilibrium with mole fraction CS_2 in vapour (y) are given below.

Equilibrium data:

x	2.96	11.06	25.8	53.18	66.3	75.75	86.04
y	8.23	26.6	49.5	74.7	83.0	88.0	93.2

19. A feed containing 50 mole % heptane and 50 mole % octane is fed into a pipe still through a pressure reducing value and then into a flash discharging chamber. The vapour and liquid leaving the chamber are assumed to be in

equilibrium. If the fraction of feed converted to vapour is 0.5, find the composition of the top and bottom plates. The following table gives VLE data:

x, mole fraction of heptane in vapour phase	1.0	0.69	0.4	0.192	0.045	0.0
y, mole fraction of heptane in vapour phase	1.0	0.932	0.78	0.538	0.178	0.0

(**Ans:** $x_w = 0.31$ $y_D = 0.69$)

20. A continuous distillation column is used to separate a feed mixture containing 24 mole % acetone and 76 mole % methanol into a distillate product containing 77 mole % acetone and a residue product containing 5 mole % acetone. The feed is A saturated liquid. A reflux ratio of twice the minimum is used. The overall stage efficiency is 60%. Determine the number of plates required for the separation.

Equilibrium data:

x	0.0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	1.0
y	0.0	0.102	0.186	0.322	0.428	0.513	0.586	0.656	0.725	0.80	1.0

(x, y mole fraction of acetone in liquid and vapour phase respectively)

(**Ans:** 24 stages)

21. The enthalpy-concentration data for a binary system is given below:

x, mole fraction of A	0.0	0.25	0.407	0.62	0.839	1.0
y, mole fraction of A	0.0	0.396	0.566	0.756	0.906	1.0
H_l, kcal/kmol	280	180	145	195	260	380
Hg, kcal/kmol	1000	1030	955	895	885	880

Rest of the data could be obtained by extrapolation. A feed mixture with an initial composition of 30 mole % A is to separate into an overhead product of 95 mole % A and a 4 mole % bottom product. Determine the ideal number of stages needed if the reflux ratio is twice the minimum reflux ratio. Feed enters as a saturated liquid.

$$x_F = 0.3, \qquad x_D = 0.95, \qquad x_w = 0.04, \qquad R = 2.4$$

22. A mixture containing 50 mole % A and 50 mole % B is distilled in a packed column to yield a top product containing 94 mole % A and a bottom product containing 95 mole % B. The feed enters a saturated vapour. Estimate the height of the packing needed if the height of a transfer unit is 0.5 m. A reflux ratio of 1.5 times the minimum is to be used. The relative volatility of A with respect to B is 2.5.

(**Ans:** NTU = 11.25, Ht = 5.625 m)

10

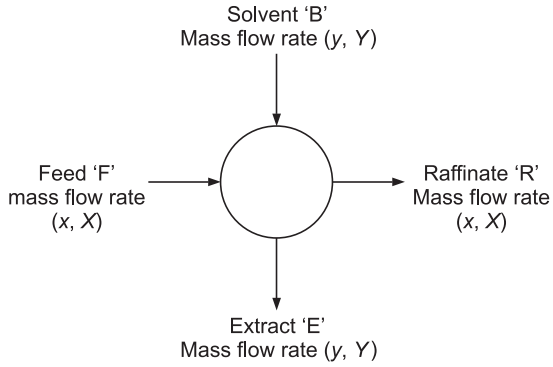
EXTRACTION

10.1 INTRODUCTION

Liquid extraction is the separation of the constituents of a liquid by contact with another insoluble liquid called *solvent*. The constituents get distributed between the two phases. The solvent rich phase is called *extract* and the residual liquid from which the solute has been removed is called *raffinate.* Some of the complicated systems may use two solvents to separate the components of a feed. A mixture of para or ortho–nitro benzoic acids can be separated by distributing them between the insoluble liquids chloroform and water. The chloroform dissolves the para isomer and water the ortho isomer. This is called *dual solvent* or *double solvent* or *fractional extraction*. Some of the components which are difficult to separate by other separation processes like distillation can effectively be separated by extraction or extraction followed by distillation, (e.g.) acetic acid – water separation. Similarly long chain fatty acids can be separated from vegetable oils economically by extraction rather than high vacuum distillation. The separation of fission products from nuclear energy process and separation of low cost metals can be effectively carried out by liquid extraction. Pharmaceutical products like penicillin are also separated by this technique. Mercaptans can be removed by using hydrocarbon oil as solvent. Phenol is extracted from coal tar using alkaline solution as solvent. Caprolactum is extracted with benzene as solvent.

10.2 EQUILIBRIA

In extraction operation generally ternary systems are involved. The solute distributes between solvent rich phase called *extract* and solvent lean phase called *raffinate*. The schematic diagram shown in Fig. 10.1 indicates the various streams involved in a typical liquid–liquid extraction operation. The equilibrium concentration of such systems can be represented in a triangular coordinate system.



x = mass fraction of solute in Feed and Raffinate stream
y = mass fraction of solute in Solvent and Extract stream
X = Mass of solute/mass of solute free components in Feed or Raffinate phase
Y = Mass of solute/mass of solute free components in Extract or Solvent phase

Fig. 10.1 Streams in extraction.

10.2.1 Equilateral–Triangular Coordinates

A mixture having a typical composition of 50% A, 30% B and 20% C is represented by point M as shown in Fig. 10.2. Now let us consider that P kg of a mixture at a point P is added to Q kg of mixture at Q, the resulting mixture is shown by point R on line PQ such that

$$\frac{P}{Q} = \frac{\text{Length of QR}}{\text{Length of PR}} = \frac{x_Q - x_R}{x_R - x_P} \tag{10.1}$$

The ternary systems usually follow any one of the two categories given below:

(i) one pair partially soluble and two pairs partially soluble
(ii) Insoluble systems.

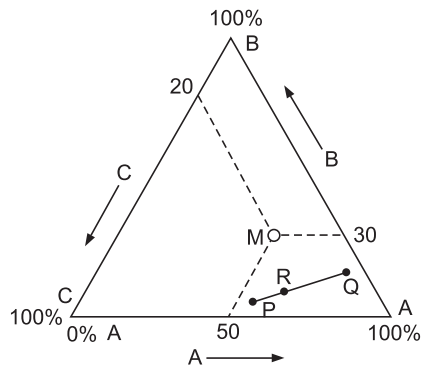


Fig. 10.2 Representation of ternary data in a triangular chart.

In all our subsequent discussions 'C' indicates the distributing solute, 'B' the solvent and 'A' the solute-free component in feed. Some of the common combinations of A, B and C are as follows:

A	*B*	*C*
Water	Chloroform	Acetone
Benzene	Water	Acetic acid

The equilibrium composition of mixtures can be represented in a triangular coordinate system. These diagrams drawn at constant temperatures are also called *isotherms*. A typical isotherm is shown in Fig. 10.3 in which 'C' is the solute which dissolves in A and B completely. A and B mutually dissolve to a limited extent. If the solubility of 'A' and 'B' is very minimal, then the points S and T will be very close to apexes A and B respectively. The curve SPQT is the *binodal solubility curve*. Any mixture outside the curve SPQT will be a homogeneous solution of the one liquid phase. Any point within the area bounded by the curve and the axis AB will form two insoluble saturated liquid phases, one rich in A phase and the other rich in B phase.

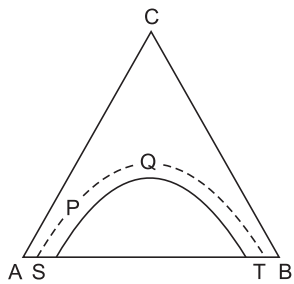


Fig. 10.3 Extraction isotherm.

10.3 SYSTEMS OF THREE LIQUIDS—ONE PAIR PARTIALLY SOLUBLE

Let us consider a ternary mixture whose effective composition is defined by point M as shown in Fig. 10.4. This mixture will form two insoluble but saturated phases. Many lines can be drawn through the point M. However, there can only be one *tie line* as indicated by the line RE passing through M. Tie line can be located by projecting the arbitrary lines passing through M to the distribution diagrams. Tie line is the one whose projections to the equilibrium distribution curve and $x = y$ (diagonal) line form a vertical line in the *xy* diagram as shown in Fig. 10.4. Whenever the distribution curve is above the diagonal line, as shown in Fig. 10.4, the extract stream will have a higher concentration of the solute than the raffinate stream. In such cases the tie line will have a positive slope as indicated by line RE. However, when the distribution curve is below the diagonal line, the raffinate will

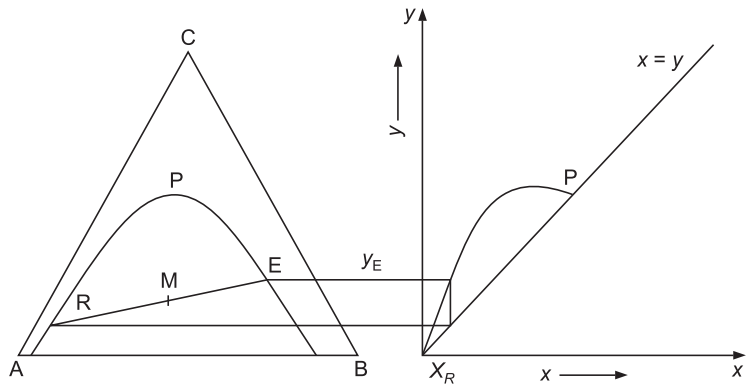


Fig. 10.4 Ternary system representation and Tie line.

have a higher concentration of solute compared to extract stream and the line RE instead of having a positive slope will have a negative slope. Occasionally, the tie lines change their slope from one direction to another and one such tie line will be horizontal. Such systems are called *solutropic systems*. When the tie line simply becomes a point 'P', it is called *Plait point* as shown in Fig. 10.4.

10.3.1 Effect of Temperature

The mutual solubility of A and B increases with increasing temperature and beyond some critical temperature, A and B are completely soluble. Thus, the heterogeneity decreases at higher temperatures. Also, the slope of tie lines and distribution curve vary with changes in temperature and it is shown in Fig. 10.5. Hence, it is preferable to operate below the critical temperature such that the heterogeneity is maintained.

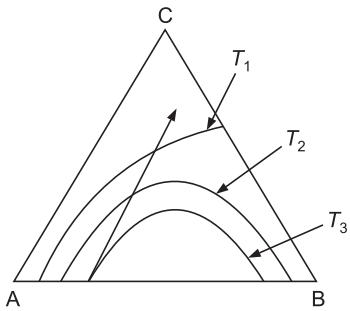


Fig. 10.5 Effect of temperature on Extraction Isotherm ($T_1 < T_2 < T_3$).

10.3.2 Effect of Pressure

Generally the effect of pressure is not very significant. It is preferable to operate above the vapour pressure of solutions.

10.4 SYSTEMS OF THREE LIQUIDS—TWO PAIRS PARTIALLY SOLUBLE

Let us assume that A and C are completely soluble, while the pairs A–B and B–C show limited solubility. A typical isotherm is shown below in Fig. 10.6. Points F and H indicate mutual solubilities of A and B and points G and J indicate those of B and C. Curves FKG is for A rich layer and HLJ is for B rich layer. The area bounded by FKGJLH indicates a heterogeneous mixture and outside this area the mixture is homogeneous. KL is a tie line which corresponds to the effective composition M. Increase in temperature usually increases the mutual solubilities and at the same time influences the slope of the tie lines.

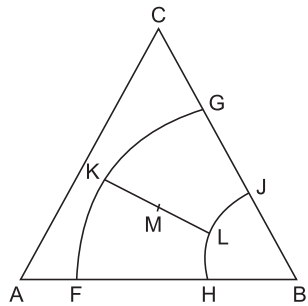


Fig. 10.6 Isotherm of system of three liquids—two pairs partially soluble.

10.5 TWO PARTIALLY SOLUBLE LIQUIDS AND ONE SOLID

When the solid does not form hydrates with the liquids, the characteristics of the isotherm will be as shown in Fig. 10.7. K and L indicate saturated solutions of C in A and B respectively. A and B are soluble only to the limited extent shown at H and J. Area bounded by HDGJ shows a heterogeneous mixture while the region KDHA and JGLB indicate homogeneous phase. RE indicates the tie line for a mixture whose effective composition is M. The region CDG consists of 3 phases,

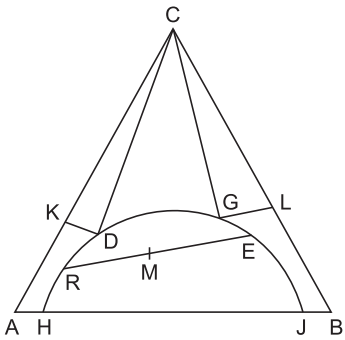


Fig. 10.7 Isotherm of system of two partially soluble liquids and one solid.

namely solid C and saturated liquid solutions at D and G. Liquid extraction is mainly confined to the heterogeneity area which is bounded by HDGJ. Temperature has a significant effect on the shape of the curve HDGJ.

10.6 OTHER COORDINATES

The equilibrium concentrations of ternary systems can also be expressed in rectangular coordinates. This is done by taking the concentration of B along *x*- axis and that of the concentrations of C in A rich phase, denoted conventionally as *x* and B rich phase, denoted conventionally as *y*, both on *y*-axis in rectangular coordinates. It will be more convenient to solve problems using graphical procedure with rectangular coordinate system. Rectangular coordinate system has been used in the worked examples presented in this chapter.

10.7 FACTORS INFLUENCING CHOICE OF SOLVENT

1. *Selectivity, β:* The effectiveness of solvent B for separating a solution of A and C into its components is measured by comparing the ratio of C to A in the B–rich phase to that in the A–rich phase at equilibrium and is called *selectivity* or *separation factor*. This is also analogous to relative volatility in distillation and it is defined as

$$\beta = \frac{[(\text{weight fraction of C})/(\text{weight fraction of A})]_{\text{Extract}}}{[(\text{weight fraction of C})/(\text{weight fraction of A})]_{\text{Raffinate}}}$$

$$= \left(\frac{y_E^*}{x_R}\right)\left(\frac{\text{weight fraction of A in raffinate}}{\text{weight fraction of A in extract}}\right) \qquad (10.2)$$

   It is preferable to choose a solvent with selectivity higher than unity. Selectivity also varies with concentration and in some systems it will vary from high values through unity to fractional values. Such systems are analogous to azeotropes.
2. *Distribution coefficient:* It is defined as the concentration of solute in extract(y) to that in raffinate(x). It is preferable to have a higher ratio of y/x as it results in the use of lesser quantity of solvent.
3. *Recoverability of solvent:* The solvent has to be recovered from extract phase for reuse. This is normally done by distillation. Hence, one should ensure that the mixture does not form an azeotrope which has a higher relative volatility and its latent heat of vaporization shall be low so that lesser energy is spent during vaporization.
4. *Density:* A larger difference in densities is necessary both for stagewise and continuous contact operations as it will help in easier separation of phases. However, at plait point the density difference is zero.
5. *Interfacial tension:* If the interfacial tension of solvent is large, more readily the coalescence of droplets or emulsions will occur but

the dispersion of one liquid in the other will be difficult. Since coalescence is usually of greater importance in extraction operation, the interfacial tension should therefore be high. It is zero at plait point.

6. *Chemical reactivity:* Solvent should be thermally stable and chemically inert towards the other components of the system and also towards the material of construction.
7. *Other properties:* Viscosity, vapour pressure and freezing point should be low for ease in handling and storage. They should also be non-toxic, non-flammable and of low cost.

10.8 OPERATIONS

Extraction operations can be carried out either as a single stage or as a multistage operation. Again the multistage operation could be either a cross-current or a counter-current operation. The leaving streams, viz. the extract and raffinate from each stage is always in equilibrium. A combination of mixer-settler is said to constitute a stage and in a multistage operation they are arranged in cascades.

10.8.1 Single Stage Operation

A typical flow diagram of a single stage extraction operation is shown in Fig. 10.8.

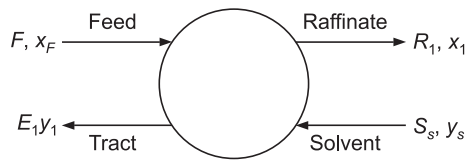


Fig. 10.8 Streams in a single stage operation.

F, R_1, E_1, and S are either the flow rates or quantities of different streams such as feed, raffinate, extract and solvent respectively and x_F, x_1, y_1, y_S are all weight fractions of solute in their respective streams.

The material balance gives

$$F + S = M_1 = E_1 + R_1 \tag{10.3}$$

where M_1 is the total weight of mixture (Feed + solvent or extract + raffinate)

A solute balance yields

$$Fx_F + Sy_S = M_1x_{M1} = E_1y_1 + R_1x_1 \tag{10.4}$$

where x_{M1} is the effective solute concentration in the extractor.

Eliminating M_1 from Eqs. (10.3) and (10.4), we get

$$\frac{S}{F} = \frac{x_F - x_{M_1}}{x_{M_1} - y_S} \tag{10.5}$$

The quantities of extract and raffinate can be computed from mixture rule given by Eq. (10.1) or by material balance given in Eq. (10.4)

$$E_1 y_1 + R_1 x_1 = M_1 x_{M_1} \tag{10.6}$$

$$E_1 y_1 + (M_1 - E_1) x_1 = M_1 x_{M_1} \tag{10.7}$$

$$E_1 = M_1 \left(\frac{x_{M_1} - x_1}{y_1 - x_1} \right) \tag{10.8}$$

Let us now try to use the phase diagram and distribution diagram to determine the product composition as shown in Fig. 10.9.

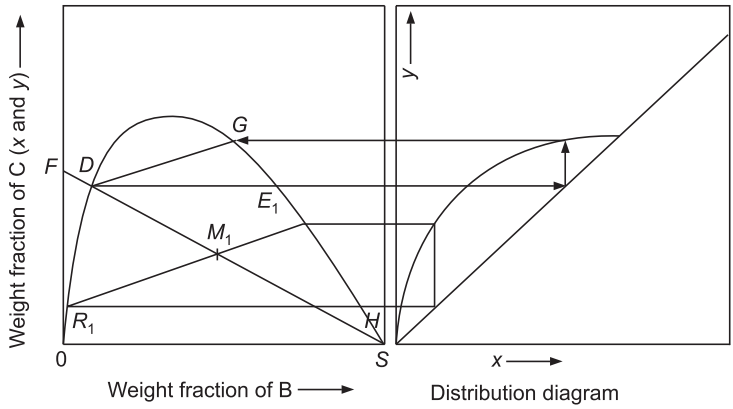


Fig. 10.9 Determination of minimum and maximum solvent.

The point *F* corresponds to feed mixture and *S*, the solvent. Once the feed and solvent are mixed, the mixture has an effective solute concentration of x_{M1} and is located as M_1 which lies on the line joining *F* and *S*. Thus the point M_1 lies within the curve. However, on settling, the mixture forms the two phases E_1 and R_1 and the line joining the points E_1 and R_1 intersects the feed line *FS* which is M_1. Though many lines can be drawn through the point M_1, only one line could be the tie line which will correspond to the equilibrium composition of extract and raffinate phases. The tie line could be located by a trial and error procedure using the equilibrium curve as shown.

10.8.1.1 Minimum solvent requirement

If the point M_1 lies on the point of intersection of curve (of solvent lean phase side) with *FS* (the point *D*) as shown in Fig. 10.9, then the corresponding amount of solvent is the minimum solvent needed and it provides an infinitesimal amount of extract as indicated by *G*.

10.8.1.2 Maximum solvent requirement

If the point M_1 lies on H (solvent rich phase side), then the amount of solvent used becomes the maximum and the corresponding raffinate concentration K obtained by the tie line indicates the infinitesimal amount of raffinate.

10.8.1.3 Steps involved in the estimation of extract and raffinate quantities

1. Plot the ternary data and equilibrium curve.
2. Locate the feed point '*F*' and solvent point '*S*' on the ternary data plot.
3. Join *FS* and locate M_1 $\left[M_1 \text{ corresponds to } x_{M_1} = \dfrac{(Fx_F + Sy_s)}{(F + S)}\right]$.
4. Draw a suitable tie line through M_1 with the help of equilibrium curve.
5. Locate the points of intersection of this tie line on the ternary data curve as E_1 and R_1 on solvent rich layer and solvent lean layer respectively and find y_1 and x_1 values corresponding to these points.
6. The quantity of extract layer, $E_1 = M_1\left(\dfrac{x_{M_1} - x_1}{y_1 - x_1}\right)$ and that of raffinate layer, $R_1 = F + S - E_1$ can be determined.

10.8.2 Multistage Cross-current Operation

A typical flow diagram of a multistage cross-current operation is shown in Fig. 10.10

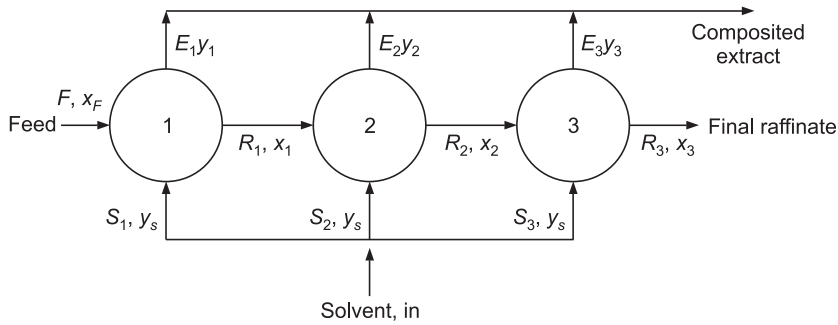


Fig. 10.10 A three stage cross-current extraction operation.

Consider a three-stage cross-current extraction process as shown in Fig. 10.10. The feed enters the first stage and the raffinate successively passes from stage (1) to (2) and (2) to (3) and finally leaves the system. Fresh or recovered solvent enters each stage. The solvent used could be of different concentrations but generally it will have the same value as it enters either fresh or after recovery from extract. The values of M_i, x_{Mi}, x_i and y_i, where i stands for the ith stage, can be computed as indicated in the single stage operation using material balances and tie lines. From these values the quantities of extract and raffinate from each stage can be computed.

Material balance across stage (1) gives

$$F + S_1 = R_1 + E_1 = M_1 \quad \text{(say)} \tag{10.9}$$

Component balance gives

$$Fx_F + S_1y_s = R_1x_1 + E_1y_1 = M_1x_{M1} \tag{10.10}$$

$$\therefore \quad x_{M_1} = \frac{(Fx_F + S_1y_s)}{(F + S_1)} \tag{10.11}$$

Similarly for any stage i

$$x_{M_i} = \frac{(R_{i-1}x_{i-1} + S_iy_s)}{(R_{i-1} + S_i)} \tag{10.12}$$

10.8.2.1 Steps

1. Plot the ternary data and equilibrium curve.
2. Locate the feed point F and solvent point S_1 on the ternary data plot.
3. Join FS_1 and locate M_1. {M_1 corresponds to x_{M1} and is given by

$$x_{M_1} = \frac{(Fx_F + S_1y_s)}{(F + S_1)}$$ }

4. Draw a suitable tie line passing through M_1.
5. Locate the points of intersection of tie line on the ternary data as E_1 and y_1 on B – rich layer and R_1 on solvent lean layer respectively. Estimate y_1 and x_1 corresponding to these points.

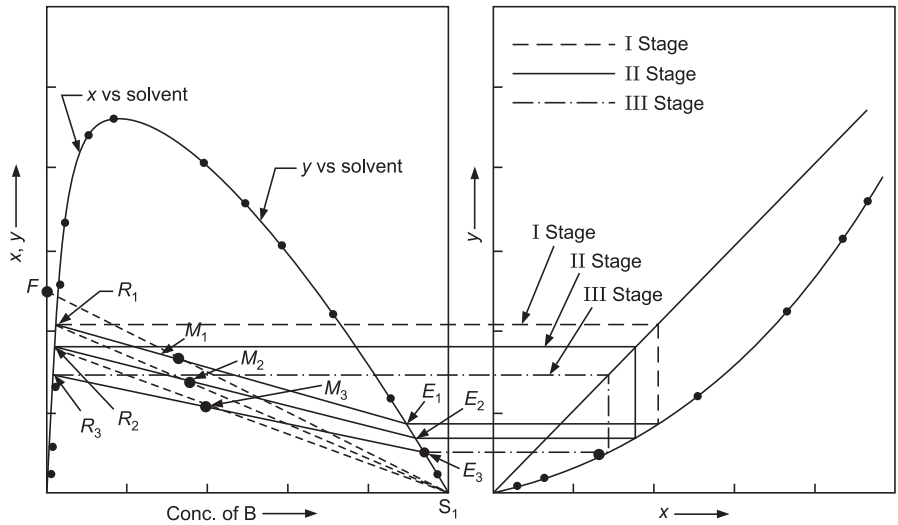


Fig. 10.11 Three stage cross-current operation.

6. The quantity of extract layer is given by $M_1\left(\frac{x_{M_1} - x_1}{y_1 - x_1}\right)$ and that of raffinate layer is given by $R_1 = F + S_1 - E_1$.

7. Join R_1S_2 and locate M_2. {M_2 corresponds to x_{M_2} and is given by

$$x_{M_2} = \frac{(R_1x_1 + S_2y_s)}{(R_1 + S_2)}\}$$

8. Draw a suitable tie line passing through M_2 to estimate y_2 and x_2 from graph.
9. The quantity of extract and raffinate E_2 and R_2 leading second stage are given by $E_2 = M_2\left(\frac{x_{M_2} - x_2}{y_2 - x_2}\right)$ and $R_2 = R_1 + S_2 - E_2$
10. Repeat the procedure for stage 3 as mentioned in steps (7) and (8) and obtain E_3, R_3, y_3 and x_3.

10.8.3 Multistage Countercurrent Extraction

A typical flow diagram of a multistage countercurrent operation is shown in Fig. 10.12.

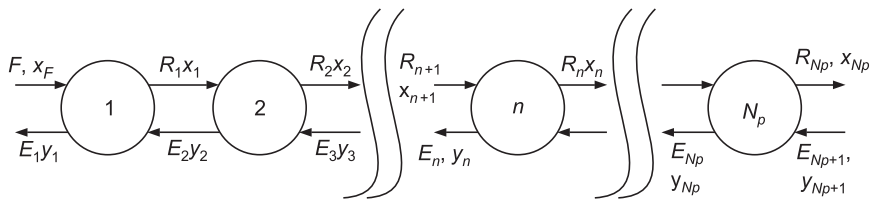


Fig. 10.12 Multistage countercurrent extraction operation.

Material balance for the system gives

$$F + E_{Np+1} = E_1 + R_{Np} \quad (10.13)$$

i.e.
$$F - E_1 = R_{Np} - E_{Np+1} \quad (10.14)$$

A component balance gives,

$$Fx_F + E_{Np+1}y_{Np+1} = E_1y_1 + R_{Np} \cdot x_{Np} \quad (10.15)$$

i.e.
$$Fx_F - E_1y_1 = R_{Np} \cdot x_{Np} - E_{Np+1} \cdot y_{Np+1} \quad (10.16)$$

A material balance from 1 to n stages gives

$$F + E_{n+1} = E_1 + R_n \quad (10.17)$$

$$F - E_1 = R_n - E_{n+1} \quad (10.18)$$

Hence, from Eqs. (10.14) and (10.18) , we get

$$F - E_1 = R_{Np} - E_{Np+1} = R_n - E_{n+1} \quad (10.19)$$

By substituting for n as 1, 2, 3, … we can show that

$$F - E_1 = R_1 - E_2 = R_2 - E_3 = \Delta_R$$

Here Δ_R, which is defined as a difference point, is the net flow outward not only at the last stage but also between any two adjacent stages and it remains constant. In other words, any line joining FE_1, R_1E_2, R_2E_3, … and extended must pass through the point Δ_R as shown in Fig. 10.13.

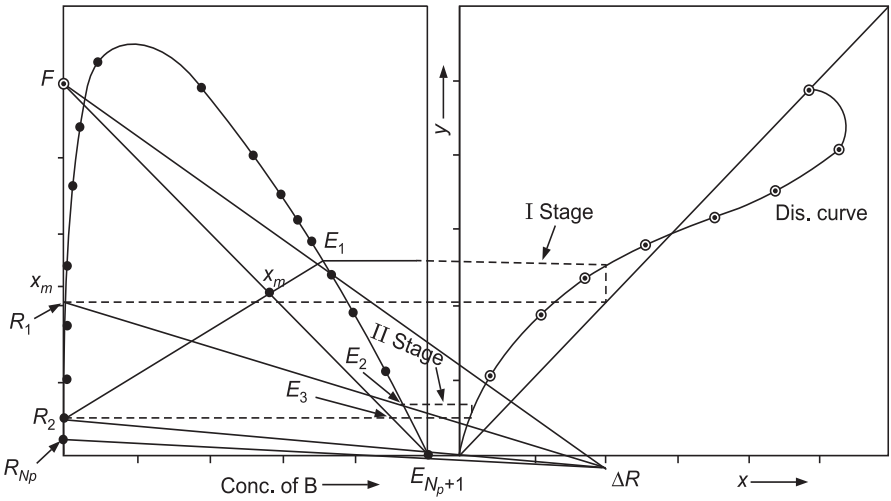


Fig. 10.13 Countercurrent operation—graphical representation of stages.

10.8.3.1 Steps involved in the determination of number of stages

1. Plot ternary data and draw the distribution curve adjacent to the ternary data in rectangular co-ordinates as shown in Fig. 10.13.
2. Locate the feed point (F), solvent point (E_{Np+1}) and the raffinate point (R_{Np}) leaving the systems based on their composition.
3. Join FS and locate x_m where, $x_m = \left[\dfrac{FX_F + (E_{N_p+1})(y_{N_p+1})}{F + E_{N_p+1}}\right]$.
4. Join R_{Np} and x_m and extend it to intersect the binodal curve which gives E_1.
5. Join F and E_1. Similarly join R_{Np} and E_{Np+1}.
6. Lines FE_1 and $R_{Np}\,E_{Np+1}$ are extended to meet and the meeting point is Δ_R.
7. Through E_1 and with the help of distribution curve, locate R_1 on solvent lean layer.
8. Join R_1 with Δ_R and extend the line to obtain E_2 on the solvent rich layer part of the ternary data plot.
9. Through E_2 and with the distribution curve, obtain R_2.
10. Proceed similarly till R_{Np} is crossed, thus number of stages needed for a specific operation is obtained.

However, if the number of stages are specified, there are two possible questions that arise.

(a) *For a specified amount of solvent, what will be the raffinate concentration?*

(b) *For a specified raffinate concentration, what is the amount of solvent to be used?*

Both need trial and error technique.

For situation (a), assume R_{Np} and proceed as discussed earlier. As soon as the specified stages are completed, check whether the assumed R_{Np} value also matches with the theoretical value obtained. If not, make another assumption of R_{Np} and proceed as earlier till the assumed R_{Np} value and the number of stages coincides with the specified values.

For situation (b), assume the quantity of solvent, estimate x_m and proceed as earlier. Check whether the specified R_{Np} value is reached for the given number of stages. If not, assume a new value for the solvent quantity again and proceed as earlier till the R_{Np} value and the number of stages match.

10.8.3.2 Minimum solvent requirement

The minimum solvent needed is fixed by the tie line which passes through the point of intersection of line FE_{mm} and solvent lean layer curve (the corresponding point of intersection is F'). The procedure to determine the minimum amount of solvent is given below and shown in Fig. 10.14.

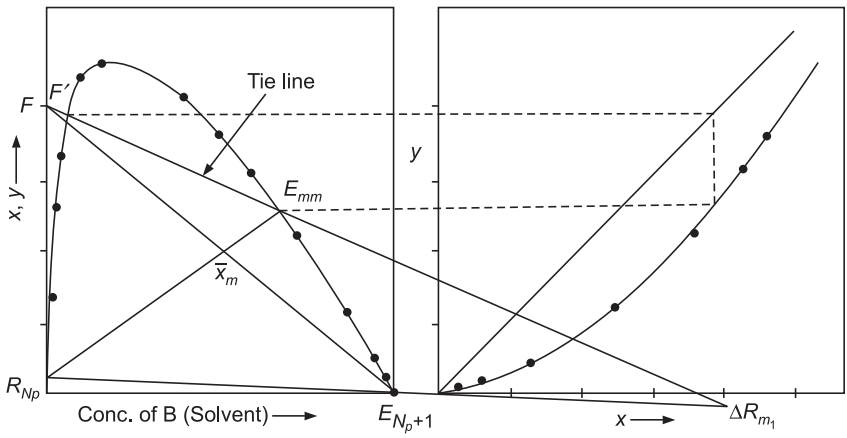


Fig. 10.14 Countercurrent operation—determination of minimum solvent.

Steps

1. Plot the ternary data and draw the distribution curve.
2. Locate F, E_{Np+1} and R_{Np}.
3. Arbitrarily draw the line $R_{Np}\ E_{mm}$ and check with the help of x-y plot whether the points F' and E_{mm} correspond to a tie line. If not, by trial and error locate a suitable $R_{Np}\ E_{mm}$ line which will ultimately correspond to tie line.
4. Join $F\ E_{Np+1}$ and $E_{mm}\ R_{Np}$ to find the intersection of these lines, $\overline{x_{mm}}$.
5. Since, $\overline{x_{mm}} = \dfrac{Fx_F + (E_{Np+1})(y_{Np+1})}{F + E_{Np+1}}$, the determined E_{Np+1} will be the minimum solvent required. (Since, all the other quantities are known.)

10.9 INSOLUBLE SYSTEMS (IMMISCIBLE SYSTEMS)

10.9.1 Cross-current Operation

In insoluble systems, the solvent (B) and the non-solute component in feed solution (A) are insoluble and remain so at all solute concentrations. Since A and B are insoluble, the amount of A and B both in their feed streams and the leaving streams remain constant. If X is the solute concentration in feed stream or raffinate stream expressed in mass ratio (kg of C/kg of A) and Y is the solute concentration in solvent or extract stream expressed in mass ratio (kg of C / kg of B), then a mass balance around stage n with reference to Fig. 10.15 yields

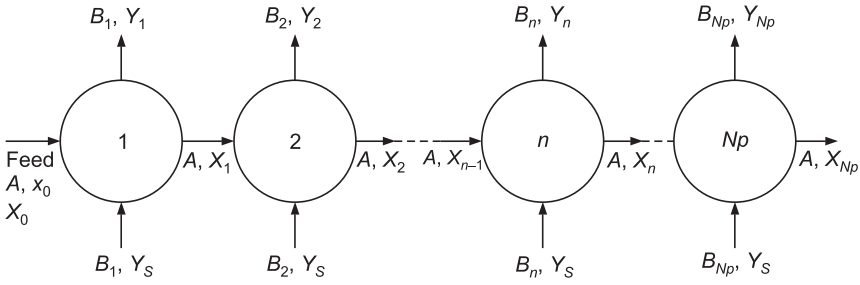


Fig. 10.15 Multistage cross-current operation for an insoluble system.

$$A \cdot X_{n-1} + B_n \cdot Y_S = B_n \cdot Y_n + A \cdot X_n \tag{10.20}$$

$$A\,[X_{n-1} - X_n] = B_n\,[Y_n - Y_S] \tag{10.21}$$

$$-\frac{A}{B_n} = \frac{(Y_n - Y_s)}{(X_n - X_{n-1})} \tag{10.22}$$

where A is the non-solute component in feed and B_n is the quantity of pure solvent used in nth stage, $-A/B_n$ is the slope of the operating line for stage n. For a typical three stage cross-current operation the construction of operating lines and the determination of final concentration of raffinate is shown in Fig. 10.16.

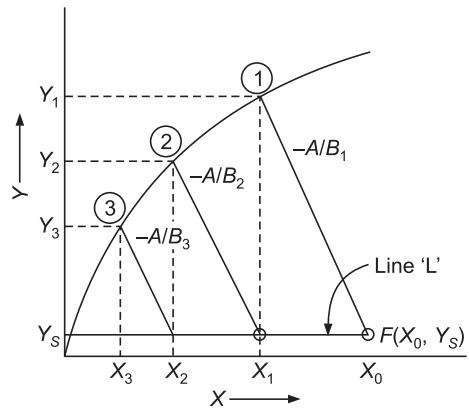


Fig. 10.16 Determination of number of stages in a cross-current operation.

10.9.1.1 Steps involved (Fig. 10.16)

1. Draw the equilibrium curve (X vs Y) on mass ratio basis.
2. Locate F (X_0, Y_S) and also draw a horizontal line L at $Y = Y_S$.
3. Draw a line with the slope ($-A/B_1$) and allow it to intersect the curve at ①.
4. Draw a vertical line from ① to the horizontal line L and the point of intersection corresponds to (X_1, Y_S).
5. From (X_1, Y_S) draw a line with a slope of ($-A/B_2$) to intersect curve at ②.
6. The vertical line drawn from ② to the horizontal line L gives the coordinates (X_2, Y_S).
7. Similarly proceed till X_{Np} is crossed and determine the number of stages needed or for the given number of stages, determine the X_{Np} value and hence the percentage extraction.

10.9.2 Countercurrent Operation

The flow of various streams in a countercurrent immiscible system with their compositions in a multistage operation is shown in Fig. 10.17.

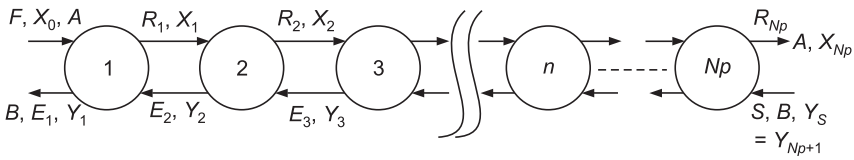


Fig. 10.17 Multistage countercurrent extraction operation for an insoluble system.

The material balance based on solute is given below:

$$A \cdot X_0 + B \cdot Y_{Np+1} = BY_1 + A \cdot X_{Np} \tag{10.23}$$

$$A\,[X_0 - X_{Np}] = B\,[Y_1 - Y_{Np+1}] \tag{10.24}$$

i.e.
$$\frac{A}{B} = \frac{(Y_1 - Y_{Np+1})}{(X_0 - X_{Np})} \tag{10.25}$$

i.e.
$$\frac{A}{B} = \frac{(Y_{\text{out}} - Y_{\text{in}})}{(X_{\text{in}} - X_{\text{out}})} \tag{10.26}$$

The operating line will have a slope of A/B and also pass through the points (X_0, Y_1) and (X_{Np}, Y_{Np+1}).

Once the operating line is constructed, the number of stages needed either for a specified percentage recovery or the exit concentration of raffinate stream can be found.

Sometimes the percentage recovery and the number of stages will be specified. The objective will be to fix the amount of solvent needed for the operation. This can be done by fixing the operating line by trial and error,

which will exactly yield both the exit concentration of raffinate and the specified number of stages.

Minimum solvent requirement is estimated by drawing either a tangent to the equilibrium curve or based on the equilibrium solute concentration in the solvent rich layer for the exit concentration of raffinate. The slope of the tangent gives the slope of operating line under minimum solvent conditions. In the later case, it is estimated by the slope of the line joining the terminal conditions.

When the equilibrium curve is of constant slope, say m', then $m' = (Y^*/X)$.

The number of stages N_p can be estimated by

$$\frac{(X_F - X_{Np})}{X_F - \dfrac{(Y_{Np+1})}{m'}} = \frac{\left(\dfrac{m'B}{A}\right)^{Np+1} - \left(\dfrac{m'B}{A}\right)}{\left(\dfrac{m'B}{A}\right)^{Np+1} - 1} \tag{10.27}$$

where, $\dfrac{m'B}{A}$ is called the *extraction factor*.

10.9.2.1 Steps involved (Fig. 10.18)

1. Draw the equilibrium curve.
2. Locate X_0, X_{Np} and Y_{Np+1}.
3. From the point (X_{Np}, Y_{Np+1}) draw a tangent to the equilibrium curve which will give slope of the operating line at minimum solvent condition, $(A/B)_{min}$.
4. If B_{actual} in terms of B_{min} is known, then we can determine $(A/B)_{actual}$ and draw the actual operating line. Otherwise, if the quantity of B is given, draw the operating line directly.

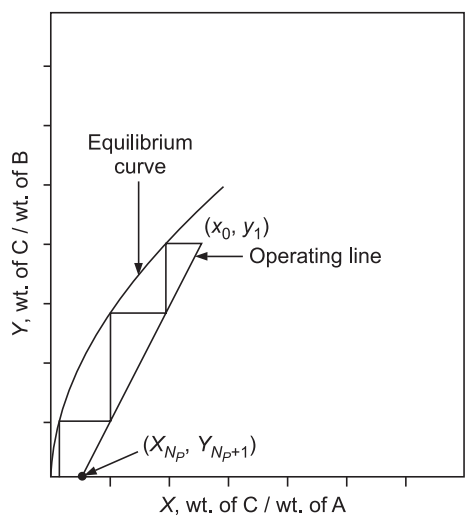


Fig. 10.18 Stages for countercurrent extraction.

5. At X_0 from the operating line draw a horizontal line to equilibrium curve which will give Y_1, the concentration of solute in final extract.
6. By stepwise construction from (X_0, Y_1), determine the number of stages needed to cross X_{Np}.
7. However, if the number of stages are prescribed, X_{Np} will have to be fixed by trial and error and checked for the prescribed number of stages.
8. In case the amount of solvent used is not given and X_{Np} along with the stages are known then the operating line has to be fixed by trial and error to ensure that both the prescribed X_{Np} and the number of stages are reached. From the slope of the operating line so fixed, we can estimate the solvent needed for the operation.

10.10 CONTINUOUS COUNTERCURRENT EXTRACTION WITH REFLUX

In a normal countercurrent extraction operation, the extract obtained will at the most be in equilibrium with the feed solution. However, the use of reflux at the extract end of the plant can provide a product even richer, as in the case of the rectifying section of a distillation column. Reflux is not used for the raffinate stream. A typical flow diagram of a countercurrent extraction with reflux is shown in Fig. 10.19.

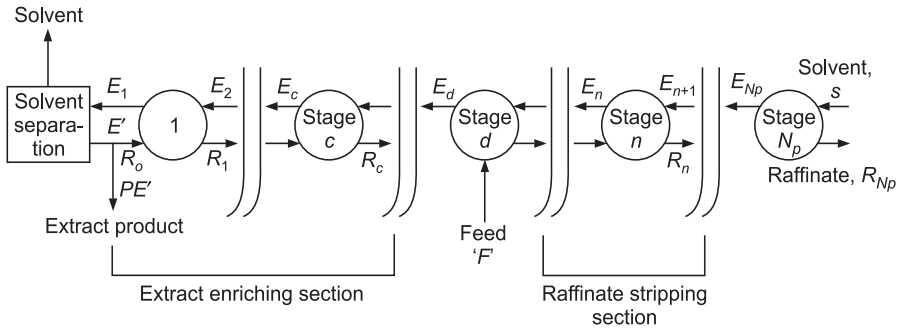


Fig. 10.19 Countercurrent extraction with reflux.

$$E_1' = E' = R_0 + P_E' \tag{10.28}$$

(The prime indicates the flow rate of solvent-free streams)

The procedure for determining the number of stages is quite similar to Ponchon–Savarit method discussed under distillation in Chapter 9.

Let us define 'N' as $\frac{B_{\text{out}}}{(A+C)_{\text{out}}}$ and X and Y as $\frac{C}{(A+C)}$ in raffinate and extract streams respectively.

Let Δ_E represent the net flow outwards from the enriching section.

i.e. $$\Delta_E' = P_E' \tag{10.29}$$

A component balance for solute indicates, $X_{\Delta E} = X_{PE}$

Balance for solvent B gives

$$B_E = \Delta_E' \cdot N_{\Delta E} \tag{10.30}$$

For all stages up to c, a balance for $A + C$ gives

$$E'_{c+1} = P_E' + R_c' = \Delta_E' + R_c' \tag{10.31}$$

i.e.
$$\Delta_E' = E'_{c+1} - R_c' \tag{10.32}$$

The component balance for A and C is

$$\Delta_E' X_{\Delta E} = E'_{c+1} Y_{c+1} - R_c' X_{Rc} \tag{10.33}$$

Similarly, a balance for B gives

$$\Delta_E' N_{\Delta E} = E'_{c+1} N'_{Ec+1} - R_c' N_{Rc} \tag{10.34}$$

Since c represents any stage, all lines radiating from point Δ_E represent extract and raffinate flowing between any two successive stages.

Solving Eq. (10.31) with Eqs. (10.33) and (10.34), we get the expression, for internal reflux ratio.

$$\frac{R_c'}{E'_{c+1}} = \frac{(N_{\Delta E} - N_{E,c+1})}{(N_{\Delta E} - N_{Rc})} = \frac{(X_{\Delta E} - Y_{c+1})}{(X_{\Delta E} - X_{Rc})} \tag{10.35}$$

$$= \frac{\text{line } \Delta_E E_{c+1}}{\text{line } \Delta_E R_c} \tag{10.36}$$

$$\text{External reflux ratio } \frac{R_o'}{P_E'} = \frac{R_o}{P_E} = \frac{(N_{\Delta E} - N_{E_1})}{(N_{E_1})} \tag{10.37}$$

and this can be used to locate Δ_E point which will have coordinates as ($X_{\Delta E}$, $N_{\Delta E}$). Similarly, we can show that $R'_{NP} - S' = R'_{n-1} - E_n' = \Delta_R'$ (for a general stage n in stripping section) and hence all operating lines will pass through Δ_R' in stripping section. A material balance for the entire plant, on solvent-free basis, gives

$$F' + S' = P_E' + R'_{NP} \tag{10.38}$$

$$F' = P_E' + R'_{Np} - S' = \Delta_E' + \Delta_R' \tag{10.39}$$

Hence, the feed point F will lie on the line joining Δ_E' and Δ_R'.

The minimum reflux ratio occurs when the line radiating either from Δ'_E or Δ'_R coincides with a tie line and also pass through feed point F.

10.10.1 Steps

The procedure for determining the number of stages in continuous countercurrent with reflux is shown in Fig. 10.20.

1. Convert the data to solvent free basis and estimate N, X, Y.
2. Plot N vs X and Y.
3. Draw the X vs Y diagram and locate $X'_{P'_E}$ and $X'_{R'_{Np}}$.
4. Locate $X'_{P'_E}$ and $X'_{R'_{Np}}$ and draw vertical lines in N vs X, N vs Y plot.

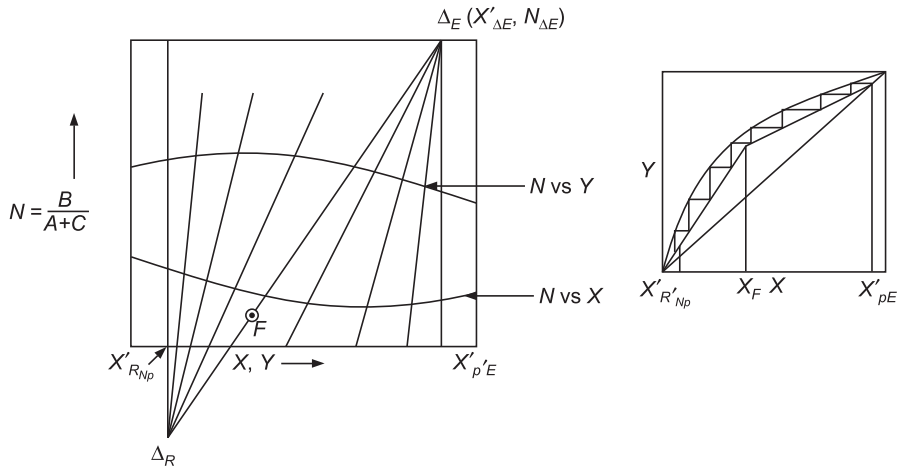


Fig. 10.20 Procedure to determine the number of stages in countercurrent extraction operation with reflux.

5. For the given reflux ratio estimate $N_{\Delta E}$ and plot ($X'_{\Delta E}$, $N_{\Delta \mathbf{E}}$) point and call it Δ_E.
6. Locate feed point F (X_F, N_F).
7. Join Δ_E and F and produce it to cut the vertical line drawn at $X'_{R'_{Np}}$ to obtain Δ_R.
8. Draw arbitrary lines from Δ_E and Δ_R point to N vs X and N vs Y plot and obtain the coordinates of the operating line.
9. Plot the coordinates of operating line in X vs Y diagram.
10. By stepwise construction starting from $X'_{P'_E}$ determine the stages needed up to $X'_{R'_{Np}}$. The stage which crosses the feed point (corresponding to X_F), gives the location of feed point.

10.11 FRACTIONAL EXTRACTION

When a solution contains two solutes, both of which can be extracted by countercurrent extraction with a suitable solvent, then any great degree of separation of the solutes by this method is difficult, unless their distribution coefficients are very large. By using partially miscible solvents, separation can be achieved.

10.12 MULTICOMPONENT EXTRACTION

For systems containing more than four components, presentation of equilibrium data and the computation of stages are very difficult (as in the extraction of petroleum lubricating oils). In such cases the number of stages needed are determined experimentally in the laboratory.

10.13 CONTINUOUS CONTACT EXTRACTORS

In these extractors liquid flows countercurrently through a single piece of equipment and one extractor is equivalent to many theoretical stages. The flow is produced by virtue of the variation in densities of the liquids. Whenever the motivating force is gravity it has a vertical orientation and if the motivating force is centrifugal force, it is horizontal in nature. Flooding is one of the common problems encountered in the operation of these devices. They are also subjected to axial mixing which severely reduces the extraction rates. The tower design procedure is similar to the design of packed absorption tower. Raffinate stream corresponds to gas stream and extract stream corresponds to the liquid stream.

$$Z = H_{tR} \cdot N_{tR} \tag{10.40}$$

where

$$H_{tR} = \frac{R}{k_R a (1-x)_{im}} \tag{10.41}$$

$$N_{tR} = \int_{x_2}^{x_1} \frac{(1-x)_{im}\, dx}{(1-x)(1-x_i)} = \int_{x_2}^{x_1} \frac{dx}{(x-x_i)} + \frac{1}{2} \ln \frac{(1-x_2)}{(1-x_1)} \tag{10.42}$$

x_i = Interface concentration of solvents

k_R = mass transfer coefficient for raffinate phase

H_{tR} = height of raffinate transfer unit

N_{tR} = Number of raffinate transfer units

$(1 - x)_{im}$ = logarithmic mean of $(1 - x)$ and $(1 - x_i)$

The height of the tower can also be estimated using the overall mass transfer coefficients as in the case of absorption in which case,

$$Z = H_{toR} \cdot N_{toR} = H_{toE} \cdot N_{toE} \tag{10.43}$$

where,

$$H_{toR} = \frac{R}{K_R a (1-x)^*_m}; \; H_{toE} = \frac{E}{K_E a (1-y)^*_m} \tag{10.44}$$

$$N_{toR} = \int_{x_2}^{x_1} \frac{(1-x)^*_M\, dx}{(1-x)(x-x^*)} = \int_{x_2}^{x_1} \frac{dx}{(x-x^*)} + \frac{1}{2} \ln \frac{(1-x_2)}{(1-x_1)} \tag{10.45}$$

$$N_{toE} = \int_{x_2}^{x_1} \frac{(1-y)^*_M\, dy}{(1-y)(y^*-y)} = \int_{x_2}^{x_1} \frac{dy}{(y^*-y)} + \frac{1}{2} \ln \frac{(1-y_1)}{(1-y_2)} \tag{10.46}$$

$$(1-x)^*_M = \frac{(1-x^*) - (1-x)}{\ln\left(\dfrac{(1-x^*)}{(1-x)}\right)} \tag{10.47}$$

$$(1-y)^*_M = \frac{(1-y) - (1-y^*)}{\ln\left(\dfrac{(1-y)}{(1-y^*)}\right)} \tag{10.48}$$

where x^* is the concentration in equilibrium with y and y^* is the concentration in equilibrium with x.

10.14 DILUTE SOLUTIONS

For dilute solutions and whenever the equilibrium curve and operating curve are linear in the operating range,

$$N_{toR} = \frac{(x_1 - x_2)}{(x - x^*)_m} \quad \text{and} \quad N_{toE} = \frac{(y_1 - y_2)}{(y^* - y)_m} \tag{10.49}$$

If the equilibrium relationship is given by $m = y^*/x$, similar to Henry's law, then

$$N_{toR} = \frac{\ln\left[\left(\frac{x_1 - \left(\frac{y_2}{m}\right)}{x_2 - \left(\frac{y_2}{m}\right)}\right)\left(1 - \frac{R}{mE}\right) + \frac{R}{mE}\right]}{1 - \left(\frac{R}{mE}\right)} \tag{10.50}$$

$$N_{toE} = \frac{\ln\left[\left(\frac{y_2 - mx_1}{y_1 - mx_1}\right)\left(1 - \frac{mE}{R}\right) + \frac{mE}{R}\right]}{1 - \left(\frac{mE}{R}\right)} \tag{10.51}$$

Though the above expressions can be used in the design of continuous contactors, it is always advisable to go in for pilot plant studies at nearly the expected operating conditions to enable the design of extractors as lot of parameters influence extraction. These include physical properties of liquids, its flow rate, solubility of solute and the presence of surface active agents. The equipment also has its own impact on the extraction performance. The factors such as type of agitator and its size, the size of extractor, the presence of baffles and type of agitation have an influence on the performance of extraction.

10.15 EQUIPMENT

The equipment used for liquid-liquid extraction operations are classified as:

Single stage mixer settler
A multistage cascade of single stage mixer settler
Continuous contactors.

10.15.1 Mixer-settler

A single stage mixer-settler is a simple arrangement with two units. In the first unit, called 'mixer', mixing of two phases takes place which leads to transfer of mass and in the second unit, called 'settler', separation of phase takes place (Fig. 10.21). In a multistage operation, several such combinations are used.

The degree of dispersion depends on the type of contactor/mixer and liquid characteristics. The liquid phases can also be mixed by the use of different types of impellers such as marine impeller, flat blade turbine, etc. The normal ratio of impeller to tank diameter is 0.25 to 0.33. The dispersion can also be achieved by the dispersion of one liquid through another liquid in the form of fine droplets with the help of nozzles. In a multistage cascade arrangement, feed after entering the first mixer (subsequently raffinate) flows from the first settler to the next mixer-settler combination till it leaves from the last settler as final raffinate. The solvent enters the last mixer and from the last settler it passes on to the next mixer before it finally leaves as the concentrated extract from the first settler. The flow, thus, is countercurrent.

However, large towers are used when large volumes of liquid have to be handled and that too on continuous basis. In these towers, the liquids flow countercurrently. The heavy phase is introduced at the top and it flows downwards. The lighter phase is introduced at the bottom and this phase flows upwards. Some of the commonly used towers for this operation are briefly described here.

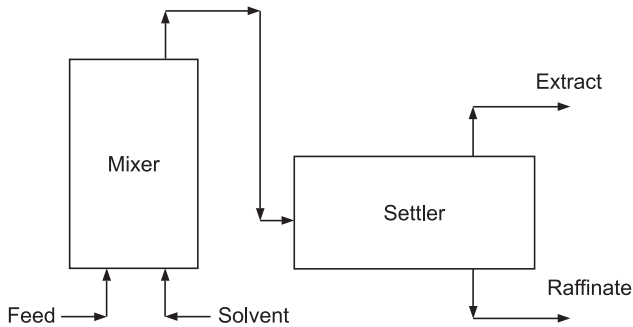


Fig. 10.21 A mixer-settler combination.

10.15.2 Mechanically Agitated Tower

These towers are provided with agitators which are mechanically agitated. The efficiency of separation increases due to agitation of the liquid streams. The agitators are of different configurations.

10.15.3 Oldshue-Rhuston Extractor

In this case the extractor is provided with flat blade disc turbine impeller for dispersing and mixing and horizontal compartmental plates which are provided to reduce axial mixing. This is a very old type of extractor.

10.15.4 Rotating Disc Contactor (RDC)

The schematic diagram of RDC is shown in Fig. 10.22. It comprises a tall vertical tower provided with inlets for both feed and solvent streams to enter and outlets

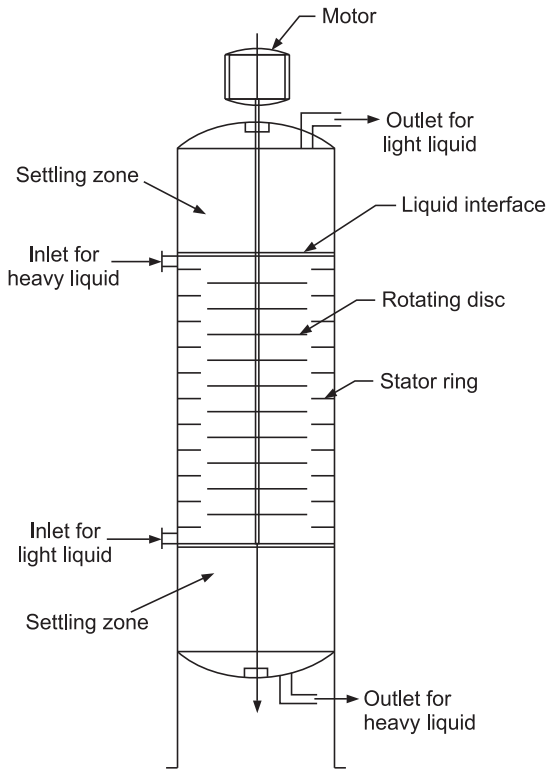


Fig. 10.22 Rotating disc contactor (RDC).

for both product (Raffinate and Extract) streams. It has a central shaft attached with rotor discs and is driven by a motor. Stators of centrally hollowed rings attached to the wall of the tower alternate position to the rotors. This can be operated at high speeds and it finds wide application in petroleum industries.

10.15.5 York-Scheibel Column

This column is provided with mixing and horizontal packed compartments arranged alternately as shown in Fig. 10.23. Mixing is done by a turbine impeller which is attached to a mechanically driven central shaft. The packed compartments are provided with wire mesh to reduce axial mixing.

10.15.6 Pulsed Column Extractor

In this extractor, there is no moving device. A reciprocating pulse input is hydraulically transmitted into the column due to which there is thorough contact between liquid streams. Column is provided with perforated plates attached as shown in Fig. 10.24. Due to the pulse input, the light and heavy liquids move upward and downward throughout the tower through the perforations. Since it has no moving parts, it finds extensive use in handling.

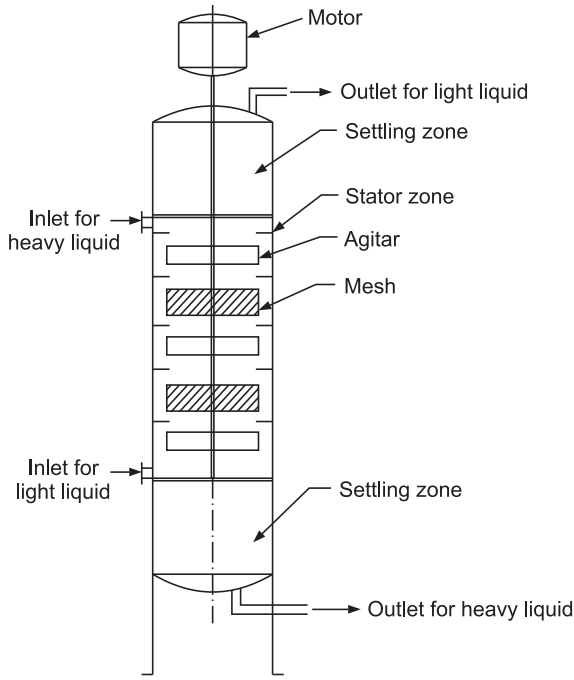


Fig. 10.23 York scheibel column.

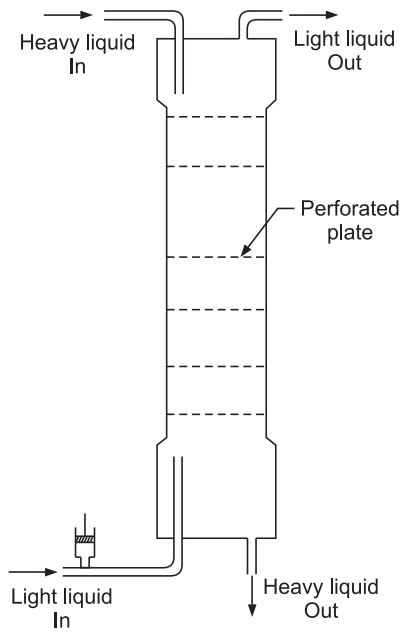


Fig. 10.24 Pulsed column extractor.

10.15.7 Other Extractors

Apart from these we have conventional packed towers, spray towers and for lesser density difference systems the centrifugal extractors.

WORKED EXAMPLES

1. A 5% (by weight) solution of acetaldehyde in toluene is extracted with water in a three stage cross-current unit. If 100 kg of water is used per stage for 500 kg of feed, calculate (using graphical method) the percentage extraction of acetaldehyde and the weights of final raffinate and mixed extract. The equilibrium relationship is given by the equation, $Y = 2.3\ X$ where Y = kg acetaldehyde/kg water and X = kg acetaldehyde/kg toluene. Assume that toluene and water are immiscible with each other.

Solution.

A : toluene, B : water, C : acetaldehyde,

F = 500 kg, x_F = 0.05, $Y = 2.3X$

B = 100 kg water/stage

Three stage cross-current operation

Assume solvent to be pure, i.e. $y_s = 0$

F = 500 kg, A = 475 kg, and C = 25 kg

Slope = $(-A/B)$

So $(-A/B)$ for each stage = $(-475/100) = (-4.75)$

Draw the operating line with a slope of – 4.75 for each stage

$$X_F = \frac{x_F}{(1 - x_F)} = \frac{0.05}{1 - 0.95} = 0.0526$$

X (kg acetaldehyde/ kg toluene)	0	0.01	0.02	0.03	0.04	0.05	0.06
Y (kg acetaldehyde/ kg water)	0	0.023	0.046	0.069	0.092	0.115	0.138

Since system is immiscible, the whole of solvent goes in extract. The feed introduced in 1st stage just passes through all stages and comes out as final raffinate:

A plot between X and Y is drawn. The operating line is drawn with a slope of – 4.75 for each of the three stages.

Weight of A in final raffinate = A = 475 kg

Final raffinate contains X_3 = 0.0161 kg C/kg A (from graph)

Amount of C in raffinate = 475 × 0.016 = 7.6 kg

Total weight of raffinate = 475 + 7.6 = 482.6 kg

Total C extracted = $(Y_1 + Y_2 + Y_3) \times 100$

= 100 × (0.082 + 0.055 + 0.037) = 17.4 kg

In extract, the amount of B = 100 kg (in each stage)

Y_3 = 0.037 kg C/kg B (from graph)

Amount of C in final stage extract = 0.037 × 100 = 3.7 kg

Total weight of extract = 300 + 17.4 = 317.4 kg

% Extraction = (17.4/25) × 100 = 69.6%

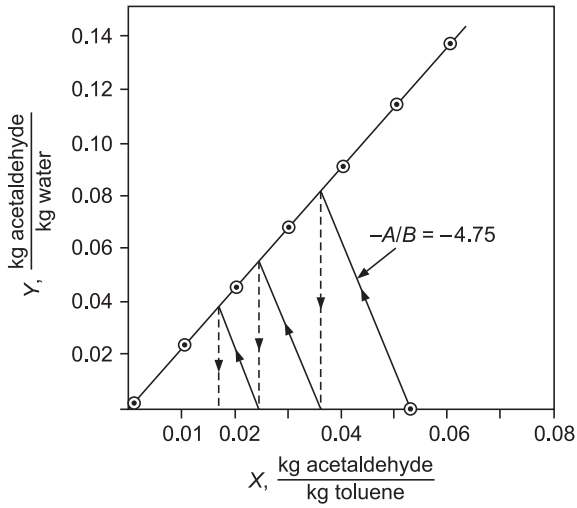


Fig. 10.25 Example 1.

2. 100 kg of a solution containing acetic acid and water containing 25% acid by weight is to be extracted with isopropyl ether at 20°C. The total solvent used for extraction is 100 kg. Determine the compositions and quantities of various streams if,

(i) The extraction is carried out in single stage

(ii) The extraction is carried out in two stages with 50 kg of solvent in each stage.

Equilibrium data:

Water layer (wt. %)		*Ether layer* (wt. %)	
Acid (*x*)	*Water* (*A*)	*Acid* (*y*)	*Water* (*A*)
0.69	98.1	0.18	0.5
1.41	97.1	0.37	0.7
2.9	95.5	0.79	0.8
6.42	91.7	1.93	1.0
13.3	84.4	4.82	1.9
25.5	71.1	11.4	3.9
36.7	58.9	21.6	6.9
44.3	45.1	31.1	10.8
46.4	37.1	36.2	15.1

Solution.

$A \rightarrow$ water, $B \rightarrow$ isopropyl ether, $C \rightarrow$ Acetic acid,

F = 100 kg, A = 75 kg, and C = 25 kg, x_F = 0.25

Total solvent used = 100 kg = B

B	0.0121	0.0149	0.016	0.0188	0.023	0.034	0.044	0.106	0.165
x	0.0069	0.0141	0.029	0.0642	0.133	0.255	0.367	0.443	0.464
B	0.9932	0.9893	0.9841	0.9707	0.9328	0.847	0.715	0.581	0.487
y	0.0018	0.0037	0.0079	0.0193	0.0482	0.114	0.216	0.311	0.362

(i) Single stage operation:

By total and component material balances,

$$F + S = M_1$$

$$100 + 100 = M_1 = 200 \text{ kg}$$

$$x_{M_1} = \frac{Fx_y + Sy_s}{F + S} = \frac{100 \times 0.25 + 100 \times 0}{100 + 100} = 0.125$$

Locate M_1 on the FS line corresponding to x_{M_1}. By trial and error, a tie line is drawn which passes through M_1.

The co-ordinates (x_1, y_1) obtained are (0.18, 0.075).

By material balance,

$$R_1x_1 + y_1E_1 = M_1x_{M_1}$$

$$R_1 + E_1 = M_1$$

$$R_1 \times 0.18 + 0.075E_1 = 200 \times 0.125$$

$$R_1 + E_1 = 200$$

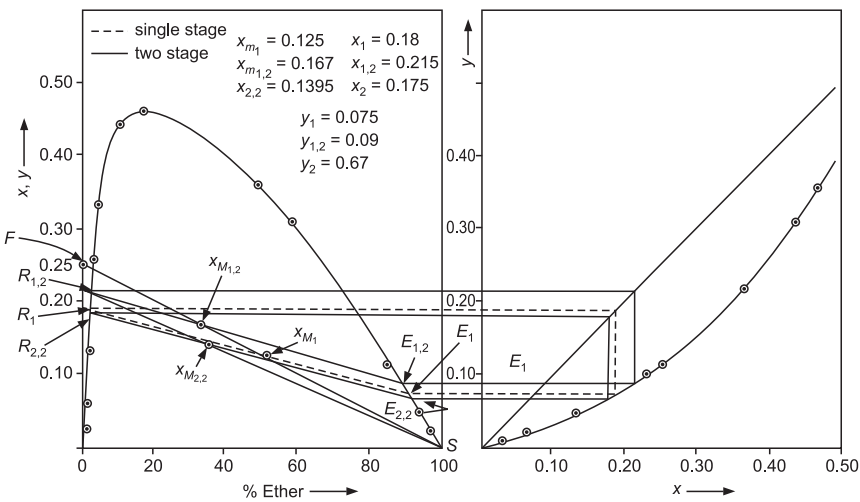


Fig. 10.26 Example 2.

Solving we get, $$E_1 = M_1\left(\frac{x_{M_1} - x_1}{y_1 - x_1}\right)$$

Quantities of product streams are

$$E_1 = 104.76 \text{ kg}$$

$$R_1 = 95.24 \text{ kg}$$

(ii) Two-stage operation:

$$F = 100 \text{ kg}, \quad S = 50 \text{ kg}$$

$$S + F = M_1$$

$$x_{M_{1,2}} = \frac{Fx_F + Sy_s}{F + S} = \frac{100 \times 0.25 + 50 \times 0}{100 + 50} = 0.167$$

$$M_1 = 50 + 100 = 150 \text{ kg}$$

Locate $M_{1,2}$ on the *Fs* line corresponding to $x_{M_{1,2}}$. By trial and error, a tie line is drawn which passes through M_{12}.

The co-ordinates (x_{12}, y_{22}) obtained are (0.215, 0.09)

By following the same procedure mentioned above and solving, we get

$$E_{12} = M_{12}\left(\frac{x_{M_{12}} - x_{12}}{y_{12} - x_{12}}\right) = 150\left(\frac{0.167 - 0.215}{0.09 - 0.215}\right) = 57.6 \text{ kg}$$

$$R_{12} = 150 - 57.6 = 92.4 \text{ kg}$$

Similarly for II stage, $x_{M22} = 0.1395$, $M_2 = 92.4 + 50 = 142.4$ kg

$x_2 = 0.175$ and $y_2 = 0.07$ (from tie line)

$E_2 = 48.14$ kg

$R_2 = 94.26$ kg

Percentage recovery = $\frac{(25 - 94.26 \times 0.175)}{25} \times 100 = 34.02\%$

3. 1000 kg/hr of an acetone-water mixture containing 20% by weight of acetone is to be counter-currently extracted with trichloroethane. The recovered solvent to be used is free from acetone. The water and trichloroethane are insoluble. If 90% recovery of acetone is desired estimate the number of stages required if 1.5 times the minimum solvent is used. The equilibrium relationship is given by $y = 1.65x$, where x and y are weight fractions of acetone in water and trichloroethane respectively.

Solution.

$$X_F = 0.2/(1 - 0.2) = 0.25$$

$$X_{N_P} = 0.25 \times 0.1 = 0.025$$

$$y_1 = 1.65 \times 0.2 = 0.33$$

$$Y_1 = 0.33/0.67 = 0.49$$

$$Y_s = 0 \text{ (Pure solvent)}$$

(the same value is got from plotting of the graph also)

X	0.05	0.1	0.175	0.25	0.325
$x = \frac{x}{1+x}$	0.0476	0.0909	0.149	0.20	0.245
$y = 1.65x$	0.0785	0.15	0.246	0.33	0.404
$Y = \frac{y}{1-y}$	0.085	0.176	0.326	0.493	0.678

$$\frac{A}{B_{\min}} = \frac{Y_1 - Y_s}{X_F - X_{Np}}$$

$$\frac{800}{B_{\min}} = \frac{0.49 - 0}{0.25 - 0.025}$$

$$B_{\min} = 367.35 \text{ kg}$$

$$B_{\text{act}} = 1.5 \times B_{\min} = 1.5 \times 367.35 = 551.025 \text{ kg}$$

$$\frac{A}{B_{\text{act}}} = \frac{Y_{1,\text{act}} - Y_s}{X_F - X_{Np}}$$

$$\frac{800}{551.025} = \frac{Y_{1,\text{act}} - 0}{0.25 - 0.025} = 1.452$$

$$Y_{1,\text{act}} = 0.327$$

An operating line with a slope of 1.452 is drawn and by stepwise construction the number of stages is determined as 5.

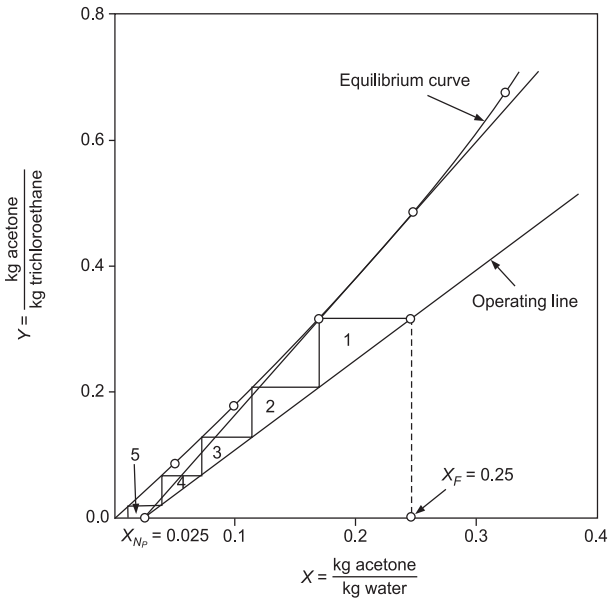


Fig. 10.27 Example 3.

4. Water–dioxane solution is to be separated by extraction process using benzene as solvent. At 25°C the equilibrium distribution of dioxane between water and benzene is as follows:

Weight % of dioxane in water	5.1	18.9	25.2
Weight % of dioxane in benzene	5.2	22.5	32.0

At these concentrations water and benzene are substantially insoluble. 1000 kg of a 25% dioxane water solution is to be extracted to remove 95% of dioxane. The benzene is dioxane free.

(i) Calculate the benzene requirement for a single batch operation.

(ii) Calculate the benzene requirement for a five-stage cross-current operation with 600 kg of solvent used in each stage.

Solution.

Solvent = amount of feed or raffinate in each stage

(B) = (F) or (R)

x	0.051	0.189	0.252
y	0.052	0.225	0.32
$X = x/(1 - x)$	0.054	0.233	0.337
$Y = y/(1 - y)$	0.05485	0.29	0.471

(i) F = 1000 kg (A = 750 kg, C = 250 kg)

x_F = 0.25, X_F = 0.25/0.75 = 0.333

$X_{R_{Np}}$ = 0.05 × 0.333 = 0.01665

Y_{in} = 0

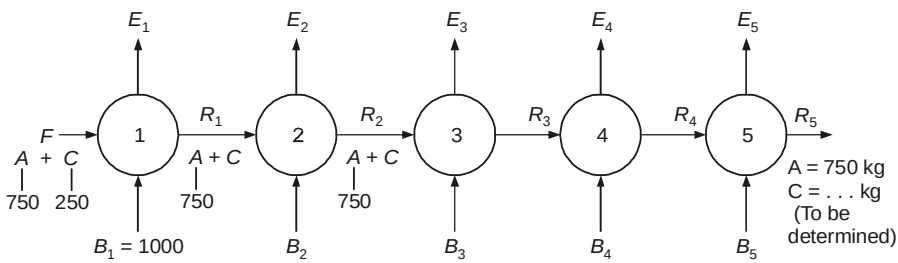
Y_1 = 0.0175 (From graph)

$$\frac{A}{B} = \frac{Y_1 - Y_s}{X_F - X_{Np}}$$

$$\frac{750}{B} = \frac{0.0175 - 0}{0.333 - 0.01665}$$

B = 13557.86 kg

(ii) Five-stage cross-current operation



Amount of solvent used is 600 kg

$$\frac{A}{B} = \frac{750}{600} = 1.25$$

Draw operating lines with a slope of –1.25 and determine the raffinate concentration.

$$X_{\text{final}} = 0.0175$$

$$\% \text{ recovery} = \frac{(0.333 - 0.0175) \times 100}{0.333} = 94.75\%$$

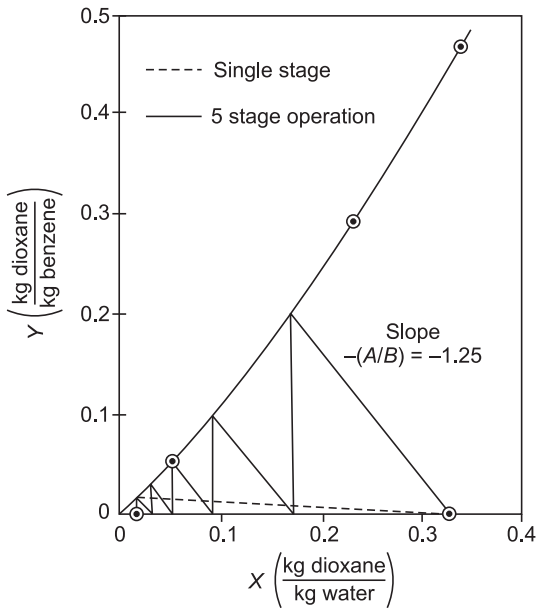


Fig. 10.28 Example 4.

5. 1000 kg per hour of a solution of *C* in *A* containing 20% *C* by weight is to be countercurrently extracted with 400 kg per hour of solvent *B*. The components *A* and *B* are insoluble. The equilibrium distribution of component *C* between *A* and *B* are as follows;

Wt. of *C*/Wt. of *A*	0.05	0.20	0.30	0.45	0.50	0.54
Wt. of *C*/Wt. of *B*	0.25	0.40	0.50	0.65	0.70	0.74

How many theoretical stages will be required to reduce the concentration of *C* to 5% in effluent?

Solution.

F = 1000 kg/h, (A = 800 kg/h, C = 200 kg/h)

x_F = 0.2, $x_{R_{Np}}$ = 0.05

Assume solvent to be pure, then

countercurrent extraction $y_s = Y_s = 0$

solvent = B = 400 kg/h

A and B are insoluble

$X_F = 0.2/(1 - 0.2) = 0.25, \qquad X_{R_{Np}} = 0.05/(1 - 0.05) = 0.0526$

$$\frac{A}{B} = \frac{Y_1 - Y_s}{X_F - X_{Np}}$$

$$\text{Slope} = \frac{A}{B} = \frac{800}{400} = 2$$

$$\frac{A}{B} = \frac{Y_1 - 0}{0.25 - 0.0526}$$

$$Y_1 = 0.395$$

Plot X vs Y to obtain the equilibrium curve.
Draw an operating line between ($X_{R_{Np}}$, Y_s) and (X_F, Y_1) and determine the number of stages by stepwise construction.
Number of stages obtained = 3.

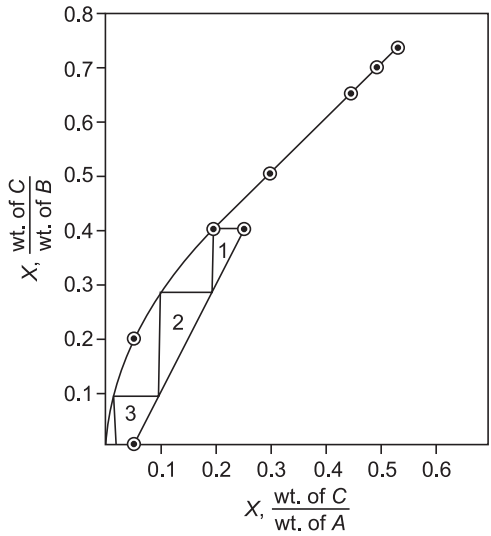


Fig. 10.29 Example 5.

6. Water–dioxane solution is to be separated by extraction process using benzene as solvent. At 25°C the equilibrium distribution of dioxane between water and benzene is as follows:

wt. % of dioxane in water	5.1	18.9	25.2
wt. % of dioxane in benzene	5.2	22.5	32.0

At these concentrations water and benzene are substantially insoluble. 1000 kg of a 25% dioxane water solution is to be extracted to remove 95% of dioxane. The benzene is dioxane free. Calculate minimum solvent

required in kg/h if the extraction is done in countercurrent fashion. Estimate the number of stages needed if 1.5 times the minimum amount of solvent is used.

Solution.

Benzene: B Water: A Dioxane: C

F = 1000 kg (A = 750 kg, C = 250 kg)

x	0.051	0.189	0.252
y	0.052	0.225	0.32
$X = x/(1 - x)$	0.054	0.233	0.337
$Y = y/(1 - y)$	0.05485	0.29	0.471

$$x_F = 0.25,$$

$$X_F = 0.25/0.75 = 0.333$$

$$X_{R_{Np}} = 0.05 \times 0.333 = 0.01665$$

$$\frac{A}{B_{\min}} = \frac{Y_{Np+1} - Y_1}{X_{Np} - X_F}$$

$$\frac{A}{B_{\min}} = \frac{Y_{Np+1} - Y_1}{X_{Np} - X_F} = \frac{0 - 0.365}{0.01665 - 0.333} = 1.1154$$

$$B_{\min} = 650 \text{ kg}$$

$$B_{\text{act}} = 1.5 \times 650 = 975 \text{ kg}$$

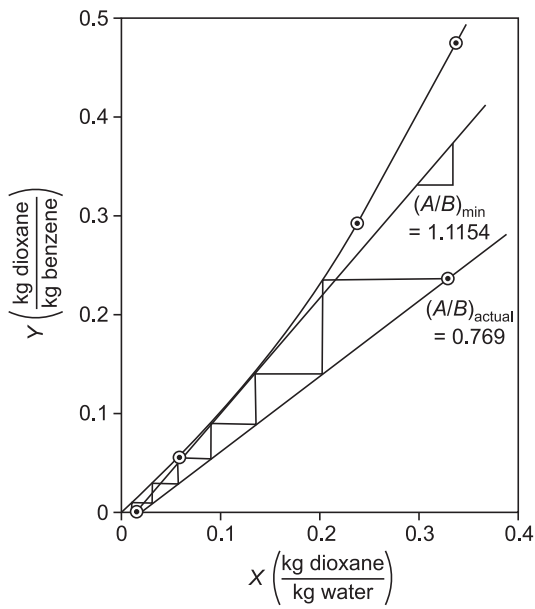


Fig. 10.30 Example 6.

$$\frac{A}{B_{\text{act}}} = \frac{Y_{Np+1} - Y_{1,\text{act}}}{X_{Np} - X_F}$$

$$\frac{750}{975} = \frac{0 - Y_{1,\text{act}}}{0.01665 - 0.333}$$

$$Y_{1,\text{act}} = 0.243$$

By stepwise construction, the number of stages can be determined as 6.

7. Nicotine in a water solution containing 1% nicotine is to be extracted once with kerosene at 20°C. Kerosene and water are insoluble.

Determine the percentage extraction if 1000 kg of feed solution is extracted once with 1500 kg solvent. What will be the extraction if three ideal stages are used with 500 kg solvent in each stage?

Equilibrium data:

X	0	0.00101	0.00246	0.00502	0.00751	0.00998	0.0204
Y	0	0.00081	0.001962	0.00456	0.00686	0.00913	0.0187

where X is kg Nicotine/kg water and Y is kg Nicotine/kg kerosene.

Solution.

Water: A, Kerosene: B, Nicotine: C

$x_F = 0.01$ $\qquad X_F = X_0 = \dfrac{0.01}{(1 - 0.01)} = 0.0101$

$F = 1000$ kg, $\quad (C = 10$ kg, $A = 990$ kg), $\quad B = 1500$ kg

$$-\frac{A}{B_n} = \frac{(Y_n - Y_s)}{(X_n - X_{n-1})} \quad \text{when } n = 1, \; -\frac{A}{B} = \frac{Y_1 - Y_s}{X_1 - X_0}$$

$$\frac{990}{1500} = \frac{Y_1 - 0}{0.0101 - X_1}$$

A line with a slope of -0.66 is drawn from (0.0101, 0) to obtain X_1 and Y_1.

$$Y_1 = 0.66\ [(0.0101) - X_1]$$

$$Y_1 = 0.0037 \text{ (From graph)}$$

$$X_1 = 0.0045$$

Amount of nicotine in extract = $0.0037 \times 1500 = 5.55$ kg

% extraction = $(5.55/10) \times 100 = 55.5\%$

For three stages

$(-A/B)$ = A $990/500 = -1.98$.

3 lines with a slope of -1.98 each are drawn starting from (0.0101,0)

$$X_3 = 0.0035, \qquad Y_3 = 0.003$$

Amount of nicotine in final extract = $0.003 \times 500 = 1.5$ kg

Total C extracted = $(Y_1 + Y_2 + Y_3) \times 500$

$= (0.0061 + 0.0037 + 0.003) \times 500 = 6.4$ kg

% extraction = $(6.4/10) \times 100 = 64\%$

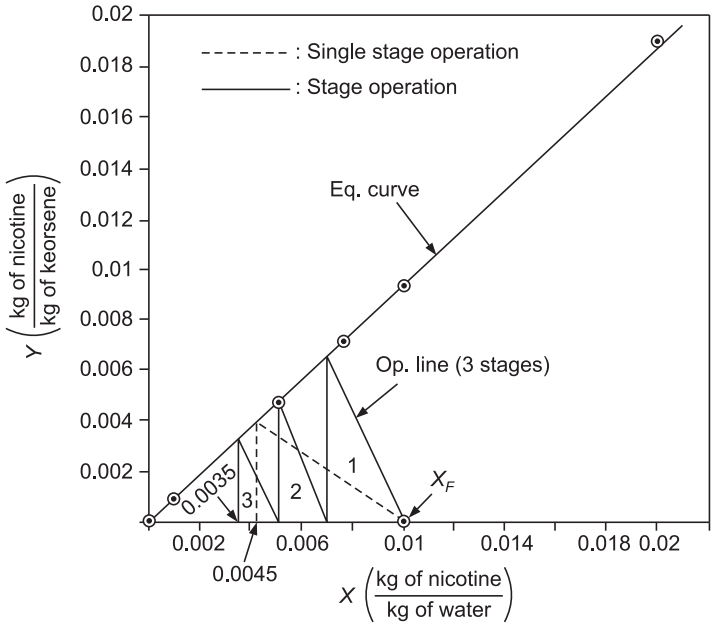


Fig. 10.31 Example 7.

8. 1000 kg/h of a water–dioxane solution containing 20% dioxane is to be continuously and counter-currently extracted with benzene at 25°C to recover 80% dioxane. Water and benzene are essentially insoluble and the equilibrium distribution of dioxane between them are as follows:

Dioxane in water wt. %	5.1	18.9	25.2
Dioxane in benzene wt. %	5.2	22.5	32.0

Determine the number of theoretical stages if the solvent rate is 1.5 times the minimum.

Solution.

Water: A Dioxane: C Benzene: B

x	0.051	0.189	0.252
y	0.052	0.225	0.32
$X = x/(1 - x)$	0.054	0.233	0.337
$Y = y/(1 - y)$	0.05485	0.29	0.471

F = 1000 kg/h

x_F = 0.2, $X_F = X_0 = 0.2/0.8 = 0.25$

Countercurrent extraction

$$X_{Np} = 0.2 \times 0.25 = 0.05$$

$$\frac{A}{B_{\min}} = \frac{Y_{Np+1} - Y_1}{X_{Np} - X_F}$$

$$\frac{800}{B_{\min}} = \frac{0 - 0.3075}{0.05 - 0.25} \quad \text{(From Graph)}$$

$$B_{\min} = 520.33 \text{ kg}$$

$$B_{\text{act}} = 1.5 B_{\min} = 1.5 \times 520.33 = 780.5 \text{ kg}$$

$$\frac{A}{B_{\text{act}}} = \frac{800}{780.5} = 1.025$$

Draw the operating line with a slope of 1.025 from (0.05,0) and by stepwise construction determine the number of stages.

No. of stages = 4

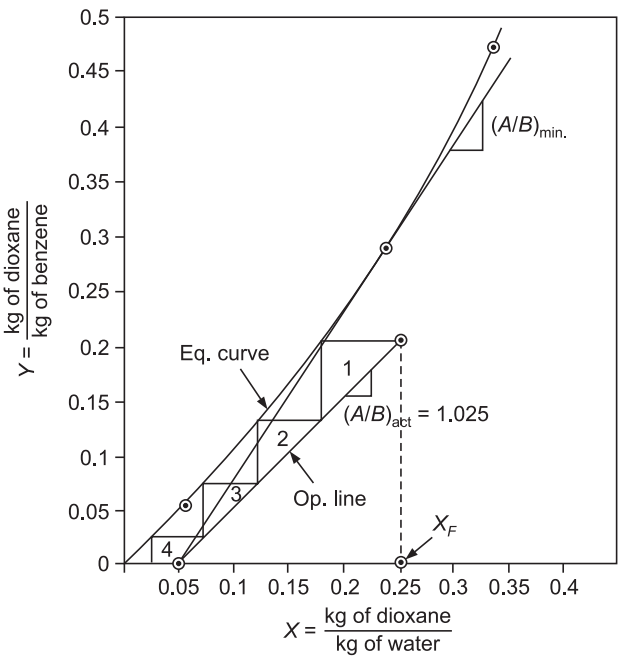


Fig. 10.32 Example 8.

EXERCISES

1. A 25% (weight) solution of dioxane in water is to be continuously extracted with 300 kg/hr of pure benzene in each stage in a cross-current extraction battery. The feed rate is 100 kg/h and if the extraction is carried out in 3 stages, estimate the % recovery.

Equilibrium data:

Dioxane in water wt. %	5.1	18.9	25.2
Dioxane in benzene wt. %	5.2	22.5	32.0

2. Repeat the above problem for a counter current extraction process using 1.5 times the minimum solvent and determine the number of stages needed to recover 90% of dioxane for a feed rate of 100 kg/h.

3. 1000 kg/hr of an acetone-water mixture containing 10% by weight of acetone is to be countercurrently extracted with trichloroethane. The recovered solvent to be used is free from acetone. The water and trichloroethane are insoluble. If 95% recovery of acetone is desired, estimate the number of stages required if 1.5 times the minimum solvent is used. The equilibrium relationship is given by $y = 1.65x$, where x and y are weight fractions of acetone in water and trichloroethane respectively.

4. Repeat problem 3 for a 4-stage cross-current operation using 300 kg/h of solvent in each stage and determine the % of recovery.

5. A 10% (by weight) solution of acetaldehyde in toluene is extracted with water in a countercurrent unit. For a 500 kg of feed, calculate the number of stages needed for reducing the acetaldehyde to 0.5% using 1.5 times the minimum amount of solvent. The equilibrium relationship is given by the equation, $Y = 2.3X$ where Y = kg acetaldehyde/kg water and X = kg acetaldehyde/kg toluene. Assume that toluene and water are immiscible with each other.

6. 500 kg/h of an aqueous solution containing 8% acetone is to be countercurrently extracted using monochlorobenzene to reduce the acetone content to 4% of its initial value. Water and monochlorobenzene are immiscible with each other. (i) Determine the minimum solvent rate, and (ii) the number of theoretical stages required if 1.3 times the minimum solvent rate is used. The equilibrium data is as follows:

kg acetone/kg water	0.030	0.074	0.161	0.210
kg acetone/kg monochlorobenzene	0.029	0.071	0.158	0.204

7. 150 kg of a solution containing acetic acid and water containing 20% acid by weight is to be extracted with isopropyl ether at 20°C. The total solvent used for extraction is 200 kg. Determine the composition and quantities of various streams if,

(i) The extraction is carried out in single stage,

(ii) The extraction is carried out in two stages with 100 kg of solvent in each stage.

Equilibrium data:

Water layer (wt. %)		*Ether layer* (wt. %)	
Acid	*Water*	*Acid*	*Water*
0.69	98.1	0.18	0.5
1.41	97.1	0.37	0.7
2.9	95.5	0.79	0.8
6.42	91.7	1.93	1
13.3	84.4	4.82	1.9
25.5	71.1	11.4	3.9
36.7	58.9	21.6	6.9
44.3	45.1	31.1	10.8
46.4	37.1	36.2	15.1

8. Repeat the Problem 7 for a countercurrent operation using 1.5 times the minimum solvent. Determine the percentage recovery after two stages.

9. 1000 kg/h of a pyridine water solution containing 50% pyridine is to be reduced to 10% by using Chlorobenzene in a countercurrent extraction battery. (i) Determine the minimum solvent requirement. By using twice the minimum rate of solvent, estimate the number of stages needed.

Chlorobenzene layer		*Water layer*	
Pyridine	*Chlorobenzene*	*Pyridine*	*Chlorobenzene*
0	99.5	0	0.08
11.05	88.28	5.02	0.16
18.95	79.9	11.05	0.24
24.1	74.28	18.9	0.38
28.6	69.15	25.5	0.58
31.55	65.58	36.1	1.85
35.08	61	44.95	4.18
40.6	53	53.2	8.9
49	37.8	49	37.8

10. Repeat problem 9 for a cross-current operation using solvent equivalent to the amount of Raffinate/feed entering each stage and estimate the number of stages needed.

11. 1000 kg/h of a solution of *C* in *A* containing 10% *C* by weight is to be countercurrently extracted with 500 kg/hr of solvent *B*. The components *A* and *B* are insoluble. The equilibrium distribution of component *C* between *A* and *B* are as follows:

Wt. of *C*/Wt. of *A*	0.05	0.20	0.30	0.45	0.50	0.54
Wt. of *C*/Wt. of *B*	0.25	0.40	0.50	0.65	0.70	0.74

How many theoretical stages will be required to reduce the concentration of *C* in A to 2%?

12. Acetone is to be recovered from dilute aqueous solutions by liquid–liquid extraction using toluene as solvent. The acetone concentration in the feed solution is 0.05 kmol/m^3 and 90% of this acetone is to be extracted by countercurrent operation. The flow rate of aqueous phase is 1.5 m^3/min. The equilibrium distribution ratio of acetone in the solvent and in the aqueous phase could be described by the relation, $y = 1.5x$, where x and y are weight fraction units.

13. Nicotine in a water solution containing 1% nicotine is to be extracted with kerosene at 20°C. Kerosene and water are insoluble. Determine the number of stage needed if 100 kilogram of feed solution is extracted once with 1.6 times the minimum amount of solvent to recover 95% nicotine.

Equilibrium data:

X	0	0.00101	0.00246	0.00502	0.00751	0.00998	0.0204
Y	0	0.00081	0.001962	0.00456	0.00686	0.00913	0.0187

where *X* is kg nicotine/kg water and *Y* is kg nicotine/kg kerosene.

14. 100 kg/h of a water–dioxane solution containing 15% dioxane is to be continuously and countercurrently extracted with benzene at 25°C to recover 95% dioxane. Water and benzene are essentially insoluble and the equilibrium distribution of dioxane between them are as follows:

Dioxane in water wt. %	5.1	18.9	25.2
Dioxane in benzene wt. %	5.2	22.5	32.0

(i) Determine the number of theoretical stages if the solvent rate is 1.5 times the minimum. (ii) If the same operation is done in a three-stage cross-current battery with 60 kg of solvent in each stage, estimate the required number of stages.

11

LEACHING

11.1 INTRODUCTION

Leaching is one of the oldest operations in chemical industries which involves the use of a solvent to remove a solute from a solid mixture. Though originally it was referred to the percolation of liquid through a bed of solids, it is now used to refer the operations by other contacting means also. *Lixiviation* is used for the leaching of alkali from wood ashes. *Decoction* refers to the operation where the solvent at its boiling is used. Whenever the solute material is present largely on the surface of an insoluble solid and is merely washed off by the solvent, the operation is called *elutriation or elution.*

It is one of the most important operations in metallurgical industries for the extraction of metals from ores of Al, Ni, Co, Mn and Zn. It is also used for the extraction of sugar from sugar beets with hot water, extraction of oil from oil seeds using organic solvents, removal of tannin from various tree barks by leaching with water, preparation of tea and coffee and extraction of many pharmaceutical products from plant roots and leaves.

The success of this operation depends on the proper preparation of the given solid. Depending on the nature of solid, the solid is crushed and ground to desired size to accelerate the leaching action. For example, a certain copper takes about 6 hours if crushed to – 60 mesh size and about 5 days for a size of 6 mm. Gold is sparsely distributed in its ore. Hence it is crushed to – 100 mesh size to have an effective leaching. Sugar beets are cut into thin wedge shaped slices called cassettes before leaching to enable the solvent water to reach the individual plant cells. In the manufacture of pharmaceutical products from plants they are dried in order to rupture the cell walls so that solvent can reach the solute easily. Vegetable seeds when used for the extraction of oil are crushed to a size of 0.15 to 0.5 mm to enable easier extraction. However, when the solid is present on the surface, no grinding or crushing is necessary and the particles can be washed directly.
To summarize, the leaching action depends on:

- The nature of solid/cell structure.

- Diffusion of solute from the material to surface and then to the bulk of the solution.
- Particle size and its distribution.
- Solubility of solute in solvent and the temperature of operation.

11.2 UNSTEADY STATE OPERATION

These operations are carried out batchwise or semibatchwise.

11.2.1 In Place (in-situ) Leaching

This operation is also called *solution mining* which refers to the percolation leaching of minerals in place at a mine, by circulation of the solvent over and through the ore bed. This technique is adopted for the leaching of low-grade copper. In these operations, the solvent/reagent is injected continuously through one set of pipes drilled down to the ore and the resulting solution is pumped out through another set of pipes. Alternatively, the solvent/reagent can be pumped into the ore bed intermittently and withdrawn through the same well. In this technique crushing and grinding of ore are avoided. In place leaching also called in-situ leaching is shown in Fig. 11.1.

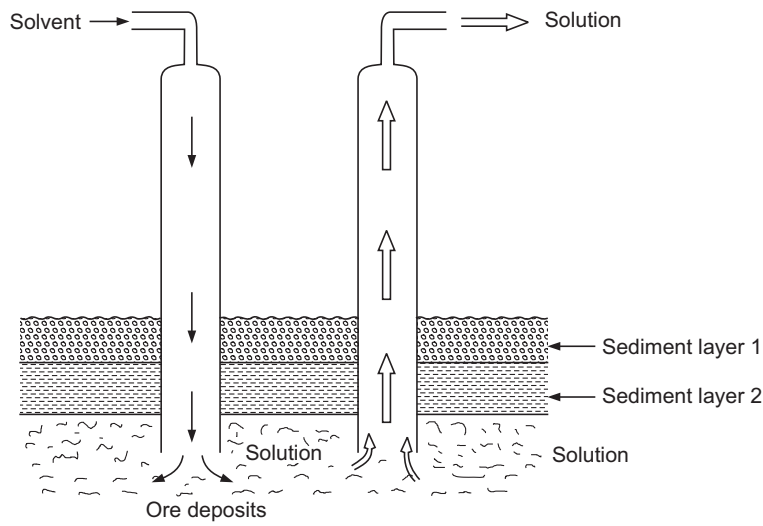


Fig. 11.1 In situ leaching.

11.2.2 Heap Leaching

Low-grade ores can be easily leached by this technique where the ore is gathered as a heap upon impervious ground. The leach liquor is pumped over the ore, which percolates through the heap and collected as it drains from the heap. This technique is used for the extraction of copper and uranium from their low grade ores. Heap leaching is shown in Fig. 11.2.

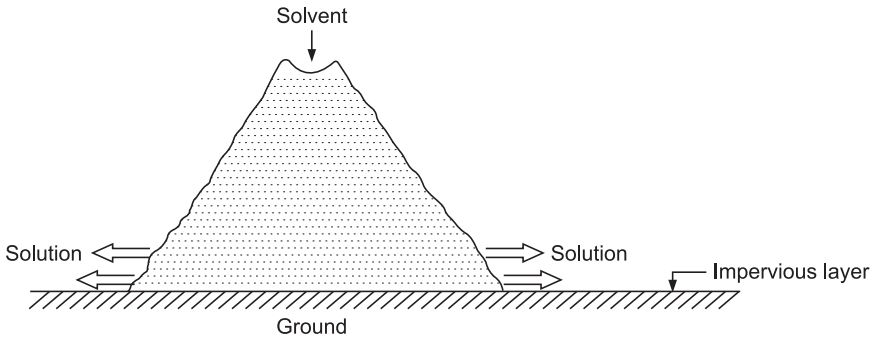


Fig. 11.2 Heap leaching.

11.2.3 Percolation Tank

Whenever small tanks are to be used, they can be made of metal or wood. The solid particles to be leached, rest on a false bottom which could be made of wood strips and may be covered by a coconut matting and a tightly stretched canvas filter cloth. The leach liquor flows to a collection pipe leading from the bottom of the tank. A very large percolation tanks are made of reinforced concrete and lined with lead or bituminous mastic. Small tanks may be provided with side doors near the bottom for removing the leached solid while the large tanks are emptied by excavating from the top. It is always preferable to fill the tanks with particles of uniform size so that voids will be more and the pressure drop required for flow of leaching liquid is least. This also leads to uniform leaching of individual particles.

For these operations the crushed solids may be filled in the tank initially and then the solvent is allowed to enter in. The solid and solvent may remain in contact with each other for a specified amount of time and then drained. During the process, if necessary, the liquid can also be circulated through the bed. The liquid can also be allowed to enter in continuously and also drained continuously. The liquid from the exit can also be recirculated, if necessary. The flow of liquid could either be downwards or upwards with proper distribution of liquid. Percolation tank is shown in Fig. 11.3.

11.2.4 Countercurrent Contact

At times, one is interested in getting a strong solution, which can be obtained by a countercurrent operation. This arrangement, also called *Shanks system*, contains number of tanks as shown in Fig. 11.4. The number of tanks generally vary from 6 to 16. In a typical system with 8 tanks at a particular time, tank 8 is empty and tanks 1 to 7 contain solids. Fresh solvent enters tank 1, where the solid has spent maximum amount of time and the material in tank 2, 3, 4, 5, 6 and 7 have progressively spent lesser time. The material in 7th tank has spent the least amount of time. The solution withdrawn from the 7th tank has the maximum solute concentration because the solution comes after contact with fresh solids. The solution withdrawn from tank 1 goes to tank 2, from tank 2 to tank 3, …, tank 6

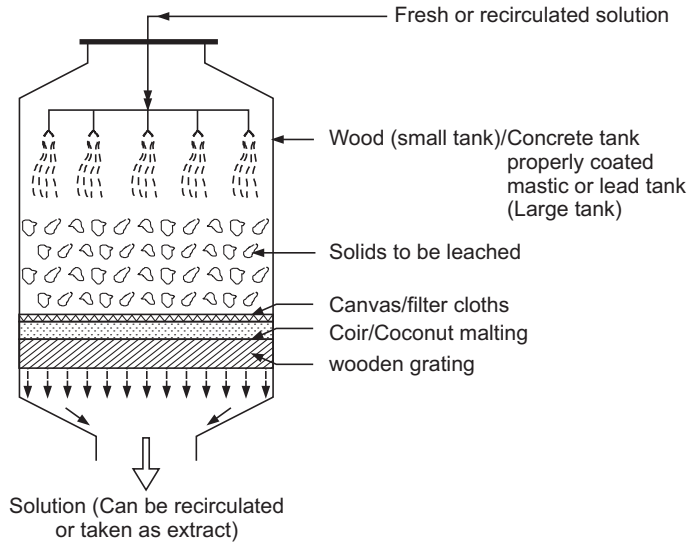


Fig. 11.3 Percolation tank.

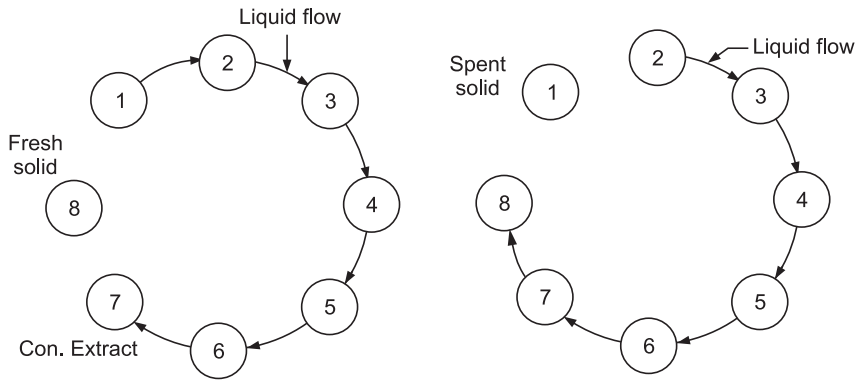


Fig. 11.4 Countercurrent system–shanks system.

to tank 7. The leached solid is discarded from tank 1 and fresh solid is now added in tank 8. The solution is transferred from tank 7 to 8, 6 to 7, 5 to 6, …, 2 to 3. Here the fresh solvent is added in tank 2, and the solid from tank 2 is finally discarded. The solution now obtained from tank '8' will have maximum solute concentration. Tank 1 which is now empty will be loaded with a fresh batch of solids. This is nothing but advancing the tanks by one. The operation is continued in this manner by keeping successive tanks as the first tank in which the fresh solvent enters. The solids move counter currently to liquid flow.

11.2.5 Percolations in Closed Vessels

At times the pressure drop for flow of liquids by gravity is high or the solvent is highly volatile. Under such circumstances the liquid is pumped through the bed of

solids in vessels called diffusers. The main advantage of these units is the prevention of evaporation losses of solvent, when they are operated above the boiling point of solvent (e.g. leaching of tannins using water at 120°C, 345 kN/m^2).

11.2.6 Filter–Press Leaching

When the solids are in finely divided form, percolation tanks are not suitable. Under such circumstances, solids can be filtered and leached in the filter press by pumping the solvent through the press cake. This is also a common feature while washing the filtered cakes.

11.2.7 Agitated Vessels

These are either vertical or horizontal closed cylindrical vessels with power-driven paddles or stirrers on vertical or horizontal shafts. They have a provision at the bottom for the withdrawal of leach solution at the end of the operation. In some of the designs, the horizontal drum is the extraction vessel, and the solid and liquid are tumbled about inside by rotation of the drum on rollers. They are operated on batch basis and each one is a single leaching stage. They can also be used in series for a multistage operation. For the leaching of finely divided solids, Pachuca tank is used. This finds extensive use in metallurgical industries. These tanks are constructed with wood, metal or concrete and lined with suitable material depending on the nature of leaching liquid. Agitation is accomplished by air lift. The bubbles rising through the central tube cause the upward flow of liquid and suspended solid in the tube and hence circulation of the mixture. Conventional mechanical agitators are also used for this purpose. Once the desired leaching is achieved, the agitation is stopped, the solids are allowed to settle and the clear supernatant liquid is decanted by siphoning over the top of the tank or by withdrawal through discharge pipes placed at appropriate level in the side of the tank. Whenever, the solids settled form a compressible sludge, the solution retained will be more and generally the last traces of solute in such cases are recovered in countercurrent manner.

11.2.8 Features of Percolation and Agitation Techniques

If a solid is in the form of big lumps, the question that arises is whether one should go in for percolation technique or agitation–settling technique. The problem is quite complicated due to the diverse leaching characteristics of the various solids and the value of solute. However, the following points are worth considering.

Though fine grinding is more costly and provides more rapid and possibly more thorough leaching, the quantity of liquid associated with the settled solid is very large. Hence, one may have to use large quantity of solvent to recover as much solute as possible. The composited extract thus obtained could be dilute. Coarsely ground particles, on the other hand, leach more slowly and possibly less thoroughly and may retain lesser quantity of solution. They may also require lesser

washing and hence the extract could be a concentrated one due to the use of lesser quantity of solvent.

Practical results have shown that leaching in an agitated vessel is more effective than by percolation for a fibrous solid like sugar cane.

Hence, one may have to decide based on the economy and the case of operation.

11.3 STEADY STATE OPERATIONS

They are classified as stagewise or continuous contact operations. Stagewise equipment are sometimes assembled in multiple units to produce multistage effects, whereas, continuous contact equipment provide the equivalent of many stages in a single unit. Some of the solids may also require grinding in order to make the soluble portions accessible to the leaching solvents. In fact, wet grinding is an operation during which some leaching could be accomplished. For example, 50 to 75% of the soluble gold may be dissolved by grinding the ore in the presence of cyanide solution. Castor oil is also extracted suitably in an attrition mill with solvent.

11.3.1 Agitated Vessels

Finely ground solids which can be readily suspended in liquids by agitation can be handled in agitated vessels. These must be arranged for continuous flow of both liquid and solid in and out of the tank. Care must be taken to ensure that no accumulation of solid occurs. Due to thorough mixing, equilibrium is always there between the solid and liquid. The agitated vessels discussed earlier can also be used.

The average holding time can be estimated both for solids and liquids separately in an agitated vessel by dividing the vessel contents by the rate of flow of solids and liquids. The average holding time should be adequate to provide the required leaching action. Short circuiting is a disadvantage encountered which can be eliminated by passing the solid–liquid mixture through a series of smaller agitated vessels such that the cumulative holding time is the required leach time. The effluent from continuous agitators are sent to a filter for separating liquid from solid upon which the solid may be washed free of dissolved solids, or to a series of thickeners for countercurrent washing.

11.3.2 Thickeners

There are mechanical devices which are meant for increasing the ratio of solid to liquid in a suspension of finely-sized particles by settling and decanting, thus producing a thickened sludge and a clear supernatant liquid. They are generally installed before any filter to minimize the filtering costs. Since both effluents can be pumped and transported, thickeners are frequently used to wash leached solids and chemical precipitates free of adhering solution in a continuous multistage countercurrent arrangement and hence worth their use in leaching operations also.

The liquid content in the sludge varies from 15 to 75% and is greatly dependent on the nature of the solids and liquids and upon the time allowed for settling. This is shown in Fig. 11.5.

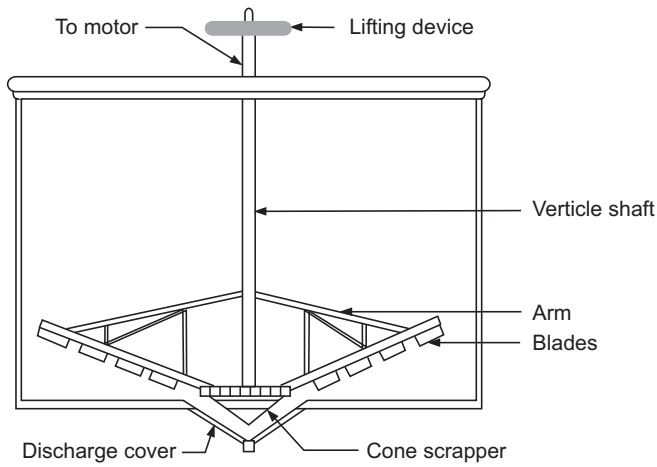


Fig. 11.5 Thickeners.

11.3.3 Continuous Countercurrent Decantation

It is an arrangement involving both the thickeners and agitators/grinders. The solids enter the first set of agitators/grinders and are mixed with overflow liquid from the 2nd thickener. Then the contents after through agitation/grinding enter the 1st thickener. The agitators along with thickener constitute the first stage. The sludge from the first thickener passes on to the 2nd thickener where it is mixed with overflow from the 3rd thickener and the sludge is then transferred to 3rd thickener where it is mixed with overflow liquid from 4th. Fresh solvent enters the last thickener. The overflow liquid taken out from the first thickener will have the maximum concentration of solute. If necessary the sludge from each stage can be thoroughly agitated with the solvent in order to effect better separation. This is shown in Fig. 11.6.

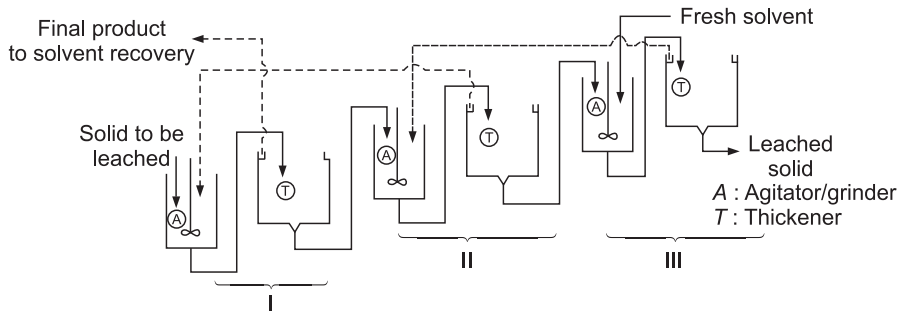


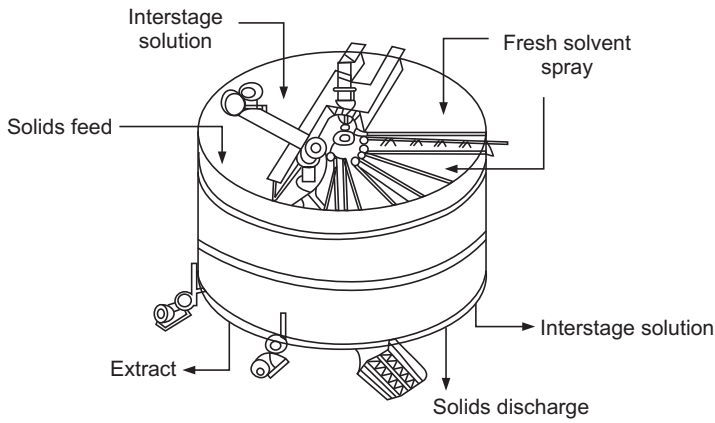
Fig. 11.6 Continuous countercurrent decantation.

11.3.4 Leaching of Vegetable Seeds

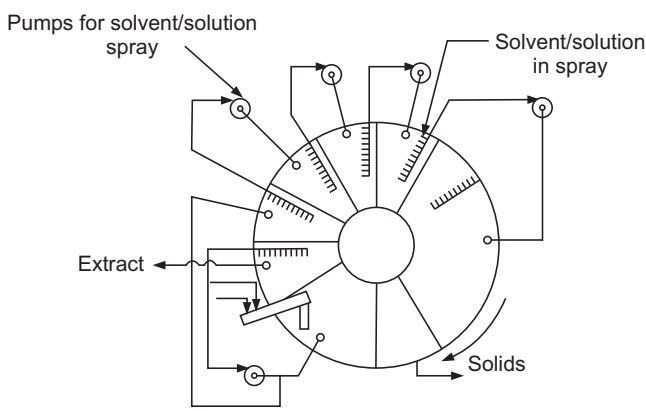
Soya beans, cotton seeds, rice bran and castor seeds are some of the products regularly leached with an organic solvent for removing the oil present in them. The process involves dehulling, precooking, adjustment of water content and flaking. In some cases solvent extraction of oil is preceded by mechanical expression of oil from oil seeds. Leaching solvents are generally petroleum fractions. Chlorinated hydrocarbons leave the residue meal a toxic one. The oil-solvent solution containing a small amount of finely divided suspended solids is called *miscella* and the leached solid is called *marc*.

11.3.4.1 Rotocel extractor

It is a modification of shanks system wherein the leaching tanks are continuously moved, permitting a continuous introduction and discharge of solids. It is shown in Fig. 11.7(a) and (b).



(a) Rotocel extractor (front view).



(b) Rotocel extractor (top view).

Fig. 11.7

It consists of a circular shell partitioned into several cells each fitted with a hinged screen bottom for supporting the solids. This shell slowly revolves above a stationary compartmented tank. As the rotor revolves, each cell passes in turn under the prepared solids feeder and then under a series of sprays by which the contents in each cell is periodically drenched with solvent for leaching. By the time one rotation is completed, when the leaching is expected to be completed, the leached solids of each cell are automatically dumped into one of the lower stationary compartments, from which they are continuously conveyed away. The solvent sprayed over each cell filled with solids, percolates downward through the solid and supporting screen into the appropriate compartment of the lower tank from which it is pumped to the next spray. The leaching is countercurrent, and the strongest solution comes from the cell which is filled with fresh solid. It is essential to maintain the equipment properly to ensure smooth operation. It is also enclosed in a vapour tight housing to prevent the escaping of solvent vapours.

11.3.4.2 Kennedy extractor

A schematic arrangement is shown in Fig. 11.8. It is a stagewise device, originally used for leaching tannins from tan bark. The solids are leached in a series of tubs and are pushed from one to next in the cascade by perforated paddles, while the solvent flows in countercurrent direction. Perforations in paddles permit drainage of liquid from solids between stages, and the solids are scrapped from each paddle as shown in Fig. 11.8. The number of tubs depends on the nature of solid, solvent and the level of extraction desired. Since it has a horizontal orientation, more floor space is required.

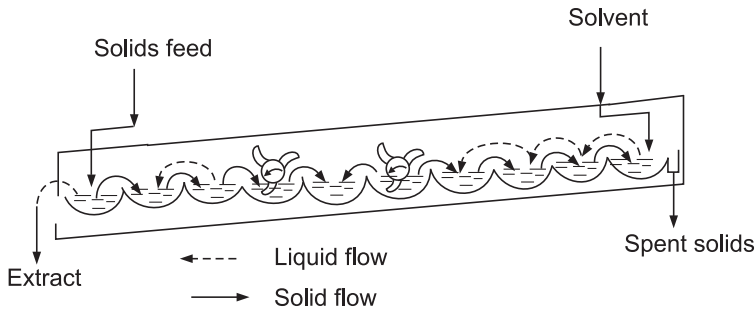


Fig. 11.8 Kennedy extractor.

11.3.4.3 Bollman extractor

It has a vertical orientation and has several perforated baskets attached to a chain conveyor for conveying solids. As the chain descends, the solids are leached in parallel flow by a dilute solvent – oil solution, called *half miscella*, pumped from the bottom of the vessel and sprayed over the baskets at the top. The liquid percolates through the solids from basket to basket and collects at the bottom as a final strong solution called *full miscella* and is withdrawn. On the ascent, the solids are leached countercurrently by a spray of fresh solvent and the product is called *half miscella*. A short drainage time is provided before leached solid in the baskets are dumped at the top. A schematic arrangement is shown in Fig. 11.9.

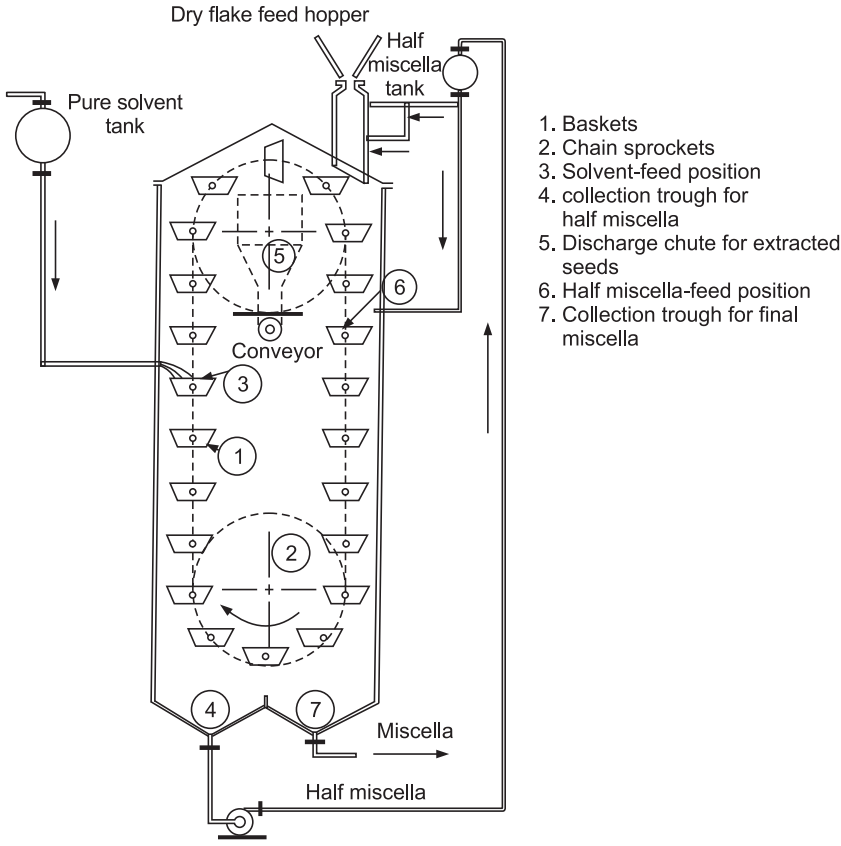


Fig. 11.9 Bollman extractor.

11.3.4.4 Continuous horizontal filter

A schematic arrangement of continuous horizontal filter is shown in Fig. 11.10. The filter in the form of a circular wheel is divided into a number of sectors and revolves in the horizontal plane. Here prepared seeds are slurried with solvent which has already been used for leaching, and the slurry is sent to the filter. The

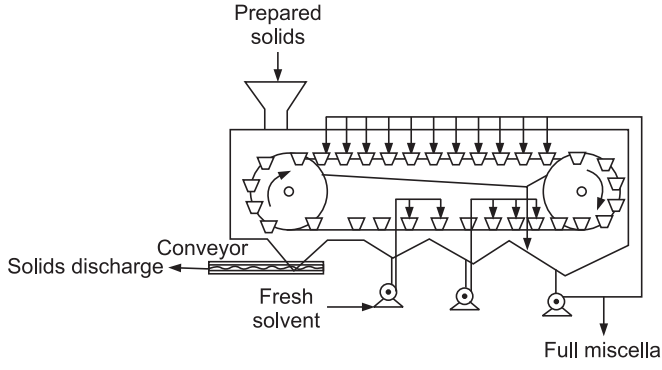


Fig. 11.10 Continuous horizontal filter.

first filtrate is passed again through the filter cake to remove finely divided solids (polishing) before being discharged as miscella. The principle behind the operation is quite similar to Rotocel extractor.

11.3.4.5 Recovery of oil

The recovery of solvent from both the miscella and leached solids is an essential feature in these operations. Recovery of oil in miscella is accomplished by evaporation of solvent and if necessary by further stripping in a tray column to remove the solvent–free oil. The oil in solid is removed by steaming and subsequent cooling. Vent gas from condensers can be sent to an absorber and scrubbed with petroleum white oil and the resulting mixture can be stripped to recover the solvent.

11.4 DEFINITIONS

Let, B = insoluble solid or inert solid (kg),
C = soluble solute (kg),
A = pure solvent (kg),

$x = \frac{C}{A+C}$; Weight fraction of solute in effluent solution (on B free basis)

$y = \frac{C}{A+C}$; Weight fraction of solute in the solid or slurry or sludge (on B free basis)

$N = \frac{B}{A+C}$; (in each phase)

The variation of N, x and y under different conditions are as follows:

(a) For a dry solid (free from solvent) $N = \frac{B}{C}$ ($\because$ $A = 0$)

$y = 1.0$

(b) Solid free from solvent and solute $N = \infty$ ($\because$ $A = 0$; $C = 0$)

(c) Pure solvent $x = 0$, $N = 0$ ($\because$ $B = 0$; $C = 0$)

11.5 DIFFERENT TYPES OF EQUILIBRIUM DIAGRAMS

11.5.1 Type 1

A typical trend of N vs x, y and equilibrium relationship is shown in Fig. 11.11(a) characteristics of such systems are:

- Preferential adsorption of the solute occurs on solid.
- Solute is soluble in the solid B and distributes unequally between liquid and solid phases at equilibrium.
- Insufficient contact time between solute and solvent.
- EF is a tie line.

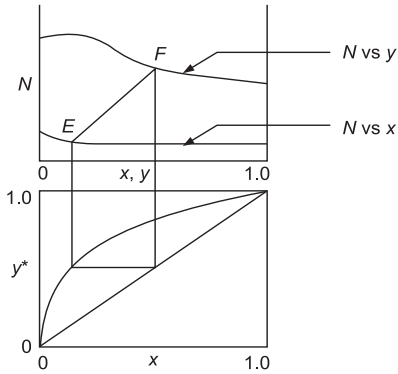


Fig. 11.11(a) Type I equilibrium.

11.5.2 Type II

A typical trend of N vs x, y and equilibrium relationship is shown in Fig. 11.11(b). The characteristics of such systems are:

- No adsorption of solute occurs.
- Solution withdrawn and the solution associated with the solid have the same composition.
- Tie lines are vertical.
- The distribution coefficient is unity.
- Solids are drained to the same extent at all solute concentrations and such a condition is known as *constant underflow condition.*
- No B is present in solution either dissolved or suspended.

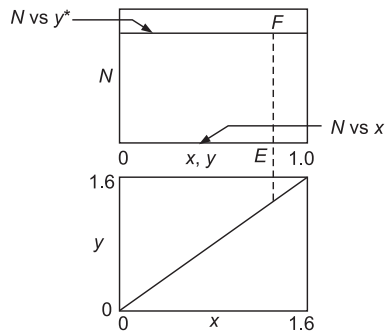


Fig. 11.11(b) Type II rquilibrium.

11.5.3 Type III

A typical trend of N vs x, y and equilibrium relationship is shown in Fig. 11.11(c). The characteristics of such systems are:

- Solute C has a limited solubility x_S in solvent A and one can never have a clear (leach) solution stronger than x_S.
- Tie lines joining slurry and saturated solution converge as shown.
- Till the concentration of x_S is reached, the solution retained in the solid and

the clear solution have some concentration and hence the distribution coefficient is unity, i.e. up to the tie line FE. No adsorption of solute occurs.

- The tie lines to the right of FE indicate the same solute concentration in clear solution but a different solute concentration in slurry as indicated by points G, H.

In practice we come across situations which will fall in any one of the above three types.

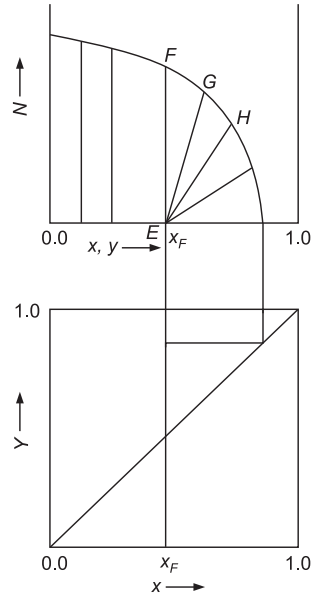


Fig. 11.11(c) Type III equilibrium.

11.6 SINGLE STAGE OPERATION

A typical single stage operation is shown in Fig. 11.2. The characteristics of various streams flowing into and out of the system are also shown. The flow rate of streams are on B free basis.

(All the streams are on absolute mass basis)

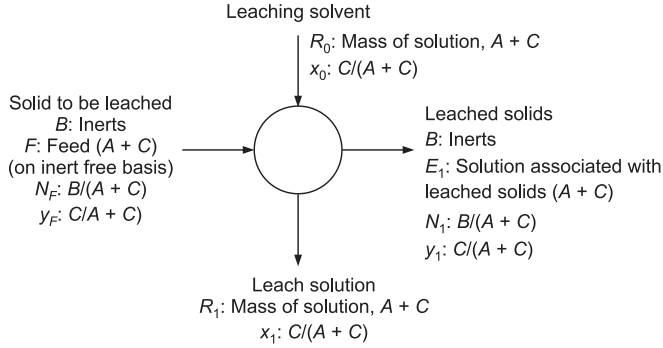


Fig. 11.12 Streams in a leaching operation.

$$N_F = \frac{B}{A+C} = \frac{B}{F}$$

$$N_1 = \frac{B}{A+C} = \frac{B}{E_1}$$

$$\therefore \qquad B = N_F \cdot F = E_1 N_1$$

Total material balance gives,

$$F + R_0 = E_1 + R_1 = M_1$$

Solute balance gives,

$$Fy_F + R_0 x_0 = E_1 y_1 + R_1 x_1$$

Solvent balance gives

$$F\ (1 - y_F) + R_0\ (1 - x_0) = E_1\ (1 - y_1) + R_1\ (1 - x_1)$$

When the solids and the solvent are mixed together in a stage (say, stage 1), the effective value of '*N*', called N_{M_1}, will be given by

$$N_{M_1} = \frac{B}{F + R_0} = \frac{B}{M_1}$$

Similarly the concentration of solute after thorough mixing in the stages is given by,

$$y_{M_1} = \frac{y_F F + R_0 X_0}{F + R_0}$$

Using the values of y_{M_1}, N_{M_1} and *N* vs. *x*, *y* diagram, one can determine the concentration and flow rates of leaving streams as indicated below.

(The co-ordinates (y_{M_1}, N_{M_1}) can be represented as shown in Fig. 11.13 in *N* vs *x*, *y* diagram).

Steps

(i) Draw the *N* vs *x*, *y* diagram.

(ii) Draw the distribution curve.

(iii) Locate *F* (y_F, N_F) and R_0 (x_0, N_0).

(iv) Join R_0 *F*.

(v) Locate M_1 (y_{M_1}, N_{M_1}) in R_0F line.

(vi) Draw the tie line R_1E_1 passing through M_1 with the help of distribution curve and read N_1 from *N* vs *y* curve.

(vii) $E_1 = B/N_1$ (weight of solution associated with sludge)

We know that,

$$F + R_0 = E_1 + R_1$$

Hence, R_1, the weight of clear solution can be estimated.

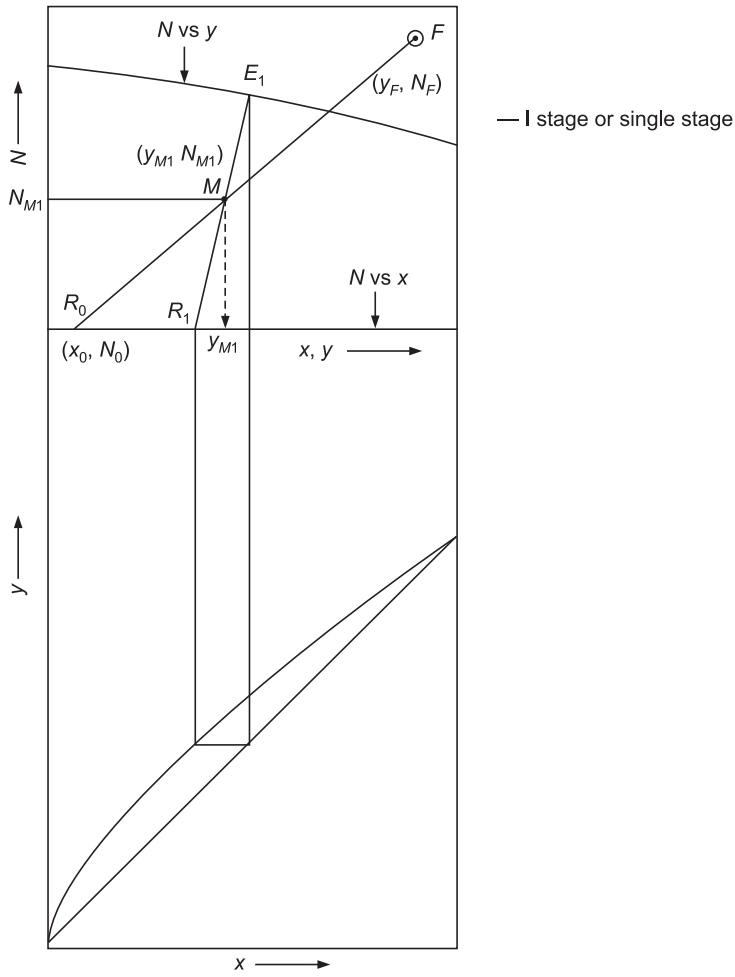


Fig. 11.13 Single stage operation.

11.7 MULTISTAGE CROSS-CURRENT LEACHING

In a multistage cross-current leaching E_1 stream from the I stage becomes the feed stream for the II stage and the E_2 stream from the II stage becomes the feed stream for the III stage. In each stage the mixture is contacted with a fresh stream of solvent.

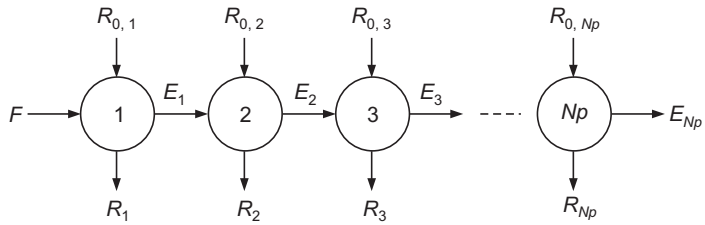


Fig. 11.14 A typical multistage cross-current operation.

Steps

(i) Proceed as per the procedure mentioned in steps (i) to (vii) of single stage operation of section 11.6.

(ii) Join E_1 with R_0 and locate M_2 (y_{M_2}, N_{M_2}). Generally R_{01}, R_{02}, R_{03}, ..., $R_{0,NP}$ are all same $R_{01} = R_{02} = R_{03} \ldots = R_{0,NP} = R_0$.

(iii) Draw the tie line E_2R_2 passing through M_2 and locate N_2.

(iv) $E_2 = \dfrac{B}{N_2}$

(v) We know from material balance, $E_1 + R_0 = E_2 + R_2$.

(vi) Hence, the unknown quantity R_2 (weight of clear solution) can be determined since the remaining quantities (E_1, R_0 and E_2) are all known.

(vii) Proceed in the same manner for other stages also.

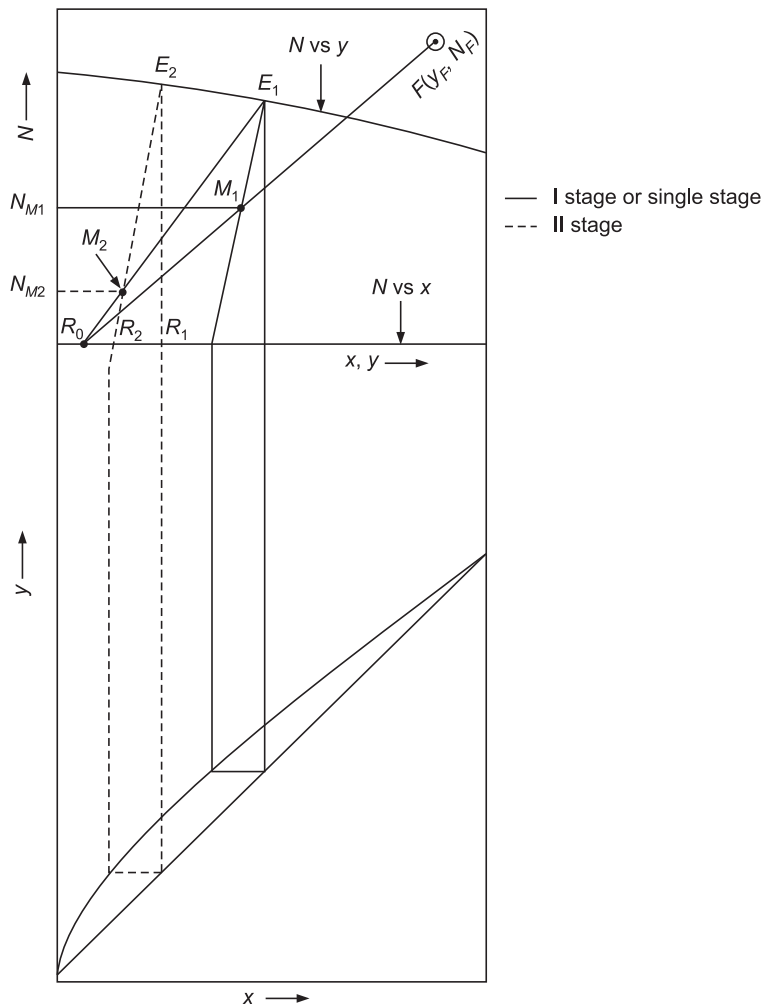


Fig. 11.15 Multistage cross-current operation.

11.8 MULTISTAGE COUNTERCURRENT OPERATION

Solution balance for the system as a whole gives,

$$F + R_{N_p+1} = R_1 + E_{N_p} = M$$

where M is the total mass of B (inert) free mixture.

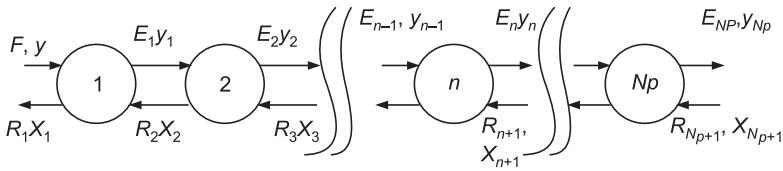


Fig. 11.16 A typical multistage countercurrent operation.

Solute balance gives,

$$F \cdot y_F + R_{N_p+1} \cdot X_{N_p+1} = R_1 x_1 + E_{N_p} \cdot y_{N_p} = M \cdot y_M$$

where,

$$N_M = \frac{B}{F + R_{N_p+1}}$$

$$y_M = \frac{Fy_F + R_{N_p+1} X_{N_p+1}}{F + R_{N_p+1}}$$

$$F - R_1 = E_{N_p} - R_{N_p+1} = \Delta_R$$

A solution balance for the first two stages gives

$$F + R_2 = R_1 + E_1$$

i.e.

$$F - R_1 = E_1 - R_2 = \Delta_R.$$

Similarly a solution balance for the first two stages yields

$$F - R_1 = E_2 - R_3 = \Delta_R.$$

It clearly indicates that the difference in flow between streams at either ends in each stage remains constant.

In a typical operation, the number of stages (for a given recovery and a given amount of solvent) or concentration of solute in the leaving stream (for a given number of stages and solvent used) or the amount of solvent (for a given number of stages and percentage recovery) will be needed.

11.8.1 Analysis of Variable Underflow System

11.8.1.1 Case I

Determination of stages for a specified recovery or final concentration:

Steps

1. Draw N vs x and N vs y and the distribution curve.
2. Locate the points F, E_{N_p} and R_{N_p+1}.

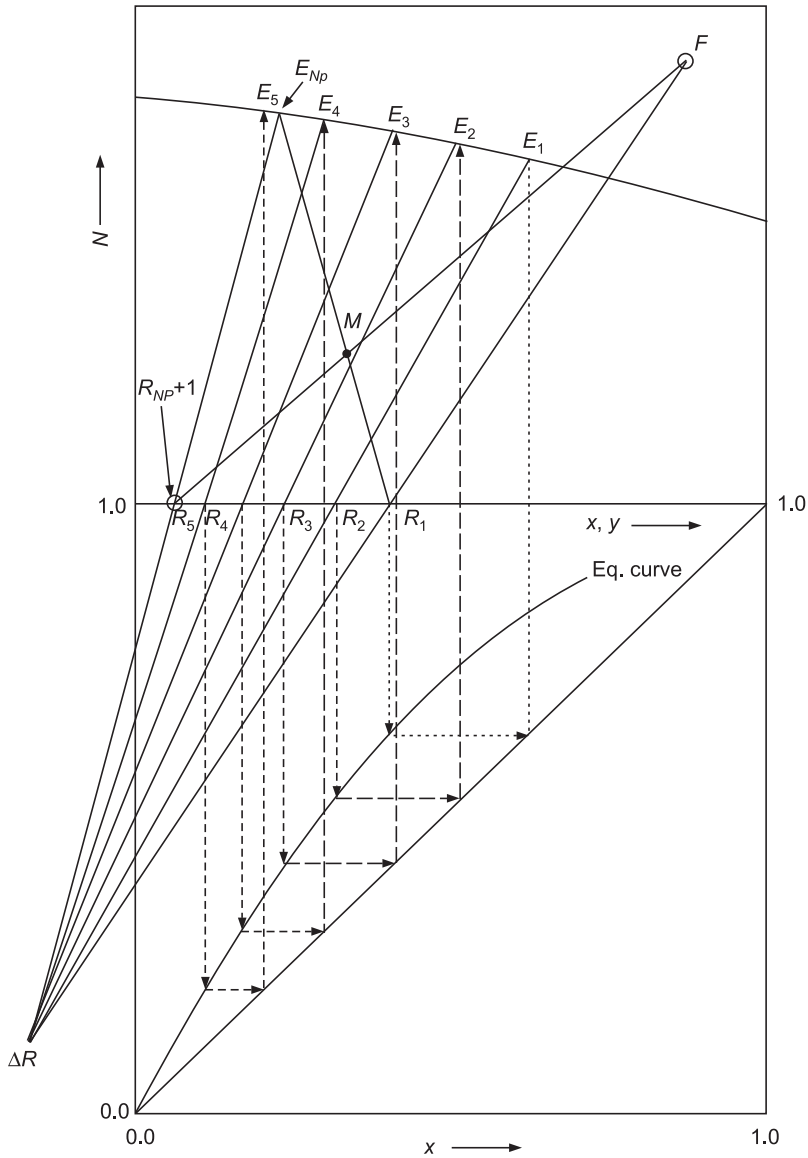


Fig. 11.17 Multistage countercurrent operation.

3. Estimate M (y_M, N_M) and locate it on FR_{N_p+1} line.
4. Join E_{N_p} with M and extend it to cut N vs x curve at R_1.
5. Join E_{N_p} with R_{N_p+1} and extend it.
6. Join F with R_1 and extend it to cut the $E_{N_p}R_{N_p+1}$ line and call the point of intersection as Δ_R.
7. Using R_1 and equilibrium curve, locate E_1. This corresponds to stage 1.
8. Join E_1 with Δ_R and this line cuts N vs x curve at R_2.

9. Using R_2 and equilibrium curve, locate E_2. This corresponds to stage 2.
10. Proceed in this manner till E_{N_p} is reached or crossed.
11. From this, the number of stages N_p can be determined.

11.8.1.2 Case II

If final concentration is needed or percentage recovery is needed for a given number of stages:

Assume some E_{N_p} value and proceed as mentioned above and verify whether the assumed E_{N_p} matches the given number of stages. If it does not match, assume a new value for E_{N_p} and proceed till the given number of stages and the assumed E_{N_p} value match.

11.8.1.3 Case III

If the solvent amount is needed:

Assume some solvent flow rate and check whether the E_{N_p} and stages match. If they do not match, assume a different value and proceed till the assumed E_{N_p} value and the given stages match.

11.8.1.4 Case IV

Minimum solvent requirement:

It is a specific solvent quantity at which the operating line becomes a tie line, i.e. FR_1 or E_1R_2 or E_2R_3, ..., becomes a tie line.

11.8.2 Number of Stages for a Constant Underflow System

The number of stages can be determined easily for constant underflow systems as the slope is constant ($m = y/x$) and operating line is straight, by using the Kremser, Brown and Souder's equation.

$$\frac{y_F - y_{N_p}}{y_1 - mx_{N_p+1}} = \frac{\left(\frac{R}{mE}\right)^{N_p+1} - \left(\frac{R}{mE}\right)}{\left(\frac{R}{mE}\right)^{N_p+1} - 1}$$

WORKED EXAMPLES

1. Oil is to be extracted from halibut liver in a countercurrent extraction battery. The entrainment of solution by the granulated liver mass is given below.

kg solution retained/ kg of exhausted liver	0.035	0.042	0.05	0.058	0.068	0.081	0.099	0.12
kg of oil/ kg of solution	0	0.1	0.2	0.3	0.4	0.5	0.6	0.68

In the extraction battery change is to be 100 kg based on completely exhausted liver. The unextracted liver contains 0.043 kg of oil/kg of exhausted material. 95% recovery is desired. The final extract is to contain 0.65 kg oil/kg of extract. The ether used as solvent is free from oil. How many kg of ether is needed per kg of liver? How many extractors are needed?

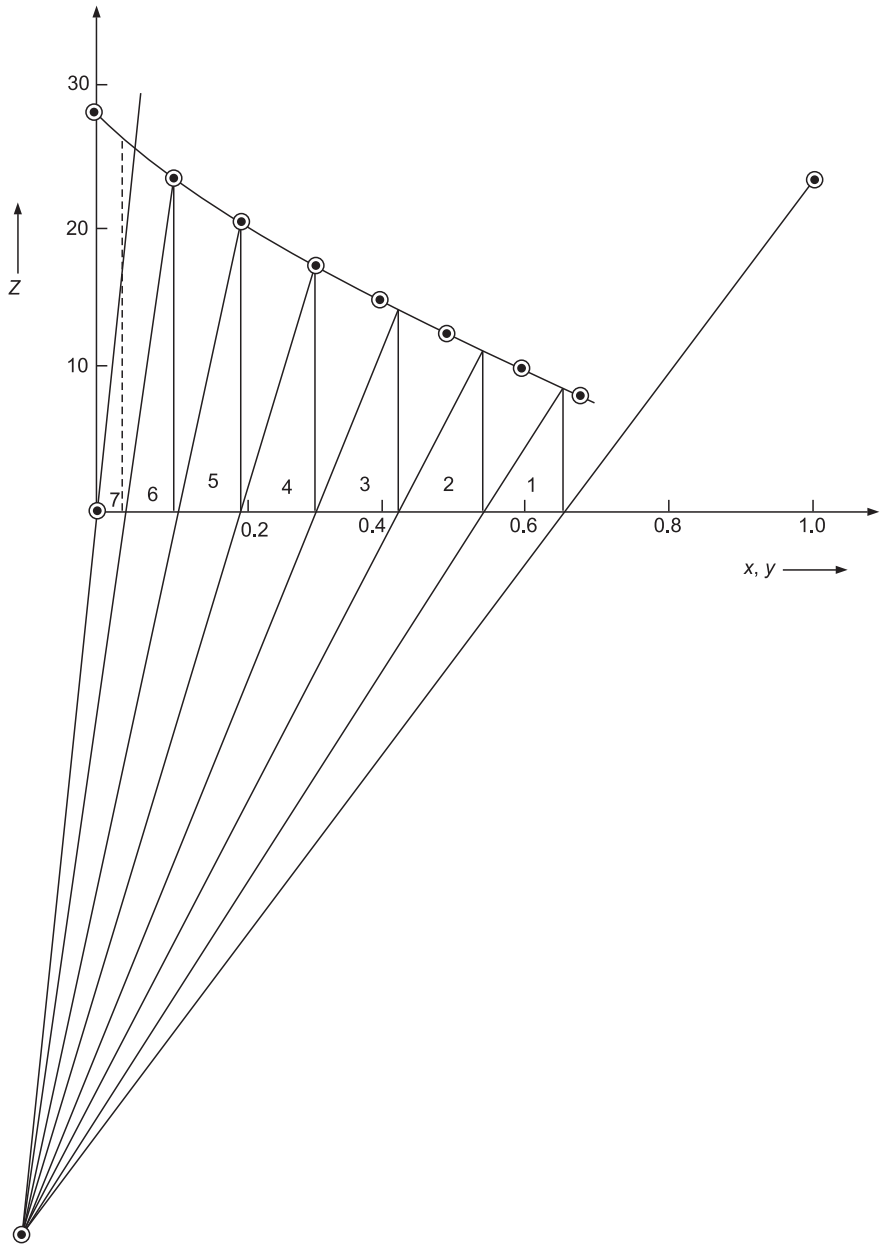


Fig. 11.18 Example 1.

Solution.

$\frac{\text{kg solution retained}}{\text{kg exhausted liver}}, 1/N$	0.035	0.042	0.05	0.058	0.068	0.081	0.099	0.12
$\frac{\text{kg oil}}{\text{kg solution}}, x, y$	0	0.1	0.2	0.3	0.4	0.5	0.6	0.68
$N = \frac{\text{kg exhausted liver}}{\text{kg solution retained}}$	28.6	23.8	20.0	17.25	14.7	12.35	10.1	8.3

Basis: 100 kg of exhausted liver

i.e. $B = 100$ kg

C (oil) $= 100 \times 0.043 = 4.3$ kg

$F = A + C$

$A = 0$ (solvent is not present)

$$N_F = \frac{B}{A+C} = \frac{100}{4.3} = 23.26$$

$$y_F = \frac{C}{A+C} = \frac{4.3}{0+4.3} = 1.0$$

$\therefore$ Feed point F is given by $(N_F, y_F) = (23.26, 1.0)$

The final extract contains 0.65 kg of oil/kg of extract

$\therefore$ R_1 is given by $(N_1, x_1) = (0, 0.65)$

R_{N_p+1} is given by $(N_{N_p+1}, x_{N_p+1}) = (0, 0)$

95% recovery of oil is to be achieved.

$\therefore$ 5% oil leaves with the liver.

i.e. oil leaving is $4.3 \times 0.05 = 0.215$ kg

$\therefore$
$$y_{N_p} = \frac{0.215}{A+0.215}$$

$\therefore$ E_{N_p} is given by (N_{Np}, y_{Np})

$$N_{Np} = \frac{B}{A+C} = \frac{100}{A+0.215}$$

$\therefore$ Slope of operating line $\left(\frac{B}{A+C}\right) \Big/ \left(\frac{C}{A+C}\right) = \frac{100}{0.215} = 465$

From the plot $E_{N_p} = (25.5, 0.055)$

Stages needed = 7.

$$N_{N_p} = 25.5 = \frac{B}{A+C}$$

$\therefore$
$$A + C = \frac{B}{25.5} = \frac{100}{25.5} = 3.92$$

$$y_{N_p} = 0.055 = \frac{C}{A+C}$$

$$A + C = 3.92$$

$\therefore$ $A = 3.92 - C = 3.92 - 0.215 = 3.705$ kg

i.e. amount of solvent in liver = 3.705 kg

Quantity of ether used

Extract contains 0.65 kg oil/kg extract

i.e. Extract contains 0.35 kg ether/kg extract

$$R_1 = \frac{C}{A + C} = 0.65$$

But C, Oil in extract = Total oil fed – Oil in exhausted liver
= 4.3 – 0.215 = 4.085 kg

$$\therefore \qquad R_1 = 0.65 = \frac{4.085}{A + 4.085}$$

$$\therefore \qquad A = 2.2 \text{ kg}$$

∴ Total ether used = Amount in extract + Amount in exhausted liver
= 2.2 + 3.705 = 5.905 kg.

2. 10 tonnes/hour of day seashore sand containing 1% by weight of salt is to be washed with 10 tonnes/hour of fresh water running countercurrent to the sand through two classifiers in series. Assume perfect mixing of sand and water occurs in each classifier and that the sand discharged from each classifier contain one part of water for every two parts of sand by weight. If the washed sand is dried in kiln, what % of salt will it retain? What wash rate is required in a single classifier in order to wash the sand equally well?

Solution.

Let x be the fraction of salt in the underflow discharge from stage 1.

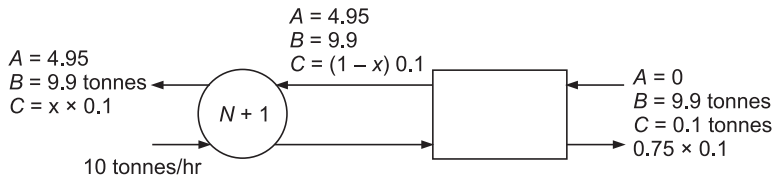


Fig. 11.19 Example 2.

Sand entering 9.9 tonnes/hour. Salt entering = 0.1 tonnes/hour. 1 part of sand discharged associated with 0.5 parts of water.

9.9 tonnes of sand leaving will be associated with 9.9/2 = 4.95 tonnes of water each stage.

By Coulson–Richardson method,

$$\frac{S_{n+1}}{S_1} = \frac{R - 1}{R^{n+1} - 1}$$

where S_1 = Quantity of solute in the sludge coming out from stage 1,
S_{n+1} = Quantity of solute in the sludge coming out from stage $n + 1$.

$$R = \frac{\text{Quantity of solution in overflow (solute or solvent)}}{\text{Quantity of solution in underflow (solute or solvent)}}$$

$$R = \frac{\text{weight of solution in overflow}}{\text{weight of soluton in underflow}} = \frac{10}{4.95} = 2.02$$

$$S_{n+1} = (x) \times (0.1)$$

$$S_1 = (1 - x) \times (0.1)$$

$$\therefore \quad = \frac{2.02 - 1}{(2.02)^2 - 1} = \frac{x}{(1-x)} = \frac{1.02}{3.08}$$

$$\therefore \quad 3.02 = \frac{1-x}{x}$$

$$\therefore \quad x = 0.249$$

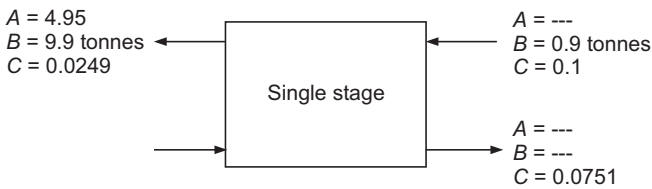


Fig. 11.20 Example 3.

Concentration in underflow

$$\therefore \quad x_1 = \frac{C}{A+C} = \frac{0.0249}{4.95 + 0.0249} = 0.5 \times 10^{-2}$$

x_1 in overflow (same as underflow) = $\dfrac{0.0751}{A + 0.0751} = 0.5 \times 10^{-2}$

$\therefore$ A = 14.93 (amount of water with extract)

Amount of water with sand = 4.95

$\therefore$ Total feed water = Water in extract + Water in sand

= 14.93 + 4.95 = 19.88

3. 100 tonnes of underflow feed containing 20 tonnes of solute. 2 tonnes of H_2O, 78 tonnes of inerts are to be leached with water to give an overflow of concentration, 15% solute. 95% recovery is desired. The underflow from each stage carries 0.5 kg of solution/kg of inert. Estimate the number of stages needed.

Solution.

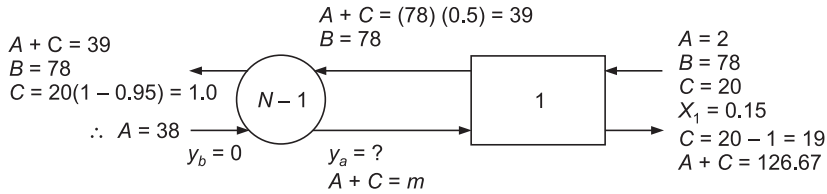


Fig. 11.21 Example 3.

$$y_b^* = \frac{C}{A+C} = \frac{1}{39} = 0.0256$$

x_1 (Desired outlet concentration of overflow) = 0.15

i.e. $$\frac{C}{A+C} = 0.15 = \frac{19}{A+19}$$

$\therefore$ $A + C = 126.67$ tonnes

Let us make a mass balance around stage 1.
Entering liquid = Leaving liquid

$$22 + m = 126.67 + 39$$

$\therefore$ $m = 143.67$ tonnes

Similarly making a solute balance, we get

$$20 + 143.67\, y_a = 19 + 39 \times 0.15$$

$\therefore$ $y_a = 0.034$

Solving by McCabe's method, we get

$y_b = 0$; $y_b^* = 0.0256$; $y_a = 0.034$; $y_a^* = 0.15$

($\because$ the leaving streams are in equilibrium)

$$(N-1) = \frac{\log\left[\dfrac{y_b - y_b^*}{y_a - y_a^*}\right]}{\log\left[\dfrac{y_b - y_a}{y_b^* - y_a^*}\right]} = \frac{-0.6562}{-0.5633} = 1.165$$

$N = 2.165$

Baker's method:

$$\frac{R^{n+1} - 1}{R - 1} = \frac{S_1}{S_{n+1}}$$

$S_1 = 39 \times 0.15 = 5.85$; $S_{n+1} = 1.0$

$$R = \frac{\text{solution/solute or solvent in overflow}}{\text{solution/solute or solvent in underflow}} \frac{143.7}{39} = 3.685$$

$$\frac{3.685^{n+1} - 1}{2.685} = \frac{5.85}{1};\ n + 1 = 2.159 \text{ stages}$$

4. A plant produces 100 tonnes/day of TiO_2 pigment which must be 99.9% pure when dried. The pigment produces by precipitation and the material as prepared is contaminated with 1 ton of salt solution containing 0.55 ton of salt/ton of pigment. The material is washed countercurrently with water in a number of thickeners arranged in series. How many thickeners will be required if water is added at the rate of 200 tonnes/day and the solid discharged from each thickener removed 0.5 ton of solvent/ton of pigment.

What will be the number of thickeners if the amount of solution removed in association with pigment varies in the following way with the concentration of the solution in the thickeners.

x	0	0.1	0.2	0.3	0.4	0.5
N	3.333	3.125	2.94	2.78	2.63	2.5

Solution.

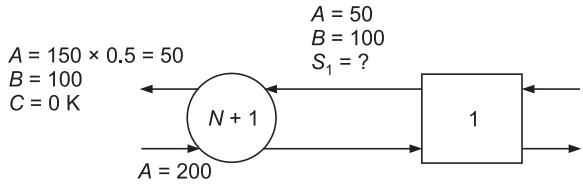


Fig. 11.22 Example 4.

Concentrated wash liquor is fed with the feed top concentrator = 1

$$A + C = 100;\ A = 45;\ B = 100$$

$$C = 0.55 \times 100 = 55$$

$$N_F = \frac{B}{A+C} = 1.0$$

$$y_F = \frac{C}{A+C} = 0.55$$

$$C = 55 - 0.1 = 54.9$$

$$A = 200 + 45 - 50 = 195$$

$$y = \frac{C}{A+C} = \frac{54.9}{249.9} = 0.22$$

$$x_1 = 0.22 = \frac{C}{A+C}$$

$$\frac{R^{n+1}-1}{R-1} = \frac{S_1}{S_{n+1}}$$

$$R = \frac{200}{50} = 4$$

$$S_1 = 14.1$$

$$S_{n+1} = 0.1$$

$$\frac{4^{n+1}-1}{4-1} = \frac{14.1}{0.1}$$

$$\therefore \quad n + 1 = 4.36$$

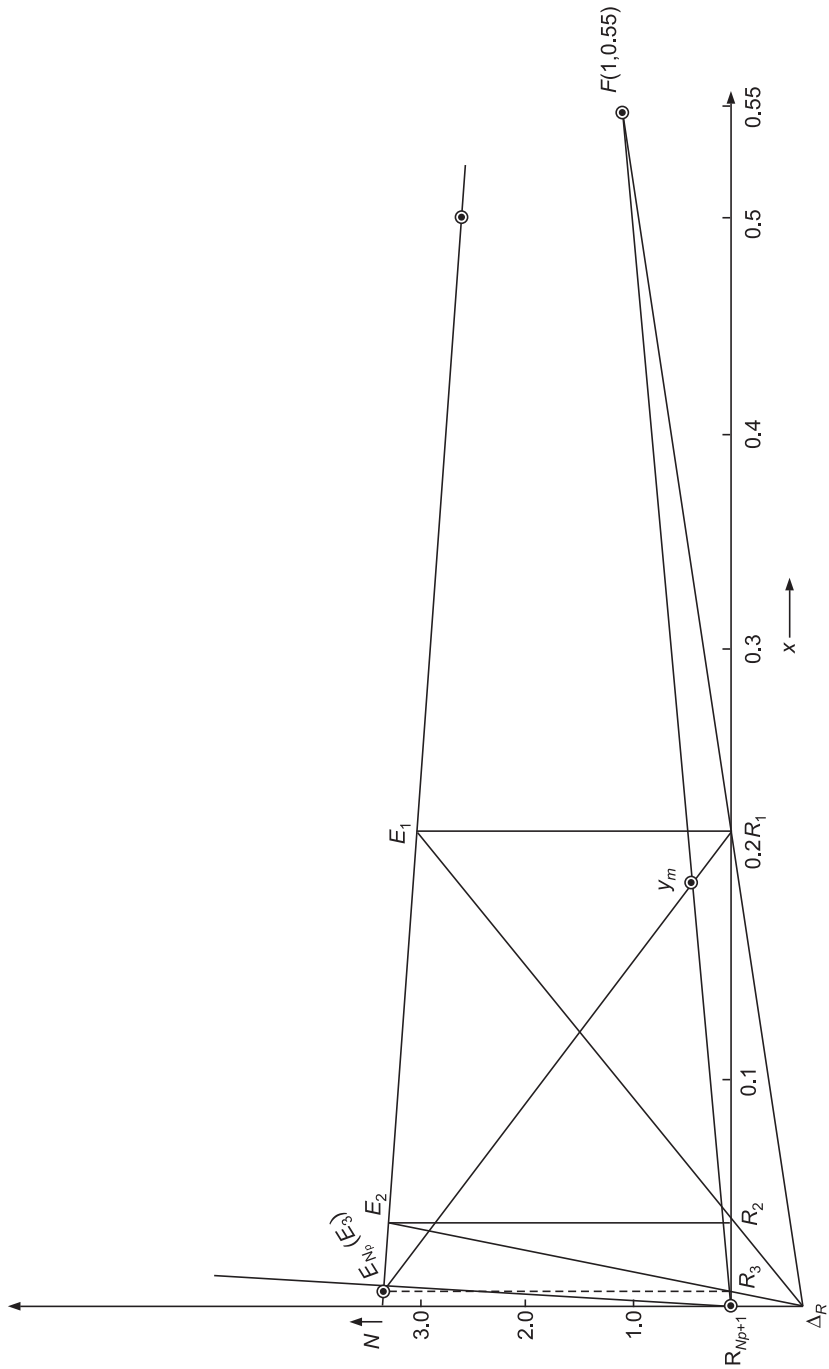


Fig. 11.23 Example 4.

(ii) Feed point F, $(N_F, y_F) = (1, 0.55)$
Leached solids leaving, E_{N_p} $(N_{N_p}, y_{N_p}) = (?, ?)$
Solvent entering, R_{N_p+1} $(N_{p+1}, x_{N_p+1}) = (0, 0)$
Solution leaving, R_1 $(N_1, x_1) = (0, ?)$

$$\bar{y} = \frac{F \cdot y_F + R_{N_p}\ x_{N_p+1}}{F + R_{N_p+1}} = \frac{(100)\,(0.55) + 0}{100 + 200} = 0.1833$$

$$\bar{N} = \frac{B}{A + C} = \frac{100}{300} = 0.333$$

Join F and R_{N_p+1} Locate $\bar{m}\,(\bar{N}, \bar{y})$.

By stagewise construction, the stages are estimated to be: 4

5. By extraction with kerosene two tonnes of waxed paper per day is to be dewaxed in a continuous countercurrent extraction system. The waxed paper contains 25% paraffin wax by weight and 75% paper pulp. The pulp which retains the unextracted wax must not contain over 0.2 kg of wax/100 kg of wax free pulp. The kerosene used for extraction contains 0.05 kg of wax/100 kg wax free kerosene, experiments show that pulp retains 2 kg of kerosene per kg of wax free pulp. The extract from battery contains 5 kg of wax/100 kg of wax free kerosene. How many stages are needed?

Solution.
Basis: 100 kg of wax and kerosene free pulp

Wax in the pulp = $100 \times \frac{25}{75} = 33.33$ kg
Wax in the solvent = 0.0005 kg of wax/kg of kerosene
Let s be the weight of solvent used.
$\therefore$ Total wax entering = wax from pulp + wax from kerosene = 33.33 + 0.0005s
Wax in the exiting pulp = 100 × 0.002 = 0.2 kg
Wax in the solution leaving
[Solvent entering – solvent carried away in leaving pulp] [Weight ratio of wax to solvent in leaving solution]

$$= [s - (2)\ (100)]\ [0.05] = (0.05s - 10) \text{ kg}$$

$\therefore$ Total wax output = $(0.05s - 10) + (0.2) = (0.05s - 9.8)$

Wax input = wax output

i.e. $33.33 + 0.0005s = 0.05s - 9.8$

$$s = 871.3 \text{ kg}$$

Kerosene in the exhausted pulp = 2 × 100 = 200
Kerosene in the extract (overflow) solution = 871.3 – 200 = 671.3 kg

i.e. wax in the extract (overflow) solution = $671.3 \times \frac{0.5}{100} = 33.565$ kg

Concentration in underflow in II unit = Concentration in overflow from I stage

Wax in underflow leaving I solution = Weight of kerosene in underflow × wax concentration

$$= (200)\left(\frac{5}{100}\right) = 10 \text{ kg}$$

The wax in the overflow from II cell to I cell by wax balance

[Wax in underflow leaving I + wax in overflow solution leaving I – wax in pulp entering I]

$$10 + 33.565 - 33.33 = 10.235 \text{ kg}$$

Concentration of this solution is $\frac{10.235}{871.3} = 0.0117$

$$x_a = y_a^* = 0.05 \text{ and } y_a = 0.0117$$

$$x_b = y_b^* = \frac{0.2}{200} = 0.001,\ y_b = 0.0005$$

$$N - 1 = \frac{\log\left[\frac{0.0005 - 0.001}{0.0117 - 0.05}\right]}{\log\left[\frac{0.0005 - 0.0117}{0.001 - 0.05}\right]} = \frac{-1.88423}{-0.641} = 2.94$$

$\therefore$ $N = 3.94$ stage; $N \approx 4$ stage

6. A five-stage countercurrent extraction battery is to be used to extract the sludge from the reaction

$$Na_2CO_3 + CaO + H_2O \longrightarrow CaCO_3 + 2NaOH$$

The $CaCO_3$ leaving each carries with it 1.5 times its weight the solution, in flowing from one unit to other. It is desired to recover 98% of NaOH. The products from the reaction enter the first unit with no excess reactant but with 6.5 kg of water/kg of $CaCO_3$.

(i) How much wastewater must be used for 1 kg of $CaCO_3$?

(ii) What is the concentration of leaving solution assuming $CaCO_3$ is insoluble?

(iii) Using the same quantity of wastewater, how many units must be employed to recover 99.5% of NaOH.

Solution.

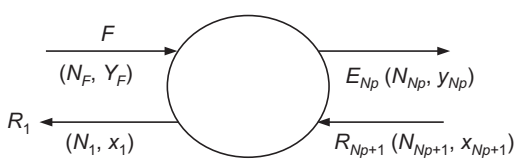


Fig. 11.24 Example 6.

Basis:

100 kg $CaCO_3$ formed

B (Inert) : 100 kg

A (Solvent) : 650 kg

C (Solute) : 80 kg (from stoichiometry)

$$\therefore \quad N_F = \frac{B}{A+C} = \frac{100}{650+80} = 0.137$$

$$y_F = \frac{C}{A+C} = \frac{80}{730} = 0.1096$$

$$\therefore \quad F\ (0.137,\ 0.1096)$$

$$N_{N_p} = \frac{1}{1.5} = 0.667$$

Recovery of NaOH is 987 = 78.4 kg

$\therefore$ NaOH in leaving stream = 1.6 kg

$$\therefore \quad y_{N_p} = \frac{C}{A+C} = \frac{C}{E_{N_p}}$$

$$E_{N_p} = \frac{B}{N_{N_p}} = \frac{100}{0.667} = 150$$

$$\therefore \quad y_{N_p} = \frac{1.6}{150} = 0.0107$$

Point E_{N_p} is (N_{N_p}, y_{N_p}) = (0.667, 0.0107)

Assume x_1 and hence locate R_1. (0, x_1) locate E_{N_p} (0.667, 0.0107), *F* (0.137, 0.1096) and R_{N_p+1} (0, 0)

Join E_{N_p}, R_{N_p+1} and *F*, R_1 and produce them to cut at Δ_R.

By stepwise construction check whether both five stages and E_{N_P} (assumed) match.

If not, make a fresh assumption of x_1 and proceed till the stages and x_1 match.

By trial and error x_1 = 0.1

Total amount of wastewater

(i) Water in sludge (*A* + *C*) = Weight of solution in sludge – weight of solute = 150 – 1.6 = 148.4 kg

(ii) Weight of water in overflow

Concentration in overflow = $x_1 = \frac{C}{A+C} = \frac{78.4}{A+C} = 0.1$

Weight of solution *A* + *C* = 784 kg

$\therefore$ Weight of solvent (*A*) = 784 – 78.4 = 705.6 kg

$\therefore$ Total weight of water added = 148.4 + 705.6 – 650 = 204 kg

Concentration of leaving solution from each stage:

$x_1 = 0.1$; $x_2 = 0.068$; $x_3 = 0.044$; $x_4 = 0.026$; $x_5 = 0.0107$

(iii) For 99.5% recovery:

Concentration of NaOH leaving = $0.995 \times 80 = 0.14$.

$$\frac{F}{R}(N_F, y_F) = F(0.137, 0.1096)$$

$$R_1(N_1, x_1) = R_1(0, 0.1)$$

$$y_{N_p} = \frac{C}{A+C} = \frac{C}{E_{N_p}}$$

$$\therefore \quad E_{N_p} = \frac{B}{N_{N_p}} = 150$$

$$\therefore \quad y_{N_p} = \frac{0.4}{150} = 2.667 \times 10^{-3} = 0.002667$$

$$E_{N_p}(0.667, 2.667 \times 10^{-3})$$

$$R_{N_p+1}(0, 0)$$

By stagewise construction, we find the number of stages as 5.

In the previous problem worked out, it is found that the sludge retains the solution varying with the concentration as follows:

NaOH	0	5	10	15	20
$\frac{\text{kg of solution}}{\text{kg of } CaCO_3}, \frac{1}{N}$	1.5	1.75	2.2	2.7	3.6
N	0.667	0.571	0.455	0.370	0.278

It is desired to produce a 10% solution of NaOH. How many stages must be used to recover 99.5% of NaOH?

Recovered NaOH = 99.5%

i.e. $99.5 \times 80 = 79.6$ kg

Solute = 0.4 kg

$$x = \frac{C}{A+C} = \frac{79.6}{A+79.6} = 0.1$$

$$\therefore \quad A = 716.4 \text{ kg}$$

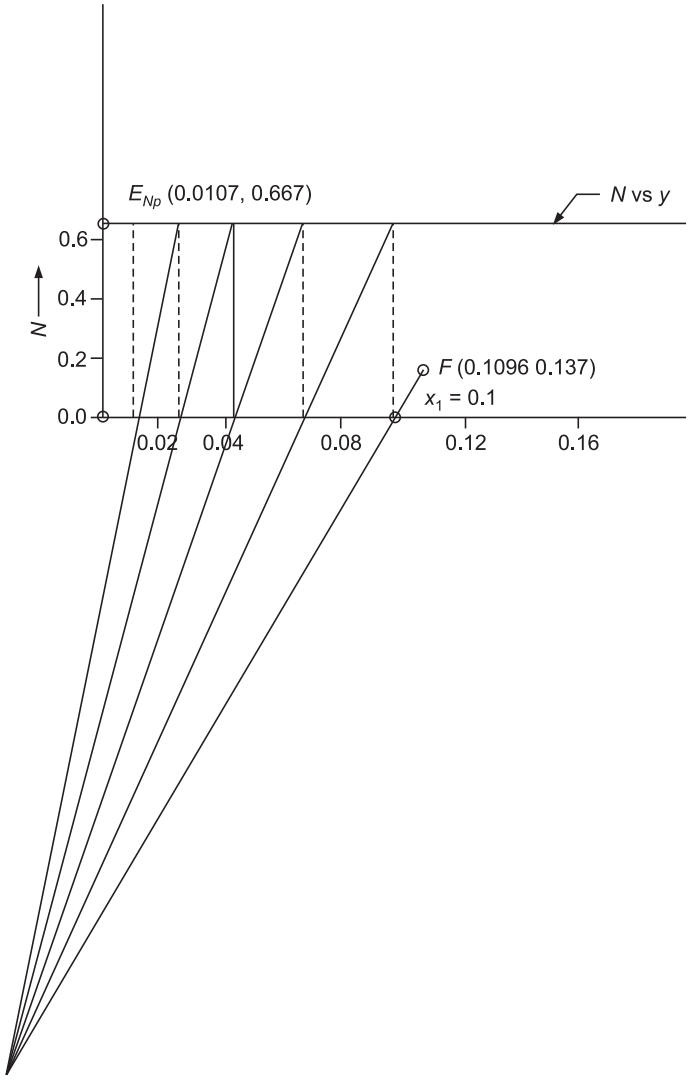


Fig. 11.25 Example 6.

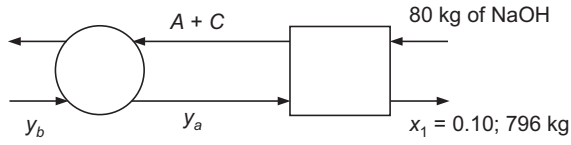


Fig. 11.26 Example 6.

EXERCISES

1. Seeds containing 25% oil by weight are to be extracted in a countercurrent plant and 95% of the oil is recovered in a solution containing 60% oil by weight. If the seeds are contacted with fresh solvent and 1 kg of solution is removed in the underflow in association with every 2 kg of insoluble matter, determine the theoretical stages required.

2. Crushed oil seeds containing 55% oil by weight are to be extracted at the rate of 5000 kg/hr using 8000 kg/hr of hexane containing 5% oil by weight as the solvent. A countercurrent two-stage extraction system is used. The oil seeds retain 1 kg of solution per kg of oil-free cake. Calculate the percent recovery of oil (based on original feed) obtained under the above conditions.

3. Seeds containing 20% oil by weight is extracted countercurrently with oil-free hexane as a solvent. Calculate the number of theoretical stages required is 90% of the oil is recovered in extract with 40% oil by weight and the amount of liquid (solvent + oil) in the underflow from each stage is 0.60 kg per kg of insoluble matter.

Use triangular coordinates or rectangular coordinates.

4. In a lime-soda process a slurry containing 10 kg water, 1 kg sodium hydroxide (NaOH) and 1 kg calcium carbonate particles. The slurry is washed countercurrently with water in four stages. The solid discharged from each stage contains 3 kg water per kg calcium carbonate. Calculate the amount of wash water needed when the discharged calcium carbonate after drying contains a maximum of 0.01 kg sodium hydroxide per kg calcium carbonate.

12

ADSORPTION

12.1 INTRODUCTION

Adsorption operation involves contact of solids with either liquids or gases in which the mass transfer is towards solids. The reverse of this operation is called *Desorption*. Adsorption operations exploit the ability of certain solids to concentrate specific substances from fluid on to their surfaces. The adsorbed substance is called *adsorbate* and the solid substance is called *adsorbent*.

Typical applications of this solid–liquid operation are as follows:

- removal of moisture dissolved in gasoline
- de-colorization of petroleum products and sugar solutions
- removal of objectionable taste and odour from water.

The solid–gas operations include:

- dehumidification of air and gases
- removal of objectionable odours and impurities from gases
- recovery of valuable solvent vapours from dilute gas mixtures
- to fractionate mixtures of hydrocarbon gases such as methane, ethane and propane.

12.2 TYPES OF ADSORPTION

The two types of adsorption are *physical adsorption* or *physi-sorption* (van der Waals adsorption) and *chemi-sorption* (activated adsorption).

Physical adsorption is a readily reversible phenomenon, which results from the intermolecular forces of attraction between a solid and the substance adsorbed.

Chemi-sorption is the result of chemical interaction, generally stronger than physi-sorption between the solid and the adsorbed substance. This process is irreversible. It has importance in catalysis.

12.3 NATURE OF ADSORBENTS

Adsorbents are usually in granular form with their size ranging from 0.5 mm to 12 mm. They must neither offer high pressure drop nor get carried away by flowing stream. They must not loose their shape and size while handling. They must have larger surface area per unit mass and also lot of pores.

Some of the commonly used adsorbents, their sources and applications are given below:

Sl. No.	*Adsorbent*	*Source*	*Application*
1.	Fuller's earth	Naturally occurring clay is heated and dried to get a porous structure.	De-colorizing, drying of lubricating oils, kerosene and engine oils.
2.	Activated clay	Bentonite or other activated clay which are activated by treatment with sulfuric acid and further washing, drying and crushing.	Used for de-colorizing petroleum products.
3.	Bauxite	A naturally occurring hydrated alumina, activated by heating at 230–815°C.	Used for de-colorizing petroleum products and for drying gases.
4.	Alumina	A hard hydrated aluminium oxide, which is activated by heating to drive off the moisture and then crushed to desired size.	Used as desiccant.
5.	Bone-char	Obtained by destructive distillation of crushed bones at 600–900°C.	Used for refining sugar and can be reused after washing and burning.
6.	Activated carbon	(i) Vegetable matter is mixed with calcium chloride, carbonized and finally the inorganic compounds are leached away. (ii) Organic matter is mixed with porous pumice stones and then heated and carbonized to deposit the carbonaceous matter throughout the porous particle. (iii) Carbonizing substances like wood, sawdust, coconut shells, fruit pits, coal, lignite and subsequent activation with hot air steam. It is available in granular or pellated form.	De-colorizing of sugar solutions, chemicals, drugs, water purification, refining of vegetable and animal oils, recovery of gold and silver from cyanide ore-leach solution, recovery of solvent vapour from gas-mixtures, collection of gasoline hydro-carbons from natural gas, fractionation of hydrocarbon gases.

(*Contd.*)

(*Contd.*)

Sl. No.	*Adsorbent*	*Source*	*Application*
7.	Silica gel	A hard granular and porous product obtained from sodium silicate solution after treatment with acid. Normally has 4 to 7% water in the product.	Used for de-hydration of air and other gases, fractionation of hydrocarbons.
8.	Molecular sieves	These are porous synthetic zeolite crystals, metal alumino-silicates.	Dehydration of gases and liquids, and separation of gas–liquid hydrocarbon mixture.

12.4 ADSORPTION EQUILIBRIA

Different gases and vapours are adsorbed to different extent under comparable conditions as shown in Fig. 12.1.

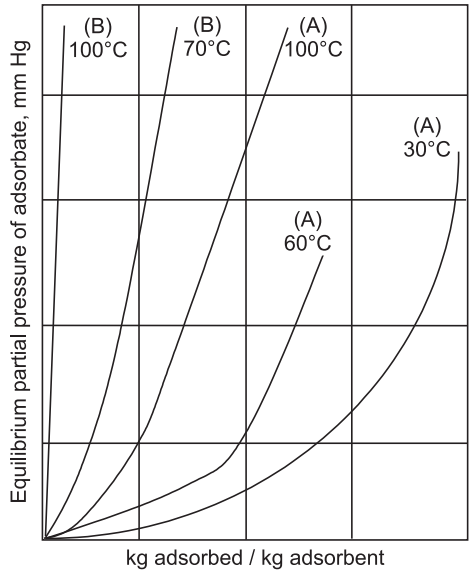


Fig. 12.1 Equilibrium adsorption on activated carbon.

As a general rule, vapours and gases with higher molecular weight and lower critical temperature are more readily adsorbed. To some extent, level of saturation also influences the degree of adsorption. The adsorption isotherms are generally concave to pressure axis. However, other shapes are also exhibited as shown in Fig. 12.2.

Repeated adsorption and desorption studies on a particular adsorbent will change the shape of isotherms due to gradual change in pore-structure. Further, adsorption is an exothermic process and hence the concentration of adsorbed gas

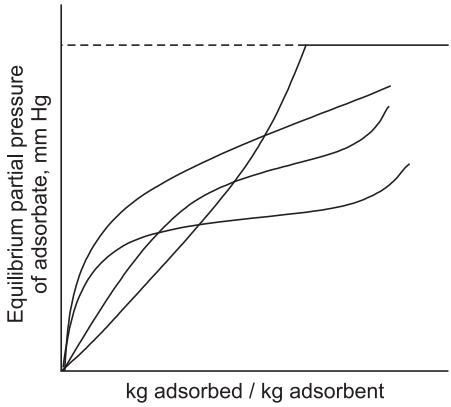


Fig. 12.2 Adsorption isotherms.

decreases with an increase in temperature at a constant pressure. Similarly an increase in pressure increases the concentration of adsorbed gas in the adsorbent at a constant temperature. There are three commonly used mathematical expressions to describe vapour adsorption equilibria, viz. Langmuir, Brunauer-Emmett-Teller (BET) and Freundlich isotherms. The first two are derived from theory whereas the last one is derived by a fit technique from the experimental data.

12.5 ADSORPTION HYSTERESIS

The adsorption and desorption operations exhibit different equilibrium phenomena as shown in Fig. 12.3 and is called adsorption hysteresis.

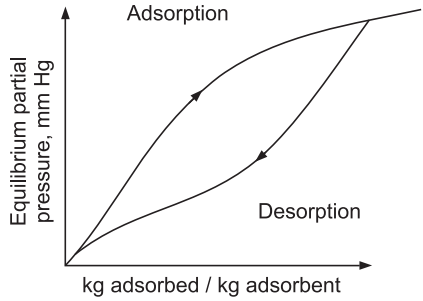


Fig. 12.3 Adsorption hysteresis.

This may be due to the shape of the openings to the capillaries and pores of the solid or due to the complex phenomena of wetting of the solid by the adsorbate. Whenever hysteresis is observed, the desorption curve is below the adsorption curve.

12.6 HEAT OF ADSORPTION

The differential heat of adsorption ($-H$) is defined as the heat liberated at constant temperature when unit quantity of vapour is adsorbed on a large quantity of solid

already containing adsorbate. Solid so used is in such a large quantity that the adsorbate concentration remains unchanged.

The integral heat of adsorption, (ΔH) at any concentration X is defined as the enthalpy of the adsorbate–adsorbent combination minus the sum of the enthalpies of unit weight of pure solid adsorbent and sufficient pure adsorbed substance (before adsorption) to provide the required concentration X, at the same temperature.

The differential heat of adsorption and integral heat of adsorption are functions of temperature and adsorbate concentration.

12.7 EFFECT OF TEMPERATURE

Increase of temperature at constant pressure decreases the amount of solute adsorbed from a mixture. However, a generalization of the result is not easy. Figure 12.1 also indicates the effect of temperature.

12.8 EFFECT OF PRESSURE

Generally lowering of pressure reduces the amount of adsorbate adsorbed upon the adsorbent. However, the relative adsorption of paraffin hydrocarbon on carbon decreases at increased pressures.

12.9 LIQUIDS

The impurities are present both at low and high concentrations in liquids. These are normally removed by adsorption technique. The characteristics of adsorption of low and high concentration impurities are different. They are discussed below.

12.9.1 Adsorption of Solute from Dilute Solutions

Whenever a mixture of solute and solvent is adsorbed using an adsorbent, both the solvent and solute are adsorbed. Due to this, only relative or apparent adsorption of solute can alone be determined.

Hence, it is a normal practice to treat a known volume of solution of original concentration C_0, with a known weight of adsorbent. Let C^* be the final equilibrium concentration of solute in the solution.

If v is the volume of solution per unit mass of adsorbent (cc/g) and C_0 and C^* are the initial and equilibrium concentrations (g/cc) of the solute, then the apparent adsorption of solute per unit mass of adsorbent, neglecting any change in volume is $v(C_0 - C^*)$, (g/g). This expression is mainly applicable to dilute solutions. When the fraction of the original solvent which can be adsorbed is small, the C^* value depends on the temperature, nature and properties of adsorbent.

In the case of dilute solutions and over a small concentration range, *Freundich adsorption Isotherm* describes the adsorption phenomena,

$$C^* = K\,[v(C_0 - C^*)]^n \qquad (12.1)$$

Freundlich adsorption equation is also quite useful in cases where the actual identity of the solute is not known, e.g. removal of colouring substance from sugar solutions, oils etc. The colour content in the solution can easily be measured using spectrophotometer or colorimeter. The interpretation of this data is illustrated in worked example 2. If the value of *n* is high, say 2 to 10, adsorption is good. If it lies between 1 and 2, moderately difficult and less than 1 indicates poor adsorption characteristics. A typical adsorption isothermal for the adsorption of various adsorbates *A, B* and *C* in dilute solution at the same temperature for the same adsorbent is shown in Fig. 12.4.

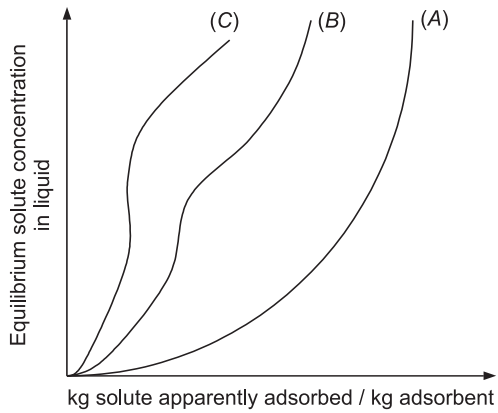


Fig. 12.4 Adsorption isotherms for various adsorbates.

12.9.2 Adsorption from Concentrated Solution

When the apparent adsorption of solute is determined over the entire range of concentrations from pure solvent (0% solute concentration) to pure solute (100% solute concentration), curves as shown in Fig. 12.5 will occur. Curve '1' occurs when the solute is more strongly adsorbed in comparison to solvent at all solute concentration. Whenever both solute and solvent are adsorbed to nearly the same extent, the 'S' shaped curve '2'occurs. In the range *PQ* solute is more strongly adsorbed than solvent. At point *Q* both are equally well adsorbed. In the range *QR* solvent is more strongly adsorbed.

12.9.3 Other Adsorption Isotherms

12.9.3.1 Langmuir adsorption isotherm

The theory proposed by *Langmuir* postulates that gases being adsorbed by a solid surface cannot form a layer more than a simple molecule in depth. His theory visualizes adsorption as a process consisting two opposite actions, a condensation of molecules from the gas phase on to the surface and an evaporation of molecules from the surface back into the body of the gas. When adsorption starts, every molecule colliding with the surface may condense on it. However, as adsorption

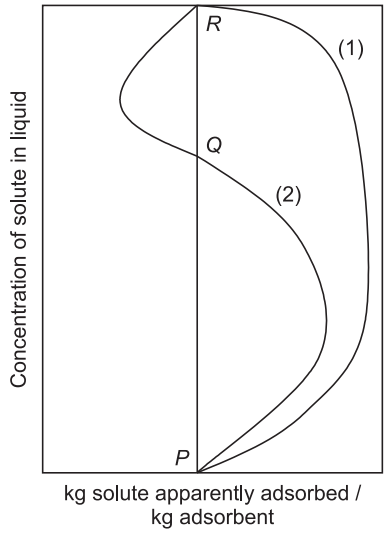


Fig. 12.5 Adsorption of solute in concentrated solutions.

proceeds, only those molecules which strike the uncovered area surface can be adsorbed. Due to this, the initial rate of condensation of molecules on the surface is very high and decreases as the time progresses. The molecules attached to the surface also get detached by factors like thermal agitation. The rate at which desorption occurs depends on the amount of surface covered by molecules and will increase as the surface becomes more fully saturated. When the rate of adsorption and desorption become equal, adsorption equilibrium is said to be reached. If 'θ' is the fraction of surface covered by adsorbed molecules at any instant, the fractional area available for adsorption is $(1 - \theta)$. The rate at which the molecules strike the unit area of surface is proportional to pressure.

Therefore the rate of condensation

$$= k_1(1 - \theta)P$$

where, k_1 is a constant.

Similarly, the rate of evaporation

$$\propto k_2\theta$$

where, k_2 is a constant.

Under adsorption equilibrium,

$$k_1(1 - \theta)P = k_2\theta$$

i.e.
$$\theta = \frac{k_1(1-\theta)P}{k_2 + k_1P}$$

$$= \frac{bP}{1+bP} \quad \text{where,} \quad b = \frac{k_1}{k_2} \tag{12.2}$$

Now the gas adsorbed per unit area or unit mass of adsorbent, y, must obviously be proportional to the fraction of surface covered. Hence,

$$y = k\theta = k\left\{\frac{bP}{1+bP}\right\} = \frac{aP}{1+bP} \tag{12.3}$$

where, a and b are constants. This is Langmuir adsorption Isotherm

12.9.3.2 BET adsorption isotherm

This postulates that the adsorption phenomenon involves the formation of many multilayers on the surface rather than a single one. Based on this, Brunauer, Emmett and Teller derived the following adsorption isotherm popularly known as BET adsorption isotherm.

$$\frac{P}{[V(P^o - P)]} = \frac{1}{[V_m C]} + \left[\frac{(C-1)}{V_m C}\right]\left[\frac{P}{P^o}\right] \tag{12.4}$$

where, V is the volume, reduced to standard conditions of gas adsorbed at pressure P and temperature T, $P°$ is the saturated vapour pressure of the adsorbate at temperature T, V_m is the volume of gas reduced to standard conditions, adsorbed when the surface is covered with a unimolecular layer, C is a constant at any given temperature given by exp $[(E_1 - E_2)/RT]$, where E_1 is the heat of adsorption for the first layer and E_2 is that for the second and higher layers.

12.10 TYPES OF OPERATION

Adsorption operations are carried out either on batch or continuous basis. Batch process is not very much used. However, a batch operation is quite useful in obtaining equilibrium data. Much widely used continuous operation can either be a single stage or a multistage operation. The multistage operation could once again either be a cross-current operation or a countercurrent operation.

12.10.1 Single Stage Operation

A schematic arrangement for a single stage operation is shown in Fig. 12.6.

The concentration of solute increases in the adsorbent from X_0 to X_1 (g/g) and the concentration of solute in the solution decreases from Y_0 to Y_1 (g/g).

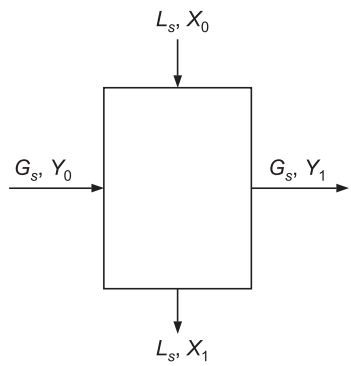


Fig. 12.6 Single stage operation.

The mass balance for solute gives

$$G_s\,[Y_0 - Y_1] = L_S\,[X_1 - X_0] \tag{12.5}$$

i.e.
$$-\frac{L_S}{G_S} = \frac{(Y_0 - Y_1)}{(X_0 - X_1)} \tag{12.6}$$

where (L_S/G_S) indicates the slope of the operating line passing through the points (X_0, Y_0) and (X_1, Y_1). If the leaving streams are in perfect equilibrium, then the point (X^*_1, Y^*_1) will lie on the equilibrium adsorption isotherm. If the equilibrium is not reached due to factors like poor contacting, then the point P represents the conditions of leaving streams as shown in Fig. 12.7.

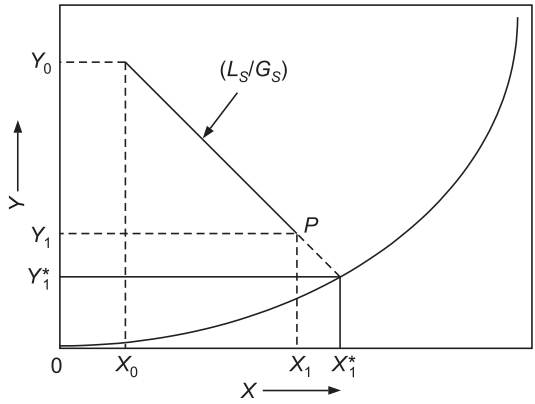


Fig. 12.7 Adsorption isotherm and operating line for a single stage operation.

Assuming the validity of Freundlich equation, especially when a low concentration of solute is involved, the equation can be written as

$$Y^* = mx^n \tag{12.7}$$

and at the final equilibrium conditions,

$$X_1 = \left(\frac{Y_1}{m}\right)^{1/n} \tag{12.8}$$

when the pure adsorbent is used, i.e. $X_0 = 0$.

Equation (12.8) yields

$$\frac{L_S}{G_S} = \frac{(Y_0 - Y_1)}{\left(\frac{Y_1}{m}\right)^{1/n}} \tag{12.9}$$

12.10.2 Multistage Cross-current Operation

A schematic arrangement of multistage cross-current operation is shown in Fig. 12.8.

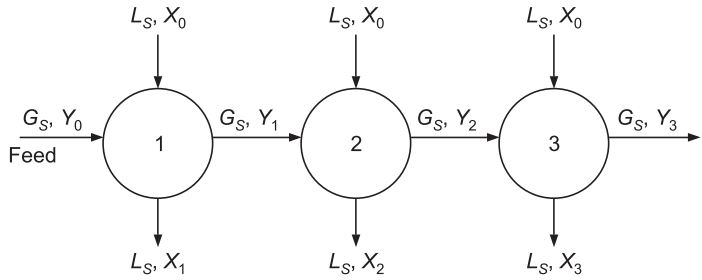


Fig. 12.8 Multistage cross-current operation.

Making a material balance of solute for stage 1 and use of Freundlich equation for the entry of pure adsorbent gives

$$G_S(Y_0 - Y_1) = L_{S1}(X_1 - X_0) \tag{12.10}$$

According to Eq. (12.9),

$$\frac{L_{S_1}}{G_S} = \frac{(Y_0 - Y_1)}{\left(\frac{Y_1}{m}\right)^{1/n}} \tag{12.11}$$

A material balance of solute for stage 2 yields,

$$G_S(Y_1 - Y_2) = L_{S_2}(X_2 - X_0) \tag{12.12}$$

Use of Freundlich equation for the entry of pure adsorbent gives

$$\frac{L_{S_2}}{G_S} = \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}} \tag{12.13}$$

A similar material balance for stage p yields

$$G_S(Y_{p-1} - Y_p) = L_{Sp}(X_p - X_0) \tag{12.14}$$

Using Freundlich equation as before gives

$$\frac{L_{Sp}}{G_S} = \frac{(Y_{p-1} - Y_p)}{\left(\frac{Y_p}{m}\right)^{1/n}} \tag{12.15}$$

This operation is represented graphically as shown in Fig. 12.9.

12.10.2.1 Steps involved in the determination of number of stages needed for a cross-current adsorption process

1. Draw the equilibrium curve (X vs Y).
2. Locate the point (X_0, Y_0) and draw the operating line with a slope ($-L_{S_1}/G_S$).
3. The intersection of operating line and equilibrium curve yields (X_1, Y_1) – the conditions of stream leaving from stage I.

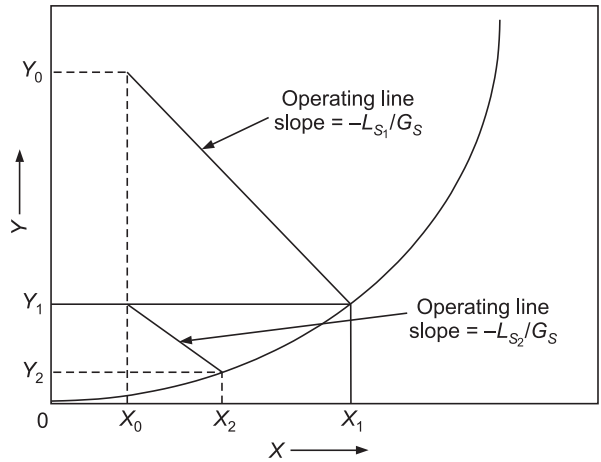


Fig. 12.9 Adsorption isotherm and operating line for a two-stage cross-current operation.

4. Locate (X_0, Y_1) and draw the operating line with a slope of ($-L_{S_2}/G_S$) (since X_0 remains constant for adsorbent for II stage).
5. Intersection of operating line and equilibrium curve yields (X_2, Y_2) — the conditions of leaving stream from stage II.
6. Proceed in the same way till the X_{Np} point is crossed and count the number of stages for the use of specified amount of adsorbent in each stage.

12.10.2.2 Optimisation of a two-stage cross-current operation

In a typical two-stage operation, the concentrations of solute both in the inlet solution stream and the outlet solution stream are fixed along with the feed rate of solution. The objective will be to use the minimum amount of adsorbent for this. If the quantity of the adsorbent is changed, the exit concentration of solution from each stage will also vary. However, the terminal conditions are always fixed and only the intermediate concentration is a variable. Hence, with one particular intermediate value, if the amounts of adsorbent used in both the stages are estimated, it will result in the minimum amount of adsorbent being used.

For the schematic arrangement shown in Fig. 12.10, the material balance equations for stages I and II are obtained from Eqs. (12.8) and (12.9) as

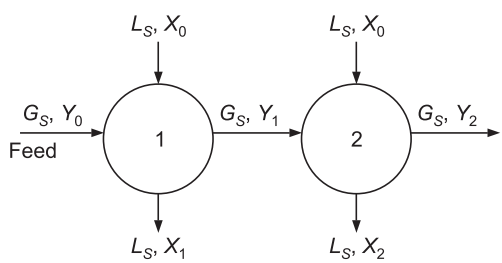


Fig. 12.10 Two-stage cross-current operation.

$$\frac{L_{S_1}}{G_S} = \frac{(Y_0 - Y_1)}{\left(\frac{Y_1}{m}\right)^{1/n}}$$

$$\frac{L_{S_2}}{G_S} = \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}}$$

Adding Eqs. (12.11) and (12.12), we get

$$\left(\frac{L_{S_1} + L_{S_2}}{G_s}\right) = \frac{(Y_0 - Y_1)}{\left(\frac{Y_1}{m}\right)^{1/n}} + \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}} \tag{12.16}$$

The total amount of adsorbent used can be optimised with respect to Y_1 (the intermediate concentration), the only variable on the R.H.S. of Eq. (12.16). The other parameters Y_0, Y_2, m and n are all fixed for a specified operation involving a specific adsorbent.

i.e. $$\frac{d}{dY_1}\left[\frac{L_{S_1} + L_{S_2}}{G_S}\right] = \frac{d}{dY_1}\left[\frac{(Y_0 - Y_1)}{\left(\frac{Y_1}{m}\right)^{1/n}} + \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}}\right] \tag{12.17}$$

$$= m^{1/n}\frac{d}{dY_1}\left[\frac{Y_0 - Y_1}{(Y_1)^{1/n}} + \frac{(Y_1 - Y_2)}{(Y_2)^{1/n}}\right]$$

i.e. $$= m^{1/n}\frac{d}{dY_1}\left[\frac{(Y_0 - Y_1)Y_2^{1/n} + (Y_1 - Y_2)Y_1^{1/n}}{[(Y_2)^{1/n}(Y_1)^{1/n}]}\right]$$

$$m^{1/n}(Y_2)^{-1/n}\frac{d}{dY_1}\left[\frac{[(Y_0Y_2^{1/n}) - (Y_1Y_2^{1/n})] + [Y_1^{(1+1/n)} - Y_2Y_1^{1/n}]}{Y_1^{1/n}}\right] \tag{12.18}$$

$$m^{1/n}(Y_2)^{-1/n}\frac{d}{dY_1}[Y_0Y_2^{1/n}Y_1^{-1/n} - Y_2^{1/n}\cdot Y_1^{(1-1/n)} + Y_1 - Y_2] \tag{12.19}$$

$$m^{1/n}(Y_2)^{-1/n}[Y_0\cdot Y_2^{1/n}(-1/n)Y_1^{-1-1/n} - Y_2^{1/n}(1-1/n)Y_1^{-1/n} + 1 - 0] \tag{12.20}$$

For minimum adsorbent R.H.S. of (12.20) should be zero.

i.e. $$Y_0Y_2^{1/n}(-1/n)Y_1^{-1-1/n} - (1-1/n)Y_2^{1/n}Y_1^{-1/n} + 1 - 0 = 0 \tag{12.21}$$

(since m, n and Y_2 have definite values)

Dividing by $\left(\frac{Y_2}{Y_1}\right)^{1/n}$, we get

$$\left(\frac{Y_0}{Y_1}\right)\left(\frac{-1}{n}\right) - \left(1 - \frac{1}{n}\right) + \left(\frac{Y_1}{Y_2}\right)^{1/n} = 0$$

i.e.
$$\left(\frac{Y_1}{Y_2}\right)^{1/n} - \left(\frac{1}{n}\right)\left(\frac{Y_0}{Y_1}\right) = 1 - \left(\frac{1}{n}\right) \tag{12.22}$$

Equation (12.22) can be solved by trial and error to get the intermediate concentration Y_1 which will optimise the total quantity of adsorbent to be used. However, also using the chart as shown in Fig. 12.11, we can get the intermediate concentration.

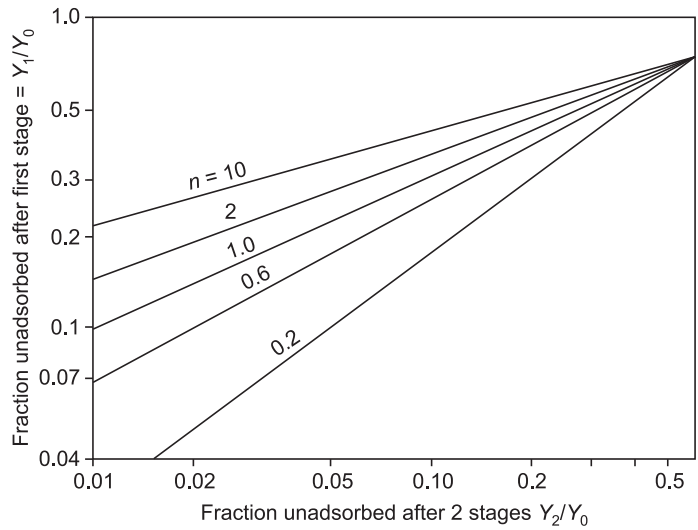


Fig. 12.11 Minimum total adsorbent two-stage cross-current Eq. (12.22).

12.10.3 Multistage Countercurrent Adsorption

The schematic arrangement as shown in Fig. 12.12 represents a multistage countercurrent operation.

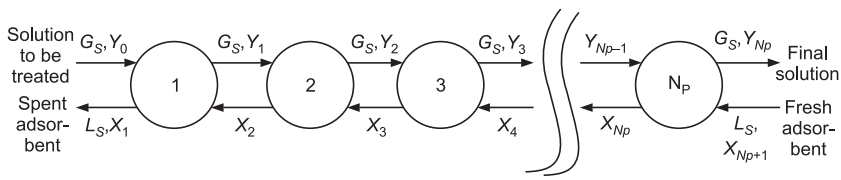


Fig. 12.12 Multistage countercurrent operation.

Solute balance for the system as a whole gives

$$G_S(Y_0 - Y_{Np}) = L_S(X_1 - X_{N_{p+1}}) \tag{12.23}$$

i.e.
$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_0 - Y_{Np})}{(X_1 - X_{Np+1})} \tag{12.24}$$

Equation (12.24) gives the slope of the operating line passing through the terminal conditions (X_1, Y_0) and (X_{Np+1}, Y_{Np}). By conventional stepwise construction starting

from the point (X_1, Y_0), the number of theoretical stages are estimated. This operation is represented graphically as shown in Fig. 12.13.

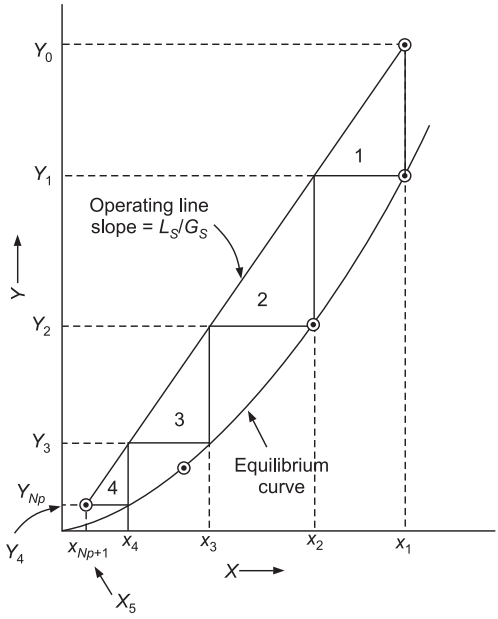


Fig. 12.13 Countercurrent multistage adsorption.

In order to determine the minimum amount of adsorbent for the process, draw a line from the point P (X_{Np+1}, Y_{Np}) which could be a tangent to the equilibrium curve. In such cases, the slope of the line gives the ratio of $(L_S/G_S)_{\min}$. However, in the case of equilibrium curve being a straight line or concave upwards, draw a horizontal line from Y_0 to intersect the equilibrium curve, (or line) at a point by Q and then join PQ which gives the slope of the operating line $(L_S/G_S)_{\min}$. The above two cases have been shown graphically in Figs. 12.14(a) and 12.14(b) respectively.

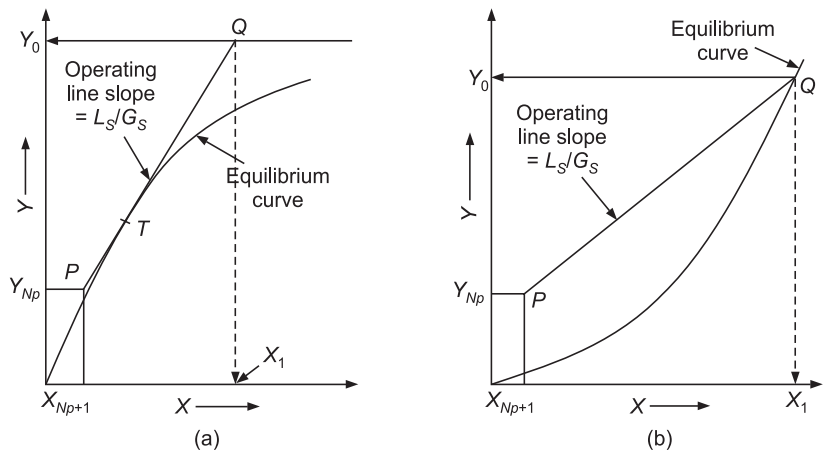


Fig. 12.14 Operating line and minimum adsorbent/solvent ratio for infinite stages.

12.10.3.1 Steps involved in determining the number of stages in a multistage countercurrent operation

1. Draw the equilibrium curve.
2. Locate the point P (X_{Np+1}, Y_{Np}).
3. Draw a line with a slope of (L_S/G_S), where L_S is the mass flow rate of solute free adsorbent and G_S is the mass flow rate of solution on solute free basis.
4. Starting from (X_1, Y_0) by stepwise construction, estimate the number of stages till the point (X_{Np+1}, Y_{Np}) is crossed. The operation is graphically shown in Fig. 12.14.
5. If it is desired to determine the amount of adsorbent needed for a specified level of solute removal from a solution stream with a specified number of stages, draw the operating line of different slopes by trial and error and choose the one which gives exactly the same number of specified stages and the specified concentration in the liquid stream. From the slope of the operating line, thus chosen, determine the amount of adsorbent to be used and the solute concentration in the adsorbent.

12.10.3.2 Optimization of two-stage countercurrent adsorption

A typical two-stage countercurrent operation is shown schematically in Fig. 12.15.

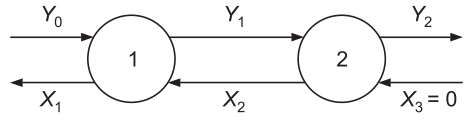


Fig. 12.15 Two-stage countercurrent adsorption.

Solute balance for the system as a whole with pure adsorbent yields

$$L_S(X_1 - 0) = G_S[Y_0 - Y_2] \tag{12.25}$$

Applying Eq. (12.8), we get

$$L_S\left(\frac{Y_1}{m}\right)^{1/n} = G_S(Y_0 - Y_2) \tag{12.26}$$

$$\frac{L_S}{G_S} = \frac{(Y_0 - Y_2)}{\left(\frac{Y_1}{m}\right)^{1/n}} \tag{12.27}$$

Applying a similar balance for stage 2, we get

$$G_S(Y_1 - Y_2) = L_S(X_2 - 0) = L_S\left(\frac{Y_2}{m}\right)^{1/n} \tag{12.28}$$

$$\left(\frac{L_S}{G_S}\right) = \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}} \tag{12.29}$$

Equating Eqs. (12.27) and (12.29), we get

$$\frac{(Y_0 - Y_2)}{\left(\frac{Y_1}{m}\right)^{1/n}} = \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}}$$

$$\frac{(Y_0 - Y_2)}{Y_2} = \frac{(Y_1 - Y_2)}{Y_2}\left[\left(\frac{Y_1}{m}\right)^{1/n} \times \left(\frac{Y_2}{m}\right)^{-1/n}\right]$$

$$\left(\frac{Y_0}{Y_2}\right) - 1 = \left(\frac{Y_1}{Y_2}\right)^{1/n}\left[\left(\frac{Y_1}{Y_2}\right) - 1\right] \tag{12.30}$$

Since Y_0, Y_2, n are all specific values for a specified level of adsorption and also a specific adsorbent, the only unknown Y_1, can be estimated by trial and error. Alternately, Y_1, can be estimated by the following chart as shown in Fig. 12.16.

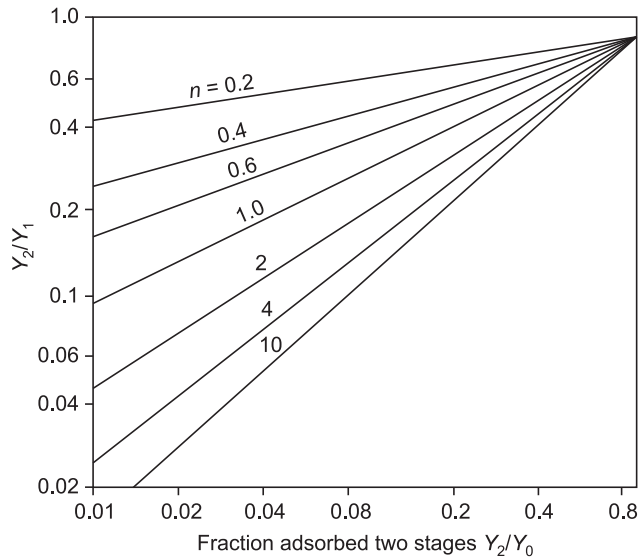


Fig. 12.16 Two-stage countercurrent adsorption Eq. (12.27).

12.11 CONTINUOUS ADSORPTION

In these adsorbers, the fluid and adsorbent are in continuous contact without any separation of the phases. This is quite analogous to gas absorption with the solid adsorbent replacing the liquid solvent. The operation can be carried out in strictly continuous, steady state fashion with both fluid and solid moving at constant rate

and the composition remains constant at a particular point. It can also be operated on semi-continuous basis with solid particles remaining stationary and fluid in moving condition. Such operations constitute unsteady adsorption process.

12.11.1 Steady State Adsorption

Continuous differential contact tower is schematically represented in Fig. 12.17.

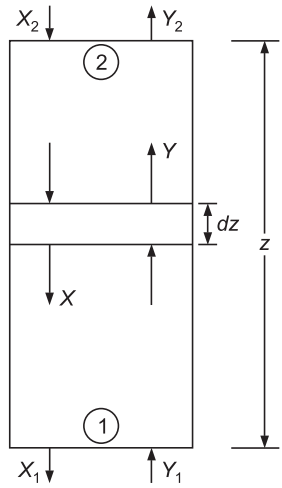


Fig. 12.17 Continuous differential contact tower.

Solute balance for the entire tower is

$$G_S(Y_1 - Y_2) = L_S(X_1 - X_2) \tag{12.31}$$

Solute balance for the upper part of the tower is

$$G_S(Y - Y_2) = L_S(X - X_2) \tag{12.32}$$

Using Eq. (12.31), one can draw the operating line and Eq. (12.32) gives us the concentration of the two phases at any point in the tower.

Making a solute balance across the element of thickness dZ,

$$L_S\,(dX) = G_S\,dY = K_y a\,(Y - Y^*)\,dZ \tag{12.33}$$

where K_y a is mass transfer coefficient based on the outside surface area a of particles, kg/m^3.s.(ΔY) and Y^* is the equilibrium concentration of the fluid corresponding to its concentration X.

Equation (12.33) on integration yields

$$N_{toG} = \int_{Y_2}^{Y_2} \frac{dY}{(Y - Y^*)} = \frac{K_y a}{G_S} \int_0^Z dZ = \frac{Z}{H_{toG}} \tag{12.34}$$

where $H_{toG} = G_S/K_y a$

N_{toG} can be determined graphically as usual.

12.11.2 Unsteady State Adsorbers

When a fluid mixture is passed through a stationary bed of adsorbent, the adsorbent adsorbs solute continuously and it results in an unsteady state operation. Ultimately, the bed may get saturated and no further adsorption results. The change in concentration of effluent stream is shown in Fig. 12.18. The system indicates an exit concentration varying from C, to a final concentration very close to inlet concentration. The point *A* indicates break point. The portion from *A* to *B* is termed the break through curve. Beyond this, very little adsorption takes place, indicating that the system has more or less reached equilibrium or saturation.

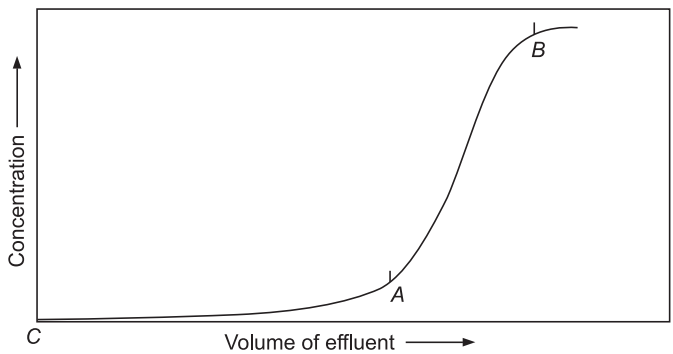


Fig. 12.18 The adsorption wave.

12.12 EQUIPMENT FOR ADSORPTION

Equipment are available for adsorption of a solute from a gaseous or a liquid stream. When the solute (which could be colouring matter, odorous substances, valuable solutes etc.) is strongly adsorbed from a liquid stream, one can use contact filtration equipment which can be operated as batch units, semi-continuous or as continuous ones. Continuous ones can be realized by fluidized bed techniques. These are similar to mixer-settler units used in extraction operations. Generally gases are treated with fluidized bed techniques.

12.12.1 Contact Filtration Equipment

The equipment consists of a mixing tank in which the liquid to be treated and the adsorbent are thoroughly mixed at the operating temperature and for a specified duration of time. In some cases like ion exchange sparging is done with air. Subsequently the slurry is filtered off to separate the solids from the solution. The filtration is done in a filter press or centrifuge or in a continuous rotary filter. Multi stage operations could easily be done by providing a number of tanks and filter combinations. The filter cake is usually washed to displace the solution. If the adsorbate is the desired product, then it can be desorbed by contact with a solvent other than the one which constituted the solution and the one in which the solute is more readily soluble. When the solute is more volatile, it can be removed by the

passage of steam or warm air through the solid. Whenever the adsorbent is activated carbon, care must be taken so that the adsorbent does not burn away at high temperatures of desorption operation. Adsorbent can also be regenerated by burning away the adsorbate.

12.12.2 Fluidised Beds

When a mixtures of gases are to be treated on a continuous basis, it is preferable to use fluidized beds. This is done by passing the gases at high velocities through a bed of granular solids in which the adsorption occurs. The beds of solids remain in suspended condition throughout the operation. The bed can be regenerated by passing steam/air at high temperatures. To improve the effectiveness of operation, one can go in for the multistage counter operation with regeneration. In these operations one has to take care to minimize or prevent the carry over of solids.

12.12.3 Steady-state Moving Bed Adsorbers

In this category of adsorbers both the solids and fluid move continuously. The composition at any particular point is independent of time. They are operated with the solids moving downwards and the liquid in upward direction. The flow of solids is plug flow in nature and it is not in fluidized state.

The Higgins contactor developed for ion exchange is an excellent facility for adsorption. Figure 12.19 indicates the arrangement. This consists of two sections. In the top section to start with adsorption takes place. Simultaneously the bottom section of the bed undergoes regeneration. After some pre-calculated duration of operation, the flow of liquids is stopped and the positions of the valves are changed

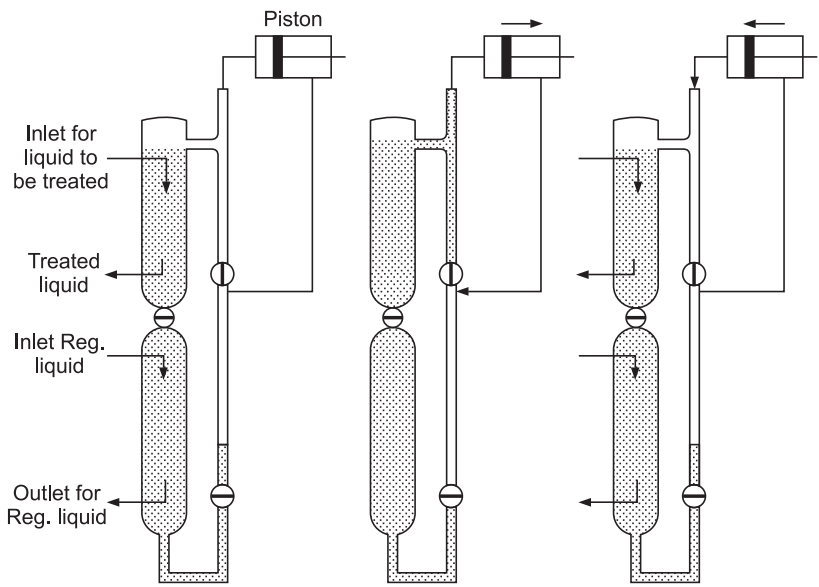


Fig. 12.19 Higgins contactor.

as indicated. The liquid-filled piston pump is moved and this leads to the clockwise movement of solids. Once again the valves are moved to their original position and the movement of solid also stops. The adsorption cycle once again starts in the top section of the unit and desorption at the bottom section.

WORKED EXAMPLES

1. One litre flask is containing air and acetone at 1 atm and 303 K with a relative humidity of 35% of acetone. 2 g of fresh activated carbon is introduced and the flask is sealed. Compute the final vapour composition and final pressure neglecting adsorption of air.

Equilibrium data:

g adsorbed/g carbon	0	0.1	0.2	0.3	0.35
Partial pressure of acetone, mm Hg	0	2	12	42	92

Vapour pressure of acetone at 30°C is 283 mm Hg.

Solution.

Let us convert the data from partial pressure to concentration in terms of g acetone/g of air.

i.e. $$\frac{2}{(760-2)} \times \frac{58}{28.84} = 5.28 \times 10^{-3} \text{ g acetone/g air}$$

Likewise the other values can also be converted to concentration in terms of mass ratios.

Hence,

X, g adsorbed/g carbon	0	0.1	0.2	0.3	0.35
Y, g acetone/g air	0	5.28×10^{-3}	32.1×10^{-3}	117×10^{-3}	276×10^{-3}

Originally the feed contains 35% RH acetone

i.e. Partial pressure/vapour pressure = 0.35

$\therefore$ Partial pressure of acetone = $283 \times 0.35 = 99$ mm Hg.

Partial pressure of air = 661 mm Hg.

$\therefore$ $$Y_0 = \frac{99}{(760-99)} \times \frac{58}{28.84} = 0.301 \text{ g of acetone/g of air}$$

$L_S = 2$ g

The feed point is $(X_0, Y_0) = (0.0, 0.301)$

Volume fraction of air in the original mixture = $\frac{(760-99)}{760} = 0.87$ liters.

i.e. volume of air = 0.87 l (At 1 atm and 303 K)

i.e. moles of air = $\frac{(0.87 \times 1)}{303} \times \frac{273}{1} \times \frac{1}{22.414} = 0.03496$ g moles

i.e. mass of air in the original mixture = $0.03496 \times 28.84 = 1.008$ g

$$\therefore \quad \frac{L_S}{G_S} = \frac{2}{1.008} = 1.984$$

Y_1 (from graph) = 13×10^{-3} g acetone/g air

Grams of acetone left behind after adsorption, per gram of air = 0.013

i.e. $$\frac{\text{Partial pressure of acetone}}{\text{Partial pressure of air}} \times \frac{58}{28.84} = \text{g acetone/g air}$$

i.e. $$\frac{\text{Partial pressure of acetone}}{661} \times \frac{58}{28.84} = 0.013 \text{ g acetone/g air}$$

$\therefore$ Partial pressure of acetone in flask after adsorption = 4.27 mm Hg

$\therefore$ Total pressure = 661 + 4.27 = 665.27 mm Hg.

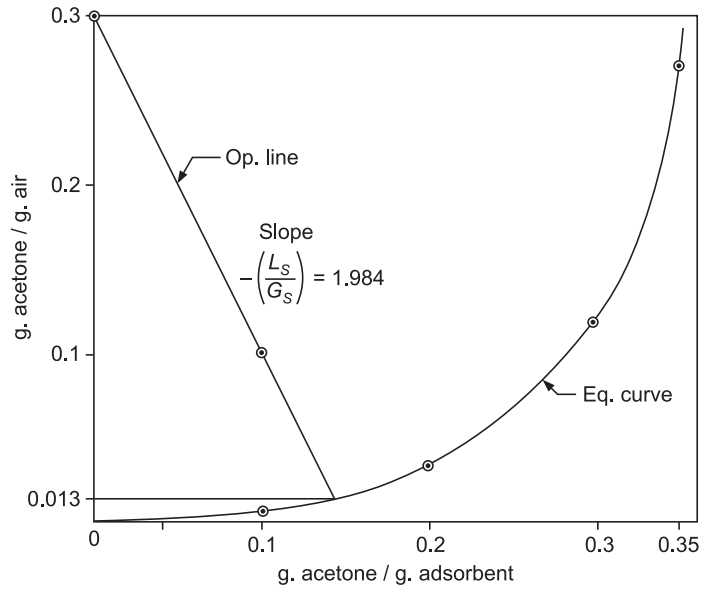


Fig. 12.20 Example 1.

2. A solid adsorbent is used to remove colour impurity from an aqueous solution. The original value of colour on an arbitrary scale is 48. It is required to reduce this to 10% of its original value. Using the following data, find the quantity of fresh adsorbent used for 1000 kg of solution for (a) a single stage and (b) a two-stage cross-current operation when the intermediate colour value is 24.

Equilibrium data:

kg adsorbent/kg of solution	0	0.001	0.004	0.008	0.02	0.04
Equilibrium colour (Y)	48	43	31.5	21.5	8.5	3.5

Solution.

(a) The given data will be converted to enable us to handle it more easily.

The initial values are X_o = units of colour/kg adsorbent = 0

Y_o = units of colour/kg solution = 48

When 0.001 kg of adsorbent is added to1 kg of solution, the colour reduces from 48 units to 43 units. These 5 units of colour are thus transferred to 0.001 kg adsorbent.

$$\therefore \qquad X, \frac{\text{units of colour}}{\text{kg adsorbent}} = \frac{(48-43)}{0.001} = 5000$$

Similarly, by adding 0.004 kg adsorbent, colour drops by 16.5 units.

$$\text{i.e.} \qquad X = \frac{16.5}{0.004} = 4125$$

X, colour adsorbent/kg adsorbent	0	5000	4125	3312.5	1975	1112.5
Y, colour/kg solution	48	43	31.5	21.5	8.5	3.5

The final solution has 4.8 units of colour

(a) *Single stage operation:*

$$\text{Slope} = -\frac{L_S}{G_S} = -0.030 \text{ (from graph)}$$

G_S is 1000 kg of solution.

$\therefore$ Dosage of carbon = 0.03 × 1000 = 30 kg

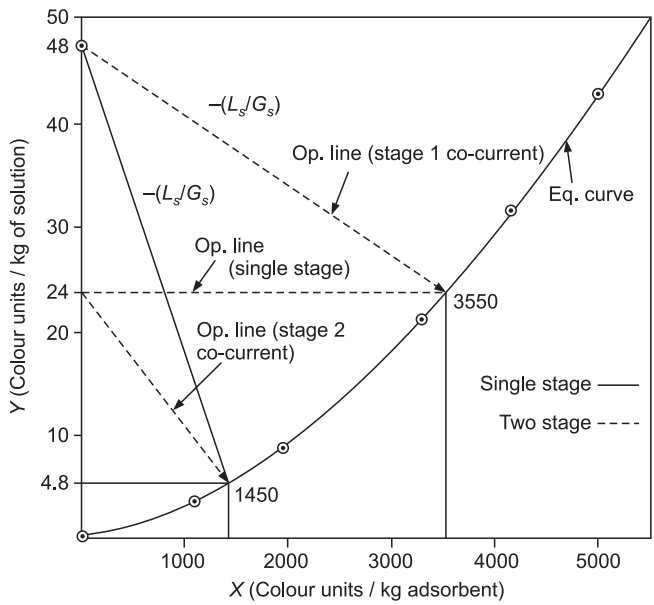


Fig. 12.21 Example 2.

(b) A two-stage cross-current operation

$$-\left(\frac{L_S}{G_S}\right)_1 = +\frac{(48-24)}{(0-3550)} = -\frac{24}{3550} = -6.76\times10^{-3}$$

$$-\left(\frac{L_S}{G_S}\right)_2 = +\frac{(24-4.8)}{(-1450+0)} = -0.01324$$

G_S is 1000 kg of solution

$\therefore$ $L_{S_1} + L_{S_2} = 6.76 + 13.24 = 20.00$ kg

3. The equilibrium decolourisation data for a certain system using activated carbon is given by the equation,

$$Y = 0.004X^2$$

where Y is g colouring impurity/kg impurity free solution and X is g colouring impurity/kg pure activated carbon.

Calculate the amount of activated carbon required per 1000 kg of impurity free solution to reduce the impurity concentration from 1.2 to 0.2 g/kg of impurity free solution using (i) a single stage operation and (ii) a two-stage cross-current operation with intermediate composition of 0.5 g. of colouring impurity per kg of impurity free solution.

$$Y = 0.004X^2$$

Solution.

Feed, G_S = 1000 kg of impurity free solution

(i) Y_0 = 1.2 g/kg of impurity free solution

Y_1 = 0.2 g/kg of impurity free solution

X_0 = 0

$$X_1 = \left(\frac{Y_1}{0.004}\right)^{0.5} = \left(\frac{0.2}{0.004}\right)^{0.5} = 7.07$$

$$-\frac{L_S}{G_S} = +\frac{(Y_0 - Y_1)}{(X_0 - X_1)} = +\frac{(1.2-0.2)}{(0-7.07)} = -\frac{1}{7.07} = -0.1414$$

$\therefore$ $L_S = 0.1414 \times 1000 = 141.4$ kg of adsorbent

(ii) Intermediate colour concentration is 0.5 g/kg of impurity free solution

$\therefore$ $$X_1 = \left(\frac{0.5}{0.004}\right)^{0.5} = 11.18$$

$\therefore$ $$-\left(\frac{L_S}{G_S}\right)_1 = \frac{(1.2-0.5)}{(0-11.18)} = -\frac{0.7}{11.18} = -0.06261$$

$$X_2 = X_{\text{final}} = 7.07$$

$$-\left(\frac{L_S}{G_S}\right)_2 = \frac{(0.5-0.2)}{(0-7.07)} = -0.04243$$

$$\therefore \qquad \left(\frac{L_S}{G_S}\right)_{\text{total}} = 0.06261 + 0.04243 = 0.10504$$

The adsorbent needed, L_S = 105.04 kg of adsorbent.

4. A solution of washed raw cane sugar of 48% sucrose by weight is coloured by the presence of small quantities of impurities. It is to be decolourised at 80°C by treatment with an adsorptive carbon in a contact filtration plant. The data for an equilibrium adsorption isotherm were obtained by adding various amounts of the carbon to separate batches of the original solution and observing the equilibrium colour reached in each case. The data with the quantity of carbon expressed on the basis of the sugar content of the solution are as follows:

$\frac{\text{kg carbon}}{\text{kg dry sugar}}$	0	0.005	0.01	0.015	0.02	0.03
% colour removed	0	47	70	83	90	95

The original solution has a colour concentration of 20 measured on an arbitrary scale and it is desired to reduce the colour to 2.5% of its original value.

(i) Convert the equilibrium data to Y and X.

(ii) Calculate the amount of carbon required for a single stage process for a feed of 1000 kg solution.

(iii) Estimate the amount of carbon needed for a feed of 1000 kg solution in a two-stage countercurrent process.

Solution.

Feed solution contains 48% sucrose.

$\frac{\text{kg carbon}}{\text{kg dry sugar}}$	0	0.005	0.01	0.015	0.02	0.03
% colour removed	0	47	70	83	90	95
$\frac{\text{kg carbon}}{\text{kg dry solution}}$	0	0.0024	0.0048	0.0072	0.0096	0.0144
$Y, \frac{\text{colour}}{\text{kg of solution}}$	20	$0.53\times20 = 10.6$	$0.30\times20 = 6$	$0.17\times20 = 3.4$	$0.10\times20 = 2.0$	$0.05\times20 = 1.0$
$X, \frac{\text{colour}}{\text{kg carbon}}$	–	$\frac{(20-10.6)}{0.0024} = 3916.7$	$\frac{(20-6)}{0.0048} = 2916.7$	$\frac{(20-3.4)}{0.0072} = 2305.6$	$\frac{(20-2)}{0.0096} = 1875$	$\frac{(20-1)}{0.0144} = 1319.4$

Feed is $(X_0, Y_0) = (0, 20)$

Final product is to have 2.5% original colour, i.e. 0.5 units = Y_1

$$\therefore \quad -\left(\frac{L_S}{G_S}\right) = \frac{(20-0.5)}{(0-100)} = -\frac{19.5}{1000} = -0.0195$$

$$\therefore \quad L_S = G_S \times 0.0195 = 19.5 \text{ kg}$$

(ii) The operating line is fixed by trial and error for exactly two stages.

$$\left(\frac{L_S}{G_S}\right) = \frac{19.5}{3175} = 6.142 \times 10^{-3}$$

$$\therefore \quad L_S = 6.142 \text{ kg}$$

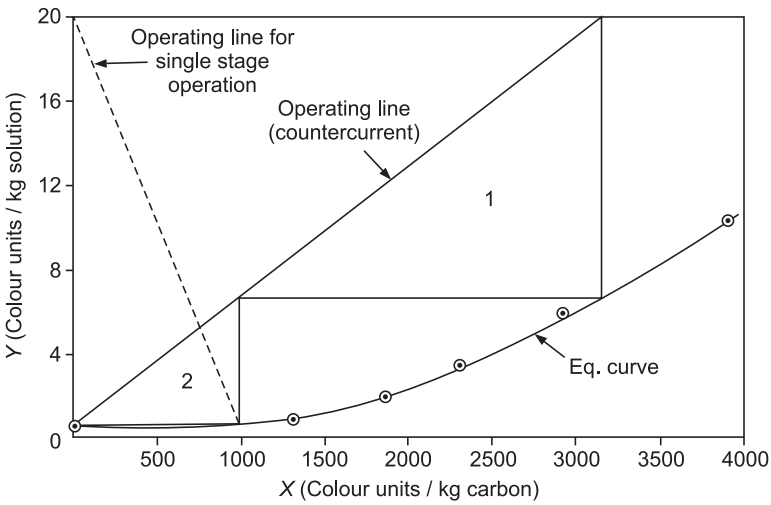


Fig. 12.22 Example 4.

5. NO_2 produced by a thermal process for fixation of nitrogen is to be removed from a dilute mixture with air by adsorption on silica gel in a continuous countercurrent adsorber. The gas entering at the rate of 0.126 kg/s contains 1.5% of NO_2 by volume and 90% of NO_2 is to be removed. Operation is isothermal at 25°C and 1 atm pressure. The entering gel will be free of NO_2.

Partial pressure of NO_2, mm Hg	0	2	4	6	8	10	12
kg NO_2/100 kg gel	0	0.4	0.9	1.65	2.6	3.65	4.85

(a) Calculate the minimum weight of gel required/h.

(b) For twice the minimum gel rate, calculate the number of stages required.

Solution.

Entering gas rate : 450 kg/hr = 0.126 kg/s

NO_2 present : 1.5 % by volume

Temperature : 25°C

Pressure : 1 std. atm.

Partial pressure of NO_2, mm Hg	0	2	4	6	8	10	12
kg NO_2/100 kg gel	0	0.4	0.9	1.65	2.6	3.65	4.85
Kg NO_2/kg gel, X	0	0.004	0.009	0.0165	0.026	0.0365	0.0485
(pp NO_2/pp air) × (46/28.84), Y, (kg/kg)	0	0.0042	0.00844	0.01269	0.01697	0.02127	0.0256

Y_{in} = 1.5%

$$Y_{in} = (\text{kg/kg}) = \frac{1.5}{98.5} \times \frac{46}{28.84} = 0.0243$$

$$Y_{in}\ (\text{kg/kg of mixture})\ \frac{0.0243}{1.0243} = 0.0237$$

G_S = 450 (1 – 0.0237) = 439.3 kg/h

90% of NO_2 is to be recovered

$$Y_{out} = \frac{1.5 \times 0.1}{98.5} \times \frac{46}{28.84} = 0.00243$$

$$\left(\frac{L_S}{G_S}\right)_{min} = \frac{0.025}{0.0375} = 0.667$$

Weight of absorbent required, L_S = 0.667 × 439.3 = 291.1 kg/h

Number of stages needed for twice the adsorbent rate

$$\therefore \qquad \left(\frac{L_S}{G_S}\right)_{act} = \frac{582.2}{439.3} = 1.334$$

No. of stages = 3

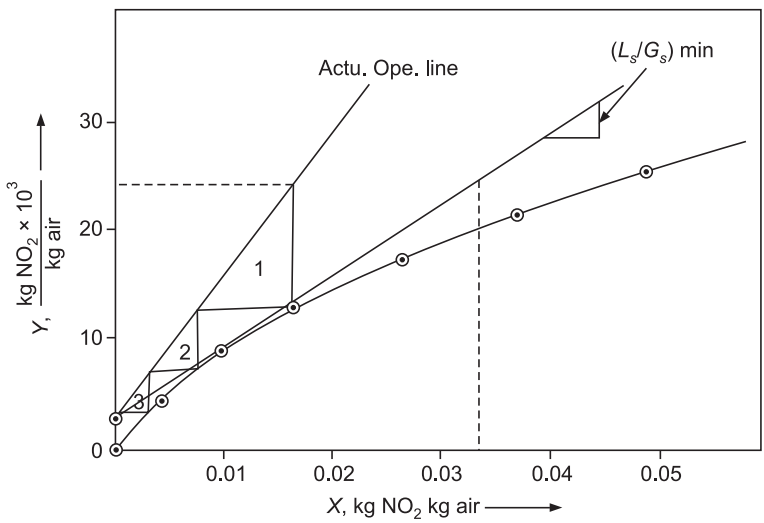


Fig. 12.23 Example 5.

6. 500 kg/min of dry air at 20°C and carrying 5 kg of water vapour/min. is to be dehumidified with silica gel to 0.001 kg of water vapour/kg of dry air. The operation has to be carried out isothermally and countercurrently with 25 kg/min. of dry silica gel. How many theoretical stages are required and what will be the water content in the silica gel leaving the last stage?

kg. of water vapour/ kg of dry silica gel, X	0	0.05	0.10	0.15	0.20
kg of water vapour/ kg of dry air, Y	0	0.0018	0.0036	0.0050	0.0062

Solution.

Quantity of dry air entering at 20°C, G_S = 500 kg/min

Quantity of water vapour entering = 5 kg/min

$$Y_1 = \frac{5}{500} = 0.01 \text{ kg water vapour/kg dry air}$$

Concentration of water vapour in leaving air, Y_2 = 0.001 kg water vapour/ kg dry air

Quantity of silica gel entering, L_S = 25 kg/min

$$X_2 = 0$$

i.e. $$\left(\frac{L_S}{G_S}\right) = \frac{25}{500} = 0.05$$

Making a material balance

$$L_S\,[X_1 - X_2] = G_S\,[Y_1 - Y_2]$$

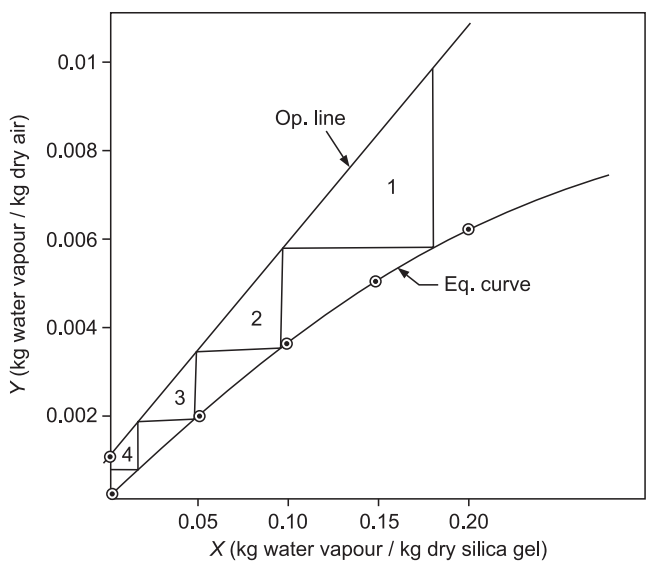


Fig. 12.24 Example 6.

$$\therefore \qquad X_1 = \frac{500 \times 0.009}{25} = 0.18$$

Total number of stages needed = 4

EXERCISES

1. The equilibrium relationship for the adsorption of colour from a carrier gas is given by $y = 0.57x^{0.5}$, where y is the gram of coloured substance removed per gram of adsorbent and x is the gram of colour/100 grams of colour free carrier. If 100 kg of the carrier containing 1 part of colour per 3 parts of total carrier is contacted with 25 kg of adsorbent, calculate the percent of colour removed by (i) single contact, (ii) two-stage cross-current contact dividing the adsorbent equally per contact.

2. Experiments on decolourisation of oil yielded the following relationship $y = 0.5x^{0.5}$, where y is the gram of colour removed per gram of adsorbent and x is the gram of colour/1000 grams of colour free oil. If 1000 kg of oil containing 1 part of colour per 3 parts of colour free oil is contacted with 250 kg of adsorbent, calculate the percent of colour removed by (i) single stage process, (ii) two-stage cross-current contact process using 125 kg adsorbent in each stage.

3. A solid adsorbent is used to remove colour impurity from an aqueous solution. The original value of colour on an arbitrary scale is 48. It is required to reduce this to 10% of its original value. Using the following data, find the number of stages needed for 1000 kg of a solution in a countercurrent operation if 1.5 times the minimum adsorbent needed is used.

Equilibrium data:

kg adsorbent/kg of solution	0	0.001	0.004	0.008	0.02	0.04
Equilibrium colour (Y)	48	43	31.5	21.5	8.5	3.5

4. An aqueous solution containing valuable solute is coloured by the presence of small amounts of impurity. It is decolourised using activated carbon adsorbent. The equilibrium relationship is $Y = 0.00009X^{1.7}$, where X = colour units/kg carbon and Y = colour units/kg solution. If 1000 kg of the original solution has 9.8 colour units/kg solution, calculate (i) the amount of colour removed by using 30 kg of adsorbent in a single stage operation and (ii) the amount of carbon needed for a two-stage countercurrent operation if the final colour in the solution is to be 10% of the original value and the solution leaving the first stage has 4 times the final colour of the solution.

5. The adsorption of moisture using silica gel varies with moisture content as follows: $Y = 0.035X^{1.05}$, where X = kg water adsorbed/kg dry gel and Y = humidity of air, kg moisture/kg dry air. 1 kg of silica gel containing 2% (dry basis) of moisture is placed in a vessel of volume 5 m^3 containing moist air. The partial pressure of water is 15 mm Hg. The total pressure and temperature are 1 atm. and 298 K respectively. What is the amount of water picked up by silica gel from the moist air in the vessel? Estimate the final partial pressure of moisture and final total pressure in the vessel.

Appendix I

IMPORTANT CONVERSION FACTORS

Quantity	*To convert*		*Multiply by*
	from	*to*	
Length	in	m	0.0254
	ft	m	0.3048
	cm	m	0.01
	Angstron	m	10^{-10}
	microns	m	10^{-6}
Area	in^2	m^2	6.452×10^{-4}
	ft^2	m^2	0.0929
	cm^2	m^2	10^{-4}
Volume	ft^3	m^3	0.02832
	cm^3	m^3	10^{-6}
	liter	m^3	10^{-3}
	Gallons (UK)	m^3	4.546×10^{-3}
	Gallons (US)	m^3	3.285×10^{-3}
Mass	Pounds (lb)	kg	0.4536
	Gram	kg	10^{-3}
Density	lb/ft^3	kg/m^3	16.019
	g/lit	kg/m^3	1.0
	g/cm^3	kg/m^3	1000
Force	lb_f	N	4.448
	kg_f	N	9.807
	Pa	N	980.7
	dyn	N	10^{-5}
Pressure	lb_f/ft^2	N/m^2 = Pa	47.88
	lb_f/in^2 (psi)	N/m^2 = Pa	6895

(*Contd.*)

(*Contd.*)

Quantity	*To convert*		*Multiply by*
	from	*to*	
	in Hg	$N/m^2 = Pa$	3386
	in water	$N/m^2 = Pa$	249.1
	mm Hg	$N/m^2 = Pa$	133.3
	atm	$N/m^2 = Pa$	1.0133×10^5
	torr	$N/m^2 = Pa$	133.3
	bar	$N/m^2 = Pa$	10^5
	kg_f/cm^2	$N/m^2 = Pa$	9.807×10^4
Heat or Energy	Btu	$J = N \cdot m$	1055
	erg	$J = N \cdot m$	10^{-7}
	cal	$J = N \cdot m$	4.187
	kcal	$J = N \cdot m$	4187
	kW·h	$J = N \cdot m$	3.6×10^6
Volumetric flow rate	ft^3/s	m^3/s	0.02832
	ft^3/h	m^3/s	7.867×10^{-6}
	cm^3/s	m^3/s	10^{-6}
	Lit/h	m^3/s	2.777×10^{-7}
Mass flow rate (Mass flux)	$lb/ft^2 \cdot h$	kg/m^2s	1.356×10^{-3}
	g/cm^2s	kg/m^2s	10
Molar flow rate (Molar flux)	$lb\ mol/ft^2 \cdot h$	$k\ mol/m^2s$	1.356×10^{-3}
	$g\ mol/cm^2s$	$kmolg/m^2s$	10
Enthalpy	Btu/lb	$J/kg = N \cdot m/kg$	2326
	cal/g = kcal/kg	$J/kg = N \cdot m/kg$	4187
Heat capacity (Holds good for molal heat capacity also)	Btu/lb. °F	$N \cdot m/kg\ K = J/kg\ K$	4187
	cal/g.°C	$N \cdot m/kg\ K = J/kg\ K$	4187

Appendix II

ATOMIC WEIGHTS AND ATOMIC NUMBERS OF ELEMENTS

Element	*Symbol*	*Atomic Number*	*Atomic weight*
Actinium	Ac	89	227.00
Aluminum	Al	13	26.98
Americium	Am	95	243.00
Antimony	Sb	51	121.76
Argon	A	18	39.94
Arsenic	As	33	74.91
Astatine	At	85	210.00
Barium	Ba	56	137.36
Berkelium	Bk	97	245.00
Beryllium	Be	4	9.01
Bismuth	Bi	83	209.00
Boron	B	5	10.82
Bromine	Br	35	79.92
Cadmium	Cd	48	112.41
Calcium	Ca	20	40.08
Californium	Cf	98	246.00
Carbon	C	6	12.01
Cerium	Ce	58	140.13
Cesium	Cs	55	132.91
Chlorine	Cl	17	35.46
Chromium	Cr	24	52.01
Cobalt	Co	27	58.94
Columbium	Nb	41	92.91
Copper	Cu	29	63.54
Curium	Cm	96	243.00
Dysprosium	Dy	66	162.46
Erbium	Er	68	167.20

(*Contd.*)

(*Contd.*)

Element	*Symbol*	*Atomic Number*	*Atomic weight*
Europium	Eu	63	152.00
Fluorine	F	9	19.00
Francium	Fr	87	223.00
Gadolinium	Gd	64	156.90
Gallium	Ga	31	69.72
Germanium	Ge	32	72.60
Gold	Au	79	197.20
Hafnium	Hf	72	178.60
Helium	He	2	4.00
Holmium	Ho	67	164.94
Hydrogen	H	1	1.00
Indium	In	49	114.76
Iodine	I	53	126.91
Iridium	Ir	77	193.10
Iron	Fe	26	55.85
Krypton	Kr	36	83.80
Lanthanum	La	57	138.92
Lead	Pb	82	207.21
Lithium	Li	3	6.94
Lutetium	Lu	71	174.99
Magnesium	Mg	12	24.32
Manganese	Mn	25	54.93
Mercury	Hg	80	200.61
Molybdenum	Mo	42	95.95
Neodymium	Nd	60	144.27
Neptunium	Np	93	237.00
Neon	Ne	10	20.18
Nickel	Ni	28	58.69
Niobium	Nb	41	92.91
Nitrogen	N	7	14.01
Osmium	Os	76	190.20
Oxygen	O	8	16.00
Palladium	Pd	46	106.70
Phosphorus	P	15	30.98
Platinum	Pt	78	195.23
Plutonium	Pu	94	242.00
Polonium	Po	84	210.00
Potassium	K	19	39.10
Praseodymium	Pr	59	140.92
Promethium	Pm	61	145.00
Protactinium	Pa	91	231.00
Radium	Ra	88	226.05
Radon	Rn	86	222.00
Rhenium	Re	75	186.31
Rhodium	Rh	45	102.91
Rubidium	Rb	37	85.48

(*Contd.*)

(*Contd.*)

Element	*Symbol*	*Atomic Number*	*Atomic weight*
Ruthenium	Ru	44	101.70
Samarium	Sm	62	150.43
Scandium	Sc	21	44.96
Selenium	Se	34	78.96
Silicon	Si	14	28.09
Silver	Ag	47	107.88
Sodium	Na	11	23.00
Strontium	Sr	38	87.63
Sulphur	S	16	32.07
Tantalum	Ta	73	180.88
Technetium	Tc	43	99.00
Tellurium	Te	52	127.61
Terbium	Tb	65	159.20
Thallium	Tl	81	204.39
Thorium	Th	90	232.12
Thulium	Tm	69	169.40
Tin	Sn	50	118.70
Titanium	Ti	22	47.90
Tungsten	W	74	183.92
Uranium	U	92	238.07
Vanadium	V	23	50.95
Xenon	Xe	54	131.30
Ytterbium	Yb	70	173.04
Yttrium	Y	39	88.92
Zinc	Zn	30	65.38
Zirconium	Zr	40	91.22

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Mass Transfer
Theory and Practice

N. Anantharaman
K.M. Meera Sheriffa Begum

Mass transfer operations are of great importance in a process industry as it has a direct impact on the cost of the final product. A chemical/process engineer therefore should have sound knowledge of the basics of mass transfer and its applications. This book is designed to equip the reader with sufficient knowledge of mass transfer operations and face the challenges ahead.

The objective of this textbook is to teach a budding chemical engineer the principles involved in analyzing a process and apply the desired mass transfer operation to separate the components involved. It deals with operations involving diffusion, interphase mass transfer, humidification, drying, crystallization, absorption, distillation, extraction, leaching and adsorption. The principles and equipment used for different mass transfer operations have been lucidly explained.

Designed for a two-semester course, this text is primarily intended for the undergraduate students of chemical, pharmaceutical, petrochemical engineering as well as biotechnology and industrial biotechnology. It will also be useful to plant engineers and design professionals.

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2. Illustrates the application of theory through worked-out numerical examples.
3. Provides exercise problems with answers at the end of each chapter for practice.

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